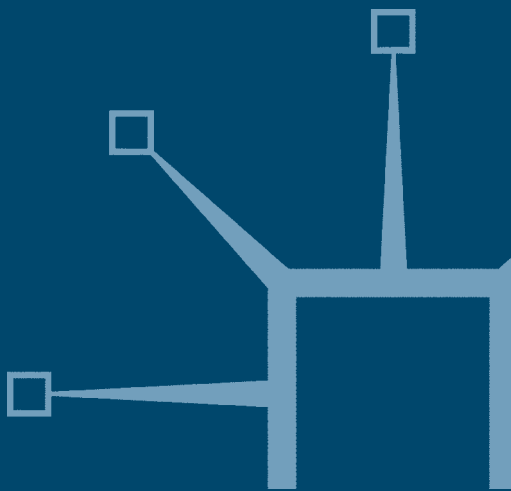


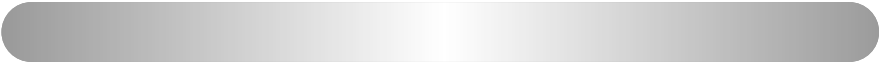
QUANTITATIVE PORTFOLIO OPTIMISATION, ASSET ALLOCATION AND RISK MANAGEMENT

Mikkel Rasmussen



Quantitative Portfolio Optimisation, Asset Allocation and Risk Management

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Mikkel Rasmussen

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PART I

A BASIS FOR QUANTITATIVE
MANAGEMENT AND
ANALYSIS

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ASSET MANAGEMENT BASICS

INTRODUCTION

The basis of any investment is the desire to obtain a return on that investment. Since there is no such thing as a free lunch, the investor or asset manager must accept some amount of risk in order to obtain the return. In other words, the *risk* taken on by the investor is the price paid for the opportunity for a positive return, and the desired level of return thus determines the exact amount of risk taken on by the investor. This is a fundamental investment relationship, which investors must consider when deciding whether to invest in either a single asset or a portfolio of assets.

So what does it take to be a successful asset manager? It is really quite simple. The manager must be *knowledgeable*, *decisive*, and *right*. This obviously looks true on paper, but is hard to achieve in the real world, and the track record for the active management industry speaks for itself. On average, approximately only half of all active equity portfolio managers outperform their benchmark, and the number of managers who outperform consistently is so low that it is hardly statistically significant.

Investment return and risk can be modelled quantitatively on both the asset and the portfolio levels. The modelling of risk in particular is inherently complex, since in reality risk is not just a single number but rather an infinite number of possible future states, each with its own implications for the investors' expected cash flow. Therefore we first need to establish what an asset *is*, and what characterises its return and risk. The same must be done for portfolios. Having done this, we can then move on to considering optimal portfolios with respect to return and risk, and eventually to optimal allocations over the entire risk spectrum.

ASSET MANAGEMENT OBJECTIVES

Asset or portfolio managers are hired to ensure the achievement of the financial goals of their clients, whatever those may be. A portfolio management contract is basically a principal–agent relationship, whereby the *principal* (the client) establishes a set of guidelines (the investment mandate) for the management of his or her assets, which the *agent* (the portfolio manager) then follows. This involves rules and guidelines for the management of assets, but also requires the establishment of adequate *incentives* for the agent in order that the agent will manage the assets properly and in accordance with client needs.

The objective of the portfolio manager is basically to generate a cash flow that matches the desired cash flow of each client as closely as possible. This implies that, contrary to popular belief, the objective of the manager is not necessarily constrained to the maximisation of return on assets. The objective is more detailed than that as it relates return to risk over an agreed period of time. The client (principal) may be a wealthy individual desiring a higher return on her assets and thus wanting to establish a portfolio of risky assets; or the client may be an elderly person wishing to invest in government bonds for his upcoming retirement; or the principal may be an institutional investor with many differing preferences for the management of different asset classes. A portfolio management contract thus needs to take into account all the factors that are important to the client in terms of preferences for portfolio return, risk, and distribution of cash flows over time.

THE CASE FOR QUANTITATIVE MANAGEMENT

Information overload

In order to generate the return required by the client, the asset manager needs a structured approach to investing. Traditionally, when selecting stocks to add to a portfolio, portfolio managers have focused heavily on company fundamentals. Managers typically focus on a limited part of the investable universe and then try to pick out winners – stocks that will perform ‘well’ over the medium to long term. They do this by spending their time examining financial statements, reading analysts’ research reports, investigating and evaluating management capabilities, evaluating products and markets, and so on. All this work must necessarily be carried out on a relative basis, so managers also need to examine whole sectors in order to be able to compare a company with its peers.

Since there are literally tens of thousands of securities to choose from, it seems quite intuitive that managers should focus on a limited number of stocks. This leads to an obvious conclusion: managers cannot possibly keep informed

about every single asset in the investable universe all the time. Even if the manager is handed a mandate to focus on US large capitalisation stocks, chances are that she will only be able to follow about a hundred of these properly, which may be 20 per cent of the total number of eligible stocks. Obviously this is inefficient, and the basic problem is the inability of a single human mind to grasp, interpret and correlate such vast amounts of information in a structured way. A manager can only consider a limited number of variables at any given time, and in addition the information received is quickly outdated. It follows that such an approach may limit the efficiency of the asset management process, possibly resulting in managers omitting stocks from the portfolio that would have helped to provide superior (risk-adjusted) portfolio returns.

Consider this simple example: suppose that we have a portfolio manager who is responsible for US large cap stocks. He is now given a mandate to create a portfolio of 25 large cap stocks. Assuming that he is actually able to form an opinion on all 500 eligible stocks and add 25 to his portfolio, two questions still remain. What weights should be assigned to each stock in the portfolio, and how is this to be determined? If the client has specified some preference in terms of sector exposure, possibly relative to a predetermined benchmark, how should the sectors be weighted relative to each other and relative to the benchmark?

Intuitively, the performance of a portfolio depends on the expected return and risk of each asset, and also on the correlation structure among asset or asset class (sector) returns. It is thus clear that even if the manager were able to form an opinion on each individual stock based on its particular earnings forecasts, balance sheet and management quality, products and markets, and even if he could correctly estimate expected return and risk for each stock, it would simply be impossible for him to grasp the correlation structure among the stocks (or sectors) and to create a portfolio that exploited these asset characteristics properly. This multidimensional structure is inherently too complex for a human mind to comprehend and work with.

This simple argument about information overload, which is a very real problem for portfolio managers, clearly suggests that some other form of decision-making tool might usefully be added to the manager's toolbox. Specifically, the vast amount of complex and ever-changing information being provided to the manager suggests that a more quantitative approach may be warranted, at least in some aspects of the asset management and investment decision making process.

Human nature and behavioural finance

In addition to the natural limits to the capabilities of the human brain in terms of memory and information processing, we also suffer from cognitive biases

when presented with a set of information. For example, we tend to overvalue recent information relative to older information, even though this may not be warranted. We also tend to be overconfident, remembering our successes better than our failures. Humans are apt to engage in herding behaviour, where following the leader becomes a goal in itself, for fear of performing worse than the competition. Biases in evaluations of companies are not easily erased, resulting in managers sticking with bad stocks for far longer than is justified, sometimes because of the recommendations of analysts who suffer from the same problems. Reluctance to realise losses made on these stocks, in the hope that what had been lost will surely be regained, accentuates this tendency. The list of behavioural caveats, of which we are often unaware, goes on. Such cognitive biases are being investigated, and this area of research is, not surprisingly, called 'behavioural finance'.¹ We will not deal with these research findings explicitly in this book, but merely use their existence here as a further justification for an increased emphasis on a quantitative approach to asset management in general, and to optimal asset allocation in particular.

Discipline

Finally, most traditional asset managers are almost solely concerned with return on assets: they do not attempt to control risk in a structured fashion. Often risk is treated as a sometimes unfortunate residual of the returns generated. This approach yields large variations in returns over time (higher risk), but average returns that are not correspondingly higher, as one would expect them to be. Again, quantitative analysis can help in designing portfolios that control risk in a systematic and structured fashion.

Thus the basic premise of this book is an attempt to provide a structured quantitative approach to portfolio management and asset allocation, as well as to portfolio risk management.

STRUCTURE OF THIS BOOK

This book is divided into four parts, each focusing on integral aspects of the asset management process. Most chapters are supplemented with extensive examples, enabling readers to perform similar exercises themselves. When possible these examples are continuously developed through each chapter in an attempt to give the reader a more coherent reading experience. Parts and chapters are designed with a steadily increasing level of complexity, ranging from very simple to highly complex. The aim of the book is to present a comprehensive quantitative approach to asset and portfolio characterisation,

portfolio optimisation, quantitative asset allocation and portfolio risk management, all within a unified framework.

Part 1 establishes a framework for quantitative investment analysis, with a bias towards asset allocation and risk management. In this section we define such fundamental building blocks as investment return and risk. These building blocks are essential to more advanced portfolio analysis. We have also included a brief chapter on asset pricing, since the ability to price an asset is necessary in order to determine whether this asset presents a good investment opportunity, with an adequate level of expected return relative to an acceptable level of risk.

In Part 2 we use our newly created building blocks to establish a unified portfolio analysis tool centred on mean-variance analysis. We begin by demonstrating how to characterise a portfolio of assets in terms of return and risk, going into detail about portfolio return and risk as well as examining closely the correlation structure of the portfolio – the extent to which the returns on individual assets or asset classes in the portfolio are correlated with each other. We use this information to demonstrate how risk can be diversified and reduced using quantitative optimisation. We go into some mathematical detail on this subject, as it is essential to the topics that will be covered in the remainder of this book. We then introduce the concept of quantitative asset allocation based on portfolio optimisation. Finally, having established a framework for quantitative optimisation of portfolios – that is, constructing portfolios that have optimal levels of return for a given level of risk – we discuss how to estimate model parameters. The process of estimation of inputs for a quantitative model, particularly one that employs optimisation, can turn out to be quite decisive.

Part 3 is perhaps the most interesting, since it extends the traditional mean-variance approach to asset allocation into one that includes more sophisticated techniques such as *Monte Carlo Simulation*. After a brief discussion of investment objectives, we move on to discuss various ways of constructing portfolios, and specifically why there are benefits to applying quantitative techniques to this process. We introduce a *Global Sector Asset Allocation Model* using Morgan Stanley's MSCI Equity indices; this will serve as a point of reference, as well as our main analysis tool, for the remainder of this book. We then go into more detail about the nature of Monte Carlo Simulation and how it can be applied to the asset allocation process. We also discuss extensively how one may usefully adjust the model inputs for greater reliability of results. Having established the quantitative asset allocation framework employing Monte Carlo Simulation in an absolute return context, we take this one step further and discuss strategic versus tactical asset allocation by adapting the model to handle benchmark-relative investment requirements. Finally we apply the asset allocation technique to a *Sector Rotation Model* that attempts to construct

suitable asset allocations for each stage in the business cycle, taking into account changing levels of returns, risks and correlations between sectors.

Part 4 focuses on risk management and the integration of this into the investment process. We begin by defining and discussing *tracking error*, perhaps one of the most common measures of risk, apart from simple volatility measures. We then move on to establishing a *Global Sector Risk Model* which builds on the asset allocation model presented earlier in the book. This model allows us to characterise the risk inherent in the portfolio, in terms both of absolute and of benchmark-relative levels of risk, but more importantly it enables us to determine the effect on portfolio risk of prospective changes to the portfolio. This is potentially a very powerful tool for asset managers who attempt to include in the investment decision-making process expectations about both return and risk. We then move on to discuss the concept of *Value-at-Risk*, which has been around for many years but has yet to establish itself as a mainstream risk-management tool. As an extension of this discussion we also examine the concept of *Extreme Value Theory*, which is essentially a refinement of Value-at-Risk, and we argue why tracking error, Value-at-Risk and Extreme Value Theory are best employed together as parts of a coherent risk-management framework.

Note

- 1 A large body of research is continuously developing on this subject.

ASSET RETURNS

DEFINING INVESTMENT RETURNS

In order to properly evaluate investments we need a measure of return on those investments. These asset returns also form the basis for all other calculations to follow in this book, in relation to assets and portfolios. We are thus not interested in the asset prices as such, but rather in the *returns* on those assets; and later on we shall turn our attention to how *risky* those returns are – that is, how much they vary over time.

The return calculations presented here are very simple and apply both to assets and to portfolios of assets. As we shall see later on, the calculation of risk differs from the asset level to the portfolio level, but for now we can safely assume that the calculations presented below are valid for single assets as well as for asset classes and portfolios. We use the *geometric return calculation* method (explained below), since this is the most intuitively appealing method for the multi-period investment framework we need to establish. This is due to the fact that this method implicitly assumes automatic reinvestment of asset returns over time.

In this book we shall focus primarily on equity return and risk, on both the stock and market levels. We do not explicitly consider the effects of dividends and stock splits, since these only make the calculations more complicated and serve no particular purpose. In an efficient market dividends will be reflected accurately in asset prices, so whether we explicitly include them or not has no effect on the end result. Furthermore there is no case for integrating these events if we simply assume that they have no real impact on the value of the stock, which in perfectly efficient markets they ought not to have. The

assumptions imply that investors would not be concerned whether dividends were paid out or reinvested in the company, and so there would be no effect on the company's stock. Stock splits do not by themselves serve any real business purpose, but may have an impact on the stock price due to certain signalling or small-investor effects. Again in this book we choose to ignore such effects since they have no material influence on our results.¹

Calculating asset returns – one period

The simple one-period return on an asset is measured as the increase (or decrease) in the value of the asset over a given time period. Define $r_{A,t}$ as the rate of return on an asset A in time period t , and $MV_{A,t}$ as the market value of asset A at the end of that period. The rate of return on Asset A over period t , $r_{A,t}$, is thus calculated by subtracting the market value of the asset at time $t-1$ from the market value of the asset at time t , dividing by the market value of the asset at time $t-1$, and then subtracting 1:

$$r_{A,t} = \frac{MV_{A,t}}{MV_{A,t-1}} - 1 \quad [\text{Eq. 2.1}]$$

For an asset A with a beginning price of \$87 and an ending price of \$103, the rate of return is thus:

$$r_A = \frac{103}{87} - 1 = 1.161 - 1 = 16.1\%$$

In the event that the asset (such as a stock) pays dividends or has other payments between t and $t-1$, these must naturally be included in the denominator in order to precisely reflect the increase in value. The method of return calculation presented in Equation 2.1 is the method that will be employed throughout this book.²

Calculating asset returns – multiple periods

Calculating returns on assets over more than one period requires the use of geometric compounding. The geometric return calculations imply a continuous compounding of returns over time, so the return on asset A over n time periods is thus:

$$r_{A,t,T} = \prod_{n=1}^T \frac{MV_{A,t}}{MV_{A,t-1}} - 1 \quad [\text{Eq. 2.2}]$$

T denotes the total number of periods over which the compounded return is calculated. With an initial market value of asset A of \$87, and three subsequent periods with values of \$91, \$96 and \$101 respectively, we obtain a compounded return of:

$$\begin{aligned} r_{A,n,3} &= \frac{MV_{A,1}}{MV_{A,0}} \cdot \frac{MV_{A,2}}{MV_{A,1}} \cdot \frac{MV_{A,3}}{MV_{A,2}} \\ &= \left(\frac{91}{87} \cdot \frac{96}{91} \cdot \frac{101}{96} \right) - 1 = \frac{101}{87} - 1 \\ &= (1.046 \cdot 1.055 \cdot 1.052) - 1 = 16.1\% \end{aligned}$$

Note how the return over the entire period can be calculated by simply using the first and the last observation, and also that the total compounded return over the three periods can be decomposed into its constituent one-period component returns. Illustrated in a chart, the individual one-period returns and the total geometrically compounded return over the three periods appears as in Figure 2.1.

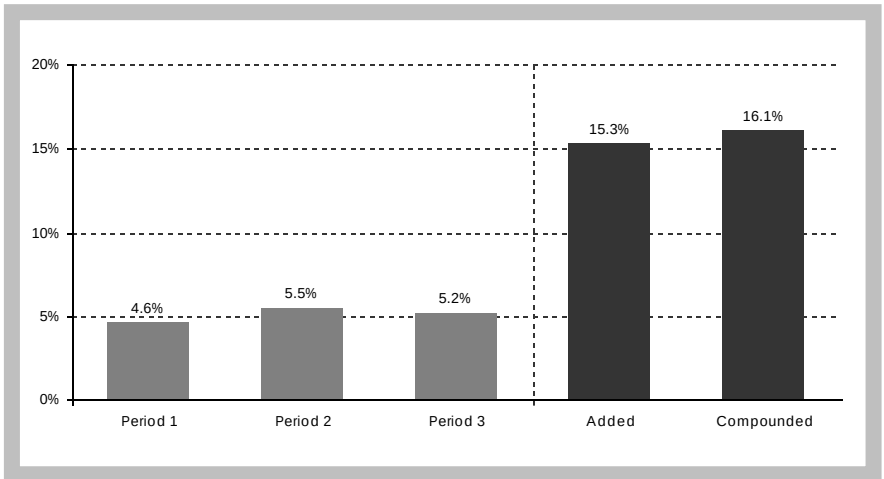


Figure 2.1 The compounding effect

Note that the simple sum of the individual returns is 15.3%, which is less than the actually observed total return over the three periods. This is the compounding effect at work, and it operates under the implicit assumption that returns on investments are reinvested after each period. So what happens is that the initial \$87 dollars grew by 4.6% to \$91, which then grew by 5.5% to \$96, which finally grew by 5.2% to \$101.

EXAMPLES FROM THE REAL WORLD

Different assets and asset classes have widely varying returns, and the degree to which their prices vary over time (the assets risk) also varies. This will become apparent in the following chapter. In order to give an indication of the average size of stock returns over time, we examine three of the most widely published equity market indices in the world: the Dow Jones Industrial Average, the S&P500 Index and the NASDAQ Composite Index. We examine how these indices performed from January 1995 to April 2002, thereby illustrating an entire economic cycle. Common wisdom has it that a complete economic cycle often lasts around seven years, and this seems to be the case for the latest cycle. Please note that this period is hardly representative of equity markets in general, since it included what has since been widely termed an IT-bubble. Furthermore, all three indices are from the United States, so they are obviously quite highly correlated.

We begin by looking at how the indices performed between 1995 and 2002 – see Figure 2.2. As is quite evident from this chart, the US equity market was highly risky during these seven years.³ From late 1998 the tech-bubble started to emerge as investors eyed what seemed to be endless opportunities and profits in the information technology sector, thus bidding up tremendously the stocks of the companies exposed to this perceived virtual goldmine. At one point the NASDAQ Composite had outperformed both the Dow Jones and the S&P500 by several hundred per cent, but in early 2002 all three indices actually stood at levels last seen some four years earlier in mid 1998. In essence,

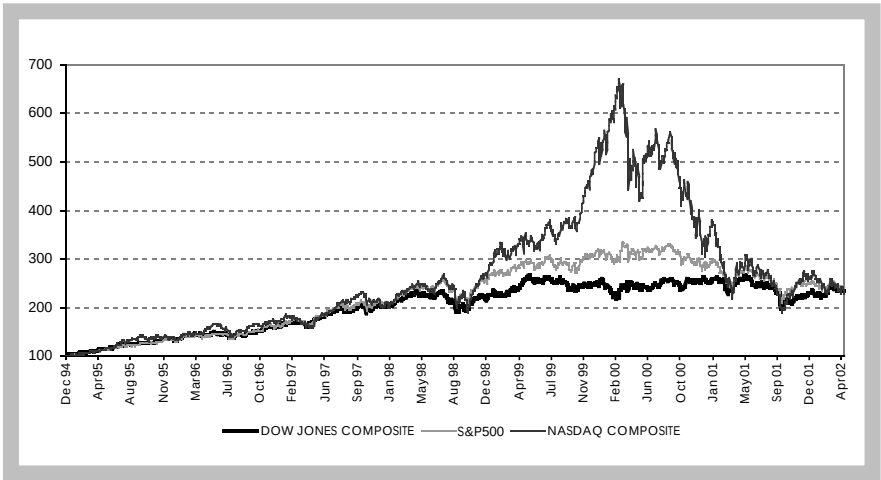


Figure 2.2 Performance of the DJIA, the S&P500 and the NASDAQ Composite Index, 1995/1–2002/4 (daily observations)

investors who might have bought a US index fund in 1998 would not be any wealthier four years later.

There are many explanations of why it went this way, but at least to some extent these can be summed up by saying that it was an unfortunate combination of bad production and inventory management on the part of technology manufacturers, a cyclical downturn in the global economy that materialised at the peak of the equity market bubble, and 'irrational exuberance' on the part of equity market analysts and investors. In short, the technology revolution did actually happen more or less the way and at the speed that most people had been predicting. But apparently people had forgotten that all products become low profit margin commodities after a while, and by the time the technology revolution was well under way and penetration rates for many of the new devices such as mobile phones and PDAs (Personal Digital Assistants) were as high as people had hoped they would be, most of these products had become commodities not very different from radios and TVs.

In the following chapters we are going to use the data from world equity markets to examine asset and portfolio risk in an asset allocation context. It might be argued that using data from such a turbulent and speculative period with such abnormal levels of returns might end up producing misleading results. However, we are not particularly interested in the levels of equity market returns, but rather in the relative levels of *risk* of these markets, as well as the extent to which these returns have been correlated over time.

We can examine the returns on the indices over time by plotting daily returns, as calculated using Equation 2.1. For the Dow Jones Industrial Average these appear in Figure 2.3. It is evident that most monthly returns lie in the +2% to -2% range, but there are a few returns that are much larger

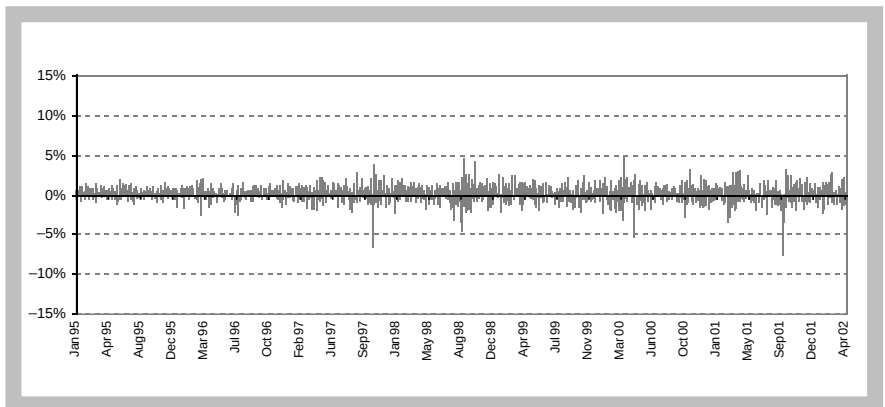


Figure 2.3 Daily returns on the Dow Jones Industrial Average, 1995/1–2002/4

than average (positive or negative). Most recently the events on 11 September 2001 sent shock waves around the world, including the financial markets. Note however that this large negative return was followed by a series of large positive returns. We have deliberately chosen a scale – on the left-hand side of the chart – that exceeds the largest return values (positive and negative), in order to demonstrate the significant differences in return variation of the three indices.

A similar chart for the S&P500 appears in Figure 2.4. Note that the basic return pattern is the same, but the degree to which returns vary over time is higher for the S&P500 than for the Dow Jones Industrial.

Finally we look at the NASDAQ Composite (Figure 2.5). Again we see that the returns follow roughly the same pattern over time, but for the NASDAQ

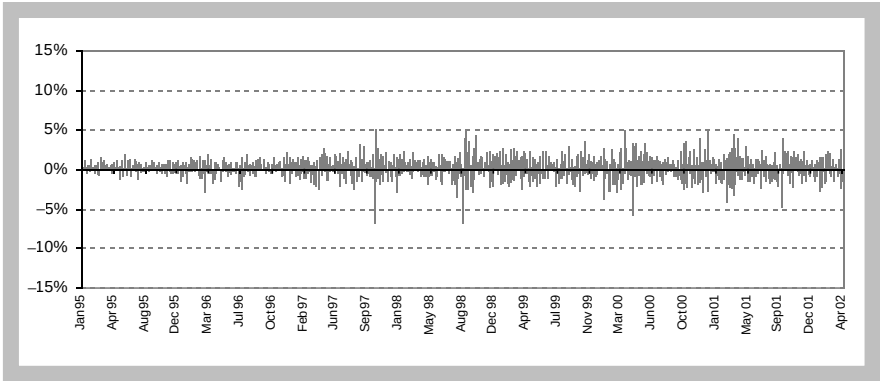


Figure 2.4 Daily returns on the S&P500, 1995/1–2002/4

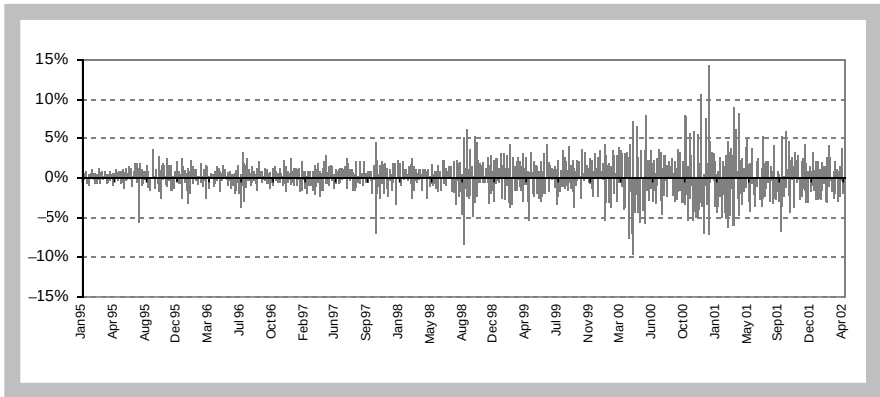


Figure 2.5 Daily returns on the NASDAQ Composite, 1995/1–2002/4

Composite they are generally much higher than is the case for either the Dow Jones Industrial Average or the S&P500.

Figures 2.3–2.5 go a long way to demonstrate what we mean when we talk about asset – and in this case stock market – returns. The variation in returns over time is essentially what we mean when we talk about *asset risk*, but we shall examine this in a quantitative fashion in the following chapter.

The charts presented in Figures 2.3–2.5 are interesting when one wants to gauge the average size and variation of daily equity market returns, but we can obtain a better sense of the magnitude of the compounded rates of returns over time by simply annualising the daily returns on all three indices from January 1995 to December 2001. This is illustrated in Figure 2.6.

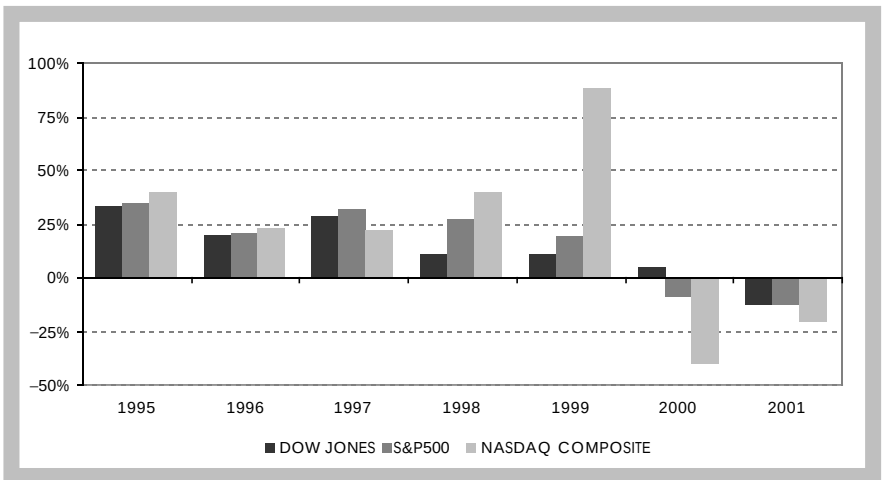


Figure 2.6 Annualised returns on the Dow Jones Industrial Average, S&P500 Index and NASDAQ Composite Index, 1995–2001

On average, stocks have done very well indeed during the 1990s. Interestingly, however, none of the indices has delivered positive returns every single year, but the years in which negative or low returns materialised were not necessarily the same for all the indices. These differences in returns on asset classes hint at the long-run opportunities for diversification of risk, which we shall discuss extensively in subsequent chapters. In the US equity market 1995 to 1999 were good years for investors. The technology-heavy NASDAQ in particular performed extremely well in 1999, but considering that the long-term average of equities is in the 12 per cent to 15 per cent annual return range, it is probably fair to say that 1999 was the result of a bubble. The year 2000 was a horrific year for most equity markets, and for technology investors in particular, who lost some 40 per cent. Although the Dow Jones managed to end up with a small gain, the subsequent year was

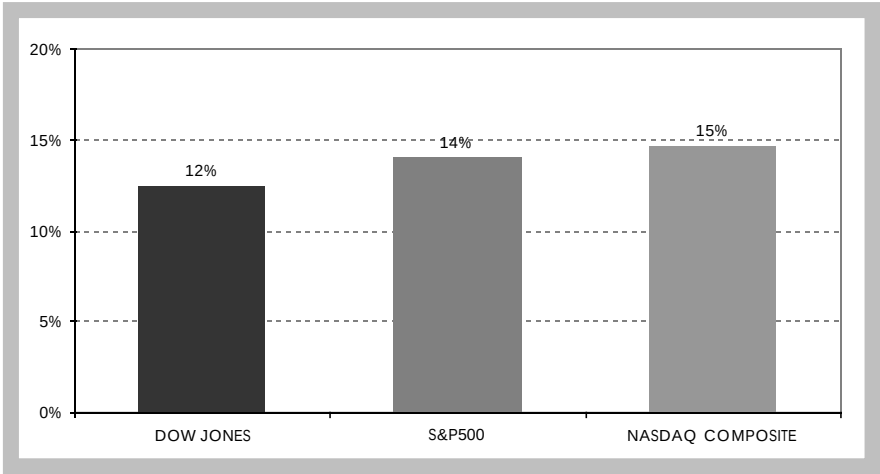


Figure 2.7 Average annual returns on the Dow Jones Industrial Average, S&P500 and the NASDAQ Composite Index, 1995–2001

bad for all three indices. The average annual returns during the period were as shown in Figure 2.7.

Although they are depicted in Figures 2.3–2.5, the first few months of 2002 are not included in Figure 2.7. So over the seven-year period the NASDAQ produced a gain slightly larger than that produced by the S&P500, and somewhat larger than that produced by the Dow Jones Industrial Average. To get a better feel for these numbers in an international context, Table 2.1 shows the returns on the three indices presented above, as well as on a range of international indices.

Interestingly, the average return of the Dow Jones Industrial Average is actually the highest during this period, whereas the worst performance is exhibited by the Hang Seng. However, considering the number of years we are examining it is remarkable how little difference there is between the annual returns over this entire period.

Figure 2.8 illustrates the performance of the four non-US indices presented in Table 2.1. We intentionally use the same scale – on the left-hand side of the chart – as for the US indices, to emphasise the marked difference in performance during 1999. The German DAX30 has consistently performed better than both the FTSE and the two Asian markets. Note how the remarkable 250+ per cent gain of the DAX30 between January 1995 and early 2000 seems small compared to the NASDAQ. Asia fared somewhat worse during this period, the Japanese market in particular being an altogether unhappy story.

Examining Figure 2.8 and Table 2.1 more closely over the whole period, we see that even though returns seem to some extent to move up and down

Table 2.1 Index returns, 1995–2001

Index	1995	1996	1997	1998	1999	2000	2001	Average
Dow Jones	33%	20%	29%	10%	10%	5%	–13%	–13%
S&P500	34%	20%	31%	27%	18%	–9%	–13%	–13%
NASDAQ	40%	23%	22%	40%	88%	–40%	–21%	–21%
FTSE100	20%	12%	25%	15%	18%	–10%	–16%	–16%
DAX30	6%	26%	45%	16%	33%	–6%	–21%	–21%
Tokyo SE	1%	–7%	–20%	–7%	58%	–25%	–20%	–20%
Hang Seng	23%	34%	–20%	–6%	73%	–13%	–24%	–24%

in sync – both with each other, and also with the US indices – there are still significant differences between equity market returns across different markets. Historically US investors in particular have been very much focused on their own equity markets, although much less so now than a decade ago. They have thus foregone significant opportunities for diversification of portfolio risk. Throughout the remainder of this book we shall use information such as this in a quantitative way, as we explore the opportunities for asset diversification across markets and sectors at various stages.

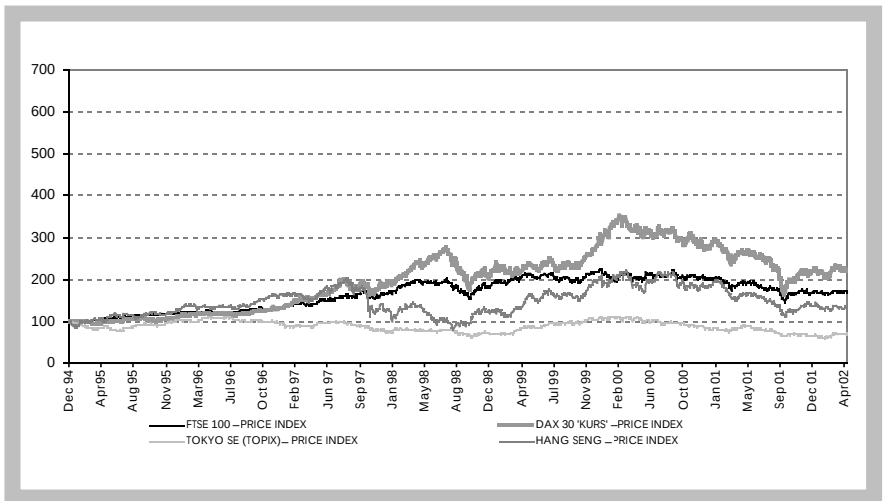


Figure 2.8 Average annual returns on the FTSE100, DAX30, Tokyo SE and Hang Seng Indices, 1995/1–2002/5

EXCESS RETURNS AND RISK-FREE RATES

In this section we briefly take a slightly more theoretical approach to asset returns and look at the so-called ‘excess returns’. The term *excess return* is used to describe the amount of investment return above the risk-free rate of return.⁴ For an asset *A* at time *t*, this excess return is defined as:

$$r_A^\circ = r_A - r_f$$

[Eq. 2.3]

where r_f is the risk-free rate of return. The risk-free rate is in turn defined as the return on a riskless asset. It is important at this stage to note that we do not explicitly consider the existence of such an asset, but simply assume that it exists. It is customary simply to use as a proxy the return on the most liquid government bond with a long duration. There are other ways of estimating the risk-free rate, but regardless of the estimation technique and the assumptions made, the investment manager should realise that risk-free rates, as calculated on the basis of actual market data, often tend to vary quite significantly from one market to the next. This is due to the fact that the real risk-free rate is to a large extent determined by the growth rate of the economy and the prevailing inflation rate. These are in turn determined by changes in productivity, labour market conditions, and changes in fiscal and monetary policies within the particular country in question. Estimated nominal risk-free rates for six countries in 1997 appear in Table 2.2.

One could argue that the risk-free rate for global portfolios ought to be calculated on the basis of every market for every good available, but this is obviously impractical – hence the convention of simply substituting the rate of return on long government bonds for the actual (but unknown) risk-free rate. In this book we simply assume that the risk-free rate is well defined and not subject to variations over time.

Table 2.2 Estimated risk-free rates of return

Country	Real GDP Growth	Expected Inflation	Nominal r_f
United States	2.9%	3.4%	6.4%
Japan	1.6%	1.0%	2.6%
Germany	2.2%	2.6%	4.9%
France	2.8%	1.7%	4.5%
United Kingdom	3.2%	2.5%	5.8%
Italy	2.0%	3.9%	6.0%

RESIDUAL/ABNORMAL RETURNS

Residual or *abnormal returns* – used in this book interchangeably – are the component of returns that is not due to systematic influences or general market movements, hence the term ‘residual’. Alternatively they can be defined as the return above that predicted by the movement of the broad stock market, hence the term ‘abnormal’.

The returns can be estimated by plotting excess market returns against excess asset returns, and then performing a simple univariate regression analysis of market excess return on asset excess return. The intercept of the regression line with the vertical axis is the abnormal return on the asset. Assuming a risk-free rate of 6% and regressing the excess returns on IBM’s stock on the excess returns on the S&P500 using monthly observations from 1995–2001, produces the plot shown in Figure 2.9. From this figure it becomes clear that, relative to the S&P500, IBM’s stock has produced abnormal returns in the order of 1.05% per month over the period from 1995/1 to 2002/4, corresponding to an annual residual or abnormal return of 13.4%. Again it must be emphasised that the period over which this analysis has been performed is relatively short and is not necessarily representative of the long-term relationship between IBM and the S&P500, and certainly no indication of what might happen in the future.

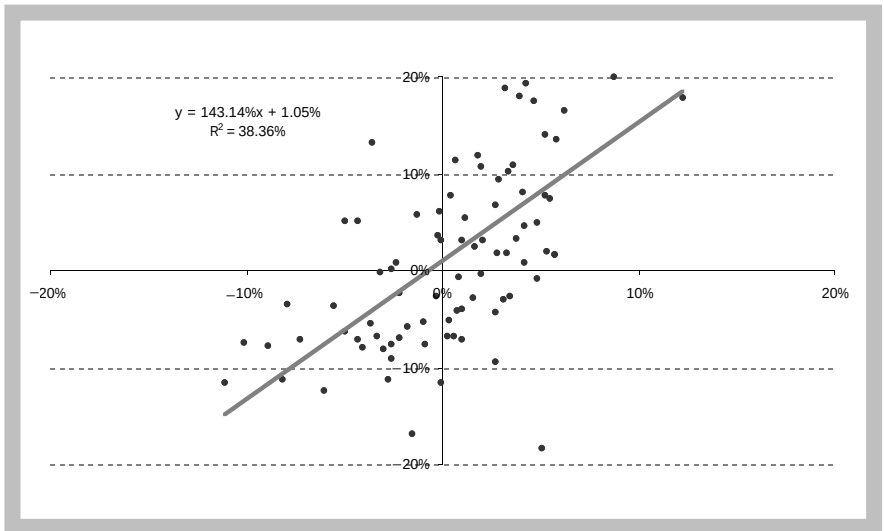


Figure 2.9 Monthly excess returns on IBM vs. the S&P500, 1995/1–2002/4

In conjunction with risk management in general and active risk management in particular, we shall take this subject up at a later stage in this book, not least because active portfolio management has as its primary goal to create abnormal or residual portfolio returns on a consistent basis, without incurring a disproportional amount of risk.

TIME-WEIGHTED RETURNS (TWR)

Another form of geometrically linked or compounded returns is *time-weighted returns* (TWR), widely used in the investment industry. TWR is not in itself a form of return but a method of calculating returns over time while attempting to ensure inter-temporal consistency. The US standard-setting body is the Association for Investment Management Research (AIMR): this body’s *GIPS* standards (*Global Investment Performance Standard*) require the calculation of time-weighted returns using quarterly data as a minimum. The calculations are furthermore required to be geometrically linked, covering interim returns between measurement periods. A specification of TWR can be found in the appendix at the end of this chapter. However, in this book we shall assume that there is no need for weighting schemes, and we shall thus focus on returns as defined in Equations 2.1 and 2.2.

SUMMARY

In order to proceed we have defined asset returns as they will appear in the remainder of this book. We use simple return calculations and geometric compounding to take account of multiple return periods. Although other calculation methods might be more precise in certain instances (excess returns and abnormal returns, as well as the TWR-methods), we have merely demonstrated these but will not apply them unless strictly necessary.

APPENDIX

Time-weighted return

TWR for an asset over T periods, with returns calculated after each transaction cash flow occurs, is given by:

$$TWR = \frac{MV_1}{C_0 + MV_0} + \frac{MV_2}{C_2 + MV_2} + \frac{MV_T}{C_{T-1} + MV_{T-1}}$$

where MV_t indicates market value, and C_t indicates cash flow for period n . Ideally, return calculations should be performed following every transaction. However this is impractical for most asset managers. Therefore the GIPS standards accept approximation methods such as the *Modified Dietz* method, whereby each cash flow is weighted by the amount of time it has been held in the portfolio. It is also assumed that within each period there exists a constant rate of return on the asset. Thus there is no need to perform a portfolio valuation each time a transaction has occurred, since each cash flow is simply assigned a weight within each period. Letting T denote the number of sub-periods and cash flows within a period, the (intra-day) Modified Dietz TWR is given by:

$$TWR_{\text{Mod.Dietz}_{0,t}} = \frac{MV_t + \sum_{t=0}^T (w_t - 1)C_t}{MV_0 + \sum_{t=0}^T w_t C_t}$$

where $w_t = \frac{t - q}{t}$ is the cash flow weight determined by the proportion of time the cash flow was in or out of the portfolio. C_t indicates the corresponding cash flow, and q denotes the time of day. If a trade occurs at midday, the weight w is 0.5, and the cash flow is deducted by 50% from the (end of day) denominator and added by 50% to the (beginning of day) numerator, since the asset traded has contributed with only half a day's worth of return on the day of purchase.

Cross overs

Given assets (long positions) that retain a positive value over time, the return calculations are straightforward, since the return over a single day is given by:

$$R = (1 + TWR_{\text{Mod.Dietz}})$$

In the case of so called *cross overs* from long to short positions (for example derivatives near their strike price), where returns are numerically negative but positive in a return sense, the most intuitive way of augmenting the performance calculations is:

$$R = \begin{cases} 1 + TWR_{\text{Mod.Dietz}}, & \forall TWR \geq 0 \\ 2 - TWR_{\text{Mod.Dietz}}, & \forall TWR < 0 \end{cases}$$

This ensures a positive return value even if the position goes short in the interim periods, and thus the principle of geometric linking can be maintained without calculation errors.

Notes

- 1 For general considerations of signalling effects, see Ross [1977], Leland & Pyle [1977], Myers & Majluf [1984] and Myers [1984].
- 2 Alternatively we could calculate logarithmic returns, but the method presented above is the most intuitive. If it is applied in a consistent manner, as it will be in this book, the geometric approach is quite adequate.
- 3 In the following chapter we shall examine the nature of asset risk more closely.
- 4 We shall discuss this further in subsequent chapters.

ASSET RISK

RISK IS NOT JUST A FOUR-LETTER WORD

Some definitions

Risk can be argued to be the result of lacking or incomplete information, since if we all had perfect foresight about everything, there would be no risk in anything. So what is the nature of risk? Let us start out by looking at a set of definitions of risk taken from various different contexts.

Risk is ...

- 'The quality or state of being in doubt, or a lack of sureness'
- 'The possibility that a future event may cause harm'
- 'A specified adverse event or consequence'
- 'Measurable uncertainty'
- 'The possible variation in outcomes'
- 'Deviations from the expected'

As the reader may have noted, the definitions presented above progress from the immeasurable to the measurable, and from qualitative to quantitative. This is of course intentional, and as we shall see we will end up using the last of the definitions presented here in a quantitative sense.

The written Chinese character for risk is a combination of the character for danger and the character for opportunities, and this is probably the best way of thinking about risk, particularly in an investment context. Risk, in a quantitative investment sense, is synonymous with variations in asset return, which can by definition be both good and bad for the investor. Any activity that has inherent dangers often has rewards to compensate for that danger. So we would expect there to be a positive relationship between risk and reward, or more precisely between risk and return. To the extent that risk is actually just uncertainty about specific events, it is clear that this uncertainty can also be an opportunity. Assuming that the uncertainty or risk is 'global' in the sense that we all have the same appreciation of future events, we can obtain an 'edge' by being able to forecast future events better than our competitors. This underscores the point made earlier: that the absence of return equates with the absence of risk; and similarly, without taking on risk we cannot obtain a return.¹

What constitutes risk and why is it important?

So how do we specify our measure of risk in a quantitative fashion? First we need to identify three factors that comprise our measure. The first factor we need to look at is *frequency* – how often is a specific event expected to happen? The second factor is the *magnitude* or *severity* of the event – what will the consequences most likely be, and to what extent can they be described in quantitative financial terms? The third and final factor is the *confidence* we have in our estimate of risk, which in turn depends on the *quality of the information* available – how certain are we about the information we have? Is the data reliable? Do we really understand it?

These three factors – frequency, magnitude and confidence – are essential to any risk analysis. Going back to the Chinese definition of risk, what is often missing here is the appreciation of the *opportunity* latent in every 'deviation from the expected'. In other words, we are often worried about potential harm or loss to the extent that we actually miss out on an opportunity for gain.

The desire to introduce structured risk analysis tools into the asset management process is solely due to the fact that we simply do not know with any significant degree of certainty the events of tomorrow or next year. Therefore, we need to be able to *estimate*, *assess* and *manage* risk within the asset management process, using any tools available. In other words, we may not be able to perfectly predict investment outcomes, but we have a good chance of assessing the nature of the *variability* of those outcomes. Although investment returns usually receive the majority of the attention given to the asset management industry in general and to portfolio managers in particular, they are really only half the story when it comes to evaluating investments and invest-

ment performance. Too often asset managers and their clients are surprisingly one-sided in their approach, focusing almost exclusively on the size of investment returns and only rarely on the risk that has been incurred – on behalf of the client – in the process of generating these returns. Risk is most often regarded as an unpleasant residual that one must simply learn to live with.

Individual portfolio managers may not be too concerned with this, particularly as – to the extent that they are paid in accordance with the performance of their portfolios – the focus is exclusively on the returns generated. Given this incentive scheme it is not very surprising that individual managers pay little attention to the trade-off between potential portfolio return and potential portfolio risk. However, from the perspective of an asset management firm this is potentially quite hazardous. The risk incurred in order to obtain the realised or expected returns is an important factor that also needs to be considered closely. Naturally, at the end of the day obtaining a return is the main reason why people invest. Without returns being generated, people would not invest. But in order to deliver high returns on a consistent basis, asset managers need to manage risk as actively as returns. Consistency is the key word here: a track record that exhibits a high degree of variability in investment returns over time is a lot less attractive than one that exhibits less variability.

Consider the following hypothetical example. Two asset management firms are trying to market similar investment products – that is, similar portfolios such as global equity funds. Over a three-year period the two portfolios have both produced a return of 21%, so at first glance they seem to be level in terms of performance. The chart in Figure 3.1 presents the monthly returns on the two funds.

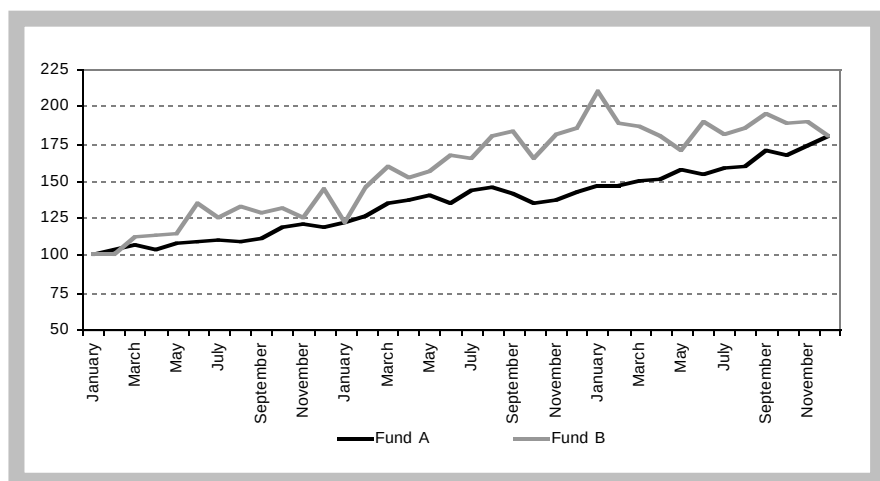


Figure 3.1 Hypothetical monthly performance of two global equity funds

Since most people are risk-averse, in theory all investors would, if given the choice, decide to invest in Fund A over Fund B, even though they have produced the same rate of return over the three-year period. Below we shall demonstrate how to quantify this return variability for a single asset and in subsequent chapters we shall focus only on portfolio return and risk. The example presented above demonstrates why the variability of returns is important to investors, asset managers and clients alike.

Bearing in mind that most asset managers employ some form of structured mean-variance-based technique for portfolio analysis,² it may even be argued that return and risk are inseparable – their interaction is by now internalised in almost all quantitative analysis systems. For this reason it is vital that portfolio managers understand the risk dynamics of their portfolios, specifically relative to their benchmark. Failure to do so may lead to inefficient asset allocation, unintentional risk-taking, loss of control over the cash flow generated by the portfolio, and ultimately the undermining of the quality of services provided to clients.

As mentioned above, the issue of risk pertains for now only to a single asset, but as we shall see later in this book, the issue of risk is particularly pertinent to *portfolios* of assets, since portfolio risk is very complex to describe and manage, and even more difficult to forecast with a particularly high degree of accuracy. As argued above, risk management is a vital and challenging component of portfolio management, presenting dangers as well as opportunities every day. Change is continuous and inevitable, but how we respond to it will to a large extent determine the future performance of our portfolios. So what we need now is a quantifiable *measure* of the risk or variability of investment returns.

DEFINING RISK

In order to be able to work with the concept of risk in the remainder of this book we need a way of quantifying this risk. On a single asset basis, financial risk is commonly quantified by some measure of variance of asset returns. This reverts to the definition presented at the beginning of this chapter, that risk is 'deviation from the expected'. We are interested in the variation of asset returns over time, and since we need some sort of reference point relative to which returns can be measured, we apply a measure of risk that relates every observation to the *average* or *mean* of all observations available.

Variance and volatility

Variance is a simple measure of variation around an average. It is a measure of dispersion of a set of observations around their mean value. Alternatively it

can be defined as the mathematical expectation of the average squared deviations from the mean.³

In order to be able to calculate the variance, we first need to find the mean of asset returns over T time periods or observations. This mean is determined by:

$$\bar{r}_A = \frac{\sum_{t=1}^T r_{A,t}}{T} \quad [\text{Eq. 3.1}]$$

where $r_{A,t}$ is the return on the asset A at time t , and $\bar{r}_A$ is the mean of asset returns over T periods.

Mathematically, variance is defined as the average of the sum of the squared differences between the returns and the mean of the returns.⁴ The expression for the variance of sample asset returns for an asset A over T time periods is thus given by:

$$\text{Var}_A = \frac{1}{T-1} \sum_{t=1}^T \left(r_{A,t} - \bar{r}_A \right)^2 \quad [\text{Eq. 3.2}]$$

We can illustrate the calculation of the variance by considering a hypothetical set of returns on asset A over six months. In Figure 3.2, the monthly returns are represented by black dots dispersed around the mean of returns, indicated by the dotted line. The variance of the returns is the average of the sum of all the shaded areas above and below the mean.⁵ Equation 3.2 thus

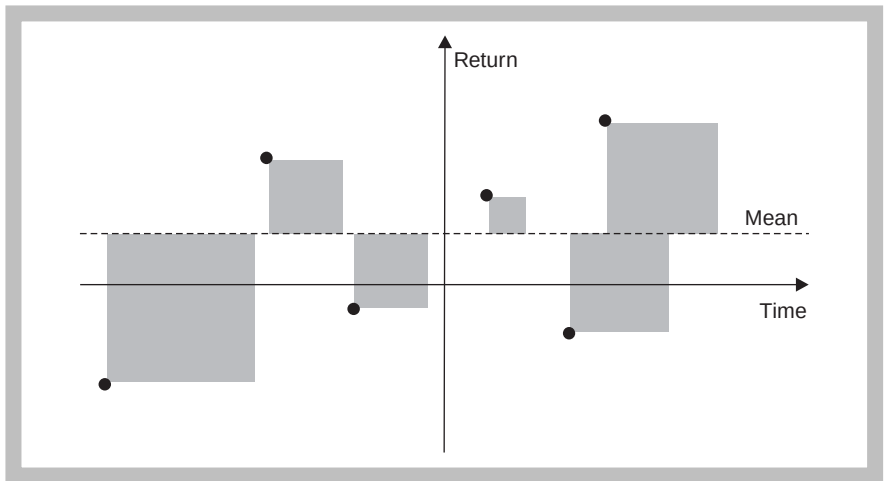


Figure 3.2 Variance (average of the sum of squared deviations from the mean)

gives us an indication of the degree to which our observed returns are dispersed around the mean. But since variance is a squared term, it is not directly comparable to the mean itself – attempting to relate the variance to the mean would be like comparing apples and oranges. We therefore need to take the square root of the variance to generate a figure comparable to the mean of returns. The *standard deviation* of returns is therefore defined as the square root of the variance of returns:

$$StDev_A = \sqrt{Var(r_A)} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)^2} \quad [\text{Eq. 3.3}]$$

Bearing this in mind, we move on to determining the *volatility* of returns, which is what we want to end up with. Assessing financial risk in absolute terms is usually done by calculating the volatility of asset returns, which is defined as the annualised standard deviation of returns. The volatility of returns is thus derived by multiplying the standard deviation of returns – Equation 3.3 – with the square root of the sample frequency. The sample frequency is simply the number of observations per year, so that daily observations yield approximately 260 (since this is the number of trading days in a year), weekly observations yield 52, and monthly observations yield 12.⁶ With a set of monthly observations, the standard deviation must be multiplied by the square root of 12. Similarly a set of daily observations requires multiplication by the square root of 261, assuming 261 trading days in a year.

To see why we must perform this task, consider the variance of monthly returns over a year. It is defined as 12 times the monthly variance:

$$Var_Y = 12 \cdot Var_M = 12 \cdot \frac{1}{T-1} \sum_{t=0}^T (r_{A,t} - \bar{r}_A)^2 = \sigma_Y^2 = 12 \cdot \sigma_M^2$$

where Var_Y is the variance of annual returns and Var_M is the variance of monthly returns. Taking the square root on both sides of the equation yields the standard deviation of annual returns, which is precisely the definition of volatility:

$$\sigma_Y = \sqrt{12} \cdot \sigma_M$$

The volatility of the returns on an asset is thus defined as:

$$\sigma_A = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)^2} \cdot \sqrt{\hat{T}} \quad [\text{Eq. 3.4}]$$

where $\hat{T}$ is the sample frequency within one year.

Let us illustrate this further with a simple example. Consider an asset *A* over a period of 12 months. For each month, we calculate the return on the asset. We then calculate the deviations from the mean and square them. We sum the squared deviations, and divide this total by 11 (subtracting 1 from 12 to correct for the loss of one degree of freedom). This sum is the variance of the assets return over the 12 months. To obtain the *volatility* of the assets return, we first take the square root of the variance of returns to obtain the standard deviation. We then multiply this by the square root of 12, since we are using monthly observations. This process is illustrated in Table 3.1, where we calculate the volatility of the return on a hypothetical stock.

Table 3.1 Variance, standard deviation and volatility using monthly returns

Month	Asset Price	Return	Squared Deviations
January	35		
February	36	0.0343	0.0000
March	39	0.0773	0.0022
April	39	0.0000	0.0009
May	36	−0.0769	0.0116
June	38	0.0556	0.0006
July	41	0.0789	0.0023
August	44	0.0732	0.0018
September	42	−0.0455	0.0058
October	41	−0.0238	0.0030
November	45	0.0976	0.0045
December	48	0.0667	0.0013
Mean		0.0307	
Variance			0.0031
Volatility			19.3%

Although this is a hypothetical stock price, the volatility of its monthly returns is comparable to the volatility of the return on a utility company stock. Compared to an average stock during this period, its risk or volatility is thus relatively low. From this point onwards, whenever we consider risk on a single asset or at a portfolio level, we will be operating only with volatilities calculated as the annualised standard deviation of returns.

Some examples

As is the case with returns – and partly as a function thereof – different assets and asset classes have widely varying volatilities, and the degree to which they themselves vary over time also differs from asset to asset and from asset class to asset class. In Table 3.2 we illustrate this point by examining the volatility levels of several of the major global equity indices during the period of 1995 to 2001. For each year we calculate the volatility of each index based on daily observations.

Table 3.2 12-month averages of volatilities (3 months of daily observations)

	Dow Jones	S&P500	NASDAQ	FTSE100	DAX30	Tokyo SE	Hang Seng
1995	9%	8%	14%	10%	13%	18%	18%
1996	11%	11%	15%	9%	12%	11%	15%
1997	15%	17%	17%	14%	22%	20%	31%
1998	16%	18%	24%	20%	27%	21%	42%
1999	14%	18%	27%	18%	21%	18%	26%
2000	17%	21%	45%	18%	23%	21%	31%
2001	18%	21%	42%	21%	27%	23%	26%

The variations in volatility levels are quite apparent from Table 3.2. In particular, the level to which the volatility of the NASDAQ index shot up during 2000 and 2001 seems quite extraordinary. As was alluded to in the previous chapter, this may principally be explained by the fact that the global technology sector in general, and this equity market index in particular, experienced a bubble during this period, both in terms of cash injections into the technology companies themselves and in terms of the level of valuations at which the stock traded. Seeing no end to the profit growth many of these companies would potentially achieve in the future, where investors would traditionally have paid little more than 30 or 40 times prospective earnings per share for a single share in a company, investors were suddenly willing to pay 100 times prospective earnings per share, or more.

In addition, during those two years the proliferation of so-called *day traders* – individuals sitting at home, some of them quitting their jobs to do so, in order to trade on their own accounts directly via the Internet – was quite significant, particularly in the United States. The participation of these many thousands of investors, with little or no experience with equity markets and no formal investment training, may well have exacerbated the volatility of the markets during those two years. This point is underscored by the fact that although the other

indices also experienced a rise in average volatilities, those increases were far from the level to which the volatility of the NASDAQ rose. The rise in the volatility of the other indices was probably caused to some extent by the same type of phenomenon as that which was the case for the NASDAQ, as well as simple contagion from that index.

To illustrate the manner in which volatilities typically move and the way in which they may change over time, Figure 3.3 presents three-month moving averages of volatilities based on three months of daily observations for a set of indices over the period 1995 to 2001. As is quite evident from Figure 3.3, there are substantial variations in volatilities over time. Note also that volatilities of the two markets tend to be correlated, and increasingly so from 1997. This makes good sense: factors that affect one equity market almost always affect other equity markets as well. Such factors are mainly changes in expectations about company earnings, interest rates or other economic variables, and also political changes. In addition, the increased correlation between the volatilities of these stock indices may to some extent reflect the increasing globalisation of equity markets.

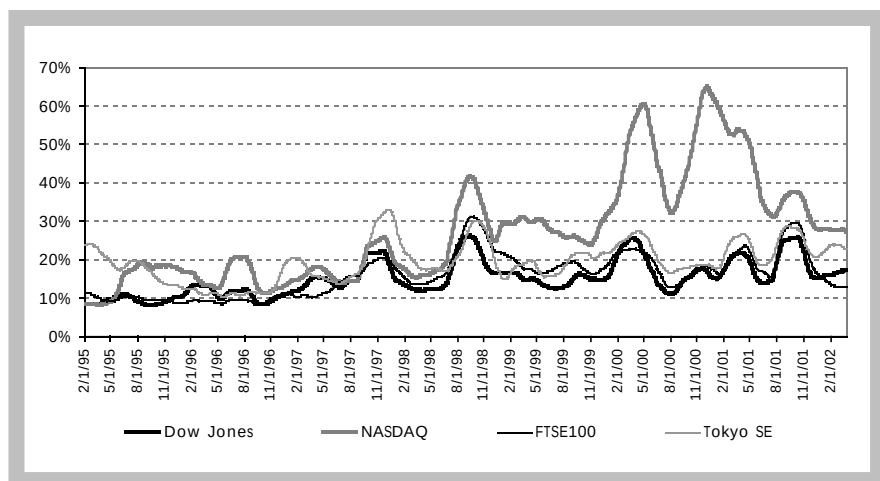


Figure 3.3 Three-month moving average of three-month volatilities, based on daily observations 1995/1–2001/5

The advantages of using a statistical measure of risk such as volatility are that it is mathematically well defined, and that it can be implemented as a parameter in quantitative portfolio models (as we shall demonstrate in subsequent chapters). There are several other risk measures that are applied in the field of financial economics and investment management. They all have their weaknesses and strengths, and as a rule the investment manager should attempt to

draw on as many of these as possible as each can provide useful information. In addition to volatility, three of the most widely applied measures for quantifying risk, both in a single asset and in a portfolio context, are Beta, shortfall risk, and downside risk. Below we briefly comment on each in turn.

Beta as a measure of asset risk

When measuring risk we most often need to do this in a portfolio context. While the subject of portfolio return and risk calculations will be introduced formally in subsequent chapters of this book, we will present here the idea of risk relative to the market (or to a chosen benchmark). For now we introduce *Beta* (β) as a relative measure of risk, and define it as:

$$\beta_A = \frac{Cov(r_A, r_M)}{Var(r_M)} \quad [\text{Eq. 3.5}]$$

Beta is thus the covariance of an asset's return with the return on the market, divided by the variance of the asset. In other words, Beta is a measure of the degree to which the returns on an asset tend to move with the return on the market, relative to the magnitude of the market's return variation. The *covariance* of an asset with the market is defined similarly to the *variance* of the return on an asset. Remember that from Equation 3.2 the variance of the return on an asset A is defined as:

$$Var(r_A) = \frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)^2 = \frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)(r_{A,t} - \bar{r}_A)$$

Instead of measuring the sum of the squared deviations from the mean of returns on a single asset, the covariance of an asset's return with the market's return measures the sum of the product of the deviations in returns on the asset and the market from their respective means. The covariance of an asset with the market is thus defined as:

$$Cov(r_A, r_M) = \frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)(r_{M,t} - \bar{r}_M) \quad [\text{Eq. 3.6}]$$

where $r_{A,t}$ is the return on the asset A at time t , and $\bar{r}_A$ is the mean of asset returns over T periods. $r_{M,t}$ is the return on the market at time t , and $\bar{r}_M$ is the mean of market returns over T periods.

Equation 3.6 reveals in a simple mathematical fashion that the closer the two returns are – simultaneously – to their respective means, the smaller are the deviations from those means, and the lower is the covariance. Similarly,

large simultaneous deviations from their respective means result in a higher covariance between the two returns. Note that the covariance of an asset's return with itself equals its variance, since Equation 3.6 would then be identical to Equation 3.2. Formally this can be demonstrated as:

$$\text{Cov}(r_A, r_A) = \frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)(r_{A,t} - \bar{r}_A) = \text{Var}(r_A) \quad [\text{Eq. 3.7}]$$

As we shall see, the measure of covariance is central to the term Beta. Again, Equation 3.5 shows that Beta indicates the risk of an asset relative to the risk of the market, in that it measures the covariation of asset *A*'s return with the return on the market, relative to the variance of the return on the market. The market is not specified here but it will often be a broad market index such as the S&P500 or perhaps the MSCI World, depending on the stock to be evaluated.

We might call Beta a standardised measure of covariation, where the standardisation factor is the market variance. From Equation 3.6 we also see that if the covariance of the asset with the market is equal to the variance of the market, then Beta equals 1. So if Beta is above 1, this indicates that the asset will on average move more than the market on any given day. If Beta is 1.10 and the market rises 10%, we would then expect asset *A* to rise by 11%. In the event of a market fall we would expect asset *A* to fall by more than the market. Similarly, if Beta is below 1 then we would expect asset *A* to move less than the market. The reader may by now have noted that Beta is in fact equal to the slope of the regression line in Figure 2.9 in Chapter 2, since it indicates to what extent the return on an asset is usually higher or lower than the return on the market on any given day. In the case of IBM, the stock of that company appears to have had a Beta of around 1.4 over the period examined.

By definition, Beta of the market is equal to 1, since the covariance of an asset with itself is equal to its variance. Similarly the covariance of the market with itself is equal to the variance of the market, which divided by itself equals 1. This leads us to the following diversification argument. By adding a large number of assets to a portfolio we exploit a diversification effect whereby the total portfolio risk is reduced. Taking this argument further leads us to the result that the inclusion of all available assets into our portfolio will eliminate all systematic risk – that is, the risk that is inherent in the market. This can theoretically be done without costs, so the only risk that we are willing to pay to eliminate is the *unsystematic* risk, since this is the only kind left after we have eliminated the systematic risk. The notion of relative risk between assets and the market will prove useful later on when we describe portfolio risk, and so Beta will be introduced formally later in this book, in a portfolio risk context.

Shortfall risk

As the name indicates, *shortfall risk* is a risk measure designed to capture the risk of falling short of some target level. For asset or portfolio returns, the shortfall risk is measured relative to a target return, for example a one-month market return defined by the investor. Portfolio risk is then assessed or considered as the possibility or risk of *not* reaching the specified target level of returns. Special cases of shortfall risk measures are the *shortfall probability*, the *shortfall expectation* and the *shortfall variance*. We shall not employ shortfall risk as a measure of risk in this book.⁷

Downside risk

The risk measures such as volatility presented above could be argued to be imprecise in that they consider any deviation from the mean of returns as risk. It may be argued that investors ought only to be concerned with the variation that is due to returns that lie *below* the mean, since investors with standard risk aversion have no qualms about receiving above-average returns. Measures of downside risk are designed to capture only this ‘negative’ variation – the return observations that lie either below the mean or below some target return level.

Semi-variance is one such measure as it calculates risk based only on ‘negative’ variation around the mean. So it is apparent that shortfall risk is actually a special case of downside risk, since it relates actual returns to some specified target return. This approach makes good sense: however there are a few problems associated with such risk measures, most prominent of which is the fact that return distributions are not symmetrical around the mean. This may result in the downside risk estimate possibly over- or underestimating the actual risk of the assets return. Later on we shall examine how returns are typically distributed around their means, as well as the extent to which they are asymmetrical in nature.

Another downside risk measure is *lower partial moment (LPM)*, which is basically an enhanced semi-variance measure. In every measure of variation lies an implicit assumption of risk aversion. The exact nature of the risk aversion of any given investor is virtually impossible to identify. Usually, attempts are made to quantify risk aversion using a utility function, since in the calculation of variance and standard deviation the deviations from the mean are all squared. This, however, limits us to just one utility function (namely the quadratic utility function), even though the investor may have many. The LPM augments this by introducing a multitude of possible utility functions, in that deviations need not necessarily be squared but can be lifted to the d th

power, with d being any number not constrained to being a predefined integer. This gives us a potentially much more flexible but also arbitrary risk measurement tool. We shall, however, not employ these measures in this book either. Although intuitively appealing, they will become virtually impractical later on as we move to the topics of quantitative asset allocation and Monte Carlo Simulation.

A BRIEF NOTE ON NORMALITY

Although measures of risk such as volatility are widely used, they do however have a number of drawbacks. First and foremost, volatility – as measured by annualised standard deviation of returns – assumes that asset returns are distributed symmetrically around the mean of returns. In fact, to the extent that standard deviations of returns are employed in the test of some statistical hypothesis, it is almost always assumed that returns follow a normal distribution as well. Yet this is in fact not usually the case. Most often returns are skewed to one side or the other, and furthermore they often exhibit more extreme market events than is predicted by the normal distribution, which gives rise to so-called *kurtosis* or ‘fat tails’. Return distributions often appear ‘skewed’, but with a higher than expected number of observations further away from the mean, relative to the normal distribution. Such skewed and kurtotic return distributions could appear as shown in Figure 3.4.

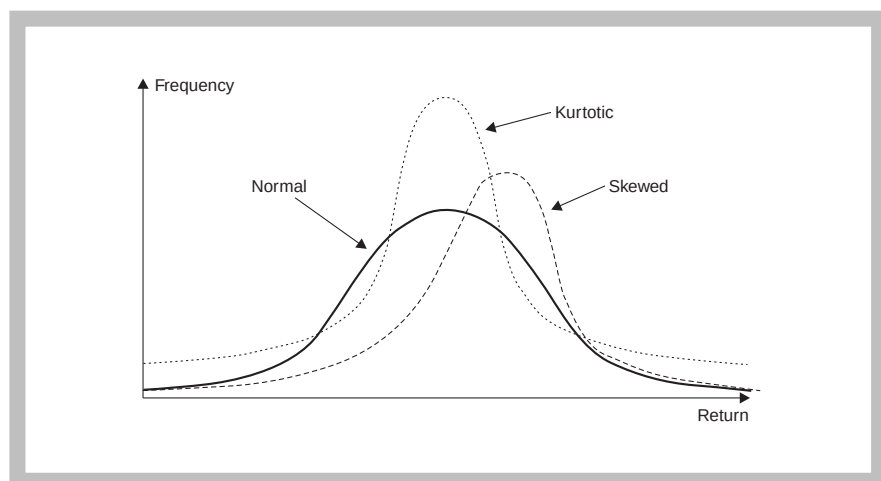


Figure 3.4 Normal, skewed and kurtotic return distributions

Skew

Mathematically, *skew* is defined as the third moment about the mean of returns, divided by the cube of the standard deviation of returns:

$$\frac{\frac{1}{T} \sum_{t=1}^T (r_{A,t} - \bar{r})^3}{\sigma_A^3} \quad [\text{Eq. 3.8}]$$

If the distribution is not skewed at all, it will have a *skewness coefficient* of 0. Positive values indicate positive skew (skewed to the right) and negative values indicate negative skew (skewed to the left). In terms of magnitude, moderate skew lies in the region of -0.5 to $+0.5$, while the skewness coefficient of highly skewed distributions lie beyond -1 and $+1$. We shall investigate this return characteristic later in this book.

Kurtosis

Kurtosis, also known as the ‘peakedness’ of a distribution, is defined as the fourth moment about the mean of returns, divided by the quadruple of the standard deviation:

$$\frac{\frac{1}{T} \sum_{t=1}^T (r_{A,t} - \bar{r})^4}{\sigma_A^4} \quad [\text{Eq. 3.9}]$$

The standard reference for kurtosis in return distribution is the normal distribution, which has a *kurtosis coefficient* of 3. The higher the kurtosis coefficient, the higher the level of kurtosis, so a kurtosis coefficient of 4 would indicate a relatively peaked return distribution, while a kurtosis coefficient of 2 would indicate a relatively flat return distribution. We shall investigate this return characteristic later in this book. Please note that the kurtosis coefficient says nothing about whether or not the distribution is symmetrical.

Implications

The implications of the non-normal return distribution are particularly strongly felt in the areas of quantitative asset allocation and portfolio risk management, and so we shall revisit this topic several times more in subsequent chapters. In particular, risk-management techniques that assume normally distributed

returns will tend to underestimate severely the risk of extreme events, which might lead to firms holding lower than required capital reserves. It could also result in asset managers taking on more risk than they realise, simply because they are working under the assumption that portfolio returns are normally distributed.

There are ways of dealing with these issues, and we shall cover these later on in this book. For now, though, we make the standard simplifying assumption that investment returns are normally distributed over time, such that $r_A \sim N(\bar{r}_A, \sigma_A)$.

SUMMARY

Volatility, defined as the annualised standard deviation of returns, is an easy-to-use tool for quantitatively assessing risk. It thus provides a unified framework for evaluating risk, with regard both to single assets and to portfolios of assets. In the remainder of this book we shall use only volatility and Beta as our risk measures. They rest on essentially the same foundation, in that they both depend on the assumption that risk can be expressed adequately as a statistical measure of variation of returns around the mean of returns. Volatility is also the measure of risk most commonly employed by asset managers and analysts, as it is relatively easy to calculate and in addition goes well with the conventional Markowitz-inspired mean-variance optimisation framework that most asset managers employ today. We shall go into significantly more detail about mean-variance analysis in subsequent chapters.

Notes

- 1 This assertion naturally assumes that the risk-free rate is zero, which is seldom the case; but the argument holds in a benchmark-relative context, to which we will return in subsequent chapters.
- 2 We shall cover mean-variance analysis in much greater detail in subsequent chapters.
- 3 Calculating variance according to this definition requires a specification of probabilities of all expected outcomes, which is not possible here.
- 4 We correct for the loss of one degree of freedom by subtracting 1 from the denominator in Equation 3.2.
- 5 This is incidentally the same principle as that which is applied in Ordinary Least Squares (OLS) regression analysis, which minimises the sum of the squared errors/deviations (shaded areas). Minimising the sum of the shaded areas results by definition in the least amount of variance and thus the best fit for a straight line, which is essentially the aim of any regression analysis.
- 6 This is known as the 'Square Root of Time' rule.
- 7 For a comprehensive text on shortfall risk, see Lawrence N. Bader, *Return Targets and Shortfall Risks*.

ASSET PRICING

PRICING AND VALUATION

Introduction

Estimating asset return and risk is the basis for most quantitative asset allocation and portfolio risk management. As demonstrated above, we can calculate these from the observed past market prices. These prices indicate the market consensus of what the asset is actually worth, based on the range of pricing methodologies that exist and the expectations of individual analysts and portfolio managers. In order to proceed, we therefore need an understanding of how the market performs its valuation of individual securities.

Discounting

The basic premise for most valuation techniques is the discounting of a future stream of payments back to the present. The basic argument for this is that investors would be able to invest \$100 today at a risk-free rate of say 3%, and then receive \$103 in a year. The investment would be risk free and this would be a so-called free lunch, whose existence economists are unwilling to accept. Therefore the \$100 must be worth less in one year than they are today, so the \$100 dollars are discounted by a discount rate equal to the risk-free rate. That leaves the \$100 expected to be received in one year worth only $\$97.01(1/1.03 \cdot 100)$.

When pricing securities, whether they are stocks or bonds or other types of assets, the relevant payments to be discounted are *free cash flows (FCF)* gener-

ated by the asset. More precisely, for bonds the payments are just the regular payments of interest and payments on the loan principal. For stocks the investor should focus on a company's free cash flow and not its profits, since cash flows more accurately describe the actual increase in value of the stock. Since there is an implicit assumption of compounding interest rates over each period, the general formula for discounting the expected FCF over N periods with the discount rate i is as follows:

$$\begin{aligned}
 PV &= \frac{FCF_1}{(1+i)} + \frac{FCF_2}{(1+i)^2} + \dots + \frac{FCF_n}{(1+i)^n} + \dots + \frac{FCF_N}{(1+i)^N} \\
 &= \sum_{n=1}^N \frac{FCF_n}{(1+i)^n}
 \end{aligned}
 \tag{Eq. 4.1}$$

PV is the *present value* of the future free cash flows. In Equation 4.1 we assume for simplicity a constant risk-free rate. This is not usually the case, in fact, since the term structure of interest rates is very rarely flat. This means that for each expected free cash flow we must assign a discount rate to that specific point in time. Most often the term structure of interest rates is constructed from a number of bonds with varying maturities, with extrapolation or bootstrapping being used to fill in the gaps in maturity where no bond exist. One of the strengths of *discounted cash flow analysis* is that it allows the investor to consider the timing and size of cash flows over the life of an investment. The analysis is applicable to a wide variety of investments, including but not limited to stocks, bonds, condominium apartments, and income-producing real estate with a fluctuating income over the projected holding period. Analysis can be performed on a before-tax or after-tax basis, depending on whether the analysis is for investment analysis or appraisal purposes. The discount rate, which reflects the time value of money, will usually be adjusted to consider the risk incurred as a result of an investment. Adjustments for risk are best made by considering the potential changes in income or expenses or the potential changes in the time value of money over the holding period. A sensitivity analysis, based on changes to all model parameters, can help to indicate the impact of changes in the free cash flows. Such measures are thus highly recommended as an addendum to any pricing of securities based on the discounted cash flow approach.

DETERMINING THE DISCOUNT RATE

Discounting the future cash flows back to the present at an appropriate discount rate is essential. As we shall see, relatively small changes in the discount rate can have large impacts on the present value of a cash flow stream. As noted, the vast majority of future cash flow streams have some risk attached to them, which is

why discounting with the risk-free rate is inappropriate, particularly for companies. The discount rate must reflect the risks inherent in the financing of company activities, which is why discount rates are typically referred to as the *weighted average cost of capital* (WACC). As the name implies, this discount rate is calculated as a weighted average of all costs (required rates of return) on the different sources of financing employed by the company. So the WACC is the weighted average of the required rates of return on each source of capital.

To see why this is so, consider a company that needs financing for an entirely new investment project. The company plans to finance the project with 40% corporate bonds and 60% new equity. Corporate bondholders require say 9% in return, and equity investors require say 13% in return in order to enter into this financing scheme. The WACC needed to finance this project is found thus:

$$WACC = 0.40 \cdot 9.0\% + 0.60 \cdot 13\% = 11.4\% \quad [\text{Eq. 4.2}]$$

The appropriate rate for discounting the future stream of free cash flows from this company would then be 11.4%. From Equation 4.2 it follows that for a company financed just by equity and debt, the general definition of the weighted average cost of capital is:

$$WACC = w_E \cdot k_E + w_D \cdot k_D(1 - t) \quad [\text{Eq. 4.3}]$$

where w_E is the weight of equity in the company's capital structure, k_E is the cost of equity capital, w_D is the weight of debt in the capital structure, k_D is the cost of debt and t is the marginal tax rate.

The debt component is multiplied by 1 minus the marginal tax rate, because interest on this debt is assumed to be tax-deductible. We use the marginal tax rate instead of the average tax rate, since we assume that we are dealing with new financing. The cost of the debt component will typically be the marginal lending rate of the company; while the cost of the equity component can be determined in a number of ways, among which the Capital Asset Pricing Model (CAPM) is a favourite among financial analysts.¹

Typically, the debt and equity components are divided into sub-components, each with its own cost of capital, in order to provide as accurate a picture as possible of the company's actual cost of capital. For example, the equity component would be split into preferred stock and common stock, with preferred stock having a lower cost due to the fact that common stock is subordinate to preferred stock in the event of bankruptcy and therefore has a lower (credit) risk. Similar arguments can be made for retained earnings relative to preferred stock and common stock, and for collateralised versus un-collateralised debt, and so on.

When discounting with the WACC, we are in essence testing whether the expected future cash flows are sufficient to cover the initial investment outlay by each capital contributor. If the *net present value* (NPV: the sum of all

discounted future net free cash flows) of a new investment is equal to zero, then the investment project will just break even – the investors will all have recovered their initial investment outlays, but nothing more than that. Any net present value *above* zero indicates that profits will have been generated for capital contributors relative to their required rates of return.

The principle demonstrated above applies to practically any type of investment, and can thus also enable stock market investors to determine the profitability of an investment scheme. A company can be argued to be simply a portfolio of investment projects, so when all the expected free cash flows of all the company's investment projects are discounted using WACC, the resulting value is the value of the company. Discounting the cash flow stream thus produces a value that represents a liquid investment by a majority shareholder. This value assumes that the shareholder could liquidate his or her investment in a timely and cost-effective manner, and that this shareholder has the ability to make decisions that ultimately impact the magnitude of cash flows.

THE DIVIDEND DISCOUNT MODEL (DDM)

The basic DDM

The most simple and widely applied valuation model based on the discounted cash flow principle is the Dividend Discount Model (DDM). This states that the price of a stock is equal to the present value of all future dividend payments. According to the DDM, the price of a stock *A* is thus formally defined as:

$$P_A = \frac{D_1}{(1+i)} + \frac{D_2}{(1+i)^2} + \frac{D_3}{(1+i)^3} + \dots = \sum_{t=1}^{\infty} \frac{D_t}{(1+i)^t} \quad [\text{Eq. 4.4}]$$

where D_t is the dividend payment from the stock at time t . Note that the dividend payments are assumed to continue indefinitely. The model is dependent on analysts' ability to forecast accurately all future dividend payments until the end of time, which of course is not very realistic. However, the model leaves us with the opportunity to vary forecast dividend payments from one year to the next.

The DDM with constant growth

The simple dividend discount model assumes perpetuity of dividends, and some models even assume constant dividend payments. In other words they assume that the company does not grow and as a result that its dividend

payments do not grow either. This is not realistic, which is why the formula can be augmented to include a growth term.

Consider a company that grows at an annual rate of g . The price of the company's stock is then defined as:

$$P_A = \frac{D_1}{(1+i)} + \frac{D_1(1+g)}{(1+i)^2} + \frac{D_1(1+g)^2}{(1+i)^3} + \dots$$

$$= \sum_{t=1}^{\infty} \frac{D_1(1+g)^{t-1}}{(1+i)^t} \quad [\text{Eq. 4.5}]$$

Using the geometric properties of an infinite series of payments, Equation 4.5 can be reduced to:

$$P_A = \frac{D_1}{i-g} \quad [\text{Eq. 4.6}]$$

Equation 4.6 suggests that the price of the stock depends only on the size of the initial dividend, the company's expected growth rate g , and the appropriate discount rate i . Interestingly we can link the price of the stock to the character of its future investments through the incorporation of the *retention rate*, the percentage of earnings that are reinvested into the company. Suppose a company that is 100% equity-financed maintains a retention rate of b . For a given return on equity, ROE , its future dividend growth rate will be equal to $ROE \cdot b$. This growth rate is often referred to as the company's *sustainable growth rate*, since this is the growth that the company will be able to sustain without having to resort to external financing.

The DDM with periodic growth

The DDM with constant growth is appealing because of its simplicity. However, it is difficult to accept that a company's entire future can be described by constant rates of growth. Furthermore, the assumption of constant growth leaves the model susceptible to huge swings in estimates of fair value, and thereby in share price, due to minor changes in the discount rate. For example, according to the simple model, a company expected to grow at 10% per year until infinity, that has a discount rate of 12% and that initially pays \$2 in dividends, should be priced at \$100. However, if the discount rate were changed to 11% then according to the model we should be willing to pay \$200 for the stock. In an attempt to remedy this, we can assume that a higher growth rate is expected to persist for a limited number of years, after which the growth rate is expected to fall to some arbitrary long-run level. More precisely we could assume that a growth rate of g_S is expected to persist for M years,

after which the company's growth would revert to the long-term level of g_L . Using these assumptions, the price of the stock would then be determined by:

$$P_A = \sum_{t=1}^M \frac{D_1(1 + g_S)^{t-1}}{(1 + i)^t} + D_M \sum_{t=M+1}^{\infty} \frac{(1 + g_L)^{t-M}}{(1 + i)^t} \quad [\text{Eq. 4.7}]$$

where $D_M = D_1(1 + g_S)^S$ is the dividend rate reached at the beginning of the period of long-term growth. This model can be expanded to include an infinite number of periods, each with its specific expected growth rate.

In the section on implied growth rates later on in this chapter we shall employ a similar multi-stage framework in reverse. The idea is simply that we can estimate the short-term (first stage) growth rates implied by the current share price, and then decide whether such a growth rate is realistic or not.

The DDM with growth and inflation

To the extent that there is inflation present in the economy in which the company operates, we must also try to correct for the loss in real value of future dividends that this causes. We do this simply by subtracting the expected inflation rate from the expected growth rate, to arrive at an expected *real* perpetual growth rate. The nature of this change is independent of whether we operate with a simple perpetual growth model or a model with periodic growth. Whether we use the constant growth assumption or the periodic growth assumption, the end result is still a relatively crude estimate of the correct price for a stock. Ideally we would like to be able to predict every variable that might affect the company, and thereby the fair price of the stock, but this task is virtually impossible since it would ultimately require that we forecast every single incident that might affect the company in the future. What we thus need is a reasonable approximation of exact forecasts, and cash flow analysis is one such tool.

THE DISCOUNTED CASH FLOW MODEL (DCF)

Cash flow analysis

As indicated, the basis for proper use of the discounted cash flow model is partially to correctly determine the free cash flow provided by the asset in question. Bonds are straightforward since the sizes of their payments are known and only interest rate uncertainty remains. For stocks, however, things are not that simple. Ideally we would estimate the free cash flow for each year to infinity, and then discount the resultant expected free cash flows back to the

present to obtain a fair value for the company. Obviously this is unattainable in practice, so we use an approximation.

Estimating the free cash flows

First, though, we consider how to obtain the free cash flow. Almost all the methods used for finding this cash flow rely on accounting information. The two most prominent are the so-called *direct method* and the so-called *indirect method*, and we shall quickly demonstrate the latter.

Using the indirect method involves splitting the company’s financial activities into three parts:

- Operating activities
- Investing activities
- Financing activities.

For each activity a separate cash flow is determined, and all three are then added to obtain the company’s total free cash flow. Table 4.1 shows the income statements of International Business Machines (IBM) from 1995 to 1999.

Table 4.1 Annual income statement, IBM 1995–1999

\$ Millions	Dec-99	Dec-98	Dec-97	Dec-96	Dec-95
Sales	87,548	81,667	78,508	75,947	71,940
Cost of Sales	49,034	45,803	42,898	40,396	35,971
Gross Operating Profit	38,514	35,864	35,610	35,551	35,969
Selling, General and Admin.	20,002	21,708	21,511	21,943	22,776
Other Taxes	0	0	0	0	0
EBITDA (Earnings Before Interest, Taxes, Depreciation and Amortisation)	18,512	14,156	14,099	13,608	13,193
Depreciation and Amortisation	6,585	4,992	5,001	5,012	5,602
EBIT (Earnings Before Interest and Taxes)	11,927	9,164	9,098	8,596	7,591
Other Income, Net	557	589	657	707	947
Total Income	12,484	9,753	9,755	9,303	8,538
Interest Expense	727	713	728	716	725

Table 4.1 *continued*

\$ Millions	Dec-99	Dec-98	Dec-97	Dec-96	Dec-95
Minority Interest	0	0	0	0	0
Pre-tax Income	11,757	9,040	9,027	8,587	7,813
Income Taxes	4,045	2,712	2,934	3,158	3,635
Special Income/Charges	0	0	0	0	0
Net Income from Cont. Op.	7,712	6,328	6,093	5,429	4,178
Net Income from Discont. Op.	0	0	0	0	0
Net Income from Total Op.	7,712	6,328	6,093	5,429	4,178
Total Net Income	7,712	6,328	6,093	5,429	4,178

The example above is just for 5 years, but the analysis can be carried forward for consecutive years simply by increasing the analyst's forecast horizon. The standard methodology entails estimating as many elements as possible of the income statement directly, say 5 or 10 years into the future. Using this information the analyst can then deduce the expected balance sheets and cash flow statements over that future period. IBM's balance sheet appears in Table 4.2, and the cash flow statement appears in Table 4.3. We will not comment on or perform an analysis of the income statement, balance sheet or cash flow statement, but simply assume that they convey the information correctly and in accordance with proper accounting principles.

Table 4.2 Annual balance sheet, IBM 1995–1999

\$ Millions	Dec-99	Dec-98	Dec-97	Dec-96	Dec-95
<i>Assets</i>					
Current Assets					
Cash and Equivalents	5,043	5,375	7,106	7,687	7,259
Receivables	27,618	26,781	23,826	23,167	23,402
Inventories	4,868	5,200	5,139	5,870	6,323
Other Current Assets	5,626	5,004	4,347	3,971	3,707
Total Current Assets	43,155	42,360	40,418	40,695	40,691

(cont'd)

Table 4.2 *continued*

\$ Millions	Dec-99	Dec-98	Dec-97	Dec-96	Dec-95
Non-current Assets					
Property, Plant and Equip.	39,616	44,870	42,133	41,893	43,981
Accum. Depr. and Depl.	22,026	25,239	23,786	24,486	27,402
Property, Plant and Equip., Net	17,590	19,631	18,347	17,407	16,579
Intangibles	0	945	0	0	0
Other Non-current Assets	26,750	23,164	22,734	23,030	23,022
Total Non-current Assets	44,340	43,740	41,081	40,437	39,601
Total Assets	87,495	86,100	81,499	81,132	80,292
<i>Liabilities and Equity</i>					
Current Liabilities					
Accounts Payable	6,400	6,252	5,215	4,767	4,511
Short-term Debt	14,230	13,905	13,230	12,957	11,569
Other Current Liabilities	18,948	16,670	15,062	16,276	15,568
Total Current Liabilities	39,578	36,827	33,507	34,000	31,648
Non-current Liabilities					
Long-term Debt	14,124	15,508	13,696	9,872	10,060
Deferred Income Taxes	1,354	1,514	1,487	1,627	1,807
Other Non-current Liabilities	11,928	12,818	12,993	14,005	14,354
Minority Interest	0	0	0	0	0
Total Non-current Liabilities	27,406	29,840	28,176	25,504	26,221
Total Liabilities	66,984	66,667	61,683	59,504	57,869
Shareholder's Equity					
Preferred Stock Equity	247	247	252	253	253
Common Stock Equity	20,264	19,186	19,564	21,375	22,170
Retained Earnings	16,878	10,141	11,010	11,189	11,630
Total Equity	20,511	19,433	19,816	21,628	22,423
Total Liabilities and Stock Equity	87,495	86,100	81,499	81,132	80,292

Table 4.3 Annual cash flow statement, IBM 1995–1999

\$ Millions	Dec-99	Dec-98	Dec-97	Dec-96	Dec-95
Cash Flow from Op. Activities					
Net Income (Loss)	7,712	6,328	6,093	5,429	4,178
Depreciation and Amortisation	6,585	4,992	5,001	5,012	5,602
Deferred Income Taxes	−713	−606	358	11	1,392
Operating (Gains) Losses	−4,791	−616	−718	−1,356	−618
Extraordinary (Gains) Losses	0	0	0	0	0
Change in Working Capital					
Change in Receivables	−1,677	−2,736	−3,727	−650	−530
Change in Inventories	301	73	432	196	107
Change in Other Curr. Assets	−130	880	−1,087	−980	−1,100
Change in Payables	−3	362	699	319	659
Change in Other Curr. Liabs.	2,827	596	1,814	2,294	1,018
Other Non-cash Items	0	0	0	0	0
Net Cash from Cont. Op.	10,111	9,273	8,865	10,275	10,708
Net Cash from Discont. Op.	0	0	0	0	0
Net Cash from Op. Activities	10,111	9,273	8,865	10,275	10,708
Cash Flow from Investing					
Cash Flow Provided by:					
Sale of Property, Plant, Equip.	6,087	905	1,130	1,314	1,561
Sale of Short-term Investments	2,616	3,945	1,439	1,470	3,149
Cash Used to Purchase:					
Property, Plant, Equipmt.	−5,959	−6,520	−6,793	−5,883	−4,744
Short-term Investments	−3,949	−4,211	−1,617	−1,613	−1,315
Other Investing Changes Net	−464	−250	−314	−1,011	−3,703
Net Cash from Investing	−1,669	−6,131	−6,155	−5,723	−5,052
Cash Flow from Financing					
Cash Flow Provided by:					
Issuance of Debt	6,409	8,066	9,142	7,670	9,193
Issuance of Capital Stock	0	0	0	0	0

(cont'd)

Table 4.3 *continued*

\$ Millions	Dec-99	Dec-98	Dec-97	Dec-96	Dec-95
Cash Used for:					
Repayment of Debt	-7,510	-5,942	-5,198	-5,911	-9,460
Repurchase of Capital Stock	-6,645	-6,283	-6,251	-5,005	-5,526
Payment of Cash Dividends	-879	-834	-783	-706	-591
Other Financing Charges, Net	0	0	0	0	0
Net Cash from Financing	-8,625	-4,993	-3,090	-3,952	-6,384
Effect of Exchange Rate Changes	-149	120	-201	-172	65
Net Change in Cash and Cash Equivalents	-332	-1,731	-581	428	-663
Cash at Beginning of Period	5,375	7,106	7,687	7,259	7,922
Free Cash Flow	3,273	1,919	1,289	3,686	5,373

The end results of this rather laborious process are the free cash flows for each fiscal year. Using estimates of future earnings and balance sheet structure, the analyst can then extend this framework into the future, and thereby attempt to estimate future free cash flows. These future cash flows are then discounted back to the present in order to assess the ‘true’ current expected value of the company and thereby of its shares.

Estimating 5 or 10 years into the future may not seem like a very long time, especially when what we are really interested in is free cash flows from here to eternity. But the practicalities of determining the future competitive landscape for a company, its product developments and that of its competitors, its costs, earnings and financial structure, and ultimately the state of the global economy for all time, make this a daunting task to say the least – the process is very far from being an exact science. In addition, it turns out that it does not take a very high discount rate to have cash flows more than 10 years out become relatively insignificant, which is rather fortunate. Assuming a constant 30-year stream of future cash flows of \$1,000,000 each and a discount rate of 5%, 10% and 15% respectively, the value of each future payment discounted back to the present value appears as shown in Figure 4.1.

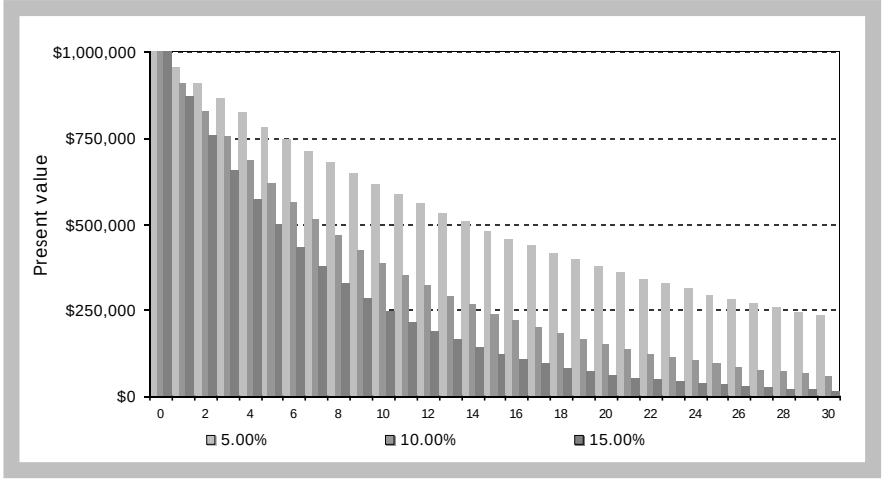


Figure 4.1 Discounting \$1,000,000 at different discount rates over 30 years

From Figure 4.1 it is apparent that when the discount rate becomes moderate in size, the discounting effect is quite large, due to the compounded interest effect. In fact at 15% the value of a future cash flow 10 years from now is reduced to less than a quarter of its initial value. And being promised to receive \$1,000,000 in 30 years is actually equivalent to receiving only \$15,003 today. Even for a discount rate of 10%, a reduction in present value to around half of the principal takes just 7 years. As far as the explicit cash flow estimates made by the analyst are concerned, it is thus clear that the biggest impact on present value stems from the most immediate future, even for moderate discount rates.

The cash flows stemming from the period that lies beyond the analyst's estimation period are regarded as a perpetuity, and the present value of the sum of these cash flows is called the *terminal value*. As a rule of thumb, typically half of the discounted cash flows pertain to the specific estimation period and the other half come from the cash flow that lies after this period. This point, combined with the increasing uncertainty about future cash flows as the estimation horizon is increased, is the reason why most analysts go no further than 5 to 10 years into the future in their specific forecasts. Because of the great uncertainties about cash flows that lie beyond say 10 years, it is customary simply to assume that all the elements of the company's income statement will continue to grow at a fixed rate, resulting in monotone development of free cash flows in the future. Using the results from the Dividend Discount Model the terminal value is determined by:

$$TV_A = \frac{FCF_T}{i - g}$$

[Eq. 4.8]

where TV_A is the terminal value of a stock A , based on the discounted value of the expected perpetual stream of free cash flows FCF_T . The sum of the net present value of free cash flows from within the forecast horizon plus the terminal value, divided by the total number of shares outstanding, thus yields the stock's fair value or target share price.

The discussion above suggests that it would be unwise to pour huge efforts into forecasting free cash flows very far into the future. Intuitively the analysts should instead focus on accurate forecasting of earnings and cash flows closer to the present, and then attempt to make a conservative estimate about the perpetual growth rate of the company used for calculation of the terminal value. This, however, is only if the analyst expects a low growth rate. The higher the expected growth rate, the more of the company's free cash flow will lie further into the future and so the higher the terminal value will be. By the same token, the more of the company's free cash flow that lies further into the future, the more sensitive to rate hikes the company's value will be. And finally, the higher the expected perpetual growth rate (assumed to be constant after the analyst's forecast period), the higher the terminal value.

OLD VS. NEW ECONOMY – A VALUATION EXAMPLE

As a simple example of the discounted cash flow approach to valuation presented above, assume that we have two listed companies. One is an 'Old Economy' airline company called Company A , and the other is a 'New Economy' computer networking company called Company B . For expositional purposes we assume that the two companies A and B each have 100 million dollars in free cash flow (FCF) in the present accounting year. We extend our forecasting horizon to 10 years. We then carry out forecast of the two companies' free cash flows. The FCF of Company A is expected to grow at a constant rate of 10% annually over the next 10 years. The FCF of Company B is expected to grow at a constant rate of 35% annually over the next 10 years. After the forecast period the FCF of both companies is expected to grow at an annual rate of 3%. For the sake of simplicity the discount rate for both companies is assumed to be 12%. In reality they would have different discount rates and it would be considerably more difficult to estimate free cash flows from Company B than from Company A . Furthermore, because of the high growth rate of Company B , its free cash flow will be more sensitive to small changes in discount rates than Company A , warranting a more structured sensitivity analysis.

Valuation and sensitivity analysis – Company A

Using the principles of discounting a series of payments and determining the terminal value thereafter, a simple DCF valuation of Company A yields the results shown in Table 4.4. The free cash flows grow at a constant 15% per year for the first 10 years, and as we can see the discounting effect causes the \$352 million in free cash flow after 10 years to be worth only \$127 million in present-day currency. The sum of the discounted free cash flows that have been explicitly forecast is only \$1.13 billion, while the terminal value is \$3.909 billion for a total ‘fair’ company value of \$5.038 billion. We can illustrate this point further by plotting the information from Table 4.4 in a chart (see Figure 4.2).

Note that even with a relatively low FCF growth rate of 15%, the vast majority of this company’s value is contained in its terminal value, again underscoring the significance of applying an appropriate discount rate for terminal value calculations. The analysts may have spent hours tweaking sales

Table 4.4 Valuation – Company A (‘Old Economy’)

Discounted Cash Flow, \$ Million		
Discount Rate		12.00%
Short-term Growth Rate		15.00%
Perpetual Growth Rate		3.00%
Year	FCF	Present Value of FCF
1	\$100	\$100
2	\$115	\$103
3	\$132	\$105
4	\$152	\$108
5	\$175	\$111
6	\$201	\$114
7	\$231	\$117
8	\$266	\$120
9	\$306	\$124
10	\$352	\$127
Total Present Value of FCF		\$1,130
Terminal Value		\$3,909
Company Value		\$5,038

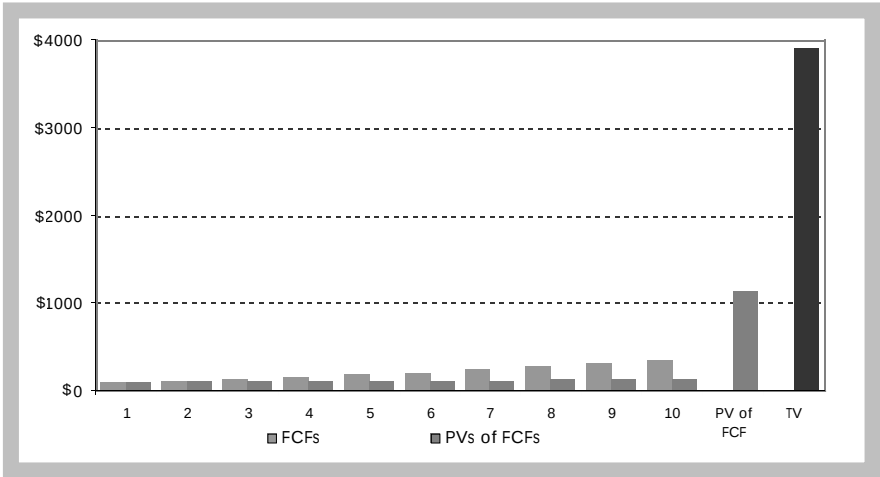


Figure 4.2 Company A: FCFs, present value of FCFs and terminal value

and profit forecasts for the company, but at the end of the day the size of the present value of free cash flows is less than one-third the size of the terminal value, even with a relatively low perpetual growth rate of 3% per year.

To gain some insight into the sensitivity of this valuation approach to changes in assumptions, we can perform a *sensitivity analysis* with respect to the discount rate, the short-term growth rate and the perpetual growth rate. Figure 4.3 illustrates the value of Company A in millions of dollars for

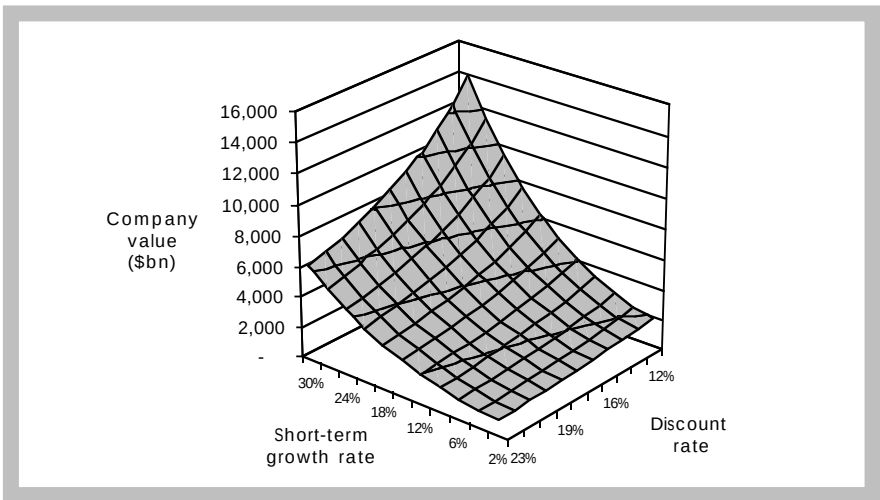


Figure 4.3 Sensitivity – Company A: short-term growth and discount rate

different combinations of short-term growth rates and discount rates, while holding the perpetual growth rate constant at 3%.

From Figure 4.3 it appears that the company valuation is more sensitive to changes in the short-term growth rate than to changes in the discount rate. Not surprisingly, there is an exponential relationship between company value and rates, because of the compounding effect mentioned earlier. The sensitivity to changes in the discount rate and the perpetual growth rate appears as shown in Figure 4.4.

From Figure 4.4 it appears that the company's value is less sensitive to changes in the perpetual growth rate than to changes in the discount rate, and also to changes in the short-term growth rate. This is because the short-term growth rate is held at a relatively low level of 15%, thereby limiting the compounding effect beyond the forecasting horizon, since the expected FCF after 10 years is relatively low.

Note how the effect of a higher perpetual growth rate is dampened for high discount rates, and thus how higher perpetual growth rates only have a significant effect on the company's value when the discount rate is lowered. This is due to the fact that for high discount rates the compounding effect on the free cash flows occurring after the analyst's forecast horizon is all but eliminated by the high discount rate. The same effect can be observed when high levels on a savings account are eliminated by high levels of inflation. The levels for the discount rate may not be very realistic. In fact one could argue that even the 12% to 15% range is quite high, but it does demonstrate clearly the interdependence between the model parameters, as well as their combined effect on valuation.

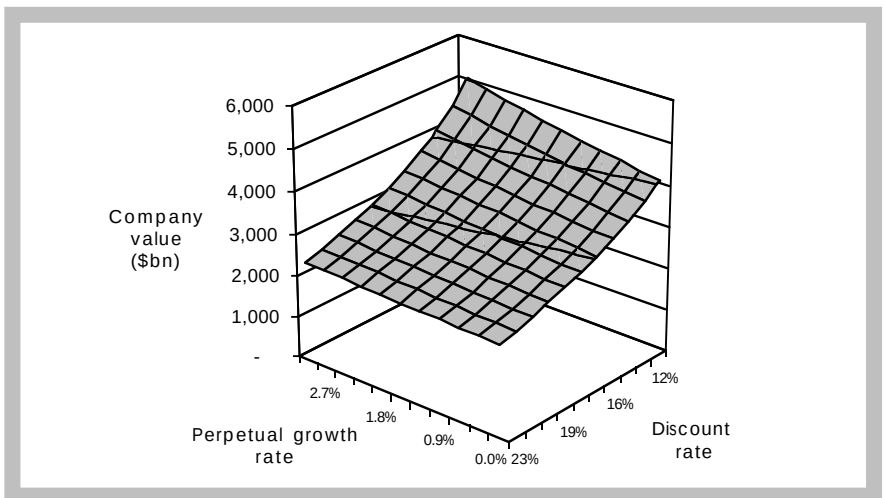


Figure 4.4 Sensitivity – Company A: perpetual growth and discount rate

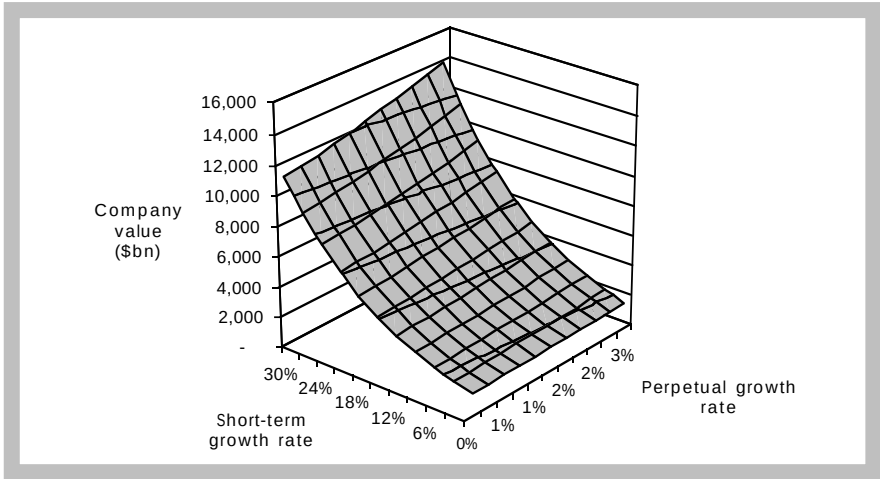


Figure 4.5 Sensitivity – Company A: short-term growth and perpetual growth

We now continue our sensitivity analysis by holding the discount rate constant at 12% and investigating Company A’s value when varying both the short-term growth rate and the perpetual growth rate (see Figure 4.5). From Figure 4.5 it is apparent that the company’s value is much more sensitive to changes in the short-term growth rate than to changes in the perpetual growth rate. While this may seem a bit surprising, the argument is essentially the same as that presented above. The compounding effect of the perpetual growth rate only comes forcefully into play if the short-term growth rate is of a reasonable size, since the free cash flow will then have reached a significant size after the forecasting period of 10 years. Once the compounding effect of a rising short-term growth rate begins to kick in at around 15%, the exponential effect of a higher perpetual growth rate becomes more pronounced.

Valuation and sensitivity analysis – Company B

Company B is the so-called ‘New Economy’ company, and has a much higher short-term growth rate than Company A. However, the higher growth rate is assumed to last only for a relatively short period. The reasoning behind this is that a company can only grow at a much faster pace than the rest of the economy for a limited number of years, after which its competitive advantage slowly disappears. Clearly a company would not be able to grow faster than the rest of the economy into infinity, since if it did the company would become

so large that eventually it would end up actually *being* the entire economy. Therefore the construction of scenarios that imply growth rates higher than those assumed for the rest of the economy must necessarily be limited to shorter time periods. Whether these are 5, 10 or 15 years is not very important. Performing a valuation of Company *B* equivalent to that performed on Company *A* we obtain the results shown in Table 4.5.

Table 4.5 Valuation – Company *B* ('New Economy')

Discounted Cash Flow, \$ Million		
Discount Rate		12.00%
Short-term Growth Rate		35.00%
Perpetual Growth Rate		3.00%
Year	FCF	Present Value of FCF
0	\$100	\$100
1	\$135	\$121
2	\$182	\$145
3	\$246	\$175
4	\$332	\$211
5	\$448	\$254
6	\$605	\$307
7	\$817	\$370
8	\$1,103	\$446
9	\$1,489	\$537
Total Present Value of FCF		\$2,665
Terminal Value		\$16,549
Company Value		\$19,214

As we can see, the 'New Economy' company has a much higher short-term growth rate, resulting in a much higher valuation. After 10 years the initial free cash flow of \$100 million has grown to almost \$1.5 billion, although after discounting this represents only a present value of \$537 million.

The discount rate is the same as for Company *A* so the present value of the last free cash flow in the 10-year period is as much as five times higher than that of Company *A*. This makes for a present value of discounted cash flows from the 10-year forecast period that is more than double that of Company *A*. The relative amount of the total company's value that is due to terminal value,

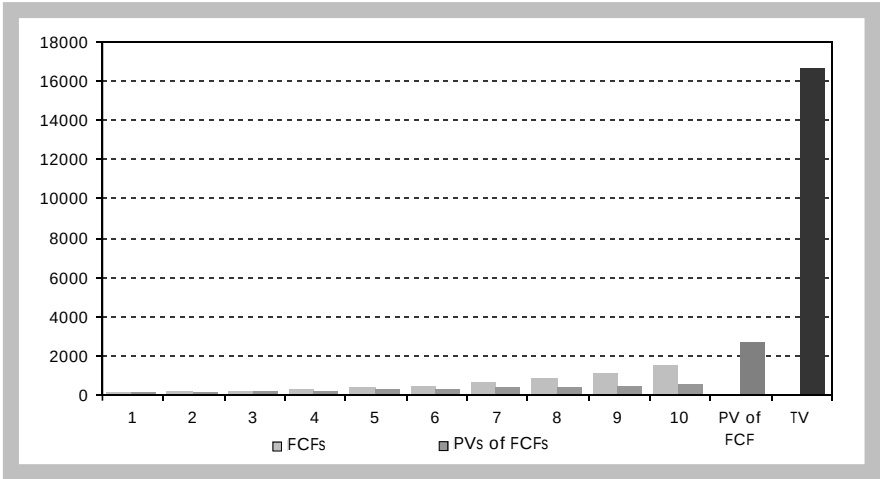


Figure 4.6 Company B: FCFs, present value of FCFs and terminal value

though, is even larger than was the case for Company A. Of the \$19.214 billion that this valuation procedure suggests as being the company's fair value, only \$2.665 billion or about 14% pertains to free cash flows that have been generated within the explicit forecasting horizon. Illustrated in a chart this becomes even more apparent (see Figure 4.6).

Again, the vast majority of the company's value thus lies beyond the forecast horizon, making the forecast quite uncertain. It can therefore be argued that even though the forecasting horizon is the same for both companies, the inherent uncertainty in valuation estimates is much higher for Company B than for Company A as a much larger amount of the value of the company lies far into the future. So the higher the short-term growth rates – and thereby the higher the apparent potential for investment return – the higher the inherent risk. In other words, the prospect of a much higher return on one investment relative to another investment naturally entails that we must incur a significantly larger amount of risk.

These results give a good indication of the generally thankless task faced by analysts in trying to forecast and value companies that are expected to experience high levels of growth in the short term. Several attempts at alternative valuation models have been put forward in recent years, but none of them seems to have taken hold within the analyst community – possibly because several of those models have been launched more for the purposes of publicity than anything else. Figure 4.7 presents a sensitivity analysis on Company B yields, similar to those presented above. As before we firstly examine what happens to the expected value of Company B as we vary the short-term

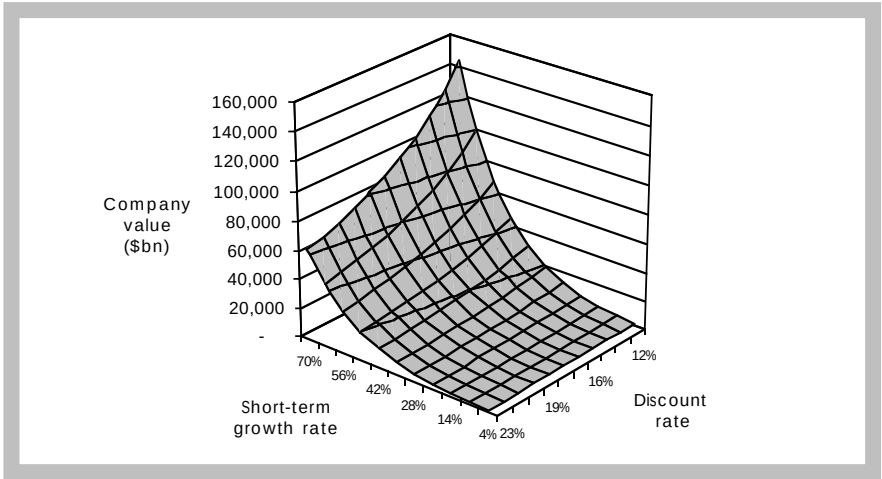


Figure 4.7 Sensitivity – Company B:
short-term growth and discount rate

growth rate and the discount rate, while holding the long-term growth rate constant at 3% (see Figure 4.7).

The first thing to notice is the much higher overall level of company value now that the short-term growth rate is set to 35%. This creates very large compounding effects within the 10-year forecast horizon, a much larger expected free cash flow after 10 years, and consequently a very much higher terminal value. As before, the sensitivity of the company's value towards changes in the short-term growth rate is greater than the sensitivity towards changes in the discount rate. This is to some extent due also to the fact that the absolute level of short-term growth is much higher than was the case for Company A, giving the compounding effect more room to explode. The sensitivity to simultaneous changes in the discount rate and the perpetual growth rate, holding the short-term growth rate constant at 35%, appears as in Figure 4.8.

As before, the company's value is less sensitive to changes in the perpetual growth rate than to changes in the discount rate, and also to changes in the short-term growth rate. In addition, the absolute value of the company is much lower in this sensitivity analysis than in the previous analysis. Even for a high level of short-term growth we do not observe a high company value, unless the discount rate is sufficiently low to allow for the compounding effect of higher short-term growth rates. Only at a discount rate level of around 15% does the increase in the perpetual growth rate and the short-term growth of 35% begin to have a significant effect. But the overall variability or sensitivity to changes in the two variables is still relatively low.

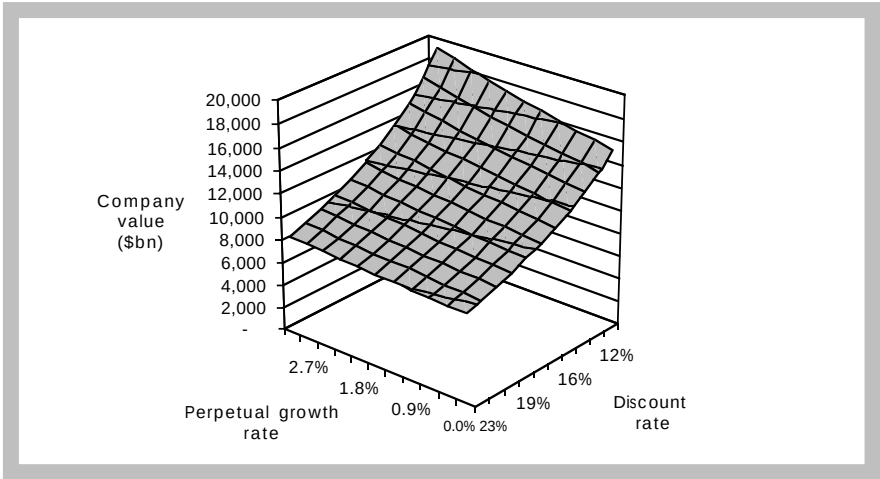


Figure 4.8 Sensitivity – Company *B*: discount rate and perpetual growth

Holding the discount rate constant at 12% and valuing the company while changing short-term growth rate and the perpetual growth rate yields the result shown in Figure 4.9. The general picture remains the same. Company value is very sensitive to changes in short-term growth rates and much less so to changes in the perpetual growth rate, mostly because of compounding effects.

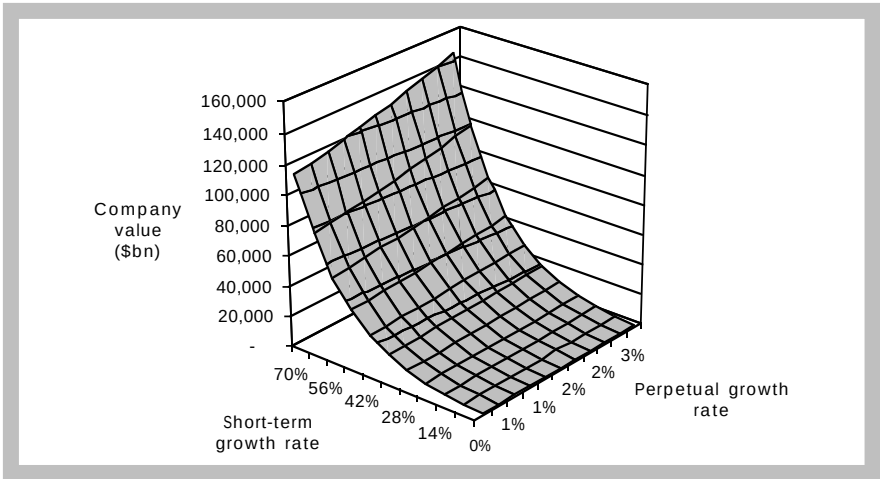


Figure 4.9 Sensitivity – Company *B*: short-term growth and perpetual growth

Independent sensitive analysis such as this could potentially provide investors with a useful supplement to standard company analyses published by brokerage firms. One could argue that a company analysis and valuation is only complete if it provides details of the assumptions made and of the sensitivity in estimates to changes in these assumptions. Having performed sensitivity analysis on fundamental model parameters such as the discount rate and growth rates, the thorough analyst would then also perform sensitivity tests on individual elements on the income statement, such as revenues, interest payments, and costs related to energy prices.

IMPLIED GROWTH RATES

Instead of working out the present value of an asset such as a stock from its expected future earnings or cash flows discounted at an appropriate rate, we can attempt to determine whether or not the growth the market is currently pricing into the stock is reasonable. If the stock price is high compared to its expected earnings per share, the investors are evidently pricing considerable growth into the stock – that is, they are discounting the ability of the company to grow its earnings per share considerably in the future. We can determine the exact earnings growth that is priced into a given stock by looking at its present earnings per share, then finding the earnings growth rate that will result in the current share price, if we grow present earnings at that particular rate over a fixed time period and then discount those earnings back to the present. The growth rate that results in the discounted value per share equal to the current share price is the growth rate that is currently being implied by that price. This can be further clarified by seeing that at that particular ‘equilibrium’ growth rate, the current market capitalisation of the company is equal to the sum of discounted future earnings of the company.

For expositional purposes we examine a 10-year period of constant earnings growth, and then a 5-year fade period to a *long run* or *perpetual* earnings growth rate equal to the growth rate of the entire economy. Again, we apply a perpetual growth rate lower than that of the economy, to avoid the asset actually *becoming* the economy. The pattern of high short-term growth, fading medium-term growth and low perpetual growth appears as in Figure 4.10.

Given the current level of earnings per share, an assumed discount rate and an assumed and fixed perpetual growth rate, we then calculate the short-term (10-year) earnings growth rate – with the resultant medium-term fading growth – that will ensure that the sum of all the discounted future earnings per share is equal to the current share price. That growth rate is then by definition the average 10-year earnings growth rate that the market is currently pricing into the stock. Having performed this calculation, the portfolio manager can

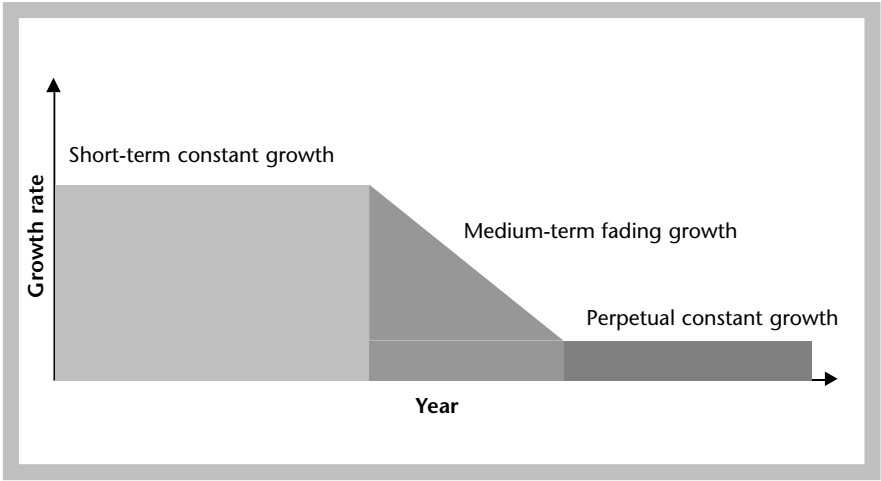


Figure 4.10 Framework for calculating implied growth rates

then decide for herself whether she believes this growth rate to be attainable. If the portfolio manager believes that the company can grow its earnings at that particular rate or higher over the next 10 years, then she should probably hold onto the stock or buy it if she does not already have it in her portfolio. If she believes the market is being too optimistic about the outlook for the company, then she should probably sell the stock.

This procedure is obviously very simplified, but the general idea is sound and a practical and easy way of assessing whether a stock is too expensive relative to its peers. The growth scheme presented in Figure 4.10 is of course completely arbitrary, and clearly there is no single correct growth scheme. One can construct schemes with many stages and many ways to fade in and out of growth phases. However, it is probably wise only to construct schemes that reflect the level and quality of relevant information available. For a semiconductor company, for instance, designing an elaborate multi-stage growth scheme covering many years may turn out to cause more confusion than enlightenment – the semiconductor sector is notoriously volatile, and only rarely do analysts have very robust demand and pricing forecasts out to or beyond two years. Keeping it simple may prove the best way forward, particularly since implied growth rates are often most useful for peer group valuation analysis of a group of similar stock.

Let us look at an example. On 1 May 2002, Sony’s shares closed in Tokyo at ¥6910, and the consensus or average analyst forecast for the earnings per share in fiscal year 2003 was ¥184. Assuming a perpetual growth rate of 2.5%, and a discount rate of 6% per year, the short-term implied growth rate that equates

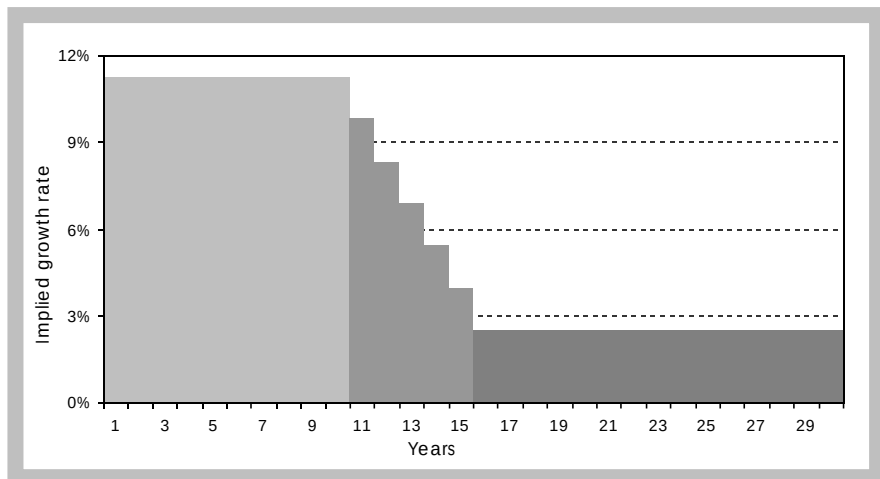


Figure 4.11 Implied 10-year growth rate, Sony Corp.

the sum of all the discounted future earnings per share to the current share price was 10.6%. The growth scheme thus appears as in Figure 4.11.

The annual earnings growth rate is 10.6% for 10 years, and then falls by around 1.45% for 5 years until it reaches the long-term constant level of 2.5% where it remains as we go towards infinity. The above outlined growth scheme yields future earnings and discounted future earnings as shown in Figure 4.12.

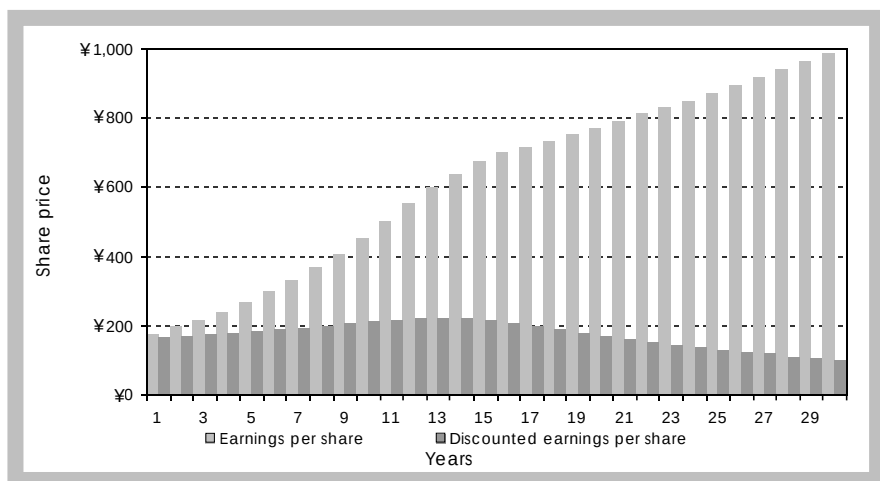


Figure 4.12 Discounted earnings per share, Sony Corp.

Figures 4.11 and 4.12 include the first 30 years of the growth scheme, but of course the scheme extends out to infinity and the discounted future earnings indicated by the grey bars trend towards zero. The sum of all the discounted future earnings from year 1 onwards yields exactly ¥6910, which was the share price on 1 May 2002. So with the assumptions made about the form of the growth scheme, the discount rate and the long-term growth rate, the markets appeared to be pricing in 10.6% earnings growth over the subsequent 10 years, which would appear to be realistically achievable for Sony. The company's shares therefore did not seem to be overvalued.

Note how earnings per share increase at an increasing rate. This is due to the compounding effect caused by the constant 10.6% growth rate for the first 10 years. Discounted earnings also rise because the short-term earnings growth rate is higher than the discount rate. After the short-term high growth period, growth fades and discounted earnings peak out and start to drop off, as the medium-term growth rate trends towards, and finally below, the discount rate. After 15 years the discounted earnings fall every year as the discount rate is now significantly higher than the earnings growth rate, causing a progressively lower present value of future earnings.

As already demonstrated, valuation analysis employing discounting in any form is usually very sensitive to changes in assumptions. The same is true for this example, and Figure 4.13 illustrates a sensitivity analysis carried out on the implied growth rate example above. From Figure 4.13 it is evident that even relatively small changes to assumptions, and to the discount rate in particular, have potentially very large effects on the fair value estimate. Hence

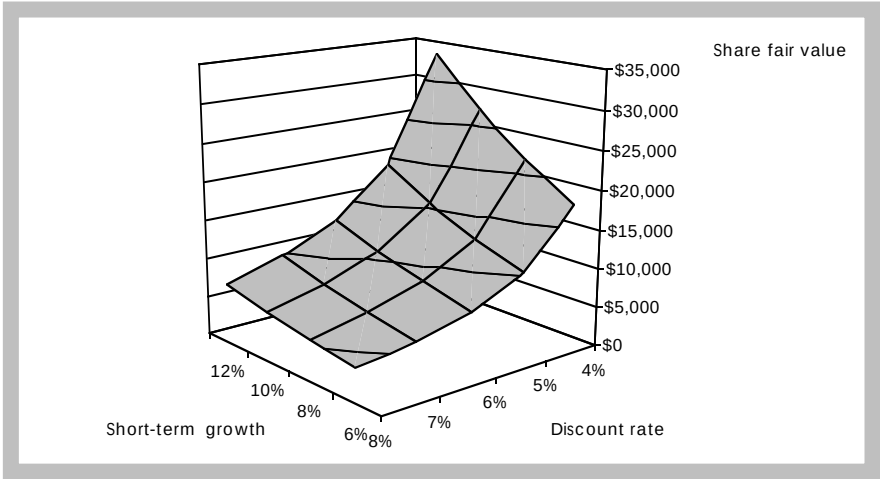


Figure 4.13 Sensitivity analysis – Sony Corp.

the need for a sensitivity analysis, which can be an important supplement to any valuation estimate using some form of discounting.

THE CAPITAL ASSET PRICING MODEL (CAPM)

Pricing based on the market

We now return briefly to a slightly more traditional tool. The Dividend Discount Model and the Discounted Cash Flow Model rely on explicit forecasting of potentially thousands of company-specific characteristics, and many of these estimates will be inherently subjective in nature. Furthermore, working with these models requires a large amount of resources, which it may be difficult to allocate, especially when dealing both with portfolios of existing stocks and with potential stocks to include. As asset managers we would therefore like a pricing framework that is simple yet robust and that applies to all assets regardless of the nature of the underlying activities.

One such theory, which has proven extremely robust and rugged since its birth in 1964, is the *Capital Asset Pricing Model* (CAPM) which we have already touched upon. It basically proposes that an asset's return can be described completely by a combination of a market return and the asset's co-variation with that market. Its logic is simple. The idea is that investors are compensated for taking on necessary risk but not for taking on unnecessary risk. It provides a framework for separating risk into necessary (systematic or market-related) risk, and unnecessary (unsystematic, asset-specific or residual) risk. The CAPM simply postulates that a linear relationship exists between the return on an asset and the return on the market, and that asset returns can thus be explained by a single factor, namely the market return.

THE SECURITY MARKET LINE (SML)

We have already introduced an asset's Beta as a risk measure. The fact that in a portfolio context the only risk we are interested in is the asset's contribution to *systematic* risk – since its *unsystematic* risk can be diversified away by adding ever more assets to the portfolio – leaves Beta as an attractive yet simple tool. Bearing in mind that the only explanatory variable in the CAPM model is the market return, we can illustrate this by defining the return on asset A as:

$$r_{A,n} = a_n + b_{A,n} \cdot r_{M,n} + \quad [\text{Eq. 4.9}]$$

where $r_{A,n}$ is the return on asset A at time n , a_n is a constant term for the linear relationship between asset return and market return, $r_{M,n}$ is the return on the

market at time n , $b_{A,n}$ is the slope of the linear relationship at time n , and $\epsilon_{A,n}$ is the error term capturing deviations from that relationship. The variance of the return on asset A is thus:

$$\begin{aligned} \text{Var}(r_{A,n}) &= \text{Var}(a_A + b_A \cdot r_{M,n} + \epsilon_{A,n}) \\ &= \text{Var}(a_A) + \text{Var}(b_A \cdot r_{M,n}) + \text{Var}(\epsilon_{A,n}) \end{aligned}$$

But since a_A is a constant, its variance is zero, so we obtain:

$$\text{Var}(r_{A,n}) = \text{Var}(b_A \cdot r_{M,n}) + \text{Var}(\epsilon_{A,n}) \quad [\text{Eq. 4.10}]$$

This shows, perhaps not surprisingly, that the risk of an asset can be divided into *systematic* (market-related) risk and *unsystematic* (residual or asset-specific) risk:

$$\text{Var}(r_{A,n}) = \text{Systematic Risk} + \text{Unsystematic Risk} \quad [\text{Eq. 4.11}]$$

We shall use this result later on when we examine risk and a portfolio versus benchmark context. The market-related risk (systematic risk) is inescapable and must be borne by all investors. The non-market-related risk (residual risk) can be diversified away, and as a consequence – to the extent that it exists in a portfolio – it is self-imposed by investors who expect a returns above that of the market.

A risk-free asset will yield the risk-free rate of return, while a risky asset will yield the risk-free rate of return *plus* some risk premium caused by a systematic and thus non-diversifiable amount of risk determined by the co-variation of the risky asset with the market. As introduced earlier, the straight line that indicates the assumed linear relationship between asset return r_A and its covariance with the market is called the *Security Market Line* (SML): it is shown in Figure 4.14.

Figure 4.14 shows how a risk-free asset will yield the risk-free rate of return, and as the covariance of the asset with the market increases, so does its expected return. Formally this relationship is expressed by:

$$r_A^x = r_f + \frac{r_M - r_f}{\sigma_M^2} \cdot \text{Cov}(r_A, r_M) \quad [\text{Eq. 4.12}]$$

Remember that Beta is defined thus:

$$\beta_A = \frac{\text{Cov}(r_A, r_M)}{\text{Var}(r_M)} = \frac{\text{Cov}(r_A, r_M)}{\sigma_M^2} \quad [\text{Eq. 4.13}]$$

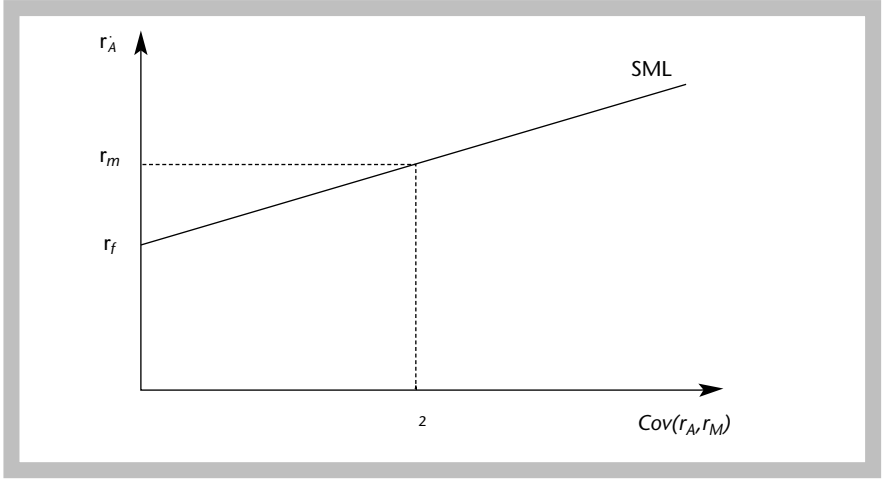


Figure 4.14 The Security Market Line

Rearranging terms and denoting the slope of the SML β_A yields the expression for the expected return on asset A as given by the single-factor CAPM model:

$$r_A^{\times} = r_f + \beta_A \cdot (r_M - r_f) \quad [\text{Eq. 4.14}]$$

An asset whose covariance with the market is high – one that has a high level of systematic and thus non-diversifiable risk – will need to deliver a higher rate of return in order for investors to buy it. This risk premium above the risk-free rate depends on the asset's Beta, and the CAPM thus postulates that the return on any asset is proportional to its risk relative to the market (quantified by Beta), and that expected residual returns are equal to zero. The revised SML from Equation 4.13 with Beta along the x -axis appears in Figure 4.15. This figure basically illustrates the same Security Market Line as before, except that instead of covariance we use the 'risk-standardised covariance term' Beta along the x -axis. Please note that the market has a Beta of 1 and that the risk-free asset has a Beta of 0. If the CAPM holds, all assets will lie on the SML, since any deviations will be arbitrated away – in a perfectly efficient market, at least. To see this, consider the following example. Suppose that for an asset A we observe that it has a return r_A that is above the SML – that is, its actual observed return is above its expected return based on the residual or inherent risk of that asset. Arbitrageurs will then exploit this discrepancy by buying the asset, because its risk-adjusted return is high. This will push up the price of the asset, resulting in a lower expected rate of return for additional

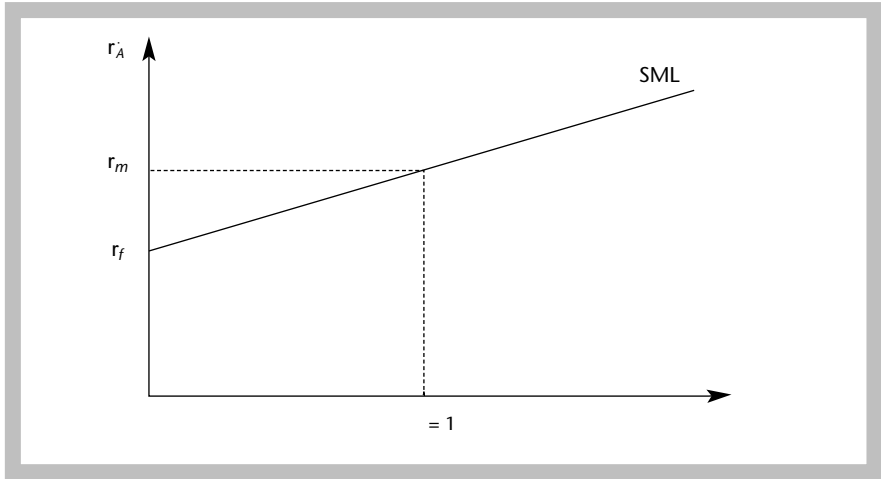


Figure 4.15 The revised Security Market Line

potential buyers. This process of arbitrage will continue until the asset has been forced down to the SML.

Figure 4.15 also hints at the fact that the CAPM is basically a model to explain *expected* returns. Ex ante we use the CAPM to try to predict expected returns, and the results the model delivers are expected asset returns that all lie exactly on the SML.

An interesting point can be made concerning this issue. If we observe returns ex post of individual stocks, we find that they do *not* in fact lie on the SML, but rather that they are scattered around the line; and the deviations from the line are captured in the error term of the regression in Equation 4.9. One could argue that this does not mean that the CAPM does not hold, but simply that our forecasts of the market returns may be inaccurate. But in order to investigate whether the market returns have been accurately forecast, we need to forecast the returns on the market’s constituent assets, which we cannot do without the CAPM and the market returns. We are thus faced with a ‘Catch 22’ which leads us to the conclusion that the CAPM is impossible to disprove. Furthermore in order to predict asset returns using the CAPM, we are dependent on our ability to forecast Betas, which in itself is not an easy task.

THE CHARACTERISTIC LINE (CL)

An estimate of the exact empirical relation between an asset and the market is often called the *Characteristic Line* (CL), and is produced via univariate regres-

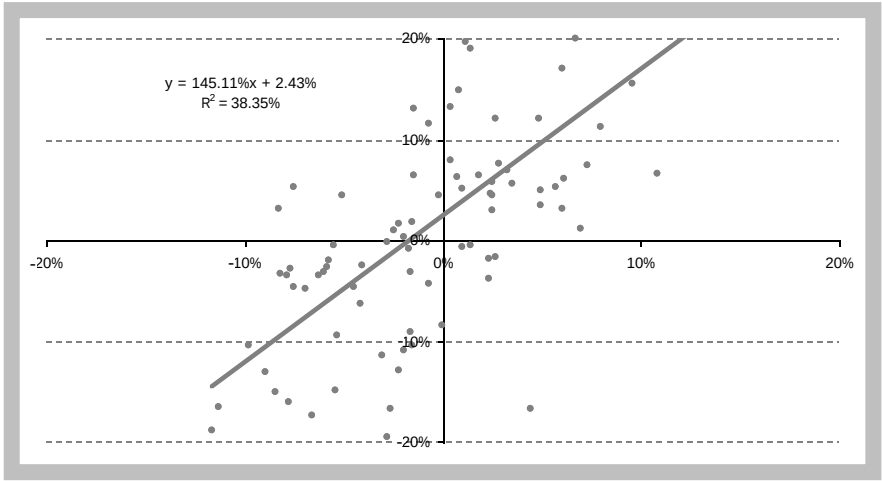


Figure 4.16 The Characteristic Line – Sony vs. TOPIX, 1995/1–2002/5

sion analysis. Because of the simplifying assumption of just one explanatory variable in the CAPM-like Characteristic Line, there is usually a wide dispersion among the estimated parameters. In addition, since there is no rule for the choice of market to be used in the regression, a wide range of asset Betas can be estimated – though obviously it seems wise to use the benchmark against which the portfolio manager measures himself. In any event, the most important thing is if possible to use the same benchmark for all estimates, thus ensuring some level of consistency in the estimates. In Figure 4.16 we illustrate the Characteristic Line (regressions results) of the share price of Sony Corp. listed in Japan relative to the Tokyo Stock Exchange TOPIX Index. We use monthly returns from January 1995 to April 2002.

Note that the chart in Figure 4.16 is essentially identical in nature to the chart of the returns on the stock of IBM, presented in Figure 2.9. The estimated Beta of Sony is 1.45, indicating that the stock is more risky than the market. Note however that the returns are quite widely dispersed around the regression line, indicating a rather poor fit, which is also borne out by an R^2 of just 0.38.²

The relatively low explanatory power of these types of regressions is not unusual for the CAPM, since it is a single-factor model and thus not able to capture all of the variation in asset returns. If we want to be able to price an asset more accurately by being able to explain more of the assets variation over time, we need a model that considers more factors than just the market return.

THE ARBITRAGE PRICING THEORY (APT)

Enhancing the CAPM

The limits to the explanatory power of the CAPM of course stem from the fact that just one explanatory variable is used in the model. As we shall see later, this – along with the implicit assumptions of the CAPM – will facilitate the establishment of a solid portfolio analysis framework.

For now, however, we are interested in explanatory power, and the only way to increase this in regression analysis is by adding more explanatory variables – that is, by going from univariate to multivariate analysis. This is in fact the basic idea behind the *Arbitrage Pricing Theory* (APT) which was developed by Ross (1976–77). It can be thought of as a generalised CAPM, in that it considers a potentially infinite number of variables, whereas the CAPM considers only one, the market return. It is a significantly more practical theory than the CAPM, primarily because it is very flexible but also because it is essentially statistical in nature. Contrary to the CAPM it does not rely on strong assumptions, but simply attempts to characterise and predict future asset returns from a number of factors. The CAPM with its implicit assumption of just one market works when considering the entire world market of not only equities but bonds, cash, art, real estate, forests, golf courses, and so on. But when working with portfolios that are restricted to a certain benchmark, such as the S&P500, the CAPM is clearly not adequate.

The Arbitrage Pricing Theory states that the expected excess return on an asset is approximately a linear function of the *risk premiums*³ on systematic factors in the economy. The basic Arbitrage Pricing Theory assumes that the excess return on a asset is related to or explained by the returns on K factors such that:

$$r_i^{\times} = \beta_{i,1} \cdot (f_1 - r_f) + \beta_{i,2} \cdot (f_2 - r_f) + \dots + \beta_{i,k} \cdot (f_k - r_f) \quad [\text{Eq. 4.15}]$$

where

$r_i^{\times}$ is the expected excess return on asset i

r_f is the risk-free rate of return

$\beta_{i,k}$ is the sensitivity of asset i to risk factors k

f_k is the return on risk factor k

$(f_k - r_f)$ is the risk premium (excess return) associated with risk factor k

Equation 4.15 simply states that the excess return on any asset can be explained by the excess returns on a number of factors, multiplied by their

respective sensitivity coefficients with regard to the asset in question. Similarly to the Capital Asset Pricing Model, the Arbitrage Pricing Theory is a normative equilibrium theory. If all assets returns behaved according to Equation 4.15, then we would observe the empirical relation:

$$r_{i,t} = \alpha_i + \beta_{i,1} \cdot (f_{1,t} - r_{f,t}) + \beta_{i,2} \cdot (f_{2,t} - r_{f,t}) + \dots + \beta_{i,K} \cdot (f_{K,t} - r_{f,t}) + \epsilon_{i,t} \quad [\text{Eq. 4.16}]$$

where

$r_{i,t}^\times$ is the expected excess return on asset i at time t

α_i is the non-factor-related or asset-specific return on asset i

$r_{f,t}$ is the risk-free rate of return at time t

$\beta_{i,k}$ is the sensitivity of asset i to risk factor k

$f_{k,t}$ is the return on risk factor k at time t

$(f_{k,t} - r_{f,t})$ is the risk premium associated with risk factor k at time t

$\epsilon_{i,t}$ is the residual return on asset i at time t

Both the CAPM and the APT indicate that it is possible to forecast expected stock returns, which is one of the elements we shall need when constructing a quantitative asset allocation and portfolio risk-management framework. However, contrary to the CAPM the APT does not tell us exactly how this should be done. The choice of factors to include in the model (and thereby also the ones to leave out) is left completely to the analyst to decide, and for this reason the model has sometimes been referred to as the *Arbitrary* Pricing Theory.

It has been said that the CAPM comes with a user's manual, whereas the APT leaves everything to the analyst. What this implies is the fact that there simply are no ex ante guidelines for selecting explanatory factors, and since we can identify statistical relations between almost any set of variables, the choice naturally becomes very subjective and difficult. Therefore one may argue that the application of multi-factor models such as the APT is more an art than a science, as there simply is no one correct answer.

The APT does offer large advantages to simpler models such as the CAPM, but it is argued here that both models are very valid, but in different situations. The CAPM may be used to establish a framework for asset management and risk relative to a market or benchmark. As we will see later, this can prove valuable for the process of active portfolio management. In a forecasting context, however, the APT is clearly the preferable model of the two, as it can produce forecasts that are fairly accurate due to the inclusion of many variables: it thus has a high level of explanatory power.

SUMMARY

In order to construct a workable asset allocation and risk-management framework that is based on quantitative techniques, we need estimates of both return and risk. Chapters 3 and 4 have attempted to provide a basis for generating these estimates. In this chapter the return estimates are based firstly on explicit valuation techniques such as the Dividend Discount Model and the Discounted Cash Flow Model. Secondly we have demonstrated how the discounted cash flow approach can be applied backwards in estimating the level of short-term earnings growth that is currently being priced into a given stock. Thirdly we have presented models that can be used to estimate both risk and return directly by relating a particular asset to the market as a whole or to a large number of explanatory factors.

Having thus established a framework for evaluating and characterising single assets, in the following chapters we move on to considering return and risk in a portfolio context, where many assets need to be considered. As we shall see, this becomes rather more complicated, but the investment insights that can be derived from such an exercise can be considerable.

Notes

- 1 We shall return to the CAPM later in this chapter.
- 2 The R^2 of a regression is expressed as 1 minus the variance of the error term divided by the variance of the asset return, thus indicating the degree to which the explanatory variable actually explains the asset return. It follows from the definition that an R^2 of 1 indicates a perfect linear relation between the market return and the asset return.
- 3 Conceptually the risk premium is the extra return that the investor demands for holding a risky asset over the risk-free asset. In pricing models this implies that the risk premium of an asset is equal to the return on that asset minus the risk-free rate of return, which is in fact the asset's *excess return*.

PART II

MODERN PORTFOLIO THEORY

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PORTFOLIO CHARACTERISATION

INTRODUCTION

The basis of investing is the desire to obtain (excess) returns on the investment. This excess return above the risk free rate necessarily implies taking on risk. We therefore need to be able quantify both return and risk on the portfolio level. We have already introduced return and risk in a *single-asset* context, so we now move on to *portfolios* of assets. In this chapter we characterise portfolios in terms of return and risk, and we will find that it is a rather more complicated matter than is the case for single assets, particularly with regards to risk. This stems from the fact that asset returns are correlated, and for this reason we need to formally introduce the *correlation coefficient* to quantify precisely the correlation among the returns on portfolio assets. These coefficients will then enable us to describe the combined returns on the portfolio's assets, and thereby the risk of the portfolio.

The tool that we shall employ extensively in this and the following chapters is the well-known *Modern Portfolio Theory*, which forms the basis of virtually all quantitative portfolio management and theory today. The theory originally dates back to 1952, when Harry Markowitz published his article on what he called 'portfolio selection'. In this article he established a framework for describing portfolios of assets in terms of the means on their returns, the variance of their returns, and the correlation between the returns on assets. For this reason the approach is also known as *mean-variance analysis*. Since its

formulation half a century ago it has been seized on by the investment industry as a workable tool for investment and risk management, in particular because of its simplicity and intuitive appeal, and it remains one of the cornerstones in the foundation on which today's asset management industry rests. There are, however, several potentially significant caveats that one must be aware of when employing modern portfolio theory, and we shall return to those later in this book. For now, however, we shall focus on characterising portfolio return and risk.

PORTFOLIO RETURN – THE SUM OF ITS PARTS

Introduction

Portfolio return is fairly easy to calculate: it is simply the weighted sum of the returns on the portfolio's constituent assets. For a two-asset portfolio with returns on asset *A* and asset *B* of 8% and 12% respectively, and with both assets having equal weights in the portfolio, the return on the portfolio is simply 50% of 8% plus 50% of 12%, which equals 10%.

In general, the portfolio return r_P given N assets with portfolio weights w_n and returns r_n can thus be expressed as:

$$r_P = \sum_{n=1}^N w_n \cdot r_n = (w_1 \cdot r_1) + (w_2 \cdot r_2) + .. + (w_n \cdot r_n) + + (w_N \cdot r_N)$$

[Eq. 5.1]

The return on each asset in the portfolio is thus simply multiplied by the proportion or weight with which that particular asset is held in the portfolio. For large portfolios it is more appropriate to use matrix notation, both for reasons of computational ease and in order to simplify derivations of portfolio risk and return equations later on.¹

Portfolio return – matrix notation

The use of matrix notation and matrix calculus enables us to work much more easily with portfolio characterisation, particularly with regard to portfolio risk, so we convert the above equation to matrix notation. Let $\vec{r}$ be the vector of annual returns on the portfolio's assets, $\vec{w}$ the vector of asset weights and $\vec{w}^T$ the transposed weight vector. Arranging the data in this manner enables simultaneous multiplication and addition of asset weights and returns, yielding the total return on the portfolio consisting of N assets. Using matrix notation, Equation 5.1 is thus rewritten:

$$r_p = \vec{w}^T \cdot \vec{r} = \begin{bmatrix} w_1 \\ \vdots \\ w_n \\ \vdots \\ w_N \end{bmatrix}^T \cdot \begin{bmatrix} r_1 \\ \vdots \\ r_n \\ \vdots \\ r_N \end{bmatrix} = [w_1 \dots w_n \dots w_N] \cdot \begin{bmatrix} r_1 \\ \vdots \\ r_n \\ \vdots \\ r_N \end{bmatrix} \quad [\text{Eq. 5.2}]$$

From this it follows that portfolio returns are linearly related to both individual asset weights and returns. However, the characterisation of a portfolio, perhaps for performance measurement purposes, in terms of portfolio return only is only half the story. The other half is the risk incurred in order to obtain that return, so we also need to describe portfolio risk quantitatively.

PORTFOLIO RISK – LESS THAN THE SUM OF ITS PARTS

Introduction

The *risk* of an asset has already been characterised as the variation over time of its returns relative to the mean of those returns, and this is quantified as *volatility* – the annualised standard deviation of asset returns. The calculations required for that task were relatively simple and straightforward. However, when considering portfolios of more than one asset, things become a bit more complicated since then – unlike when calculating portfolio return – the portfolio's risk is not just a simple (weighted) sum of its parts. That is, the nature of the portfolio – which is essentially just a number of assets – can yield aggregate portfolio characteristics that differ significantly from what one would expect, given individual asset characteristics. This is due to the variation over time of the individual assets in the portfolio relative to one another – their *co-variation*. As we demonstrated earlier, we can describe the co-variation over time between two assets relative to each other by calculating covariance. We now revisit this concept.

Covariance and correlation

As indicated earlier when introducing Beta as a measure of risk relative to the market, the co-variation between the return on an asset and the return on the market can be described mathematically by the term *covariance*:

$$\text{Cov}(r_A, r_M) = \frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)(r_{M,t} - \bar{r}_M) \quad [\text{Eq. 5.3}]$$

The market can be considered an asset with its own expected return and volatility, so it follows that Equation 5.3 can be generalised to describe the covariance between any two assets A and B by:

$$\text{Cov}(r_A, r_B) = \frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)(r_{B,t} - \bar{r}_B) \quad [\text{Eq. 5.4}]$$

The covariance of asset returns is an *absolute* measure of co-variation between two assets. This means that it does not consider the magnitude of the deviations from the means. We can therefore usefully standardise the covariance term by adjusting for the magnitude of deviations from the assets' respective means. This magnitude is quantified by the standard deviation of returns on the two assets, so to standardise the covariance term we simply divide it by the multiple of the two assets' standard deviations, which yields the *correlation coefficient*. The correlation coefficient ρ_{AB} between the returns on asset A and the returns on asset B is thus expressed by:

$$\begin{aligned} \text{Corr}(r_A, r_B) &= \frac{\frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)(r_{B,t} - \bar{r}_B)}{\sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)^2} \cdot \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_{B,t} - \bar{r}_B)^2}} \\ \text{Corr}(r_A, r_B) &= \rho_{AB} = \frac{\text{Cov}(r_A, r_B)}{\sigma_A \cdot \sigma_B} = \frac{\sigma_{A,B}}{\sigma_A \cdot \sigma_B} \quad [\text{Eq. 5.5}] \end{aligned}$$

We have already pointed out that the covariance of an asset's return with itself is equal to its variance. Examining Equation 5.5 we see that this implies that the correlation coefficient always lies between -1 and $+1$.

Please note that we are using asset return *standard deviations*, not volatilities, in the denominator, because the magnitude of covariance and correlation is not dependent on the length of the time period over which these are estimated. Therefore the standard deviations do not need be annualised.

The concept of diversification

Before we formally introduce volatility in the portfolio context, we first look briefly at three two-stock portfolios to get a sense of the nature of the diversification effects that can exist between asset returns.

As mentioned above, the risk of a portfolio, quantified by its volatility, is heavily dependent on the exact nature and magnitude of the covariance or correlation between asset returns. If the returns on assets in the portfolio are correlated, there may exist opportunities for reducing the level of total portfolio risk by selecting appropriate assets and asset weights, in an attempt to offset individual asset risks against each other. In other words we attempt, in a structured manner, to exploit the fact that asset returns often move in somewhat consistent patterns relative to each other. For example, if demand for autos is brisk and the stocks of auto manufacturers move up, so will the stocks of suppliers of auto parts, because they both depend on the end demand for autos. At least, that is what would happen in an ideal world. The returns on the two stocks are positively correlated, which means there is no significant benefit from diversification by holding both types of stock. However, holding auto stocks along with stocks of utilities companies – which are somewhat more defensive in nature, and thus tend to move down whenever consumer-oriented stocks such as autos move up – will most likely result in reduced variation in total portfolio return, simply because the returns on the two types of stocks tend to cancel each other out. This is the essence of diversification.

Imagine a simple two-asset portfolio with weights of 50% on both assets. Figures 5.1–5.3 illustrate the importance of considering the correlation between asset returns when considering portfolio risk. Figure 5.1 shows the prices of two stocks over 12 time periods beginning at 0 and ending at 12.

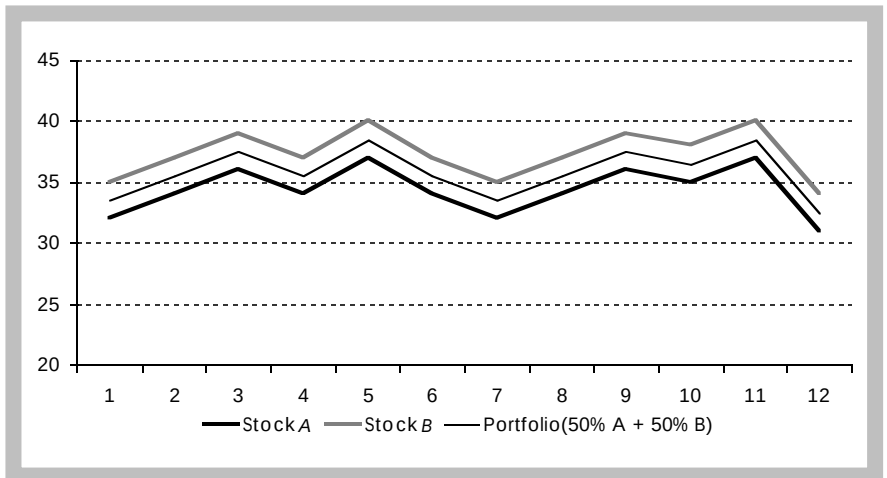


Figure 5.1 Correlation coefficient of +1

The two stocks are in perfect *directional sync* in the sense that when one stock rises the other stock rises as well, and so their prices exhibit perfect correlation.

When the price of Stock A rises so does the price of Stock B, and when the price of Stock A falls the price of Stock B falls. In general this implies a positive relationship between the return on the two assets, and in this particular case the correlation coefficient is actually $+1$, indicating that there is perfect correlation between the two stocks. However, this says nothing about the *magnitude* of the respective moves, and neither does it say anything about the *causality* between the two stocks. In other words, we cannot conclude from Figure 5.1 for example that Stock B follows Stock A or vice versa. The correlation coefficient simply tells us that the two stocks are *directionally* correlated.

Now consider a portfolio consisting of Stock A and Stock C. Stock C behaves differently from Stock B, and as a consequence it also behaves differently relative to Stock A. This means that a portfolio of these two stocks will exhibit different characteristics than the portfolio of Stock A and Stock B. The portfolio of stocks A and B appears over time as shown in Figure 5.2.

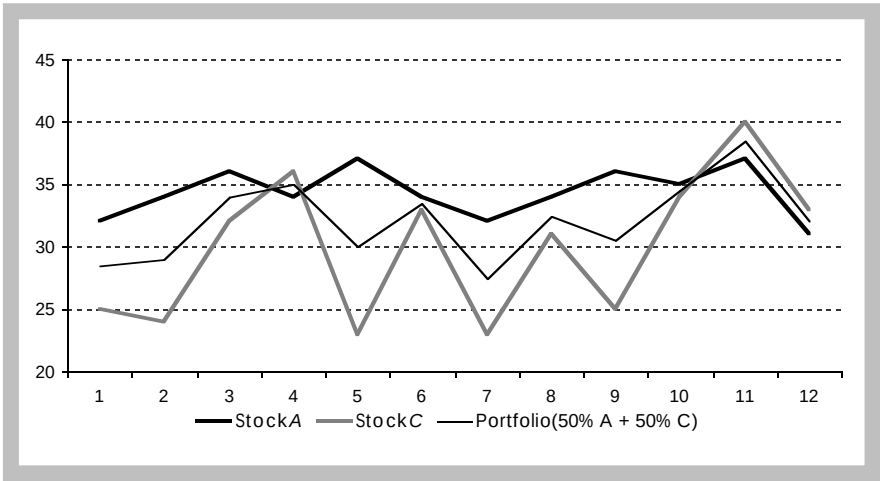


Figure 5.2 Correlation coefficient of 0

As can be seen from Figure 5.2, the prices of the two stocks apparently move independently of each other. There appears to be no relation between them over time, and in fact their correlation coefficient is zero, indicating that mathematically speaking they exhibit no correlation. Both stocks, however, are clearly risky since they move up and down over time, and this has an effect on the portfolio as well. Please note that this should *not* be taken to mean that the two stocks always move independently of one another. This point is underscored by the fact that there are times when Stock A and Stock C move in opposite directions, thus reducing portfolio risk; but there are also times when they move in the same

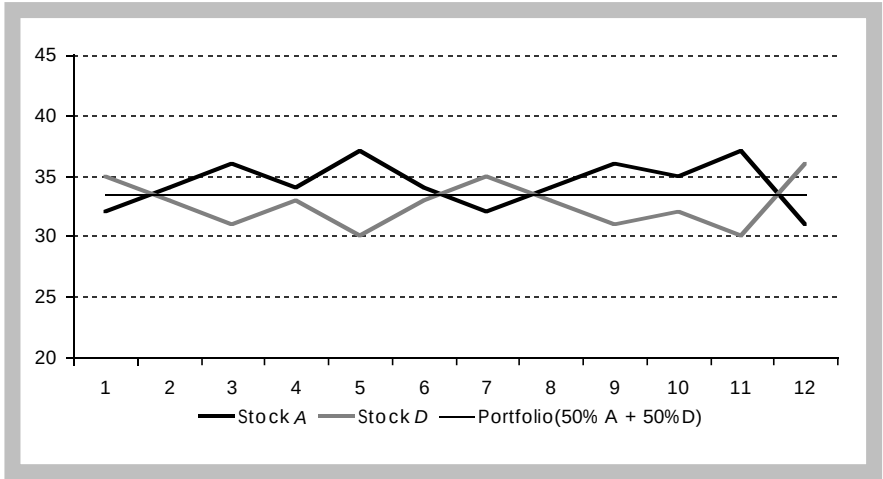


Figure 5.3 Correlation coefficient of -1

direction, increasing portfolio risk significantly. Stock C is much more volatile than Stock A, but the combined portfolio is significantly less volatile than Stock C. Much of the risk in Stock C returns are thus diversified away by Stock A.

Finally, consider a portfolio of Stock A and Stock D, whose returns have a correlation coefficient of -1 . The chart of their respective prices during the 12 periods is shown in Figure 5.3. Note how Stock D always moves in the opposite direction from Stock A. And not only that, but the two stocks also move by exactly the same amount, resulting in a completely riskless portfolio. Both stocks exhibit both up and down moves over time, implying individual stock risk, but since the portfolio contains exactly 50% of each stock and since their correlation coefficient is -1 , the portfolio risk ends up being exactly zero. We have essentially eliminated all risk from the portfolio by selecting asset weights that ensure that the negative correlation characteristics of the assets in the portfolio are fully taken advantage of. This may not be particularly realistic, but it does demonstrate the principle of diversification quite nicely. Below we shall go into detail about how to quantitatively assess the level of risk of a portfolio.

Portfolio volatility – two assets

As demonstrated above, when dealing with risk in a portfolio context the most important aspect to consider is the way in which the assets in the portfolio move relative to each other – that is, how their returns are correlated. We therefore need to quantify how and to what extent the movement of each asset relates to the movement of every other asset in the portfolio. This potentially

gives the portfolio manager the opportunity to diversify the portfolio and to reduce risk, since holding similar amounts of two assets that generally move in opposite directions will tend to eliminate or at least to reduce the risk of the portfolio as a whole.

Using the foundation established in Chapter 3, the variance of the return on a portfolio P of assets A and B can be expressed as:

$$Var_P = \frac{1}{T-1} \sum_{t=1}^T (r_{P,t} - \bar{r}_P)^2 \quad [\text{Eq. 5.6}]$$

which is identical to the definition of the variance of a single asset, except that here, instead of using the returns on a single asset, we calculate risk on the basis of the returns on the entire portfolio r_P . Equation 5.6 can be used for determining the volatility of a portfolio, provided we have the total portfolio returns, but here we are interested in decomposing portfolio volatility. Assuming that the portfolio P holds just two assets A and B , the variance expression in Equation 5.6 can be expanded to specify the variance of the two-assets portfolio, with asset weights at time t equal to $w_{A,t}$ and $w_{B,t}$ respectively:

$$\begin{aligned} Var_P &= \frac{1}{T-1} \sum_{t=1}^T \left[(w_{A,t} \cdot r_{A,t} + w_{B,t} \cdot r_{B,t}) - (w_{A,t} \cdot \bar{r}_A + w_{B,t} \cdot \bar{r}_B) \right]^2 \\ &= \frac{1}{T-1} \sum_{t=1}^T \left[(w_{A,t} \cdot r_{A,t} - w_{A,t} \cdot \bar{r}_A) + (w_{B,t} \cdot r_{B,t} - w_{B,t} \cdot \bar{r}_B) \right]^2 \\ &= \frac{1}{T-1} \sum_{t=1}^T \left[(w_{A,t} \cdot r_{A,t} - \bar{r}_A) + (w_{B,t} \cdot r_{B,t} - \bar{r}_B) \right]^2 \\ &= \frac{1}{T-1} \sum_{t=1}^T (w_{A,t}^2 \cdot Var_A + w_{B,t}^2 \cdot Var_B + 2 \cdot w_{A,t} \cdot w_{B,t} \cdot Cov_{A,B}) \quad [\text{Eq. 5.7}] \end{aligned}$$

From this we see that portfolio risk depends on individual asset weights and on individual asset risk plus some measure of the co-variation between asset returns expressed by the covariance terms, which in turn are composed of the correlation coefficients between the assets and their respective standard deviations. Note that the third term on the right-hand side of Equation 5.7 can be negative, implying the potential for decreasing portfolio volatility even though asset volatilities are positive. This term is the quantifiable measure of diversification that we are looking to exploit in quantitative asset allocation and risk management. It quantifies the extent to which portfolio risk is reduced as a result of less than perfect (linear) co-variation between asset returns.

Following the conventions on risk established above, the standard deviation of the portfolio return is thus defined as the square root of the variance of portfolio returns:

$$StDev(r_p) = \sqrt{\frac{1}{T-1} \sum_{t=1}^T \left(w_{A,t}^2 \cdot Var_A + w_{B,t}^2 \cdot Var_B + 2 \cdot w_{A,t} \cdot w_{B,t} \cdot Cov_{A,B} \right)}$$

which in turn implies that portfolio volatility is expressed as:

$$\sigma_p = StDev(r_p) \cdot \sqrt{\hat{T}} \quad [\text{Eq. 5.8}]$$

where $\hat{T}$ is the sample frequency of returns.

Portfolio volatility – three or more assets

As is evident from Equation 5.7, determining volatility for a two-asset portfolio is more complex than for a single asset, but it is still manageable. We now turn to portfolios of three or more assets.

The method for calculating risk in the form of volatility for a portfolio of three or more assets is identical to the case of two assets, in that it involves both the variances of returns and the covariance between returns. We first consider a portfolio of three stocks A , B and C . Again, the variance of the portfolio over T time periods is generally defined by:

$$Var_p = \frac{1}{T-1} \sum_{t=1}^T (r_{p,t} - \bar{r}_p)^2 \quad [\text{Eq. 5.9}]$$

Expanding the expression for portfolio return variance in Equation 5.9 to include three assets yields:

$$\begin{aligned} Var_p &= \frac{1}{T-1} \sum_{t=1}^T \left[(w_{A,t} \cdot r_{A,t} + w_{B,t} \cdot r_{B,t} + w_{C,t} \cdot r_{C,t}) \right. \\ &\quad \left. - (w_{A,t} \cdot \bar{r}_A + w_{B,t} \cdot \bar{r}_B + w_{C,t} \cdot \bar{r}_C) \right]^2 \\ &= \frac{1}{T-1} \sum_{t=1}^T \left[(w_{A,t} \cdot r_{A,t} - w_{A,t} \cdot \bar{r}_A) + (w_{B,t} \cdot r_{B,t} - w_{B,t} \cdot \bar{r}_B) \right. \\ &\quad \left. + (w_{C,t} \cdot r_{C,t} - w_{C,t} \cdot \bar{r}_C) \right]^2 \\ &= \frac{1}{T-1} \sum_{t=1}^T \left[w_{A,t}^2 (r_{A,t} - \bar{r}_A)^2 + w_{B,t}^2 (r_{B,t} - \bar{r}_B)^2 + w_{C,t}^2 (r_{C,t} - \bar{r}_C)^2 \right. \\ &\quad + 2w_{A,t}w_{B,t} (r_{A,t} - \bar{r}_A)(r_{B,t} - \bar{r}_B) \\ &\quad \left. + 2w_{A,t}w_{C,t} (r_{A,t} - \bar{r}_A)(r_{C,t} - \bar{r}_C) \right. \\ &\quad \left. + 2w_{B,t}w_{C,t} (r_{B,t} - \bar{r}_B)(r_{C,t} - \bar{r}_C) \right] \end{aligned}$$

$$\begin{aligned}
 & + 2w_{B,t}w_{C,t}(r_{B,t} - \bar{r}_B)(r_{C,t} - \bar{r}_C) \\
 & = \frac{1}{T-1} \sum_{t=1}^T \left[w_{A,t}^2 \cdot \text{Var}(r_A) + w_{B,t}^2 \cdot \text{Var}(r_B) + w_{C,t}^2 \cdot \text{Var}(r_C) \right. \\
 & \quad \left. + 2w_{A,t}w_{B,t}\text{Cov}(r_A, r_B) + 2w_{A,t}w_{C,t}\text{Cov}(r_A, r_C) + 2w_{B,t}w_{C,t}\text{Cov}(r_B, r_C) \right] \\
 & \hspace{15em} [\text{Eq. 5.10}]
 \end{aligned}$$

The standard deviation of the portfolio's return is then given by:

$$\begin{aligned}
 \text{StDev}(r_p) = & \frac{1}{T-1} \sum_{t=1}^T \left[w_{A,t}^2 \cdot \text{Var}(r_A) + w_{B,t}^2 \cdot \text{Var}(r_B) + w_{C,t}^2 \cdot \text{Var}(r_C) \right. \\
 & + 2w_{A,t}w_{B,t}\text{Cov}(r_A, r_B) \\
 & + 2w_{A,t}w_{C,t}\text{Cov}(r_A, r_C) \\
 & \left. + 2w_{B,t}w_{C,t}\text{Cov}(r_B, r_C) \right] \hspace{2em} [\text{Eq. 5.11}]
 \end{aligned}$$

which in turn implies that the portfolio volatility as before can be expressed as:

$$\sigma_p = \text{StDev}(r_p) \cdot \bar{T} \hspace{15em} [\text{Eq. 5.12}]$$

where once again $\hat{T}$ is the sample frequency of returns. Ignoring individual time periods, and assuming that all asset standard deviations have been annualised, thereby turning them into volatilities, Equation 5.10 can be generalised to yield an expression for risk of a portfolio of N assets:

$$\text{Var}_p = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \text{Cov}(r_i, r_j) \hspace{5em} [\text{Eq. 5.13}]$$

which means that portfolio volatility is:

$$\begin{aligned}
 \sigma_p &= \sqrt{\text{Var}_p} \\
 &= \sqrt{\sum_{i=1}^N \sum_{j=1}^N w_i w_j \text{Cov}(r_i, r_j)} \\
 &= \sqrt{\sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_i \sigma_j \rho_{ij}} \hspace{5em} [\text{Eq. 5.14}]
 \end{aligned}$$

Equation 5.14 is the most compact expression of portfolio volatility that we can obtain using the standard mathematical notation, revealing quite clearly the complexity of portfolio volatility since it is determined by a weighted average of assets' return covariances, which in turn are determined by asset return

variances and correlations. In order to see how portfolio volatility is a combination of both individual asset variation and co-variation between assets, we can divide Equation 5.14 into two components – the variance component and the covariance component:

$$\sigma_p = \sqrt{\sum_{i=1}^N w_i^2 s_i^2 + \sum_{i=1}^N \sum_{j=1}^N w_i w_j \text{Cov}(r_i, r_j)} \cdot \bar{T}$$

It is equally clear that the complexity of calculations, and thus the number of terms in the expression of portfolio volatility, increase dramatically when the number of assets in the portfolio increases. This is because the number of permutations between asset covariances increases exponentially with the number of assets included in the portfolio. In portfolios actively managed by institutional investors it is not unusual to observe anywhere from 20 to 100 or even 500 individual assets in a single equity portfolio. It thus becomes evident that a more workable method for calculating portfolio volatility is required, and that this method needs to be relatively simple and practical. *Matrix calculus* provides an excellent tool for performing such complex portfolio calculations.

Portfolio volatility – matrix notation

Equation 5.14 shows that the components of portfolio volatility are the asset weights, the asset volatilities and the correlations between all assets in the portfolio. So we begin by introducing the $N \times N$ correlation matrix:

$$P = \begin{bmatrix} \text{Corr}(r_1, r_1) & \dots & \text{Corr}(r_1, r_N) & \dots & \text{Corr}(r_1, r_N) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Corr}(r_1, r_N) & \dots & \text{Corr}(r_N, r_N) & \dots & \text{Corr}(r_N, r_N) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Corr}(r_1, r_N) & \dots & \text{Corr}(r_N, r_N) & \dots & \text{Corr}(r_N, r_N) \end{bmatrix} \quad [\text{Eq. 5.15}]$$

This is designated the Greek letter ‘Rho’ to signify a matrix of correlation coefficients between individual pairs of portfolio assets:

$$P = \begin{bmatrix} 1 & \dots & \rho_{n,1} & \dots & \rho_{N,1} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \rho_{1,n} & \dots & 1 & \dots & \rho_{N,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \rho_{1,N} & \dots & \rho_{n,N} & \dots & 1 \end{bmatrix} \quad [\text{Eq. 5.16}]$$

This matrix thus describes the correlations between all N assets in a portfolio. ρ_{ij} is the correlation coefficient between asset i and asset j . Note that ρ_{ij} is equal to ρ_{ji} , since asset i 's correlation with asset j is obviously identical to asset j 's correlation with asset i .

The concept of the correlation matrix is similar to the concept of the correlation coefficients themselves. Each value in the matrix describes the correlation between two assets in the portfolio when looked at in isolation (that is, without considering the remaining portfolio assets). Following the definition given in Chapter 3, all correlation coefficients lie in the range from -1 to $+1$. Note also that the diagonal of the correlation matrix is constituted of 1s, since by definition the return on an asset is perfectly correlated with itself.

The relation between correlations and covariances for two assets A and B has already been defined as:

$$\text{Corr}(r_A, r_B) = \rho_{A,B} = \frac{\text{Cov}(r_A, r_B)}{\sqrt{\text{Var}(r_A) \cdot \text{Var}(r_B)}} = \frac{\sigma_{A,B}}{\sigma_A \cdot \sigma_B}$$

Accordingly, since the covariance between the two assets can be expressed as:

$$\text{Cov}(r_A, r_B) = \sigma_A \sigma_B \rho_{A,B} \quad [\text{Eq. 5.17}]$$

it then follows that for portfolios with many assets, the corresponding *variance–covariance matrix* Σ can be stated in terms of the correlation matrix P , the *volatility matrix* σ and the transposed volatility matrix σ^T . The latter two take the following form:

$$\sigma = \sigma^T = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 \\ 0 & 0 & \sigma_n & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \sigma_N \end{bmatrix} \quad [\text{Eq. 5.18}]$$

Using the volatility matrices, the variance–covariance matrix Σ for N assets can be expressed simply as:

$$\Sigma = \sigma^T \cdot P \cdot \sigma \quad [\text{Eq. 5.19}]$$

This formulation is analogous to Equation 5.17 for N assets and is obviously much easier to work with when we consider a very large number of assets. Using Equation 5.16 and Equation 5.18 to expand Equation 5.19 yields:

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 \\ 0 & 0 & \sigma_n & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \sigma_N \end{bmatrix}^T \bullet \begin{bmatrix} 1 & \dots & \rho_{n,1} & \dots & \rho_{N,1} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \rho_{1,n} & \dots & 1 & \dots & \rho_{n,N} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \rho_{1,N} & \dots & \rho_{n,N} & \dots & 1 \end{bmatrix}$$

$$\bullet \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 \\ 0 & 0 & \sigma_n & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \sigma_N \end{bmatrix} \quad [\text{Eq. 5.20}]$$

Equation 5.20 thus specifies all the calculations that generate the variance–covariance matrix. Performing the pre- and post-multiplications of $\mathbf{P}$ by σ^T and σ respectively, we obtain the expression for the variance–covariance matrix $\mathbf{\Sigma}$:

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1 \sigma_1 & \dots & \sigma_n \rho_{n,1} \sigma_1 & \dots & \sigma_N \rho_{N,1} \sigma_1 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sigma_1 \rho_{1,n} \sigma_n & \dots & \sigma_n \sigma_n & \dots & \sigma_N \rho_{N,n} \sigma_n \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sigma_1 \rho_{1,N} \sigma_N & \dots & \sigma_n \rho_{n,N} \sigma_N & \dots & \sigma_N \sigma_N \end{bmatrix}$$

$$= \begin{bmatrix} \text{Var}(r_1) & \dots & \text{Cov}(r_{n,1}) & \dots & \text{Cov}(r_{N,1}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Cov}(r_{1,n}) & \dots & \text{Var}(r_n) & \dots & \text{Cov}(r_{N,n}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Cov}(r_{1,N}) & \dots & \text{Cov}(r_{n,N}) & \dots & \text{Var}(r_N) \end{bmatrix} \quad [\text{Eq. 5.21}]$$

This is the matrix that summarises all the return co-relations between individual asset returns in the entire portfolio, as it describes the covariances between the returns on all assets in a portfolio. Note that the diagonal of the matrix is constituted of variances of returns on each individual asset. As pointed out earlier, this is due to the fact that an asset's covariance with itself is equal to its variance.

Alternatively, the variance–covariance matrix can be constructed from the raw data set of historical asset returns. Given an $N \cdot T$ matrix $\mathbf{H}$ of N assets and T return observations, $\mathbf{\Sigma}$ is given by:

$$\mathbf{\Sigma} = \frac{1}{T-1} \mathbf{H} \bullet \left(\mathbf{I} - \frac{1}{T} \mathbf{1} \bullet \mathbf{1}^T \right) \mathbf{H}^T$$

where $\mathbf{I}$ is the conformable *identity matrix*², and $\mathbf{1}$ is a conformable vector of 1s. This formulation is analogous to equations presented above expressing variance and covariance.

Recall from Equation 5.5 that the volatility of an entire portfolio of N assets could be expressed as:

$$\sigma_P = \sqrt{\sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_i \sigma_j \rho_{ij}}$$

Letting $\vec{w}$ and $\vec{w}^T$ denote the weight vector and the transposed weight vector respectively, letting Σ denote the variance–covariance matrix, and using the matrix form results above, the volatility of the entire portfolio is then given by:

$$\begin{aligned} \sigma_P &= \sqrt{\begin{bmatrix} w_1 \\ \vdots \\ w_n \\ \vdots \\ w_N \end{bmatrix}^T \begin{bmatrix} \text{Var}(r_1) & \dots & \text{Cov}(r_{n,1}) & \dots & \text{Cov}(r_{N,1}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Cov}(r_{1,n}) & \dots & \text{Var}(r_n) & \dots & \text{Cov}(r_{N,n}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \text{Cov}(r_{1,N}) & \dots & \text{Cov}(r_{n,N}) & \dots & \text{Var}(r_N) \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_n \\ \vdots \\ w_N \end{bmatrix}} \\ &= \sqrt{\vec{w}^T \Sigma \vec{w}} \end{aligned} \quad [\text{Eq. 5.22}]$$

The very simple expression in Equation 5.22 comprises all the information we need in order to calculate portfolio volatility, and compared to Equation 5.10 and Equation 5.14 it thus enables us to work with very large portfolios relatively easily. Particularly when we need to derive specific solutions for portfolio optimisation problems, the matrix notation will prove a very useful tool indeed. We shall demonstrate this shortly.

Returning to the example of a portfolio of three assets A , B and C , the variance–covariance matrix and the portfolio volatility are thus constructed as follows. The correlation matrix P is pre- and post-multiplied by the transposed volatility matrix σ^T and the volatility matrix σ respectively, to obtain the variance–covariance matrix:

$$\Sigma = \begin{bmatrix} \sigma_A & 0 & 0 \\ 0 & \sigma_B & 0 \\ 0 & 0 & \sigma_C \end{bmatrix}^T \begin{bmatrix} 1 & \rho_{B,A} & \rho_{C,A} \\ \rho_{A,B} & 1 & \rho_{C,B} \\ \rho_{A,C} & \rho_{B,C} & 1 \end{bmatrix} \begin{bmatrix} \sigma_A & 0 & 0 \\ 0 & \sigma_B & 0 \\ 0 & 0 & \sigma_C \end{bmatrix} \quad [\text{Eq. 5.23}]$$

Performing these pre- and post-multiplications yields:

$$\Sigma = \begin{bmatrix} \sigma_A^2 & \sigma_B \rho_{B,A} \sigma_A & \sigma_C \rho_{C,A} \sigma_A \\ \sigma_A \rho_{A,B} \sigma_B & \sigma_B^2 & \sigma_C \rho_{C,B} \sigma_B \\ \sigma_A \rho_{A,C} \sigma_C & \sigma_B \rho_{B,C} \sigma_C & \sigma_C^2 \end{bmatrix} \quad [\text{Eq. 5.24}]$$

Again, to see exactly what this expression means, we can rewrite it using variance and covariance terms:

$$\Sigma = \begin{bmatrix} \text{Var}(r_A) & \text{Cov}(r_B, r_A) & \text{Cov}(r_C, r_A) \\ \text{Cov}(r_A, r_B) & \text{Var}(r_B) & \text{Cov}(r_C, r_B) \\ \text{Cov}(r_A, r_C) & \text{Cov}(r_B, r_C) & \text{Var}(r_C) \end{bmatrix} \quad [\text{Eq. 5.25}]$$

From Equation 5.22 recall that $\sigma_P = \sqrt{\vec{w}^T \Sigma \vec{w}}$. Expanding the weight vectors and the variance–covariance matrix of this expression, we obtain:

$$\sigma_P = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}^T \begin{bmatrix} \text{Var}(r_A, r_A) & \text{Cov}(r_B, r_A) & \text{Cov}(r_C, r_A) \\ \text{Cov}(r_A, r_B) & \text{Var}(r_B, r_B) & \text{Cov}(r_C, r_B) \\ \text{Cov}(r_A, r_C) & \text{Cov}(r_B, r_C) & \text{Var}(r_C, r_C) \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \quad [\text{Eq. 5.26}]$$

We then perform vector and matrix multiplication on Equation 5.25 to obtain an expression for the volatility of the three-asset portfolio:

$$\sigma_P = \begin{aligned} & w_A^2 \text{Var}(r_A) + w_B^2 \text{Var}(r_B) + w_C^2 \text{Var}(r_C) \\ & + 2w_A w_B \text{Cov}(r_A, r_B) \\ & + 2w_A w_C \text{Cov}(r_A, r_C) \\ & + 2w_B w_C \text{Cov}(r_B, r_C) \end{aligned} \quad [\text{Eq. 5.27}]$$

Not surprisingly, Equation 5.27 is exactly equivalent to the previously presented three-asset portfolio volatility expression in Equation 5.11. Again, from Equation 5.27 it is evident that, portfolio volatility is determined, more than anything else, by asset return covariances. Note again the way in which the volatility expression expands as we add more assets, and how the matrix notation enables us to work more easily with large portfolios.

Having thus provided a methodology for determining portfolio return and volatility, we now move on to quantifying potential diversification effects between assets.

THE NATURE OF DIVERSIFICATION

Quantifying portfolio diversification

As demonstrated above, a portfolio's volatility depends on asset weights, volatilities and covariances (correlations). A more basic determinant of

portfolio volatility is the *number* of assets in a portfolio. When considering adding assets to a portfolio, the important factor to look at is not the risk of an asset by itself (quantified by its volatility), but rather the risk it adds to the entire portfolio given the correlation of its returns with the returns on each of the assets already in the portfolio. In other words, the relevant factor to watch is the asset’s contribution to risk as measured by its covariance with the assets already in the portfolio. The basic concept is identical to that underlying the asset Betas, and it is in fact the essence of portfolio diversification.

The diversification qualities of a portfolio depend on the way in which each asset is correlated with the other assets in different states of the world – that is, when markets go up, down or sideways, when general volatility levels change, when the macro economic environment changes, when interest rates change, and so on.

Figure 5.4 illustrates the diversification effects in a portfolio of two assets A and B for different correlation coefficients. The two assets (represented by the black dots) have expected returns of 4% and 18%, and volatilities of 11% and 26%, respectively. Each line connecting the two assets, for varying correlation coefficients, shows the expected portfolio return at a given level of volatility for a continuous range of different weight combinations, spanning from 100% in asset A and 0% in asset B, to 0% in asset A and 100% in asset B. Alternatively we can interpret the chart as indicating how low a level of portfolio volatility we can obtain for a given level of expected return.

For a correlation coefficient of +1 we see that there are no diversification benefits to be gained from having two assets rather than just one in the portfolio. Both return and volatility are linear combinations of the returns and

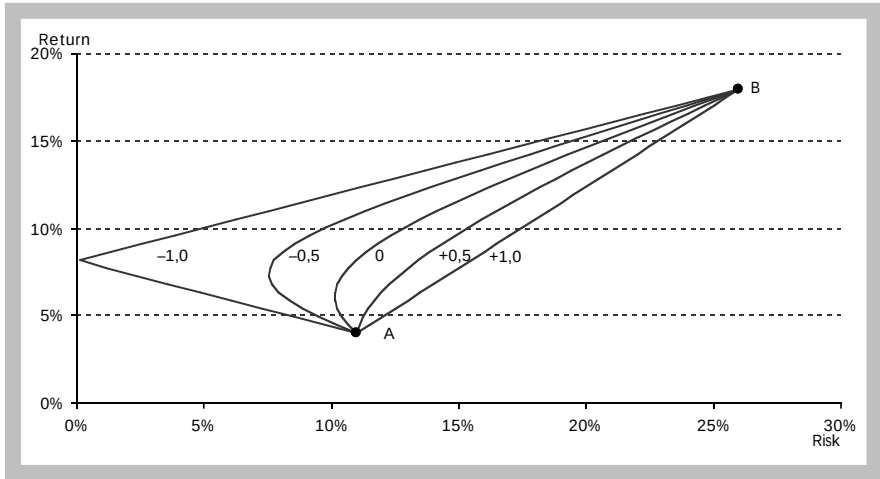


Figure 5.4 Diversification at work – 2-asset portfolio

volatilities of asset *A* and asset *B*. Having two assets instead of one generally provides us with an opportunity to choose any portfolio on the straight black line, thus obtaining a desired level of return and risk, but when the correlation coefficient is $+1$ we are not gaining anything in terms of portfolio risk reduction by having both assets in the portfolio. The lowest attainable level of portfolio volatility is obtained when holding only asset *A* in the portfolio.

A correlation coefficient of 0 provides clear diversification benefits: it is now possible to obtain volatility levels for a given level of return that are clearly below those of the portfolio with a correlation between asset returns of $+1$. In fact, for certain combinations of asset *A* and asset *B* we find that we can achieve a portfolio volatility that is lower than both volatilities of the individual assets. The lowest attainable portfolio volatility, 10.1%, is now obtained by allocating 85% to asset *A* and 15% to asset *B*.

For a portfolio of two assets with correlation coefficient -1 we observe very large diversification effects, since the two assets now move in opposite directions at all times. In fact, portfolio volatility can now be reduced to zero, given the 'right' mix of asset *A* and asset *B*. Exactly what constitutes the 'right' mix for a given level of return or risk and how to identify this mix will be discussed in the following chapter.

The diversification dynamics that we attempt to show in Figure 5.4 are analogous to Figures 5.1, 5.2 and 5.3, since asset returns with correlation coefficients below 1 will affect a reduction in overall portfolio volatility. Clearly, we can extend this analysis to include many more assets and thereby gain insight into portfolio return and risk characteristics of large portfolios. Having demonstrated the mechanics of quantitative portfolio diversification, the obvious question then becomes: how many assets does a portfolio need to hold in order to obtain a satisfactory level of diversification?

A 'satisfactory' level of diversification is of course arbitrary, but the tool provided by Modern Portfolio Theory gives us the opportunity to calculate portfolio risk for any number of assets with any characteristics as described by their individual returns, volatilities and correlations. Intuitively, the satisfactory number of assets depends on the exact types of assets that are held, their individual weightings in the portfolio, the current state of the world and the portfolio manager's resultant appetite for risk, and of course most importantly to what extent the returns on the assets are correlated with each other.

It is obvious that the more alike the stocks in the portfolio are, the more highly correlated they will tend to be, and thus the lower the potential diversification benefits, as all assets in the portfolio will then tend to move in the same direction most of the time. Being 'alike' can pertain in particular to their size, the industry within which they operate, the actual products or services they sell, their currency exposures, and so forth.

How much is enough?

An illustrative approach to the issue of how many assets make a diversified portfolio can be made by a procedure of selecting stocks at random. We begin by selecting a single stock (Stock #1) from a broad index (here the MSCI World). We then calculate the level of risk (volatility) for that particular stock, using daily observations over an appropriate period. This stock is the only stock in our portfolio, and so portfolio volatility is of course equal to its volatility since there are no other stocks with which to obtain a diversification effect.

We then select another stock (Stock #2) at random, and add that to the portfolio so that we now hold 50% of each stock in the portfolio. Using the techniques derived from Modern Portfolio Theory presented above, we again calculate portfolio volatility, which now also involves calculating the covariance between the returns on the two stocks. Automatically some degree of diversification will now begin to present itself. We then pick yet another stock (Stock #3) and add this to the portfolio, which now comprises three stocks each with a weighting of 33.33%, and calculate portfolio volatility.

This process of picking stocks at random, adding them to the portfolio and calculating portfolio volatility is continued until a fairly stable level of portfolio volatility appears. In other words we continue to diversify until the marginal diversification effect goes towards zero. Figure 5.5 illustrates the resultant port-

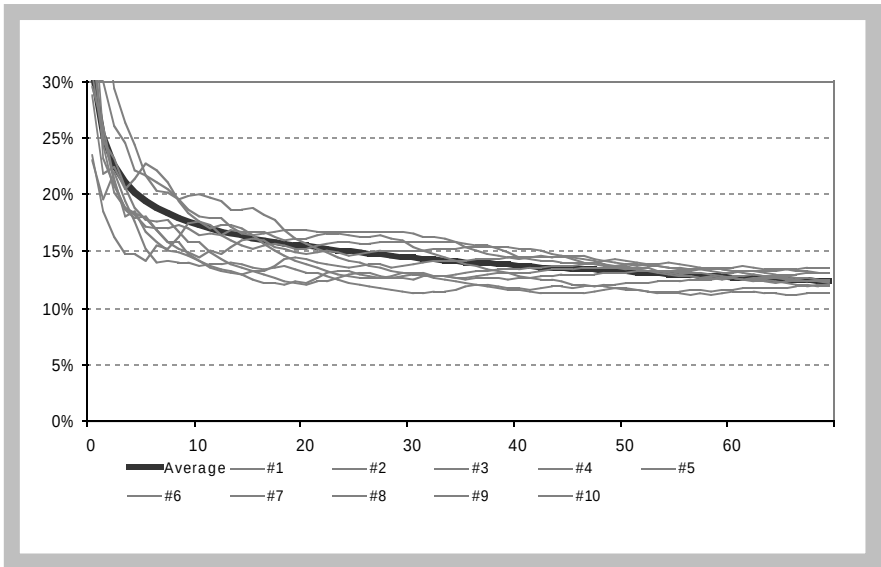


Figure 5.5 Portfolio volatility as 70 MSCI world stocks are successively added

folio volatility from randomly picking 70 stocks from the MSCI World Index and successively adding them to the portfolio in random order. The exercise is based on daily observations from 1995 to 2001, and is completed 100 times, after which an average of the results is calculated. We have added the first 10 of the 70 tries to the chart, to illustrate how portfolio volatility is gradually reduced as more and more assets are added to the portfolio.

Figure 5.5 illustrates that the diversification effect that we seek appears immediately as we add more stocks to the portfolio, and that after having added between 15 and 20 stocks, the marginal diversification effect becomes quite small. Obviously, continuing to add assets to the portfolio will reduce portfolio volatility further, eventually reaching the market's volatility (as we finally end up with all available assets included in the portfolio), but the incremental value of diversification is much larger at the beginning of the process.

Here it is important to point out that we obtain a low level of portfolio volatility by adding ever more assets, but at a price. For each asset we add to the portfolio we will suffer a loss of total return in the form of transaction costs. It is therefore obvious that at some point the costs of adding new assets to the portfolio will outweigh the benefits, but since the trade-off depends on individual investor risk aversion, the exact point is virtually impossible to determine.

By showing how the volatility of portfolios with successively larger numbers of equal-weighted assets will tend toward the volatility of the market, the approach demonstrated above thus provides an indication of the potential benefits to portfolio diversification. However, since we are working with a limited set of randomly chosen stocks, the results presented in Figure 5.5 do not mean that we cannot do better than this. There is always the possibility of adding certain stocks from the investment universe of stocks to the portfolio while removing others, and thereby obtaining even lower levels of risk for the same level of return.

As we shall see in the next chapter, we can precisely determine an appropriate set of asset weights that will generate high levels of diversification and a lower level of risk than by mindlessly pouring assets into the portfolio and thereby incurring large transaction costs.

SUMMARY

In this chapter we have characterised portfolios of two, three and N stocks to demonstrate that portfolio returns are a linear function of the weights and returns on their constituent assets. Portfolio risk is a more complex quantity that requires determination of asset-by-asset correlations, as well as of individual asset volatilities. Calculating portfolio volatility using standard notation results in virtually unmanageable algebraic expressions. Turning to matrix

notation provides an easy method for calculating portfolio volatilities, even for portfolios composed of very large numbers of assets.

The core of a portfolio's volatility is the variance–covariance matrix, which expresses all asset-by-asset covariances. This matrix is generated by pre- and post-multiplying the correlation matrix with the volatility matrix. Portfolio volatility is then determined by pre- and post-multiplying the variance–covariance matrix with the portfolio's weight vector. The exact determination of portfolio volatility, given a set of assets and their individual characteristics, enables us to investigate the nature of portfolio diversification. With appropriate use of correlations between individual assets, it is in some cases possible to reduce portfolio volatility to levels below individual asset volatilities. We demonstrate that adding stocks to a portfolio will reduce its volatility, and as the number of assets goes towards infinity we obtain the market volatility. However, this procedure is associated with very large transaction costs. We also note that the marginal diversification effect (the reduction in portfolio volatility of adding assets to a portfolio) diminishes quite quickly, so that the vast majority of the diversifiable risk can be eliminated by adding approximately 15–20 equal-weighted assets to the portfolio. This is the case irrespective of the choice of stocks, as this result obtains in a consistent manner even when choosing assets at random. This suggests the possibility of choosing assets systematically, and thereby reducing volatility levels for a given level of expected return to levels even lower than those obtained by randomly picking assets and adding them to the portfolio.

APPENDIX

Matrix calculus

Definition of a vector

A *vector* is a one-dimensional matrix of the form $N \cdot 1$ or $1 \cdot N$. For example, a $3 \cdot 1$ vector appears as follows:

$$\vec{a} = \begin{bmatrix} 5 \\ 2 \\ 7 \end{bmatrix}$$

whereas a $1 \cdot 3$ vector appears as follows:

$$\vec{b} = [3 \quad 4 \quad 8]$$

Definition of a Matrix

A *matrix* is a rectangular array of numbers of the form $N \cdot K$, with N rows and K columns. Each number in the matrix is called an *element*. For example, a $5 \cdot 4$ matrix with 20 elements appears as follows:

$$A = \begin{bmatrix} 1 & 4 & 7 & 4 \\ 3 & 7 & 9 & 8 \\ 5 & 6 & 2 & 7 \\ 3 & 1 & 3 & 6 \\ 2 & 5 & 2 & 5 \end{bmatrix}$$

A *quadratic matrix* is a similar array, with the number of rows being equal to the number of columns. For example, a $4 \cdot 4$ matrix appears as follows:

$$B = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 4 & 9 & 5 & 7 \\ 1 & 6 & 2 & 4 \\ 8 & 1 & 7 & 5 \end{bmatrix}$$

Vector and matrix transposition

Transposition of vectors and matrices is necessary in order to carry out portfolio calculations. The basic idea is turning rows into columns and columns into rows. Thus, transposing the vector $\vec{a}$ given above results in:

$$\vec{a}^T = [5 \quad 2 \quad 7]$$

Transposing matrices is slightly more complicated, but the basic principle remains the same. Rows are converted into columns and columns are converted into rows. For a $3 \cdot 3$ matrix, or any other quadratic matrix, it is helpful to think of the transposition as a rotation around the diagonal. Thus, in the case of the quadratic matrix B presented above, the transposed matrix appears as follows:

$$B^T = \begin{bmatrix} 2 & 4 & 1 & 8 \\ 3 & 9 & 6 & 1 \\ 4 & 5 & 2 & 7 \\ 1 & 7 & 4 & 5 \end{bmatrix}$$

Adding and subtracting matrices

Addition and subtraction operations are very simple to perform, but are only possible if the two matrices are of the same dimension. For example, we

cannot add a 3·3 matrix and a 3·4 matrix, but we can add a 3·3 and 3·3. To add two matrices, simply add each element in one to the corresponding element in the other as demonstrated in the following example:

$$A = \begin{bmatrix} 2 & 7 & 3 \\ 4 & 1 & 9 \\ 1 & 6 & 8 \end{bmatrix} \quad B = \begin{bmatrix} 7 & 1 & 2 \\ 3 & 4 & 9 \\ 5 & 8 & 0 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 2 & 7 & 3 \\ 4 & 1 & 9 \\ 1 & 6 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 1 & 2 \\ 3 & 4 & 9 \\ 5 & 8 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 5 \\ 7 & 5 & 18 \\ 6 & 14 & 8 \end{bmatrix}$$

The same principle applies for subtraction, where we simply subtract each element in one matrix from each corresponding element in the other:

$$A - B = \begin{bmatrix} 2 & 7 & 3 \\ 4 & 1 & 9 \\ 1 & 6 & 8 \end{bmatrix} - \begin{bmatrix} 7 & 1 & 2 \\ 3 & 4 & 9 \\ 5 & 8 & 0 \end{bmatrix} = \begin{bmatrix} -5 & 6 & -1 \\ 1 & -3 & 0 \\ -4 & -2 & 8 \end{bmatrix}$$

Adding and subtracting vectors is analogous to adding and subtracting matrices. Each element in the vector is added to or subtracted from its corresponding element in the other vector.

Multiplying matrices

Multiplication of matrices is more complicated. In order to be able to multiply two matrices A and B , the number of columns in the first matrix A must be equal to the number of rows in the second matrix B . For example it is possible to multiply a 3·4 matrix with a 4·3 matrix, but it is not possible to multiply a 3·4 matrix with a 5·4 matrix. Multiplication is done by multiplying the first element in the first row of the first matrix with the first element in the first column of the second matrix, and then adding to that the multiplication of the second element in the first row of the first matrix and the second element in the first column of the second matrix. This process is carried out for all the elements in the first row and first column, and adding all the results we obtain the first element of the resultant new matrix. The process is repeated for all rows and columns until the new matrix is complete. This seems complex at first but in practice it is quite simple, and most spreadsheets and statistics software suites will carry this out as a matter of routine. Multiplication of the two matrices A and B presented above looks like this:

$$\begin{aligned}
 A \cdot B &= \begin{bmatrix} 2 & 7 & 3 \\ 4 & 1 & 9 \\ 1 & 6 & 8 \end{bmatrix} \cdot \begin{bmatrix} 7 & 1 & 2 \\ 3 & 4 & 9 \\ 5 & 8 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} (2 \cdot 7 + 7 \cdot 3 + 3 \cdot 5) & (2 \cdot 1 + 7 \cdot 4 + 3 \cdot 8) & (2 \cdot 2 + 7 \cdot 9 + 3 \cdot 0) \\ (4 \cdot 7 + 1 \cdot 3 + 9 \cdot 5) & (4 \cdot 1 + 1 \cdot 4 + 9 \cdot 8) & (4 \cdot 2 + 1 \cdot 9 + 9 \cdot 0) \\ (1 \cdot 7 + 6 \cdot 3 + 8 \cdot 5) & (1 \cdot 1 + 6 \cdot 4 + 8 \cdot 8) & (1 \cdot 2 + 6 \cdot 9 + 8 \cdot 0) \end{bmatrix} \\
 &= \begin{bmatrix} 50 & 54 & 67 \\ 76 & 80 & 17 \\ 65 & 89 & 56 \end{bmatrix}
 \end{aligned}$$

Multiplication of vectors and matrices

Multiplication of vectors and matrices is similar in principle to the multiplication of matrices, but also requires that the number of columns in the transposed vector be equal to the number of rows in the matrix. Pre-multiplying a 3×1 vector $\vec{v}$ with a 3×3 matrix A appears as follows:

$$\begin{aligned}
 \vec{v} &= \begin{bmatrix} 6 \\ 2 \\ 5 \end{bmatrix}, \quad A = \begin{bmatrix} 7 & 1 & 2 \\ 3 & 4 & 9 \\ 5 & 8 & 0 \end{bmatrix} \\
 \vec{v}^T \cdot A &= [6 \quad 2 \quad 5] \cdot \begin{bmatrix} 7 & 1 & 2 \\ 3 & 4 & 9 \\ 5 & 8 & 0 \end{bmatrix} \\
 &= (6 \cdot 7 + 2 \cdot 3 + 5 \cdot 5 \quad 6 \cdot 1 + 2 \cdot 4 + 5 \cdot 8 \quad 6 \cdot 2 + 2 \cdot 9 + 5 \cdot 0) \\
 &= [73 \quad 54 \quad 30]
 \end{aligned}$$

Should we wish to post-multiply the resultant vector with another vector, the result would be a single number:

$$[73 \quad 54 \quad 30] \cdot \begin{bmatrix} 3 \\ 7 \\ 4 \end{bmatrix} = (73 \cdot 3 + 54 \cdot 7 + 30 \cdot 4) = 717$$

Factor or scalar multiplication

Factor or scalar multiplication is done by multiplying every element in the matrix by the factor or scalar. There are no restrictions: regarding dimensions for example, multiplying matrix A by 2 is done as follows:

$$2A = 2 \cdot \begin{bmatrix} 2 & 7 & 3 \\ 4 & 1 & 9 \\ 1 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 & 2 \cdot 7 & 2 \cdot 3 \\ 2 \cdot 4 & 2 \cdot 1 & 2 \cdot 9 \\ 2 \cdot 1 & 2 \cdot 6 & 2 \cdot 8 \end{bmatrix} = \begin{bmatrix} 4 & 14 & 6 \\ 8 & 2 & 18 \\ 2 & 12 & 16 \end{bmatrix}$$

Special matrices

The diagonal matrix

A *diagonal matrix* is a matrix with elements only in the diagonal, and all other elements equal to zero:

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

The Identity Matrix

A special but extremely handy matrix is the *identity matrix* I , which is essentially a regular diagonal matrix, but with ones in the diagonal:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The usefulness of the identity matrix stems from its application in inverting matrices, which is an essential part of matrix calculus. To illustrate, the identity matrix has the following property:

$$A^{-1}A = I$$

This property only applies for quadratic or square matrices that are *non-singular*. For a matrix to be non-singular, it has to be true that there exists a different matrix which, when multiplied by the first matrix, results in the identity matrix. This is essentially what the equation above states.

Notes

- 1 For an introduction to vector and matrix calculus, please see the Appendix of this chapter.
- 2 The identity matrix is explained in the Appendix.

QUANTITATIVE PORTFOLIO OPTIMISATION AND EFFICIENT PORTFOLIOS

PORTFOLIO EFFICIENCY

More can be less

In the previous chapter we demonstrated that there exist natural and automatic diversification effects from adding ever more stocks to a portfolio. Because the returns on assets are only very infrequently perfectly correlated, including several assets in a portfolio will tend to reduce overall portfolio risk: and generally speaking, the more stocks are included, the lower the portfolio risk. There is, however, an obvious downside to this approach. A very large number of stocks in a portfolio will entail larger transaction costs, particularly if the portfolio is expected to be turned over a few times during a year. In addition, the more stocks the portfolio manager includes in the portfolio, the more difficult it will be for him to 'tilt' it purposefully towards certain factors or characteristics that he deems might perform better than average in the future. In other words, the ability to precisely control the characteristics of the portfolio, and thereby its expected cash flow, is degraded as the number of stocks in the portfolio becomes very large. Finally, it can become quite a task for a portfolio manager to have to keep a close eye on a large number of stocks,

so it is obvious that the risk of adverse performance from a single stock increases with a large number of stocks.

For these reasons the portfolio manager, in his quest for diversification, should attempt to exploit more precisely the characteristics of individual assets and asset classes. Instead of simply piling a huge number of assets into the portfolio, the portfolio manager should aim to identify sectors and stocks that complement each other in terms of risk. In other words, the manager should concentrate on finding stocks in sectors or countries, or preferably a combination of the two, whose returns exhibit a low and stable level of correlation with each other. By being selective in this manner, it is possible for the portfolio manager to exclude stocks that exhibit undesirable characteristics in terms of risk, and to include stocks that exhibit desirable characteristics, thereby diversifying away much of the potential risk.

In the quest for this type of *non-naïve* or *intelligent* diversification, we thus need to establish an objective function that can guide our efforts towards making a selection of stocks that exploits each stock's particular characteristics in an efficient manner. We shall do so in this chapter, but first we must specify precisely what is meant by the term 'efficiency' in a portfolio context.

Efficiency defined

In general, *efficiency* is defined as the utilisation of resources in such a manner that the maximum output or gain is generated. Implicit in this definition is the quality of *optimality*. In an optimum it is impossible to squeeze more output out of a given amount of resources, and applying this train of thought to the mean-variance framework which we have established in the previous chapters provides us with the following definition of portfolio efficiency:

In a portfolio context we define efficiency as the maximum attainable return for a given level of volatility, or alternatively, the minimum attainable volatility for a given level of return.

We thus use mean-variance efficiency as the measure of portfolio efficiency, and more precisely we designate efficient portfolios as those portfolios that cannot be improved upon in terms of the return versus risk trade-off. It is thus not possible to alter an efficient portfolio without paying a price in the form of lower return or higher volatility. As we shall see, the vast majority of attainable portfolios are not efficient in the strict mean-variance sense, which suggests that we can improve on them at no cost (in terms of return or volatility) by altering their composition. In other words, by changing the composition or weightings of the portfolios we should be able either to increase the level of

expected return without lowering the expected level of risk, or to lower the expected level of risk without incurring a lower level of expected return. In order to establish which portfolios are efficient and which are not, we need to be able to determine efficiency in a quantitative sense. We therefore turn to *quantitative portfolio optimisation*.

QUANTITATIVE PORTFOLIO OPTIMISATION

Objectives

In the Markowitz-inspired mean-variance or return-volatility framework presented in the previous chapters, optimisation and optimal portfolios are closely linked to efficiency as we have just defined it. An optimal portfolio in this context is thus a portfolio that is mean-variance efficient. In other words, in efficient portfolios we can obtain no higher return and no lower volatility without paying a price in terms of either higher volatility or lower return respectively.

Having introduced techniques for calculating return and risk on single assets as well as on portfolios, it seems obvious to take this one step further and attempt to tailor the portfolio to fit a specific return/risk profile: that is, to specify a portfolio – in terms of asset weights – that is expected to generate desired levels of either return, risk or both, while maintaining portfolio efficiency. Since every asset in the portfolio is different from every other asset, it follows intuitively that such a tailoring process should centre on varying the weight of each asset in the portfolio. Of course this should not happen in a random fashion, but rather should be performed using mathematical algorithms. The process is known as *portfolio optimisation*, and depending on the exact nature of the optimisation problem, different techniques are appropriate.

What is essential in this context is the fact that assets in a portfolio are not perfectly correlated, and this is what we can exploit with quantitative optimisation techniques. That is, when the price of one asset changes, the other assets will tend to move either by more or by less, up or down. So intuitively, and as demonstrated earlier, it is possible to reduce the total amount of risk in a portfolio by choosing to include different *amounts* of different assets (with different return/risk/correlation characteristics) in the portfolio, in order to obtain the desired combined portfolio characteristics.

In general, quantitative portfolio optimisation seeks to first determine:

- the minimum-variance portfolio

- the minimum level of volatility for a given level of return
- the maximum level of return for a given level of volatility.

Although on the surface these appear to be different goals, they essentially involve the same type of procedure. As noted, determining the asset weights that enable the creation of portfolios with the three characteristics listed above is achieved through quantitative portfolio optimisation using mathematical optimisation techniques: that is, *optimisation algorithms*. Through the remainder of this chapter we shall – among other things – demonstrate that the second and third goals in fact represent a dual problem, and that they actually yield the same result when optimisation is carried out. However, we begin by mathematically determining the minimum-variance portfolio, which will serve as a navigation or reference point of a sort throughout the remainder of this book.

The minimum-variance portfolio

The *minimum-variance portfolio* is the portfolio (that is, the combination of asset weights) that, given the particular return and risk characteristics of each asset, generates the lowest amount of risk achievable. In other words, the minimum-variance portfolio specifies the asset weights that generate the lowest possible portfolio risk, without any additional constraints on the desired level of return or on the maximum or minimum extent to which an asset can enter into the portfolio. As indicated, and in consonance with the methodology presented in previous chapters, we actually determine the minimum *volatility* of the portfolio, but it has become customary to refer to this unique portfolio as the minimum-*variance* portfolio. Since volatility is a derivative of variance, the two portfolios are in fact identical, but going forward we shall use the conventional term, the minimum-variance portfolio.

The minimum-variance portfolio is determined by performing a mathematical optimisation using a standard search algorithm, by which asset weights are changed or adjusted in order to achieve the lowest possible level of volatility. This includes the possibility that some asset weights are set to zero – that is, they are to be dropped from the portfolio altogether. The minimum-variance portfolio is important in a portfolio analysis context because it reveals the absolute lowest level of portfolio risk that is attainable from the assets available to the portfolio manager. As demonstrated earlier, this level of risk may well end up being lower than the volatility of the market, because of the diversification effects discussed above, and also because here we have the option of dropping stocks from the portfolio whose characteristics do not meet our needs for diversification. So if the inclusion of an asset does not benefit the

portfolio in terms of reduced overall portfolio volatility, we simply leave it out altogether. The minimum-variance portfolio thus serves as a measure for assessing the properties of the portfolio in terms of diversification potential, and as a starting point for additional analysis of the range of mean-variance efficient portfolios that a particular set of assets is capable of generating. Recall from the previous chapter that portfolio variance is defined by:

$$Var_p = \sum_{i=1}^N \sum_{j=1}^N w_i w_j Cov(r_i, r_j)$$

and hence that using matrix notation the portfolio volatility can be expressed as:

$$\sigma_p = \sqrt{Var(r_p)} = \sqrt{\vec{w}_p^T \Sigma \vec{w}_p}$$

where $\vec{w}_p$ and $\vec{w}_p^T$ denote the weight vector and the transposed weight vector respectively, and Σ denotes the variance–covariance matrix determined using the correlation matrix and the volatility matrix.

In this section our goal is straightforward. We wish to minimise portfolio volatility. We are not concerned with constraints on asset weights or the level of return the portfolio is expected to provide. Our only goal is to determine the portfolio that, given the expected risk and correlation characteristics of the assets available to us, minimises portfolio volatility. For this reason the only constraint on the minimisation problem is that the sum of the asset weight should be equal to 1 – that is, we want to be fully invested.

The minimisation problem can be arranged as a LaGrange problem.¹ This entails defining the objective function $f(x)$ that needs to be minimised, as well as the constraints subject to which the optimisation must be performed. Formally this is expressed as:

$$Min \quad \vec{w}_p^T \Sigma \vec{w}_p$$

Subject to:

$$\vec{w}_p^T \bullet \vec{e} = 1 \quad \text{[Eq. 6.1]}$$

where $\vec{e}$ is the unit vector consisting of a single column of 1s. As indicated, the constraint subject to which the optimisation must take place ensures that all asset weights sum to 100%. We have not restricted the portfolio from selling stocks short, since this is a common practice in modern investment management. In addition it would make the formal derivation of the solution unnecessarily complicated for these expositional purposes; and should we wish to restrict short selling of stocks, it is quite easy to construct a computer algorithm that ensures that this goal is attained.

The solution to the quantitative optimisation problem presented in Equation 6.1 requires firstly that we arrange the problem in the LaGrange form, secondly that we determine the partial derivatives of the equation and set them equal to zero, and then finally – using standard matrix algebra – that we derive an expression for the optimum, which in this case is a global minimum.² Performing this algebraic exercise yields:

$$\vec{w}_p^* = \left(\vec{e}^T \quad \Sigma^{-1} \quad \vec{e} \right)^{-1} \Sigma^{-1} \vec{e} \tag{Eq. 6.2}$$

The vector $\vec{w}_p^*$ in Equation 6.2 is thus the vector of asset weights that minimises portfolio volatility without regard for the resultant level of return, and without restrictions on short selling. The solution is mathematically well defined, and interestingly depends only on the structure of the variance–covariance matrix. Thus, the characteristics of the variance–covariance matrix alone determine the optimal combination of asset weights that will result in the minimum level of volatility. This is because the variance–covariance matrix contains all the information we need about asset volatilities and correlations.

Let us present a simple numerical example. Consider the case of a hypothetical portfolio containing three assets *A*, *B* and *C*, with the return, volatility and correlation characteristics as shown in Table 6.1.

In order to lay down a bit of guiding intuition for this exercise, we can imagine that asset *A* is a long-term government bond with a relatively low level of return and a correspondingly low level of risk. Asset *B* is a so-called ‘Old Economy’ stock, such as that of a heavy machinery or auto company; and asset *C* is a so-called ‘New Economy’ stock, such as that of a telecommunications networking company. The two stocks have a higher expected level of return than the long-term government bond, but also a significantly higher level of expected risk. Please note that these are *expected* levels of return and risk. Such expectations have been known to be wrong from time to time.

The correlation coefficient between the two stocks is relatively high, whereas the correlation coefficients between the bond and the two stocks are quite low. This makes good intuitive sense, since stocks and bonds are often treated as being two alternatives in the most basic asset allocation decision. That is, when investors want to reduce their risk they usually move out of

Table 6.1 Returns, volatilities and correlations: three-asset portfolio

Assets	Returns	Volatilities	Correlation Matrix		
A	7%	12%	1.0000	0.2500	0.0500
B	12%	21%	0.2500	1.0000	0.4500
C	18%	28%	0.0500	0.4500	1.0000

stocks and into bonds, causing the returns on these two asset classes to have a low and sometimes even negative correlation with each other. Recall from Equation 5.19 that a portfolio's variance–covariance matrix is defined by:

$$\Sigma = \sigma^T P \sigma$$

Expanding this expression for the three-asset portfolio in Table 6.1 yields the following variance–covariance matrix:

$$\begin{aligned} \Sigma &= \begin{bmatrix} 0.12 & 0 & 0 \\ 0 & 0.21 & 0 \\ 0 & 0 & 0.28 \end{bmatrix}^T \begin{bmatrix} 1 & 0.25 & 0.05 \\ 0.25 & 1 & 0.45 \\ 0.25 & 0.45 & 1 \end{bmatrix} \begin{bmatrix} 0.12 & 0 & 0 \\ 0 & 0.21 & 0 \\ 0 & 0 & 0.28 \end{bmatrix} \\ &= \begin{bmatrix} 0.0144 & 0.0063 & 0.0017 \\ 0.0063 & 0.0441 & 0.0265 \\ 0.0017 & 0.0265 & 0.0784 \end{bmatrix} \end{aligned}$$

This in turn means that the inverse of the variance–covariance matrix – which we require for our solution – is defined as:³

$$\Sigma^{-1} = \begin{bmatrix} 74.4631 & -12.1382 & 2.5010 \\ -12.1382 & 30.4122 & -10.0040 \\ 2.5010 & -10.0040 & 16.0779 \end{bmatrix}$$

We now have all the elements we need in order to determine the asset weights that deliver the minimum variance achievable for this particular three-asset portfolio. Applying Equation 6.2 to the data in Table 6.1 yields:

$$\begin{aligned} \vec{w}_p^* &= (\vec{e}^T \Sigma^{-1} \vec{e})^{-1} \Sigma^{-1} \vec{e} \\ \vec{w}_p^* &= \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}^T \begin{bmatrix} 74.4631 & -12.1382 & 2.5010 \\ -12.1382 & 30.4122 & -10.0040 \\ 2.5010 & -10.0040 & 16.0779 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right)^{-1} \\ &\quad \begin{bmatrix} 74.4631 & -12.1382 & 2.5010 \\ -12.1382 & 30.4122 & -10.0040 \\ 2.5010 & -10.0040 & 16.0779 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.794 \\ 0.101 \\ 0.105 \end{bmatrix} \end{aligned}$$

The minimum-variance portfolio for this rather limited universe of assets, consisting of just two stocks and one bond, thus suggests that the vast majority (79.4%) of our funds should be invested in government bonds. This is of course not very surprising since these have by far the lowest volatility.

However, not all of our funds are to be placed in long-term government bonds, and this is the small revelation that this simple example provides. A portfolio manager who seeks minimum volatility should *not necessarily* invest all her funds in perceived low-risk long-term government bonds. In fact, she can do even better by also buying stocks to benefit from the diversification effect that they provide in conjunction with bonds. In this example the asset manager is advised to hold 10.1% of the ‘Old Economy’ stock and 10.5% in the ‘New Economy’ stock, resulting in a total equity holding of 20.6%. This may appear surprising to some, but the result from this rather simple portfolio optimisation example clearly demonstrates the potential benefits of diversification between assets and asset classes using quantitative techniques.

Having thus determined the optimal set of asset weights for the minimum-variance portfolio, we can now examine what level of return and volatility we can expect from this particular unique portfolio composition. From Equation 5.2 we recall that portfolio return is defined as:

$$r_p = \vec{w}_p^T \vec{r} = \begin{bmatrix} w_A \\ w_B \\ w_C \end{bmatrix}^T \begin{bmatrix} r_A \\ r_B \\ r_C \end{bmatrix} = [0.794 \quad 0.101 \quad 0.105] \begin{bmatrix} 7\% \\ 12\% \\ 18\% \end{bmatrix} = 8.66\%$$

By applying vector multiplication as demonstrated in the appendix to the previous chapter, we discover that the minimum-variance portfolio is expected to deliver a return of 8.66% over the next year, which is actually slightly above the expected return on the long-term government bond. Having established that the optimised portfolio with the lowest level of risk achievable is expected to generate a level of return higher than that of bonds, all that is left is to determine what level of risk we should expect from this portfolio. From Equation 6.1 we recall that portfolio volatility is given by:

$$\sigma_p = \sqrt{\vec{w}_p^T \Sigma \vec{w}_p}$$

and from this it follows that the volatility of the minimum-variance portfolio is:

$$\sigma_p = \sqrt{\begin{bmatrix} 0.794 \\ 0.101 \\ 0.105 \end{bmatrix}^T \begin{bmatrix} 0.0144 & 0.0063 & 0.0017 \\ 0.0063 & 0.0441 & 0.0265 \\ 0.0017 & 0.0265 & 0.0784 \end{bmatrix} \begin{bmatrix} 0.794 \\ 0.101 \\ 0.105 \end{bmatrix}} = 11.07\%$$

We thus see that the portfolio is expected to incur a volatility of 11.07%, which is notably *lower* than the volatility of the supposedly safest asset, the long-term government bond. In Figure 6.1 we illustrate the level of expected volatility for the three assets and the minimum-variance portfolio.

Note that the expected volatility of the minimum-variance portfolio is significantly lower than the average of the three assets by themselves, and, as indicated, even lower than that of the long-term government bonds. This is a graphic testament to the potential of quantitative portfolio optimisation for portfolio risk management and asset allocation purposes. Note also that this result is obtained using only three assets. Increasing the number of assets available would increase the diversification effects even further.

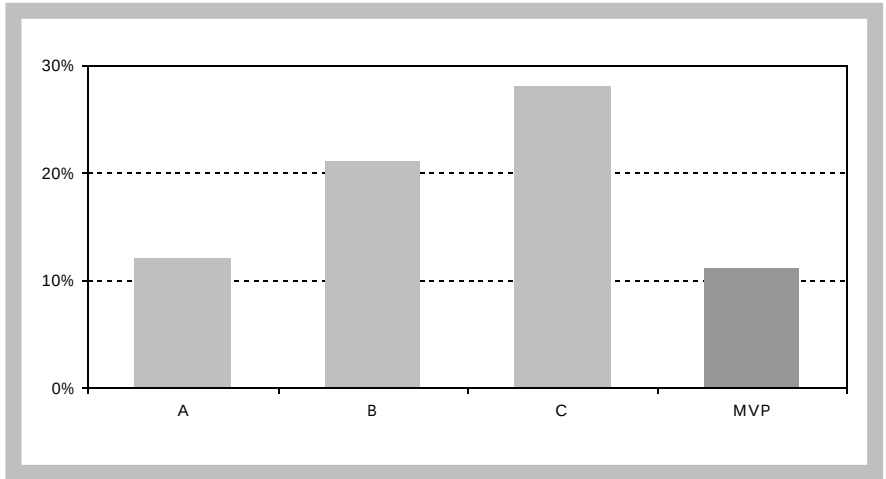


Figure 6.1 Asset and minimum-variance portfolio volatilities

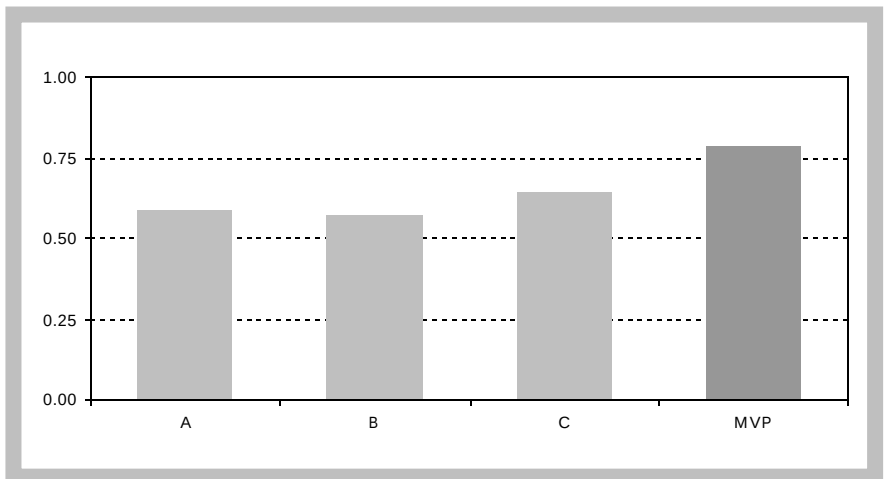


Figure 6.2 Asset and minimum-variance portfolio Sharpe Ratios

We can further illustrate the point by looking at the Sharpe Ratios – the expected return divided by the expected risk – of the three assets as well as that of the minimum-variance portfolio (Figure 6.2).

It is quite evident that the optimised portfolio delivers a much higher level of expected return per unit of expected risk. Alternatively, it delivers a much lower level of expected risk per unit of expected return. This is what quantitative diversification is all about. By quantitatively optimising the portfolio, it is possible to do away with a lot of the volatility that would otherwise be incurred by a naive investor.

Minimising volatility at some level of return

We have now determined the minimum attainable level of portfolio volatility, regardless of the expected returns generated by that particular portfolio composition. However, suppose we are interested in obtaining a very specific level of expected return that lies above that provided by the minimum-variance portfolio. In that case we need to optimise the portfolio as before, but with one additional constraint, namely that the portfolio return must be equal to some specified level $\check{r}$ determined by the portfolio manager. The optimisation problem thus appears as follows:

$$Min = \overline{\vec{w}_p^T} \Sigma \vec{w}_p \quad [\text{Eq. 6.3}]$$

Subject to:

$$\vec{w}_p^T \bullet \vec{r} = \check{r}$$

$$\vec{w}_p^T \bullet \vec{e} = 1$$

where $\vec{w}_p$ is the weight vector, $\vec{w}_p^T$ is the transposed weight vector, $\vec{e}$ is a unit vector consisting of a single column of ones, and $\check{r}$ is a number that expresses our desired level of return, the *target return*. The first constraint implies that the portfolio return must be equal to a fixed value, since calculating $\vec{w}_p^T \bullet \vec{r}$ results in a number or scalar, which is precisely what we want. If for example the target return is 12%, then $\check{r} = 0.12$.

As before, the second constraint ensures that we are fully invested. Again, the solution can be found by using the LaGrange optimisation method. This yields the following result:⁴

$$\vec{w}_p^* = \lambda_1 \Sigma^{-1} \vec{r} + \lambda_2 \Sigma^{-1} \vec{e} \quad [\text{Eq. 6.4}]/[\text{Eq. 6A.23}]$$

where the LaGrange multipliers (Lambdas) appear as follows:

$$\lambda_1 = \frac{\kappa_1^{-1} - (\check{r}\kappa_3^{-1} - \kappa_1^{-1})(\kappa_1\kappa_3^{-1} - \kappa_2\kappa_1^{-1})^{-1}\kappa_2\kappa_1^{-1}}{(\check{r}\kappa_3^{-1} - \kappa_1^{-1})(\kappa_1\kappa_3^{-1} - \kappa_2\kappa_1^{-1})^{-1}} \quad [\text{Eq. 6.5}]$$

$$\lambda_2 = \frac{(\check{r}\kappa_3^{-1} - \kappa_1^{-1})(\kappa_1\kappa_3^{-1} - \kappa_2\kappa_1^{-1})^{-1}}{(\check{r}\kappa_3^{-1} - \kappa_1^{-1})(\kappa_1\kappa_3^{-1} - \kappa_2\kappa_1^{-1})^{-1}} \quad [\text{Eq. 6.6}]$$

and the Kappas are defined as:

$$\kappa_1 = \vec{e}^T \Sigma^{-1} \vec{r} \quad [\text{Eq. 6A.14}]$$

$$\kappa_2 = \vec{e}^T \bullet \Sigma^{-1} \bullet \vec{e} \quad [\text{Eq. 6A.15}]$$

$$\kappa_3 = \vec{r}^T \bullet \Sigma^{-1} \bullet \vec{r} \quad [\text{Eq. 6A.18}]$$

$$\kappa_4 = \vec{r}^T \Sigma^{-1} \vec{e}^T \quad [\text{Eq. 6A.19}]$$

Note that the Kappas are numbers or scalars, since they each represent a $N \cdot N$ matrix pre- and post-multiplied by a $1 \cdot N$ vector and an $N \cdot 1$ vector respectively. This means that we can easily determine the asset weights that give us the minimum expected volatility for a vector of expected asset returns. Again, we denote $\vec{w}_p^*$ as the solution (a vector of asset weights) to the problem, but here it represents the vector that minimises portfolio volatility at some specified level of return. Again, there is no restriction on short selling of assets. Note from Equation 6.4 that the variance–covariance matrix no longer by itself reveals the optimal combination of assets that yields the desired solution, and that the problem has become rather more complex by adding the target return constraint.

Consider again the example presented above of a hypothetical portfolio containing three assets *A*, *B* and *C*, where we now specify that we have a desired expected return of 12%. In order to determine the optimal solution, we first need to determine the values of the four Kappas. Inverting the variance–covariance matrix and using the unit and return vectors, we obtain:

$$\kappa_1 = \vec{e}^T \Sigma^{-1} \vec{r} = 7.0737$$

$$\kappa_2 = \vec{e}^T \bullet \Sigma^{-1} \bullet \vec{e} = 81.6708$$

$$\kappa_3 = \vec{r}^T \bullet \Sigma^{-1} \bullet \vec{r} = 0.7507$$

$$\kappa_4 = \vec{r}^T \Sigma^{-1} \vec{e} = \kappa_1 = 7.0737$$

Note that Kappa-1 and Kappa-4 are identical, since the inverted variance–covariance matrix is symmetrical around the diagonal. Note also that these values, which can be said to characterise the investment universe since they describe the volatility and correlation structure of the portfolio of available assets, do not change with the desired level of return. They are simply a function of asset characteristics – that is, a characterisation of the variance–covariance matrix and the asset returns – and are as such independent of portfolio composition.

Using the target return of 12.0% and the four Kappas, we obtain the two LaGrange multipliers:

$$\lambda_1 = \kappa_1^{-1} - (\check{r}\kappa_3^{-1} - \kappa_1^{-1})(\kappa_1\kappa_3^{-1} - \kappa_2\kappa_1^{-1})^{-1}\kappa_2\kappa_1^{-1} = 0.2420$$

$$\lambda_2 = (\check{r}\kappa_3^{-1} - \kappa_1^{-1})(\kappa_1\kappa_3^{-1} - \kappa_2\kappa_1^{-1})^{-1} = -0.0087$$

Inserting the values of λ_1 and λ_2 into the solution provided in Equation 6.5 we obtain:

$$\vec{w}_p^* = 0.2420 \cdot \Sigma^{-1} \cdot \vec{r} + (-0.0087) \cdot \Sigma^{-1} \cdot \vec{e}$$

Finally we insert the inverse variance–covariance matrix, the return vector and the unit vector, to obtain the vector of asset weights that generate the portfolio with the minimum attainable level of expected volatility for an expected return of 12%:

$$\vec{w}_p^* = 0.2410 \cdot \begin{bmatrix} 4.2060 \\ 0.9991 \\ 1.8686 \end{bmatrix} + (-0.0087) \cdot \begin{bmatrix} 84.8259 \\ 8.2700 \\ 8.5749 \end{bmatrix} = \begin{bmatrix} 0.453 \\ 0.170 \\ 0.377 \end{bmatrix}$$

This simple example shows that a portfolio consisting of 45.3% long-term government bonds, 17.0% ‘Old Economy’ stocks, and 37.7% ‘New Economy’ stocks will – in theory – provide us with 12% expected return. The expected volatility of this portfolio is:

$$\sigma_p = \sqrt{\vec{w}_p^T \cdot \Sigma \cdot \vec{w}_p}$$

$$\sigma_p = \begin{bmatrix} 0.453 \\ 0.170 \\ 0.377 \end{bmatrix}^T \cdot \begin{bmatrix} 0.0144 & 0.0063 & 0.0017 \\ 0.0063 & 0.0441 & 0.0265 \\ 0.0017 & 0.0265 & 0.0784 \end{bmatrix} \cdot \begin{bmatrix} 0.453 \\ 0.170 \\ 0.377 \end{bmatrix} = 14.26\%$$

Note that the portfolio is expected to yield a return of 12%, which is equal to the expected return on the ‘Old Economy’ stocks by itself. So in terms of expected return versus expected risk, we would receive the same expected return from holding the optimised portfolio as from holding 100% ‘Old Economy’ stocks. However, the optimised portfolio’s expected volatility is significantly lower than that expected from the ‘Old Economy’ stock.

By also including both bonds and ‘New Economy’ stocks into the portfolio in a controlled and structured manner, and by taking advantage of the specific correlational characteristics of the returns on this portfolio’s assets, we can thus

obtain the same return as on the ‘Old Economy’ stocks, but at a much lower expected level of volatility. Specifically we are expected to incur a volatility of just 14.26% from the optimised portfolio, whereas a portfolio fully invested in ‘Old Economy’ stocks would imply an expected volatility of as much as 21%, which is obviously significantly higher than the expected volatility of the optimised portfolio.

Maximising return at some level of volatility

Maximisation of returns is probably the most common perspective on investing. For some investors this is the only goal that is given attention, which, ironically, is why most of these particular investors end up losing money in the stock market. More prudent investors will let risk enter into the equation in one form or another. The exact approach to this can be more or less rigorous and systematic, but taking account of expected risk ought in principle to be as important as focusing on expected return. As we have already argued, portfolio risk is a function of (the variation in) portfolio returns. Conversely there can be no potential for return without the potential for risk. In the framework presented here, we thus argue for giving equal consideration to both return and risk since they are by their very nature and definitions intertwined.

Quantitative maximisation of portfolio return must be carried out subject to two constraints. The first is that the sum of the portfolio weights must equal 1, and the second is that portfolio volatility must be equal to some pre-specified *target volatility* $\check{\sigma}$. The problem is thus analogous to the previous problem of minimising volatility for some level of return, and it can be arranged formally as follows:

$$\text{Max } f(x) = \vec{w}_p^T \bullet \vec{r} \quad [\text{Eq. 6.7}]$$

Subject to:

$$\vec{w}_p^T \bullet \Sigma \bullet \vec{w}_p = \check{\sigma}$$

$$\vec{w}_p^T \bullet \vec{e} = 1$$

Again, the optimisation problem above is solved using the LaGrange methods, but here we solve for the weight vector that maximises return for a given level of volatility. It could be argued that the volatility should be scaled in accordance with some measure of investor risk aversion⁵, but we leave that aside for ease of computation. When solving the problem we must ensure that the portfolio volatility is at a predefined level $\check{\sigma}$, and that asset weights sum to 1.

These are the two constraints defined above, under which the optimisation must be performed. From the definition of portfolio efficiency we know that minimising portfolio volatility subject to a target return constraint is equivalent to maximising portfolio return subject to a target volatility constraint. This is because both problems, when solved, result in an optimal solution such that we would be unable to produce a better trade-off between return and risk by changing the portfolio composition. In other words, we are faced with a dual problem such that we need only solve one side of it in order to obtain the result for both. In this chapter we will therefore refrain from the algebra required to explicitly determine the portfolio that maximises expected portfolio return for a given level of expected volatility. Instead we will use the fact that such an exercise is equivalent to solving the problem of determining the minimum level of expected portfolio volatility for a given level of expected portfolio return, the solution to which is provided in Equation 6.4.

Being able to maximise expected portfolio return subject to the constraint of fixing the expected portfolio volatility enables us to perform an interesting experiment on the three-asset portfolio of long-term government bonds, 'Old Economy' stocks and 'New Economy' stocks. Suppose we would like to construct a portfolio with expected volatility of 12%. In order to obtain this we could simply invest 100% of our funds in the long-term government bond to obtain an expected return of 7% and an expected volatility of 12%. Alternatively we could optimise the portfolio and determine the asset weights that would give us the maximum expected portfolio return, subject to an expected portfolio volatility of 12%. This yields the optimal weight vector:

$$\vec{w}_p^* = \begin{bmatrix} 0.618 \\ 0.136 \\ 0.2461 \end{bmatrix}$$

With these asset weights we would receive an expected portfolio return of:

$$r_p = \vec{w}_p^T \bullet \vec{r} = [0.618 \quad 0.136 \quad 0.246] \bullet \begin{bmatrix} 7\% \\ 12\% \\ 18\% \end{bmatrix} = 10.39\%$$

The expected return on the optimised (and thus optimally diversified) portfolio is now 10.39%, which is a full 3.39% higher than the expected return from investing solely in long-term government bonds, which would entail the exact same level of expected volatility.

This example illustrates that quantitative portfolio optimisation can be applied as a portfolio risk-management tool, or as a tool for enhancing expected portfolio returns, or as a combination of the two. It should be noted

that whether this expected outcome actually materialises rests squarely on the accuracy of the estimates of asset return, volatilities and correlations. If these are not estimated or rather forecasted correctly, we will not obtain the results suggested above. In fact we might end up doing worse than if we had invested solely in the long-term government bond. This underscores the crucial point that applying quantitative optimisation techniques, whether for return enhancement or risk-management purposes, does not magically produce superior portfolios that will invariably generate higher returns at lower risk or vice versa, and the use of such techniques does not remove the responsibility from the asset manager. Optimising a portfolio does not add value to the investment process unless the estimated model inputs add value. Paraphrasing an old saying, one could reasonably argue that in this case, *garbage in* leads to a whole lot of *garbage out*. This leaves us with the conclusion that the asset manager should still make the investment decisions, and quantitative investment techniques such as those presented so far in this book should merely serve as a supplement to the existing workbox of investment management tools.

However, if one wants to employ quantitative tools for investment management purposes, it obviously makes sense to develop and improve upon them as much as possible. In the following sections and chapters we shall extend the traditional quantitative portfolio optimisation approach as presented so far, and add several features that make the process more intuitive and workable, and that have the potential for making the arrived-at solutions much more robust. Before we can do that, however, we need to examine one final topic, namely that of the so-called 'efficient frontier'.

THE EFFICIENT FRONTIER

Another look at diversification

Recall how in the two-asset portfolio in Figure 5.4 we could trace out the possible or attainable portfolio return and risk combinations, by varying the weights of stocks *A* and *B* between 0% and 100% respectively. The basic principle was to illustrate the return and risk of the portfolio in the two-dimensional *return-risk space*. In Figure 6.3 the thick grey line traces out the return/risk combinations of the two assets assuming a correlation coefficient between returns of -0.5 . As in Figure 5.4, expected returns on the two assets are 4% and 18% respectively, and volatilities are 11% and 26% respectively.

Using the quantitative optimisation techniques presented in the previous section we find that the minimum-variance portfolio – indicated in the chart by an X – consists of 75.6% of asset *A* and 24.4% of asset *B*, yielding an

expected return of 7.41% and an expected volatility of 7.53%, implying strong diversification benefits from the relatively low correlation coefficient of -0.5 . The *maximum-variance portfolio* is simply the portfolio consisting of 100% of asset B, yielding a return of 18% and a volatility of 26%. Note that the part of the return/risk trade-off line that lies between A and X is dominated by the part of the line that lies between X and B, in the sense that instead of holding 100% in asset A, we could receive a higher amount of return for the same amount of risk by selecting portfolio Y instead. The same is true for all the other possible portfolios lying on the XA line. They are all dominated in terms of their return/risk ratio by the portfolios that lie on the XY line. In other words, the XB line is return/risk efficient, and the XA line is not. This result leads us to the definition of the *efficient frontier*:

The efficient frontier is the line between the minimum-variance portfolio and the maximum-variance portfolio that traces out all attainable portfolios (asset combinations) that produce optimal/efficient portfolios.

In other words, the efficient frontier is the line in return/risk space that traces out all the portfolios for which we cannot obtain a higher level of return for a given level of risk, or alternatively for which we cannot obtain a lower level of risk for a given level of return. So the line XB in Figure 6.3 is by definition the efficient frontier for this particular two-asset portfolio.

Now assume that we have many more assets available to us. In that case we would not necessarily include all stocks in the portfolio. We might elect not to

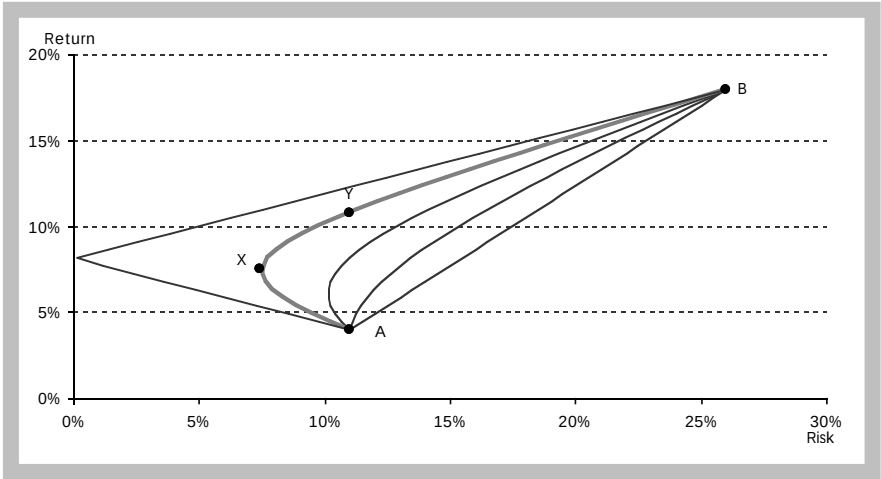


Figure 6.3 Return/risk combinations with correlation coefficient of -0.5

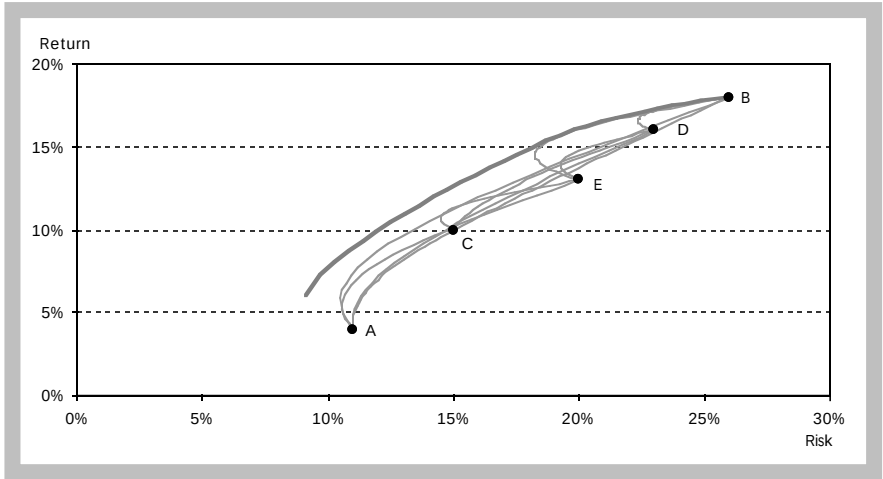


Figure 6.4 The efficient frontier for a five-asset portfolio

hold one or more of the stocks available to us, and we might elect to hold stocks in different proportions to obtain the best return/risk trade-off. Adding three more hypothetical assets to Figure 6.3, and assuming that the correlation coefficient between asset *A* and asset *B* is now 0.5, yields the asset-by-asset return/risk combinations shown in Figure 6.4, as well as the efficient frontier (the thick grey line) for the new five-asset portfolio.

The thin grey lines trace out the expected level of return and risk that result from every possible combination of weights in the two-stock portfolios, and the thick grey line traces out the efficient frontier when all five stocks are available for optimisation of the portfolio. Note that we have not included the part of the efficient frontier that is dominated by other portfolios. This implies that all of the five assets in the portfolio, except asset *B*, are inefficient investments by themselves. Only if we combine those with the other assets available do we obtain efficient portfolios. Asset *B* is the only efficient investment among the five assets, because it is impossible to obtain a higher return at the same level of risk as that provided by asset *B*. Correspondingly it is impossible to obtain a lower level of risk for the same return as that provided by asset *B*. Note also how the diversification effect between asset *B* and assets *E* are somewhat higher than for assets *C* and *E*. This is because the correlation coefficient between asset *B* and asset *E* is significantly lower than the correlation coefficient between asset *C* and asset *E*, thus providing more opportunity for diversification.

We do not actually need to calculate all the possible efficient portfolios to be able to produce the efficient frontier, which in principle contains an infinite number of different portfolios. Moving along the efficient frontier, we

will often observe ‘kinks’, which represents points on the efficient frontier where assets are either added or dropped from the portfolio. These points are called *corner portfolios*. It can be shown that every point on the efficient frontier is a linear combination of the two corner portfolios most closely adjacent to that point. Thus we only need the corner portfolios to be able to construct the entire efficient frontier. In practice we employ computerised optimisation techniques, and so we can easily obtain as many points on the efficient frontier as we like.

An equity portfolio consisting of four US indices plus cash

To introduce more intuition, let us now introduce a real-world example, using as asset classes four widely published US stock indices as well as cash:

- the Dow Jones Industrial – representing large cap stocks
- the S&P500 – representing a broad range of ‘Old Economy’ stocks
- the NASDAQ Composite – representing ‘New Economy’ stocks
- the Russel 2000 – representing small cap stocks
- cash proxy – 1 Month deposit rate for the US Dollar.

The inclusion of cash into the investable universe will enable us to liquidate our invested equity positions if we should expect the stock markets to fall.

Table 6.2 lists asset returns, volatilities and the correlation matrix specifying the correlation coefficients between the returns on the four indices for our hypothetical US-centric portfolio. Volatilities and correlations are calculated based on daily returns from 1 January to 1 May 2002. Expected returns – realistic or not – are chosen in accordance with a hypothetical asset manager’s expectations about market conditions over the coming year.

Table 6.2 Asset class characteristics – four US equity indices and cash

	Return	Volatility	Correlation Matrix				
Cash	2.2%	5.2%	1.00	0.02	0.01	−0.01	−0.01
Dow Jones Industrial	5.5%	15.5%	0.02	1.00	0.89	0.60	0.65
S&P500	6.4%	17.8%	0.01	0.89	1.00	0.82	0.77
NASDAQ Composite	8.8%	30.5%	−0.01	0.60	0.82	1.00	0.87
Russel 2000	6.5%	18.4%	−0.01	0.65	0.77	0.87	1.00

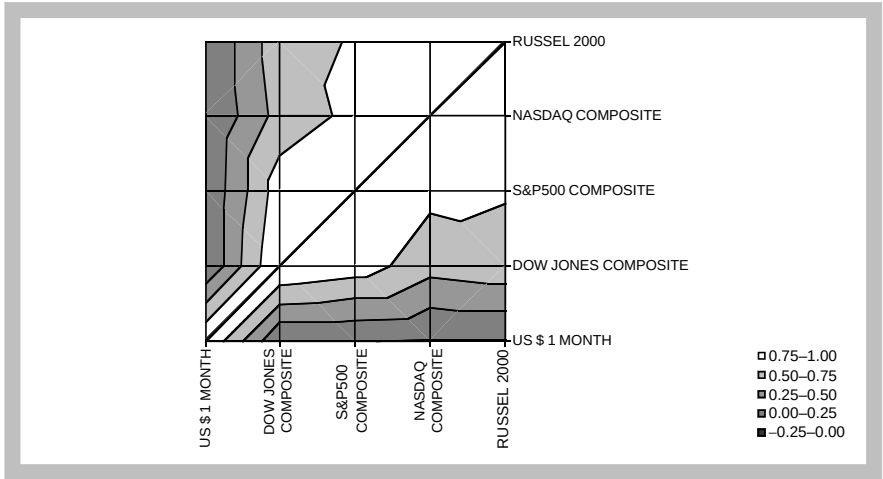


Figure 6.5 Correlation matrix – four US equity indices and cash

As we can see, the NASDAQ has by far the highest level of volatility, but has also been assigned the highest level of expected return. The lowest level of expected risk and return is expected, not surprisingly, for cash. Cash has a low correlation with the other asset classes, which in turn are highly correlated with each other. In Figure 6.5 we illustrate the correlation matrix presented in Table 6.2.

This chart underscores the fact that the correlation coefficients between cash and the other four asset classes are quite low. Thus, there would appear to be a

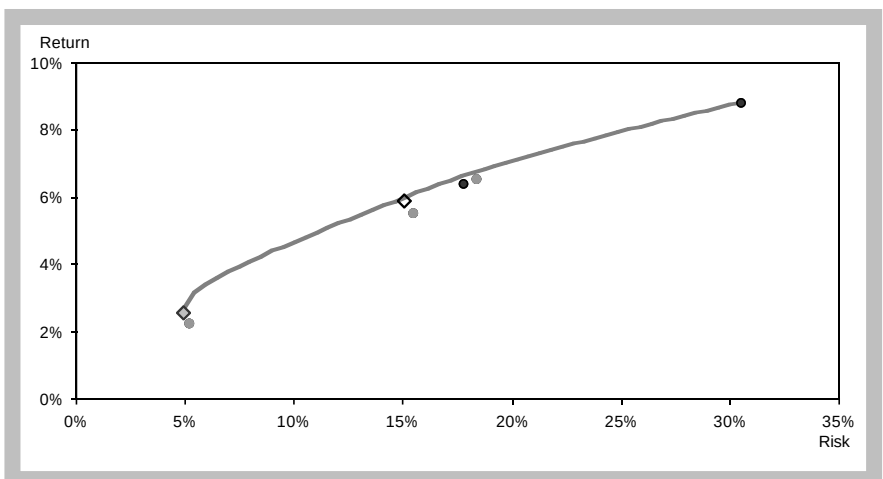


Figure 6.6 Efficient frontier – four US equity indices and cash

potential for quite significant diversification benefits. Using the information provided in Table 6.2 and carrying out the quantitative optimisation presented above, we obtain the efficient frontier shown in Figure 6.6. The large grey diamond to the far left indicates the minimum-variance portfolio. As previously pointed out, this portfolio has lower volatility than the asset class with the lowest volatility – in this case cash. The dots in the return/risk space represent the return/risk combinations of the five asset classes. The light grey dots indicate the asset classes that *are* included in the current portfolio, and the black dots indicate asset classes that are *not* included in the current portfolio.

The white diamond near the efficient frontier indicates the equal-weighted portfolio – that is, the portfolio that holds 20% in each asset class. This portfolio is obviously inefficient in the mean-variance sense, since it lies below the efficient frontier, but not by very much. This is quite often the case. Equal-weighting all assets or asset classes tends to result in portfolios that lie quite close to the efficient frontier. However, the return and risk estimates that form the basis of this portfolio calculation do not take into account the level of confidence the manager has in his estimates, so there may still be good reason to construct portfolios that are very different from the equal-weighted portfolio.

It may not be immediately evident from the chart above, but the minimum-variance portfolio only includes cash, the Dow Jones Industrial Average and the Russel 2000. The other two asset classes are left out of the portfolio altogether. Table 6.3 characterises the minimum-variance portfolio (MVP) and the equal-weighted portfolio (EWP) for this – albeit limited – investment universe.

It makes good sense that just under 90% of the portfolio has been allocated to cash, which is by far the least risky asset class, and that the remainder has been placed in stocks. But why have only two and not all four equity indices

Table 6.3 MVP and EWP – four US equity indices and cash

	Minimum-variance Portfolio	Equal-weighted Portfolio
Cash	89.60%	20.00%
Dow Jones Industrial	6.67%	20.00%
S&P500	0.00%	20.00%
NASDAQ Composite	0.00%	20.00%
Russel 2000	3.73%	20.00%
Expected Return	2.58%	5.88%
Expected Volatility	4.93%	15.02%
Expected Sharpe Ratio	0.52	0.39

been included in the minimum-variance portfolio? Intuitively one would have thought that including all four indices would provide a higher degree of diversification.

This intuition is basically sound, but the problem is that the optimisation algorithm just does what it is told and it is very good at what it does. It simply finds the optimum – that is, the portfolio with the lowest expected volatility – and since no other restrictions have been placed on it, it ends up with just two equity indices and cash in the portfolio. While optimal in a strictly mathematical sense, this solution is not very attractive from a practical investment management perspective. One way of circumventing or alleviating this problem would be to simply apply brute force and to constrain the optimisation to include all equity indices by setting minimum levels above zero for all asset weights. However, we would then be faced with the problem of choosing appropriate constraints. And even if we could somehow define such appropriate minimum levels or weights, the solution to the optimisation problem would then by definition be sub-optimal.

These issues have the potential to seriously limit the power and applicability of quantitative portfolio optimisation and asset allocation. In the following chapters we shall introduce more advanced optimisation and asset allocation techniques that solve this and several other problems that tend to plague traditional mean-variance optimisation techniques. For now, however, we continue with the example of a portfolio of equity indices and cash, but now we introduce non-US equities in order to observe what possible diversification benefits those might provide us with.

BENEFITS FROM INTERNATIONAL DIVERSIFICATION

Conventional wisdom

Traditionally, investors have been quite reluctant for some reason to invest in countries other than their own. This is particularly true for the United States, but it is also true for investors in most other countries around the world. One reason for this may be that investors look at their local equity market and think they can obtain exposure to any type of stock – that is, stream of cash flow – they would desire. They can choose between energy companies, software companies, auto companies, telecommunications companies, and media companies, and perhaps not surprisingly they see no need to broaden out into global equity markets. Another reason may be the perception that equity markets in the rest of the world are less transparent and less efficient. Finally, the perceived currency risk and the difficulty in predicting this might scare potential investors away from placing money abroad. All three points may be

valid to some extent, but over the long run they ought not to be a serious impediment to international diversification.

Even though the use of international diversification is becoming more widespread, many equity investors – pension funds and other institutional investment organisations, as well as individual retail investors – still do not take sufficient advantage of the opportunities for diversification that the equity markets in the rest of the world offer them relative to their domestic equity exposure.

We shall demonstrate the potential benefits to international diversification with a simple example. Continuing with the portfolio of four US indices plus cash presented in the previous section, we now add a new asset class representing investment opportunities outside the US. We thus include the MSCI World ex US as a proxy for international equities, and even though it is only one equity index additional to the four already present in the investable universe, it is quite evident from Table 6.4 that its inclusion has the potential to affect significant diversification benefits.

Note that the correlation coefficient between the MSCI World ex US and the other asset classes is quite low. All correlation coefficients are below 0.5, and the correlation coefficient between the MSCI World ex US and the NASDAQ is as low as 0.37. This is perhaps slightly surprising given the manner in which technology companies were very much the only game in town during much of the late 1990s. More than anything else, this was probably a result of the NASDAQ behaving independently from the rest of the world, rather than the other way round, which is another indication of the bubble-like nature of that particular index during much of the period upon which these parameters are estimated. Almost completely detached from the rest of the world’s equity markets, it continued rising based on the opportunities, at that time seemingly limitless, for profit growth in this ‘Information Age’. It could reasonably be argued that this

Table 6.4 Asset characteristics – four US equity indices, MSCI-W ex US and cash

	Return	Volatility	Correlation Matrix					
Cash	2.2%	5.2%	1.00	0.02	0.00	−0.01	−0.01	0.00
Dow Jones Industrial	5.5%	15.5%	0.02	1.00	0.89	0.60	0.65	0.39
S&P500	6.4%	17.8%	0.00	0.89	1.00	0.82	0.77	0.41
NASDAQ Composite	8.8%	30.5%	−0.01	0.60	0.82	1.00	0.87	0.37
Russel 2000	6.5%	18.4%	−0.01	0.65	0.77	0.87	1.00	0.46
MSCI World ex US	5.7%	14.5%	0.00	0.39	0.41	0.37	0.46	1.00

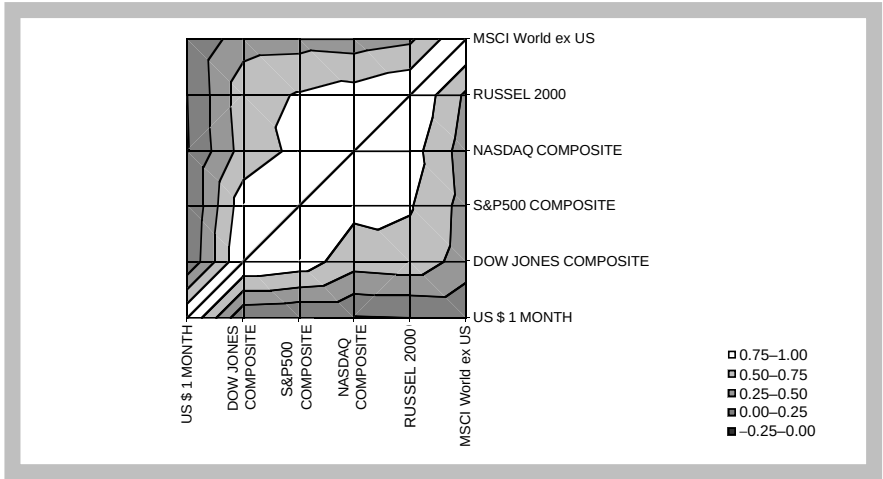


Figure 6.7 Correlation matrix – four US equity indices, MSCI-W ex US and cash

fact makes the parameters presented in the table somewhat less representative of ‘normal’ market conditions, but since this example is simply meant to demonstrate a methodology, we shall not concern ourselves with that particular issue.

Correlations between the MSCI World ex US and four other equity indices indicate a clear opportunity for diversification beyond that found in national stock markets. The correlation structure of the six asset class portfolio including the MSCI World ex US appears in Figure 6.7.

The efficient frontier including international equity

Using the correlation matrix above, together with the expected returns and volatilities of all six asset classes in the portfolio, enables us to determine the effects on the efficient frontier of adding international equity to the portfolio. The efficient frontier for the portfolio that includes the MSCI World ex US as an asset class appears as in Figure 6.8.

The figure clearly shows the effect of the new asset class, indicated here by a grey square. The previous efficient frontier is indicated by the dotted line, and the new efficient frontier making use of international equity is indicated by the solid grey line. Even though it has a low expected return compared to most of the other asset classes, the low correlation of the MSCI World ex US with the US equity indices causes the new efficient frontier to be shifted up and to the left. Thus, the asset manager can now obtain both a lower expected

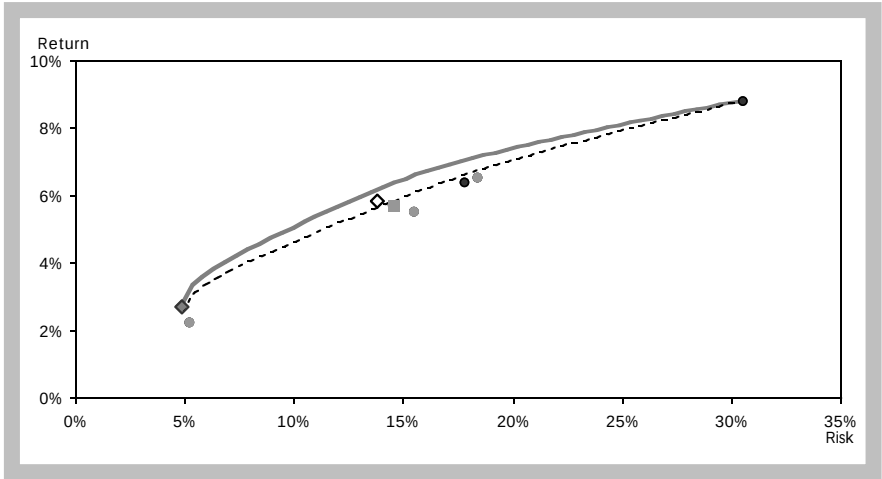


Figure 6.8 Efficient frontier – five US equity indices, MSCI-W ex US and cash

portfolio volatility for a given level of expected portfolio return, and a higher expected portfolio return for a given level of expected portfolio volatility. All the portfolios along the new, internationally diversified, efficient frontier dominate the portfolios along the old efficient frontier, except for the maximum variance portfolio which in both cases included nothing but the NASDAQ Composite.

Note that the minimum-variance portfolio, again indicated by the grey diamond to the far left of the chart, makes immediate use of the diversification opportunities provided by the inclusion of international equity into the investable universe. Note also how much less attractive the (new) equal-weighted portfolio has become, as it now lies further away from the efficient frontier. Cash, the Dow Jones Industrial Average and the Russel 2000 are still included, and the NASDAQ is still excluded completely, along with the S&P500. Again, one cannot help but think of this as slightly counterintuitive, since we would expect the inclusion of more asset classes to result in a lower level of expected volatility; as indicated, however, we shall attempt to remedy this in the following chapters. Table 6.5 characterises the new minimum-variance portfolio (MVP) as well as the equal-weighted portfolio (EWP) for the expanded investable universe.

The optimal portfolio now includes around 8% international equity, and around 15% in equities in total, compared to around 10% in total for the portfolio that did not include international equity. The holdings in the two previously included indices as well as the holding is cash are now all lower than before. This, however, is particularly true for the Russel 2000, which has

Table 6.5 MVP and EWP – five asset classes and cash

	Minimum-variance Portfolio	Equal-weighted Portfolio
Cash	85.41%	16.67%
Dow Jones Industrial	5.15%	16.67%
S&P500	0.00%	16.67%
NASDAQ Composite	0.00%	16.67%
Russel 2000	1.29%	16.67%
MSCI World ex US	8.15%	16.67%
Expected Return	2.71%	5.85%
Expected Volatility	4.81%	13.75%
Expected Sharpe Ratio	0.56	0.43

been more than halved compared to when international equity was not included in the investable universe. It thus seems that the Russel 2000 has to a large extent been directly substituted by international equity. This in turn tells us that in terms of asset class characteristics, the Russel 2000 and the MSCI World ex US are actually quite similar relative to the other four asset classes.

Note that the volatility of the minimum-variance portfolio is now down to 4.81%, from 4.93% for the portfolio that did not include international equity; and that the expected Sharpe Ratio of the minimum-variance portfolio is now

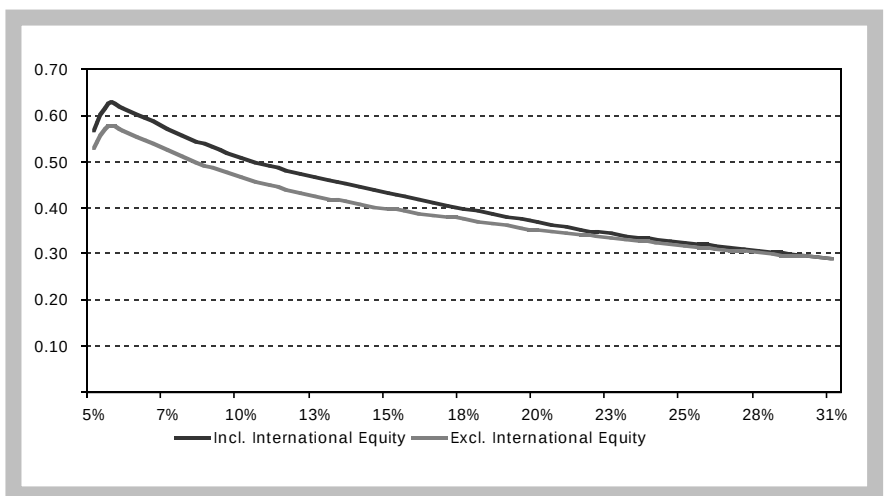


Figure 6.9 Sharpe Ratios for the two efficient frontiers

higher, at 0.56 compared to 0.52. Note also that even though the equal-weighted portfolio is further away from the new efficient frontier and thus – on a relative basis – a worse alternative than before, its Sharpe Ratio is actually higher at 0.43 than the 0.39 that it was with international equity excluded in the investable universe.

We can illustrate the improvement in the overall return/risk trade-off from the inclusion of international equity by examining the Sharpe Ratios for the two efficient frontiers along the entire range of feasible volatilities (see Figure 6.9).

Overall, it seems therefore, that there is significant potential in terms of risk reduction and/or return enhancement if one broadens the investable universe from holding just domestic assets to holding international assets also.

Further issues

Aside from this clear shift of the efficient frontier up and to the left in return/risk space, note also that the efficient frontier is now slightly further away from the equal-weighted portfolio marked by the star in Figure 6.8. Examining the equal-weighted portfolio including international equities we find that it will result in an expected return of 12.33%, whereas an optimised portfolio with the same expected volatility will result in an expected return of 12.99%. The difference of 0.66% is not really a lot, but the composition of the optimised portfolio is very different from that of the equal-weighted portfolio.

Let $\vec{w}^E$ denote the vector of equally weighted asset classes, and $\vec{w}^*$ the optimised weight vector with the same expected return. The difference between the two vectors turns out to be:

$$\vec{w}^E - \vec{w}^* = \begin{bmatrix} 16.67\% \\ 16.67\% \\ 16.67\% \\ 16.67\% \\ 16.67\% \\ 16.67\% \end{bmatrix} - \begin{bmatrix} 85.41\% \\ 5.15\% \\ 0.00\% \\ 0.00\% \\ 1.29\% \\ 8.15\% \end{bmatrix} = \begin{bmatrix} -68.75\% \\ 11.52\% \\ 16.67\% \\ 16.67\% \\ 15.38\% \\ 8.51\% \end{bmatrix}$$

These differences are very large, particularly when one considers that the expected return from the optimised portfolio in excess of that of the equal-weighted portfolio is a mere 0.13%. This very significant change in the portfolio's composition for such a small change in expected return hints at the fact that simple quantitative portfolio optimisation brings with it a problem of

unstable asset weights when changing required target volatility or return levels. This instability or lack of robustness of optimisation results could potentially render the technique useless for practical investment applications. However, there are a series of remedies to this and other problems arising from the use of quantitative optimisation, and these will be discussed in subsequent chapters.

Correlations among country Indices

To underscore further the point of benefits from international diversification, Table 6.6 presents correlations between the returns on large international equity indices over the period 1960–1984.

The table is another indication that there are potentially large benefits from diversifying portfolios using international equity markets. Correlation coefficients are not quite as low as would be expected based on the information in Table 6.2. This is due to different estimation periods. However they are undeniably much lower than correlations among domestic market indices whether they be large- or small-cap, Value or Growth, ‘Old’ or ‘New Economy’ indices.

We can conclude from this that relying solely on domestic equity for equity portfolio diversification is clearly inadequate and sub-optimal. The explicit specification of asset class characteristics such as correlation coefficients that quantitative portfolio optimisation and analysis brings with it thus reveals the opportunities available to equity managers – opportunities that might not be fully exploited or even considered by non-quantitative portfolio construction techniques.

Table 6.6 Correlations among major international equity indices

	Dow Jones	S&P500	NASDAQ	FTSE100	DAX30	Tokyo SE	Hang Seng
Dow Jones	1.00	0.89	0.60	0.39	0.38	0.11	0.12
S&P500	0.89	1.00	0.82	0.40	0.40	0.10	0.12
NASDAQ	0.60	0.82	1.00	0.34	0.38	0.09	0.13
FTSE100	0.39	0.40	0.34	1.00	0.65	0.23	0.35
DAX30	0.38	0.40	0.38	0.65	1.00	0.24	0.39
Tokyo SE	0.11	0.10	0.09	0.23	0.24	1.00	0.39
Hang Seng	0.12	0.12	0.13	0.35	0.39	0.39	1.00

The efficient surface

An extension of the efficient frontier concept is the *efficient surface*. Imagine holding a portfolio in which one or several of the asset classes is or are fixed at some percentage level of the total portfolio holding. This might be the case for pension funds holding real estate in the portfolio that cannot easily be sold off. Similarly, while having a large cash position for an extended period of time may be less risky, it can seriously impede performance as the investment opportunity set defined by the efficient frontier will be limited, and more so the larger the cash position. Using the portfolio of four US equity indices, one international equity index and cash, we demonstrate this point in Figure 6.10 by presenting an illustration of efficient portfolios for varying levels of the minimum cash position. The result is the efficient surface.

When the cash position, or any other asset class for that matter, is locked, the investment opportunity set available to the manager is clearly constrained. This can be seen from the way in which the efficient frontiers are forced downwards as the required cash position is gradually increased from 0% to 30%. Furthermore, the increasingly larger required cash position limits us from attaining return and risk levels above certain limits.

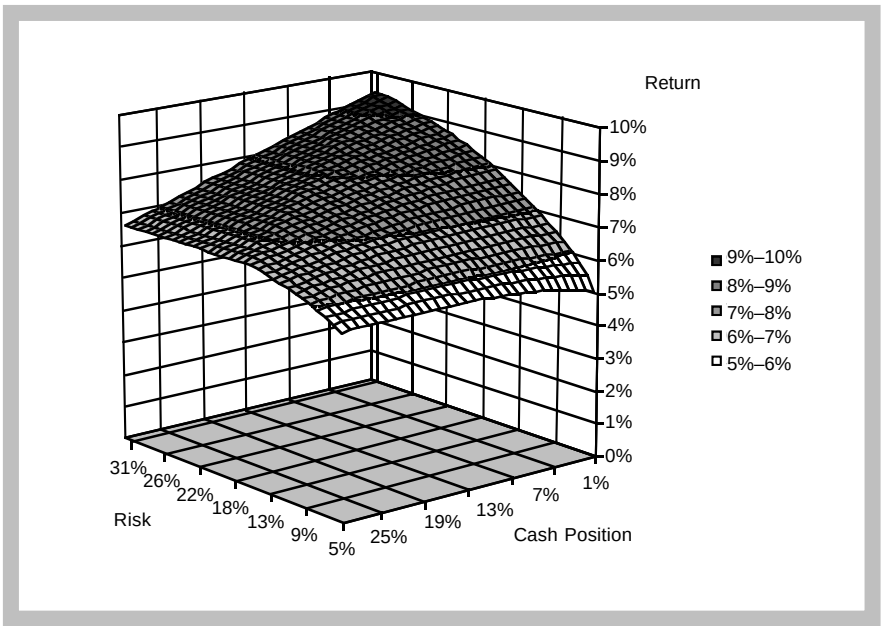


Figure 6.10 The efficient surface – varying minimum cash position

OPTIMISATION AND DIVERSIFICATION

Adding random stocks to the portfolio and optimising

In Figure 5.5 in the previous chapter we demonstrated the ‘naive’ diversification approach of simply adding stocks to a portfolio to reduce volatility. A natural extension would be to apply optimisation to the process, in order to observe its impacts. We thus repeat the procedure of adding stocks and calculating volatility, but instead of equal weighting all stocks, we determine the minimum-variance portfolio for every try. We compare the result with the result from the ‘naive’ approach. Figure 6.11 illustrates the portfolio volatilities as we successively add more stocks to the portfolio and the optimise.

The thick black line indicates the average portfolio volatilities as more and more stocks are added to the portfolio, but *after* optimisation has been carried out. The thick grey line indicates the same line but without optimisation, which is the line shown in Figure 5.5. Included also are a set of thin grey lines that illustrate portfolio volatility in 10 of the 100 trials where stocks were successively added to the portfolio. What is important to remember here is that calculating portfolio volatilities now involves optimisation, which means that the optimiser (working on behalf of the portfolio manager) now has the opportunity to include or exclude stocks as they become available. As is

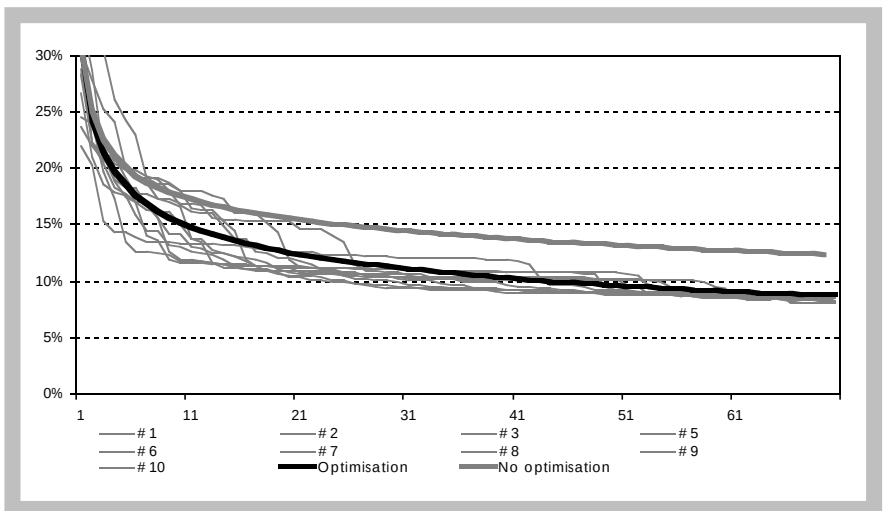


Figure 6.11 Volatility of optimised portfolio as 70 randomly chosen stocks are successively added to the portfolio

evident, this results in a significantly lower level of portfolio volatility when the investable universe is gradually increased.

Figure 6.11 clearly shows the significant effect portfolio optimisation can have on portfolio volatility as more and more stocks are added to the portfolio. For the first 5 stocks the effect is limited, but from 5 stocks up the added diversification, by 'intelligent' exclusion, inclusion and weighting of stocks according to their characteristics, is dramatic. Please note that the assets are the exact same stocks as in the case of no optimisation (the thick grey line). Having added 70 stocks, the difference in volatility between the non-optimised and the optimised portfolios is 3.1% on average and around 3.6% when 70 stocks have been included in the portfolios. These are relatively large difference that are partially due to the fact that general market volatility was high during the period used for estimating asset characteristics. But the effect of employing optimisation is still impressive, and underscores the potential of deliberate quantitative portfolio construction using optimisation techniques.

In this example the portfolio manager would be able to reach the desired level of volatility much more quickly than by using the 'naive' approach without optimisation, since she no longer relies on more or less accidental or randomly occurring diversification effects. Importantly, the return aspect has not been considered explicitly here, but it is obvious from the analysis throughout this chapter that optimising the portfolio will not only provide benefits in terms of lower expected portfolio volatility, but will in most cases also be able to deliver higher expected portfolio returns.

The number of assets in the optimised portfolios

As illustrated before, the 'intelligent' inclusion or exclusion of stocks according to their individual return, risk and correlation characteristics considerably enhances portfolio performance in terms of risk. The 'non-naive' selection of appropriate stocks for the portfolio as they become available lowers overall portfolio volatility, but it also significantly lowers the number of stocks in the portfolio. Figure 6.12 illustrates the number of stocks in the portfolio versus the number of stocks available as we gradually enlarge the investable universe.

The thick black line indicates the average number of stocks included in the optimised portfolios as the number of available stocks in the investable universe increases from 1 to 70. The dotted black line indicates the number of stocks available. The thin grey lines illustrate, for 10 of the 100 trials, the number of assets included as they became available.

The average number of stocks included clearly shows how the optimisation procedure leaves out a large number of stocks because they simply do not add any diversification effect that cannot be generated by an appropriate portfolio

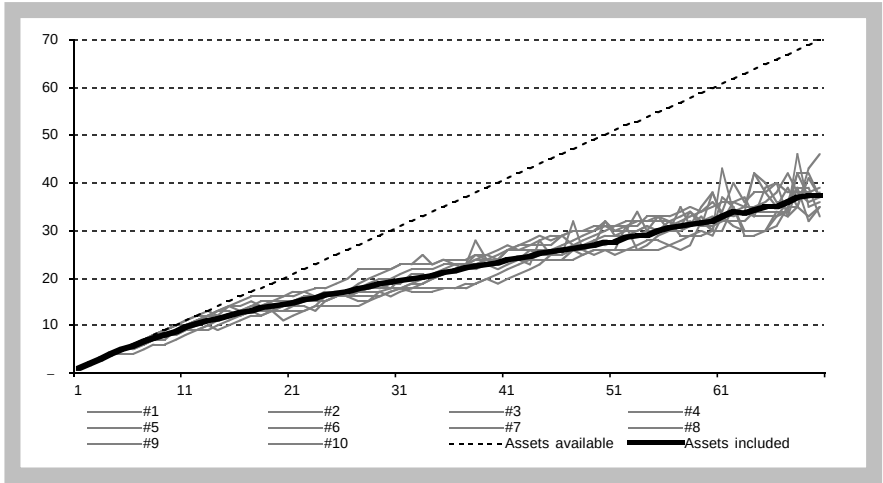


Figure 6.12 Number of stocks included in the minimum-variance portfolio as the 70 randomly chosen stocks become available

composition using other available stocks. In fact, the average number of assets included in the optimised portfolio is very close to half the number of available assets, namely 37 stocks out of 70 available. The deliberate selection and deselection of assets on the basis of their characteristics thus makes for portfolios with far fewer assets and yet with better expected return/risk properties relative to the 'naive' portfolio.

SUMMARY

This chapter has demonstrated that by using quantitative optimisation techniques we can determine an appropriate asset allocation for a portfolio based on both expected and desired levels of future asset class returns and risks. Instead of forming portfolios by 'throwing darts at the *Wall Street Journal*', the asset or asset class selection and portfolio weighting can be carried out in a quantitatively structured manner based on solid and consistent mathematical techniques. This ensures that consistency is carried over into the asset allocation and portfolio management processes.

We have also demonstrated the clear benefits from international diversification, by adding international equities to a portfolio initially consisting of only US Equity Indices. In terms of expected return, but particularly in terms of expected risk, large diversification effects were observed. For a portfolio in the middle of the risk spectrum (as measured from the

minimum variance to the maximum variance portfolio) the effect in terms of reduction of expected volatility was in the order of 1–2 per cent per annum, without any loss of expected return. It is quite important to note that there is nothing inherently special about international equity.

International equity does not hold some magical ability to decrease portfolio volatility just because they are non-US. The effect simply stems from the fact that returns on international equity are determined by many other factors in addition to the factors affecting US equities. In fact, international equity is just another asset class that has different characteristics from domestic equity, and therefore has a natural diversification effect on the portfolio. Any other asset class could in principle have affected the same increase in portfolio diversification, and it is quite likely that we might be able to find asset classes with even greater diversification effects.

The application of quantitative optimisation to the asset allocation process is discussed in detail in Chapter 9. However, before we do this we must first turn our attention to potential problems in using this type of mathematical technique. The analysis in this chapter assumes that all expected returns, volatilities and correlations are simply given and that they are perfect predictors of the future. This is of course not the case, which is why the asset manager needs to dedicate resources to estimating or forecasting these model inputs. This is the subject of the next chapter.

APPENDIX

LaGrange optimisation

LaGrange optimisation, also called problem-solving by the use of the *LaGrange multiplier*, is a mathematical method used to maximise or minimise a constrained function. There may be a number of constraints, and these can themselves take the shape of linear or non-linear functions.

Linear LaGrange problems

When referring to ‘linearity’ of constraints we hint at the fact that mathematical equality must hold for each constraint in the problem. In other words, each constraint contains an equality requirement. For example, if we wish to maximise the function $f(x)$ subject to a function $g(x)$ being some exact value, then the problem takes the following form:

$$\text{Max(Min)} f(x_1, x_2, \dots, x_n), \text{ subject to } g(x_1, x_2, \dots, x_n) = b$$

In the case of m constraints, the problem can be written:

$$\text{Max(Min)} f(x_1, x_2, \dots, x_n), \text{ subject to } \begin{cases} g_1(x_1, x_2, \dots, x_n) = b_1 \\ \vdots \\ g_m(x_1, x_2, \dots, x_n) = b_m \end{cases} \text{ with } (m < n)$$

Using vector notation we obtain:

$$\text{Max(Min)} f(\vec{x}), \text{ subject to } g_j(\vec{x}) = b_j, \text{ for } j = 1, 2, \dots, m, \text{ with } (m < n)$$

The *LaGrange problem* itself is then formulated as follows:

$$F(\vec{x}) = f(\vec{x}) - \lambda_1 g_1(\vec{x}) - \lambda_2 g_2(\vec{x}) - \dots - \lambda_m g_m(\vec{x})$$

This reduces to the formulation of the *general linear LaGrange problem*:

$$F(\vec{x}) = f(\vec{x}) - \sum_{j=1}^m \lambda_j g_j(\vec{x})$$

The basic idea behind the LaGrange method is that each constraint is assigned a unique number called the LaGrange multiplier, λ_j . The solution to the problem is then obtained by finding the first-order conditions to each constraint, setting them equal to zero, and isolating the variable in which we are interested. This is done by differentiation, whereby we derive the partial derivatives of each constraint with respect to the variable we wish to change, in order to reach an optimum; and then setting each of these partial derivatives equal to zero. This results in m equations with n variables, and since $m < n$ by design, we can solve this problem by the use of standard algebra. The mechanics of this process for a problem with m constraints appears as follows:

$$\begin{aligned} \frac{\partial F(\vec{x})}{\partial x_1} &= \frac{\partial f(\vec{x})}{\partial x_1} - \sum_{j=1}^m \lambda_j \frac{\partial g_j(\vec{x})}{\partial x_1} = 0 \\ &\vdots \\ &\vdots \\ \frac{\partial F(\vec{x})}{\partial x_n} &= \frac{\partial f(\vec{x})}{\partial x_n} - \sum_{j=1}^m \lambda_j \frac{\partial g_j(\vec{x})}{\partial x_n} = 0 \end{aligned}$$

Let us demonstrate with a simple example:

$$\text{Max(min)} = x^2 + 2y + z, \text{ subject to the constraints } \begin{cases} 2x - 3y - z = 1 \\ x + y - 5z = 4 \end{cases}$$

Note that the number of variables in the function to be maximised or minimised is less than the number of constraints. Note also that the two

constraints have already provided us with two equations for the solution of the problem. The LaGrange problem then appears as follows:

$$F(x,y,z) = x^2 + 2y + z - \lambda_1(2x - 3y - z - 1) - \lambda_2(x + y - 5z - 4)$$

The three *first-order conditions* to this problem (F.O.C. 1–3: the equations conditional on which the solution is optimal) take the following form:

$$\frac{\partial F(x,y,z)}{\partial x} = 2x - \lambda_1(2) - \lambda_2(1) = 0 \quad \text{F.O.C.1}$$

$$\frac{\partial F(x,y,z)}{\partial y} = 2 - \lambda_1(-3) - \lambda_2(1) = 0 \quad \text{F.O.C.2}$$

$$\frac{\partial F(x,y,z)}{\partial z} = 1 - \lambda_1(-1) - \lambda_2(-4) = 0 \quad \text{F.O.C.3}$$

We now have a set of 5 equations, and 5 variables x, y, z, λ_1 and λ_2 , which we can solve.⁶

Isolating λ_2 in F.O.C. 1 yields: $\lambda_2 = -0.25\lambda_1$.

Inserting this into F.O.C. 2 and isolating λ_1 yields:

$$2 - (-3)\lambda_1 - \left(-\frac{1}{4}\lambda_1\right) = 0 \quad \lambda_1 = 0.6154$$

This implies that: $\lambda_2 = -0.25 \cdot 0.6154 = 0.1538$, which in turn implies that:

$$2x - 2(0.6154) - 0.1538 = 0 \quad \underline{x = 0.6923}.$$

Inserting x into the second constraint yields:

$$0.6923 + y - 5z = 4 \quad y = 4 + 5z - 0.6923.$$

Inserting this into the first constraint yields:

$$2 \cdot 0.6923 - 3(4 + 5z - 0.6923) - z = 1 \quad \underline{z = 0.1538}.$$

Inserting again yields:

$$y = 4 + 5(0.1538) - 0.6923 \quad \underline{y = 1.3077}.$$

So the solution to the problem becomes:

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.6923 \\ 1.3077 \\ 0.1538 \end{bmatrix}$$

The LaGrange multiplier is widely used in several fields in economics, since it can handle any number of constraints. However the interpretation of the results can become quite difficult for a large number of variables and constraints, since in many cases there exists no one single solution. In fact, it is not unusual for a multiple constraint LaGrange problem to result in a large number of different solutions.

The minimum-variance portfolio

Finding the minimum-variance (volatility) portfolio entails minimising the expression for portfolio volatility, subject only to the constraint that asset weights must sum to 1 so that we are fully invested:

$$\text{Min } f(x) = \overline{\vec{w}^T \bullet \Sigma \bullet \vec{w}} \quad [\text{Eq. 6A.1}]$$

Subject to:

$$\vec{w}^T \bullet \vec{e} = 1 \quad [\text{Eq. 6A.2}]$$

where $\vec{w}$ is the weight vector, $\vec{w}^T$ is the transposed weight vector, Σ is the variance–covariance matrix, and $\vec{e}$ is the unit vector consisting of a single column of 1s. The LaGrange function subsequently appears as follows:

$$L(\vec{w}) = \vec{w}^T \bullet \Sigma \bullet \vec{w} - \lambda_1(\vec{w}^T \bullet \vec{e} - 1) \quad [\text{Eq. 6A.3}]$$

Taking the partial derivatives of $L(\vec{w})$ yields the first-order conditions:

$$\frac{\partial L(\vec{w})}{\partial \vec{w}} = 2\Sigma \bullet \vec{w} - \lambda_1 \vec{e} = 0 \quad \text{F.O.C.1}$$

$$\frac{\partial L(\vec{w})}{\partial \lambda_1} = \vec{w}^T \bullet \vec{e} - 1 = 0 \quad \text{F.O.C.2}$$

Rearranging F.O.C. 1 yields:

$$\vec{w} = \frac{\lambda_1}{2} \Sigma^{-1} \bullet \vec{e} \quad [\text{Eq. 6A.4}]$$

From F.O.C. 2 $\vec{w}^T \bullet \vec{e} = 1$ we can pre-multiply with $\vec{e}^T$ on both sides. Using this, to obtain:

$$\vec{e}^T \bullet \vec{w} = 1 = \frac{\lambda_1}{2} \vec{e}^T \bullet \Sigma^{-1} \bullet \vec{e} \quad [\text{Eq. 6A.5}]$$

Rearranging to find λ_1 leads to:

$$\lambda_1 = 2(\vec{e}^T \cdot \Sigma^{-1} \cdot \vec{e})^{-1} \quad [\text{Eq. 6A.6}]$$

which, using Equation 6A.4, yields a weight vector that specifies the solution to the general problem of minimising portfolio risk:

$$\vec{w}^* = (\vec{e}^T \cdot \Sigma^{-1} \cdot \vec{e})^{-1} \cdot \Sigma^{-1} \cdot \vec{e} \quad [\text{Eq. 6A.7}]$$

The inverse matrix (inverting a matrix)

Inverting matrices is necessary in order to solve portfolio optimisation problems using the LaGrange method. In order to invert a matrix it needs to be square or quadratic, and it needs to be *non-singular*. In order to determine whether this is in fact the case, we need to introduce the *determinants*, *minors* and *cofactors* of the matrix into the analysis.

Determinants

In order to be able to find the inverse of a matrix, the determinant of that matrix needs to be different from zero. This holds if the matrix is non-singular. For example the determinant of the square 2 · 2 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 3 & 2 \end{bmatrix}$$

is given by:

$$|A| = a_{11}a_{22} - a_{12}a_{21} = 5 \cdot 2 - 7 \cdot 3 = -11 \neq 0$$

Since the determinant of A is different from zero, the matrix is non-singular, which means that its inverse can be constructed. For larger matrices the process becomes a little more complex, although the basic principle remains the same.

We first need to define minors $|M_{ij}|$ of the matrix, which are simply sub-matrices of the original matrix, constructed by taking the i th row and the j th column out of the matrix and then calculating the determinant. For example, if we have a matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 2 & 7 & 3 \\ 4 & 1 & 2 \\ 1 & 6 & 8 \end{bmatrix}$$

then its minors are calculated by:

$$|M_{11}| = \begin{vmatrix} 1 & 2 \\ 6 & 8 \end{vmatrix} = 1 \cdot 8 - 6 \cdot 2 = -4, |M_{12}| = \begin{vmatrix} 4 & 2 \\ 1 & 8 \end{vmatrix} = 4 \cdot 8 - 1 \cdot 2 = 30$$

$$|M_{13}| = \begin{vmatrix} 4 & 1 \\ 1 & 6 \end{vmatrix} = 4 \cdot 6 - 1 \cdot 1 = 23$$

$$|M_{21}| = \begin{vmatrix} 7 & 3 \\ 6 & 8 \end{vmatrix} = 7 \cdot 8 - 6 \cdot 3 = 38, |M_{22}| = \begin{vmatrix} 2 & 3 \\ 1 & 8 \end{vmatrix} = 2 \cdot 8 - 1 \cdot 3 = 13$$

$$|M_{23}| = \begin{vmatrix} 2 & 7 \\ 1 & 6 \end{vmatrix} = 2 \cdot 6 - 1 \cdot 7 = 5$$

$$|M_{31}| = \begin{vmatrix} 7 & 3 \\ 1 & 2 \end{vmatrix} = 7 \cdot 2 - 1 \cdot 3 = 1, |M_{32}| = \begin{vmatrix} 2 & 3 \\ 4 & 2 \end{vmatrix} = 2 \cdot 2 - 4 \cdot 3 = -8$$

$$|M_{33}| = \begin{vmatrix} 2 & 7 \\ 4 & 1 \end{vmatrix} = 2 \cdot 1 - 4 \cdot 7 = -26$$

Now we can find the determinant for the entire matrix. Strictly speaking, it is not necessary to find all minors in order to find the determinant, but it will prove useful later on. To see why we only need a few minors to find the determinant we can look at the definition of a determinant, in terms of minors:

$$|A| = \sum_{j=1}^n a_{ij}(-1)^{i+j} |M_{ij}|$$

In the example given above, this would equate to:

$$\begin{aligned} |A| &= \sum_{j=1}^3 a_{ij}(-1)^{i+j} |M_{ij}| \\ &= a_{11}(-1)^{1+1} |M_{11}| + a_{12}(-1)^{1+2} |M_{12}| + a_{13}(-1)^{1+3} |M_{13}| = \\ |A| &= (2)(-1)^2(-4) + (7)(-1)^3(30) + (3)(-1)^4(23) = -149 \neq 0 \end{aligned}$$

Since the determinant of A is different from zero, the matrix is then non-singular.

Minors and cofactors

Having identified the determinant $|A|$ and the minors $|M_{ij}|$ of a matrix A , we can determine the cofactors of the matrix, which are defined as:

$$C_{ij} = (-1)^{i+j} |M_{ij}|$$

This means that $C_{ij} = M_{ij}$ when $i + j$ is an even number, and that $C_{ij} = (-1)M_{ij}$ when $i + j$ is an odd number. For the 3×3 matrix presented above, this means that the cofactors are calculated by:

$$|C_{11}| = (-1)^2(-4) = -4, |C_{12}| = (-1)^3(30) = -30, |C_{13}| = (-1)^4(23) = 23$$

$$|C_{21}| = (-1)^3(38) = -38, |C_{22}| = (-1)^4(13) = 13, |C_{23}| = (-1)^5(5) = -5$$

$$|C_{31}| = (-1)^4(11) = 11, |C_{32}| = (-1)^5(-8) = 8, |C_{33}| = (-1)^6(-26) = -26$$

Now we can move on to define the inverse matrix. Given a non-singular and square matrix A , we will define a matrix B , consisting of the cofactors of A , such that each element in B is defined as $b_{ij} = |C_{ij}|$. The matrix B appears as follows:

$$B = \begin{bmatrix} |C_{11}| & |C_{12}| & |C_{13}| \\ |C_{21}| & |C_{22}| & |C_{23}| \\ |C_{31}| & |C_{32}| & |C_{33}| \end{bmatrix} = \begin{bmatrix} -4 & -30 & 23 \\ -38 & 13 & -5 \\ 11 & 8 & -26 \end{bmatrix}$$

This matrix is then transposed, and the result is called the *adjoint* of A :

$$B^T = \begin{bmatrix} Adj A \end{bmatrix} = \begin{bmatrix} |C_{11}| & |C_{21}| & |C_{31}| \\ |C_{12}| & |C_{22}| & |C_{32}| \\ |C_{13}| & |C_{23}| & |C_{33}| \end{bmatrix} = \begin{bmatrix} -4 & -38 & 11 \\ -30 & 13 & 8 \\ 23 & -5 & -26 \end{bmatrix}$$

It can be shown that:

$$\begin{bmatrix} Adj A \end{bmatrix} A = |A| \bullet I$$

and since A is non-singular ($|A| \neq 0$), we can divide both sides of the equation by $|A|$, resulting in:

$$\frac{1}{|A|} \begin{bmatrix} Adj A \end{bmatrix} A = I$$

Since $A^{-1}A = I$, this implies that:

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} Adj A \end{bmatrix} = \begin{bmatrix} \frac{|C_{11}|}{|A|} & \frac{|C_{21}|}{|A|} & \frac{|C_{31}|}{|A|} \\ \frac{|C_{12}|}{|A|} & \frac{|C_{22}|}{|A|} & \frac{|C_{32}|}{|A|} \\ \frac{|C_{13}|}{|A|} & \frac{|C_{23}|}{|A|} & \frac{|C_{33}|}{|A|} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-4}{-149} & \frac{-38}{-149} & \frac{11}{-149} \\ \frac{-30}{-149} & \frac{13}{-149} & \frac{8}{-149} \\ \frac{23}{-149} & \frac{-5}{-149} & \frac{-26}{-149} \end{bmatrix}$$

It is easy to verify that this is in fact the inverse, since:

$$A^{-1}A = I = \begin{bmatrix} \frac{-4}{-149} & \frac{-38}{-149} & \frac{11}{-149} \\ \frac{-30}{-149} & \frac{13}{-149} & \frac{8}{-149} \\ \frac{23}{-149} & \frac{-5}{-149} & \frac{-26}{-149} \end{bmatrix} \cdot \begin{bmatrix} 2 & 7 & 3 \\ 4 & 1 & 2 \\ 1 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Minimising volatility for a given level of return

Finding the minimum volatility portfolio entails minimising the expression for portfolio volatility, subject to the constraint that asset weight must sum to 1, and the constraint that portfolio return must equal a specified level:

$$\text{Min } f(x) = \overrightarrow{w}^T \bullet \Sigma \bullet \overrightarrow{w} \quad [\text{Eq. 6A.8}]$$

Subject to:

$$\overrightarrow{w}^T \bullet \overrightarrow{r} = \check{r} \quad [\text{Eq. 6A.9}]$$

$$\overrightarrow{w}^T \bullet \overrightarrow{e} = 1 \quad [\text{Eq. 6A.10}]$$

where $\overrightarrow{w}^T$ is the transposed weight vector, and $\overrightarrow{e}$ is the $N \cdot 1$ unit vector consisting of a single column of 1s. $\check{r}$ is the *target return*, specifying the level of return for which we wish to minimise volatility.

The LaGrange function subsequently takes the following form:

$$L(\overrightarrow{w}) = \frac{1}{2} \overrightarrow{w}^T \bullet \Sigma \bullet \overrightarrow{w} - \lambda_1(\overrightarrow{w}^T \bullet \overrightarrow{r} - \check{r}) - \lambda_2(\overrightarrow{w}^T \bullet \overrightarrow{e} - 1) \quad [\text{Eq. 6A.11}]$$

Taking the partial derivatives of $L(\overrightarrow{w})$ yield the three first-order conditions:

$$\frac{\partial L(\overrightarrow{w})}{\partial \overrightarrow{w}} = \Sigma \bullet \overrightarrow{w} - \lambda_1 \overrightarrow{r} - \lambda_2 \overrightarrow{e} = 0 \quad \text{F.O.C. 1}$$

$$\frac{\partial L(\vec{w})}{\partial \lambda_1} = \vec{w}^T \bullet \vec{r} - \check{r} = 0 \quad \text{F.O.C. 2}$$

$$\frac{\partial L(\vec{w})}{\partial \lambda_1} = \vec{w}^T \bullet \vec{e} - 1 = 0 \quad \text{F.O.C. 3}$$

Rearranging F.O.C. 1 yields:

$$\vec{w} = \lambda_1 \Sigma^{-1} \bullet \vec{r} + \lambda_2 \Sigma^{-1} \bullet \vec{e} \quad [\text{Eq. 6A.12}]$$

Using the fact that F.O.C. 3 states that $\vec{w}^T \bullet \vec{e} = 1$, we pre-multiply Equation 6A.12 with $\vec{e}^T$ on both sides to obtain:

$$\vec{e}^T \bullet \vec{w} = 1 = \lambda_1 \vec{e}^T \Sigma^{-1} \bullet \vec{r} + \lambda_2 \vec{e}^T \bullet \Sigma^{-1} \bullet \vec{e} \quad [\text{Eq. 6A.13}]$$

To simplify notation, we now introduce the constants (scalars) κ_1 and κ_2 :

$$\kappa_1 = \vec{e}^T \bullet \Sigma^{-1} \bullet \vec{r} \quad [\text{Eq. 6A.14}]$$

$$\kappa_2 = \vec{e}^T \bullet \Sigma^{-1} \bullet \vec{e} \quad [\text{Eq. 6A.15}]$$

Using the two Kappas, we can write Equation 6A.13:

$$\begin{aligned} 1 &= \lambda_1 \kappa_1 + \lambda_2 \kappa_2 \\ \lambda_1 &= (1 - \lambda_2 \kappa_2) \kappa_1^{-1} \end{aligned} \quad [\text{Eq. 6A.16}]$$

Using the fact that F.O.C. 2 states that $\vec{w}^T \bullet \vec{r} = \check{r}$, we can pre-multiply Equation 6A.12 with $\vec{r}^T$ on both sides to obtain:

$$\vec{r}^T \bullet \vec{w} = \check{r} = \lambda_1 \vec{r}^T \bullet \Sigma^{-1} \bullet \vec{r} + \lambda_2 \vec{r}^T \bullet \Sigma^{-1} \bullet \vec{e} \quad [\text{Eq. 6A.17}]$$

For further notational simplicity we now introduce the constants (scalars):

$$\kappa_3 = \vec{r}^T \bullet \Sigma^{-1} \bullet \vec{r} \quad [\text{Eq. 6A.18}]$$

$$\kappa_4 = \vec{r}^T \bullet \Sigma^{-1} \bullet \vec{e} = \kappa_1 \quad [\text{Eq. 6A.19}]$$

We can now rewrite Equation 6A.17:

$$\begin{aligned} \check{r} &= \lambda_1 \kappa_3 + \lambda_2 \kappa_4 \\ \lambda_1 &= (\check{r} - \lambda_2 \kappa_4) \kappa_3^{-1} \end{aligned} \quad [\text{Eq. 6A.20}]$$

We now have two equations expressing λ_1 . Using the fact that $\kappa_1 = \kappa_4$, this enables us to rewrite λ_1 :

$$\lambda_1 = \lambda_1$$

$$\begin{aligned}
 (1 - \lambda_2 \kappa_2) \kappa_1^{-1} &= (\check{r} - \lambda_2 \kappa_4) \kappa_3^{-1} \\
 (1 - \lambda_2 \kappa_2) \kappa_1^{-1} &= (\check{r} - \lambda_2 \kappa_1) \kappa_3^{-1} \\
 \lambda_2 (\kappa_1 \kappa_3^{-1} - \kappa_2 \kappa_1^{-1}) &= \check{r} \kappa_3^{-1} - \kappa_1^{-1} \\
 \lambda_2 &= (\check{r} \kappa_3^{-1} - \kappa_1^{-1}) (\kappa_1 \kappa_3^{-1} - \kappa_2 \kappa_1^{-1})^{-1}
 \end{aligned} \tag{Eq. 6A.22}$$

Inserting this into Equation 6A.16 allows us to rewrite λ_1 in a form that includes only Kappa's and does not include λ_2 :

$$\begin{aligned}
 \lambda_1 &= (1 - \lambda_2 \kappa_2) \kappa_1^{-1} \\
 \lambda_1 &= (\kappa_1^{-1} - \lambda_2 \kappa_2 \kappa_1^{-1}) \\
 \lambda_1 &= \kappa_1^{-1} - (\check{r} \kappa_3^{-1} - \kappa_1^{-1}) (\kappa_1 \kappa_3^{-1} - \kappa_2 \kappa_1^{-1})^{-1} \kappa_2 \kappa_1^{-1}
 \end{aligned} \tag{Eq. 6A.23}$$

Using Equation 6A.17 finally enables us to define the weight vector $\vec{w}$ that minimises volatility at the specified level of return $\check{r}$:

$$\vec{w}^* = \lambda_1 \Sigma^{-1} \bullet \vec{r} + \lambda_2 \Sigma^{-1} \bullet \vec{e} \tag{Eq. 6A.24}$$

Expanding this to specify all the components, we obtain:

$$\begin{aligned}
 \vec{w}^* &= \left(\kappa_1^{-1} - (\check{r} \kappa_3^{-1} - \kappa_1^{-1}) (\kappa_1 \kappa_3^{-1} - \kappa_2 \kappa_1^{-1})^{-1} \kappa_2 \kappa_1^{-1} \right) \Sigma^{-1} \bullet \vec{r} \\
 &+ \left((\check{r} \kappa_3^{-1} - \kappa_1^{-1}) (\kappa_1 \kappa_3^{-1} - \kappa_2 \kappa_1^{-1})^{-1} \right) \Sigma^{-1} \bullet \vec{e}
 \end{aligned} \tag{Eq. 6A.25}$$

Notes

- 1 For an introduction to LaGrange optimisation, please refer to the appendix of this chapter.
- 2 For the complete derivation of the minimum-variance portfolio, please refer to the appendix of this chapter.
- 3 To see how a matrix is inverted, please refer to the appendix of this chapter.
- 4 For the derivation of the minimum volatility portfolio given a specified target return, please refer to the appendix of this chapter.
- 5 We will introduce this aspect later in this book but for now we ignore risk aversion, which is equivalent to simply assuming a risk aversion coefficient of 1 multiplied onto the volatility term.
- 6 Conceptually the first-order conditions can be interpreted as follows. Setting them equal to each other implies that the partial derivatives of F with respect to the constraints are the same. In other words the 'slopes' of the three contours defined in 3D (x, y, z) space are the same. That means that they are all tangents to each other, which implies an optimum. This is easily extendable to multidimensional space.

ESTIMATING MODEL PARAMETERS

EXPECTED RETURN AND RISK

So far we have simply assumed that asset returns and volatilities and their correlations are given. In this chapter we will introduce a framework for estimating and/or forecasting the model parameters. These parameters are needed to perform quantitative portfolio optimisation and analysis, which provides the basis for the subject of the next chapter, quantitative asset allocation.

As we have demonstrated, the optimisation procedure requires estimates of expected returns and volatilities, and also an estimated (expected) correlation structure among the investable assets. These three elements form the basis for the entire quantitative portfolio optimisation approach, and as such it is essential that they are estimated correctly and in an intuitive and consistent manner. Furthermore it can be argued that ideally we are interested not in *historical* numbers but rather in estimates of *future* realisations of returns, volatilities and correlations. Unfortunately, in order to obtain such estimates we have no better tool than to look at past realisations, and then from those try to produce forecasts or educated guesses as to the future state of those three sets of variables.

As we will demonstrate below, there exist a large number of ways in which to estimate or forecast model parameters, and the choice can potentially have a significant impact on asset allocation results. We shall begin by revisiting the CAPM, as well as a factor model such as the APT. In addition we will investigate several volatility estimators and correlation and covariance matrix estimators. The results are not to be regarded as scientific as such, but rather as

a demonstration of some of the options available to asset managers when attempting to determine appropriate variables as inputs for the quantitative portfolio optimisation and asset allocation process. Our analysis and results are thus merely designed to provide an overview of the field of parameter estimation. The decision concerning which techniques (if any) to use is as always entirely the responsibility of the investment manager.

THE CAPM REVISITED

The CAPM has become one of the mainstays of the financial industry, and despite (or perhaps because of) its simplicity it has proven extremely robust and difficult to refute. The CAPM is the epitome of normative asset-pricing models. Being a univariate model, the CAPM is extremely simple and intuitive in its nature, which makes it easy to understand and work with. This is of course the reason why it is still with us so many years after its birth. Taking a slightly humorous approach, we offer the following.

The CAPM sketch

Imagine Monty Python's famous *Parrot Sketch* as a conversation between a finance academic and a finance practitioner. John Cleese would play the part of the finance practitioner and Michael Palin the part of the finance professor.¹

John Cleese (Entering a large university): "Excuse me. I would like to register a complaint about this financial market theory which I purchased from this very boutique."

Michael Palin: "Ah, the CAPM, a remarkable theory. What's wrong with it?"

John Cleese: "I will tell you what is wrong with it. It's dead."

Michael Palin: "Nah, Nah. It's resting. It will be up and about shortly. Haven't you been reading *The Journal*?"

John Cleese: "Never mind that. I know a dead theory when I see one and I'm looking at one right now."

Michael Palin: "Nah, it's not dead. It's resting."

John Cleese (Incredulous): "Resting?"

Michael Palin: "Yeah, resting. A remarkable theory, the CAPM. Beautiful plumage."

John Cleese (Raising his voice): “Beautiful plumage? It’s stone dead.”

Michael Palin: “No, it’s just resting.”

John Cleese: “This theory is definitely deceased. This theory wouldn’t move if you put 4000 volts through it. It’s bleeding demised. It’s passed on. This theory is no more. It has ceased to be. It has expired and gone to meet its maker. This is a late theory. It’s bereft of life. It rests in peace. If you weren’t still trying to salvage it, it would be pushing up the daisies. It has rung down the curtain and joined the choir invisible. This is an ex-theory.”

Michael Palin: “Well, then we’ll have to replace it.” (After rustling around the university for a few moments, he returns.) “I’m sorry but we don’t have any more financial market theories left.”

As far-fetched as this may seem, it is in fact not too far from the truth. For as long as most people can remember, the CAPM has been the workhorse of financial theory in general, and asset pricing and valuation in particular. And although it has never really proven its investment worth in practice, at least not as a stand-alone decision-making tool capable of generating superior returns, it has – due mainly to its simplicity – formed the basis for further important developments such as the Arbitrage Pricing Theory (APT), which is basically a multivariate CAPM. We shall examine both below.

Formal derivations of the CAPM

In Chapter 4 we presented the CAPM as a normative univariate predictor of asset returns. Following Taggart (1996), we can derive the CAPM formally in a portfolio context, using the tools presented in previous chapters. Similarly to the Security Market Line, we can establish an *Asset Allocation Line*, which is the tangency line to the efficient frontier. Its intercept with the return axis is at the risk free rate – that is, if the asset has no risk, its (excess) return is equal to the risk free rate (see Figure 7.1).

Imagine a fully invested portfolio that is a mixture of the market portfolio M , and a risky asset A . The market portfolio is a hypothetical portfolio assumed to exist, consisting of all assets in correct relative proportions such that its return exactly mimics that of the whole market. The excess return on this mixed portfolio P is given by:

$$r_P^\times = \gamma \cdot r_A^\times + (1 - \gamma)r_M^\times - r_f \quad [\text{Eq. 7.1}]$$

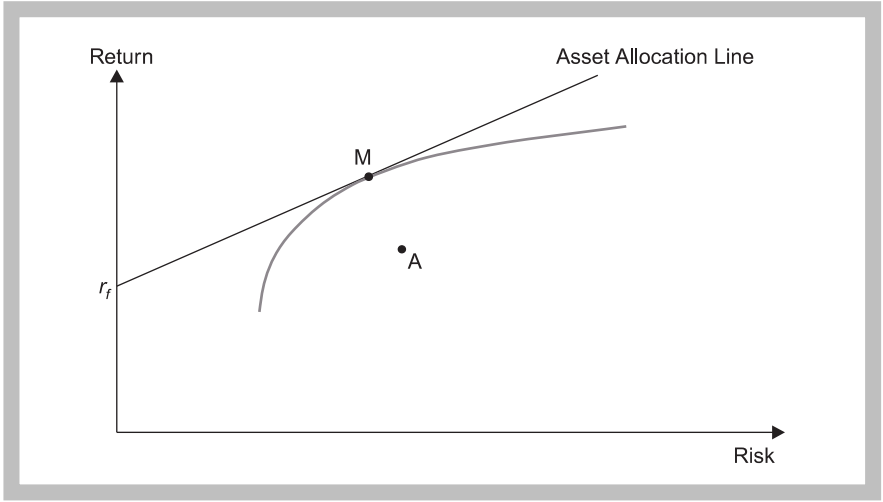


Figure 7.1 Asset Allocation Line – one risky asset A and the market portfolio M

where γ is the proportion of the portfolio invested in the risky asset A , and $(1 - \gamma)$ is the proportion invested in the market portfolio. The risk, expressed as the variation (possibly scaled to appear as the portfolios volatility) of the portfolio, is given by:

$$\sigma_P^2 = \gamma^2 \cdot \sigma_A^2 + (1 - \gamma) \cdot \sigma_M^2 + 2\gamma(1 - \gamma) \cdot \rho_{A,M} \cdot \sigma_A \cdot \sigma_M \quad [\text{Eq. 7.2}]$$

From Figure 7.1 it is clear that the slope of the Asset Allocation Line is given by:

$$\frac{r_P^\times - r_f}{\sigma_P^2} = \frac{\gamma \cdot r_A^\times + (1 - \gamma)r_M^\times - r_f}{\gamma^2 \cdot \sigma_A^2 + (1 - \gamma) \cdot \sigma_M^2 + 2\gamma(1 - \gamma) \cdot \rho_{A,M} \cdot \sigma_A \cdot \sigma_M} \quad [\text{Eq. 7.3}]$$

The portfolio that maximises return relative to risk (the Sharpe Ratio) is the portfolio that lies on the tangency point between the Asset Allocation Line and the efficient frontier. We find this by taking the derivative of the slope Equation 7.3 with respect to γ , setting it equal to zero, and solving for γ :

$$\frac{d\left(\frac{r_P^\times - r_f}{\sigma_P^2}\right)}{d\gamma} = -\frac{1}{2} [\gamma \cdot r_A^\times + (1 - \gamma) \cdot r_B^\times - r_f]$$

$$\begin{aligned} & \cdot [2 \cdot \gamma \cdot \sigma_A^2 - 2(1 - \gamma)\sigma_M^2 + 2(1 - 2\gamma) \cdot \rho_{A,M} \cdot \sigma_A \cdot \sigma_M] \cdot \sigma_P^{-3} \\ & + [r_A^\times - r_M^\times] \sigma_P^{-1} = 0 \end{aligned} \quad [\text{Eq. 7.4}]$$

By multiplying through with $-\sigma_P^{-1}$ and rearranging terms we obtain:

$$\begin{aligned} & [\gamma \cdot r_A^\times + (1 - \gamma) \cdot r_M^\times - r_f] \cdot [\gamma \cdot \sigma_A^2 - (1 - \gamma) \cdot \sigma_M^2 \\ & + (1 - 2\gamma) \cdot \rho_{A,M} \cdot \sigma_A \cdot \sigma_M] \\ & = [r_A^\times + r_M^\times] \cdot [\gamma^2 \cdot \sigma_A^2 + (1 - \gamma)^2 \cdot \sigma_M^2 + 2\gamma(1 - \gamma) \cdot \rho_{A,M} \cdot \sigma_A \cdot \sigma_M] \end{aligned} \quad [\text{Eq. 7.5}]$$

Finally we multiply through all elements of the expression on both sides of Equation 7.3, collect terms, and rearrange to obtain an expression for γ , the proportion of the portfolio invested in the risky asset A:

$$\begin{aligned} & \gamma \cdot ([r_A^\times - r_f] \cdot \sigma_M^2 + [r_M^\times - r_f] \cdot \sigma_A^2 - ([r_A^\times - r_f] + [r_M^\times - r_f]) \cdot \rho_{A,M} \cdot \sigma_A \cdot \sigma_M) \\ & = [r_A^\times - r_f] \cdot \sigma_M^2 - [r_M^\times - r_f] \rho_{A,M} \cdot \sigma_A \cdot \sigma_M \end{aligned} \quad [\text{Eq. 7.6}]$$

which lets us define:

$$\gamma = \frac{[r_A^\times - r_f] \cdot \sigma_M^2 - [r_M^\times - r_f] \rho_{A,M} \cdot \sigma_A \cdot \sigma_M}{([r_A^\times - r_f] \cdot \sigma_M^2 + [r_M^\times - r_f] \cdot \sigma_A^2 - ([r_A^\times - r_f] + [r_M^\times - r_f]) \cdot \rho_{A,M} \cdot \sigma_A \cdot \sigma_M)} \quad [\text{Eq. 7.7}]$$

This is the proportion that would be held in the risky asset if the investor were to seek the best trade-off between return and risk on the efficient frontier. But according to the CAPM, in equilibrium all investors hold only the market portfolio as their risky asset. In other words they hold all the assets in the market corresponding to their respective weight in the market portfolio. This means that no investors deviate from these market weights, which means that γ is equal to zero. All investors hold only some combination of the risk free asset and the market portfolio. This allows that, using Equation 7.1, we can obtain the expected return on asset A, which can be written:

$$r_A^\times = r_f + \frac{\rho_{A,M} \cdot \sigma_A \cdot \sigma_M}{\sigma_M^2} [r_M^\times - r_f] = r_f + \frac{\text{Cov}(r_A, r_M)}{\sigma_M^2} \cdot [r_M^\times - r_f] \quad [\text{Eq. 7.8}]$$

which by using the definition of Beta yields:

$$r_A^\times = r_f + \beta_A [r_M^\times - r_f] \quad [\text{Eq. 7.9}]$$

Equation 7.9 is of course the CAPM model as presented in Chapter 4, but here it is derived as the optimal portfolio choice given a risky asset and the market portfolio. In equilibrium a stock will be priced so that its excess return (or risk premium) is proportional to the excess return on the market portfolio, and so that the factor of proportionality is the stock's Beta, measuring the stock's systematic risk.

In essence, this means that when forecasting returns using the CAPM, the return on a stock is only dependant on its systematic risk component. No investor will be forced to take on *unsystematic* risk, because this risk can be diversified away. Thus the only risk that is relevant when pricing stocks and assessing stock returns is the magnitude of the systematic risk component. The simplicity of the CAPM is what makes it so attractive, but for asset return forecasting it must be said to be inadequate for exactly the same reason. Many more factors than just the return on the overall market have an impact on the return on an asset. This point is reflected in the next section, on the formulation of the Arbitrage Pricing Theory (APT).

FACTOR MODELS – THE APT REVISITED

Forecasting asset returns

The CAPM sketch is of course not entirely accurate, since asset managers have not completely run out of financial market theories or forecasting models. It is important to realise that the CAPM and the APT are fundamentally different, and for this reason should be applied differently. As noted above, the CAPM provides a framework for expected (consensus) returns that can be used to quantify deviations from the expectations of the market as a whole. This is useful in the active portfolio management process, in which managers wish to increase active risk by deviating from a benchmark.² However, as a means of determining or forecasting future returns the CAPM is seriously lacking, due to the fact that it only has one explanatory variable: the market (excess) return. The primary use of the APT in portfolio management is for forecasting risk. Multi-factor models such as the APT draw on information provided by many more variables to attempt to determine asset returns and risk, so not surprisingly this results in more accurate forecasts. To the extent that a manager can only outperform the market by superior information, it makes a lot of sense to devote resources to trying to improve his forecasting ability.

There are basically three types of factor models available, *fundamental*, *macroeconomic* and *statistical*, and we will discuss all of these below. For quantitative portfolio optimisation we are interested in reliable estimates of future asset returns to use as inputs into the portfolio optimisation and asset allo-

cation procedure. For this purpose the multivariate APT is probably the most obvious candidate. From Equation 4.15 we know that the basic multi-factor model appears as follows:

$$r_i^{\times} = \beta_{i,1} \cdot (f_1 - r_f) + \beta_{i,2} \cdot (f_2 - r_f) + + \beta_{i,k} \cdot (f_k - r_f)$$

where

- $r_i^{\times}$ is the expected excess return on asset i
- r_f is the risk free rate of return
- $\beta_{i,k}$ is the sensitivity of asset i to risk factor k
- f_k is the return on risk factor k
- $(f_k - r_f)$ is the risk premium (excess return) associated with risk factor k

Using matrix notation we can express the general multi-factor model as follows:

$$r_t^{\times} = B \bullet f_t + E_t \tag{Eq. 7.10}$$

- $r_t^{\times}$ is the vector of expected excess returns on assets
- B is the matrix of asset sensitivities (exposures) to risk factors
- f_t is the vector of factor excess returns at time t
- E_t is the vector of asset-specific residual returns at time t

Factor models can be used for forecasting returns, but this is not the way they are usually applied. They are more often applied to forecasting risk. Since the choice of explanatory factors is limited only by the imagination of the asset or risk manager, it follows that there are as many multivariate asset-pricing models as there are variables to use as inputs. However, three distinct types stand out: Fundamental Factor Models, Macroeconomic Factor Models and Statistical Factor Models. Each has strengths and weaknesses, which we will discuss below.

Fundamental factor models

This approach derives its name from the fact that these models are essentially a multiple cross-sectional regression of fundamental company factors or characteristics of a stock’s returns. As is the case with other factor models, stock returns that are not captured by the factor regression analysis are presumed to be stock-specific and uncorrelated.

The fundamental model assumes that the return on an asset can be explained fully by the quantifiable characteristics of individual assets or stocks. For example, an asset is characterised by its balance sheet leverage, liquidity, market capitalisation, price/earnings ratio and other valuation

metrics, interest rate sensitivity, dividend yield, earnings growth, industry classification, and the like. These asset attributes or factors are then used as proxies for B . The matrix of asset sensitivities or exposures is then determined via regression analysis performed on, for example, the monthly returns on factors.

The primary strength of such fundamental factor models is the fact that they operate on readily identifiable factors that are used and evaluated every day by portfolio managers, thus making the result intuitive and more easily interpretable. In addition, since the model operates on factors and not on stock-specific price history, any change in the characteristics of a company is immediately carried through and reflected in the model output. However, the factors themselves cannot be observed, since they are essentially a synthesis or an aggregation of company characteristics. Although we can thus indirectly calculate the return and risk of a factor, we are not able to *observe* it directly anywhere. The main point of criticism concerning the fundamental factor model approach is the fact that the large number of arbitrarily chosen factors often overlap, exhibiting so-called *multicollinearity*, which causes fundamental factor models to over-fit the data used for estimating factors. This means that in terms of forecasting, they often perform better in sample than out of sample. The practical implications of this may be that portfolio risk is seriously underestimated. This is an experience most portfolio managers have had at one time or another, particularly when markets are very volatile.

Macroeconomic Factor Models

Macroeconomic Factor Models use macroeconomic variables to try to forecast the excess return on factors f_t through a time series of macroeconomic variables. Those variables are typically economic state variables, such as changes in the rate of inflation, expected GDP growth, the spread between corporate and government bond yields, interest rates, exchange rates, and so on.

Referring back to Equation 7.10, once f_t has been estimated, B can then be determined in a regression analysis as the average of each asset's sensitivity to the excess factor returns. In practice this is done by regressing time series of the variables on each stock's return to obtain Betas for each factor with respect to the variables. Again, stock returns that are not captured by the factor regression analysis are presumed to be stock-specific and uncorrelated.

A benefit of Macroeconomic Factor Models is that they typically exhibit stable behaviour since they are tied to the real economy; they thus offer portfolio managers insight into the sensitivity of a stock to certain macroeconomic

variables about whose future state she can form an opinion. The obvious drawback is the fact that the model does not capture such essential company-specific descriptors as balance sheet structure or credit quality.

Statistical Factor Models

This type of factor model is also called *blind factor models*, since they entail estimating both f_t and B using a purely statistical approach. They start off with the full historical asset variance-covariance matrix and then decompose this matrix into a factor component and a stock-specific component. The analysis will typically yield a set of factor exposures B , which can then be used to determine factor excess returns f_t . The factor exposures and sensitivities are not specified as observable or even measurable real world factors. They are inferred from the behaviour of assets, without an attempt at explaining their nature or their behaviour. Some models use a combination of a defined fundamental model and a statistical model, where the error terms that result from regression of the fundamental factors are then decomposed, using *principal component analysis* (PCA), to obtain a set of blind factors that can explain the behaviour of the error term.

A benefit of using this approach is that since the factors are never defined, the structure of the model can evolve over time to fit the observed asset behaviour, without the manager having to specify a new set of explanatory variables every other month. The *problem* with this approach, from a practical portfolio management point of view, is that the manager never learns what the blind factors are. Even though they can be identified in terms of magnitude and even tracked over time, the manager is never able to identify what they are composed of; and what is more, the nature of the factors may change over time without the manager being able to see this. This results in a 'hands-off' feel to the whole process that is quite unpleasant for a portfolio manager.

VOLATILITY AND CORRELATION

Directions

As indicated previously, what we are looking for in a portfolio optimisation and asset allocation context are the future values of model parameters. Therefore we are only interested in past volatilities and correlations to the extent that they help us forecast future values. In this section we examine five techniques for estimating or forecasting asset or market volatility. These are the *Historical* (equally weighted observations) model, the *Exponentially Weighted Moving*

Average (EWMA) model, the *Autoregressive Conditional Heteroscedasticity* (ARCH) model, the *Generalised Autoregressive Heteroscedasticity* (GARCH) model, and the *Multivariate Density Estimation* (MDE) model. Despite these rather extravagant names, the underlying maths applied in all of them is relatively simple, and we shall explain each in turn.

Historical (equally weighted)

We have already demonstrated how asset volatilities and correlations are determined using simple historical data. This is easy to work with, but also 'naive' in the sense that it attempts to place no structure on the way the data is treated or how for example volatility might evolve over time. Historical volatility of an asset A is thus given by Equation 3.5.

$$\sigma_A = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_{A,t} - \bar{r}_A)^2} \cdot \sqrt{\hat{T}}$$

The most serious problem with using the historical volatility estimate as a forecast is the so-called *ghost feature*. This name refers to the effect an extreme observation will have on the forecast long after the event it reflects has occurred. Figure 7.2 illustrates the volatility (based on 60 daily observations) of the S&P500 from January 1996 to May 2002. Note the jumps in volatility,

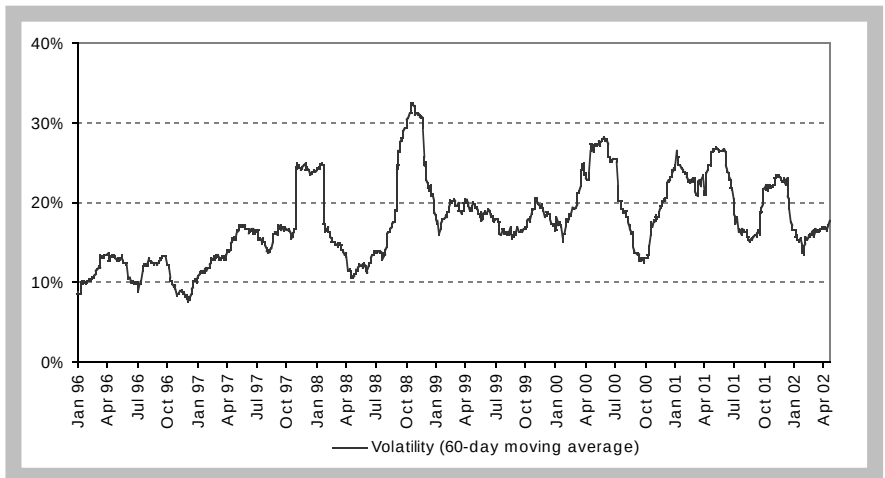


Figure 7.2 60-day moving average volatility of the S&P500, 1996/1–2002/5

particularly in late 1997. On the 27th of October the S&P500 dropped 6.87% and then regained 5.12% the next day. Such an event is very unusual and thus has a large impact on volatility, which shoots up from around 16% to just under 25%. The following days are relatively calm, so the volatility of the index flattens out, but it still *appears* to be high. That it is in fact stable and low can be seen from the fact that it appears to plummet back to some 16% around 60 days later when the extreme event drops out of the time series over which volatility is estimated. The initial jump is caused by one extreme event, but the subsequent fall does not reflect any change in the actual behaviour of the index: it is purely a function of the estimation period of 60 days, and the fact that all observations are weighted equally. How to treat such an event is a difficult issue, but the following method is often applied and with some success.

Exponentially Weighted Moving Average (EWMA)

As noted, an obvious deficiency arising from using historical volatility as a forecast of future volatility is the fact that it places equal weight on all observations. Volatility tends to cluster – that is, there is often autocorrelation present – so it follows that a method by which more weight is placed on the most recent observations and less weight is placed on observations further into the past might be warranted.³ This makes intuitive sense if one thinks of the release of news that may have an effect on markets or particular stocks. At first the reactions are significant, since perhaps only a few investors know and they are unable to immediately determine the exact effect of the news. But as the news is slowly disseminated out to all investors, the effect of the news abates until it is finally fully disseminated to all market participants.

Using the *exponentially weighted moving average* (EWMA) to some extent reduces the ghost feature problem mentioned above. Formally we can express this by:

$$\frac{x_t + \lambda \cdot x_{t-1} + \lambda^2 \cdot x_{t-2} + \dots + \lambda^n \cdot x_{t-n}}{1 + \lambda + \lambda^2 + \dots + \lambda^n} \quad [\text{Eq. 7.11}]$$

where x is the parameter to which we wish to apply the weighting scheme to, in this case volatility. Since the denominator converges to $1/(1 - \lambda)$ as the number of observations n goes to infinity, the EWMA volatility forecast for asset A may be written:

$$\begin{aligned} \sigma_{A,t} &= \frac{(1 - \lambda) \sum_{i=1}^{\infty} \lambda^{i-1} (r_{A,t+1-i} - \bar{r}_A)^2 \cdot \hat{T}}{(1 - \lambda) (r_{A,t-1} - \bar{r}_A)^2 + \lambda \sigma_{A,t-1}^2 \cdot \hat{T}} \quad [\text{Eq. 7.12}] \end{aligned}$$

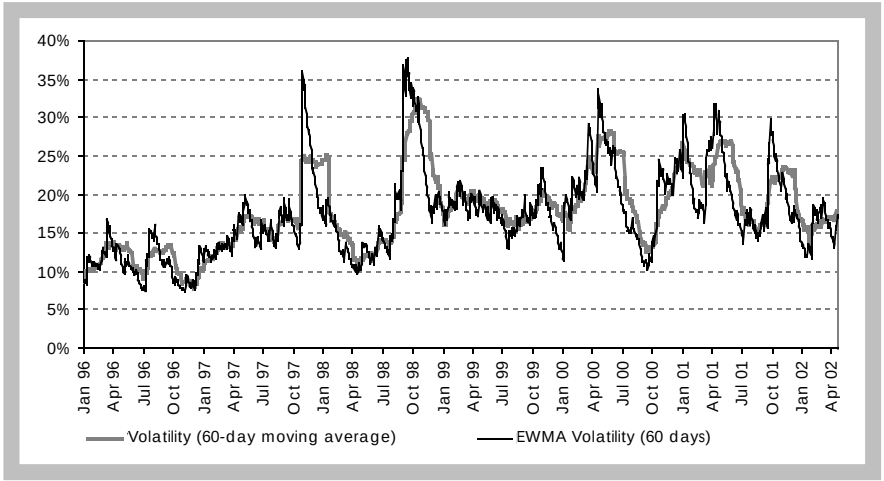


Figure 7.3 Exponentially weighted volatility of the S&P500, 1996/1–2002/5

where $\bar{r}_A$ is the average return on asset A and $\bar{T}$ is the return observation frequency. Figure 7.3 again illustrates the volatility of the S&P500 from January 1996 to May 2002, but this time with an exponentially weighted moving average (EWMA) volatility forecast included as the black line.

The EWMA forecast seems to track the historical (equally weighted) estimator, but there are a few important exceptions. Note again the event of late October 1997, where the Historical method immediately generates a high volatility that then stays high for 60 days, even though market conditions have calmed considerably. The EWMA, however, generates a volatility that shoots up (even higher than the historical estimator's result) but then quickly abates as markets calm down, and by the time the historical estimator is just about to drop the extreme event, the EWMA forecast is more than 5% below the forecast of the historical (equally weighted) forecast.

Similar extreme events are evident in early August 1998 and in mid-September 2001. Initially the EWMA volatility forecast shoots up, but then it starts to reduce significantly, even as the Historical method forecast continues to rise. In this particular case the Historical method is thus completely out of sync with actual market conditions. The same general picture is painted by other events both before and after.

In short, when using the EWMA model, we obtain an estimator or a forecast that responds more quickly to the effect of significant market events, particularly if volatility is building up over a shorter period, but also one that 'shakes off' the effect of those events much more quickly, thus more accu-

rately reflecting current market conditions. Since the EWMA is more focused on the most recent events, and is thus more responsive to recent changes, it yields a forecast that – to a much lesser degree than was the case for the historical estimator – carries through the effects of past events. The EWMA forecast thus seems to be much more in tune with actual market conditions than the Historical method.

Autoregressive Conditional Heteroscedasticity (ARCH)

Autoregressive Conditional Heteroscedasticity models were originally introduced by Engle (1982), in an attempt to estimate and forecast the variance in UK inflation. The term ‘autoregressive’ pertains to the fact that future observations are to some extent dependent upon past observations. The basic idea behind the ARCH model in the volatility estimation context is thus that volatility is persistent, but the exact form of this persistence is unknown. A simple form of the ARCH model is written:

$$\sigma_T^2 = \hat{\alpha}_0 + \sum_{i=1}^p \hat{\alpha}_i \cdot r_{T+1-i}^2 \quad [\text{Eq. 7.13}]$$

where p represents the order of the ARCH process (the number of observations included in forecasting future volatility), and $\hat{\alpha}_i$ are estimated weights put on the observations. $\hat{\alpha}_0$ is a constant. Using p past observations (most often fewer than 10), the weights $\hat{\alpha}_i$ that are put on each observation are then determined using maximum likelihood techniques, in order to produce the set of weights that provides the best estimate of current volatility.

Using this type of estimator is not entirely without problems. Primarily, the issue of fluctuations or spurious peaks in estimates, which plagues both the standard volatility estimator and the EWMA volatility estimator, still persists. This is because the order of the ARCH estimator is typically quite low, causing single observations to have a potentially large impact on the forecast volatility. On the other hand, the ARCH methodology ensures that the weighting scheme is not ‘naive’, and does not just place equal weight on all observations.

Generalised ARCH (GARCH)

The standard ARCH methodology has been generalised by Bollerslev (1986) so that it takes the form:

$$\sigma_T^2 = \hat{\alpha}_0 + \sum_{i=1}^p \hat{\alpha}_i \cdot r_{T+1-i}^2 + \sum_{i=1}^q \hat{\beta}_i \cdot \sigma_{T-i}^2 \quad [\text{Eq. 7.14}]$$

where p and q determine the order of the process. The volatility forecast is now dependent not just on the latest p return observations, but also on the latest q volatilities. The model thus has autoregressive characteristics on two parameters. This makes for a more flexible and accurate model, since both p and q can now be set to maximise explanatory power.

Multivariate Density Estimation (MDE)

As an alternative to the traditional regression-based estimation techniques, Multivariate Density Estimation (MDE) focuses on the state of the (economic) universe at the time when the forecast is being made. In other words, instead of using a fixed weighting scheme for each observation that enters into the generation of the forecast, MDE uses a range of state variables, and places higher emphasis (weight) on observations that occurred during *similar* states than on observations that occurred during *dissimilar* states.⁴

For example, if the current level of interest rates is high, then the volatility forecast for the next period is produced by putting more emphasis (weight) on similar past economic states where volatilities were high. Most often the MDE technique employs several variables to characterise and then compare the current economic state with past states. This is done by defining a current state vector $\mathbf{x}_t$ containing the current state variables that we choose to include in our model. This current state vector is then compared to a range of past state vectors, and the differences between the two determine to what extent the current state is similar to that particular past state, and using a kernel function K a weight is then assigned to that observation accordingly. Formally we can express this by:

$$\sigma_t^2 = \sum_{i=1}^s \omega(\mathbf{x}_{t-i}) \cdot r_{t+1-i}^2 \quad [\text{Eq. 7.15}]$$

where the weight put on observation i is determined by:

$$\omega(\mathbf{x}_{t-i}) = K\left(\frac{\mathbf{x}_{t-i} - \mathbf{x}_t}{h}\right) / \sum_{i=1}^s K\left(\frac{\mathbf{x}_{t-i} - \mathbf{x}_t}{h}\right) \quad [\text{Eq. 7.16}]$$

h is the ‘bandwidth’ of the kernel inputs, and it determines how variable are the final weights put on individual observations. The higher the bandwidth, the lower the variability of weights, and the closer to a uniform or equal weighting scheme the result will be. An obvious way of determining an appropriate

bandwidth is by examining the variability of the vector x and then adjusting the bandwidth accordingly. The kernel function that places relative weights on ‘near’ versus ‘far’ observations of state variables takes the form:

$$K(z) = (2\pi)^{-m/2} \cdot e^{-1/2 \cdot z^T \cdot z} \qquad \text{[Eq. 7.17]}$$

This particular kernel function is the multivariate normal density function, but does not represent a distributional assumption per se. It might take on any well-defined mathematical form.

The advantage of using this type of estimator is that the analyst can introduce any state variable he likes and any number of variables he deems necessary. The model is thus highly adaptable and flexible – but that can also be said to be its biggest drawback, since the analyst is forced to make a number of explicit and potentially significant assumptions about the state variables to include, and the kernel function and bandwidth to apply. The model is not widely used, but its intuition is very strong, and its flexibility and adaptability ought to see it become more widespread in the future.

Comparing weighting schemes

The weighting schemes of the four models presented above can be easily illustrated. Figure 7.4 illustrates the weight that each model places on past observations (in this case 180).

Standard deviation is obviously a rather dull forecasting model, as it places equal weight on all observations. EWMA forecasting is more intuitive, in that

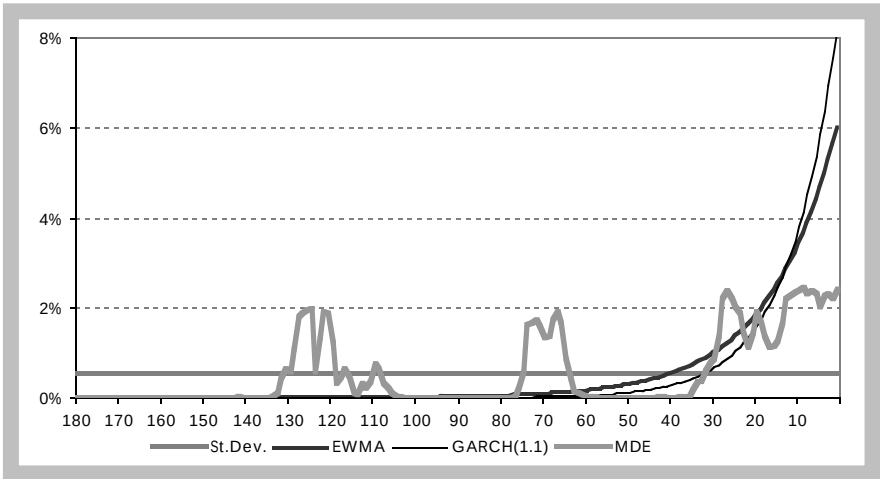


Figure 7.4 Weighting schemes (per cent) of different forecast methods

it places increasingly more weight on observations the closer they are to the present. In this case the GARCH(1.1) forecaster has the effect of ‘pulling’ the weighting scheme even closer to the present than does the EWMA forecaster.

The model that really stands out is the MDE. It is clear from Figure 7.4 that multivariate density estimation results in semi-intelligent placing of weight on past observation. It happens not in accordance with how far back in the past observations are, which would assume that ‘further away’ means ‘of less value’, but rather in accordance with how similar were the general market conditions (quantified by the state variables) to the conditions prevailing at the time the forecast is made. In this case, apart from the most recent past, there are two periods between 60 and 80 days and 105 and 135 days ago respectively, that were similar to the conditions prevailing at present time. Therefore the relative weight placed on the volatility observed during these two periods is much higher than for other periods. Note that the MDE scheme presented here is not static. It will change dynamically through time as, for each future time period, the observations upon which the model currently places high weights move further into the past. These observations will most likely continue to be important for a while and the weighting scheme will then change accordingly, keeping all relevant observations included in the forecast.

RETURN DISTRIBUTIONS (RISK CHARACTERISATION)

As we have already pointed out, the portfolio optimisation framework presented in previous chapters is very much dependent on the assumptions made about expected returns, levels of risk and correlation coefficients. In particular, the assumed level and nature of risk are important factors. We shall examine these issues in greater detail in subsequent chapters. However, in order to get a clearer sense of the character and magnitude of variation in asset returns, we briefly examine the issues here with a set of charts.

In Figure 7.5 we illustrate the return frequency distribution for the S&P500 using daily return observations from 1 January 1995 to 1 May 2002. The grey line that has been overlaid on the chart indicates the normal distribution that would describe the variability of the S&P500, if we measured this using only volatility – that is, annualised standard deviation. So Figure 7.5 shows that, far from being normally distributed, the returns on the S&P500 are exhibiting a fair amount of kurtosis. But it does seem that the distribution is nearly symmetrical. The volatility – annualised standard deviation of the index over this period – was 17.7%.

Performing a similar analysis on the NASDAQ over the same period yields the result shown in Figure 7.6. This distribution is somewhat more *peaked*, which is seen from the relatively large number of observations very close to the

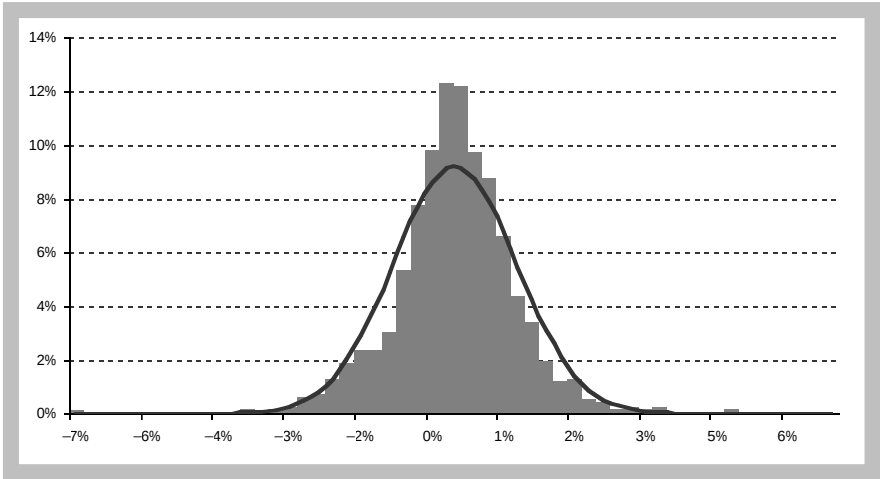


Figure 7.5 Return frequency distribution for the S&P500,1995/1–2002/5

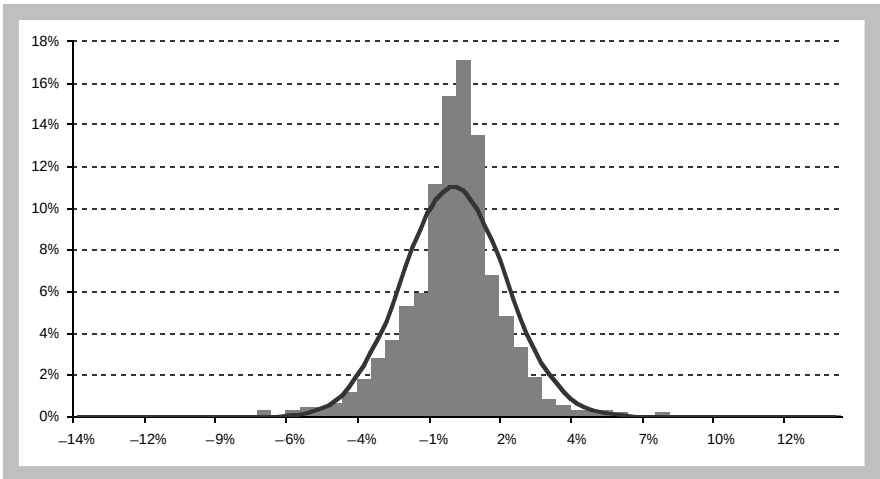


Figure 7.6 Return frequency distribution for the NASDAQ, 1995/1–2002/5

mean, and, from the relatively large number of observations very far from the mean. But the distribution also seems to exhibit skewness, although not very much. The wide range of return values indicated on the x -axis is due to the 14.2% return that occurred on 1 March 2001. However, there were several other significant moves in the index, including a number of daily gains or losses in the order of 7% to 8%. The volatility of the index over this period was 30.5%.

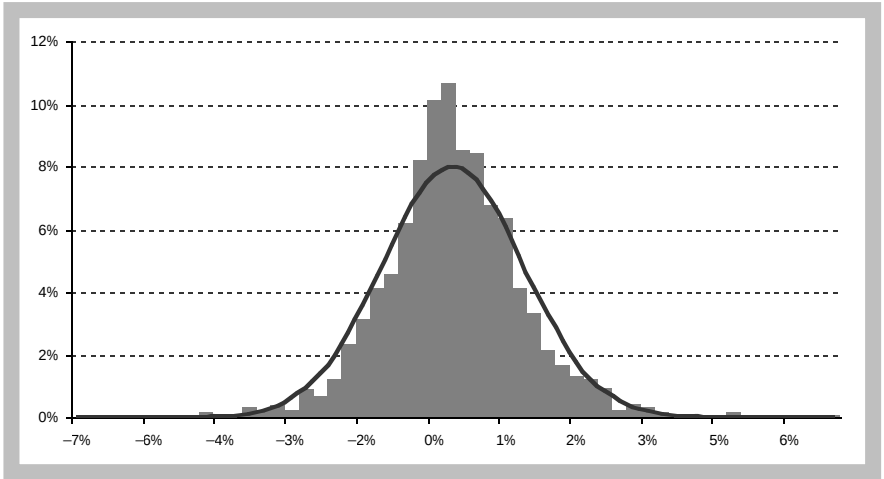


Figure 7.7 Return frequency distribution for the TOPIX, 1995/1–2002/5

Finally we examine the Tokyo Stock Exchange TOPIX over the same period (Figure 7.7). The return on this index seems to be quite close to normally distributed, and seems to exhibit very little kurtosis or skewness. This may be due to the fact that it encompasses a very large number of stocks, which will often make the daily performance of the entire index a function of many contradictory factors. The volatility of the index over this period was 20.3%.

Table 7.1 contains the estimated mean return, volatility, kurtosis and skewness of seven widely known equity indices. The numbers are estimated using daily observations from 1 January 1995 to 1 May 2002. Note that the volatility of the Hang Seng is as high as that of the NASDAQ, but that the level of

Table 7.1 Daily mean return, volatility, kurtosis and skewness of equity indices

Index	Mean Return	Volatility	Kurtosis	Skew
Dow Jones	0.05%	15.5%	5.18	−0.51
S&P500	0.05%	17.7%	3.41	−0.20
NASDAQ	0.06%	30.4%	4.41	0.13
FTSE100	0.03%	16.8%	1.60	−0.16
DAX30	0.05%	22.9%	2.78	−0.38
Tokyo SE	−0.01%	20.4%	2.32	0.14
Hang Seng	0.03%	30.6%	10.04	0.45

Table 7.2 Return distributions expressed by standard deviations

Index	0<std<1	1<std<2	2<std<3	3<std<4	4<std<5	5<std<6	6<std<7	7<std<8
Norm. Dist.	45.34%	31.78%	15.83%	5.49%	1.31%	0.22%	0.02%	0.00%
Dow Jones	74.20%	21.48%	3.60%	0.41%	0.21%	0.10%	0.00 %	0.00%
S&P500	75.46%	19.53%	3.68%	0.92%	0.41%	0.00%	0.00%	0.00%
NASDAQ	78.64%	17.13%	2.56%	0.98%	0.49%	0.10%	0.00%	0.10%
FTSE100	74.62%	19.34%	5.22%	0.61%	0.20%	0.00%	0.00%	0.00%
DAX30	75.96%	19.42%	3.82%	0.50%	0.30%	0.00%	0.00%	0.00%
Tokyo SE	73.12%	21.41%	4.10%	1.14%	0.11%	0.11%	0.00%	0.00%
Hang Seng	77.66%	17.29%	3.76%	0.86%	0.32%	0.00%	0.00%	0.11%
100 stocks	81.79%	13.51%	3.41%	0.88%	0.25%	0.09%	0.04%	0.02%

kurtosis is very much higher. It would thus seem that there were quite a few very large daily returns (positive and negative) on the Hang Seng during the estimation period. Perhaps surprisingly, the highest level of skew (positive or negative) is exhibited by the Dow Jones Industrial Average, closely followed by the Hang Seng. Note also that the average daily return on the TOPIX over this period is negative.

An alternative way of presenting this data is to examine what proportion of returns were between 0 and 1 standard deviations from the mean, what proportion of returns were between 1 and 2 standard deviations from the mean, and so on. The result of this exercise is presented in Table 7.2, and for easy comparison we have included at the top of the table the characterisation of the normal distribution. It is clear from Table 7.2 that all indices exhibit kurtosis, since the percentage of returns that are very close to the mean is higher than is the case for the normal distribution, and also because we observe several instances where returns very far from the mean are 'over-represented' relative to the normal distribution. We have also included the average of results from the analysis of 100 randomly picked individual stocks from the MSCI World. Each stock is analysed using the same technique and the estimation period is 1 January 1990 to 1 May 2002. Kurtosis is even more evident from these numbers, since the number of observations that are very close and very far from the mean is much higher than is the case for the normal distribution.

At this stage it is difficult to conclude anything very significant from this, except to say that the characterisation of indices – or any other asset or portfolio of assets – is potentially very important for risk management purposes. We shall return in subsequent chapters to the implications of non-normally distributed asset returns for portfolio optimisation, asset allocation and risk management.

Marginal return and volatility analysis

Returning to the portfolio optimisation model presented in previous chapters, it is useful to obtain a sense of the sensitivity of the model to changes in model inputs. The first and most obvious step would be to perform what we call a *marginal return and volatility analysis* (MRVA). The analysis involves marginally changing the assumed expected returns and volatility of the asset in the portfolio. A crude way of performing a marginal return analysis is to reduce all asset returns but one proportionally by 1% increase the return on the asset whose marginal return we wish to investigate, and then perform an optimisation to see the result on portfolio return and volatility. We can perform the same analysis for asset volatilities.

When performing these types of test the portfolio manager needs to be aware that these tests are *ceteris paribus* tests, and furthermore that they do not

involve assets that are *not* in the portfolio. In that sense it may be dangerous to draw very strict conclusions from the results of such tests concerning the overall asset allocation of the portfolio, but the tests can give the portfolio manager a better understanding of the characteristics and in particular the stability of his portfolio. We have already eluded to the fact that such changes in assumptions about expected return or volatility can potentially have a very large impact on portfolio optimisation results. The extent of this will become clear in subsequent chapters.

THE CORRELATION STRUCTURE

As we have already demonstrated, the correlation structure – that is, how the returns on assets in a portfolio are correlated with each other – is a significant factor in determining and exploiting the opportunities for diversification that may exist. Generally, the lower the average correlation coefficient, the less asset returns will tend to move in the same direction, and the larger the opportunities for diversification. The nature of the correlation structure and how it develops over time is thus a potentially significant issue in quantitative portfolio optimisation, asset allocation, and risk management. In this section we therefore briefly attempt to characterise the correlation coefficients between some of the equity indices examined previously in this chapter. The train of thought that this represents, as well as the actual implication for portfolio optimisation, asset allocation and risk management, will be revisited in subsequent chapters.

Correlation coefficients over time

We begin by illustrating how correlation coefficients can develop over time. Using the same daily observations for the major equity indices as above, we illustrate the 3-month moving correlation coefficients between the Dow Jones on the one hand, and the S&P500, the NASDAQ Composite, the FTSE 100 and the TOPIX index on the other. Although it is possible to use exponentially weighted correlation coefficients, applying a similar principle to that presented above for exponentially weighted volatilities, here we shall only look at correlation coefficients calculated as already demonstrated in a previous chapter (see Figure 7.8).

Note that here we only examine the correlation coefficients of the Dow Jones with a set of investment alternatives, but a similar chart could be produced for each of the remaining indices, thus providing further insight into the correlation structure of the group of asset we have chosen to examine.

What is immediately striking about Figure 7.8 is that the correlation between returns on the Dow Jones and on the S&P500 is consistently high.

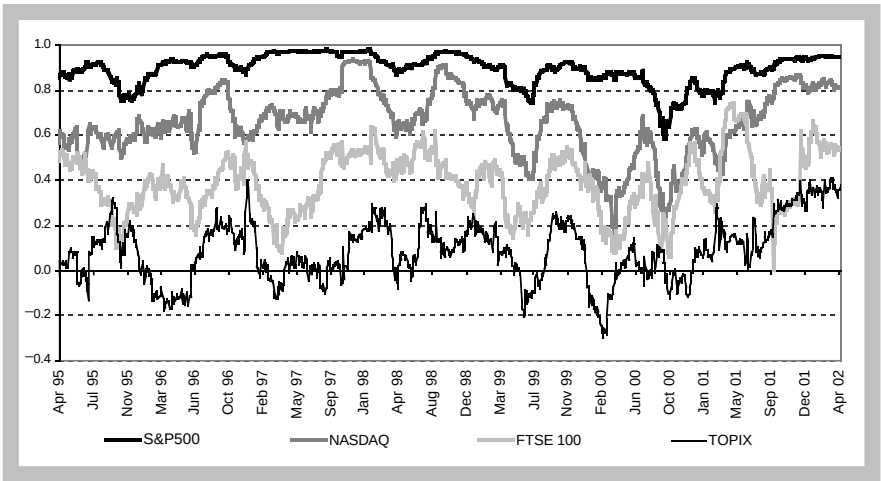


Figure 7.8 3-month correlation coefficients
for the Dow Jones, 1995/4–2002/4

Except for a few short periods, it has remained above 0.8 for the entire period, and the average coefficient is 0.89. In terms of opportunities for diversification, there is thus not very much to be gained from supplementing investments in the Dow Jones with investments in the S&P500. The Dow Jones' correlation with the NASDAQ is somewhat lower, but also tends to fluctuate considerably more than that of the S&P500. It does however remain below that of the S&P500 for the entire period; and with an average correlation coefficient of 0.65, it presents a somewhat better opportunity for diversification. Lower still is the correlation between returns on the Dow Jones and the FTSE 100. Although traditionally perceived as being closely linked, the returns on the US and UK equity markets have actually tended to exhibit a rather low correlation over the past five years. During brief spells, the coefficient has even touched on zero, but the average has been 0.38.

The correlation coefficient between the returns on Dow Jones and the Japanese TOPIX is by far the lowest among the group. It has a high of around 0.4 and a low of around -0.3 , and its average over the entire period is just 0.09, which essentially means that technically it is almost uncorrelated with the Dow Jones. However there are periods when the correlation coefficient rises, most noticeably during the latter half of 2001 and the spring of 2002. This is most likely to be because the Japanese equity market is dominated by technology-related companies, and as the bubble of 1999 and 2000 burst, the Japanese markets tumbled along with those of the rest of the world and particularly the US; and expecting that any upturn in the Japanese equity market would depend on the US economy – in terms of both corporate and consumer

spending – investors have thus bought Japanese equities when the US economy seemed to improve, and sold whenever it seemed to be deteriorating. This naturally causes a rising correlation coefficient between the two markets. During the spring of 2002, there seemed to develop a widening acceptance that Japan might be able to pull out of its economic malaise more independently than previously assumed. Should that perception continue to grow, one would suspect that the correlation coefficient between the Dow Jones and the TOPIX would tend to fall again, as Japan reverts to trading as ‘its own market’ and not as a US equity market derivative.

This simple analysis demonstrates that correlation coefficients are not just numbers that are entered into an equation or a model for portfolio optimisation. They can be used to assist managers in prospective asset allocation and risk management decisions, if interpreted and perhaps even forecast correctly. We shall return to this type of analysis in Chapter 15, on sector risk models.

Correlation coefficient distributions

In the section above, we mentioned that correlation coefficients can fluctuate over time. The exact nature and extent of this fluctuation can be easily illustrated using a frequency chart. Figure 7.9 illustrates how the correlation coefficients between the Dow Jones and the TOPIX are distributed.

In a portfolio optimisation and asset allocation context, this type of analysis provides the asset manager with an assessment of whether the correlation levels

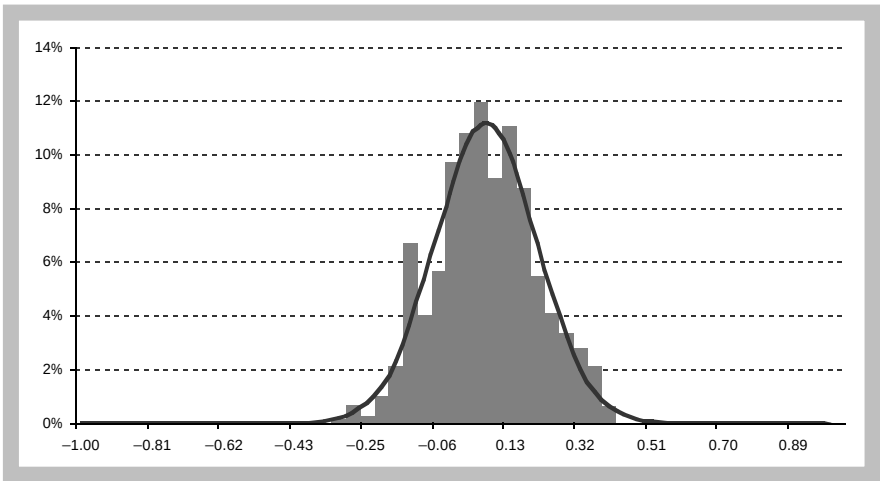


Figure 7.9 Distribution of correlation coefficients – Dow Jones and TOPIX

currently observed are 'normal', and thus whether any apparent diversification effects that seem to be presenting themselves can reasonably be expected to be sustainable over the near to medium term. From the chart we note that the correlation coefficient between the Dow Jones and the S&P500 is relatively stable over time: this is positive news for quantitative portfolio optimisation and asset allocation, since it means that we can reasonably assume that the correlation coefficient will remain in the range illustrated, which in turn means that we can have more confidence in the result of the portfolio optimisation.

Stability of the correlation structure over time

We can take this one step further and examine correlations between changes in correlation coefficients. While this may sound a little far-fetched, performing such an analysis can actually provide us with some insight into how the correlation structure changes over time, and to what extent it may benefit or obstruct our quest for structured exploitation of available diversification effects. To be more precise, if the correlation coefficient between changes in individual asset correlation coefficients is high – which it often is – it means that these individual asset correlations tend to move in the same direction. This in turn implies that falls in correlation coefficients, when they happen, tend to happen across the board, thereby increasing the opportunities for diversification among all assets or asset classes. It would thus appear that opportunities for diversification are increased for all possible prospective portfolio compositions. The opposite obviously holds true for rising asset correlations. They will tend to rise in unison, thereby reducing opportunities for diversification across the board.

A note on correlation and volatility estimation inputs

A question that is difficult to answer is whether we should use daily, weekly or monthly return observations for our estimates and forecasts. If one is a so-called long-term investor, one should not be too concerned with daily movement of the stocks in one's portfolio. Rather, keeping in mind that the existing stocks in the portfolio are intended to stay there for a long period – for tax reasons – before being sold, it would be natural to estimate on the basis of monthly returns.

Alternatively, one could make the assumption that we are not concerned with the *absolute* levels of volatility of returns, and as a consequence not too interested in obtaining very precise absolute estimates. Rather, the most important thing – if for no other reason than that correlation and volatility levels change over time – would thus seem to be the *relative* level of correl-

ations and volatilities between returns on stocks in a portfolio. In that case it would be appropriate to estimate on the basis of daily returns, since this provides the analyst with the largest number of observations per time period. To be more precise, this might partially – perhaps with the help of the EWMA or MDE methodology – eliminate the problem of having to go back a very long time (years) in order to obtain enough observations to be able to perform a statistically sound analysis of correlations and volatilities.

An argument for using weekly observations is the fact that the market participants are very focused on news events in particular weeks. Investors often look towards a particular week to reveal economic news on the economy or on companies or sectors, and thus the week interval captures this tendency of markets, or rather market sentiment, to adjust its direction weekly. Note that this will often entail no change in direction at all but just a continuation of last week's sentiment. In a correlation estimation context this is particularly important, since we wish to capture co-movements in our estimation; and since the market as a whole has a tendency to adjust direction weekly, this estimation interval will probably reveal the most information about precise co-movements among stocks and stock returns.

Ultimately the choice is up to the asset manager – and making *no* choice on the matter actually represents a choice in itself.

SUMMARY

In this chapter we have presented a range of tools for estimating or forecasting asset returns, volatilities and correlations. As is well known, forecasting asset returns is very difficult indeed, whereas forecasting volatilities and correlations is somewhat more manageable.

The CAPM has formed the basis for much work on return forecasts and also asset valuation over the past few decades, but the most widespread factor models are multi-factor models using either fundamental data, economic data, statistical decomposition and characterisation, or a combination of the three. Most commercially available multi-factor models rely on fundamental analysis.

Instead of using the standard volatility measure in a dynamic portfolio context, we have demonstrated that exponentially weighted moving averages of volatilities tend to provide a more intuitive result. ARCH, GARCH and MDE techniques have also been presented as viable alternatives to the simple volatility estimator. Of these, the MDE seems the most intuitive, but so far it has seen little use in practical investment management, principally because it requires the asset managers to make a range of explicit choices concerning model assumptions. Using data from 1 January 1995 to 1 May 2002, we have demonstrated that asset returns are not normally distributed, neither on the

index level nor on individual asset levels. The implications of this in quantitative portfolio optimisation, asset allocation and risk management will be explored in the following chapters. Finally we presented a rudimentary analysis and characterisation of correlation coefficients, and demonstrated that they tend to remain relatively stable over time, which helps us in our quest for portfolio diversification in a dynamic portfolio framework.

Notes

- 1 David Nawrocki: *The Case for the Relevancy of Downside Risk Measures*.
- 2 The topics of benchmark-relative portfolio optimisation, asset allocation and risk management will be discussed in further detail in subsequent chapters.
- 3 This approach has been implemented into such products as RiskMetrics™.
- 4 MDE estimators may take an unlimited number of forms, but the type presented here is identical to that presented in Boudoukh et al. (1997).

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PART III

ASSET ALLOCATION

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INVESTMENT OBJECTIVES AND BENCHMARK SELECTION

THE INVESTMENT POLICY STATEMENT

Basic content

It should be very clear by this point that the objective of an asset manager is to obtain a return for the client, or more precisely to do this while at the same time generating a cash flow stream that matches client needs in term of both magnitude and variation over time: that is, return and risk. Usually the asset manager and the client will together work out an *investment policy statement* that will guide the investment process. An investment policy statement details the foundation upon which investment decisions are made. The document establishes portfolio goals, clarifies investment guidelines, and identifies investment parameters and monitoring benchmarks. An investment policy statement is essentially the blueprint for the portfolio. Generally speaking, there are four main elements that must be contained in the investment policy statement.

Setting realistic objectives

An investment programme should be guided by a rational, consistent policy suited to the investor's unique circumstances. To establish clear and definable

expectations, the investment policy statement should state the investor's objectives and goals in a concise manner. It should establish risk tolerance, return requirements, income needs, liquidity requirements, investing time horizon, tax considerations, legal and regulatory issues, and unique needs and circumstances particular to the client. This written investment statement thus clarifies the overall investment plan, and clearly articulates the client's investment objectives and restrictions, thus providing a measurable basis of feedback with the client's advisor. An investment policy statement compels the investor and the investor's advisor to be more disciplined and systematic in their decision-making, which in itself should improve the odds of meeting the investment goals. Because objectives and expectations are clarified for all concerned parties, misunderstandings are less likely to arise once the investment policy statement has been agreed upon.

Defining the asset allocation policy

Once investment objectives are established, the investment policy statement should define the asset allocation strategy by specifying an acceptable long-term asset allocation mix between the various asset classes that the client would like to be invested in over the long term. It should be designed to provide the highest rate of return commensurate with an acceptable level of risk, as well as to include any other portfolio constraints. The investment policy statement should explain how the advisor will evaluate the investment vehicles used to implement long-term asset allocation strategy. The investment policy statement directs the investor and investment professionals to focus clearly on the investor's goals and objectives when considering any potential investment opportunities.

Establishing management procedures

By providing a guide for selecting, monitoring and evaluating the performance of those responsible with managing and investing the assets and making changes as appropriate, planning ahead makes it easier for everyone involved when the financial markets become turbulent. Approved decision-making and implementation procedures are specified so that everyone concerned will know what to expect. In a deliberate fashion, rather than in the 'heat of battle', decisions can be made as to how things will be done under a variety of changing circumstances and conditions.

Determining communication procedures

The investment policy statement provides a concise method of communicating the process and objectives amongst all parties involved with the investments, and assigns responsibility for implementation. It provides a ready means to communicate to advisors, beneficiaries and current and future fiduciaries how the investor proposes to go about acting upon his or her duties. In order to be most effective, an investment policy statement should not be too restrictive and should provide as much flexibility as possible. The wording of the investment policy statement should allow for changes in the following: the client's financial situation, procedures for modifying the investment policy statement, any material changes at the advisor's firm, and the authority vested in the advisor.

The investment policy statement must be based on the specific characteristics and preferences of individual clients, taking into consideration their expectations or requirements regarding return and their tolerance for risk. Here, as earlier, we simply assume that risk can be expressed fully by standard deviation of returns or volatility. While this may not be completely accurate, it is adequate and also greatly simplifying for the portfolio construction and asset allocation process. The investment policy statement will need to be very specific on the following issues.

Additional components

Amount to be invested

The size of the portfolio to some extent determines the degrees of freedom for the portfolio manager in terms of the number of assets that can reasonably be included in the portfolio. As has been demonstrated already, there are potentially very large benefits to adding a fairly high number of assets to the portfolio, but for portfolios with a relatively low net value it may be counterproductive to add too many assets to the portfolio. This is because transaction costs may erode the additional performance that adding a few extra assets to the portfolio would generate at some risk level. This is of course mainly relevant for smaller investors. For institutional investors this is of less importance, since these institutions are often able to trade in large blocks for several mandates at a time, thus reducing considerably the cost per asset traded in the portfolio.

Risk tolerance

Possibly the most important aspect of the formulation of an investment policy statement is the establishment of the client's risk tolerance. As mentioned

earlier, the portfolio manager takes on risk in order to obtain a return as a reward for this risk. Exactly how much risk the client is willing to accept must necessarily be determined and to the extent possible also quantified. This process of uncovering and describing thoroughly the client's risk tolerance is essential to the process of managing the portfolio in a way consistent with this level of tolerance. How to do so is another matter. Since investor preferences cannot easily be described in terms of the utility functions or indifference curves often applied in theoretical economics, the asset managers must often determine the client's risk tolerance by the use of examples of possible future performance of the portfolio. This kind of scenario construction can reveal to a reasonable extent the client's tolerance for risk.

Many asset management organisations – large as well as small – that cater to institutional as well as private investors often apply some form of questionnaire which poses a series of questions to the client concerning hypothetical situations: in each situation, where the client is forced to choose from a handful of options as to what action to take. For example, the client could be asked whether, after having won \$100 on the lottery, he would buy new lottery tickets using all of the \$100 or perhaps using only a part of it. Questions can also pertain to more tangible issues, such as whether the investment included the sum total of the client's portfolio (wealth) or only a small part. This questionnaire will to a large extent determine the risk aversion or tolerance of the client.

Although the questions and scenarios may appear somewhat simplistic, they are nonetheless used for this sort of client assessment. Such a series of questions will typically result in a ranking of the client on a scale from minimum risk taker (maximum risk aversion) to maximum risk taker (minimum risk aversion). Often this can lead to a quantitative proxy for risk aversion – a risk aversion coefficient. This may not be very meaningful in itself, but it can place the risk tolerance of the client in perspective and express it in relative terms, thereby enabling the asset manager to suggest a model that is placed approximately in the correct spot on the 'risk-meter'.

Investment time horizon

The investment horizon is important as it is closely linked to the risk tolerance. The longer the time horizon, the more risk a client can take on, since historically there exists a linear relationship between asset return and risk, as demonstrated for example by the CAPM in previous chapters. Greater risk entails greater probability of losing a large portion of the portfolio, but it also means that the client has a longer period of time in which to recuperate the loss and possibly to add more value to the portfolio. Thus the time horizon will be

instrumental in determining the types of assets that should be included in the portfolio to start with, since this enables the client to place limits of sorts on the expected future volatility of the portfolio. The portfolio weights that the asset allocation process result in can then be changed gradually as time passes and as the client gets closer to the time when he or she needs the money. In this way the expected volatility of a portfolio will typically be intentionally decreased as the client nears the time of liquidating the portfolio.

Liquidity needs or cash flow requirements

As indicated, the job of the portfolio manager is not necessarily to chase as much return as possible: rather it is to generate the stream of cash flows that is desired by the client. This cash flow is typically determined by the liquidity needs of the client, who may have mortgage payments or other large time-dependent expenses that should be covered. This of course means that the portfolio manager must see to it that the assets in the portfolio can be liquidated in an appropriate 'time window'. This in turn means investing in assets or asset classes that are part of markets that will have sufficient liquidity around the time when the client needs the cash transfer. Investing in futures, options or other derivatives may therefore not be productive for the overall portfolio management process unless the portfolio manager has timed these purchases correctly to coincide with the cash flow requirements of the client.

Tax considerations

Obviously, tax considerations may affect the way a portfolio is optimally managed, especially if there are certain assets whose returns are exempt from taxation altogether, or if the returns on certain assets change after a specific holding period. Given that asset management is increasingly becoming a global business, it follows that differences in taxation of capital gains in different countries may very well affect the formulation of the optimal investment strategy.

CHOOSING THE BENCHMARK

Introduction

The policy statement is the blueprint for the asset allocation decision, and thus also for the choice of benchmark. Since the investment policy statement

determines the asset classes that are to be invested in and most often also contains limits for asset class weightings in the portfolio, the subsequent benchmark choice must be consistent with the investment policy statement if it is to have any practical investment and performance evaluation value. The benchmark serves as a starting point for planning, constructing, implementing and evaluating portfolios and portfolio performance over time. This entails a set of requirements and considerations in the process of choosing an appropriate benchmark. These will be discussed below.

The choice of benchmark should be made jointly by the client and the asset managers, and the benchmark itself should ideally serve as a conveyor of information about the investment policy and process. Properly formulated benchmarks can serve as catalysts for more effective asset management; and conversely, improperly formulated benchmarks can lead to ineffective management and to disputes between clients and asset managers. Indices used as benchmarks are typically constructed by large investment and performance measurement houses such as Morgan Stanley, Wilshire, Frank Russel Company, J. P. Morgan, and Lehman Brothers.

Below is the general methodology for construction of country equity indices at Morgan Stanley Capital International (MSCI)¹:

- Define the universe of listed securities within each country.
- Sort securities into industry groups.
- Select securities until 60% of each industry's market cap, and therefore the country's market cap, is included in the index.
- Select stocks with sufficient liquidity and free float.
- Avoid cross ownership among stocks in the index.
- Apply the full market capitalisation weight to each stock.

Approaches vary between companies, but the general methods remains the same. The objective is to construct indices that are as applicable as possible to practical portfolio management.

Stability of benchmarks

In choosing a benchmark, asset managers should be aware of possible migration effects within the benchmark over time, which may change the characteristics of the benchmark without necessarily changing the characteristics of the portfolio. As time goes by, companies are added to and removed from the

indices, and this may change the market capitalisation, the overall liquidity, and also the volatility of the index. In the initial years following the introduction of the S&P400, roughly 16 companies were added and removed from the index per year. While only 4% of the names changed each year, nearly 10% of the market capitalisation was removed annually. Many companies deleted from the S&P400 were added to the S&P500 in the early 1990s. There was also a shift in the sectoral composition of the index, with the inclusion of companies from the emerging sectors.

If the portfolio manager holds in his portfolio only a limited share of stocks in the benchmark, it is probable that the relative performance of the portfolio will change over time, even though the manager sticks closely with his investment mandate and has very little turnover in his portfolio. Therefore broad, relatively stable indices with relatively low company turnover rates are preferable. Naturally the benchmark that is chosen for the portfolio must have some relevance, in terms of the manager's ability to track the benchmark and in terms of the nature of the assets in the benchmark relative to the nature of the assets in the portfolio.

Currency effects

When choosing a benchmark that contains international as well as domestic assets, the choice implicitly regards currencies as well. In other words, as well as making a decision on asset class exposures (countries and sectors), the asset manager also makes a decision on currency exposure. Since currency movements can explain a large portion of portfolio performance on internationally diversified portfolios, this is an important point both in the portfolio management process and in the benchmark choice decision. A way to control these effects is to use *hedged benchmarks*. In an unhedged benchmark, the performance relates both to asset returns and to currency returns. In a hedged benchmark the currency returns are filtered out, allowing a more direct assessment of portfolio performance.

Unhedged benchmarks are used when the investment mandate does not include a consistent hedged position, although currency activity may be allowed as a means of adding to return and/or reducing risk. They might be used, for example, by investors with relatively low allocations to foreign assets, and/or investors who actively desire currency exposure.

Hedged benchmarks might be used, for example, by investors with material allocations to foreign assets who wish to eliminate the volatility of foreign currencies. However, this should be done in close accordance with the level of the actual use of currency hedging in the portfolio. When the portfolio mandate stipulates the strategic hedging of some or all of the currency expo-

sure, then a hedged benchmark is used. Such benchmarks may eliminate all currency exposure (a fully hedged benchmark) or they may eliminate a fixed proportion (between 0% and 100%). The proportion chosen is often called the *benchmark hedge ratio*.

There is an increasing focus on the correct choice of benchmark, both from regulators and from advisory entities, which underlines the importance of this activity from the perspective of the asset managers as well as from that of the clients.

General rules

Generally, for a benchmark to be suitable for use in practical portfolio management it must as a minimum exhibit the following five qualities:

- Representative
- Investable/tradable
- Replicable
- Dividend correspondence
- Acceptable by managers as well as clients.

The benchmark must be representative of the asset class or mandate under consideration. It must be tradable or investable, in the sense that it must be possible to trade all assets that are included in the portfolio. In other words it must be a viable investment alternative. If this is not possible, it will not be possible to fulfil the second prerequisite, namely replicability. It is obvious that the portfolio manager must be able to replicate the benchmark, otherwise he would be fighting a losing battle from the start. The third prerequisite also ties into replicability since there must be correspondence between the handling of dividends in the benchmark and the portfolio that is to be managed. If the portfolio contains assets that pay dividends, then the dividends must be added back into the benchmark before performance calculations are performed, otherwise the portfolio manager would exhibit outperformance where none is present. It seems quite obvious that the benchmark should be acceptable to managers and clients. More precisely, this entails that managers see the benchmark as a neutral position. As will be discussed in the following chapters, asset managers are often measured by their performance relative to the benchmark, and any deviation in the initial strategic asset allocation as defined by the benchmark will cause the managers to perform differently than the benchmark. Therefore the manager is interested in benchmarks that

convey neutrality from an investment strategy point of view. For the client it is equally obvious that the benchmark should be a neutral position from which she can subsequently ask the manager to deviate. Given that the manager is measured by relative performance, this enables the client to control the risk (in relative terms) of the manager's bets (tactical asset allocation). This will be discussed in subsequent chapters.

The use of peer groups as benchmarks

Instead of using major indices as benchmarks, the client and the asset manager can agree to use an 'artificial' index constructed for example on the basis of a peer group of similar managers. This may be relevant in situations where the asset manager is asked to manage a portfolio of assets that belong to asset classes for which proper indices are not readily available. This might be such asset classes as real estate, private equity, venture capital, hedge funds, or very narrow definitions of sectors. For example, an asset manager managing a technology fund might be measured against the NASDAQ, but since the NASDAQ contains stocks such as drugs and biotechnology, this may not be appropriate. Alternatively, the manager might be measured against a universe of similar funds, thus giving the client as well as the asset manager some indication of the manager's relative performance.

The major advantage of using peer groups as benchmarks is the fact that these are readily available investment alternatives. Some of the drawbacks are the facts that their prices are often not available in real time, that there may not be established guidelines for which funds to include in the benchmark and with what weight, that *survivorship bias*² will tend to overstate the actual performance of the peer group thereby making it unreasonably difficult for the manager to beat the benchmark, and that it is difficult at best for the manager to move to a neutral position because this is hard to define.

SUMMARY

In this chapter we have briefly outlined the nature and purpose of the investment policy statement. We have also presented guidelines for defining or choosing an appropriate benchmark against which to evaluate both the character and the performance of portfolios.

It should be noted that this and subsequent chapters assume that the investment manager chooses or is given a benchmark to follow, and that his or her task is to track that benchmark. This in turn entails that the manager may deviate from the benchmark in order to attempt to obtain portfolio return

in excess of the benchmark. What this implies is that the manager believes that he or she has superior information relative to other market participants, and that in consequence he or she can enhance performance by taking bets on certain assets or asset classes. Essentially this is what the active versus passive portfolio management debate is all about. We shall return to this issue later in this book.

Notes

- 1 Further explanation of the methodology can be found at <http://www.msci.com/>.
- 2 Survivorship bias occurs because the best performing funds appreciate faster in value than the worst performing funds, causing the worst performing funds to continuously be filtered out from the benchmark. One could argue that this is also the case for stock indices, which is another argument for using broad indices with low levels of company turnover.

QUANTITATIVE PORTFOLIO CONSTRUCTION AND ASSET ALLOCATION

THE ASSET ALLOCATION DECISION

Ultimately the analysis of assets and asset classes must lead to some decision by the portfolio manager as to the composition of the appropriate portfolio for the circumstances. The two terms 'asset allocation' and 'portfolio composition' will be used interchangeably throughout this and following chapters. In this book we will not be going into detail about the determination of the exact impact of the initial asset allocation on ex post portfolio performance. Performance attribution is a very comprehensive topic by itself, and should require a book on its own. However, a well-known and often quoted empirical study by Brinson, Singer and Beebower (1986/91) indicates that as much as 90 per cent of a portfolio's performance can be explained by the initial asset allocation. This means that even if the asset manager has hired and paid for exceptional portfolio managers possessing consistently superior stock-picking skills, this will not automatically lead to exceptional performance. This is because the initial asset allocation may have been far off the mark in terms of forecasting where large returns would materialise, and furthermore the portfolio may be seriously constrained by asset allocation limits that do not allow the portfolio manager to fully exploit the opportunities available to him. Given the fact that such a large proportion of portfolio performance is explained by the asset allocation decision, it could be argued that *the asset allocation decision is the most important decision in the life of a portfolio*. For this reason we shall now move on

to establishing a quantitative asset allocation framework, which we can then extend and apply to various other purposes later on in this book.

Asset managers and portfolio managers in particular often spend the vast majority of their time on stock-picking, and comparatively little on the overall strategic asset allocation considerations and decisions. To the extent that the portfolio manager is tasked solely with stock-picking, which is often the case, and to the extent the asset manager has a dedicated asset allocation team, this may not present a problem. Troubles begin if asset managers, typically chief investment officers, ask their portfolio managers to construct portfolios based on asset allocations that they themselves define. The portfolio manager will tend to focus almost all her time on selecting stocks, so there is a real danger of giving too little attention to the asset allocation process itself.

The portfolio construction and asset allocation process can be made very simple or very complicated, and it can entail either very quantitative techniques or very qualitative techniques. Since qualitative techniques are often subjective in nature and dependent on the manager having to make the asset allocation decision, we shall concentrate solely on quantitative techniques. In principle, there is no single correct answer or solution to those processes, but applying quantitative techniques should in theory provide for better and more robust portfolios in terms of return relative to the risk incurred. This is particularly true for strategic asset allocation purposes, and to a lesser extent for tactical asset allocation purposes. We shall discuss both later on in this book.

Below we will briefly examine a few typical non-quantitative techniques and then move on to applying to the asset allocation process the quantitative portfolio optimisation techniques presented in Chapter 6. The latter half of this chapter will provide the basis for further analysis and development of quantitative asset allocation and risk management techniques developed and presented in this book.

TRADITIONAL PORTFOLIO CONSTRUCTION TECHNIQUES

Shooting darts at the *Wall Street Journal*

Obviously there can be varying degrees of complexity in the asset allocation process. As readers may know there have been studies – not all of them entirely scientific in nature – of monkeys selecting stock from, or having individuals throwing darts at, the *Wall Street Journal*. This random ‘portfolio construction technique’ has on occasion resulted in what is for the asset management community a slightly embarrassing number of portfolios that

outperform relative to an appropriate peer group or benchmark. Selecting stocks or other assets at random is clearly not optimal if the manager has information about the market superior to that of the average market participant, but given the lack of precise information and/or the ability to understand, interpret and use this information by portfolio managers, there may be an argument for simply selecting stocks at random and then equal-weighting them in the portfolio. In general, if the portfolio manager does not have superior information she should stick to holding the benchmark, or at least a sufficient number of the stocks in the benchmark to ensure an adequate degree of portfolio diversification and *risk control*.

This reflects a general point to which we shall return in a risk-management context. Whether it is on the individual stock level, the sector level or some factor level, the degree of deviation of a particular portfolio relative to a benchmark should always correspond to, and as far as possible be proportional to, the amount and quality of information available to the manager on that specific issue: the more information available to the manager and the higher its quality, the more confident the manager can generally be in her deviations from the benchmark. There are ways of quantifying this, and we shall do so in the chapters ahead.

Relying on (good?) experience

This approach is slightly more sophisticated than the previous approach, which was of course entirely random in nature. Often portfolio managers focus on a single area of expertise such as a particular investment style. This could be 'Growth' or 'Value', 'Small caps' or 'Large caps', or the like. Since stock markets are very dynamic entities that are impossible to describe, much less to predict perfectly in quantitative terms, there is a strong argument for relying on experience in the portfolio construction and stock selection process – the latter is particularly suited for such an approach. Highly experienced portfolio managers can often sort through the available information and extract the portion that will prove relevant and even decisive for future portfolio performance. Thus relying on this very human quality is the basis of much of the outperformance that is produced in asset management.

Having huge amounts of information available is of no use if no one can make sense of it and evaluate its relative importance. For this reason, quantitative techniques have become increasingly widespread in recent years, as the amount of available information with potential investment decision relevance has grown exponentially.

Screening approach

Screens are the first step in a semi-quantitative approach to portfolio construction. They are widely used, with varying degrees of complexity, but the basic premise is the same: use some method of ranking stocks – based on size, valuation measures, financial ratios, earnings forecasts and revisions, or something similar – and select a particular proportion of the top-ranked stocks to add to the portfolio. A typical ranking scheme may appear as follows:

- Select 200 stock from an index
- Rank them according to expected return/risk
- Select the top 30
- Equal-weight (or capitalisation-weight) them to create a portfolio.

This is probably the most basic portfolio construction technique available, and it does have several obvious benefits. It is easy to understand, to explain and to update. There is a clear link between cause and effect, in that stocks are only added to the portfolio if they fulfil certain predefined criteria, such as expected return relative to expected risk. If a stock does not fulfil this criterion, it is simply not included in the portfolio. Another advantage of screening techniques such as this is the fact that the portfolios that are generated are usually very robust in the sense that new estimates of expected return or risk would probably not result in assets being dropped from the portfolio. If stocks are in the top 30 of an equity universe based on some selected criteria, it is likely that the vast majority of them will stay there even if analysts change earnings forecasts or target prices significantly. This in turn results in a relatively low trading volume and so relatively low transaction costs for the portfolio.

In some respects the screening approach makes perfect sense, not least because it can be performed on an infinite number of different levels, thereby actively managing the vast amounts of information available. However, it does have several drawbacks. In the example presented above, all information other than expected return and risk is ignored, making it a very crude selection tool indeed. Screens are most likely biased toward certain sectors and industries. Applied to the US stock market between 1997 and 2000, the most likely outcome of the screening process presented above would have been a near 100% technology portfolio, thereby completely missing out on any opportunities even for random diversification and active portfolio risk management. Furthermore, the screening approach does not consider explicitly the possible benefits from a purposeful diversification through portfolio optimisation. For a given level of expected return, the only risk-controlling element is the diver-

sification that comes naturally from the fact that hardly any stock returns are perfectly correlated, so adding many stocks to the portfolio will automatically entail some level of diversification or risk control.

However, the 'risk control' is entirely random. In fact it can hardly be characterised as control. A more appropriate term would be 'uncontrolled risk reduction', which does not sound particularly appealing. In the portfolio construction process itself there is no way of telling whether the stocks chosen are suited to each other in the particular portfolio to which they are being added. In addition there is no explicit consideration of the correlation structure among the assets included in the portfolio. All the portfolio manager can do is to aim at some level of expected or desired return and then basically to hope for the best in terms of risk, probably relying on the notion of a reasonable or natural relation between return and risk. This is the archetypal *ex post* approach: the manager is unable to gauge the performance of the portfolio in terms of return and risk until events have occurred. *Ex ante* there is basically no risk control other than what might be termed 'accidental' risk control. The term 'risk control', as opposed to 'risk management', is used here intentionally, since there is no explicit management of risk, but rather a more or less accidental control of, or restriction on, portfolio risk.

Having accepted that the asset allocation explains the vast majority of portfolio performance, it is rather unsettling to realise that the screening approach presented here – although applied extensively throughout the industry in either this form or more comprehensive forms – does not indicate anything about the relative weights of each asset class in the portfolio. This means that the manager has nothing to go by when deciding how much to hold of each asset class, which further frustrates any attempt at more rigorous risk control. If managers insist on applying screening as their selection tool, it may be wiser simply to select the equal-weighted portfolio without any particular emphasis on individual asset classes, since there is no way of knowing exactly how the returns on individual asset classes will interact over time. This may be an oversimplified criticism, but it is not entirely off the mark.

Stratification approach

Stratification has been called 'glorified screens'. Basically this approach entails just screens with limits on asset classes and/or asset class characteristics. It provides a tool for stratifying or dividing the assets into mutually exclusive groups, and then performing simple screenings on one or more chosen criteria. It is a very flexible technique, in that the stratification can be performed on a country, a sector, a currency or a factor level, or on the basis of any other para-

meter the manager deems relevant at a particular point in time. In principle, the asset manager can add an infinite number of limits to the screening process, in order to enforce some – although still somewhat uncertain and unpredictable – form of risk control. But again, risk control in this context is really aimless risk reduction rather than anything else.

Given that stratification techniques are extensions of screening techniques, they are prone to the same problems as simple screenings. The risk control is accidental, but there is a slight improvement over the simple screening technique. Using stratification, the manager is able to more easily tie the performance of the portfolio to a benchmark, by letting the mutually exclusive groups mimic the benchmark weights to some degree, depending on the client's tolerance for risk. In relative terms, this ensures a degree of risk control that is still uncertain but more clearly defined and manageable than was the case for the simple screening process. When the portfolio is tied to the benchmark in the manner indicated above, the portfolio will tend to be less biased simply because it has to have some weight in each benchmark segment. The manager will therefore not end up with a portfolio of stocks virtually all of which belong to the same sector or industry.

Bottom-up/top-down approaches

The *bottom-up* and *top-down approaches* are techniques that aim at selecting stocks and building portfolios starting with either a fundamental company level analysis or a macroeconomic or geographical analysis.

Bottom-up managers stress individual companies and company fundamentals over macroeconomic or top-down consideration. Hard-core fundamental-analysis bottom-up managers do not consider such things as sector and geographic allocation in constructing portfolios. They build portfolios stock by stock, based on the overall attractiveness of individual stocks, their valuations (most often relative to an appropriate peer group of stocks), and the fundamentals of each company's specific business environment in its particular industry at that particular time. Such managers will typically buy stocks of what are deemed to be fundamentally sound companies with attractive valuation ratios – P/E, P/B (Price to Book), EV/EBITDA (Enterprise Value to EBITDA), and so on – sustainable future earnings growth rates, high and growing levels of free cash flows, and so forth. In addition, they will examine qualitative aspects, as well as focusing on such things as a sustainable ability to grow, based on the market or industry within which it operates. The company must also hold some competitive advantage over other companies in its particular line of business, preferably both in terms of quality of products and services provided, stemming for example from superior technology; it

must also have lower costs than its competitors. The company must have a clear business strategy and be focused on exploiting any synergies available within the company. A strong financial position must be present in the form of a strong balance sheet, a sound debt policy and a high return on equity.

Naturally, the manager will also want to select stocks that have good management. This usually means examining the track record of individual management members, and establishing whether there exists an incentive structure within the company that is able to spur management as well as employees to perform their best in all economic and market conditions. Measures such as *economic value added* (EVA) are often employed as a proxy for management quality. Several studies indicate a 'management premium', according to which investors are willing to pay a premium for companies with good management. This is particularly true for companies in fast-growing sectors such as technology and biotechnology. These are just a few of the parameters that a fundamentally oriented portfolio manager will consider when picking stocks.

Top-down managers, in contrast, will typically start out with a benchmark and then try to determine which geographical regions – or in the case of domestic equity portfolios, sectors and industries – are likely to perform better than average in the future. Having established this, the manager will then decide on which asset classes or sectors to over- and underweight, in accordance with expected returns.

Asset managers will often apply some combination of the two approaches, for example using a bottom-up approach for stock selection and a top-down approach for risk control. Alternatively they will begin with a top-down approach, identifying geographic region or particular sectors that are expected to do well, and then apply a bottom-up approach to the stock selection, possibly including some form of quantitative stock-level risk management tool to control for portfolio exposures to certain risk factors. The possibilities are endless, which is perhaps why there are so many different asset managers around.

Thematic approaches

The *thematic approach* has become increasingly popular among global investors, although perhaps a little less so after the bursting of the technology bubble. The basic idea of this approach is to identify very long-term trends in the way the global economy develops, and then to bet on the sectors or stocks that are exposed to these potential growth opportunities. An example of such a trend would be the increasing proportion of elderly in the industrialised world. This puts pressure on public care for the elderly, opening up opportunities to companies in the care sector. This point is underscored by the fact that

elderly are typically better off than younger people, and are therefore better able to pay for private care services. This pattern extends from the most basic care in the home of the elderly person to highly complicated surgery in private hospitals. Another theme would be the increase in the proliferation of information technology in industrialised as well as in developing countries. This opens opportunities to technology leaders as well as to service providers related to information technology. In the selection process for a thematic approach, most of the criteria for the actual stock selection remain the same as above. This approach is to some extent dependent on equity markets reflecting the future accurately at all times, which means that asset managers who rely on this approach may experience protracted periods of poor performance during times when markets are turbulent and perhaps not focused on long-term trends. Short-term economic and political uncertainty may thus seriously undermine this approach.

Although this approach makes good intuitive sense, it often experiences serious shortcomings over lengthy periods of time, especially those when the equity markets are driven by everything but the long-term outlook. Looking again to the bursting of the technology bubble, many technology investors made gigantic returns on their investments in 1999, and some most likely pointed to their assessment that information technology would proliferate at a tremendous rate over the following years, and in this they were right. However, because this became such an attractive business area, thousands of companies entered into the business of information technology hardware and software, which inevitably led to a huge oversupply. A few years later many of those same companies are struggling for their very survival, and indeed many of them no longer exist. So the portfolio manager or investment strategist may have been right in the assessment that information technology would revolutionise the industrialised world, but this did not mean that everyone was going to make money. In such an instance, the portfolio manager must basically ignore the long-term theme and focus his attention on short-term issues, such as whether a particular formerly high-flying technology company can obtain an extension on its credit line, or whether it may have to declare bankruptcy next week.

A fundamental problem with the way asset management organisations are set up is that they may claim, and even work towards, an investment process that seeks returns over a fairly long investment horizon. For example, the process may be based principally on the notion that by dynamically adjusting the sector allocation of the portfolio during the business cycle, the portfolio will on average outperform over the medium to long term. However, it is more than likely that the portfolio managers and investment officers responsible for the asset allocation, and thus for the decisions about when and how to change the sector allocation, are all paid according to a *bonus* or *incentive*

scheme covering one year or less. In addition, it is customary to hold portfolio evaluation meetings at least once very month or even every week, to discuss strategy and sector allocation. Portfolio performance – and portfolio managers – are thus being evaluated many times within the stated investment horizon, and it is quite conceivable that a lot of potential outperformance is being foregone or even destroyed because the portfolio managers, in their quest for short-term as well as long-term returns and thus a higher bonus at the end of the year, periodically deviate from their long-term strategy in order to ‘chase’ short-term moves. And this chasing may be caused both by the managers’ desire to obtain bonus payments, and by pressure to outperform consistently even over very short time horizons. Even though a manager may be fundamentally very right in his evaluation and assessment of stocks or sectors relative to each other, since equity markets do not behave rationally at all times – and sometimes behave very irrationally for a very long time – the manager may nevertheless end up underperforming relative to his benchmark for a sustained period of time, which in practice is unacceptable for an asset management organisation.

Putting together an incentive scheme is never an easy thing to do: there are many issues that need to be considered, ranging from the ability of the asset management organisation to hold onto its most valuable employees, to political issues over relative salary levels within the organisation. But as a minimum, the incentive scheme ought to reflect and to be commensurate with the investment process of the asset management organisation.

What is what?

With all the different investment management products currently available, there really is no clear distinction between *non-quantitative* and *quantitative* methods. All investment professionals use numbers to some extent in their decision-making. Some use simple numbers such as valuation metrics only, others use a plethora of parameters as inputs into models, perhaps without ever questioning the validity and investment intuition behind them, and without back-testing their effectiveness.

Obviously, the most sensible approach in terms of the trade-off between time and effort probably lies somewhere between the two extreme points, and it has to be said that much of the product differentiation that can be observed in the asset management community is designed to be just that: *product differentiation*. In other words, it is not unusual for an asset management organisation to develop and implement an investment framework as much for marketing purposes and the desire to differentiate the organisation from its competitors as to add investment value. Whether an (active)

asset management organisation can *actually* add value or not is one thing, but if it cannot even convince potential clients that it *could* provide added value then it is in serious trouble.

QUANTITATIVE PORTFOLIO OPTIMISATION FOR ASSET ALLOCATION

Moving up

Although the portfolio construction and asset allocation approaches presented above are merely a small part of the possible non-quantitative techniques available, they provide a good overview of certain aspects of the way in which a number of non-quantitative asset managers operate. Because they are fairly simple, these techniques typically imply a great degree of freedom to construct portfolios and to determine appropriate asset allocations, but this freedom comes at the price of poor or non-existent risk control. It is probably fair to say that when it comes to what could broadly be called the portfolio construction process, screening and stratification techniques are probably more appropriate for stock selection than for asset allocation. This means that we need to use more advanced techniques for asset allocation.

As already demonstrated, there are potentially significant advantages in applying quantitative portfolio optimisation techniques to the portfolio construction and asset allocation process. A structured and quantitatively based approach to asset allocation, taking into account the individual relationships between asset class returns expressed by the correlation matrix and the application of optimisation algorithms, enables the asset manager to exploit these asset class return correlations in an optimal way. And as we have also pointed out, even for relatively small portfolios with a limited number of assets, it is an impossible task for the portfolio manager to grasp every relationship over time between individual asset or asset class returns. If the asset manager is to be able to take advantage of the opportunities available to her with the investable assets or asset classes, therefore, some form of quantitative method must be employed. In the remainder of this chapter we will build on a previous example to demonstrate how to carry out a complete quantitative asset allocation using portfolio optimisation.

Example with four US indices, international equities plus cash

In Chapter 6 we demonstrated how quantitative portfolio optimisation can yield an optimal weight vector that provides the highest possible expected

return for a given level of expected risk. This was extended to producing an entire efficient frontier, along which the asset manager – given a fixed set of investment alternatives – would not be able to obtain a higher expected return for particular levels of expected risk. The efficient frontier for the six-asset class portfolio including cash, the Dow Jones Industrial Average, the S&P500, the NASDAQ, the Russel 2000 and non-US equity represented by the MSCI World ex US is presented in Figure 9.1, but in this chart without special indications of this or of the equal-weighted portfolio as was the case in Figure 6.8.

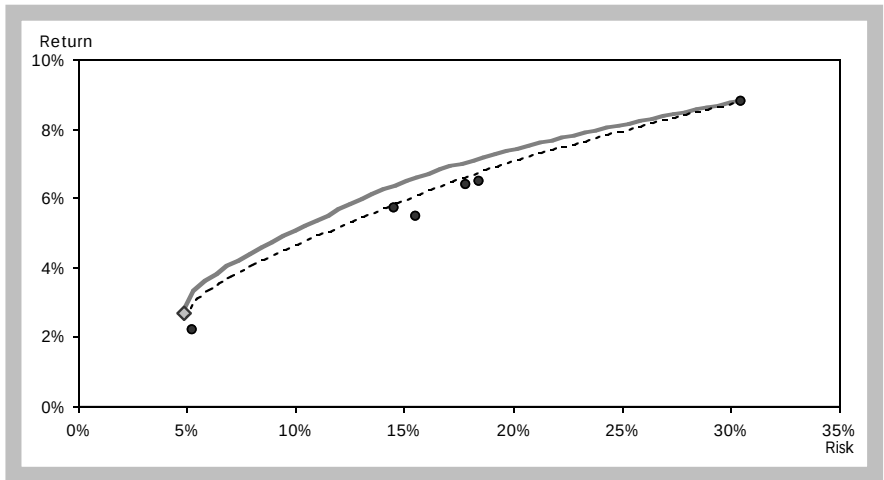


Figure 9.1 Efficient frontier – five US equity indices, MSCI-W ex US and cash

Asset Allocation Topography

As we have already demonstrated, a picture can indeed say more than a thousand words. It is possible to explain the appearance and dynamic development of the sea floor in words, but an image or even a film would be able to convey huge amounts of information almost instantly. This is true for portfolio asset allocation also. The allocation of assets with different weights in the portfolio according to their individual return and risk characteristics, combined with the complete correlations structure, is a highly complex system which cannot easily be explained with a few words. However, a graphical illustration can give a clearer picture of the portfolio's characteristics. Each point on the efficient frontier is the result of a unique portfolio produced by a unique set of asset weights. As we move along the efficient frontier from the minimum-

variance portfolio to the maximum variance portfolio, gradually increasing our desired expected level of risk, the portfolio composition changes according to asset characteristics.

In the six-asset class portfolio presented in Figure 9.1, we begin with the minimum-variance portfolio: this contains almost nothing but cash, since this is the least risky asset class. However, as was evident from Table 6.4, there are significant diversification opportunities available to the manager, so in addition to cash, the minimum-variance portfolio also contains some Dow Jones Industrial Average, some Russel 2000 and some MSCI World. More precisely, the portfolio weight vector that produces the minimum-variance portfolio appears as follows:

$$\vec{w}^* = \begin{bmatrix} 85.4\% \\ 5.1\% \\ 0.0\% \\ 0.0\% \\ 1.3\% \\ 8.2\% \end{bmatrix}$$

More than 85% of the portfolio is placed in cash, a little over 5% in the Dow Jones Industrial Average, just over 1% in the Russel 2000 and a little over 8% in the MSCI World. Note that none of the portfolio's assets are invested in the NASDAQ, since our quest is for the portfolio that provides the absolute minimum obtainable level of expected risk, regardless of expected return. But note also that the S&P500 too is left out entirely, even though its expected volatility is lower than that of the Russel 2000. Although this may seem a little surprising, it is explained by the fact that relative mainly to cash, the Russel 2000 provides a better opportunity for diversification than does the S&P500. This is also evident from the correlation matrix in Table 6.4, although the difference between the two seems to be very small.

The portfolio weight vector presented above produces an expected portfolio return of 2.71% and an expected portfolio volatility of 4.81%. Now, the interesting bit is how the optimal portfolio composition changes if we aim for a higher level of expected portfolio return or risk. Assume now that we are willing to accept a slightly higher portfolio volatility of 7.0% instead of 4.81%, as we would expect the minimum-variance portfolio to produce. Optimising the portfolio subject to this constraint yields the following optimal weight vector:

$$\vec{w}^* = \begin{bmatrix} 50.4\% \\ 4.9\% \\ 10.1\% \\ 0.0\% \\ 8.7\% \\ 25.9\% \end{bmatrix}$$

Increasing the target volatility to 7.0% thus results in an optimal portfolio that has a somewhat different composition from that of the minimum-variance portfolio. Specifically, the amount invested in cash is reduced sharply (around 15 percentage points) to 50.4%, the amount placed in the Dow Jones Industrial Average is almost unchanged, the S&P500 is now included in the portfolio with 10.1% due to our lower risk aversion, the Russel 2000 is increased significantly (around 7.5 percentage points) to 8.7%, and the MSCI World represents the biggest change with an increase of over 17 percentage points to 25.9%. This is further testament to the benefits that US – and in principle many other – investors may achieve by diversifying internationally.

The NASDAQ, however, is still excluded from the portfolio, which would probably seem a little odd to most people. Again, the technical explanation is that in terms of diversification against the major component in the asset composition, cash, the optimisation algorithm can find better alternatives, such as the MSCI World or the S&P500 at this particular level of desired expected risk. From an investment intuition standpoint, it is perhaps slightly unsettling that we are completely ignoring this particular market in our asset allocation decision. These points hint at some of the augmentations that are required in order for quantitative asset allocation to become a practical tool, and we shall introduce such tools in the following chapter.

For now, though, moving along the efficient frontier, the portfolio thus sways increasingly towards the more risky asset classes available, resulting in increasingly larger weights being assigned to these asset classes, finally resulting in a portfolio with 100% invested in the asset with the highest expected volatility and return. Assigning darker shades to higher weights and lighter shades to lower weights, the dynamic way in which the asset allocation evolves as we move along the efficient frontier can be illustrated as in Figure 9.2.

While at first glance the chart may appear difficult to interpret, it does in fact nicely present the way in which asset classes fade in and out of the optimal portfolio as we move along the efficient frontier from the minimum to the maximum-variance portfolio. As expected, the optimal portfolio composition for low target volatility portfolios consists mostly of cash, and as

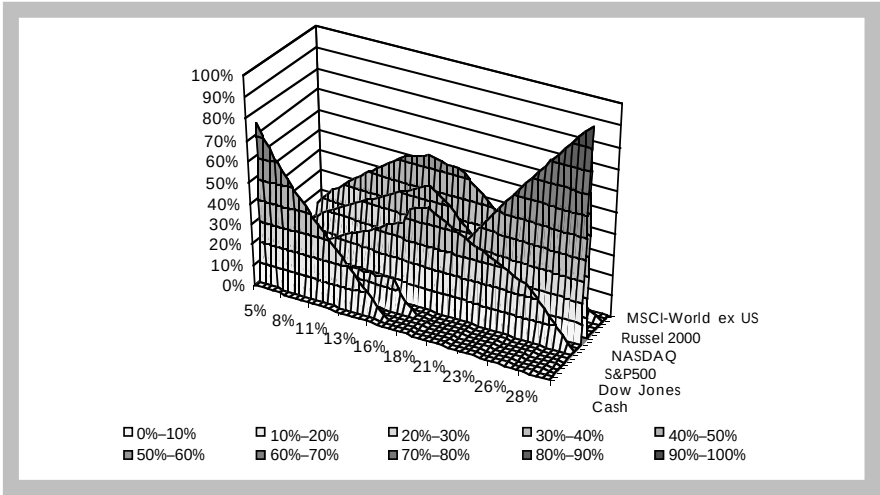


Figure 9.2 Asset allocation along the efficient frontier, 3D

we begin to increase the target volatility, more and more of the Russel 2000 and the MSCI World ex US are included in the optimal portfolios. It is also interesting to observe for how long the asset class cash stays in the optimal portfolio as we increase the desired expected level of volatility, in spite of its relatively low level of expected return. This is of course due to its very low correlation coefficient with the other asset classes. The amount of the total portfolio invested in the Russel 2000, the S&P500 and the MSCI World ex US peaks at a target volatility level of between 14% and 15%, after which it drops very quickly, only to be replaced by the NASDAQ that has the highest level of expected volatility.

Note that the Dow Jones is only included in the portfolio for a relatively short period as we move along the efficient frontier. For a target volatility level of above around 18% the optimal portfolio includes none of the Dow Jones but begins to include progressively more of the NASDAQ. As already highlighted, the significant diversification effects that are provided by the MSCI World ex US means that this asset class remains in the portfolio from start to finish as we move along the efficient frontier. The portfolio ends up with 100% placed in the NASDAQ index, as this is the most risky asset class in this rather limited investment universe.

It is interesting to note how quickly some of the asset classes enter and leave these so-called optimal portfolios. This feature can perhaps be seen more clearly in Figure 9.3, a chart of cumulative asset class weights, which essentially illustrates the same data but in a chart where each asset class is indicated by its own shade. This asset allocation chart more clearly illustrates how asset

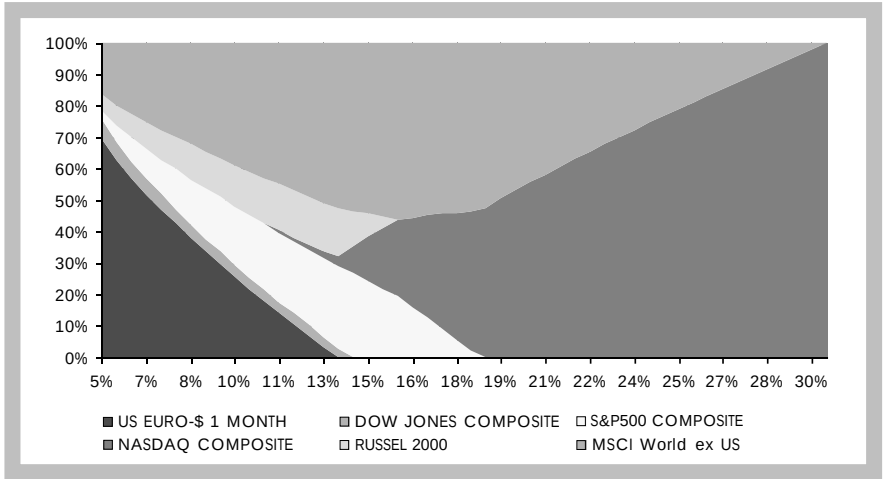


Figure 9.3 Asset allocation along the efficient frontier, cumulative percentages

classes are faded in and out, more risky asset classes replacing less risky asset classes in the portfolio as we move along the efficient frontier. But it also demonstrates the problem of swiftly changing portfolio weights as we move along the efficient frontier. Even though the fading in and out of asset classes along the efficient frontier may appear smooth at first glance, there are actually some quite puzzling things going on.

Note how quickly and suddenly the Russel 2000 is dropped from the portfolio as the target level of volatility is increased above around 14%. After building up to constituting around 15% of the entire portfolio, its share of the portfolio crashes to 0 in the space of a few percentage points of change in the target volatility level. Similarly the amount invested in the S&P500 builds up from around 0 to around 26%, and then disappears from the optimal portfolio within just a few percentage points of change in the target volatility level. Note also that this occurs just as the NASDAQ is beginning to become included in the optimal portfolio. This implies that at higher levels of expected risk, the NASDAQ is actually a superior asset class to the S&P500, given the high level of the portfolio invested in the MSCI World ex US.

Another notable feature is that the initial introduction of the NASDAQ into the optimal portfolio at a target portfolio volatility of around 11% is almost insignificant in size at first, right until the point where both cash and the Dow Jones are dropped from the optimal portfolio. At that point the amount invested in the NASDAQ shoots up at the cost of a reduction in the amount invested in the Russel 2000, because the diversification potential of the

NASDAQ is higher than that of the Russel 2000 when the portfolio cannot include either cash or the Dow Jones.

It is quite evident from this analysis that the many relations between asset class returns defined by the correlation matrix are much too complex for an asset manager to be able to grasp and apply, or even to predict ex post. The reasons for an asset class being included at a certain level of expected risk, and the causality behind the manner in which it fades in and out of the optimal portfolios along the efficient frontier, can be explained if one makes an effort to do so. But the structure of even small portfolios such as this with only six asset classes is much too complex to be managed without the use of quantitative techniques.

Instability of asset allocation results

As already noted, from Figure 9.3 we observe the violent changes to the composition of the optimal portfolios as we move along the efficient frontier from the minimum-variance portfolio to the maximum-variance portfolio. Such radical changes to the portfolio composition as we change the target volatility level only slightly would entail huge transaction costs. More importantly from a client relationship standpoint, however, it would be difficult for the asset manager to explain to the client why the long-term strategic asset allocation recommendation should change this much in the event that the client decides to seek a slightly higher level of volatility.

In fact, this point is probably most of the reason why institutional asset managers in the past have shied away from using quantitative optimisation techniques for asset allocation purposes. Instead they have stuck with more traditional rules of thumb, such as 60%/40% in equity and fixed income respectively for investors with a higher risk tolerance, and 40%/60% respectively for investors with a lower risk tolerance. The two options just mentioned are of course extremely simplistic, and most asset managers will have relatively sophisticated asset allocation tools at their disposal, but these will quite often be more qualitative than quantitative in nature. That is, the asset managers may well employ relatively sophisticated techniques for assessing equity markets in terms of valuations, momentum and the like, but for the actual asset allocation decision they will more often than not rely almost exclusively on the ability of the manager to fully understand asset class relations and dynamics. This is a task that we have demonstrated to be highly challenging at best, not least because of the very complex correlation structure that exists between asset class returns over time.

Using more sophisticated quantitative approaches often brings with it severe frustrations about correct specifications of optimisation models,

algorithms and input parameters, and even then there seems to be no guarantee that optimal portfolios will not exhibit these very erratic changes in composition. Thus, relying solely on the asset allocation suggested by the quantitative optimisation algorithm – even in the case where we are able to be reasonably successful in predicting asset class returns, correlations and volatilities – would probably make most asset managers uncomfortable. This is mainly because asset classes tend to move in and out of the optimal portfolio suddenly, and often without good investment intuition. In addition, several of the available asset classes are dropped from the optimal portfolio altogether at a very early stage as we move along the efficient frontier and increase the desired expected level of portfolio volatility. This illustrates very forcefully the point that in its most basic form the optimisation algorithm may well be too powerful for the level of information available to it.

Just when the asset manager might benefit the most from employing quantitative techniques, therefore, he shies away from them because the traditional quantitative asset allocation models that exist today have severe limitations and shortcomings, as we have already demonstrated.

Opportunities for augmentation of models and inputs

The quantitative portfolio optimisation and asset allocation technique presented so far, in this and previous chapters, is valid in principle but may be seriously flawed in practice. More precisely, the optimisation actually delivers optimal asset allocation recommendations based on the input parameters – returns, volatilities and correlations – but if the inputs are not perfect forecasts of the future, the recommended portfolios may end up causing more harm than good. In fact, if all model parameters were estimated or forecast correctly so that they were perfect predictors of the future, the asset manager employing quantitative optimisation would then hold truly optimal portfolios and would thus be able to squeeze every last drop of performance from the available assets. The problem lies not in the optimisation procedure itself, therefore, but rather in the quality of input parameters and the fact that the optimisation algorithm is too powerful for the quality of information available. Input parameters are inherently inaccurate because there is – as yet – no way of perfectly foretelling the future. Input parameters are in most cases simply historical averages of observed returns, volatilities and correlations. At other times they may be exponentially moving averages, or forecasts of future realisations of model parameters using techniques such as those briefly outlined in Chapter 7.

So from this point on it would seem that we should devote some effort to the more appropriate – that is, the more workable – specification of the opti-

misation model and algorithm, and to the process of generating model inputs that can provide us with smoother, less erratic, and more dependable asset allocation results. Monte Carlo Simulation offers such a tool, and we shall investigate that in the next chapter, along with input augmentation techniques, ultimately ending up with a very rigorous and practical asset allocation methodology that is both intuitive and practical. However, before we do so we introduce a new investment universe that we will employ in the analysis carried out throughout the remainder of this book.

INTRODUCING AN MSCI GLOBAL SECTOR MODEL

To illustrate further the points made above concerning seemingly erratic and non-intuitive changes to optimal portfolio composition along the efficient frontier, and in order to enhance investment intuition, we now change our investment universe slightly. Instead of a universe consisting mainly of broad US equity indices, we turn to a more realistic *Global Sector Model* based on the sector indices from the MSCI World data series. In a practical investment context such an investment universe is probably more appropriate, since it more accurately reflects the manner in which global portfolios are managed today.

With increasing globalisation and integration of sectors across countries, as well as closer currency ties within geo-economic areas such as in the European Union, more and more asset managers are currently moving from a regional approach to a global sector approach. So instead of having regional equity teams, each analysing the auto sector, the oil sector, the retail sector, and so forth, an increasing number of asset management organisations have set up global sector teams that focus primarily on specific global sectors. The emphasis has thus shifted from regional allocations of funds to sector allocations.

Whether this is universally the best way to proceed for the asset management industry remains to be seen, since the assumption of fully integrated global markets may be a bit ahead of its time. The theory of the global sector approach is that it will provide a more efficient management process, because – from a business model point of view – auto companies in Japan are assumed to be fundamentally similar to auto companies in the US, in the same way that power companies in the US are assumed to be fundamentally similar to power companies in Europe. It probably requires highly experienced portfolio managers in order to succeed with such a framework, since regional differences in company management traditions, business models and economic fundamentals still remain. It is therefore likely that significant experience is needed to be able to spot the differences and subsequently to determine what impact they may have on an optimal portfolio of stocks within a particular global sector.

Table 9.1 MSCI Global Sector Model data

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	9.7%	17.1%	1.00	0.64	0.60	0.41	0.36	0.31	0.55	0.33	0.15	0.46
Materials	10.3%	17.2%	0.64	1.00	0.81	0.70	0.50	0.36	0.70	0.44	0.25	0.28
Industrials	11.6%	15.2%	0.60	0.81	1.00	0.87	0.57	0.47	0.83	0.66	0.48	0.46
Cons. Disc.	12.1%	16.8%	0.41	0.70	0.87	1.00	0.40	0.35	0.73	0.78	0.67	0.28
Cons. Staples	9.7%	13.1%	0.36	0.50	0.57	0.40	1.00	0.67	0.68	0.16	0.22	0.55
Health Care	11.0%	14.6%	0.31	0.36	0.47	0.35	0.67	1.00	0.60	0.23	0.30	0.44
Financials	12.0%	18.9%	0.55	0.70	0.83	0.73	0.68	0.60	1.00	0.47	0.42	0.50
IT	16.0%	33.0%	0.33	0.44	0.66	0.78	0.16	0.23	0.47	1.00	0.70	0.13
Telecomm.	13.2%	19.6%	0.15	0.25	0.48	0.67	0.22	0.30	0.42	0.70	1.00	0.18
Utilities	7.5%	10.5%	0.46	0.28	0.46	0.28	0.55	0.44	0.50	0.13	0.18	1.00

However, from an overall asset allocation perspective, it is quite conceivable that a global sector approach may add value, simply because on the aggregate – that is, at a non-stock-specific level – there seem to be fairly consistent patterns in the way global equity sectors tend to move relative to each other. It is with that in mind that we now introduce our Global Sector Model. The Global Sector Model presented here is thus meant to reflect the current trend in the asset management industry, and hopefully it provides for more immediate investment relevance and intuition.

Model inputs are estimated using monthly return data on the ten MSCI Global Sector indices from 1 January 1995 to 1 May 2002. This actually leaves us with a relatively poor data set from which to estimate model inputs, at least in terms of statistical significance. However, this fact also allows us to demonstrate estimation error correction techniques in the following chapters. For now we begin by introducing the MSCI Global Sector Model. In Table 9.1 we present the model parameters that we have estimated for the model.

Volatilities and correlations are historically observed values. However, expected returns are more or less arbitrarily chosen for expositional purposes. Whether these expected returns are correct forecasts of the actual future returns is not important. What we are interested in here is to present a framework with which we can then proceed for the remainder of this book.

As is evident from Table 9.1, there are quite significant differences between the observed levels of volatility across the ten MSCI sectors. During the period over which correlations and volatilities have been estimated, information technology has been the most risky asset class by far, with a volatility level of 33%. But financials have also been quite risky, with a volatility level of 18.9%. The least risky asset class has been utilities, with a volatility level as low as 10.5%. This is very low, even for an index which itself can be seen as a portfolio of stocks with the benefits of diversification that this implies. Figure 9.4 illustrates the correlation matrix for this investment universe.

It is quite clear that there exist significant opportunities for diversification of risk between the global sectors. Several of the correlation coefficients are below 0.4. Usually the normal band for correlation coefficients between individual equities or equity indices ranges from 0.4 to 0.6, so the fact that several coefficients are below that level bodes well for the potential for structured portfolio risk control using quantitative portfolio optimisation. The raw data presented in Table 9.1 yield the efficient frontier shown in Figure 9.5.

As before, the minimum-variance portfolio and its constituents are marked with a grey diamond and with grey dots respectively. Again the star illustrates the position of the equal-weighted portfolio in the return/risk space. The asset allocation chart for the MSCI Global Sector Model appears in Figure 9.6.

Examining Figure 9.6, a familiar picture emerges. While the asset allocation recommendations implied by the chart may look sensible at first glance, they

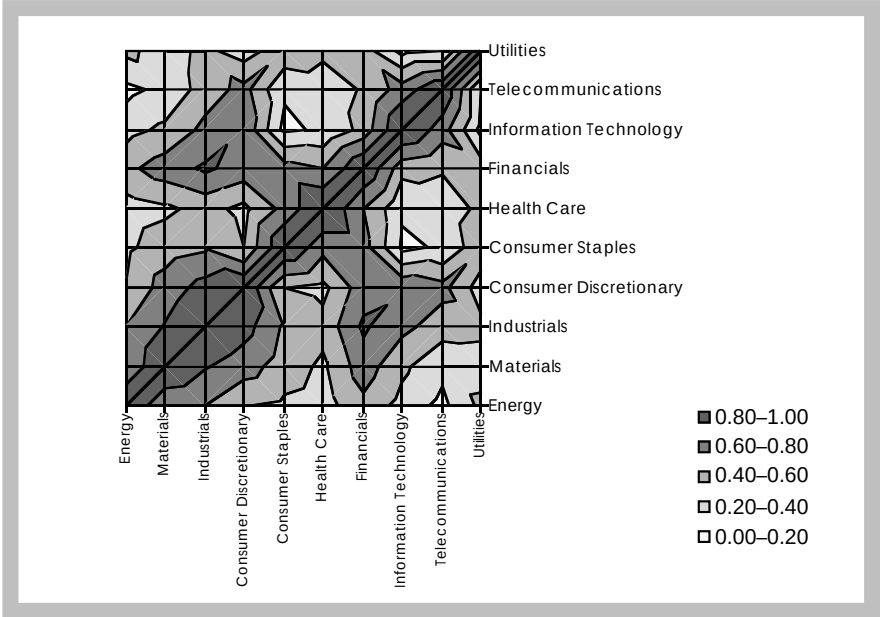


Figure 9.4 Correlation matrix – MSCI Global Sector Model

are in fact quite non-intuitive. Note how assets are most often included in the portfolio only for very small intervals in the risk spectrum from the minimum to the maximum-variance portfolio. Note how this usually occurs only very

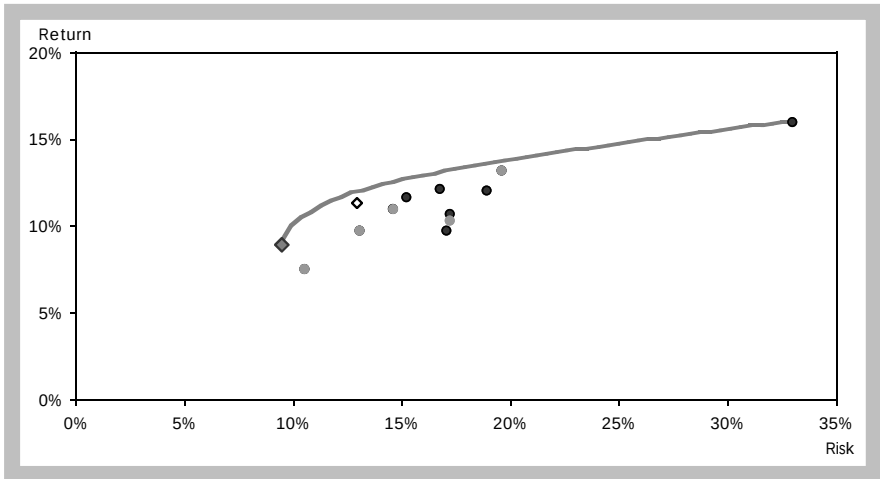


Figure 9.5 The efficient frontier – MSCI Global Sector Model

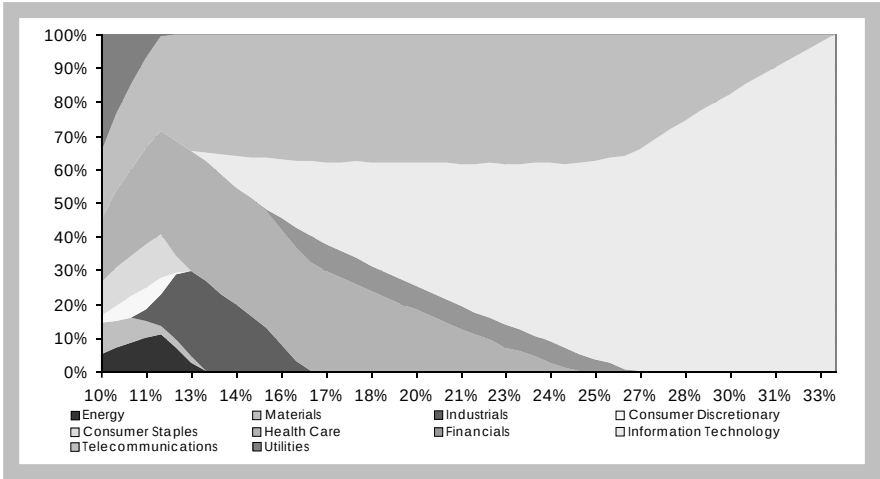


Figure 9.6 Asset allocation topography along the efficient frontier

close to the minimum-variance portfolio, how sectors tend to get dropped very suddenly from the portfolio, and how the so-called optimal portfolios are completely dominated by information technology and telecommunications once we move about a third along the risk spectrum towards the maximum-variance portfolio. In other words, the diversification opportunities available in the portfolio do not get exploited for just moderately high or even medium target volatilities. Once again, an asset allocation scheme such as that suggested in Figure 9.6 is highly non-intuitive, and is unlikely to be implemented by investment practitioners.

In terms of expected portfolio performance, how does return measure up against risk along the efficient frontier? The well-known Sharpe Ratio measures annual return divided by the annualised standard deviation or volatility of returns. For the MSCI Global Sector Model we can easily calculate these ratios along the efficient frontier, since each point on the efficient frontier represents an asset allocation or portfolio composition that is optimal in the sense that it maximises return for a given level of risk. The Sharpe Ratio for the asset allocation chart presented above appears as in Figure 9.7.

This indicates that the maximum Sharpe Ratio is found very close to the minimum-variance portfolio at a target volatility level of approximately 10%. Increasing the target volatility level beyond this level will result in an increase in expected return, but at a proportionately higher level of expected volatility. In other words the marginal benefit in terms of return versus risk, as measured by the Sharpe Ratio from increasing the target volatility level, is actually negative beyond this point. What the figure also indicates is that an asset

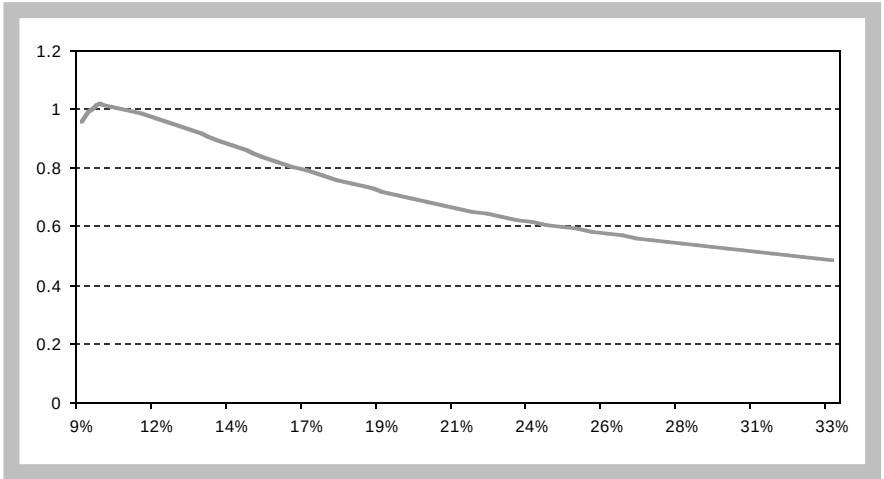


Figure 9.7 Expected Sharpe Ratio along the efficient frontier

manager or a client can quite easily change the portfolio in the pursuit of higher returns, but this will come at the price of higher risk, and disproportionately so. Increasing the expected return by 1 percentage point almost universally causes expected risk to increase by more than one percentage point, and the ratio between the two becomes less and less favourable for the asset managers the higher the desired level of expected risk.

We will build on this MSCI Global Sector Model in the following chapter, where we attempt to augment the model and to create a methodology that is more applicable in practice.

SUMMARY

In this chapter we have introduced quantitative asset allocation based on the quantitative portfolio optimisation techniques presented in previous chapters. As shown, and as experienced many times by practitioners, the results of the quantitative asset allocation process are often highly non-intuitive and unstable, thereby reducing the investment value for the asset manager both in terms of applicability in the investment process and in terms of using the asset allocation results for marketing of investment services.

However, a few important points emerge from the analysis of the application of quantitative portfolio optimisation to asset allocation. In its most basic form, an optimiser is not a delicate instrument that – based on precise input data – is capable of delivering a sophisticated analysis of, and

even recommendations on, how to allocate assets in a portfolio in an optimal way. Optimisers apply brute force to squeeze a solution out of the available data, without regard for investment intuition. In fact, the word 'optimiser' and the notion of 'optimised portfolios' may well have misled many an asset manager into thinking that his portfolio was somehow better than every other portfolio *because* it had been optimised. This is a false and potentially dangerous misperception. Quantitative optimisers and the mathematical algorithms that lie behind them are often much too powerful for the quality of the input provided for the optimisation. This results in extreme solutions that are often lacking in intuition and practical applicability.

Because of the frustrations experienced over the use of optimisers and their at times nonsensical results and recommendations, investment professionals have either shied away from quantitative optimisation all together, or alternatively placed so many constraints on the optimisation as to seriously limit the potential of the whole process. The latter has often been called the 'torturing of data' approach, the point being that if you torture the data, the data will eventually confess and tell you what you want to hear – that is, the optimiser will eventually produce the asset allocation that you find attractive in your own mind. Of course, this leaves quantitative optimisation virtually impotent, since investment managers will merely be told what they already know, right or wrong.

However, not all hope should be given up. There are ways of augmenting and refining the asset allocation process in order to avoid the instability of optimisation results that is often experienced by asset managers. This can be done without torturing the data and without imposing constraints that will ultimately eliminate the advantage that quantitative portfolio optimisation can provide. In the next chapter we will present a Monte Carlo Simulation-based asset allocation technique that is practical as well as intuitive, without eliminating the inherent benefits of quantitative portfolio optimisation.

QUASI-RANDOM MONTE CARLO SIMULATED ASSET ALLOCATION (QRMCSAA)

QUANTITATIVE OPTIMISATION AND MONTE CARLO SIMULATIONS

The nature of optimisation

As already pointed out in the previous chapter, the results of a quant-based allocation of assets can become quite erratic with large changes in asset weights for just small changes in the required portfolio risk. This is especially true for portfolios with many assets. The reason for this is that quantitative optimisation is essentially an 'error maximiser'.¹ This is due to the fact that the nature of optimisation is to search for extremes. A quantitative optimiser will search within given constraints for the optimal solution to the optimisation problem, without regard for investment intuition or common sense. It knows only those constraints that have been given to it by the asset manager, and within those it will find an extreme solution maximising return and/or minimising risk.

One might argue that the mathematical optimisation algorithm is simply too powerful for the level or quality of information fed into the procedure. A way to augment this is to impose a large number of constraints, but the price

of this is poorer performance in terms of both risk and return. For every new constraint on the optimisation problem, the potency of the algorithm is reduced, ultimately eliminating altogether the advantage of using quantitative optimisation. Additionally the search for extremes most often results in highly unstable asset allocation suggestions from portfolio optimisers. Asset managers who have attempted to produce a workable solution to quantitative portfolio optimisation have mostly become frustrated with the results, since these often proved both wildly unstable, non-intuitive at best, and investment nonsense at worst. In fact these reasons have historically resulted in asset managers simply ignoring the potential of quantitative return/risk/correlation portfolio optimisation, and have stuck to rules of thumb regarding so-called *conservative*, *neutral*, *aggressive* and *very aggressive* portfolios, with more or less predetermined asset allocations corresponding to these very broad definitions of investor risk-aversion. Usually asset managers have given up trying to optimise portfolios using expected return and risk. In addition, taking into account the restrictions on holdings stipulated by client mandates, asset managers have quite possibly missed out on significant performance both in terms of portfolio return and portfolio risk control.

Questioning optimality

As illustrated in the previous chapters, the power of optimisation algorithms, combined with the relatively low quality and reliability of estimated or forecast model inputs, can create erratic changes in recommended portfolio weights, even for relatively small changes in model inputs. As also demonstrated in the previous chapter, portfolio weights may change dramatically due to a slight change in the required level of portfolio return or risk. This means that 'optimal' portfolios may actually not be as optimal as we would like them to be. Furthermore, the nature of model inputs is that they vary over time. This implies that an expected asset class return of 10% could easily turn out in practice to be 15% or 0%. This in turn implies that optimality is 'fuzzy' and consequently that the optimised portfolios are in fact not optimal. They are more accurately described as 'guesstimates' based on the available information at the present time, and since these are not certain predictors of the future they must have variance to account for their uncertainty. An optimised portfolio is thus more like a probability distribution, and not like a single point in return/risk space. We will illustrate this below.

The remainder of this chapter will focus on various augmentations of the basic asset allocation model so far. These will prove useful tools in alleviating many of the problems that 'naive' optimisation entails, principally the erratic changes in portfolio weights as well as the 'fuzzyness' of optimised portfolios

discussed above. Having introduced these augmentations to the asset allocation model, we will then introduce methods to adjust model inputs to account for estimation error. This will yield more useful model inputs, which can further enhance the value of the quantitative portfolio optimisation and asset allocation methodology presented in this book. Since the asset allocation technique presented here relies heavily on Monte Carlo Simulation, we will begin by introducing this technique.

An introduction to Monte Carlo Simulation

The instability and high sensitivity of optimisation results that has led many asset managers to cast away quantitative portfolio optimisation can be augmented by adding another layer of quantitative methodology in the form of Monte Carlo Simulation. The name Monte Carlo alludes to the nature of the simulation procedure, which, in essence, involves drawing random numbers from a distribution, and then using the random numbers as inputs for a mathematical process, in this case portfolio optimisation. The simulation technique is particularly useful when several inputs with different characteristics are employed simultaneously in a model. Each input can be specified in terms of expected values, variation around the expectation, and even the shape of the distribution from which random numbers are drawn. It can in principle be applied to any problem that involves some element of uncertainty, and thus comes in very handy for simulating portfolio and asset class return.

Monte Carlo Simulated portfolio returns

In the case of quantitative portfolio optimisation, the Monte Carlo Simulation technique is particularly useful because it can potentially encompass a very large number of assets or asset classes, each with its own specific characteristics which cannot be described by a single number (as is the case with standard deviation or volatility, for example). The return on each asset class is simulated according to a probability distribution of returns, preferably determined by examining the actual empirical distributions of assets, or asset class returns over a sufficiently long historical period. Probability distributions can thus in principle take any form, including forms that may not be particularly easy to work with in explicit mathematical terms.

The procedure runs as follows. Single return values from each asset class return distribution are drawn randomly from their respective probability distributions. Given these simulated return values and the target volatility

level, the portfolio is then quantitatively optimised to yield an optimal portfolio for this particular simulated vector of expected returns. A simple Monte Carlo Simulation of portfolio return can be illustrated as in Figure 10.1, which illustrates how each simulated asset class return is simultaneously drawn from its own unique and individually specified probability distribution. Based on each round of random draws, the portfolio is then optimised, yielding a weight vector that produces an expected portfolio return and an expected portfolio risk level. This process is repeated many times, in accordance with the specified characteristics of each of the asset classes. This yields a distribution of optimised portfolio returns. For each simulated return vector there exists an optimal portfolio and thus an optimal asset allocation. This is of course true for all the points along the efficient frontier. In Figure 10.1 only three return distributions have been specified, but of course one exists for every asset class in the portfolio. Note how the return on asset class N is determined by values drawn from a skewed distribution, while the returns on assets 1 and n are drawn from normal distributions. This demonstrates the relative ease with which Monte Carlo Simulation can introduce asset return characteristics that cannot be expressed accurately in terms only of mean and variance, or volatility. We may wish to introduce both skew and kurtosis in the asset class characterisation, and this is easily done with Monte Carlo Simulation. In order to fully realise the potential of optimality expressed in terms of distributions of returns in an asset allocation context we will now introduce the *efficient ridge*.

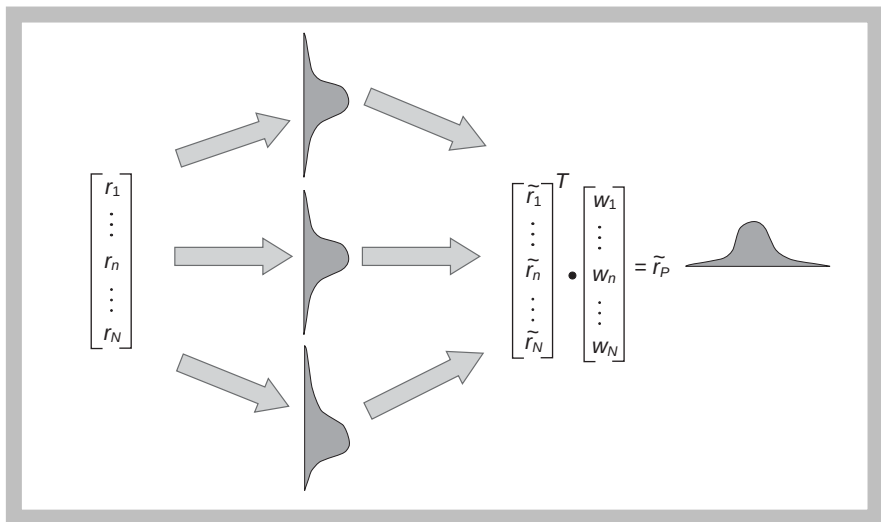


Figure 10.1 Monte Carlo Simulation of portfolio returns

THE EFFICIENT RIDGE

What was the efficient frontier, again?

The efficient frontier is the range of return/risk optimal portfolios that span from the minimum-variance portfolio to the maximum-variance portfolio. Along the efficient frontier we cannot obtain a higher return at a lower risk, or a lower risk at a higher return. But the efficient frontier is efficient only in the sense that it is optimal if inputs are perfect forecasts of the future. If model inputs are imperfect forecasts of the future, the optimised portfolios will not be optimal at all. And since crystal balls are not readily available off the shelves in hardware stores, asset managers have to make do with less than perfect estimates of future states of the economy, and in particular with inherently uncertain estimates of future stock market performance. This uncertainty translates into variability of the assets or asset classes that constitute a given portfolio's investment universe. Since every asset return entails some level of variability, it follows that the portfolio return is also variable, which in turn means that the efficient frontier as defined above is variable, and thus in fact not a line plotting optimal portfolios, but rather an 'area' of probable portfolio returns, each corresponding to a level of risk. The major insight here is that optimality is in fact an uncertain entity – and in a strict sense it does not in fact exist, as the ex ante optimised portfolios virtually always turn out to be ex post sub-optimal. Optimality in a portfolio context is an elusive quality that can best be expressed in terms of probability distributions, since these are the most accurate measures of the uncertainty with which we are faced. Optimality thus depends on the distributional characteristics of the returns on the assets or asset classes included in the portfolio's investment universe.

Characterising return distributions

Every asset class is characterised by an expected return, a volatility and a set of correlations with other asset classes. But that is not all. Each asset class's returns can be described more accurately by their distributions around their respective means. This is significant because, contrary to common wisdom (and practice), asset returns are *not* normally distributed. As we have illustrated previously, they exhibit skewness and kurtosis. This has implications for the way in which asset managers ought to think about efficient portfolios and quantitative asset allocation. Most analysis of risk in an asset management context is based on the assumption that returns are normally distributed: it follows that if this is not the case, the conclusions may be inaccurate at best, and incorrect at worst. We shall examine this point in this and subsequent chapters.

Let us return to the MSCI Global Sector Model presented in the previous chapter. The minimum-variance portfolio is the point in return/risk space that exhibits the lowest possible amount of risk attainable, regardless of the desired level of return. This means that it is the unique quantitative solution to the problem solved in Chapter 6, where an optimum was found for that particular portfolio. However, as already indicated this may not be as accurate as we would like to think, since the assets that comprise the portfolio are not fixed points in return/risk space. So the expected return and risk of each asset class are uncertain, which is why it is most accurately described, not by a single point, but by a probability distribution in return/risk space. In the MSCI Global Sector Model presented in the previous chapter, each asset class has been specified in terms of an expected return, an expected risk, and a set of correlations with the other asset classes. This was sufficient to determine the minimum-variance portfolio as well as all the points on the efficient frontier. We now add an extra dimension in the form of asset class return distributions. Based on historically observed return distributions for the ten sector indices, we assume and include into the model the set of return distributions shown in Figure 10.2.

Information technology and telecommunications are, not surprisingly, the most volatile sectors, as can be seen from the relatively flat probability distributions. Materials and utilities on the other hand exhibit relatively little volatility. We shall use these sector return probability distributions as inputs for the Monte Carlo Simulation, but it is important to keep in mind that the expected probability distributions can be based on historically observed frequency distributions as well as more or less subjective expectations about future distributions of returns. If the asset manager for some reason thinks he has special insight into the future distribution of returns, it is very easy to apply those expectations to the model by simply changing the probability distribution from which the Monte Carlo Simulation draws its expected sector return values. This is one of the strengths of the simulation procedure. The asset manager is able to specify model inputs not just in terms of returns, volatilities and correlations, but also in terms of the shape of the probability distribution, potentially yielding a much more accurate picture of portfolio risk. This is particularly true compared to the conventional approach of using normal distributions to characterise returns that are in fact not normally distributed.

The process of selecting or estimating appropriate probability distributions is completely up to the asset or risk manager responsible for assessing risk in the allocation context. It is thus clear that this choice is to some extent arbitrary, which means that a decision needs to be made by the asset manager – a decision that will inevitably have an effect on the asset allocation result. This is, however, clearly more appropriate than simply assuming or even pretending that returns are normally distributed, which everyone knows is

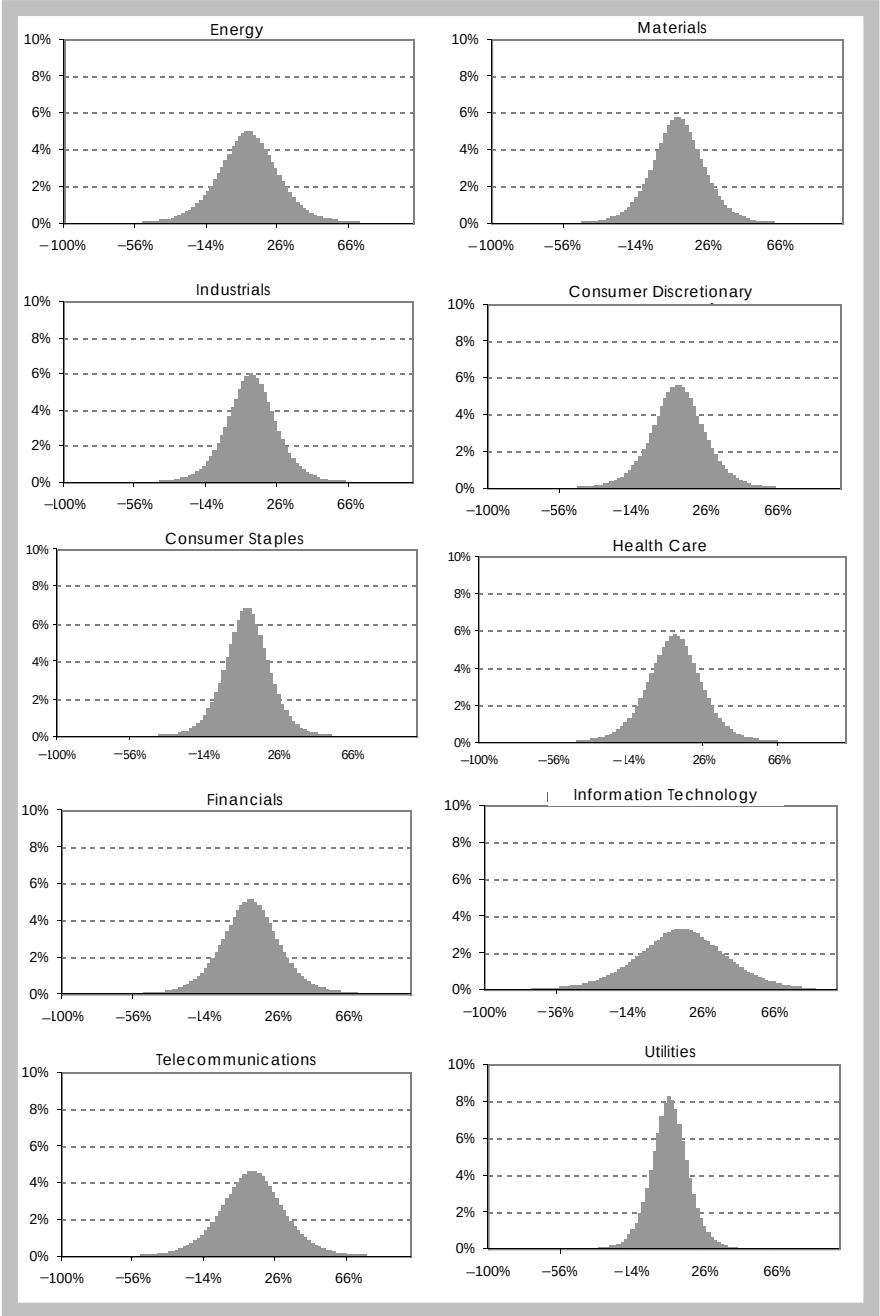


Figure 10.2 Return distributions for 10 MSCI
global sectors, 1995/1–2002/5

inaccurate at best. This way the asset manager is forced to make an informed decision on a point which may potentially be quite important for the asset allocation results. Naturally, the expected sector return distributions have an impact on the efficient frontier.

Let us first re-examine the minimum-variance portfolio. Given that a return probability distribution has been specified for each asset class, we can now run a Monte Carlo Simulation of asset class returns, in order to investigate the exact shape of various portfolios, including the minimum-variance portfolio. The minimum-variance portfolio, expressed not in terms of a single non-variable point in return/risk space but in terms of an entire probability distribution, appears as in Figure 10.3

The return distribution characterising the expectation of the returns on the minimum-variance portfolio is very ‘narrow’ – that is, concentrated around the mean, which is a result of the minimum-variance portfolio containing a large proportion of the low-risk sectors such as utilities and materials. Although the distribution appears to be *jagged*, it is in fact entirely smooth. The jaggedness in Figure 10.3 is a result of discrete frequency counting intervals. The exact sector composition of the minimum-variance portfolio is shown in Table 10.1.

The table clearly shows that most of the assets in the portfolio have been allocated to the utilities sector. The materials sector is also a relatively low-risk sector, which is why a significant proportion of the total portfolio has been allocated to it. However, the telecommunications sector – despite its relatively high level over expected risk – has been allocated 12% of the portfolio. This is

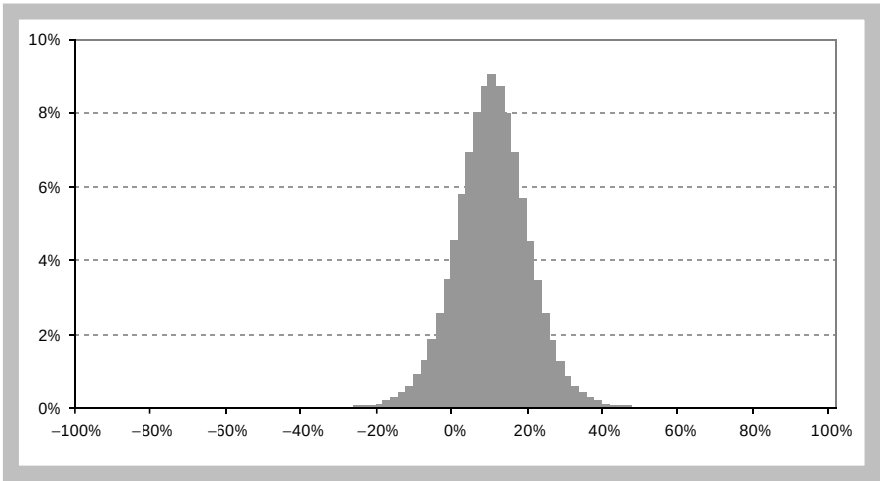


Figure 10.3 Return distribution, minimum-variance portfolio

Table 10.1 Sector composition, minimum-variance portfolio

Optimal Weights	
Energy	0.0%
Materials	10.6%
Industrials	0.0%
Consumer Discretionary	0.0%
Consumer Staples	7.3%
Health Care	8.4%
Financials	0.0%
Information Technology	0.0%
Telecommunications	12.0%
Utilities	61.6%

because the returns on this sector have historically exhibited a low correlation with returns on utilities. One can think of the return distribution illustrated in Figure 10.3 as an average of asset class return probability distributions, weighted in accordance with the minimum-variance portfolio’s weights of the respective asset classes. But the weighting process is in fact carried out indirectly by the Monte Carlo Simulation. This is one of the benefits of using this type of technique: no mathematical weighting scheme is required. The weighting process does not require a complicated mathematical exercise that describes each asset class in terms of return, variance, skewness and kurtosis. It simply runs simulations of returns a sufficiently large number of times to create a well-defined aggregate or average probability distribution. Note that the probability distribution of returns on the minimum-variance portfolio is almost identical to the distribution for the utilities asset class, and it is thus virtually normally distributed. This is because the vast majority of the portfolio has been placed in utilities, which exhibit normally distributed returns over time.

The probability distribution for the middle variance portfolio appears as in Figure 10.4. Notice that the return probability distribution is now considerably wider than before, since the expected volatility has increased, with larger proportions of the portfolio now being allocated to asset classes with higher expected volatility, and less being placed in the less volatile sectors. The exact portfolio composition is shown in Table 10.2. The utilities sector has now been dropped completely from the optimised portfolio, and instead telecommunications as well as information technology now feature prominently, supple-

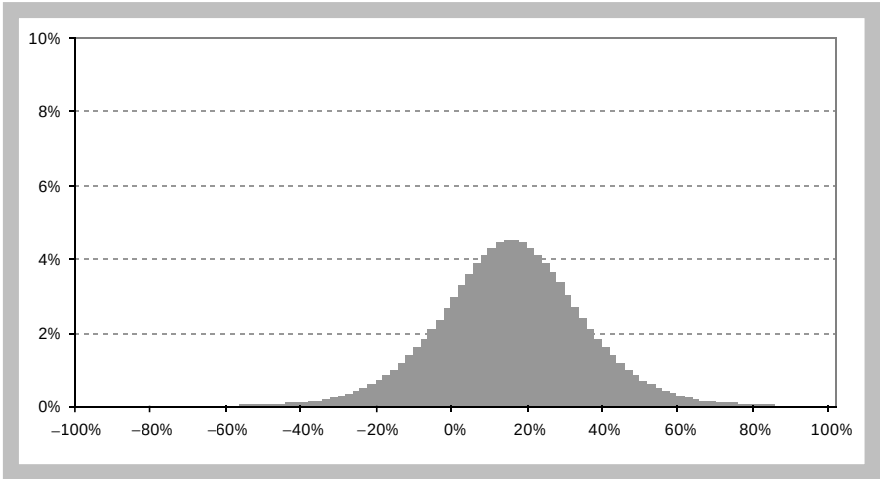


Figure 10.4 Return distribution, middle-variance portfolio

Table 10.2 Sector composition, middle-variance portfolio

	Optimal Weights
Energy	0.0%
Materials	0.0%
Industrials	0.0%
Consumer Discretionary	0.0%
Consumer Staples	0.0%
Health Care	10.8%
Financials	6.7%
Information Technology	44.0%
Telecommunications	38.4%
Utilities	0.0%

mented by the health care and financials sectors, which historically have low return correlation coefficients with the telecommunications and information technology sectors.

Lastly, we establish the expected return probability distribution for the maximum-variance portfolio. This is simple, as this portfolio consists only of the information technology sector – see Figure 10.5. Note how much flatter the

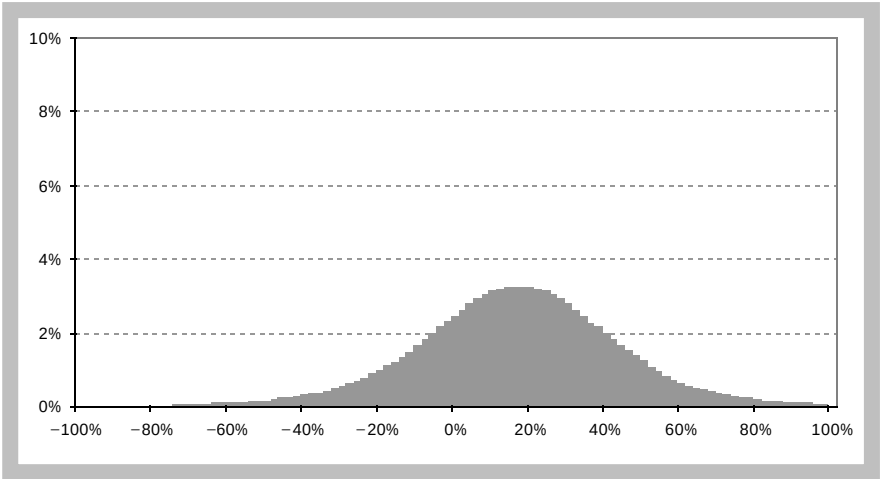


Figure 10.5 Return distribution, maximum-variance portfolio

distribution is compared to those of the minimum- and middle-variance portfolios, as portfolio returns many standard deviations from the mean are now much more likely than was the case for either the minimum- or the middle-variance portfolios. The exact portfolio composition is simple, as shown in Table 10.3.

Table 10.3 Sector composition, maximum-variance portfolio

Optimal Weights	
Energy	0.0%
Materials	0.0%
Industrials	0.0%
Consumer Discretionary	0.0%
Consumer Staples	0.0%
Health Care	0.0%
Financials	0.0%
Information Technology	100.0%
Telecommunications	0.0%
Utilities	0.0%

Clearly a similar return characterisation can be carried out for every optimal portfolio, and so we now move on to examining all the expected sector return probability distributions along the efficient frontier.

The efficient ridge

In order to broaden the analysis of return distributions in an asset allocation context, we now demonstrate how the simulation of expected sector returns for a portfolio with a particular level of desired risk can be extended to include all the points along the efficient frontier.

Reverting briefly to the asset allocation chart presented in Figure 9.6, it is clear that the composition of the optimal portfolio changes as we move along the efficient frontier. Since portfolio asset classes are characterised by different levels of expected return, volatility and by the distribution of their returns, it follows that the portfolio's returns must also change characteristics as we move along the efficient frontier. Asset classes that are less risky tend not only to have expected volatilities that are actually lower, but they tend also to exhibit returns closer to the normal distributions. Furthermore, assets that are more risky tend to exhibit higher degrees of skewness and kurtosis, although the emphasis on these two 'extra' characteristics varies from one asset class to another. So as we move along the efficient frontier, not only does the expected volatility of portfolio returns change, but the entire *distribution* of returns changes characteristics, going from nearly normally distributed to exhibiting skewness and kurtosis. However, as can be seen from Figure 10.2, all asset classes – in this case all ten MSCI World Global Sectors – are non-normally distributed. Specifically, when examining the distribution one finds that a so-called *logistic distribution* – that is, one that is heavily kurtotic – provides the best fit with the actually observed returns over the estimation period. The analysis that follows in this and subsequent chapters is therefore based on the estimated return distributions presented in Figure 10.2. Consider again the efficient frontier presented in Figure 9.5 and derived from the sector characterisation presented in Table 9.1, this time without special indications of either the minimum-variance or the equal-weighted portfolios – see Figure 10.6. We noted above that the fact that each asset class has variance, and as a consequence is best described by a probability distribution, implies that every point on the efficient frontier also has variance. Giving the efficient frontier variance and considering the distributional characteristics of the individual asset classes and of the portfolio, we can transform the efficient frontier into what might more appropriately be called an *efficient ridge*.²

The efficient ridge is determined by simulating a very large number – often several thousands of years' worth – of individual asset class return data, and

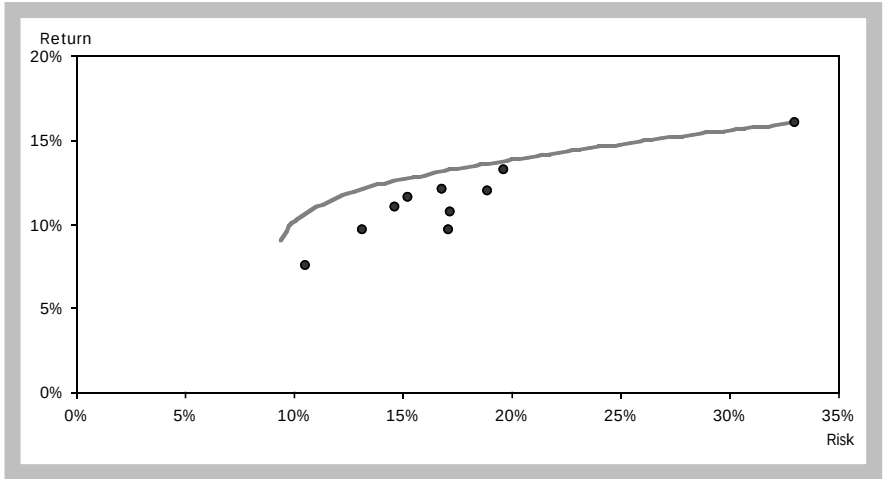


Figure 10.6 The efficient frontier – MSCI Global Sector Model

then optimising the portfolio to generate the efficient frontier. For each level of expected risk, we record the resultant level of expected return from the optimised portfolio and note the distribution of simulated returns. In fact we do not necessarily need to generate thousands of years of simulated portfolio returns as the result is statistically significant after a few hundred simulations, but in order to achieve smooth return distributions and thereby to generate a smooth efficient ridge we run a considerably higher number of simulations. This appears as in Figure 10.7.

This chart illustrates in three dimensions the expected return probability distributions along the efficient frontier for different volatilities along the entire risk spectrum. Darker shades indicate higher probabilities. Note that as we move from the minimum-variance portfolio along the risk spectrum to the maximum-variance portfolio, the return distribution of the optimised portfolio becomes 'wider' as expected volatility increases. As a result of the characterisation of sector returns whereby each sector is described by a logistic probability distribution, the efficient ridge exhibits only kurtosis and no skew.

In tune with capital market theory, this also entails a higher level of expected return. Figure 10.7 is designed to present the argument that the efficient frontier is in fact not a narrowly defined frontier as such. It is not a certain and well-defined boundary that one can cross, thereby traversing from 'efficient portfolio territory' into 'inefficient portfolio territory'. Rather, it is a fuzzy zone of probability distributions around the most likely – but not absolutely certain – efficient frontier. This also illustrates an earlier point: that the notion of 'efficient' versus 'inefficient' portfolios is actually quite

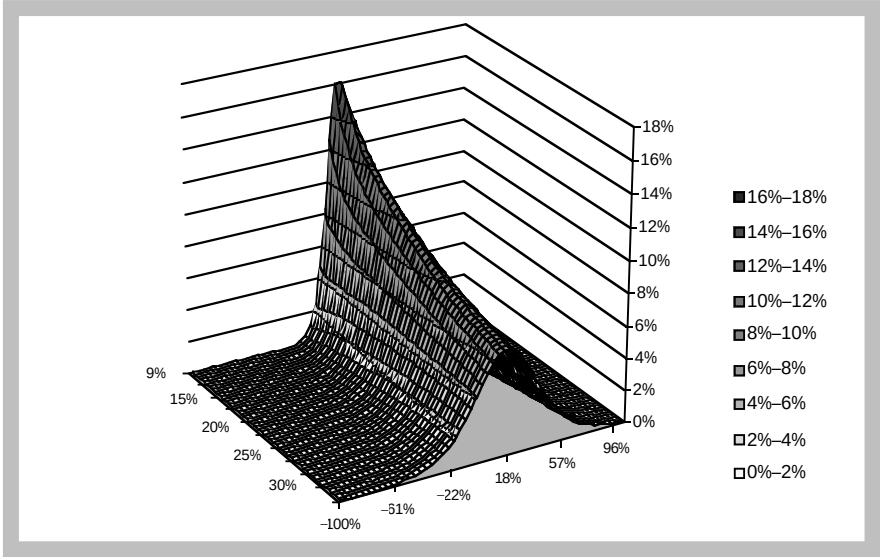


Figure 10.7 The efficient ridge: 3D

misleading since the notion of efficiency is founded on the assumption of perfect foresight, which is simply not available in investment management. This type of augmentation to the traditional efficient frontier framework represents a large improvement in efficient portfolio characterisation.

In Figure 10.8 we present the same data as in Figure 10.7, but this time in a two-dimensional view seen from ‘above’ the efficient ridge. In order to better

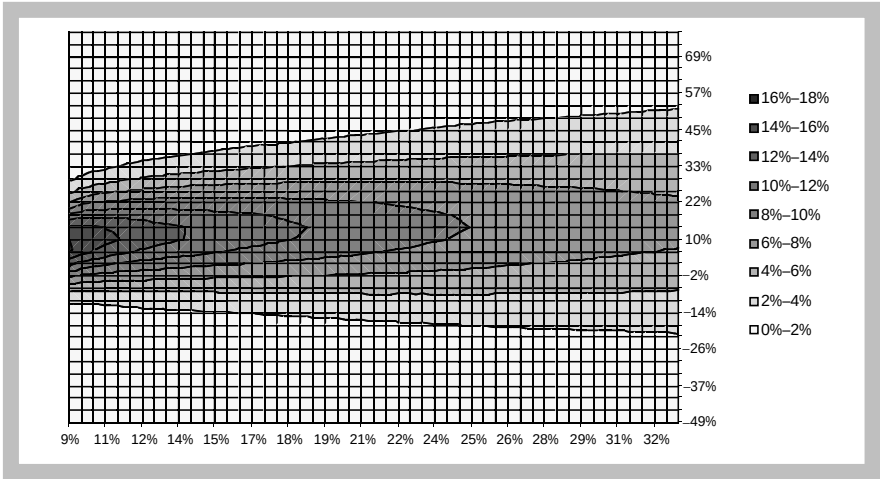


Figure 10.8 The efficient ridge: 2D

see the shape of the efficient ridge from this viewpoint, the scale on the right-hand side now spans only from -50% to 80%. This chart illustrates exactly the same efficient frontier as that in Figure 10.6, but perhaps more clearly demonstrates just how uncertain each point along the efficient frontier really is; and in particular that increased volatility naturally implies a greater degree of quantifiable uncertainty, as demonstrated by the growing 'width' of the expected probability distributions for increasing target volatilities along the efficient ridge.

Moving from the minimum-variance portfolio to the maximum-variance portfolio, the optimised portfolios include fewer and fewer of the available asset classes, and the asset classes that the optimal portfolios do contain are increasingly the riskier ones. To the far left, near the minimum-variance portfolio, most of what we could call the *probability mass* is bunched together close to the mean. This is indicated by the dark shades at the centre of the distributions. As we move further to the left, increasing the expected level of risk, more and more of the probability mass moves out and away from the mean of returns, thereby making the return outcome of the optimised portfolio more uncertain. In total – despite the diversification effects that are fully exploited by the quantitative optimisation algorithm – the expected return outcome becomes more and more uncertain, so the portfolio return probability distribution becomes wider and wider.

In this section we have provided the efficient frontier with the variance that it inherently holds, and thus transformed it into the efficient ridge, which shows in much more detail the likely expected level of return and risk of the quantitatively optimised portfolios. We now move on to further refine the asset allocation framework.

THE QUASI-RANDOM MONTE CARLO SIMULATED ASSET ALLOCATION

Going one better

The simulation approach presented above can be refined further by ensuring that asset returns are correlated as they are drawn from their respective probability distributions. This is where the asset allocation simulation goes from a simple Monte Carlo Simulation to what we will call the *Quasi-Random Monte Carlo Simulated Asset Allocation* or QRMCSAA. Logically, in the process of drawing quasi-random numbers, it is absolutely essential that asset class returns are drawn in accordance with the correlation structure – the correlation matrix – and also in accordance with the distributional characteristics of each asset class. The logic behind this is straightforward: even though many

simulation programs draw purely random numbers for Monte Carlo Simulations – for example in Monte Carlo Simulation-based option valuation – in a portfolio asset allocation context this would be very simplistic, and would most likely lead to misleading asset allocation results. Ignoring the correlations between asset class returns as we draw simulated return values from each asset class's return distribution would in essence filter out valuable information on how asset returns are correlated and distributed. Such information is essential to an accurate characterisation of the dynamics of the portfolio, in particular, the portfolio's expected volatility. Therefore, we draw quasi-random returns from asset class return distributions according to the specified correlation matrix.

QRMCSAA

The rationale behind the introduction of this structured quasi-random drawing of returns is quite straightforward. While quantitative optimisation techniques have historically produced non-intuitive portfolio recommendations, the asset manager ought for obvious reasons at least to avoid putting non-intuitive data into the model. The problem of non-intuitive results caused by the combination of a powerful optimisation algorithm and low quality inputs is bad enough as it is; the last thing asset managers would want to do is to add to the problems by inputting non-intuitive model parameters. This is probably one of the most important elements of the QRMCSAA methodology presented here. Ensuring that model inputs are drawn from their respective probability distributions in a manner that is consistent with, and accurately reflects, the actual behaviour of asset returns is crucial to the QRMCSAA process. For example, in the MSCI sector model presented above it would be somewhat unlikely to observe a negative correlation between the information technology sector and the telecommunications sector for a sustained period, simply because the earnings of the companies involved in those two sectors are often driven by the same factors – economic growth in general, and spending on telecommunications in particular. With purely random return generation, however, such a negative correlation between their respective returns would occur approximately every other time on average, causing serious misrepresentation of the actually observed return data, in addition to being highly non-intuitive and producing inaccurate results. In fact, the average correlation between the two indices, as estimated using data from January 1995 to May 2002, is 0.70, and the quasi-random returns for the two sectors should be drawn accordingly. If not, this would introduce even more errors into the model than the estimation errors already present – errors which are bad enough by them-

selves. Ignoring the problem and not specifying a correlation matrix for the Monte Carlo Simulation's random returns is equivalent to assuming no correlation or a correlation coefficient of zero, which will seriously and adversely affect the results of the simulation.

In the Quasi-Random Monte Carlo Simulated Asset Allocation methodology presented here, values are thus no longer drawn completely at random, but rather are drawn purposefully according to a set of consistent asset class assumptions. Logically, the returns must simulate accurately the observed returns in the estimation period.

Figure 10.9 is a *stylised* example of one set of 10 sector return values, drawn simultaneously from their respective return probability distributions and in accordance with the historical correlations specified by the correlation matrix. The dark lines in the return probability charts represent the set of ten quasi-random returns drawn from each distribution.

As can be seen from Table 9.1, these are more or less in accordance with the specified correlation matrix; but obviously, since there is still a strong random element in this process, some of the sets of return values may not all strictly follow the behaviour specified in the correlation matrix. However, in this stylised example we observe that for this particular period, the higher or positive returns seem to be observed in the technology-heavy and cyclical sectors, while we observe that lower or negative returns seem to occur in the defensive sectors such as utilities and consumer staples.

Each of the ten quasi-randomly drawn return values is then entered into the optimisation algorithm, and the portfolio is optimised in order to determine the optimal asset allocation. Figure 10.10 illustrates the QRMCSAA for the MSCI Global Sector Model, as determined using the methodology presented in Figure 10.1.

Figure 10.10 demonstrates a clear and significant change from the 'naive' asset allocation without adjusted inputs and without application of the QRMCSAA technique, as presented in Figure 9.6. Compared to the 'naive' asset allocation along the efficient frontier, this allocation is clearly much more intuitive and practical. Please note that this is in fact the asset allocation that – as we move from the minimum-variance portfolio to the maximum-variance portfolio – yields not the efficient *frontier* but rather the efficient *ridge*, as demonstrated earlier.

The asset allocation is significantly different from the 'naive' asset allocation, especially as we move to the right in the chart, increasing the target or expected volatility of the portfolio.³ Observing each asset class in turn we can see that no asset class is suddenly added to or dropped from the portfolio, and there are no sudden and erratic changes in the amounts allocated to the different asset classes. Transition from one asset class to another is smooth, intuitive and pleasantly uneventful.

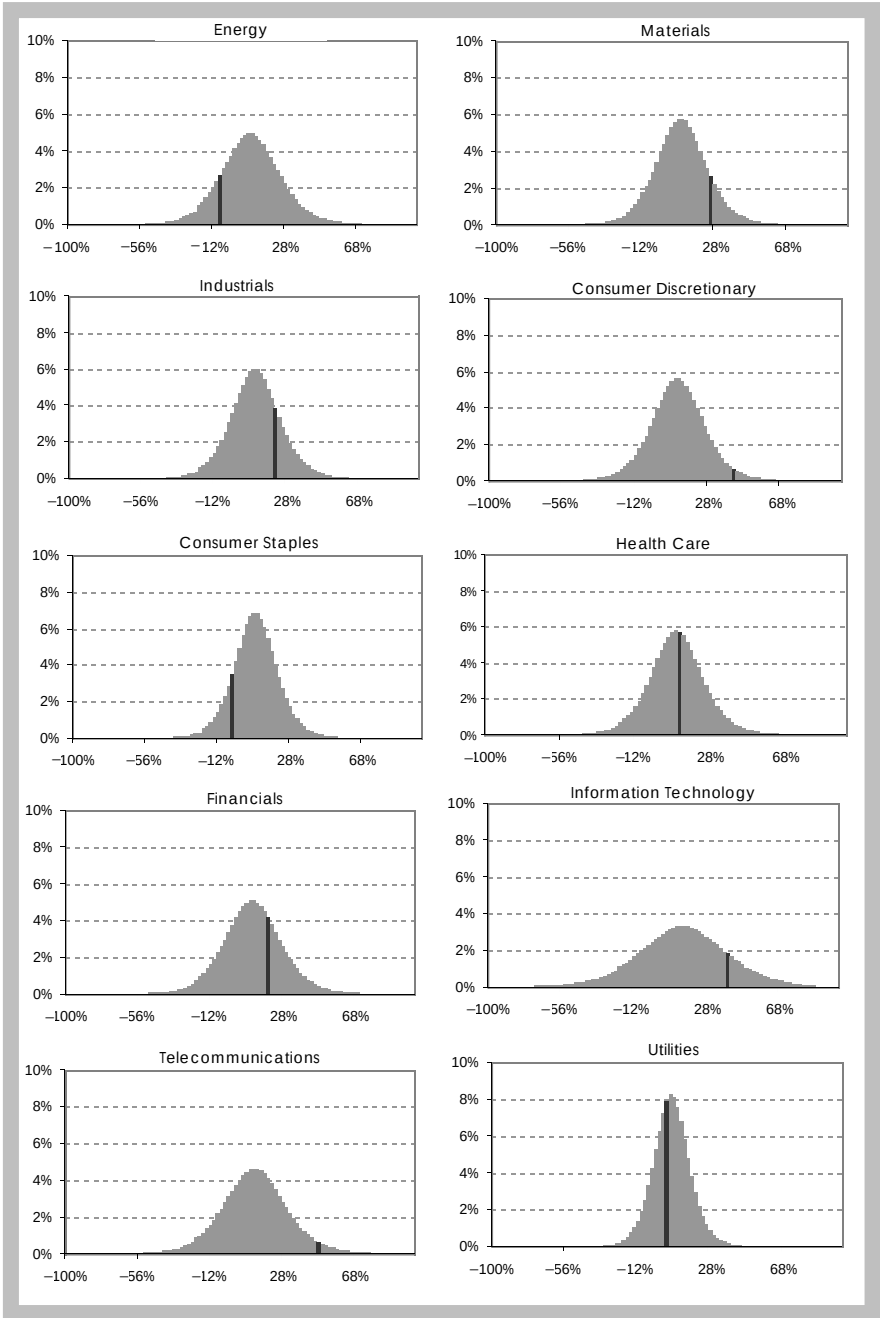


Figure 10.9 Return distributions for 10 MSCI
Global Sectors, 1995/1–2002/5

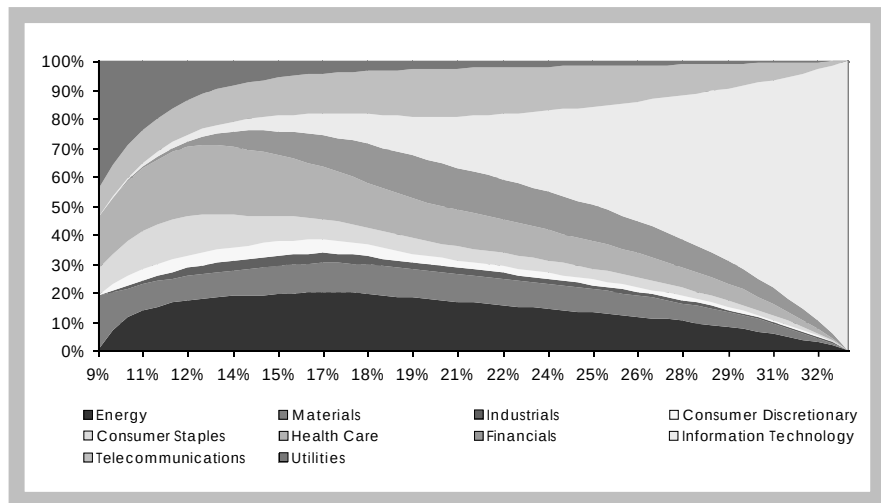


Figure 10.10 QRMCSAA – MSCI Global Sector Model

In addition to this, the inclusion of asset classes in the portfolio get ‘drawn out’. In other words, asset classes stay in the portfolio for much longer over the target volatility spectrum from the minimum-variance portfolio to the maximum-variance portfolio. This makes for a much more intuitive asset allocation that also seems to make better investment sense. All sectors are represented in all the optimal portfolios along the entire risk spectrum. This is very different from the original ‘naive’ asset allocation chart. Moving from a target volatility level of approximately 12% to approximately 15% should not result in a recommendation of a change in portfolio composition that entails turning over – selling and reinvesting – more than half of the portfolio’s assets. This was the case for the ‘naive’ asset allocation, where the most of the sectors were dropped from the optimal portfolios very early on as the target volatility was increased.

Another very notable characteristic of the QRMCSAA relative to the ‘naive’ asset allocation in Figure 9.6 is that in the latter the proportion of the portfolio invested in the telecommunications sector is virtually unchanged between target volatilities of approximately 13% and 25%. At that point financials exit from the optimal portfolios, resulting in a violent rotation between sectors. Because the correlation between financials and telecommunications is lower than the correlation between information technology and telecommunications as soon as financials exit the optimal portfolio, telecommunications are reduced rapidly and replaced by information technology. This erratic change does not occur in the QRMCSAA chart in Figure 10.10. This again demonstrates the point that an application of the QRMCSAA technique presented

above produces recommended portfolios that are intuitively explainable, contrary to those often produced by many ‘naive’ asset allocation models. Presenting this sort of asset allocation chart to a prospective client ought to instil considerably more confidence in the overall capabilities of the asset manager, in particular with regard to the asset allocation – the basis of any investment process.

Sensitivity to assumption changes

We have already alluded to the fact that quantitative portfolio optimisation asset allocation in the traditional or naive form entails letting the optimisation algorithm find extreme solutions. Logically, this leads to asset allocation results that are highly sensitive to changes in model assumptions, and the following example is designed to demonstrate this point.

Suppose the asset manager believes that, relative to the other sectors, the health care sector will see a slightly higher return than previously assumed, and therefore changes the expected return for this sector from 11.0% initially to 12.0%. This is indeed a minor change, so the impact on the recommended asset allocation ought not to be too great. Figure 10.11 illustrates the impact on the efficient frontier of raising the expected return on the health care sector by 1 percentage point. The expectation about the health care sector is indicated by the grey dot.

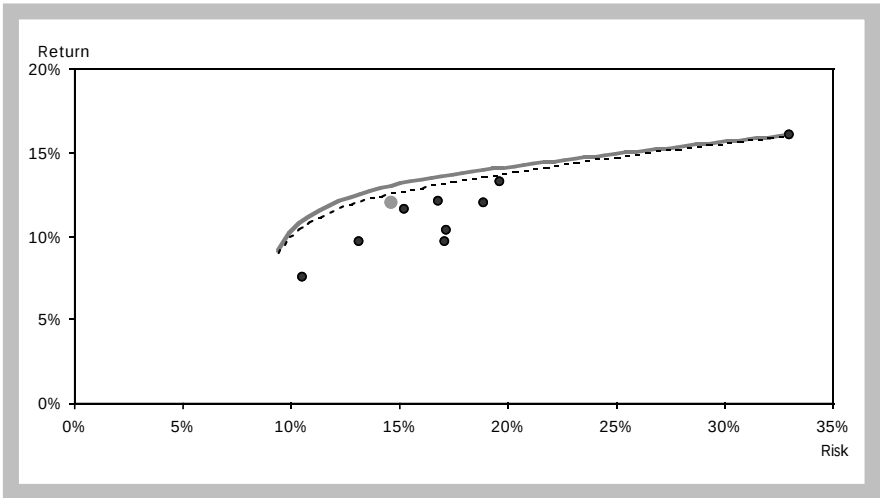


Figure 10.11 Efficient frontier – MSCI Global
Sector Model, health care: 12%

Not surprisingly, since one of the asset classes now has a higher expected return, the efficient frontier shifts upwards in the medium volatility range, but is of course still tied down in the two extremes, so the minimum and the maximum-variance portfolios are unchanged, both in terms of expected return and risk, and in terms of portfolio expansion. This already indicates that the quantitative portfolio optimisation algorithm has exploited the new return assumption to the fullest. The precise impact on the naive asset allocation can be seen from Figure 10.12. Clearly, the naive asset allocation is changed dramatically relative to Figure 9.6, as a result of the comparatively small change in the expected or assumed return on the health care sector. Most notably, the increase in the emphasis put on the health care sector across the entire risk spectrum is very large. At a target volatility level of around 14%, the recommended portfolio includes as much as 60% health care, and on average the sector features twice as prominently as before across the risk spectrum. In fact, it is now the second most prominent sector after information technology, thus replacing telecommunications in that position. Telecommunications, on the other hand, has been reduced quite significantly, and actually drops out of the optimal asset allocation altogether at a target volatility of around 25%. Note also that the financial sector has now been entirely dropped from the asset allocation.

Although somewhat logical, the changes observed from the relatively small changes in model inputs are extreme, and overall the nature and extent of the changes are non-intuitive. In addition, it would simply not be practical to

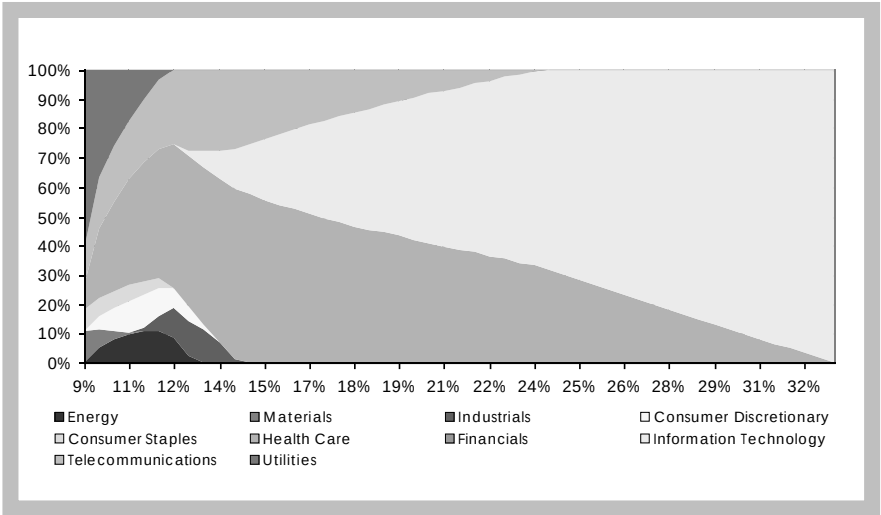


Figure 10.12 Asset allocation – MSCI Global
Sector Model, health care: 12%

adhere to the recommendations of such an asset allocation when it exhibits such a high degree of sensitivity to small changes in model assumptions: firstly because it requires a large rotation, (buying and selling between sectors almost regardless of the target volatility level), and secondly because the plausibility of such a significant change is difficult to explain to the asset manager’s clients.

Employing the QRMCSAA methodology established so far yields the asset allocation chart shown in Figure 10.13. Quite clearly the difference between this and the original Quasi-random Monte Carlo Simulated Asset Allocation presented in Figure 10.10, is much less than the difference between the asset allocation charts in Figures 9.6 and 10.11. In fact, without careful examination the impact is barely perceptible. The health care sector actually does get assigned a higher weight across the whole optimal asset allocation along the entire risk spectrum, but the impact on the recommended asset allocation is nowhere near as extreme as was the case for the naive asset allocation methodology. The increase in the weight assigned to the health care sector is principally at the cost of exposure to the financial sector. Changes to the other sectors are minor. The optimal portfolio’s average exposure to the industrials sector gets decreased a little, whereas the average exposure to energy sector gets increased a little, but the recommended asset allocation along the risk spectrum is basically unchanged.

This is all testament to the fact that the QRMCSAA methodology is significantly more robust, intuitive and practical in nature, and more specifically to the fact that the quality of information employed in the methodology now corresponds more closely to the power of the optimisation algorithm.

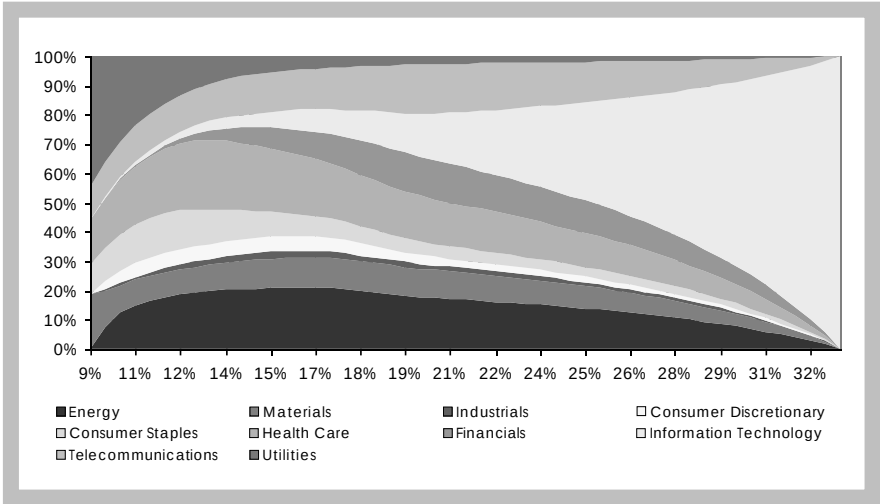


Figure 10.13 QRMCSAA – MSCI Global Sector Model, health care: 12%

We have now laid the foundation for the QRMCSAA model, presenting the intuition as well as the mechanics behind it. In the next chapter we shall refine it further by taking into account the estimation errors and model biases that may result from using historical data as model inputs.

A brief note on modelling return distributions

Here we simply assume that we have well-defined return distributions for the simulations that need to be carried out. In reality, the estimation of return distributions is somewhat complex and closely tied to the portfolio risk-management process, and we shall therefore refrain from delving deep into this subject here. It should however be noted that arguably the subject of estimation error, in determining not only expected returns and volatilities but also the return distributions themselves, deserves serious attention. We shall investigate this further in subsequent chapters.

SUMMARY

In this chapter we have presented the Quasi-Random Monte Carlo Simulated Asset Allocation methodology for allocating assets according not only to the expected return, risk and correlations, but also to the expected probability distributions of returns. The QRMCSAA is a very robust and usable method for quantitative asset allocation, since it takes the best from Modern Portfolio Theory, portfolio optimisation, Monte Carlo Simulation and statistical analysis, and combines these to deliver extremely robust and intuitive asset allocation results that are unmatched by the traditional ‘naive’ asset allocation methodologies.

A few important points emerge from the analysis of the application of quantitative optimisation to portfolio asset allocation presented in this chapter. In its most basic form an optimiser is not a delicate instrument that – based on precise input data – can deliver a sophisticated analysis of how best to allocate portfolio assets. Optimisers apply brute force to squeeze a solution out of available data, without regard for investment sense or intuition. In fact the word ‘optimiser’ and the notion of an optimised portfolio may well have misled many an asset manager into thinking that his portfolio, having been adjusted according to the suggestions of some quantitative optimiser, was somehow better than every other portfolio. This notion is false and potentially very dangerous. Because of the frustrations experienced over the use of optimisers and their at times nonsensical results and recommendations, investment professionals have either shied away from quantitative optimisation

altogether or put so many constraints on the optimised portfolios as to seriously limit the potential of the technique. The latter has often been called the 'torturing the data' approach. If one tortures the data, the data will eventually confess and tell the manager what she or he wants to hear – that is, it will produce the asset allocation that the manager found attractive in his or her own mind from the start. Of course, this excessive use of constraints leaves quantitative optimisation virtually impotent, since investment managers will merely be told what they already know.

Nonetheless, quantitative portfolio optimisation is a powerful tool in the hands of experienced investment professionals who understand the dynamics of portfolio return and risk in addition to the dynamics of quantitative optimisation techniques, including their opportunities and limitations. The problem with quantitative optimisation is not the model itself, but rather the fact that forecasts of asset class return volatilities and correlations are inaccurate. Therefore, quantitative portfolio optimisation and asset allocation should not be abandoned. Rather, asset managers should focus on developing better forecasting techniques so as to provide better forecasts as model inputs. Thereafter, asset managers should work to make sure that the asset allocation model appropriately reflects the investment philosophy, the style and the mindset of the investment organisation. As demonstrated in this chapter, establishing such a framework to support quantitative asset allocation may very well provide real value for asset managers in the form of improved risk control, and quite possibly superior returns over the longer term.

The analysis carried out in this chapter is based on the 10 MSCI sectors, but in the appendix, in Figures 10.14–10.16, we have provided the results of the QRMCSAA applied to the 23 MSCI industry groups. The 10 sectors are essentially aggregates of the 23 industry groups, so they present the same data and the same investment universe. In subsequent chapters we shall proceed with the sector model, but the three charts presented in Figures 10.14–10.16 demonstrate how the approach presented here can be extended to a more detailed level of asset class characterisation. We could also extend the analysis even further to the industry level, but that would leave us with 59 industries to be characterised and entered into the model. Although more detailed, it would not necessarily entail better results. The lower the aggregation level, the more unstable the model inputs over time; and in addition it could be argued that most asset managers would probably be content with an asset allocation methodology based on a framework with only 10 classifications and not as many as 59. But the charts in the appendix demonstrate that the model can in principle be adapted to any investment universe and to any level of aggregation required.

The premise of this chapter is the notion that single-state portfolio optimality expressed only in terms of return and volatility values is contentious at best and misleading at worst. Throughout, we make the assertion that any variable that exhibits some sort of variation over time is best described not as a single point in return/risk space, but as a probability distribution. And given that the probability distributions employed in the QRMCSAA methodology are continuous, this arguably yields an infinitely more accurate description of model inputs. This is because every simulated asset class return is statistically equivalent to the original single return/risk estimate. Every simulated return is thus just as valid as a model input as the original, regardless of whether we simulate 10 years or 10 million years of asset class returns. Whether inputs are actually infinitely better is of course difficult to ascertain, but it remains clear that this description of model inputs is superior. This feature can actually be seen directly from the asset allocation charts.

Previously the argument was made that the optimisation algorithm is too powerful for the available quality of model inputs. Simulating asset class returns using Monte Carlo Simulation means that we can describe model inputs in a far richer way. This effectively increases the quality and depths of the input information available many times. This is evident when we compare the naive asset allocation charts with the QRMCSAA charts. Examining these it becomes clear that the quality of input information now matches the power of the optimisation algorithm more equally, yielding more robust and less unstable results. This is perhaps best demonstrated by changing the expected return of a single asset class by one percentage point, and observing the resultant change in asset allocation for the conventional model and the QRMCSAA respectively. The conventional model reacts violently because the quality of information provided for the optimiser is so low, so it is easy for the optimiser to go to extremes – as we have told it to do. But for the QRMCSAA the resultant asset allocation chart reveals only smaller changes, as the quality of information in model inputs is more refined.

APPENDIX

Four Monte Carlo Simulation algorithms

Below we present four stylised algorithms for performing a Monte Carlo Simulation-based asset allocation, beginning with the simplest form of ‘resampling’ as presented in Michaud (1998), and then going on to three more advanced versions of the Quasi-Random Monte Carlo Simulation-based Asset

Allocation developed in this book. They are basically variations on the same theme, but the underlying assumptions of each method are potentially very important for the accuracy of results and the investment value they will be able to provide for the asset manager.

Method I: Resampling

Underlying assumptions

- Normally distributed returns.
- Uncorrelated returns as they are drawn from normal distributions.

Methodology

- 1 Monte Carlo Simulate 18 years of returns (based on historical observations).
- 2 Calculate model input parameters (returns and volatilities) on the basis of simulations.
- 3 Optimise to find the efficient frontier (51 points).
- 4 Repeat steps 1–3 500 times.
- 5 Find the average of the asset allocation for all 51 points.
- 6 Illustrate and evaluate the asset allocation.

Note: Stein estimators and Bayesian priors can be applied to model input parameters.

Method II: QRMCSAA hybrid

Underlying assumptions

- Non-normally distributed returns.
- Correlated returns as they are drawn from non-normal distributions.

Methodology

- 1 Monte Carlo Simulate 18 years of returns (historical or forecast), taking into account the non-normality of return distributions as well as the (historical) correlations between asset class returns.
- 2 Calculate model input parameters (returns and volatilities) on the basis of the QRMCMC simulations.
- 3 Optimise to find the efficient frontier (51 points).
- 4 Repeat steps 1–3 500 times.
- 5 Find the average of the asset allocation for all 51 points.
- 6 Illustrate and evaluate the asset allocation.

Note: Stein estimators and Bayesian priors can be applied to model input parameters.

Method III: QRMCSAA, multiple efficient frontiers

Underlying assumptions

- Non-normally distributed returns.
- Correlated returns as they are drawn from non-normal distributions.

Methodology

- 1 Estimate (historical or forecast) returns, volatilities and correlations.
- 2 Examine historical return distributions and fit non-normal distributions to each asset or asset class.
- 3 For each asset class, draw quasi-random (correlated) returns one year at a time from respective distributions, holding volatilities and correlations constant.
- 4 For each simulated year, optimise to find the entire efficient frontier, and record corresponding asset allocation (portfolio weights) for all 51 points along the frontier.
- 5 Repeat steps 3–4 500 times.

- 6 Find the average of the asset allocations for all 500 QRM-simulated efficient frontiers.
- 7 Illustrate and evaluate the asset allocation.

Note: Stein estimators and Bayesian priors can be applied to model input parameters.

Method IV: QRMCSAA, all points on the efficient frontier

Underlying assumptions

- Non-normally distributed returns.
- Correlated returns as they are drawn from non-normal distributions.

Methodology

- 1 Estimate (historical or forecast) returns, volatilities and correlations.
- 2 Examine historical return distributions and fit non-normal distributions to each asset or asset class.
- 3 Draw quasi-random (correlated) returns from respective distributions, holding volatilities and correlations constant.
- 4 Optimise to find the efficient portfolio (begin with the minimum-variance portfolio), and record the corresponding asset allocations (portfolio weights).
- 5 Repeat steps 3–4 500 times.
- 6 Find the average of the asset allocations for the single point on the efficient frontier.
- 7 Move on to the next point on the efficient frontier (minimum-variance portfolio plus one increment) and repeat the procedure for all 51 points along the efficient frontier.
- 8 Illustrate and evaluate the asset allocation.

Note: Stein estimators and Bayesian priors can be applied to model input parameters. This methodology fits well with the representation of the efficient ridge, which is also determined by simulating portfolios for each point on the efficient frontier.

Asset Allocation along the efficient frontier (Naive AA)

Energy	Materials	Indust.	Cons. Discr.	Cons. Staples	Health Care	Finan.	IT	Telecomm.	Utilities
0.00%	10.63%	0.00%	0.00%	7.32%	8.56%	0.00%	0.00%	12.21%	61.28%
6.34%	5.33%	0.00%	4.86%	11.46%	18.06%	0.00%	0.00%	19.00%	34.94%
9.36%	1.87%	0.00%	8.69%	13.22%	22.35%	0.00%	0.00%	21.15%	23.36%
11.00%	0.00%	1.26%	10.48%	14.10%	25.43%	0.00%	0.00%	23.22%	14.50%
11.35%	0.00%	6.17%	7.91%	13.82%	28.07%	0.00%	0.00%	25.69%	6.98%
11.37%	0.00%	11.03%	5.53%	13.34%	30.47%	0.00%	0.00%	27.92%	0.34%
7.68%	0.00%	20.13%	1.89%	5.07%	34.02%	0.00%	0.00%	31.21%	0.00%
3.19%	0.00%	26.98%	0.00%	0.00%	35.42%	0.00%	0.00%	34.41%	0.00%
0.00%	0.00%	26.78%	0.00%	0.00%	35.55%	0.00%	2.55%	35.12%	0.00%
0.00%	0.00%	23.09%	0.00%	0.00%	35.33%	0.00%	6.13%	35.46%	0.00%
0.00%	0.00%	19.57%	0.00%	0.00%	35.10%	0.00%	9.27%	36.07%	0.00%
0.00%	0.00%	16.36%	0.00%	0.00%	34.88%	0.00%	12.23%	36.53%	0.00%
0.00%	0.00%	12.80%	0.00%	0.00%	34.81%	0.56%	15.07%	36.77%	0.00%
0.00%	0.00%	7.71%	0.00%	0.00%	34.22%	3.49%	17.47%	37.12%	0.00%
0.00%	0.00%	3.50%	0.00%	0.00%	33.58%	5.66%	19.87%	37.40%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	32.26%	7.98%	22.09%	37.67%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	29.82%	7.95%	24.25%	37.98%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	27.88%	7.71%	26.55%	37.86%	0.00%

cont'd

Asset allocation along the efficient frontier (Naive AA) *(continued)*

Energy	Materials	Indust.	Cons. Discr.	Cons. Staples	Health Care	Finan.	IT	Telecomm.	Utilities
0.00%	0.00%	0.00%	0.00%	0.00%	25.53%	7.74%	28.53%	38.20%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	23.88%	7.43%	30.78%	37.90%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	21.89%	7.34%	32.77%	38.00%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	20.07%	7.15%	34.77%	38.02%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	18.22%	7.05%	36.71%	38.02%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	16.24%	7.03%	38.53%	38.20%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	14.55%	6.86%	40.46%	38.13%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	12.44%	6.92%	42.10%	38.53%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	10.93%	6.71%	44.05%	38.31%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	9.21%	6.60%	45.84%	38.34%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	7.44%	6.54%	47.58%	38.44%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	5.79%	6.41%	49.35%	38.44%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	4.30%	6.20%	51.18%	38.31%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	2.63%	6.10%	52.90%	38.37%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.81%	6.11%	54.51%	38.57%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.47%	56.61%	37.92%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	3.72%	58.84%	37.44%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.13%	61.11%	36.75%	0.00%

0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.63%	63.39%	35.98%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	66.00%	34.00%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	68.83%	31.17%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	71.62%	28.38%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	74.36%	25.64%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	77.05%	22.95%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	79.72%	20.28%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	82.34%	17.66%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	84.94%	15.06%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	87.51%	12.49%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	90.05%	9.95%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	92.57%	7.43%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	95.07%	4.93%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	97.54%	2.46%	0.00%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	0.00%	0.00%

Asset allocation along the efficient frontier (QRMCSAA)

Energy	Materials	Indust.	Cons. Discr.	Cons. Staples	Health Care	Finan.	IT	Telecomm.	Utilities
0.09%	15.65%	0.00%	0.00%	11.57%	14.90%	0.00%	0.00%	13.29%	44.49%
6.87%	10.39%	0.60%	3.07%	12.17%	16.89%	0.34%	0.53%	13.52%	35.63%
11.36%	7.50%	1.13%	4.70%	12.75%	18.54%	0.53%	1.01%	13.66%	28.82%
13.58%	6.84%	1.65%	4.96%	13.30%	19.96%	0.75%	1.48%	13.71%	23.76%
14.88%	6.67%	2.08%	4.99%	13.64%	21.09%	1.09%	1.87%	13.77%	19.91%
15.93%	6.62%	2.48%	5.00%	13.76%	22.14%	1.51%	2.26%	13.85%	16.45%
16.63%	6.76%	2.83%	4.94%	13.53%	23.01%	1.99%	2.67%	13.93%	13.71%
17.15%	6.98%	3.13%	4.85%	12.95%	23.61%	2.57%	3.13%	14.05%	11.59%
17.58%	7.23%	3.35%	4.76%	12.21%	23.83%	3.26%	3.62%	14.20%	9.94%
17.96%	7.49%	3.53%	4.71%	11.59%	23.62%	4.03%	4.15%	14.37%	8.65%
18.30%	7.73%	3.66%	4.71%	10.95%	23.11%	4.80%	4.73%	14.49%	7.62%
18.63%	7.99%	3.75%	4.76%	10.34%	22.39%	5.47%	5.39%	14.57%	6.80%
18.98%	8.29%	3.76%	4.87%	9.76%	21.55%	5.96%	6.14%	14.66%	6.14%
19.32%	8.59%	3.72%	5.00%	9.07%	20.67%	6.36%	6.94%	14.75%	5.59%
19.56%	8.86%	3.64%	5.06%	8.51%	19.75%	6.83%	7.79%	14.88%	5.13%
19.67%	9.14%	3.55%	4.98%	8.02%	18.84%	7.33%	8.68%	15.06%	4.73%
19.60%	9.41%	3.45%	4.78%	7.59%	17.97%	7.85%	9.64%	15.31%	4.40%
19.34%	9.61%	3.34%	4.54%	7.19%	17.19%	8.40%	10.68%	15.59%	4.12%

18.96%	9.73%	3.24%	4.28%	6.84%	16.48%	8.95%	11.76%	15.90%	3.87%
18.56%	9.75%	3.13%	4.02%	6.53%	15.82%	9.46%	12.87%	16.22%	3.65%
18.16%	9.68%	3.01%	3.77%	6.26%	15.22%	9.89%	14.01%	16.53%	3.46%
17.79%	9.56%	2.91%	3.55%	5.97%	14.72%	10.18%	15.21%	16.84%	3.29%
17.45%	9.40%	2.81%	3.35%	5.63%	14.36%	10.32%	16.49%	17.05%	3.13%
17.11%	9.24%	2.72%	3.18%	5.26%	14.13%	10.39%	17.88%	17.10%	3.00%
16.75%	9.06%	2.64%	3.03%	4.99%	13.87%	10.42%	19.35%	17.02%	2.87%
16.37%	8.89%	2.56%	2.91%	4.81%	13.54%	10.40%	20.89%	16.87%	2.76%
16.00%	8.72%	2.48%	2.80%	4.66%	13.20%	10.35%	22.49%	16.66%	2.64%
15.64%	8.52%	2.41%	2.71%	4.53%	12.82%	10.28%	24.14%	16.44%	2.52%
15.27%	8.31%	2.34%	2.61%	4.42%	12.43%	10.18%	25.84%	16.20%	2.41%
14.92%	8.08%	2.26%	2.50%	4.30%	12.02%	10.05%	27.61%	15.95%	2.30%
14.58%	7.83%	2.19%	2.39%	4.18%	11.61%	9.89%	29.45%	15.68%	2.20%
14.24%	7.58%	2.11%	2.28%	4.07%	11.19%	9.70%	31.38%	15.36%	2.11%
13.89%	7.32%	2.03%	2.16%	3.93%	10.79%	9.47%	33.39%	15.00%	2.02%
13.54%	7.05%	1.94%	2.04%	3.77%	10.39%	9.21%	35.49%	14.62%	1.94%
13.17%	6.77%	1.86%	1.91%	3.61%	9.99%	8.93%	37.69%	14.21%	1.85%
12.78%	6.48%	1.77%	1.79%	3.45%	9.56%	8.63%	39.99%	13.75%	1.77%
12.37%	6.18%	1.68%	1.68%	3.29%	9.12%	8.33%	42.41%	13.26%	1.69%
11.93%	5.86%	1.59%	1.57%	3.12%	8.66%	8.01%	44.94%	12.72%	1.60%
11.47%	5.52%	1.49%	1.47%	2.94%	8.18%	7.68%	47.60%	12.14%	1.51%

cont'd

Asset allocation along the efficient frontier (QRMCSAA) (*continued*)

Energy	Materials	Indust.	Cons. Discr.	Cons. Staples	Health Care	Finan.	IT	Telecomm.	Utilities
10.99%	5.17%	1.40%	1.35%	2.75%	7.69%	7.33%	50.40%	11.52%	1.41%
10.48%	4.82%	1.30%	1.24%	2.55%	7.16%	6.94%	53.35%	10.85%	1.31%
9.94%	4.45%	1.20%	1.12%	2.35%	6.61%	6.52%	56.49%	10.11%	1.21%
9.38%	4.07%	1.10%	1.01%	2.15%	6.02%	6.04%	59.82%	9.31%	1.10%
8.79%	3.68%	0.99%	0.90%	1.93%	5.41%	5.51%	63.38%	8.44%	0.99%
8.17%	3.26%	0.88%	0.79%	1.70%	4.77%	4.90%	67.18%	7.50%	0.87%
7.51%	2.81%	0.76%	0.68%	1.46%	4.10%	4.23%	71.23%	6.48%	0.75%
6.78%	2.34%	0.63%	0.56%	1.20%	3.38%	3.51%	75.60%	5.38%	0.61%
5.91%	1.82%	0.49%	0.44%	0.93%	2.63%	2.73%	80.38%	4.19%	0.48%
5.01%	1.30%	0.35%	0.31%	0.66%	1.87%	1.95%	85.21%	2.99%	0.34%
3.37%	0.74%	0.20%	0.18%	0.38%	1.06%	1.11%	91.06%	1.70%	0.19%
0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	0.00%	0.00%

MSCI Global Industry Group Model

In Figure 10.14 we present the efficient frontier produced by the MSCI Global Industry Group Model, with model inputs estimated using daily return observations from 1 January 1995 to 1 May 2002.

Again, the expected returns are somewhat arbitrarily chosen, but the point here is not to forecast returns, but rather to demonstrate the QRMCSAA methodology applied to an investment universe characterised in more detail than the MSCI Sector Model.

Note that there as many as 15 industry groups included in the minimum-variance portfolio, including the industry group with the highest level of expected risk.

This demonstrates that this particular asset class – the software and services industry – is correlated in such a beneficial way with the traditionally defensive industry groups that it actually appears in the portfolio designed to provide the least amount of risk possible. Note also that that the equal-weighted portfolio is some distance from the efficient frontier, simply because the high number of asset classes present in the investment universe allows for a very high degree of diversification, and thus much more potential for quantitative optimisation.

In Figure 10.15 we present the efficient ridge for the MSCI Global Industry Group Model. It is the product of individual characterisations for the 23 MSCI

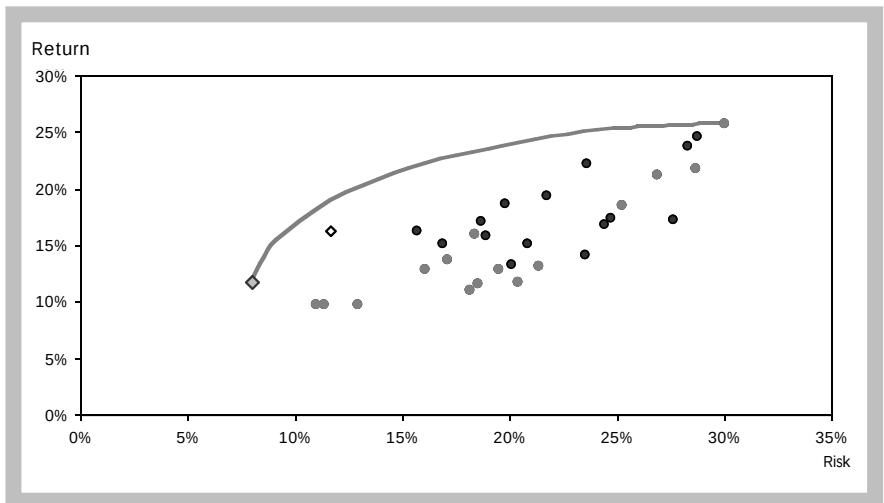


Figure 10.14 The efficient frontier – MSCI Global Industry Group Model

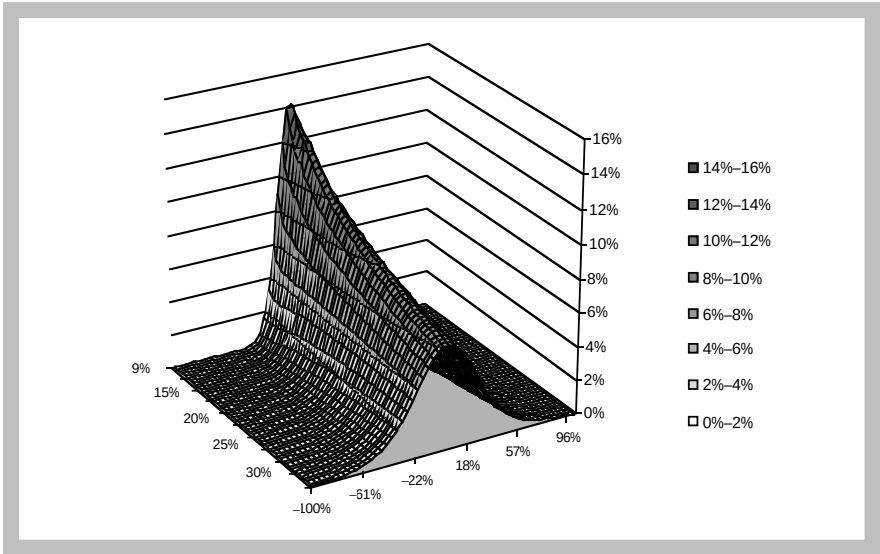


Figure 10.15 The efficient ridge: 3D

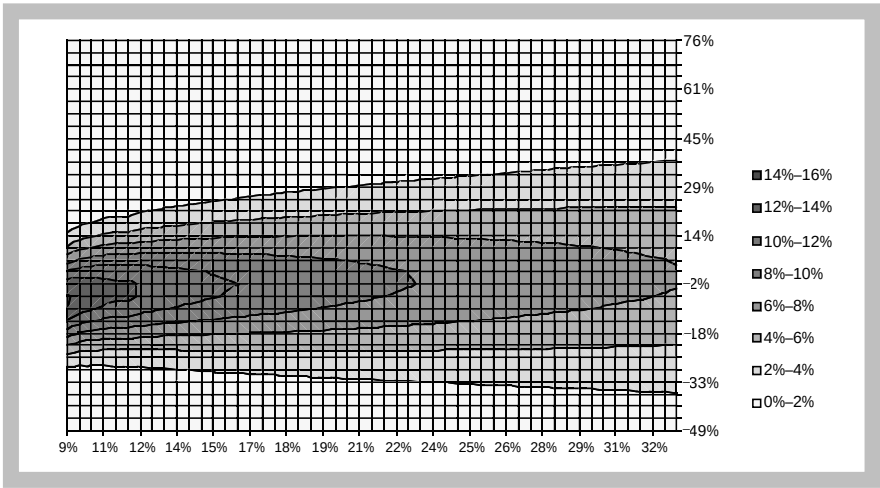


Figure 10.16 The efficient ridge: 2D

global industry groups, in terms of expected return, volatility and return distributions. Seen from above, the efficient ridge appears as in Figure 10.16.

Again it is clear that the distribution becomes wider as we move towards increasingly risky portfolios, indicating a significantly higher degree of uncertainty about portfolio returns.

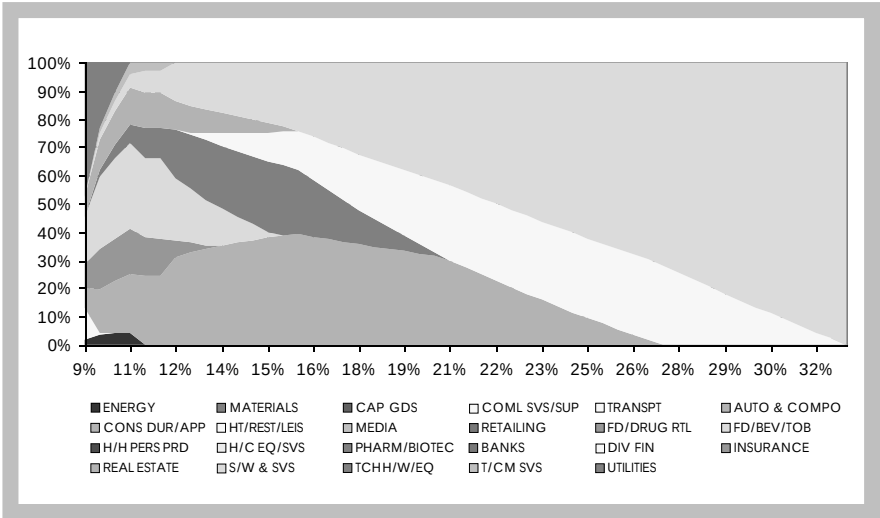


Figure 10.17 Asset allocation – MSCI Global Industry Sub-group Model

Figure 10.17 presents the naive asset allocation that is derived by using simple optimisation techniques to determine the optimal portfolios along the efficient frontier. We noted above that only about 15 of the 23 available industry groups appear in the minimum-variance portfolio. Note also that only a few of the 23 industry groups are included as we move along the risk spectrum from the minimum-variance portfolio to the maximum-variance portfolio, and also that the weight being assigned to each industry group changes rapidly as the target volatility level is increased. In fact the optimiser excludes all but 6 industry groups – or about one quarter of the number available – as soon as the target volatility level is raised above approximately 14%. A target volatility level above 16% leaves only 4, and above a target volatility level of 27% the portfolio is composed of only 2 out of 23 industry groups. Not many asset managers would follow such an asset allocation advice.

Finally, Figure 10.18 presents the asset allocation chart derived using the QRMCSAA methodology. The change from Figure 10.15 is quite dramatic. Every asset class is now included virtually all the way across the risk spectrum, and when the weights placed on individual industry groups are changed, this happens in a smooth fashion whereby one industry group is phased out as another is phased in. Again, this demonstrates the power of this methodology compared to the naive asset allocation approach, as well as the fact that results are infinitely more intuitive and applicable to real investment decision-making.

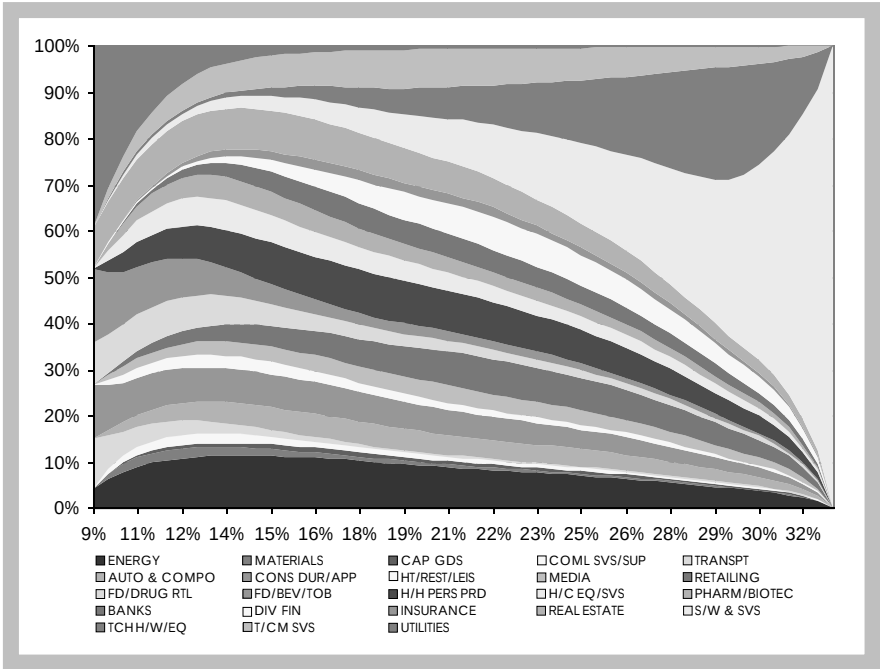


Figure 10.18 QRMCSAA – MSCI Global Industry Sub-group Model

Notes

- 1 A term originally coined by Richard Michaud (1989).
- 2 Methodologies easily adaptable to computers for constructing the efficient ridge can be found in the appendix to this chapter.
- 3 A table showing the numerical values of all asset allocations along the efficient ridge can be found in the appendix to this chapter.

REFINING THE QRMCSAA MODEL

BAYESIAN PRIORS AND STEIN ESTIMATORS

Adjusting model inputs – a slippery slope

In order to further remedy the problem of unstable asset allocations, we can apply several statistical techniques that enhance the investment value of model inputs. Some are crude and others are more advanced, but in a portfolio optimisation and asset allocation context they all share the goal of refining the inputs used in order to reduce the uncertainty of the recommended asset allocation as a result of just small changes in model inputs. In other words, they are designed to adjust model inputs in order to obtain more intuitive and applicable outputs that can better provide investment value for the asset manager.

Ultimately, the goal of any active asset allocation procedure must be to add value in the form of superior returns relative to a 'naive' approach such as the equal-weighted portfolio approach or a benchmark-mimicking portfolio. Instability of portfolio weights when inputs are changed only slightly can seriously hamper the ability of the active asset manager to provide added value, as a large part of the portfolio optimisation and asset allocation recommendations that come out of the optimisation procedure may seem nonsensical. But forcing stability upon portfolio weights as the desired level of return and volatility is changed is not automatically congruent with optimal portfolio

performance in terms of returns. In principle, asset managers must adhere to theoretically sound methodologies in the investment process, but in practice there is some justification for using less than perfectly rigorous applications, since our ability to predict or even to describe the dynamics of the investment universe is unfortunately somewhat limited. Thus there may be instances where we should be guided more by considerations about the practical applicability of the asset allocation processes than by strict theoretical adherence.

Using statistical methods to adjust model inputs for potential errors or biases is thus an attractive proposition. However, exactly how and by how much to tweak model inputs is a matter of some contention – we shall present a number of methods below. Obviously, it is possible to attempt an adjustment of returns, volatilities and correlations. Ultimately it is up to the manager to decide what, how and by how much model parameters should be adjusted. Again, it should be underscored that quantitative portfolio optimisation and asset allocation is not an exact science but rather an art. Fortunately, the validity of the results of this particular form of art are readily observable to the manager *ex post*. The research area on correct estimation and adjustments of optimisation model inputs is huge, and it would be too far-reaching to attempt to cover it in detail here. In the next section we will therefore briefly discuss general types of adjustment procedures for enhancing the optimisation inputs.

The mechanics of Stein estimators and Bayesian priors

A class of adjustment techniques that have drawn wide attention in the statistical and financial economics communities are the so-called *Bayesian* techniques. The basic idea behind these techniques is that model inputs are adjusted towards to some predefined value or *prior*. The prior can be any arbitrary value or some educated guess as to the future value of a parameter, itself perhaps based on past experience. Specifically this may be an expectation about long-term returns or volatilities for asset classes.

In general, the use of Bayesian priors imposes some measure of intuitive structure as well as an element of long-run expectations on the model inputs. In a paper on the subject, Charles Stein (1955) demonstrated how, under very general conditions, individually estimated sample means are not admissible statistics for a multivariate population,¹ since there may be information in the group statistics that is not visible in individual statistics. In the context of quantitative portfolio optimisation and asset allocation, this in turn implies that there are ways of improving simple asset class return means as inputs for an optimisation procedure. More precisely, it is necessary to apply multivariate estimation of asset return means in order to capture all of the relevant information contained in these. Simultaneous estimation of asset

returns, volatilities and correlations can, under certain conditions, contain information that is not present in univariate estimation of individual characteristics. This additional information can so to speak be *added back* to the estimates by the use of priors.

The Stein-type estimation techniques are essentially estimation error-correction techniques. They are statistical techniques designed to reduce the effect of the estimation error that is always present in sampled data. As already discussed, pure and 'naive' quantitative optimisation is designed to seek viable mathematical solutions without regard for investment sense or intuition. It scrupulously exploits the opportunities presented by the model inputs to arrive at a solution as extreme as possible. Another way of saying this is that quantitative optimisation has the effect of exacerbating any estimation errors that may be present in the data. Unfortunately for quantitative asset managers, the largest estimation errors tend to appear in the worst possible areas. This is because a quantitative portfolio optimiser will construct portfolios of assets with large expected returns, low volatilities and low correlations. And these just happen to be the characteristics that exhibit the highest degree of estimation error. The reason for this is that the number of observable observations available that can characterise these events is typically quite low, so the basis for decision-making on these issues can be quite weak. Therefore it makes sense to assume that the quality of information provided by individual observations is low, and so we adjust each observation relative to a prior whose statistical foundation is likely to be somewhat firmer.

Shrinkage

As noted, applying Stein estimators and Bayesian priors is aimed at reducing the effect of estimation errors by estimating model inputs in a statistically accurate and sound manner, while also imposing some measure of exogenous insight upon the inputs, for example in the shape of long-term average returns, volatilities and correlations that may help guide the determination of appropriate model inputs.

The basic premise in estimating model inputs from historical data is to let the data speak for itself. This is fundamentally a sound principle, provided that there is enough data to estimate inputs properly. The *cure* for the problem of insufficient amounts or low quality of data is to impose exogenous structure on the estimates. The exact type and amount of structure to be imposed should be determined by the problem at hand. In the case of asset class returns, a viable structure could be imposed by introducing very long run historical returns. Say for example that an asset manager has some sort of long-term estimates of asset class returns. She then develops short-term forecasts – or

samples – for these same asset classes, which result in wide dispersion among expected future returns.

In an attempt to control any estimation errors that may be present, and thus to avoid the optimisation procedure producing extreme results based on the short-term forecasts, the asset manager can then *shrink* the individual estimates towards the *grand mean* of long-term asset class returns. The grand mean of long-term returns for all asset classes thus acts as a prior or *anchor* which guides the determination of the expected asset class returns that are to be used as inputs for the quantitative optimisation and asset allocation procedure. Typically, the extent to which the individual returns are shrunk is determined by some measure of their distance from the grand mean. The further away they are and the more variance they exhibit, the more they are shrunk towards the grand mean. This will result in more similar expected returns – but still with a bias towards the actually estimated short-term returns – yielding a smaller risk that the optimiser may produce extreme, impractical and unusable asset allocation recommendations.

OPTIMAL RETURN SHRINKAGE

The notion of an *optimal* level of shrinkage of historical asset class mean returns towards some prior can perhaps be a bit misleading. Recall that the notion of optimal portfolios is contentious because real optimality requires perfect foresight. The same applies to the notion of optimal input shrinkage. Whether shrinkage is optimal or not can of course only be determined *ex post*, and the result will inevitably depend heavily on the selected prior, which is more or less arbitrarily chosen. However, there are several ways of shrinking model inputs towards a prior while still maintaining some form of optimality in a statistical sense. Two statistical methods are presented below. They both attempt to revise or adjust the estimated model inputs so as to cause the quantitative optimisation algorithm to act as if the estimation risk were already explicitly incorporated in the original optimisation model.

Stein-I shrinkage

The first approach – here called Stein-I – is the classic James-Stein estimator,² whose result is a vector $\vec{r}_{s1}$ of estimation-error-adjusted historical asset class returns. The technique presented below is essentially an estimation-error correction technique, which yields a vector of adjusted asset class returns, based on historical returns $\vec{r}_H$ and the grand mean of historical returns $\vec{r}_G$. Formally it is expressed as follows:

$$\vec{r}_{S1} = (1 - \xi) \cdot \vec{r}_H + \xi \cdot \vec{r}_G \bullet \vec{e} \quad [\text{Eq. 11.1}]$$

where

$$\vec{r}_H = \begin{bmatrix} \bar{r}_1 \dots \bar{r}_N \end{bmatrix} \quad [\text{Eq. 11.2}]$$

is the vector of N historical means of each asset class, so that:

$$\bar{r}_i = \frac{1}{k} \sum_{\tau=t-k}^{t-1} r_{i,\tau} \quad [\text{Eq. 11.3}]$$

where k is the number of observations in the estimation period for asset class i . e is a vector of 1s.

The grand mean of historical asset class return means in Equation 11.1 is defined by:

$$\vec{r}_G = \frac{1}{n} \sum_j^N \bar{r}_i \quad [\text{Eq. 11.4}]$$

The *shrinkage factor* ξ employed in Equation 11.1 is defined as:

$$\xi_{S1} = \text{Min} \left[1, \frac{N - 2}{k \cdot (\vec{r}_H - \vec{r}_G \bullet \vec{e})^T \bullet \Sigma^{-1} \bullet (\vec{r}_H - \vec{r}_G \bullet \vec{e})} \right] \quad [\text{Eq. 11.5}]$$

Equation 11.1 shrinks the historical means towards the grand mean by the shrinkage factor defined in Equation 11.5 above. The shrinkage factor ranges from 0 to 1, and the higher (closer to 1) the shrinkage factor is, the more the means are shrunk towards the grand mean of historical asset class returns.

The nature of the shrinkage factor ξ deserves some attention. It depends on the number of asset classes, the number of observations employed for historical return estimates, the variance–covariance matrix Σ , and the distance between individual historical asset class means and the grand mean of all asset class returns. Examining Equation 11.5 more closely, we see that the amount of shrinkage of asset class returns depends on the distance of the individual historical asset class returns from the grand mean of historical returns, and also on asset return variability. More specifically, shrinkage increases with a smaller number of observations, with a higher number of asset classes, with a larger distance between historical and grand mean returns, and with a higher overall level of covariance (that is, higher correlations and/or volatilities) in the portfolio.

The intuitions behind these mechanisms are that fewer observations mean that the risk of individually significant estimation errors is larger, and so there

is more reason to shrink estimates towards a common historical mean. Furthermore, the higher the number of asset classes, the higher the risk of estimation error, since as the number of asset classes is increased, the number of means and thus their estimation error increases linearly, while the number of covariances and thus their estimation error increases quadratically. So for large portfolios the estimation-error problem may be seriously compounded. This is why the application of these estimation-error-dampening techniques grows very important for large portfolios with a large number of assets or asset classes.

Lastly, the greater the distances between historical return means and the grand mean, the greater the likelihood of estimation error, and thus the greater the amount of shrinkage. And finally, the higher the general level of covariance – that is, the higher the levels of correlations and/or volatilities between asset classes – the larger the amount of shrinkage, since larger numerical values tend to imply higher levels of absolute estimation error. Each of these three factors is influential in determining the exact size of the appropriate shrinkage factor to be applied to the historical asset class returns. Perhaps the most important factor in determining the amount of shrinkage is the number of observations available for the estimation of the returns.

Continuing the analysis of the MSCI Global Sector Model, Figure 11.1 illustrates the relationship between the size of the appropriate shrinkage factor and the number of observations available for estimation of historical asset class returns.

Figure 11.1 demonstrates that shrinkage depends heavily on the number of observations available for estimation of returns. If only one observation is

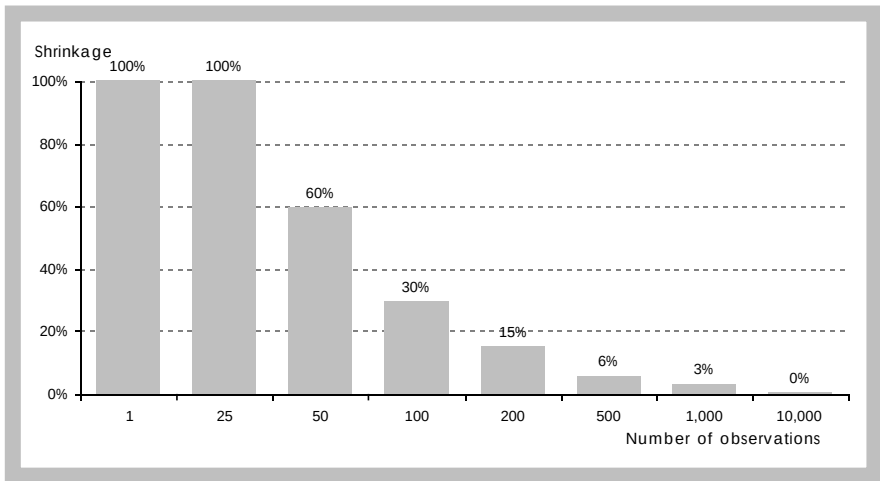


Figure 11.1 Optimal shrinkage factor – Stein-I

available we will of course end up with 100% shrinkage of historical asset class returns towards the grand mean. Even 25 observations require 100% shrinkage, but 50 observations require 60% shrinkage. As soon as we move up to a more realistic number of observations, the optimal shrinkage becomes quite a bit smaller. Using monthly observations and estimating over a 5-year period gives the asset manager 60 observations, requiring shrinkage of some 50% of historical mean returns towards the grand mean.

Using 5 years of weekly observations would yield 260 observations, and thus an optimal shrinkage factor of 11.5%. Using daily observations over 5 years of 261 trading days would yield 1305 observations, and an optimal shrinkage factor of only 2.3%. Precisely how much shrinkage is optimal for the portfolio of MSCI Global Sectors can be illustrated by plotting the optimally shrunk returns against varying numbers of available observations (Figure 11.2). With just 1 or even 25 observations all asset class returns are shrunk to the historical grand mean, because the lack of observations makes individual estimates too uncertain to include unadjusted in a quantitative optimisation algorithm. Note again how as the number of observations is increased, the individual asset class returns tend towards their respective estimated historical means, finally ending up with zero shrinkage, a 0% weighting of the grand mean, and a 100% weighting of individual historical means. Of course, typically asset managers will have a lot more data to work with.

Applying the Stein-I optimal shrinkage estimator based on 88 monthly observations from January 1995 to May 2002, the optimal shrinkage factor is

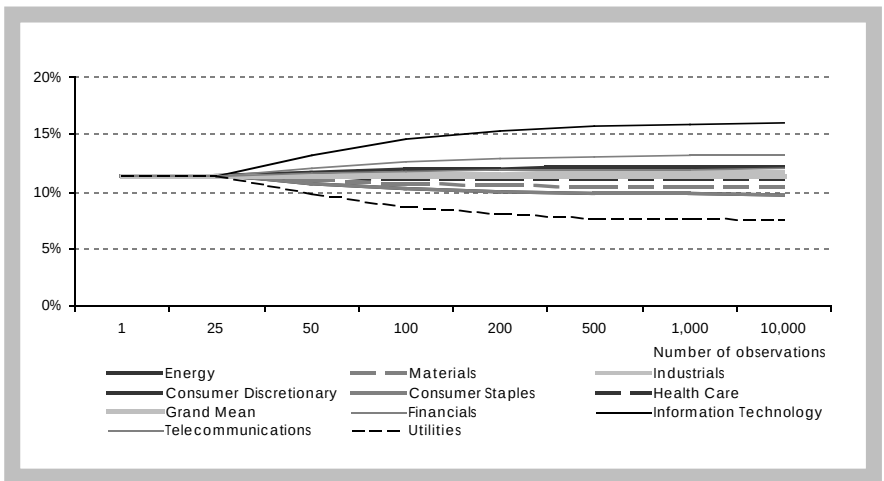


Figure 11.2 Optimally shrunk historical returns – Stein-I

Table 11.1 Stein-I optimal shrinkage adjusted returns – MSCI Global Sector

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	10.2%	17.1%	1.00	0.64	0.60	0.41	0.36	0.31	0.55	0.33	0.15	0.46
Materials	10.6%	17.2%	0.64	1.00	0.81	0.70	0.50	0.36	0.70	0.44	0.25	0.28
Industrials	11.5%	15.2%	0.60	0.81	1.00	0.87	0.57	0.47	0.83	0.66	0.48	0.46
Cons. Disc.	11.8%	16.8%	0.41	0.70	0.87	1.00	0.40	0.35	0.73	0.78	0.67	0.28
Cons. Staples	10.2%	13.1%	0.36	0.50	0.57	0.40	1.00	0.67	0.68	0.16	0.22	0.55
Health Care	11.1%	14.6%	0.31	0.36	0.47	0.35	0.67	1.00	0.60	0.23	0.30	0.44
Financials	11.8%	18.9%	0.55	0.70	0.83	0.73	0.68	0.60	1.00	0.47	0.42	0.50
IT	14.4%	33.0%	0.33	0.44	0.66	0.78	0.16	0.23	0.47	1.00	0.70	0.13
Telecomm.	12.6%	19.6%	0.15	0.25	0.48	0.67	0.22	0.30	0.42	0.70	1.00	0.18
Utilities	8.8%	10.5%	0.46	0.28	0.46	0.28	0.55	0.44	0.50	0.13	0.18	1.00

33.8%. Applying this shrinkage factor to the original expected sector returns presented in Table 9.1, yields the – estimation-error corrected – expected returns for the 10 MSCI Global Sectors shown in Table 11.1. Note that the expected returns are now much more similar after having been adjusted for estimation error. This amount of shrinkage may seem a little non-intuitive, but it simply stems from the fact that we only have 88 monthly return observations: the estimation uncertainty or expected estimation error results in the Stein-I estimator *pulling* the expected returns heavily towards the grand mean of returns. Please note that neither expected volatilities nor correlations are affected by this shrinkage, as it focuses solely on estimation error in expected returns.

The adjusted model inputs yield the new efficient frontier shown in Figure 11.3. Because the expected returns are now much more similar than before, the

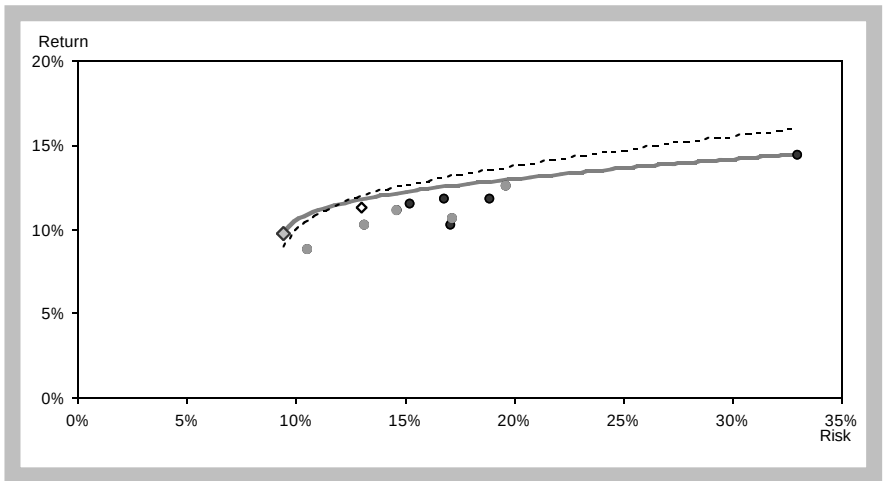


Figure 11.3 Efficient frontier – Stein-I return shrinkage

efficient frontier is somewhat flatter than was the case for the original efficient frontier. Not surprisingly, though, the efficient frontier is placed *higher* in the low-risk spectrum and *lower* in the high-risk spectrum than was the case for the original efficient frontier, which is indicated by the dotted line. This is because the returns of low-risk asset classes have been shrunk upwards towards the grand mean, and the returns of the high-risk assets have been shrunk down towards the grand mean of returns. In addition, the expected volatility of the information technology sector is the highest among the ten sectors, and so this sector's expected return is shrunk the most. All this has the effect of *twisting* the efficient frontier around the centre of the expected return spectrum, thereby flattening the efficient frontier. The resultant *naïve* asset allocation chart appears as in Figure 11.4.

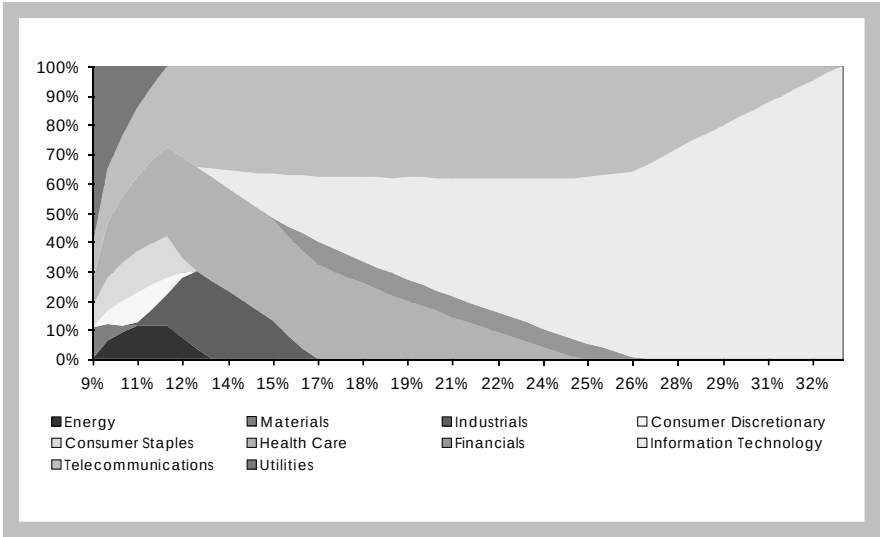


Figure 11.4 Asset allocation topography – Stein-I

Note that most of the participation of low-risk sectors occurs only very close to the minimum-variance portfolio, that the information technology sector is included just as consumer discretionary and consumer staples are dropped, and that the financials sector is included just as the industrials sector is dropped. This shows how the tug of war between asset characteristics (principally correlation coefficients) changes dynamically as we move along the risk spectrum.

As we have also seen earlier, the changes in portfolio composition occur quite rapidly and violently as we move along the risk spectrum. In addition, for target volatility levels above 15% only a handful of the available sectors are left in the recommended portfolios, which bypasses many of the opportunities for diversification presented in the portfolio. Moving beyond 26% as a target volatility, which is in fact a very common level of volatility for most managed portfolios, effectively means having just two assets in the portfolio. This is clearly unacceptable as a practical guide for asset allocation. However, when we apply the QRMCSAA methodology we obtain the asset allocation chart shown in Figure 11.5.

Compared to the original unadjusted QRMCSAA in Figure 10.10, we see that the information technology sector now plays a somewhat smaller role in the asset allocation as we move along the risk spectrum from the minimum- to the maximum-variance portfolio. Similarly, the telecommunication services sector is reduced for almost all levels of target volatility. However, the materials and industrials, as well as the two consumer sectors, are increased.

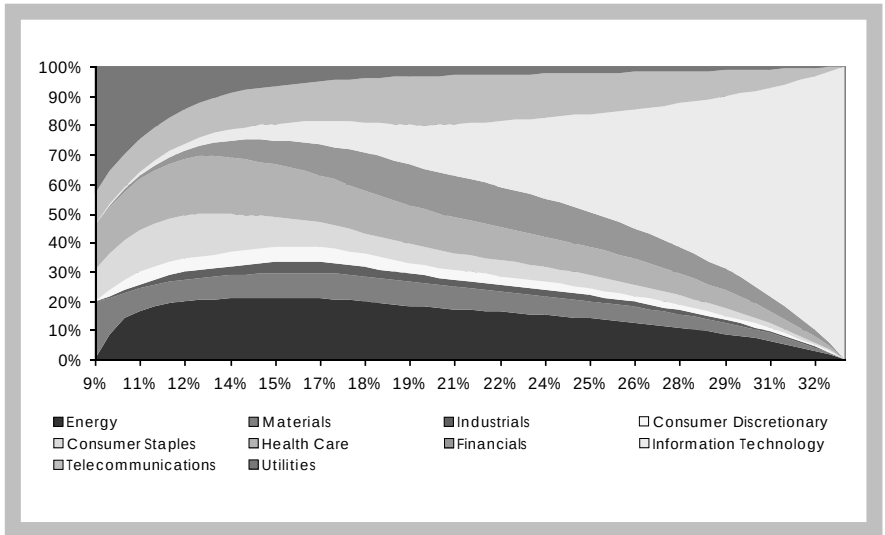


Figure 11.5 QRMCSAA – Stein-I return shrinkage

Overall, the tendency of this QRMCSAA asset allocation chart is that it is more similar to the equal-weighted portfolio, which would have had 1/10 of the assets in the portfolio placed in each sector for all levels of target volatility. This difference makes good sense – realising that you have absolutely no information at all, you would place your assets in equal amounts in all sectors. Here, however, while the level of information is low due to the relatively small number of observations, we do have some information, so the asset allocation *topography* is somewhere in between the two states of either having no information at all (in which case equal-weighting all sectors is optimal) or having perfect information. In addition, it appears that the low-risk sectors are more prominent in this asset allocation across the risk spectrum, which also makes good sense – on a relative basis, their return/risk trade-off (the Sharpe Ratio) has become more attractive. So with an unchanged correlation matrix the optimisation algorithm is more likely to include larger proportions of these sectors in the optimal portfolios.

The Stein-I estimator can be criticised on a number of issues, most importantly the fact that it shrinks expected returns without any intelligent guidance, but simply for the sake of estimation-error adjustment. Most asset managers will probably not feel comfortable with the efficient frontier it produces either, although the resultant QRMCSAA asset allocation topography makes a lot more sense than the *naïve* asset allocation. We now attempt to remedy this.

Stein-II shrinkage

The second shrinkage approach – here called Stein-II – for optimal shrinkage of expected returns, was developed by Jorion (1986/1991). It is a Bayes-Stein estimator based on the same methodology as the Stein-I estimator, but instead of shrinking returns towards the mean of asset class historical returns, the shrinkage target is the return on the minimum-variance portfolio. The rationale behind this is that since the minimum-variance portfolio is the portfolio with the universally lowest level of attainable volatility, shrinkage towards this particular portfolio will entail some level of automatic yet purposeful risk control in addition to the effects of the shrinkage itself. In other words, this type of shrinkage is intelligent and well thought out, as opposed to the Stein-I estimator which is less sophisticated.

The Stein-II estimator appears as follows:

$$\vec{r}_{S2} = (1 - \xi) \cdot \vec{r}_H + \xi \cdot \vec{r}_G \cdot \vec{e} \quad [\text{Eq. 11.6}]$$

The grand mean towards which all asset class historical returns will be shrunk is the return on the minimum-variance portfolio:

$$\bar{r}_G = \frac{\vec{e}^T \cdot \Sigma^{-1} \cdot \vec{r}_H}{\vec{e}^T \cdot \Sigma^{-1} \cdot \vec{e}} \quad [\text{Eq. 11.7}]$$

Applying this formula yields a return on the minimum-variance portfolio of 8.44%. The shrinkage factor for the Stein-II methodology is defined as:

$$\xi_{S2} = \frac{s}{(s + k)}$$

where

$$s = \frac{N + 2}{(\vec{r}_H - \vec{r}_G \cdot \vec{e})^T \cdot \Sigma^{-1} \cdot (\vec{r}_H - \vec{r}_G \cdot \vec{e})} \quad [\text{Eq. 11.8}]$$

Even though this shrinkage estimator can be said to be just as arbitrary in its choice of parameters as the previous (Stein-I) estimator, there is some logic to its formulation. Instead of just shrinking all returns towards the same mean simply because for statistical reasons they need shrinking of some magnitude, the Stein-II estimator shrinks purposefully towards the global minimum risk portfolio. In other words it *pulls* returns towards that which would result if the investor was interested in the least amount of risk possible, thus *aiming* or *tilting* the portfolio characteristics in a *risk-reducing* direction by embedding a bias towards the least risky portfolio attainable. For the MSCI Global Sector Model,

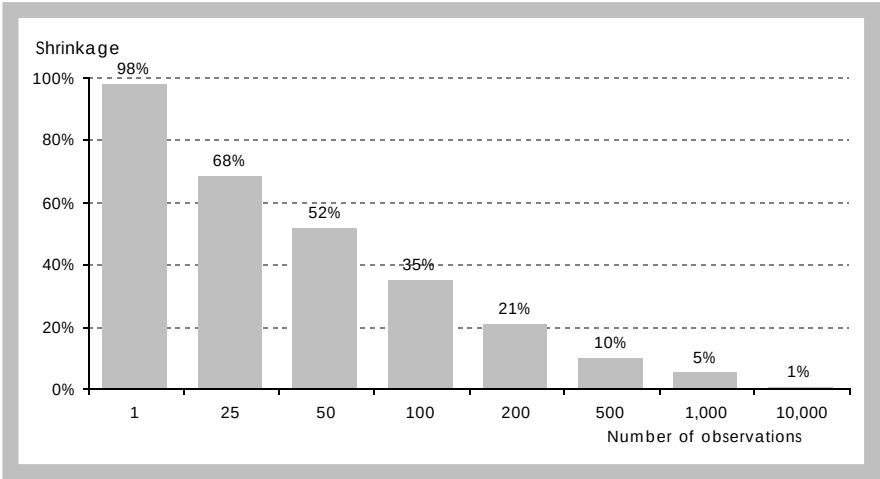


Figure 11.6 Optimal shrinkage factor – Stein-II

the return on the minimum-variance portfolio is 8.44%. Using the Stein-II technique yields optimal shrinkage factors that appear as in Figure 11.6.

As indicated, with just one observation the returns must be shrunk almost completely towards the return on the minimum-variance portfolio. Three years' worth of monthly data (that is, 36 months) implies an optimal shrinkage factor of 59%, and five years' worth of data (that is, 60 months) implies an optimal shrinkage factor of 47%. The optimally shrunk returns for varying numbers of observations appear as in Figure 11.7

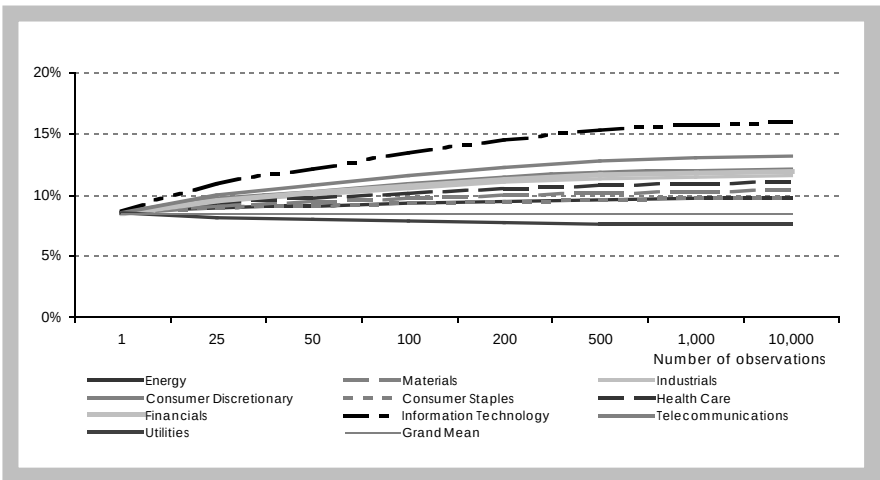


Figure 11.7 Optimally shrunk historical returns – Stein-II

Table 11.2 Stein-II optimal shrinkage-adjusted returns

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	9.1%	17.1%	1.00	0.64	0.60	0.41	0.36	0.31	0.55	0.33	0.15	0.46
Materials	9.4%	17.2%	0.64	1.00	0.81	0.70	0.50	0.36	0.70	0.44	0.25	0.28
Industrials	10.1%	15.2%	0.60	0.81	1.00	0.87	0.57	0.47	0.83	0.66	0.48	0.46
Cons. Disc.	10.4%	16.8%	0.41	0.70	0.87	1.00	0.40	0.35	0.73	0.78	0.67	0.28
Cons. Staples	9.1%	13.1%	0.36	0.50	0.57	0.40	1.00	0.67	0.68	0.16	0.22	0.55
Health Care	9.8%	14.6%	0.31	0.36	0.47	0.35	0.67	1.00	0.60	0.23	0.30	0.44
Financials	10.3%	18.9%	0.55	0.70	0.83	0.73	0.68	0.60	1.00	0.47	0.42	0.50
IT	12.4%	33.0%	0.33	0.44	0.66	0.78	0.16	0.23	0.47	1.00	0.70	0.13
Telecomm.	10.9%	19.6%	0.15	0.25	0.48	0.67	0.22	0.30	0.42	0.70	1.00	0.18
Utilities	8.0%	10.5%	0.46	0.28	0.46	0.28	0.55	0.44	0.50	0.13	0.18	1.00

Compared to the Stein-I estimator, returns are shrunk less aggressively for fewer observations, and also more smoothly for an increasing number of observations. Given the level of information provided by the 88 monthly observations, the optimally shrunk expected returns according to the Stein-II technique are as shown in Table 11.2. These expected returns are a lot lower than was the case either for the original set of returns or for the Stein-I adjusted returns. This obviously means that the expected portfolio returns will be lower across the entire risk spectrum from the minimum to the maximum-variance portfolio. One could consider adding this difference back to the adjusted returns, but in reality it makes no material difference to results. This is because the levels of sector returns are unchanged relative to the average return, and so the results of the optimisation is not affected. The Stein-II adjusted returns are more diverse than was the case for the Stein-I method, and so we obtain a slightly more intuitive-looking efficient frontier with a more appealing curvature upwards and to the right in return/risk space (Figure 11.8).

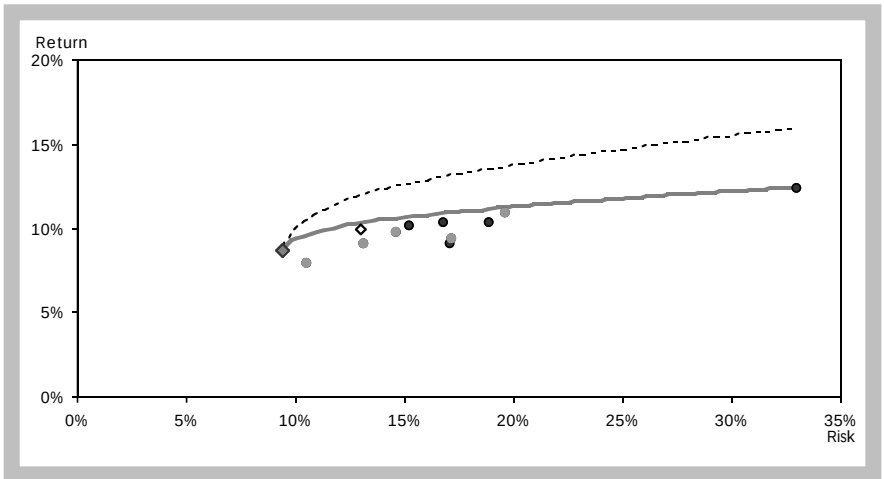


Figure 11.8 The efficient frontier – Stein-II return shrinkage

Using the Stein-II estimator yields more intuitive results than the Stein-I. We now have a more reasonable efficient frontier, but one that is quite a bit lower than the original efficient frontier indicated by the dotted line. Note how expected sector returns have been shrunk or *pulled* quite heavily towards the return on the minimum-variance portfolio, and also how close the equal-weighted portfolio is to the efficient frontier. This hints at the fact that many optimal portfolios may not be statistically significantly different from portfolios that are not situated on the efficient frontier itself.

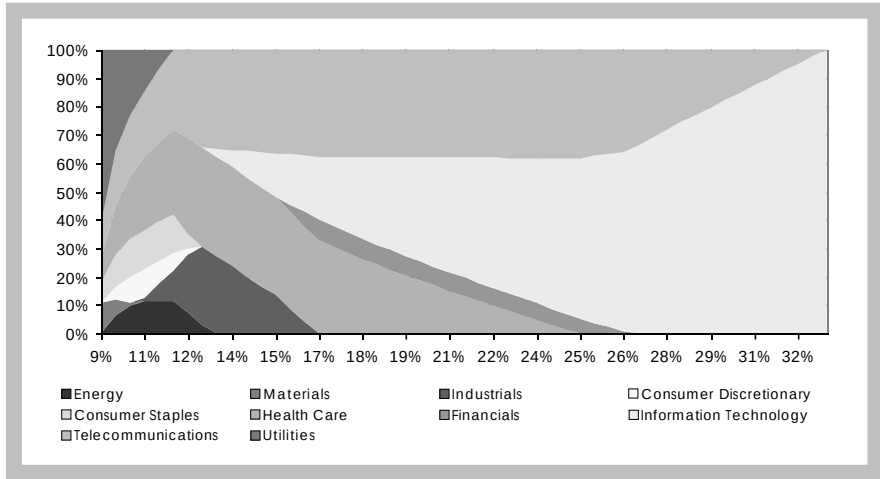


Figure 11.9 Asset allocation topography – Stein-II

The *naïve* asset allocation topography that results from these optimally shrunk returns now appears in Figure 11.9.

The similarity with Figure 11.4 may have to do with the fact that the variance–covariance matrix that enters into the shrinkage estimator is the same as that of the Stein-I estimator. The only difference is the return towards which the expected sector returns are shrunk.

Applying the QRMCSAA methodology we obtain Figure 11.10. Compared to Stein-I we again see that this asset allocation is very similar to the QRMCSAA

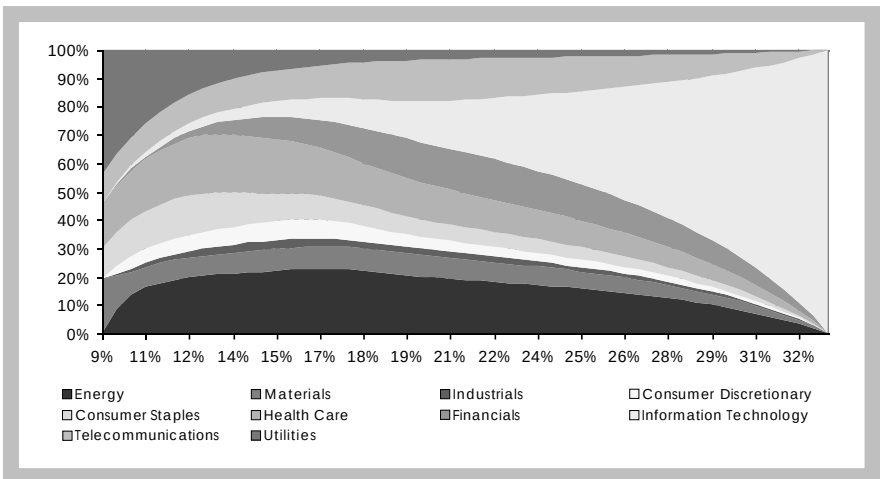


Figure 11.10 QRMCSAA – Stein-II

chart of the Stein-I estimator presented in Figure 11.5. The most notable differences from the QRMCSAA chart produced by the Stein-I technique are the decrease in the recommended weights in telecommunication services and the increase in the energy sector. Overall the QRMCSAA chart is less like that of the equal-weighted portfolio, which makes sense since the sector returns have been shrunk significantly less towards a common prior using the Stein-II technique.

Applying optimal return shrinkage as a practical tool

What we have demonstrated above is that optimal return shrinkage has an effect on the optimal QRMCSAA, and that this effect is explainable and intuitive. As we indicated at the beginning of this chapter, applying shrinkage to return estimates is more an art than a science, and whether to use Stein-I, Stein-II or some other return shrinkage estimator is very much up to the asset manager.

Stein-I shrinkage is carried out based only on estimation uncertainty or expected estimation error, whereas Stein-II shrinkage is carried out purposefully towards the return on the minimum-variance portfolio. Both shrinkage factors are in principle open to augmentations, but the overriding message that must be conveyed here is that simply ignoring these methods and not choosing any of them is in fact *also* a choice, and perhaps even the worst choice of all. Given an inadequate amount of information or a relatively low level of quality of information, the asset manager must decide on what method to apply, if any; but in the end it is a decision that must be made by the asset manager, whether he actively makes it or not. Simply sticking one's head in a bush and ignoring the subject does not make the problems and pitfalls of quantitative asset allocation go away. In this regard, not making a choice, actually constitutes a choice in itself.

OPTIMAL COVARIANCE MATRIX SHRINKAGE

The grand mean approach

Until now we have examined techniques for adjusting expected returns in order to obtain more reliable asset allocation results. However, given the impact on portfolio characteristics – both *plain vanilla* portfolios and optimised portfolios – of the correlation structure present in the investment universe, it goes without saying that statistical techniques for dealing with estimation errors and other issues must be examined for the correlation/covariance matrix as well.

We begin by examining an approach to correlation matrix adjustment that is equivalent to the return grand mean approach presented above. This type of

technique is obviously the simplest approach to adjusting expected returns, volatilities or correlations, in that it essentially stipulates that all individual returns, volatilities or correlations are to be set equal to their respective historical means. Given that volatilities are reasonably stable over time, and that some form of forecast of returns is usually present in the portfolio optimisation framework, one widely used variant of the grand mean approach is to assume that all correlations are equal to the grand mean of correlations.

In terms of imposing structure on the estimator, one could argue that this is a case of going to extremes.³ It is essentially equivalent to accepting that we have no forecasting ability whatsoever when it comes to correlations. Indeed, correlations can be very difficult to forecast, and they do exhibit substantial variation over time. Accepting this fact, all we can really do is to look at the average historical correlation coefficient between all asset classes, and then simply use that average as a substitute for every single coefficient in our variance–covariance matrix, except for those on the diagonal.

The average correlation coefficient for the MSCI Global Sector Model is 0.48, which means that the optimisation inputs now appear shown in Table 11.3.

Even though the average or *grand mean* of the correlation matrix is 0.48, the diagonal still consists of 1s since every asset class’s covariance with itself is equal to 1 regardless of whether we adjust overall correlation levels or not. Expected returns and volatilities are unchanged relative to the original MSCI Global Sector Model example presented in Table 9.1.

The efficient frontier for this portfolio now appears as shown in Figure 11.11. The dotted line again indicates the *original* efficient frontier using correlations

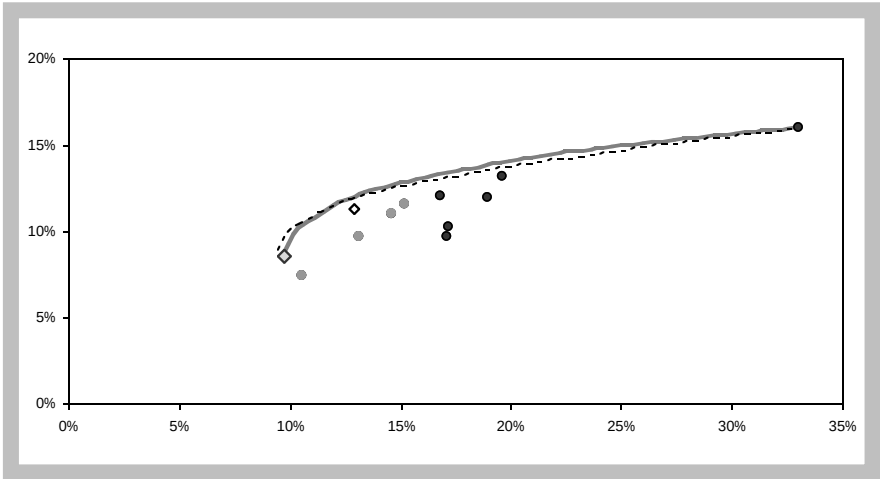


Figure 11.11 The efficient frontier – equal correlations

Table 11.3 Equal correlations – grand mean approach

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	9.7%	17.1%	1.00	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
Materials	10.3%	17.2%	0.48	1.00	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
Industrials	11.6%	15.2%	0.48	0.48	1.00	0.48	0.48	0.48	0.48	0.48	0.48	0.48
Cons. Disc.	12.1%	16.8%	0.48	0.48	0.48	1.00	0.48	0.48	0.48	0.48	0.48	0.48
Cons. Staples	9.7%	13.1%	0.48	0.48	0.48	0.48	1.00	0.48	0.48	0.48	0.48	0.48
Health Care	11.0%	14.6%	0.48	0.48	0.48	0.48	0.48	1.00	0.48	0.48	0.48	0.48
Financials	12.0%	18.9%	0.48	0.48	0.48	0.48	0.48	0.48	1.00	0.48	0.48	0.48
IT	16.0%	33.0%	0.48	0.48	0.48	0.48	0.48	0.48	0.48	1.00	0.48	0.48
Telecomm.	13.2%	19.6%	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	1.00	0.48
Utilities	7.5%	10.5%	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	1.00

estimated on the basis of historical data, and once more the grey dots indicate the assets that are present in the minimum-variance portfolio, and the white diamond indicates the expected return and risk of the equal-weighted portfolio.

From Figure 11.11 we see that compared to the original efficient frontier the new equal correlations efficient frontier indicates less opportunity for diversification in the low-risk spectrum (the left side of the chart), and more opportunity for diversification in the medium- to high-risk spectrum (the right side of the chart). In fact for target volatility levels above approximately 12%, the equal correlations MSCI Global Sector Model indicates more expected return per unit of risk (essentially a higher Sharpe Ratio) than was the case for the efficient frontier generated by the original correlation/covariance matrix. In other words, the asset manager obtains a higher expected return for a given level of high expected risk and a lower level of expected return for a lower level of low expected risk. This in turn indicates a riskier overall investment portfolio, which becomes even clearer when we examine the asset allocation.

Figure 11.12 illustrates how the asset allocation along the efficient frontier is expected to look based on the equal correlations approach. The chart does indeed indicate that the resultant optimal portfolios that are recommended based on the equal correlations approach may well turn out to be riskier than the original portfolios. Note that less risky asset classes are kept in the portfolio for longer than before, that the energy sector is hardly present at all, that the health care sector is severely reduced, that the materials sector is now completely excluded, and that the erratic behaviour of portfolio weights is still present but has *shifted* a bit further out into the risk spectrum.

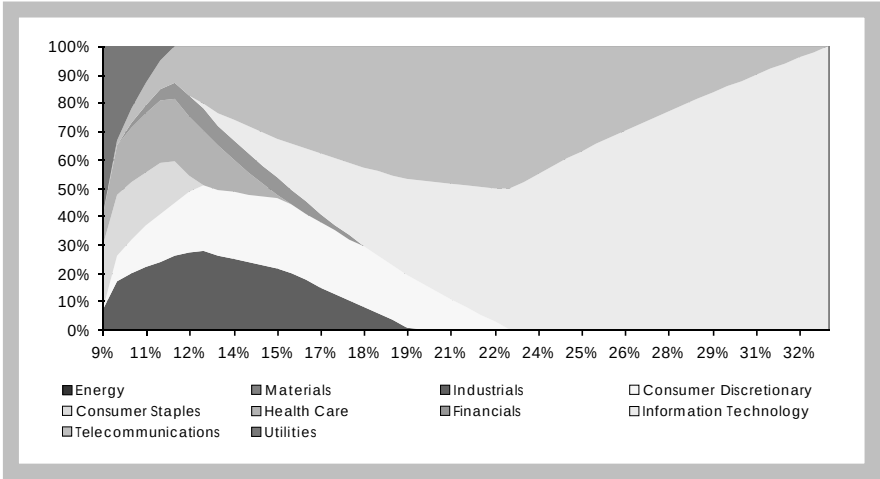


Figure 11.12 Asset allocation – equal correlations

Note in particular the *kink* in the proportion of the portfolio invested in both the telecommunication services and the information technology sectors around a volatility level of approximately 21%. The proportion of the portfolio invested in the telecommunication services sector drops from 50% to 0% within just 10 percentage points along the horizontal axis of target volatility. The reason for health care being so overlooked as we move along the risk spectrum is that its correlation coefficients with the other asset classes have changed dramatically, seriously weakening its ability to provide diversification in the portfolio relative to the other assets. Traditionally, health care and pharmaceuticals as well as biotechnology stocks have had a low correlation with technology and telecommunications stocks, and the effect of a change in that correlation coefficient is clear from the asset allocation topography above.

Applying the QRMCSAA methodology yields the asset allocation chart shown in Figure 11.13. Not surprisingly, the asset allocation chart shows recommended asset allocations across the risk spectrum that are more like the equal-weighted portfolio – that is, the portfolio in which all sectors are assigned the same weight across the entire risk spectrum. Overall the major difference between this asset allocation chart and the original unadjusted asset allocation chart presented in Figure 10.10 is thus that sectors tend to be much more equally represented across the risk spectrum, owing to the fact that the covariances (the products of volatilities and correlations between sectors) are more similar under the assumption of equal correlations. There is thus less opportunity for the quantitative optimiser to seek diversification among sectors.

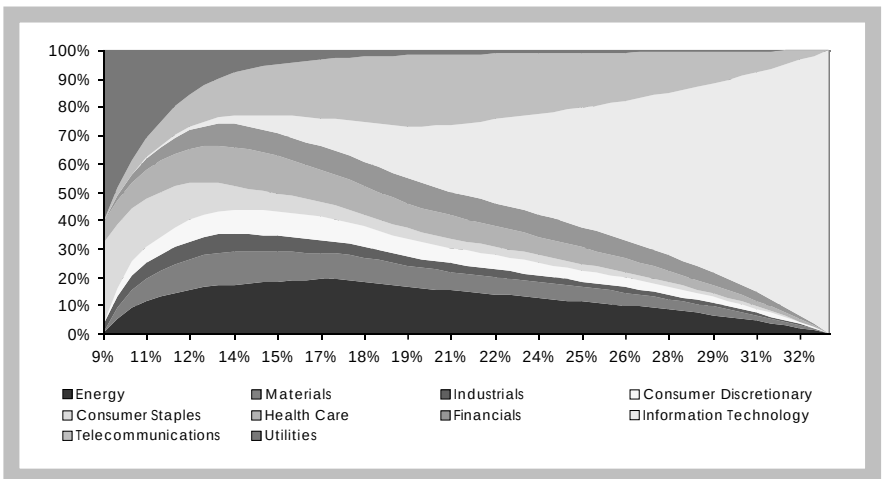


Figure 11.13 QRMCSAA topography – equal correlations

Although the equal correlations approach is applied in some asset management contexts, it is not very clear what the precise justification for it might be. Presumably it is an attempt to dampen erratic portfolio asset allocation behaviour, but as demonstrated above this may not be possible through such a technique alone. The equal correlations approach may seem like a rather naive approach to optimisation, and indeed the assumption that all correlations are identical somewhat undermines the whole idea of attempting to quantitatively exploit the opportunities for diversification that particular asset class characteristics present. One might even say that applying equal correlations to the asset allocation process is the ultimate surrender to the gods of inaccurate input estimation. It is equivalent to accepting that one has no ability whatsoever to forecast asset class correlations, and not many asset managers will concede that point.

Optimal covariance matrix shrinkage

As already pointed out, the problem of estimation error may be severe for returns but is likely to be much worse for covariances and correlations, particularly in large portfolios since the number of returns increases linearly with the number of assets, whereas the number of covariance or correlation terms increases quadratically with the number of assets. This is why estimation error problems are particularly serious for the variance–covariance matrix, because it consists of so many elements, and also why the observed near-exclusive interest in and emphasis on return forecasts by asset allocators may be somewhat misguided. At the end of the day, returns matter to clients, but return generation without sufficient understanding and appreciation of the risks incurred in bringing these returns about may become the downfall of many an asset manager during periods when risk aversion is particularly high and markets are extremely volatile.

There exist a plethora of variance–covariance matrix estimators, but few employ shrinkage of the matrix to correct for potential estimation error. Typical shrinkage targets are the *equal correlations* matrix, the *identity matrix*,⁴ the *pseudo-inverse* or *Moore–Penrose inverse*,⁵ the *Principal Component Analysis* (PCA) based matrix,⁶ and the *factor model based matrix*.⁷

Traditional Bayes–Stein estimators, including the two return shrinkage estimators presented above, are often inherently problem-ridden in at least two important respects. Firstly, they break down when the number of variables (asset classes) is less than the number of observations available for the estimation of inputs. This is also known as *ill-conditioning*, and leads to the variance–covariance matrix not being invertible, which in turn is a necessary condition for the application of both Stein-I and Stein-II.⁸ Secondly, they do not exploit the *a priori* know-

ledge that stock return correlations are almost always positive, which provides an intuitive prior for the asset allocator. A third and quite serious problem in the formulation and use of shrinkage estimators is a problem of logical inconsistency which seems to persist in most shrinkage estimators, and that may result in an impaired value of the Bayes-Stein-type estimators. It results from the fact that the prior is often estimated from the available sample data, but at the same time it is assumed to be independent from that same sample data.

Ledoit (1999) deals with these problems, firstly by introducing a covariance estimator that determines optimal shrinkage based on the minimisation of a loss function that does not involve the inverse of the variance–covariance matrix. Secondly it introduces as its prior the variance–covariance matrix of the market index containing all equities,⁹ thus imposing structure on the estimator while at the same time exploiting the knowledge that equity correlations tend to lie in the middle of the positive range of values – that is, between 0 and 1. Thirdly, the estimation includes a correlation term which measures correlation between the estimation errors on the sample covariance matrix and the estimation error on the shrinkage target covariance matrix. This requires a bit of explaining. Intuitively, if the two estimation errors that may be present in the sample covariance matrix and the shrinkage target covariance matrix are positively correlated, then potential benefits from combining the information they contain will be smaller. Conversely for negatively correlated estimation errors, which result in larger benefits from combining the information contained in the two matrices. Therefore this correlation term must be accounted for.

The estimator essentially uses the same shrinkage methodology as the two Stein estimators of returns, but it shrinks the historical sample covariance matrix towards the covariance matrix of the *market index*. The two matrices are characterised by their respective levels of bias and estimation error. The bias is towards the in-sample model parameters resulting from a heavily specified market index model. The more parameters are present, the more explanatory power there is in-sample but also the less explanatory power out-of-sample, due to *over-fitting* of the model. The market index covariance matrix is generated by performing individual regressions of the returns on every component of the market M , on the return of the market. The regressions appear as follows:

$$r_{it} = \alpha_i + \beta_i r_{Mt} + \varepsilon_{it} \quad [\text{Eq. 11.9}]$$

α_{it} is the abnormal return on asset i at time t . β_i is the exposure of asset i to the market index M , and ε_{it} is the standard OLS regression residual left unexplained by the market index which is obviously the only explanatory variable employed here.¹⁰ Such regressions are performed for every asset class. Expressing this in matrix terms yields:

$$\vec{r} = \vec{\alpha} + \vec{\beta} \cdot r_M + \vec{\varepsilon} \quad [\text{Eq. 11.10}]$$

The covariance matrix implied by this model is consequently:

$$\Phi = \sigma_M^2 \cdot \vec{\beta}^T \cdot \vec{\beta} + \Delta \quad [\text{Eq. 11.11}]$$

where σ_M^2 is the variance of the market returns, and Δ is the diagonal matrix of asset class-specific variances of regression residuals that cannot be explained by the market index.

The objective of the optimal shrinkage procedure is now to find the optimal trade-off between the sample covariance matrix Σ (based on observed asset class returns), and the market index covariance matrix Φ (based on individual regressions of asset classes). Since the required shrinkage tends to zero as the number of observations N goes towards infinity, the standard shrinkage factor will be of the general form 'Constant'/ N . Consequently the goal is to determine the constant, here called φ . The optimally shrunk covariance matrix is thus given by:

$$\tilde{\Sigma} = \frac{\varphi}{N} \cdot \Phi + \left(1 - \frac{\varphi}{N}\right) \cdot \Sigma \quad [\text{Eq. 11.12}]$$

Given this goal of determining the constant φ that minimises the distance between the true and the estimated covariance matrix based on the *Frobenius Norm*,¹¹ this objective can be expressed formally by the quadratic loss function:

$$\min_{\varphi} \text{AsyE} \left[\left\| \frac{\varphi}{N} \cdot \Phi + \left(1 - \frac{\varphi}{N}\right) \cdot \Sigma - \Omega \right\|^2 N \right] \quad [\text{Eq. 11.13}]$$

This is necessarily an asymptotic function since the true covariance matrix here called Ω is unknown. Now let Θ be the loss function:

$$\Theta = \text{AsyE} \left[\left\| \frac{\varphi}{N} \cdot \Phi + \left(1 - \frac{\varphi}{N}\right) \cdot \Sigma - \Omega \right\|^2 \cdot N \right] \quad [\text{Eq. 11.14}]$$

which we can expand to:

$$\begin{aligned} &= \text{AsyE} \left[\left\{ \frac{\varphi^2}{N^2} \cdot \|\Phi\|^2 + \left(1 - \frac{\varphi}{N}\right)^2 \cdot \|\Sigma\|^2 + \|\Omega\|^2 \right. \right. \\ &\quad \left. \left. + 2 \frac{\varphi}{N} \left(1 - \frac{\varphi}{N}\right) \cdot \Phi \cdot \Sigma - 2 \frac{\varphi}{N} \cdot \Phi \cdot \Omega - 2 \left(1 - \frac{\varphi}{N}\right) \cdot \Sigma \cdot \Omega \right\} \right] \end{aligned}$$

The optimal shrinkage constant φ that minimises this function by setting $d\Theta/d\varphi = 0$ is thus given by:

$$\begin{aligned}
 \frac{d\Theta}{d\varphi} &= 2AsyE \left[\frac{\varphi}{N} \cdot \|\Phi\|^2 + \left(1 - \frac{\varphi}{N}\right) \cdot \|\Sigma\|^2 + \left(1 - 2\frac{\varphi}{N}\right) \cdot \Phi \bullet \Sigma \right. \\
 &\quad \left. - \Phi \bullet \Sigma + \Sigma \bullet \Omega = 0 \right] \\
 &= 2AsyE \left[\frac{\varphi}{N} \cdot \left\{ \|\Phi\|^2 + \|\Sigma\|^2 - 2 \cdot \Phi \bullet \Sigma \right\} \right. \\
 &\quad \left. - \left\{ \|\Sigma\|^2 - \Phi \bullet \Sigma + \Phi \bullet \Omega - \Sigma \bullet \Omega \right\} \right] = 0
 \end{aligned}
 \tag{Eq. 11.15}$$

Isolating φ we obtain:

$$\varphi = \frac{AsyE \left[\left\{ \|\Sigma\|^2 - \Phi \bullet \Sigma + \Phi \bullet \Omega - \Sigma \bullet \Omega \right\} \cdot N \right]}{AsyE \left[\|\Phi\|^2 + \|\Sigma\|^2 - 2 \cdot \Phi \bullet \Sigma \right]}$$

Rearranging terms we obtain:

$$\begin{aligned}
 \varphi &= \frac{AsyE \left[\left(\Sigma - \Phi \right) \bullet \left(\Sigma - \Omega \right) \cdot N \right]}{AsyE \left[\|\Phi - \Sigma\|^2 \right]} \\
 &= \frac{AsyE \left[\|\Sigma - \Omega\|^2 \cdot T \right] - AsyE \left[\left(\Phi - F \right) \bullet \left(\Sigma - \Omega \right) \cdot N \right]}{\|\Phi - \Omega\|^2} \\
 \varphi &= \frac{\pi - \rho}{\gamma}
 \end{aligned}
 \tag{Eq. 11.16}$$

The matrix F is the true and unobservable factor model covariance matrix. Given that φ thus incorporates the true but unobservable covariance matrices for both the sample covariance matrix and the factor model covariance matrix, we obviously need to find appropriate substitutes or proxies for these unobservable matrices. This is the reason why we employ asymptotic variables. We will explain further below.

First, though, looking at the shrinkage factor in Equation 11.16 we see that this equation states that the shrinkage factor that determines the optimal trade-off between the estimation error of the observed sample covariance matrix and the estimation bias of the market index covariance matrix is determined by the sum of asymptotic errors on entries in the observed sample covariance matrix, minus the sum of the asymptotic *covariances* between errors on entries in the market index covariance matrix and the observed sample covariance matrix. This is then divided by the mis-specification of the market

index covariance matrix, expressed as the estimated factor covariance matrix minus the true but unobservable factor covariance matrix.

So what does all this mean in practice? Obviously we cannot determine the true and unobservable matrices, but what we can do is determine the variability (variance and covariance) in the estimates of the sample covariance matrix and the factor-based covariance matrix respectively. In other words, we can find out how uncertain the estimates of the covariance matrices are, and also to what extent these uncertainties (errors in the matrix estimates) are correlated with each other by simply determining the covariance between these errors.¹² This asymptotically approaches the differences between the true unobservable matrices and the estimated matrices, which is exactly what we require in order to determine φ in Equation 11.16.

The insight that is applied in this estimator is the fact that the market index covariance matrix has a lot of bias due to its stringent specification, but not much in the way of estimation error, at least not in-sample. On the other hand, the historical sample covariance matrix is unbiased but potentially has a lot of estimation error. The optimal level of shrinkage is therefore determined by a trade-off between the estimation bias of the market index covariance matrix, and the estimation error of the historical sample covariance matrix. While this may seem complicated, the intuition is quite clear. Statistical methods that apply Bayesian priors often run into the problem of estimating the prior from the same dataset as the sample covariance matrix. This logical inconsistency is elegantly dealt with here by subtracting the covariance between the estimation errors of the sample matrix and the market index matrix from the estimation errors of the sample matrix. The division by the mis-specification of the market index covariance matrix yields the ratio between estimation error and estimation bias of the two matrices. The optimal shrinkage factor is thus the constant that minimises this ratio. This trade-off can be illustrated as shown in Figure 11.14.

In Figure 11.14 we see how no shrinkage at all results in a covariance matrix that has a lot of estimation error but no estimation bias, and conversely how a shrinkage of 100% eliminates the estimation error, but introduces a sizeable amount of estimation bias. So naturally it follows that for a given level of shrinkage – depending on the number of observations available for estimation, the variance and covariance between matrix estimates, and also the number of assets in the portfolio – there exists an optimal shrinkage level that minimises the combined errors of the covariance matrix estimate. The really clever bit about this covariance shrinkage estimator is the fact that the determination of the size of the shrinkage factor can be performed *a priori* on the basis of historical observations and the hypothesised market in the shape of the market index. We are thus able to determine to what extent we can expect the uncertainty in the covariance matrix estimate to impact our asset allocation result, and then correct for this expected impact. Another inventive character-

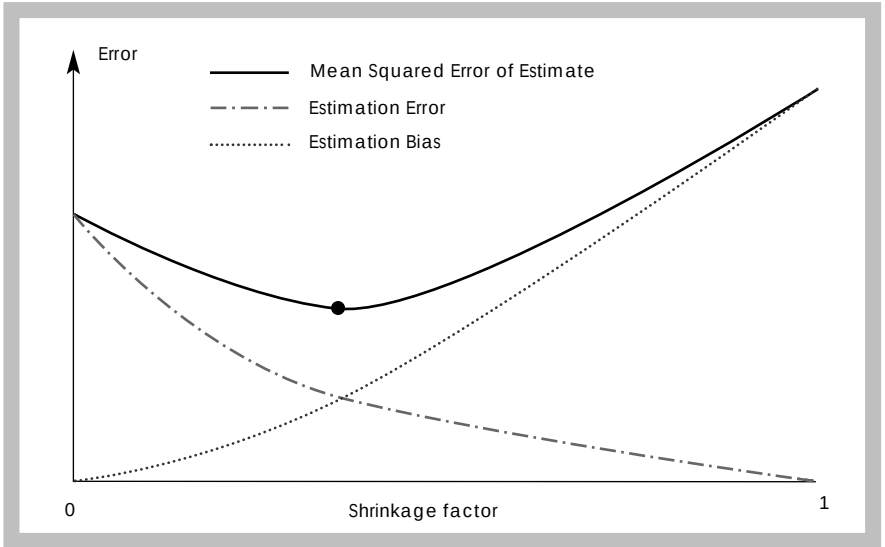


Figure 11.14 Optimal covariance shrinkage

istic of the estimator is the introduction of a correlation term describing the correlation between the estimation error and the estimation bias. In statistical theory an ever-present problem is presented by the fact that the prior applied is determined from the same dataset as the sample. This is intuitively unappealing, but using the technique presented above we escape from this problem by explicitly taking into account the covariance between errors in estimates of the sample covariance matrix and the factor-based covariance matrix.

To briefly recap, in the original *naïve* model with no shrinkage of any of the inputs, the expected sector returns and volatilities as well as the *covariance* matrix appear as shown in Table 11.4. As for the amount of shrinkage of the covariance matrix for the MSCI Global Sector Model where inputs are estimated based on no more than 88 observations, the optimal shrinkage factor is approximately 13%. This means that the optimal covariance matrix to be employed in the portfolio optimisation and asset allocation consists of 13% factor-based covariance matrix, and 87% sampled covariance matrix. This results in the optimally shrunk covariance matrix shown in Table 11.5.

At first glance there appears to be no significant difference between the two covariance matrices. The biggest positive deviation from the original unadjusted covariance matrix is in the order of approximately 4%, and the largest negative deviation is in the order of approximately 3%. The changes to the individual covariance terms appear minimal, and in fact the effect on the shape of the efficient frontier is hardly noticeable (Figure 11.15).

Table 11.4 No shrinkage variance–covariance matrix – ‘naive’ approach

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	9.7%	17.1%	0.029	0.019	0.016	0.012	0.008	0.008	0.018	0.019	0.005	0.008
Materials	10.3%	17.2%	0.019	0.030	0.021	0.020	0.011	0.009	0.023	0.025	0.009	0.005
Industrials	11.6%	15.2%	0.016	0.021	0.023	0.022	0.011	0.010	0.024	0.033	0.014	0.007
Cons. Disc.	12.1%	16.8%	0.012	0.020	0.022	0.028	0.009	0.008	0.023	0.043	0.022	0.005
Cons. Staples	9.7%	13.1%	0.008	0.011	0.011	0.009	0.017	0.013	0.017	0.007	0.006	0.008
Health Care	11.0%	14.6%	0.008	0.009	0.010	0.008	0.013	0.021	0.017	0.011	0.008	0.007
Financials	12.0%	18.9%	0.018	0.023	0.024	0.023	0.017	0.017	0.036	0.029	0.016	0.010
IT	16.0%	33.0%	0.019	0.025	0.033	0.043	0.007	0.011	0.029	0.109	0.045	0.004
Telecomm.	13.2%	19.6%	0.005	0.009	0.014	0.022	0.006	0.008	0.016	0.045	0.038	0.004
Utilities	7.5%	10.5%	0.008	0.005	0.007	0.005	0.008	0.007	0.010	0.004	0.004	0.011

Table 11.5 Optimal covariance shrinkage

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	9.7%	17.1%	0.029	0.019	0.014	0.010	0.008	0.008	0.017	0.015	0.004	0.007
Materials	10.3%	17.2%	0.019	0.030	0.021	0.019	0.011	0.009	0.023	0.021	0.010	0.004
Industrials	11.6%	15.2%	0.014	0.021	0.023	0.021	0.012	0.011	0.024	0.030	0.015	0.006
Cons. Disc.	12.1%	16.8%	0.010	0.019	0.021	0.028	0.009	0.009	0.023	0.041	0.024	0.003
Cons. Staples	9.7%	13.1%	0.008	0.011	0.012	0.009	0.017	0.013	0.017	0.005	0.007	0.008
Health Care	11.0%	14.6%	0.008	0.009	0.011	0.009	0.013	0.021	0.017	0.009	0.007	0.007
Financials	12.0%	18.9%	0.017	0.023	0.024	0.023	0.017	0.017	0.036	0.025	0.015	0.009
IT	16.0%	33.0%	0.015	0.021	0.030	0.041	0.005	0.009	0.025	0.109	0.047	0.001
Telecomm.	13.2%	19.6%	0.004	0.010	0.015	0.024	0.007	0.007	0.015	0.047	0.038	0.003
Utilities	7.5%	10.5%	0.007	0.004	0.006	0.003	0.008	0.007	0.009	0.001	0.003	0.011

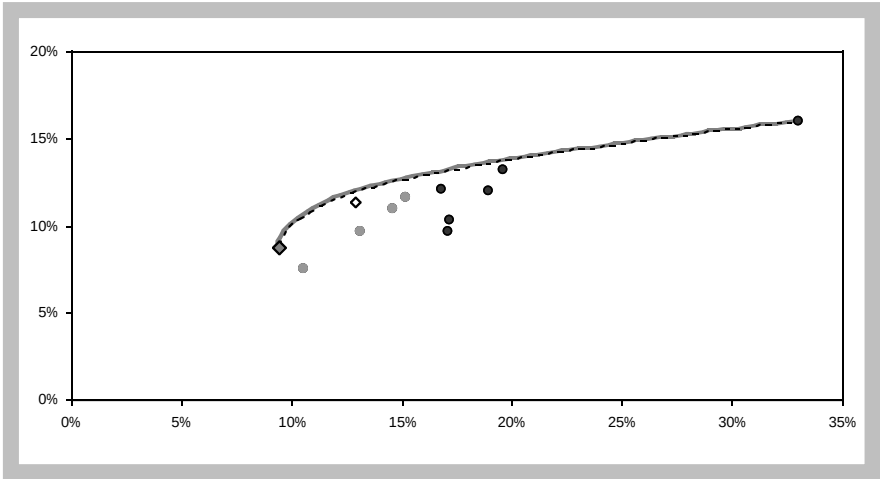


Figure 11.15 The efficient frontier – optimal covariance shrinkage

Surprisingly, perhaps, there are now only four sectors included in the minimum-variance portfolio, and some sectors that are included in the optimally shrunk covariance matrix framework are not included in the original framework. There is no very sensible explanation for this, except that the quantitative optimiser in its most basic form brutally exploits any and every opportunity for diversification, without regard for investment intuition. The asset allocation chart in Figure 11.16 presents the *naïve* asset allocation along the risk

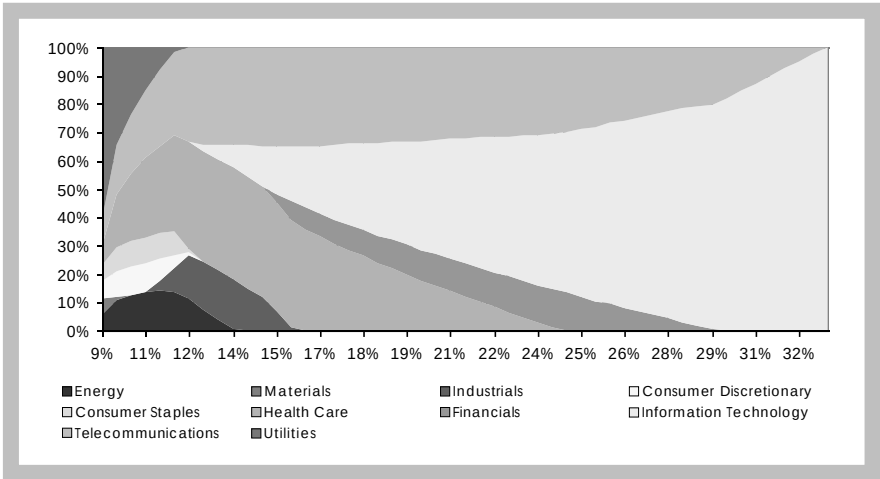


Figure 11.16 Asset allocation topography – optimal covariance shrinkage

spectrum. Compared to the original asset allocation chart for the unadjusted covariance matrix presented in Figure 9.6, the differences are relatively minor. The telecommunications sector fades out a little more quickly as we move along the risk spectrum, and also the consumer discretionary and the energy sectors play a bigger role for low-risk portfolios. In addition, the role of materials has been reduced somewhat under the optimal covariance shrinkage framework. However, this is not a fair way to evaluate the optimally shrunk covariance matrix. The optimisation technique is still the same as before, which means that in the 'naive' approach we obtain asset allocation charts that exhibit violent changes in portfolio weight for varying levels of target volatilities.

To see the full effect of the optimally shrunk covariance matrix, we must employ the QRMCSAA technique. So, letting the QRMCSAA methodology grind away at these new model inputs yields the asset allocation chart shown in Figure 11.17. Compared to the QRMCSAA chart produced by the original and unadjusted covariance matrix, we see that in the high-risk spectrum the telecommunication services sector is reduced in importance and more or less replaced by the information technology sector. In the low-risk spectrum energy is reduced and more or less replaced by the health care sector, which has now taken up a more prominent role across the entire risk spectrum. Overall it seems that the optimally shrunk covariance matrix (the covariance matrix that is optimally weighted between the sampled and the market index regression-based covariance matrix) yields asset allocations that are a bit less like the equal-weighted portfolio. This implies that the factor-based covariance matrix is liable to yield asset allocations that more heavily exploit opportunities for diversification.

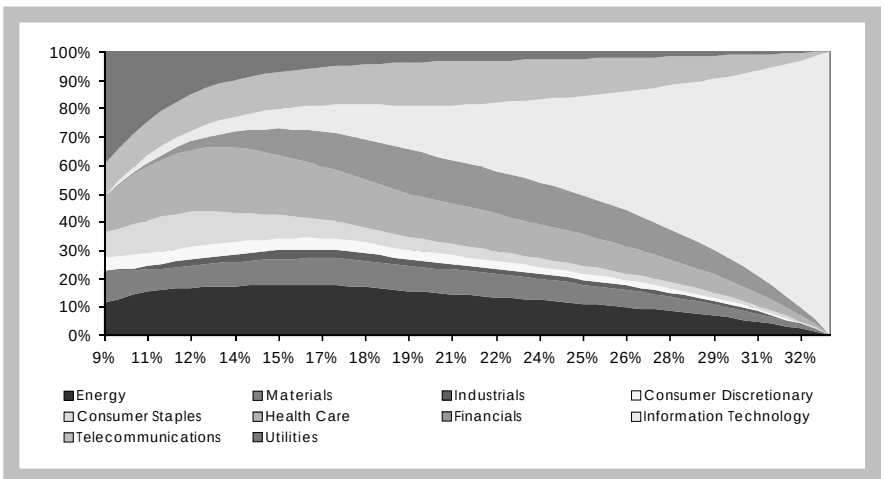


Figure 11.17 QRMCSAA topography – optimal covariance shrinkage

In general, the optimally shrunk covariance matrix incorporates the intuitive argument that if we have uncertain information about a range of variables that essentially express the same type of information, and if we then enter these into the model in an equal fashion, we might be better off assuming smaller differences between them – that is, shrinking them – than we would be by simply accepting their estimated values. The argument is similar to that of the other shrinkage estimators.

A note on purely subjective forecasts

Whether rules of thumb or quantitative optimisation are used, at the end of the day the asset manager will base her asset allocation on expected asset or asset class performance in terms of expected return, volatility and correlations. Therefore, a sound argument exists for using a correlation matrix based on historical returns (possibly adjusted using Bayes and Stein techniques), and then making explicit forecasts for expected returns and volatilities. Extending this train of thought further, the argument can be made that since asset class volatilities seem to be relatively stable over time, volatilities used in the model should be based on historical (possibly exponentially weighted) moving averages, and therefore that asset managers should focus on forecasting asset class returns. This would greatly simplify the quantitative portfolio optimisation analysis, and provide for a reasonably workable application of quantitative asset allocation.

SUMMARY

In this chapter we have further refined the QRMCSAA model in order to deal with issues that may adversely affect the basic QRMCSAA model. The augmentations are founded in the notion that optimised portfolios are not really optimal, even if Monte Carlo Simulation techniques are employed, and there are essentially three reasons why this may be the case:

- uncertainty about the future (no crystal ball for asset class return and risk)
- estimation error
- model bias.

These three factors must be traded off against each other to arrive at an asset allocation methodology that is applicable, useful, practical, sensible, intuitive and robust in use, specifically with regard to changes in model inputs. The only reason for using Stein estimators, the Bayesian techniques, Monte Carlo

Simulations and the like is that forecasts are imperfect. In fact, if forecasts were perfect, all of the above techniques would be detrimental to investment success in that they would effectively force the asset manager away from truly optimal solutions, as he or she would try to correct for or reduce the effects of estimation errors that were not there at all.

As long as the information provided by model inputs is perfect in the sense that the forecasts of asset characteristics are 100% accurate, there is nothing wrong per se with very sensitive optimisers such as those frequently termed 'naive' in this book. However, the argument can be made that since optimisation in a portfolio context is often effectively error maximisation, and in addition accepting the contention that the optimisation algorithm is much too powerful for the level of information available, the above-mentioned techniques can be considered valid augmentation procedures for the portfolio optimisation and asset allocation process.

In this book we have borrowed heavily from previous work on asset allocation, Monte Carlo Simulation and also statistics. Authors such as Markowitz, Sharpe, Ledoit, Jobson and Korkie, and Michaud have all performed extensive work in the areas of Monte Carlo Simulation and portfolio optimisation, although in most cases borrowing bits and pieces from each other. This field of research – and practice – is very much an area of ongoing development, and efforts are constantly being made in each of the above-mentioned areas of research in order to improve on the existing quantitative asset allocation framework. In this chapter we have attempted to bring together the best from all disciplines and combine them to yield a practical asset allocation methodology.

In previous work on asset allocation in particular, consideration has apparently not been given to any large extent to such essential items as an appropriate specification of non-normal asset return probability distributions, and the need for the values in the Monte Carlo Simulation to be drawn quasi-randomly in accordance with the specified correlation matrix. As noted, those elements are included in the QRMCSAA methodology presented in this book and must be considered crucial for dependable asset allocation results. In addition, the conventional way of drawing random numbers has been to draw a large number of random returns, to identify their mean and variance, and then to input these averages into the model. In the QRMCSAA methodology we do not average out the randomly drawn returns. Rather, in addition to the added refinement of drawing in accordance with the correlation matrix, we let the results of each individual return draw enter directly into the model, in that we optimise the portfolio after each individual draw. This yields more realistic and intuitive model inputs, but crucially also yields the opportunity of having the correlation structure of the portfolio being reflected directly in the optimised asset allocation results. Finally, the inclusion of model input

augmentation in the form of optimally shrunk expected returns and in particular an optimally shrunk covariance matrix serves to provide more credible, less uncertain and more robust asset allocation results.

Consequently the QRMCSAA methodology is, in this author's humble opinion, very close to what might be called 'state of the art' in quantitative asset allocation. The methodology takes the best of all disciplines and combines it to yield a model that is arguably much better than conventional mean-variance models. Perhaps slightly pushing the point, one might say that diversification among sciences and techniques in different areas of research has yielded a result that is much better than the simple sum of the applied techniques. By no means does this imply that that further improvements cannot be made, but the QRMCSAA methodology presented here seems to be a sound and robust basis for further development.

Notes

- 1 Admissibility in a statistical context is defined in terms of relative performance. If no other estimator is uniformly better, then the statistic is admissible for use in the model.
- 2 Stein (1955).
- 3 We note that results by Frost & Savarin (1986) and Jobson & Korkie (1985) suggest that this approach may improve out-of-sample performance of optimised portfolios.
- 4 The variance-covariance matrix that contains only 0s except for the diagonal which contains 1s.
- 5 This pseudo-inverse variance-covariance matrix compensates for the lack of data and the resultant inability to invert the matrix by constructing an 'artificial' matrix.
- 6 PCA involves decomposing the covariance matrix into *Eigenvectors* and *-values* which are employed as proxies for explanatory factors. A notable characteristic of this approach is the fact that the factors cannot be identified explicitly and so the economic intuition behind their use is unclear. However, they can successfully be employed to explain very large proportions of the covariances with very (if unidentifiable) few factors.
- 7 This is the type of approach companies such as BARRA employ for quantitative portfolio analysis.
- 8 Ledoit (1996) demonstrates that unless the number of observations is at least 10 times as large as the number of variables, serious 'small sample problems' occur for the estimation of the covariance matrix.
- 9 The 'market index' used is the Centre for Research in Security Prices (CRSP) equity database containing between 909 and 1314 stocks for the 120 months of data upon which mode inputs are estimated.
- 10 More advanced factor models can easily be employed to replace the Single Index Model.
- 11 The Frobenius Norm is a statistical measure for determining the solution that yields the smallest error in the *least-squares* sense.
- 12 For a thorough explanation of the determination of asymptotic variables that enter into the shrinkage factor expression in Equation 11.16 please refer to Ledoit (1999).

STRATEGIC AND TACTICAL ASSET ALLOCATION

INTRODUCTION

Strategic asset allocation (SAA) is closely related to the decision concerning benchmarks and benchmark choice. The strategic asset allocation decision – as opposed to the *tactical asset allocation* (TAA) decision – is the long-term investment allocation decision, and as such is widely believed to be the single most important decision in the investment process. The decision essentially involves determining an appropriate long-term allocation of funds according to long-term expectations about the future risk and return of assets or asset classes, as well as about the expected correlation structure between the assets. It is not so much a question of attempting to determine the return that one can expect from an asset class, and subsequently the weight to assign it over the long run. Rather it involves taking advantage of the correlation structure of the investable universe, in order to achieve the desired return/risk profile that is desired over the long term.

Most asset management clients have developed – either by themselves or with the help of asset managers or investment consultants – a quantitative framework for determining an appropriate long-term asset allocation policy that must subsequently be adhered to by the asset manager. This long-term asset allocation policy, which is usually incorporated and expressed explicitly in detail in client investment policy statements, serves as a road map both for the client and for the asset manager. And in theory at least, it ensures that the

client is invested in certain asset classes by certain amounts at any given time, thus receiving the desired investment return (cash flow) as stipulated in the investment policy statement.

Up until this point we have only considered portfolios in isolation from any external reference point. Now that we have established a framework for quantitative portfolio optimisation and asset allocation, we can move on to a more realistic scenario that includes *benchmarks*. With benchmarks against which to evaluate and structure the portfolio we can work with tactical short-term deviations from the long-run asset allocation. In this chapter we will thus introduce benchmarks as tools used by asset managers as external reference points for portfolios, established in order to control the relative performance of the portfolio both in terms of return and risk. The establishment of a benchmark against which we can measure portfolio performance allows us to quantify the tactical short-term deviations from the benchmark and to attempt to determine their impact on expected portfolio return and risk. In this chapter we will thus provide a benchmark-relative asset allocation framework that applies the QRMCSAA technique presented earlier to the strategic asset allocation decision, and we will demonstrate how an appropriate tactical asset allocation decision can be made, based on quantitative information about the investment universe.

SAA VS. TAA – THEORY

The strategic asset allocation (SAA) represents the very long-term expectations regarding asset class returns, risk, and also the correlations between individual asset class returns. This can be seen as a starting point for the asset allocation that will eventually be selected within the asset management organisation. Another way of looking at this is by imagining having to invest an amount of money in a set of assets which were then to be left untouched for 100 years. How would we as asset managers invest these assets in order to obtain an appropriate trade-off between expected return and expected risk, given the correlation structure of the investable universe, and given our risk aversion? The asset allocation we ended up with could rightfully be regarded as the long-term *strategic* asset allocation. As active managers we have the opportunity to rotate between asset classes over time, based on superior information regarding market movements. These shorter term decisions represent the manager's tactical decisions, designed to outperform the benchmark over the short term. The asset allocation that results from these decisions to temporarily deviate from the long-run strategic asset allocation is the *tactical* asset allocation.

The consensus efficient frontier

The argument could be made that working under the assumption of informationally efficient markets over the long run, we would expect there to be no arbitrage opportunities available, and so in this idealised world the CAPM would hold. Over the long run we would expect to observe a perfectly linear relationship between risk and return, so that all assets and asset classes would plot exactly on the Security Market Line (SML). In a quantitative portfolio optimisation and asset allocation context, this would mean that there should always be a fixed linear relationship between the returns and risks of all assets in the portfolio. Therefore the efficient frontier – used as a basis for strategic asset allocation – in return/risk space, would be constructed from a linear string of assets plotted in return/risk space.

As mentioned earlier, this adoption of the CAPM implies a consensus expected return for each asset, based on that asset's Beta relative to the market, which in turn becomes the appropriate benchmark. This means that we can construct a *consensus efficient frontier* based on the linear string of expected return and risk combinations for each asset class. Assuming for expositional purposes a highly simplistic 1:1.5 relationship between return and risk for the MSCI sector model presented in previous chapters – so that 10% return is expected to cost 15% volatility – the consensus efficient frontier would appear as in Figure 12.1. Note that some of the sectors plot exactly on top of each other, since they have the same initial expected return, and thus – in this simplistic CAPM-inspired framework – the same expected risk. The resultant naive asset allocation appears as shown in Figure 12.2.

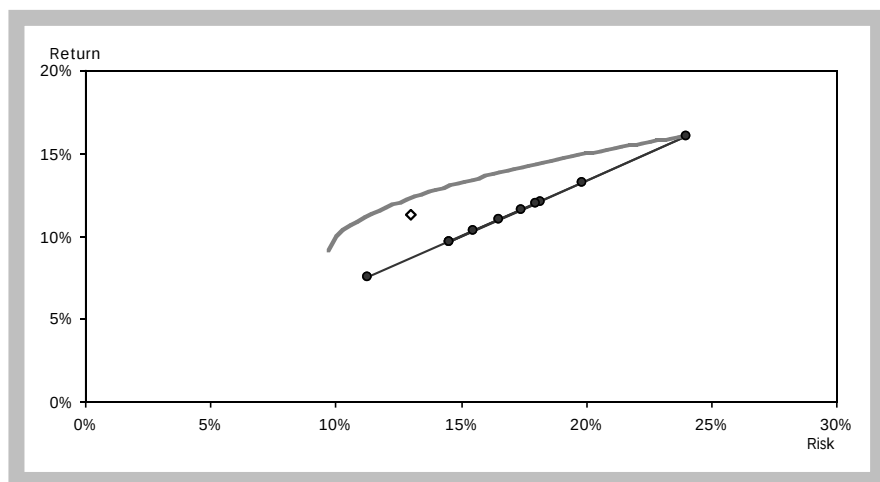


Figure 12.1 Long-run consensus efficient frontier

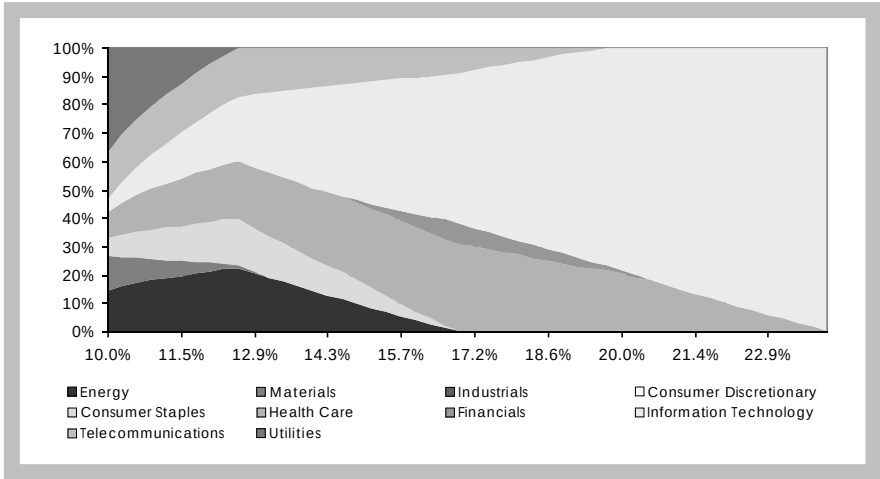


Figure 12.2 Asset allocation – long-run consensus efficient frontier

As we have seen many times already, the asset allocation chart provides unsatisfactory input for real-world asset allocation decisions. The industrials and the consumer discretionary sectors are not included in the asset allocation at any level of desired volatility, and sectors drop in and out erratically. The asset allocation generated using the QRMCSAA methodology is shown in Figure 12.3. Compared to Figure 10.10, the chart implies a much reduced emphasis on the information technology sector, and an increased emphasis on the telecommunications sector. Stated briefly, this is due to the fact that in this

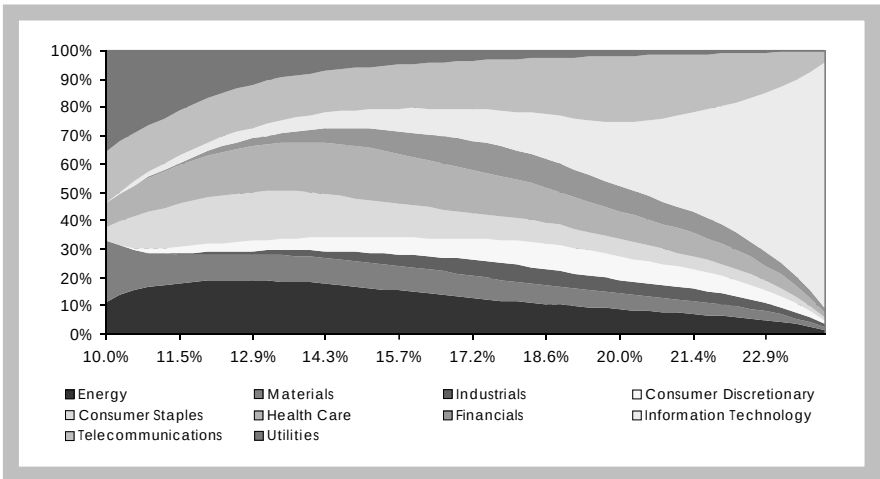


Figure 12.3 QRMCSAA – long-run consensus efficient frontier

simplified framework expected risk is now linearly proportional to expected return, which means that relative to the original inputs in Chapter 10, the return/risk trade-off (the Sharpe Ratio) of the telecommunications sector now appears more attractive than that of the information technology sector.

The implication of Figures 12.1–12.3 is that a manager's deviation from this consensus efficient frontier can only be caused by expectations of asset return and risk that are different from the consensus expectations. If the manager's expectations regarding asset class return and risk were exactly similar to those of the market, he would generate an expected efficient frontier exactly identical to the consensus efficient frontier. However, if the manager really did have superior information regarding asset class return and risk, his *personal* expected efficient frontier would deviate from the *consensus* expected efficient frontier. Specifically, for the same expected level of risk he would then choose an asset allocation different from that of the consensus efficient frontier asset allocation. This in turn implies that the objective of asset managers is to acquire, to develop and then to apply information relative to the market or benchmark which is superior to that represented by the consensus efficient frontier. In this theoretical set-up, tactical asset allocation can thus be defined as deviations in expectations of individual asset class return and risk based on perceived superior information: as a consequence of this information, the asset manager will deviate from the long-term strategic asset allocation (or benchmark) and take active bets on certain countries, sectors, stocks, and so on. We demonstrate below.

The manager-specific efficient frontier

Suppose the asset manager has a different set of expectations in particular regarding the return and risk of the consumer staples sector. In particular, if the asset manager believes that he has superior information about the expected return on the consumer staples sector, such that he believes it will be 11.5% as opposed to the original 9.7%, the implied efficient frontier now appears as shown in Figure 12.4. The manager now has higher expectations regarding the returns on the consumer staples sector, but at the same levels of expected risk. Compared to the original consensus efficient frontier indicated by the dotted line, this results in an asset manager-specific efficient frontier that is *pushed* upwards. In terms of the asset allocation along the efficient frontier, this naturally results in a much larger weight being placed on the consumer staples sector as we move along the risk spectrum (Figure 12.5). Not surprisingly, the consumer staples and the information technology sectors now completely dominate the asset allocation chart. Applying the QRMCSAA methodology yields the results shown in Figure 12.6. It is evident from Figure 12.5 that the consumer staples sector now features much more prominently in the asset

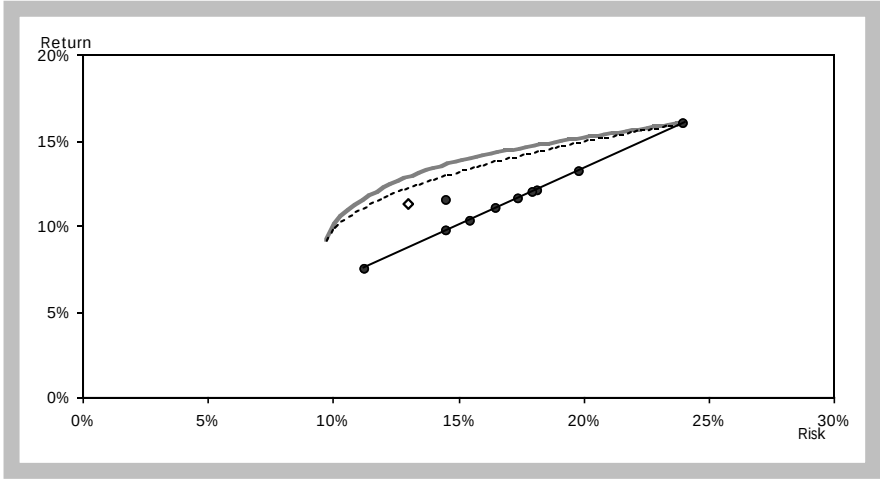


Figure 12.4 Efficient frontier – manager with superior information

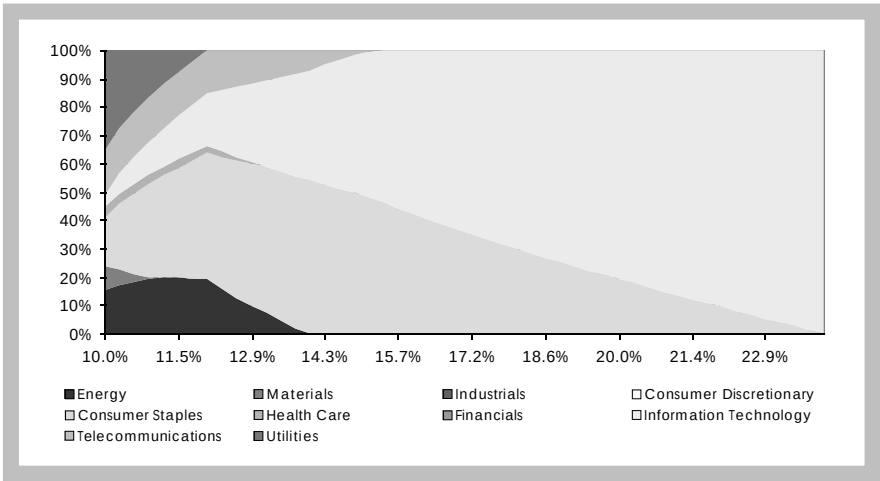


Figure 12.5 Asset allocation – manager with superior information

allocation chart, mainly at the expense of the health care sector but also at the expense of the energy sector. This is because the return correlations between these three and the remaining seven sectors are somewhat similar, which induces the quantitative optimisation algorithm to substitute one for the other, without drastically changing the emphasis put on other sectors.

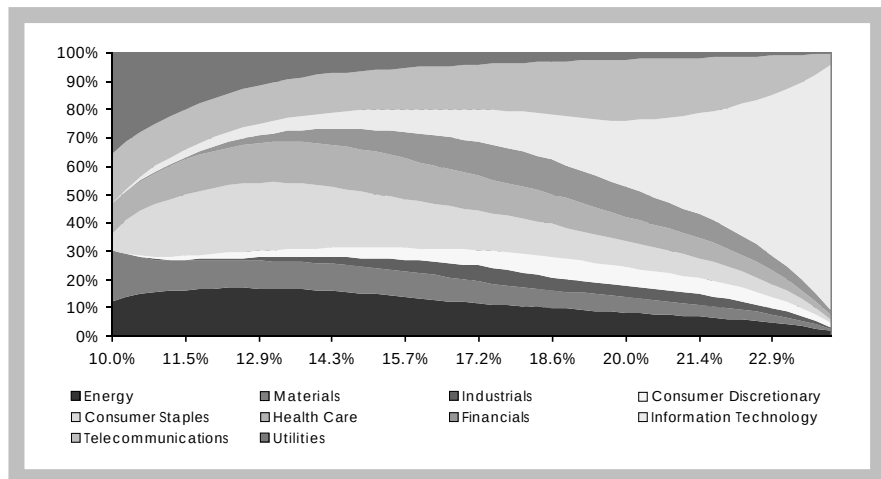


Figure 12.6 QRMCSAA – manager with superior information

So far we have demonstrated the impact on asset allocation of the asset manager having return expectations that deviate from the consensus. As with previous chapters, this was carried out in a total portfolio volatility framework. However, most often when using a benchmark for structuring a portfolio, the asset manager will have agreed with the client to focus on *active risk* or *tracking error*. So before moving on to quantitative asset allocation in a benchmark-relative context, we briefly introduce the concept of active risk.

Introduction to active risk

Given that managers are most often evaluated against some benchmark, the most common tool applied in order to quantify the degree to which portfolios track their benchmark is the so-called *tracking error*, which is simply the annualised standard deviation – that is, volatility of returns in excess of the return on the benchmark, also termed *active returns*.¹ These active returns are of course a result of the portfolio having deviated from the benchmark on either a sector level or a stock level or both.

In order to describe tracking error quantitatively, we first define the *active portfolio* as the difference between the portfolio weight vector and the benchmark weight vector:

$$\vec{w}_{PA} = \vec{w}_P - \vec{w}_B \quad [\text{Eq. 12.1}]$$

$\vec{w}_{PA}$ is thus the active portfolio weight vector, $\vec{w}_p$ is the portfolio weight vector and $\vec{w}_B$ is the benchmark weight vector. Previously we demonstrated that portfolio volatility can be expressed as:

$$\sigma_p = \sqrt{\vec{w}_p^T \bullet \Sigma \bullet \vec{w}_p} \quad [\text{Eq. 12.2}]$$

where Σ is the covariance matrix. It therefore follows that the portfolio's active risk (tracking error) relative to the benchmark can be expressed as:

$$\Psi_p = \sqrt{\vec{w}_{PA}^T \bullet \Sigma \bullet \vec{w}_{PA}} \quad [\text{Eq. 12.3}]$$

This measure of benchmark-relative risk enables the client to evaluate the asset manager in terms of his ability to track a benchmark. By the same token it can be used as an initial guideline for the management of assets, in that the client can stipulate the maximum level of expected active risk. Subsequently it becomes the asset manager's primary portfolio risk-control tool. We shall explore this topic in much greater detail in subsequent chapters.

Quantitative active risk minimisation and asset allocation

In Chapter 6 we demonstrated how a portfolio can be quantitatively optimised either by maximising expected return for some level of expected volatility, or by solving the dual problem of minimisation of the expected volatility for some level of expected return. The object here is to apply the same quantitative optimisation technique to the problem of asset allocation, but in an active risk context. In order to do this we must first stipulate the limitations we wish to impose on the optimisation. As before, we need the asset weights to sum to 100%. Secondly, we need active weights to sum to zero. Finally, assuming a tracking error minimisation framework, we need to specify the level of expected active return for which we wish to minimise tracking error.

$$\text{Min} = \sqrt{\vec{w}_{PA}^T \bullet \Sigma \bullet \vec{w}_{PA}} \quad [\text{Eq. 12.4}]$$

Subject to:

$$\vec{w}_{PA} = \vec{w}_p - \vec{w}_B$$

$$\vec{w}_{PA}^T \bullet \vec{r} = \check{r}_{PA}$$

$$\vec{w}_p^T \bullet \vec{e} = 1$$

$$\vec{w}_{PA}^T \bullet \vec{e} = 0$$

where $\check{r}_{PA}$ is the required expected level of portfolio active return, and $\vec{e}$ is the unit vector. This problem can be solved with a heavy dose of matrix algebra, which we shall refrain from presenting here – not least because any calculations

with the aim of determining the optimal asset allocation in an active risk framework are most easily carried out using a computerised optimisation algorithm.

SAA VS. TAA – PRACTICE

The SAA is the benchmark

Selecting a benchmark implies selecting a market for observing consensus expected returns. This has implications for the active management. In practice, asset managers are most often evaluated relative to some benchmark, which can be one of the MSCI or FT indices, or any other appropriate index. This of course does not exclude the establishment of a very long-term asset allocation, from which managers can deviate tactically over shorter periods of time. However in the remainder of this book we will for simplicity assume that the very long-term asset allocation is identical to the benchmark allocation. This enables us to more precisely address the issues of active management in an asset allocation as well as in a risk-management context. We thus have only one benchmark per portfolio, and this benchmark represents our strategic long-term asset allocation. Short-term tactical deviations from this benchmark can then be made if the manager believes she has superior information that can support such a deviation. The duration of the deviation is up to the manager to decide.

Returning to the optimisation problem in an active risk context presented in Equation 12.4, we see that the solution to the optimisation problem does not take into account the need for asset class weights to be non-negative. This means that we have to specify the maximum and minimum levels of portfolio weights. Here we stipulate that the maximum difference between the individual asset class weights and the weight of that asset class in the benchmark must be 100% on the downside, and 300% on the upside. This means that the minimum position is zero percentage points invested in any given sector, and the maximum position is three times that of the benchmark weight. This adds considerably to the complexity of the solution of the optimisation problem, and therefore in practice we employ the optimisation algorithm of a fast computer to calculate the optimal portfolios, subject to the constraints presented above. Even so, many hours' worth of calculation is required in order to complete an asset allocation chart.

We now adapt the portfolio optimisation and asset allocation framework from previous chapters to the active return vs. active risk context. The first step is to determine an appropriate maximum level of tracking error. In this example we employ a maximum tracking error of 6%, but its size is not important. However, between 3% and 8% is not unusual for equity portfolios, depending on their precise nature and on the inherent level of risk of the underlying stocks. Once this is done, we divide the active risk spectrum into 51 intervals, for which

we maximise expected active return. For each interval the optimal portfolio is the portfolio that generates the maximum level of active return given the specified level of active risk. For the MSCI Global Sector Model that was used as an example in Chapters 9–11, the model inputs appear as in Table 12.1A.

These are exactly the same basic inputs that enter into the original total return vs. total risk model in Chapter 9. However, in addition to these inputs we specify the MSCI benchmark weights and the minimum and maximum limits for the active sector weigh – that is, each sector’s permitted deviation from the benchmark weight. Also included is the resultant maximum and minimum total sector weights. So the additional limits needed for active return asset allocation appear as in Table 12.1B.

We can construct a new augmented efficient frontier that depicts that active return versus the active risk that is required in order to obtain that level of active return. The interpretation is analogous to that of the traditional total return efficient frontier – see Figure 12.7. Note that the minimum variance portfolio in this context is the minimum active risk portfolio – that is, the portfolio that generates a tracking error of zero. This is of course the portfolio that has no active weights, implying a perfect track of the index. Of course, this also means that the portfolio generates zero active return. Not surprisingly, the active return efficient frontier exhibits falling marginal increases in expected active return as the expected active risk is increased. This simply means that the active return per unit of active risk falls as we increase the desired tracking error. This can be seen from the curvature up and to the right of the active return efficient frontier.

Under the current set of assumptions we have limited the weight of the sectors to three times that of the benchmark, which is the MSCI World. If we

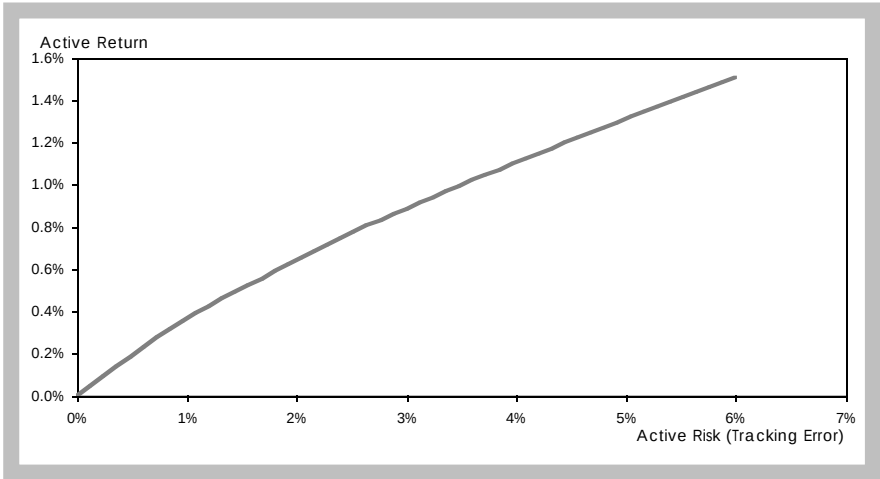


Figure 12.7 Active return efficient frontier

Table 12.1A MSCI Global Sector Model

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	9.7%	17.1%	1.00	0.64	0.60	0.41	0.36	0.31	0.55	0.33	0.15	0.46
Materials	10.3%	17.2%	0.64	1.00	0.81	0.70	0.50	0.36	0.70	0.44	0.25	0.28
Industrials	11.6%	15.2%	0.60	0.81	1.00	0.87	0.57	0.47	0.83	0.66	0.48	0.46
Cons. Disc.	12.1%	16.8%	0.41	0.70	0.87	1.00	0.40	0.35	0.73	0.78	0.67	0.28
Cons. Staples	9.7%	13.1%	0.36	0.50	0.57	0.40	1.00	0.67	0.68	0.16	0.22	0.55
Health Care	11.0%	14.6%	0.31	0.36	0.47	0.35	0.67	1.00	0.60	0.23	0.30	0.44
Financials	12.0%	18.9%	0.55	0.70	0.83	0.73	0.68	0.60	1.00	0.47	0.42	0.50
IT	16.0%	33.0%	0.33	0.44	0.66	0.78	0.16	0.23	0.47	1.00	0.70	0.13
Telecomm.	13.2%	19.6%	0.15	0.25	0.48	0.67	0.22	0.30	0.42	0.70	1.00	0.18
Utilities	7.5%	10.5%	0.46	0.28	0.46	0.28	0.55	0.44	0.50	0.13	0.18	1.00

Table 12.1B Weight limits – MSCI Global Sector Model

	MSCI World	Minimum weight	Maximum weight	Minimum active weight	Maximum active weight
Energy	4.15%	0.00%	12.46%	−4.15%	8.31%
Materials	6.63%	0.00%	19.89%	−6.63%	13.26%
Industrials	13.29%	0.00%	39.87%	−13.29%	26.58%
Cons. Disc.	17.03%	0.00%	51.09%	−17.03%	34.06%
Cons. Staples	7.94%	0.00%	23.81%	−7.94%	15.88%
Health Care	7.82%	0.00%	23.45%	−7.82%	15.63%
Financials	15.11%	0.00%	45.34%	−15.11%	30.23%
IT	14.16%	0.00%	42.48%	−14.16%	28.32%
Telecomm.	7.98%	0.00%	23.95%	−7.98%	15.97%
Utilities	5.89%	0.00%	17.67%	−5.89%	11.78%

were to allow overweighting sectors by more than that, we would be able to achieve an even higher tracking error, even to the point of decreasing active return. That is, we would be able to force the optimiser to find asset allocations that would result in a very high tracking error: but with the other existing restrictions on the optimisation, this would lead to lower expected active return. Thus, from an investment standpoint it would be foolish to pursue a tracking error of more than 6% given the model assumptions, as doing so would actually reduce our expected active returns.

Figure 12.8 illustrates the asset allocation chart for the benchmark-relative active return approach, generated using the naive optimisation approach. Over the active risk spectrum spanning from 0% to 6%, this is the asset allocation that will generate the maximum level of active return per unit of active risk or tracking error. Note how the composition of the portfolio changes as we move from a tracking error of around 1% to one of around 2%. Several sectors drop out completely from the asset allocation, but compared to the total risk framework presented in earlier chapters, the transition seems less erratic. This however is merely due to the fact that the risk spectrum over which we observe the asset allocation is much narrower here, giving the impression that changes in portfolio composition are smoother. However, the old problems of this naive approach persist, even in this framework.

At this point it is important to realise that there is no direct link between this asset allocation chart and that produced in Chapter 11 for total portfolio return and risk. The optimal portfolios presented here are not designed to do anything but secure a certain level of active return and active risk, regardless

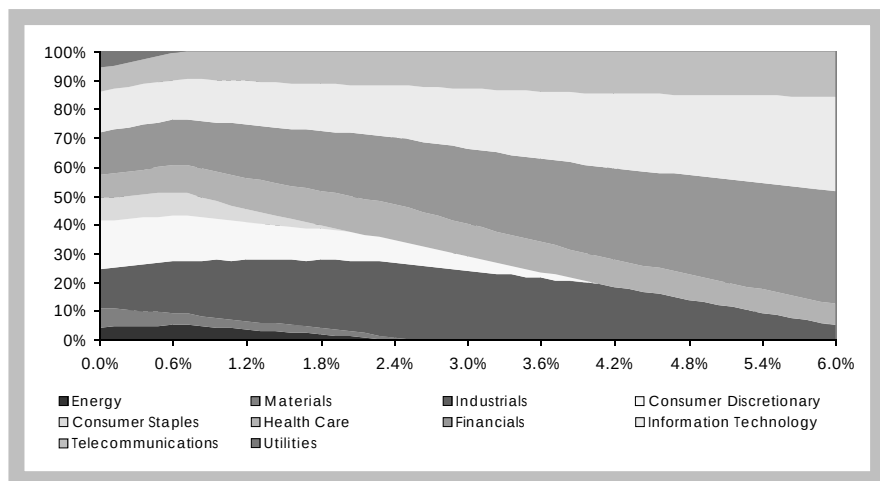


Figure 12.8 Active return asset allocation

of the absolute levels of total portfolio return and risk. Therefore one should not be surprised by the fact that they look very different.

Another way of observing what is actually happening is by plotting the active bets relative to the benchmark as we move along the active return efficient frontier (Figure 12.9). This depicts the over- and underweighting of sectors as we move along the active return efficient frontier. Note that at any given point along the active risk spectrum, the active weights sum to zero, as they should. At very low levels of active risk only six sector weights deviate from the benchmark – that is, their active bets become non-zero. As the desired level of active risk is increased, all sector weights begin to deviate from their respective benchmark weights, and quite quickly the active weights of several sectors *bump into* their respective maximum holding limits. One example is the utilities sector, which quickly moves to its maximum negative deviation of -5.9% (no portfolio weight). This also occurs for the consumer discretionary sector at a tracking error of around 1.9% , and to the consumer staples sector at a tracking error of around 3.8% .

The active weights of the financials and the information technology sectors gradually increase over the active risk spectrum, indicating that under these assumptions they, together with the consumer discretionary sector, account for most of the contributions to tracking error.² However, since we intentionally restrict the tracking error from exceeding 6% , the active weights for each of these sectors do not reach their respective limits of 30.23% and 28.32% . The active weight assigned to the industrials sector rises steadily until a desired level of tracking error of around 2.4% is reached. At this point it starts to fall

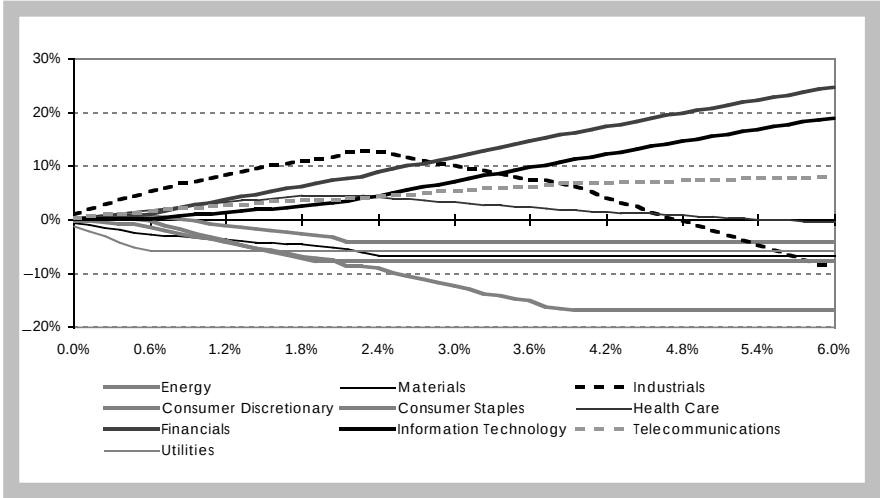


Figure 12.9 Active bets along the active return efficient frontier

and eventually crosses over into negative territory, implying that it goes from a significant positive bet on the sector to a significant negative bet on the sector, depending on the desired level of tracking error. Note that the reversal occurs around a level of desired tracking error where several active weights reach their respective limits, thus forcing a new active structure on the portfolio, and this is what is causing the reversal in the active weight assigned to the industrials sector. Depending on the active weight assigned to the other sectors, the active weights of the industrials sector thus appear somewhat unstable.

Insightful as such a mathematical analysis may be, we still seem to have the all too familiar problems of erratic changes in portfolio composition as we move along the efficient frontier. Although at first glance they appear slightly less violent than was the case for the total return vs. total risk asset allocation topography, one should keep in mind that we are looking at just over 6 percentage points on the active risk *axis*. This yet again puts into perspective the problem of having an optimisation algorithm that is too powerful for the level of quality of information provided for it to work with.

Tactical QRMCSAA

In most asset management organisation the assets are managed in accordance with specifications as to the appropriate benchmark to track and an appropriate tracking error. This means that an asset management organisation must also take account of active return and risk when considering its prospective asset

allocation decision. Of course, this type of methodology is purely concerned with the future and is as such inherently uncertain, but ignoring the potential of a tool such as this may mean that the asset manager is foregoing valuable information about the dynamics of the portfolio and of the investment universe.

The QRMCSAA technique can relatively easily be adapted to handle the active portfolio context that is necessary in order to perform asset allocation based on tracking error requirements. The nature of the simulation technique is exactly the same as for the total return/risk framework. We draw quasi-random asset returns from pre-specified sector return distributions, and optimise the expected active return for each level of expected active risk along the active return efficient frontier. As before, this is repeated many times, each time with statistically equivalent results in terms of significance, and finally a simple average of the resultant asset allocations is derived. The only difference here is that we now depict the asset allocation that is expected to generate the desired level of *active* return and risk, instead of that depicting the asset allocation that is expected to generate the desired level of *total* return and risk.

In Figure 12.10 we illustrate the QRMCSAA topography for the tactical asset allocation framework in which we quantitatively optimise the portfolio to yield the maximum level of expected active return, for the range of expected active risk that is attainable given the specific constraints put onto this particular tactical asset allocation example. As in the total risk example for the MSCI Global Sector Model presented in the previous chapter, with the QRMCSAA technique applied we obtain considerably smoother, more intuitive and more applicable solutions to the asset allocation problem. But here we operate in a tactical benchmark-relative asset allocation framework. Not surprisingly, the

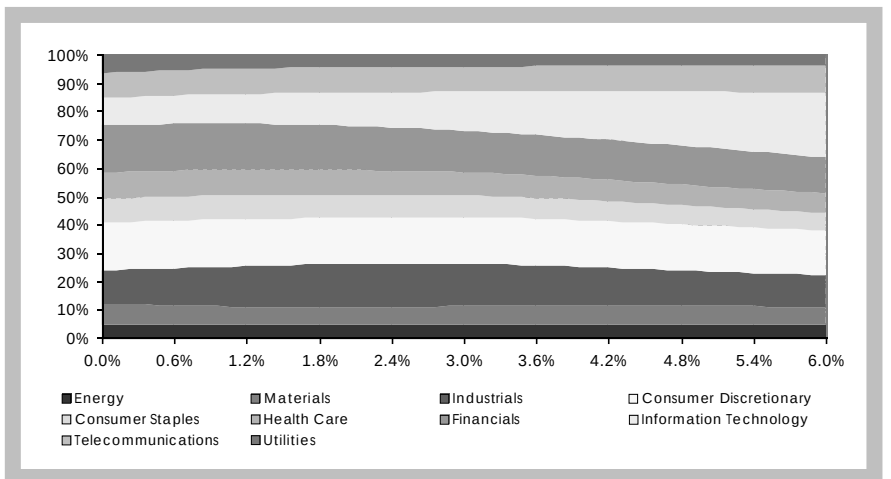


Figure 12.10 QRMCSAA – tactical asset allocation

information technology sector again features prominently in the asset allocation chart, along with the consumer discretionary sector. The changes in total portfolio weights as we move along the active risk spectrum appear quite small, but considering that the benchmark weights are constant, the resultant active weights actually vary quite a bit. We demonstrate this in Figure 12.11.

Given the benchmark weights presented in Table 12.1B, the active weights – the bets relative to the benchmark as we move along the active return efficient frontier – appear as in Figure 12.11. Not surprisingly, the active weights change more smoothly than was the case for the simple optimisation without the QRMCSAA technique applied. Note in particular how the industrials sector exhibits a similar pattern to that of the naive asset allocation, but how in this framework it starts out in the minimum tracking error portfolio with a small negative bet, then moves to a positive bet, and then reverts back to a negative bet. The basic pattern is the same as previously, but the magnitude is far smaller and the changes in active weight occur much more smoothly. Notice also how this happens in conjunction with first an increasing positive bet and then a fairly large negative bet in the financials sector. The most notable feature of the asset allocation chart is the way in which the information technology sector starts out in the minimum tracking error portfolio with a large negative bet, which is then increased to a large positive bet at a tracking error of 6%. Not surprisingly, this is to some extent offset by the bets in the utilities sector, which we already know has a low correlation with the information technology sector, providing good opportunities for diversification both in the total risk and the active framework. For high levels of desired tracking error, most sector

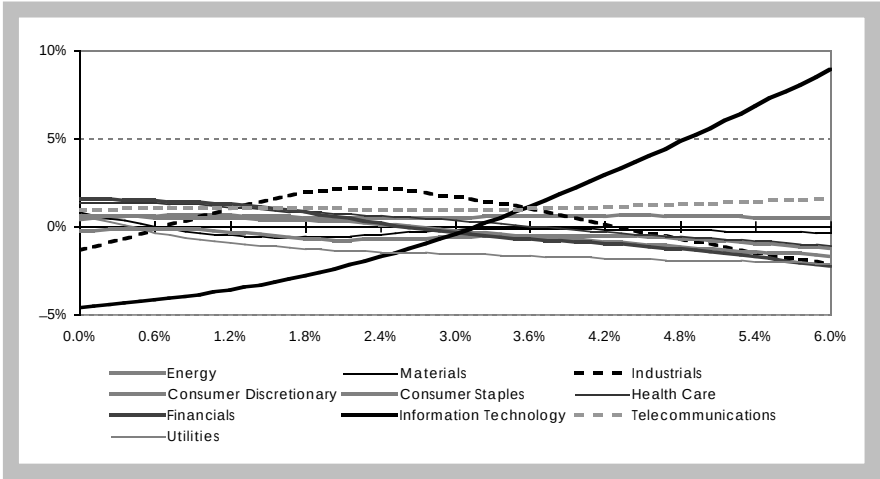


Figure 12.11 QRMCSAA – active bets along the active return

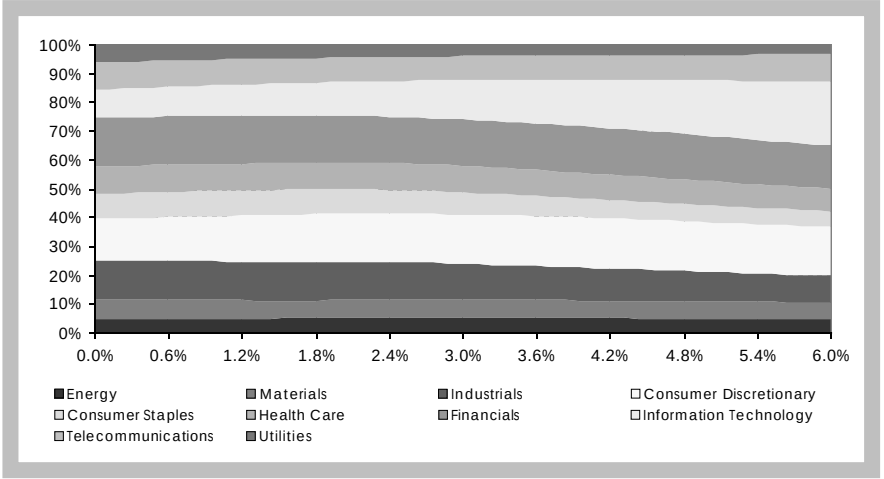


Figure 12.12 QRMCSAA – tactical asset allocation topography: MSCI Global Sector Model, Ledoit

bets are negative to offset the information technology sector, which is clearly the largest and most potent contributor to tracking error.

Application of the Ledoit shrinkage technique presented earlier yields the asset allocation shown in Figure 12.12. The active weights (or bets) along the active return efficient frontier appear as in Figure 12.13.

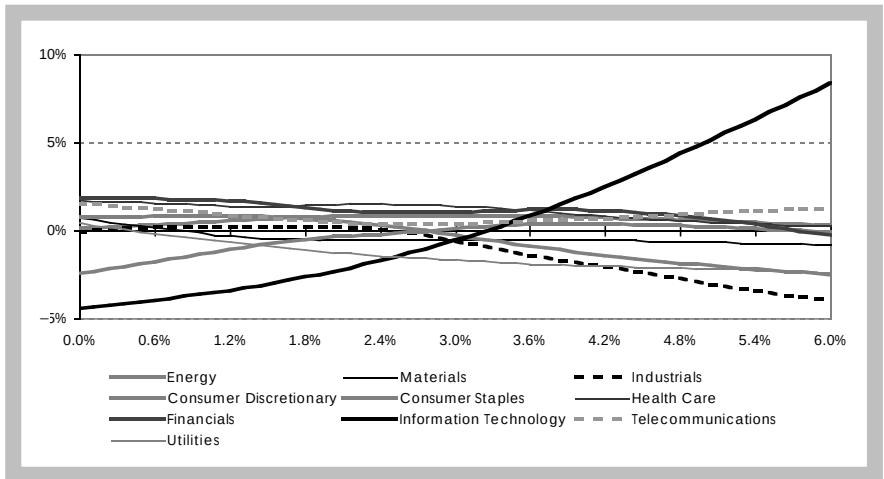


Figure 12.13 QRMCSAA – active bets along the active return efficient frontier: MSCI Global Sector Model, Ledoit

The chart is similar to the original QRMCSAA chart in Figure 12.11, except for the industrials sector, for which the active weights – both positive and negative – are now smaller. Also, the consumer discretionary sector now starts out with a larger negative bet than previously. On average, the deviations from the benchmark for each sector are now slightly larger, which is only logical given that the input has been shrunk towards a common mean, thus forcing them to appear more alike.

SUMMARY

In this chapter we have demonstrated how the QRMCSAA methodology can be applied to a benchmark-relative investment framework. Instead of focusing on *absolute* return and risk we have focused on *active* return and risk, establishing first a longer term tactical asset allocation (based on long-term expectations to return and risk of the MSCI sectors), and subsequently a shorter term tactical asset allocation, constructed in accordance with our short-term expectations. By illustrating the active weight on the sectors across the active risk spectrum we can examine how the optimal deviations from the benchmark evolve as we change the desired level of tracking error. Applying the QRMCSAA yields results that are somewhat different, but in general more intuitive and practically applicable than when applying the naive optimisation approach. Performing the analysis using the optimally shrunk covariance matrix yields similar results, but in general the active weights across the risk spectrum are smaller and of similar size, as the characteristics of the ten sectors are now more similar.

Note again that this is still all within the established quantitative optimisation framework. Combined with estimates of future long-run expected asset class risk and return, the QRMCSAA technique presented above and the statistical risk evaluation of the optimal portfolios, we are approaching a workable solution to quantitative asset allocation.

Notes

- 1 In Chapter 14 we provide an extensive discussion of the nature of tracking error in a risk management context.
- 2 The topic of marginal contributions to total risk as well as active risk – that is, tracking error – will be examined in extensive detail in Chapter 14.

SECTOR ROTATION

THE SECTOR ROTATION FRAMEWORK

The business cycle

Fluctuations that occur over the long run, in GDP, industrial production, business investments and other real economic variables, are sometimes referred to as the *business cycle*. This gives the impression that the economy and consequently the equity markets follow a clearly defined cycle which alternates neatly between favourable and unfavourable economic and market conditions, with the latter usually being six to nine months ahead of the former. The reality is more complex than that, but this initial conceptual starting point does provide us with the basis for a *Sector Rotation Model* in a quantitative asset allocation framework.

The intuition behind a quantitative Sector Rotation Model is thus that the equity markets go through peaks and troughs according to the expectations that market participants have regarding the business cycle. Since some sectors are thought of by investors as *offensive* (high Beta) and others as *defensive* (low Beta), it follows that in terms of the asset allocation there are times when it is beneficial to be overweighted relative to the benchmark in offensive sectors (such as the information technology, telecommunications and to some extent the industrials sectors), while being underweighted in defensive sectors (such as the energy, utilities and to some extent pharmaceuticals sectors). Widely used terms such as *cyclical* and *non-cyclical* also apply in principle in this regard. By the same token it follows that there are other times when it is beneficial to be overweighted in the defensive sectors and underweighted in offensive sectors.

Asset allocation using sector rotation

To the extent that the real business cycle is reasonably predictable, and also that the stock market reacts accordingly without too much *irrational exuberance* one way or the other, the asset manager can in principle attempt to outperform the benchmark purely by alternately over- and underweighting offensive and defensive sectors respectively, depending on the current and expected state of the economy and the resultant expected returns on equity in different sectors. This concept is nothing new: asset managers perform this kind of assessment and subsequently implement it in real-world portfolios all the time. Usually, though, this happens in a very 'non-quantitative' way, simply because there is little reliable data to support quantitatively based asset allocation decisions over the short to medium term. In other words, applying quantitative asset allocation techniques for tactical short- to medium-term changes to the portfolio is not very practical in the traditional quantitative portfolio optimisation framework. Partly this has to do with the erratic changes to the recommended asset allocation, or in this case sector weightings, that occur whenever we introduce even small changes to the underlying model assumptions in the traditional quantitative asset allocation framework. This has been demonstrated extensively in the previous chapters. While the asset manager may often appear to be on shaky ground in this particular respect, relying primarily on qualitative information, it actually does make some sense to let qualitative variables enter into the asset allocation decision. Qualitative variables – such as market sentiment, expectations concerning interest rate changes, political uncertainty, current overall risk aversion among investors, temporary liquidity issues in certain markets, prospective changes to the composition of benchmarks, possible politically motivated changes to exchange rates, and so on – are indeed very valid reasons for not simply accepting the asset allocation that has been suggested by the quantitative asset allocation model. The above-mentioned variables are extremely difficult to predict and to quantify, and so their respective impacts on a short- to medium-term asset allocation are most likely best left for a discussion in asset allocation committees.

Having said that, there are also strong arguments for applying quantitative asset allocation techniques whenever possible, since the correlation or covariance structure of an entire portfolio is impossible for any one person to grasp. As argued earlier, this means that the manager is simply unable to adequately exploit the opportunities for diversification that a given portfolio presents, because the interrelations between asset returns are simply too complex. As demonstrated, this applies both to the total risk and to the active risk framework. The only problem has so far been the difficulty in using quantitative asset allocation models for short- to medium-term tactical deviations from the strategic asset allocation, because of unstable asset allocation results.

However, the QRMCSAA technique presented in this book provides a much more intuitive and applicable solution to short- to medium-term changes in model assumptions, since changing the model inputs results in very intuitive changes to the recommended asset allocation – changes that actually make investment sense. Furthermore, this type of tactical asset allocation would focus on medium-term tactical deviations from the long-run strategic asset allocation. Thus it would not be beneficial for deviations designed to last (say) a month, but would be more appropriate for deviations intended to last more than six months.

CONCEPTUAL FRAMEWORK

Defining the phases of the cycle

The following is an attempt to define four phases in the economic cycle and to link them to equity market performance in terms of investment returns, volatilities and correlations. It represents the typical picture that can be observed, but there may be significant deviations from this pattern due to exogenous shocks to the economy. Additionally there may be significant differences in the manner in which the real macroeconomy and the equity markets interact in different geographical regions and in different economic regimes. The determination of the four phases in this chapter is thus a stylised attempt to quantify market characteristics for asset allocation purposes in a fairly simple manner.

We define these four phases of the economic/business cycle:

- Phase 1: Macro bottoming – earnings expansion – rising equity markets.
- Phase 2: Macro improving – earnings peaking – equity markets peaking.
- Phase 3: Macro peaking – earnings peaked out – equity markets falling.
- Phase 4: Macro declining – earnings trough – equity markets bottoming.

The phases are designed so that they comprise a full cycle. Using the simplistic terms ‘cyclical vs. non-cyclical’ and ‘growth vs. value’ the cycle appears as shown in Figure 13.1.

Below we will discuss in very simple terms the relations between the economy, the company earnings and the equity markets. This will set the stage for a formal modelling of each phase of the cycle, and thus for the different sets of assumptions that will form the basis for the quantitative asset allocations.

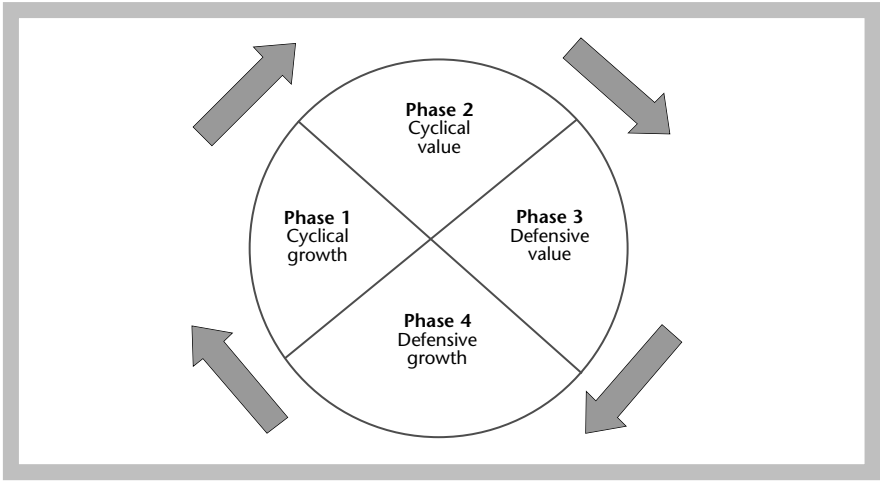


Figure 13.1 A stylised economic cycle for macro, earnings and equities

The economy, company earnings and equity markets

Figure 13.2 is a stylised illustration of how the economic cycle progresses over time, with the thick black line indicating economic fundamentals, the dotted grey line indicating company earnings, and the thick grey line indicating equity markets. Note that in this set-up company earnings are assumed to lead the

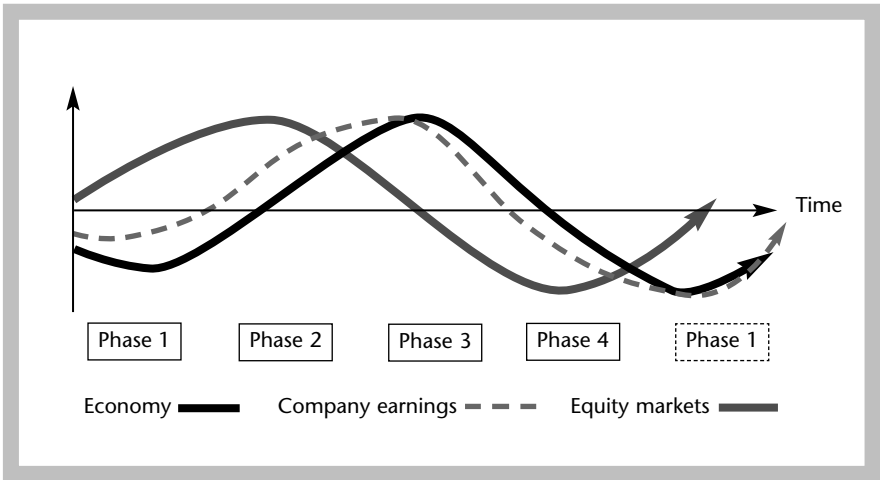


Figure 13.2 The equity market cycle

fundamentals of the economy. The rationale is that increased business activity feeds through to the (aggregate) economy with a short lag. In other words, businesses are the first to feel the improvement after an economic downturn.

Equity markets are assumed to lead company earnings. This is based on the assumption that investors and equity analysts are able to obtain information about business fundamentals before these fundamentals change. This is not always the case, although for obvious reasons both investors and analysts will always contend that it is. Note also that the cycle is presented as proceeding very smoothly, and that the equity market neatly and calmly anticipates changes to economic fundamentals in advance of them actually occurring. This is rather optimistic: equity markets do as a rule predict changes to the real economy, but this usually happens in a much more erratic way. However over the long run this is what has tended to occur in the past.

We stress that the framework established here is an approximation of the real world. It is highly simplified and is not an attempt at establishing a complex econometric model. It merely outlines the generally accepted way in which an economy works. Table 13.1 summarises the economic cycle.

A similar, stylised chart can be drawn for sector returns, volatilities and correlations (Figure 13.3). The thick black line represents equity market returns, the thick grey line indicates sector volatilities, and the dotted grey line indicates sector correlations. We summarise this information in Table 13.2, and we shall discuss it further below.

We will now discuss the dynamics of and causalities between these variables as we move through the phases in the economic cycle.

Table 13.1 The economic cycle

	Phase 1	Phase 2	Phase 3	Phase 4
Macro	Low	Medium	High	Medium
Earnings	Low/Trough	Rising	High/Peak	Falling
Equity markets	Rising	High/Peak	Falling	Low/Trough

Table 13.2 Market dynamics during the economic cycle

	Phase 1	Phase 2	Phase 3	Phase 4
Return	High	Medium	Low	Medium
Volatilities	Low	High	Medium	High
Correlations	High	Medium	Low	Medium

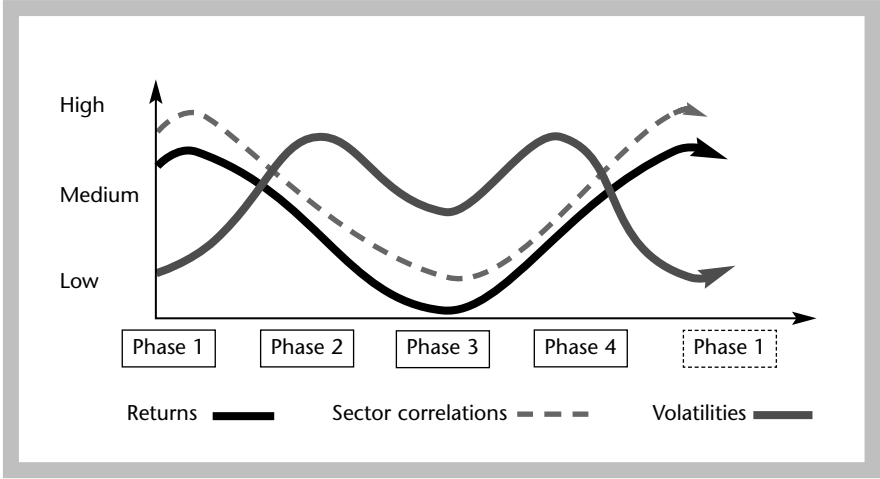


Figure 13.3 Equity market characteristics

Phase 1: earnings expansion – rising equity markets

Phase 1 is characterised by a low /trough in macroeconomic conditions. After a period of contraction, the economy has bottomed out and economic activity is now picking up again. This is reflected in company earnings, which have started growing as business investments increase and demand starts picking up, reducing inventories at companies and thus further increasing demand for new products. Since the stock market is usually several months *ahead* – conventional wisdom has it that this is six to nine months on average – of the economy, often leading most macroeconomic leading indicators, this is already being reflected in share prices which will have begun to rise. This means that equity returns are relatively high, as investors discount the expected and/or observed rise in company earnings and profits, and as analysts raise their earnings estimates and target prices for individual stocks in accordance with the expectations of improved business conditions. This will make stocks look cheaper on earnings-based valuation metrics such as P/E, P/CF, and EV/EBITDA, driving up equity prices. At this point returns are thus relatively high and the market is usually on an up trend.

Typically, asset managers have already allocated their assets so that they are exposed to the sectors they expect to deliver the highest returns, based on expectations regarding earnings growth, profitability and share price valuation. This means that only very little sector rotation goes on, and so correlations are high, since most sectors move mostly up together, without great changes to portfolio compositions being made by asset managers. Since

earnings are strong, visibility high and sentiment positive, there is relatively little volatility. These market characteristics are reflected in Figure 13.3.

Phase 2: earnings peaking – equity markets flat

In *Phase 2* the economy has now picked up serious momentum, and economic growth is high. Business investments are impacting demand more broadly in the economy, and growth rates are therefore rising. Company earnings are approaching a peak, and the increasing demand allows companies to expand margins and thereby to increase profits as inventories are being worked down. Since profitability and growth are both high, valuations are also high. The equity markets are thus at very high levels, but unlikely to rise further for the simple reason that it would be difficult for fundamentals or the news flow to improve. During this stage in the economic cycle the central bank will often raise interest rates in order to pre-empt rising inflation. These factors combined mean that equity markets are in the process of peaking out.

For this reason equity returns are at a medium level. Anticipating the peak in company earnings and therefore also equity markets, asset managers will at this point begin to consider changing their asset allocations to something less aggressive. As we move closer and closer to the earnings peak, this will result in considerable sector rotation, where asset managers move out of high-Beta growth sectors such as the information technology and telecommunications sector, and into more *defensive* or so-called *value* sectors. This rotation implies selling off sizeable chunks of equity in one sector and then buying equity in another sector. This implies that correlations between sectors are relatively low, and lower than in Phase 1 as all sectors no longer move up together, due to the increasing uncertainty about the future earnings outlook. Sector returns are therefore low during this phase. However, because of the intensive rotation and the large volume being moved around in the markets, and because of the indications that economic activity is worsening and inventories at companies are starting to rise, volatilities are high and sectors can move up or down considerably in a relatively short amount of time, leading to violent share price movements within sectors and across the whole equity market.

Phase 3: earnings peaked out – equity markets falling

Phase 3 is characterised by an economy that has peaked. Growth in the economy is not rising any further, and some sectors are experiencing contracting demand. Company earnings are beginning a down trend as earnings level out and then start to fall, profit margins are shrinking, and company

profits drop significantly. Equity markets are now falling; and this is true for most sectors, as earnings visibility is poor, and uncertainty levels are high. This means that equity market returns are low or perhaps even negative for a while, if the equity market levels require a correction to more accurately reflect fundamentals and prospective company earnings.

Correlations between sectors are now low. Although one might expect higher correlations due to the overall falling market, they are in fact often lower due to the sector rotation that managers perform searching for a *safe haven*. Since such (perceived) safe havens could change during the downturn, asset managers will often switch from one haven to another, depending on the news flow on both the economy and corporate earnings, causing correlations between sectors to drop further. Volatilities are at a medium level, as markets move downwards in a relatively uneventful fashion: most investors realise that fundamentals are deteriorating, and that equity markets must therefore go through a correction.

Phase 4: earnings trough

Phase 4 represents the trough for the equity markets. The economy is approaching another trough and company earnings will be bottoming out within the foreseeable future, possibly partly due to interest rate cuts that often occur at this stage in the economic cycle. Visibility is thus improving for company fundamentals, even though earnings may still be depressed at this point. These positive signals result in more money coming into the equity markets, and returns thus begin to increase to medium levels as markets slowly begin to level out and rise, discounting the improving fundamentals.

Seeing the light at the end of the tunnel, asset managers now begin to stir and increasingly to change their asset allocations in an attempt to position themselves for growth and a pick-up in company profits. This repositioning leads to large sector rotations once more, resulting in correlations being low, but still higher than in Phase 3 as there is now a consensus that the economy is heading for an upturn, which will benefit all sectors. Volatilities rise to high levels because of the increasing bullishness on equity markets, and the asset manager dashes to pick up the stocks that she deems to benefit the most from improving fundamentals, based on expectations of a possible increase in market share during the expansion, margin expansion or simply high levels of revenue growth.

The information summarised in Table 13.2 allows us to build a simple Sector Rotation Model, based on the QRMCSAA technique presented earlier in this book.

A NOTE ON DETERMINING APPROPRIATE MODEL INPUTS

Expected correlations

As a rule correlations are higher when markets move more violently. That is, when markets rise they rise together, and when they fall they fall together. During periods of relative equity market calm, correlations are usually somewhat lower. As indicated earlier in this book, equity market correlations typically lie in the middle of the positive range. From looking at data for the major international markets it appears that correlations are approximately half their maximum level in periods of more violent market moves. So to assume that correlations are high during Phase 1, medium during Phase 3, and low (about half the highs) during Phase 2 and Phase 4 seems to be a reasonable proposition.

Interestingly, correlations between changes in the correlation coefficients are quite high. The average correlation coefficient between the correlation coefficients between the ten MSCI sector indices from 1995 to 2002, measured by daily observations, is around 0.5. This implies that the correlation coefficients themselves generally move in the same direction when they change. This makes constructing a Sector Rotation Model considerably easier, since we can relatively safely assume that when correlation coefficients change they all change in more or less the same direction, making the introduction of separate correlation matrices for individual periods tractable.

The fact that correlations are relatively stable over the medium term can be illustrated by examining the average of each MSCI sector's correlation with the market index, the MSCI World. This essentially gives us an indication of the level of a sector's correlation with the other sectors (Figure 13.4). From the chart it becomes apparent that over the period examined here the average sector correlation with the MSCI World has been relatively stable, except for a period from late 1999 to mid 2001, where the average coefficient dropped to around 0.40 and then rose back up to around 0.7. To what extent this is a function of the equity market bubble that developed and revolved around the information technology sector is difficult to assess, but it does appear that as the bubble burst, deflated and gradually spread from *dot-com companies* to internet infrastructure companies to telecommunications companies, the average correlation with the MSCI World seems to have returned to more normal levels in the 0.6 to 0.9 range.

Even over this period of time with its market abnormalities, the correlation coefficients remained in a fairly narrow range. This is illustrated further in Figure 13.5. Except for the abnormal period already pointed out, the correlation coefficients seem to lie nicely in the 0.6 to 0.9 range. Assuming that future correlation levels are similar to past levels, we can tentatively

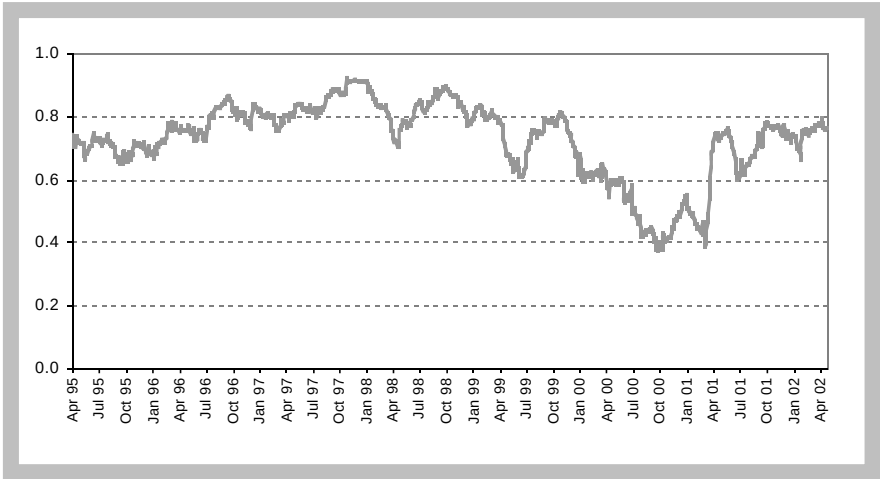


Figure 13.4 Average sector correlation with the MSCI World Index

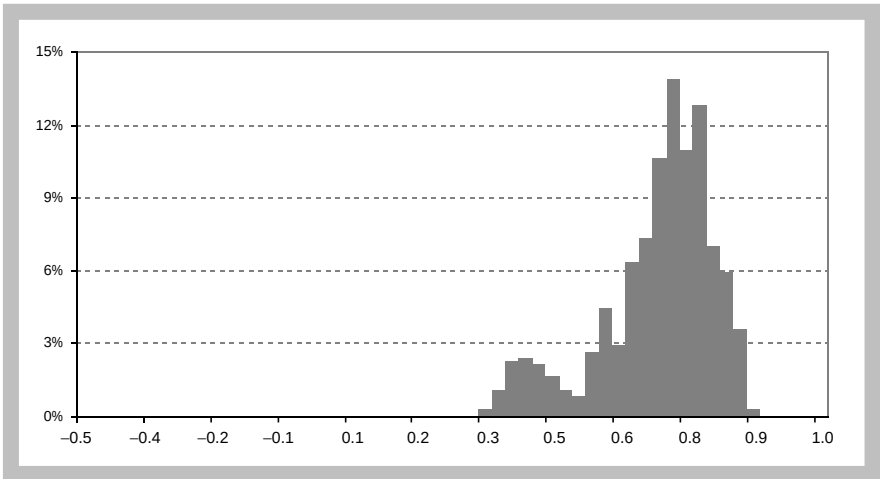


Figure 13.5 Correlation coefficient frequency chart – 10 MSCI sectors

treat Figure 13.5 as a correlation coefficient probability distribution. The chart clearly demonstrates that even though correlation coefficients change over time, they stay in a relatively narrow range. The majority of the time the coefficients tend to stay fairly close to the average. As noted above, this implies that any quantitative asset allocation model that revolves around a covariance matrix (based on estimated correlations and volatilities) can be

assumed to be relatively stable over time. We shall use this characteristic in Chapter 14, where we develop a quantitative sector risk model.

Expected volatilities

Volatilities are naturally higher in rising and falling markets than in markets that move sideways. In fact they seem to follow the direction of correlations. Figure 13.6 illustrates the volatility level (as defined by the annualised standard deviation of daily returns based on 60 daily observations) for the complete MSCI World Index and for the information technology and utilities sectors. The chart suggests that in certain periods volatilities tend to stay above levels that are considerably higher than the average volatility level. To a certain extent this is caused by the *ghost feature* of volatilities discussed in previous chapters, but even so there seem to exist periods of sustained high volatilities. For the information technology sector the volatility level remained at between 50% and 60% for a considerable amount of time, which is very high indeed. It has since come back to around 30%, but even this is very high compared to the average of the MSCI World Index, which is around 15%. The average volatility level for the ten MSCI sectors tends to move between 15% and 25%. Empirically there is strong evidence that correlations and volatilities move together, particularly in market downturns.¹ We shall exploit this feature later on.

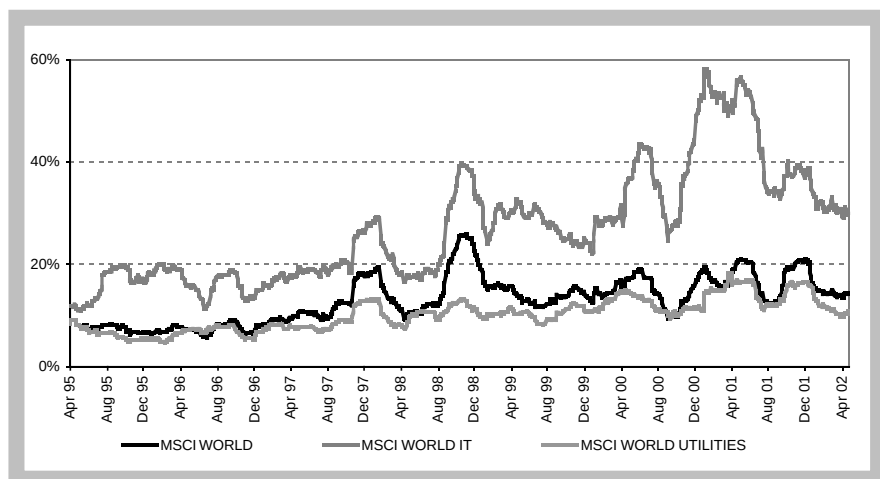


Figure 13.6 Selected volatility levels over time

Expected returns

When generating expected returns for this sort of experiment, it is not crucial what the absolute level of expected return is. What matters is the level of return for each sector relative to the other sectors. The optimiser does not care if the expected return for information technology is 35% or 10%. It cares only about whether that level is higher or lower than that of the other. The important aspect to focus on is thus the relative return levels between the sectors in the model. One might consider using the same returns for all economic phases or scenarios in order to let the volatilities and correlations determine the outcome, accepting that these are easier to predict than returns. However, it is a fact that there are large and fairly consistent patterns in sector returns during the economic cycle, and so we must include this specifically in the model. Figure 13.7 illustrates how the daily returns on the MSCI World were distributed from January 1995 to May 2002.

Generally speaking, one can investigate the characteristics and levels of sector returns during the economic cycle, but ultimately it is the asset manager’s job to determine the relative input levels that are to be used in the model. So again, quantitative models may assist the asset manager in his investment decision-making process, but at the end of the day the model is only as good as its inputs.

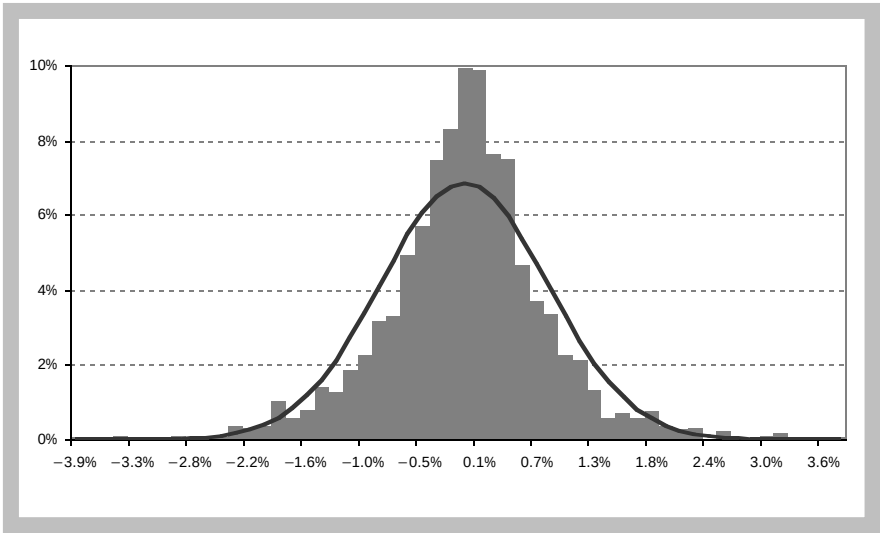


Figure 13.7 Average MSCI sector return frequency chart

ASSET ALLOCATION THROUGH THE BUSINESS CYCLE

In this section we construct a four-phase asset allocation scheme for the four stages of the business cycle, based on the QRMCSAA technique developed through this book. For each phase in the economic cycle we construct a quantitative asset allocation model using the QRMCSAA methodology.

Examining the period from January 1995 to May 2002, we have designated four sub-periods to represent the four phases. We use this segmentation to estimate four correlation matrices, one for each phase. The four phases are shown in Table 13.3.

Table 13.3 The economic cycle

	Phase 1	Phase 2	Phase 3	Phase 4
From:	01-01-95	01-01-00	01-09-00	01-10-01
To:	31-12-99	31-08-00	31-09-01	01-05-02

The levels of return and volatility assigned to each sector in each of the four economic phases have been modelled in accordance with reasonable assumptions as to which sectors may outperform other sectors during each of the four phases. Over the long run, all sectors should deliver the same risk-adjusted return, so we have simply assumed a simple 1:1.5 relationship between expected return and expected risk, so that we obtain a long-term linear relation between return and risk, similar to that of the CAPM. The problem of which inputs to use is an important point. As already pointed out, the results generated by quantitatively based asset allocation techniques are only as good as the assumptions put into the model. It is thus the task of the asset manager to produce good-quality estimates and forecasts of prospective sector returns, volatilities and correlations. The Sector Rotation Model presented here is thus merely intended to demonstrate the potential and limitations of the QRMCSAA technique when applied to short- to medium-term tactical asset allocation issues.

Phase 1: bullish asset allocation

In Phase 1, returns – particularly the returns on high-Beta sectors – are high, as are the correlations between sectors as discussed previously. Volatilities, on the other hand, are relatively low compared to the other three phases. See Table 13.4.

Table 13.4 Phase 1 – model inputs

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	4.4%	6.6%	1.00	0.54	0.57	0.52	0.59	0.52	0.53	0.41	0.53	0.55
Materials	6.6%	9.9%	0.54	1.00	0.86	0.82	0.64	0.57	0.81	0.55	0.62	0.64
Industrials	7.5%	11.3%	0.57	0.86	1.00	0.91	0.76	0.70	0.89	0.67	0.69	0.68
Cons. Disc.	7.7%	11.6%	0.52	0.82	0.91	1.00	0.72	0.68	0.85	0.65	0.69	0.65
Cons. Staples	5.3%	8.0%	0.59	0.64	0.76	0.72	1.00	0.77	0.71	0.56	0.68	0.66
Health Care	7.2%	10.8%	0.52	0.57	0.70	0.68	0.77	1.00	0.66	0.60	0.61	0.59
Financials	7.7%	11.6%	0.53	0.81	0.89	0.85	0.71	0.66	1.00	0.58	0.70	0.69
IT	9.6%	14.4%	0.41	0.55	0.67	0.65	0.56	0.60	0.58	1.00	0.50	0.40
Telecomm.	8.4%	12.6%	0.53	0.62	0.69	0.69	0.68	0.61	0.70	0.50	1.00	0.68
Utilities	5.1%	7.7%	0.55	0.64	0.68	0.65	0.66	0.59	0.69	0.40	0.68	1.00

The naive asset allocation chart appears in Figure 13.8. Not surprisingly, this asset allocation is quite *aggressive* in the sense that recommended portfolios are dominated by the information technology sector, although the telecommunications sector also features heavily in this phase of the economic cycle. As always when using the naive approach, some sectors are completely excluded from the suggested portfolios, while others make only very brief appearances as we move from the minimum-variance portfolio to the maximum-variance portfolio. The industrials sector as well as the consumer discretionary sector are not included, and the consumer staples sector only makes a very brief appearance in low-risk portfolios. It does make some sense that these sectors are forced into the background by the riskier sectors, but would we comply with such an asset allocation and completely exclude two sectors, and only for very low-risk portfolios include another? Most asset managers – as eager as they are to generate superior returns – would probably shy away from such an asset allocation, benchmark or no benchmark.

We now repeat the QRMCSAA process presented earlier. As in previous chapters, the returns on each sector are modelled using a probability distribution, and correlations between sectors are also explicitly taken into account when sector returns are quasi-randomly drawn. The methodology for generating an optimal asset allocation based on these model inputs is identical to that presented in earlier chapters. Sector returns are simulated according to predetermined probability distributions and correlation coefficients, and for each draw of simulated sector returns, a set of optimal asset allocations along the efficient frontier is generated. This procedure is repeated many times in

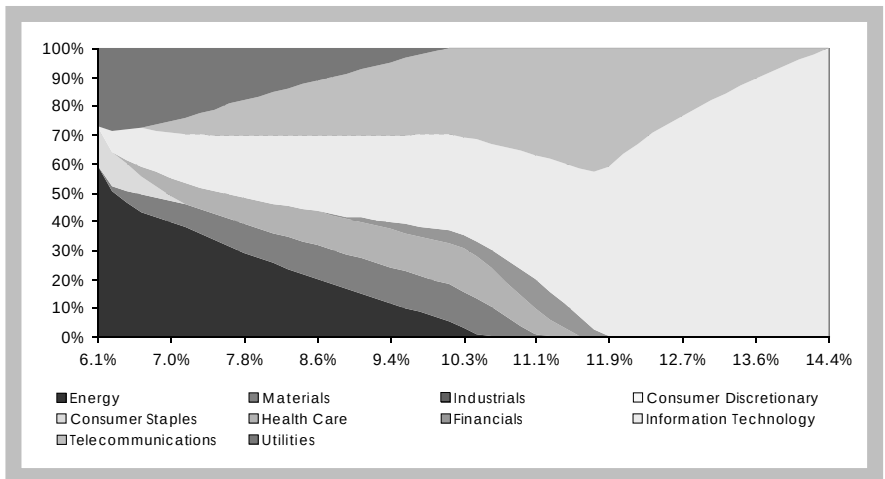


Figure 13.8 Asset allocation – Phase 1: MSCI Global Sector Model

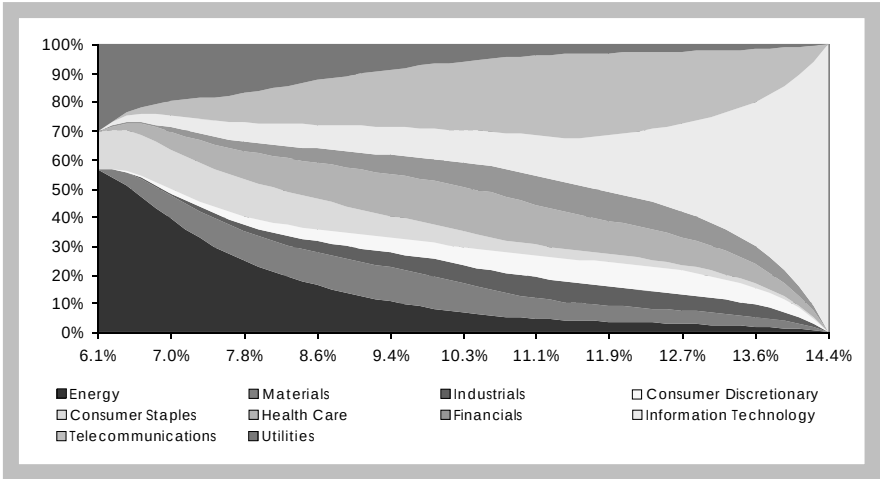


Figure 13.9 QRMCSAA – Phase 1: MSCI Global Sector Model

order to generate an average of all the simulations. The asset allocation for a Phase 1 scenario appears in Figure 13.9

The information technology sector still features prominently in this chart, as do the telecommunications sector and to some extent the health care sector. But the two previously excluded sectors, industrials and consumer discretionary, are now present, although with relatively small proportions allocated to them. Given that this asset allocation is designed to reflect an optimal allocation in Phase 1 of the business cycle, it makes good sense that the information technology and the telecommunications sectors, which tend to trade in tandem as illustrated by the correlation matrix, make up more than a third of the entire portfolio as we approach the middle-variance portfolio. From hereon the amount invested in these two sectors increases further until the portfolio is composed exclusively of the information technology sector. Note how the utilities sector is also kept in the portfolio across the entire risk spectrum, due to its highly desirable diversification characteristics relative to the information technology and the telecommunications sectors. Again, this clearly demonstrates that when we take a more refined view of sector characteristics we may get significantly different and more sensible results than when using the *naïve* approach.

Phase 2: moderately defensive asset allocation

In Phase 2 returns are more subdued, as are correlations. But because of increased nervousness about the future of company earnings, volatility is high. The model inputs appear in Table 13.5. The naive asset allocation that results

Table 13.5 Phase 2 – model inputs

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	3.9%	5.9%	1.00	0.39	0.25	0.17	0.23	0.20	0.31	0.02	0.02	0.32
Materials	5.4%	8.1%	0.39	1.00	0.56	0.40	0.46	0.34	0.52	0.08	0.16	0.37
Industrials	5.1%	7.7%	0.25	0.56	1.00	0.75	0.50	0.53	0.71	0.55	0.56	0.53
Cons. Disc.	6.9%	10.4%	0.17	0.40	0.75	1.00	0.36	0.48	0.67	0.62	0.63	0.46
Cons. Staples	5.6%	8.4%	0.23	0.46	0.50	0.36	1.00	0.64	0.56	0.13	0.14	0.44
Health Care	7.6%	11.4%	0.20	0.34	0.53	0.48	0.64	1.00	0.58	0.28	0.27	0.42
Financials	8.2%	12.3%	0.31	0.52	0.71	0.67	0.56	0.58	1.00	0.39	0.50	0.57
IT	6.2%	9.3%	0.02	0.08	0.55	0.62	0.13	0.28	0.39	1.00	0.60	0.27
Telecomm.	6.6%	9.9%	0.02	0.16	0.56	0.63	0.14	0.27	0.50	0.60	1.00	0.41
Utilities	6.4%	9.6%	0.32	0.37	0.53	0.46	0.44	0.42	0.57	0.27	0.41	1.00

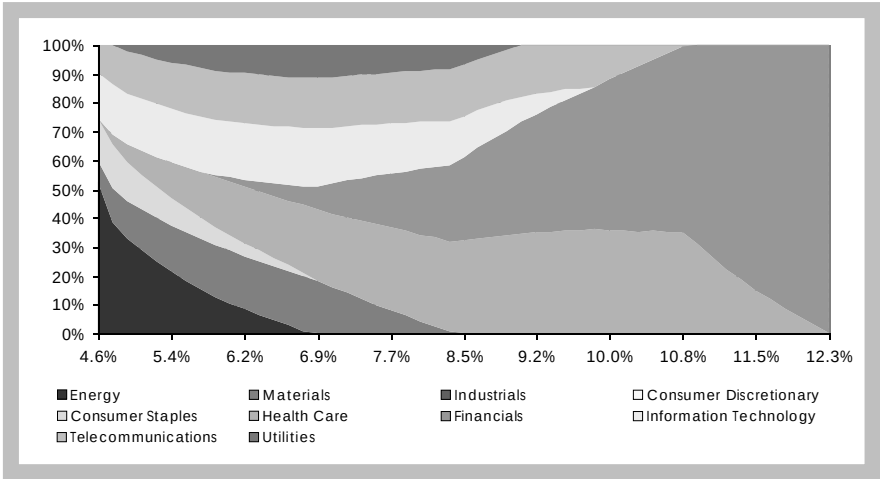


Figure 13.10 Asset allocation – Phase 2: MSCI Global Sector Model

from these assumptions appears in Figure 13.10. We now have the financials and the health care sector dominating the asset allocation. Again, one sector is completely excluded and three others make only brief appearances.

The QRMCSAA chart for the sector characteristics presented in Table 13.5 appears in Figure 13.11. Compared to the situation in Figure 11.9, the infor-

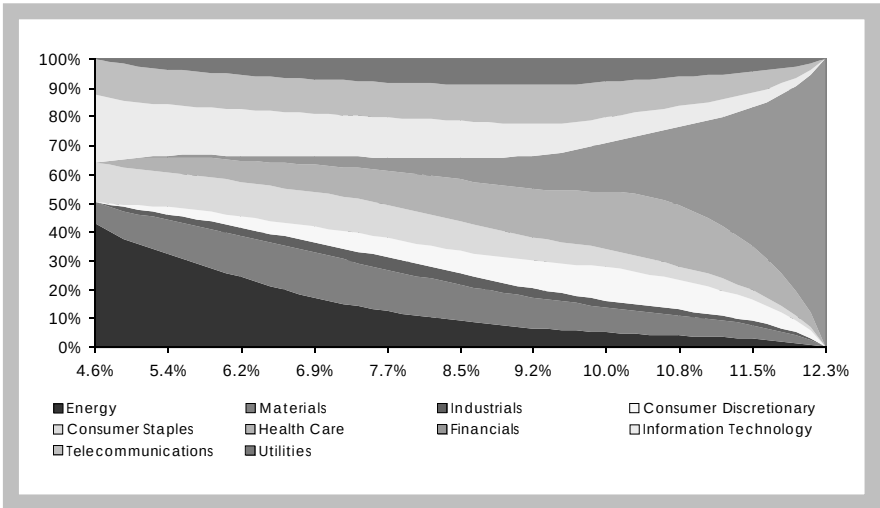


Figure 13.11 QRMCSAA – Phase 2: MSCI Global Sector Model

mation technology and telecommunications sectors now feature much less prominently, just as one would expect in a period of slowing earnings growth and rising uncertainty about the economic outlook. Note also how more defensive sectors feature more prominently in the asset allocation.

Phase 3: defensive/bearish asset allocation

In Phase 3 returns are low, as are correlations, and volatilities have reduced to a medium level. The model inputs are presented in Table 13.6. The resultant naive asset allocation appears in Figure 13.12. Defensive sectors are more prominent here than in Phases 2 and 3, and offensive sectors are relatively subdued.

Application of the QRMCSAA yields the allocations shown in Figure 13.13. Using the QRMCSAA methodology yields an asset allocation chart in which defensive sectors are even more prominently featured than was the case in the naive framework. The information technology and telecommunications sectors are at their lowest levels in any of the four phases. Note again how the health care sector is very prominent now, because of its relatively high return and its beneficially low correlation coefficients with information technology and telecommunications in particular. More economically sensitive sectors, such as the industrials sector and the consumer discretionary sector are now confined to relatively brief appearances at the low end of the risk spectrum. The more defensive energy and consumer staples sectors now feature more prominently than in any of the other three phases.

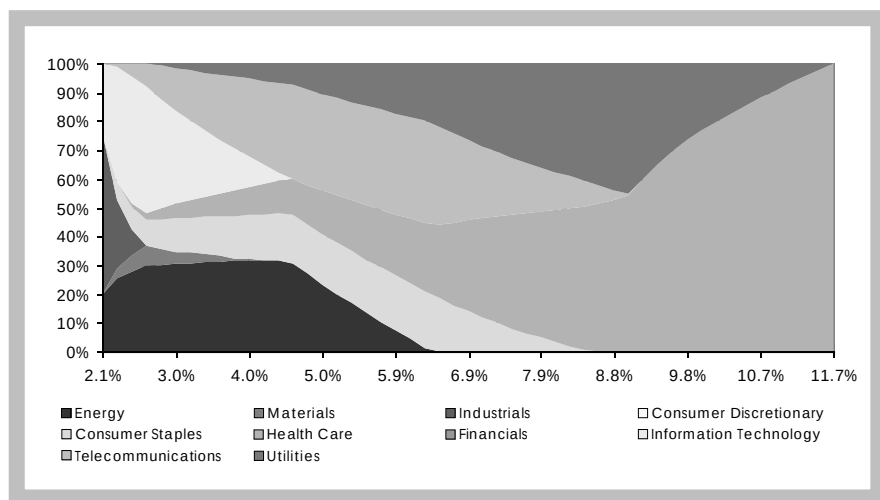


Figure 13.12 Asset allocation – Phase 3: MSCI Global Sector Model

Table 13.6 Phase 3 – model inputs

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	3.1%	4.7%	1.00	0.23	0.08	0.01	0.17	0.20	0.16	−0.12	−0.13	0.30
Materials	4.1%	6.2%	0.23	1.00	0.52	0.40	0.51	0.42	0.57	−0.02	0.15	0.33
Industrials	1.6%	2.4%	0.08	0.52	1.00	0.71	0.37	0.40	0.67	0.46	0.53	0.44
Cons. Disc.	2.4%	3.6%	0.01	0.40	0.71	1.00	0.16	0.24	0.62	0.56	0.62	0.36
Cons. Staples	6.4%	9.6%	0.17	0.51	0.37	0.16	1.00	0.60	0.46	−0.13	−0.04	0.31
Health Care	7.8%	11.7%	0.20	0.42	0.40	0.24	0.60	1.00	0.41	−0.02	0.04	0.28
Financials	4.2%	6.3%	0.16	0.57	0.67	0.62	0.46	0.41	1.00	0.30	0.44	0.42
IT	2.2%	3.3%	−0.12	−0.02	0.46	0.56	−0.13	−0.02	0.30	1.00	0.57	0.13
Telecomm.	5.3%	8.0%	−0.13	0.15	0.53	0.62	−0.04	0.04	0.44	0.57	1.00	0.32
Utilities	7.2%	10.8%	0.30	0.33	0.44	0.36	0.31	0.28	0.42	0.13	0.32	1.00

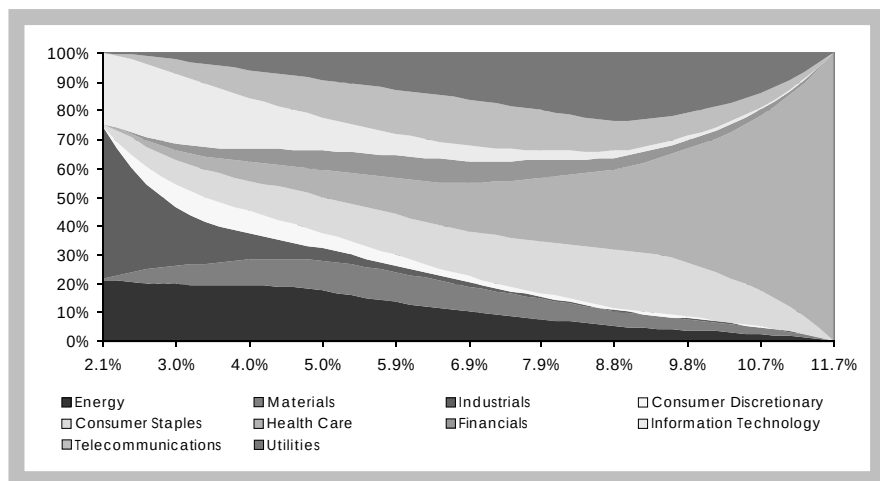


Figure 13.13 QRMCSAA – Phase 3: MSCI Global Sector Model

Overall this seems to fit very nicely into the traditional ideas of which sector to overweight and which to underweight during an economic downturn. The ‘value-added’ gain generated here stems from the fact that an optimisation algorithm has fully exploited opportunities from diversification between sectors, something that an asset manager could never do to the same degree of perfection.

Phase 4: moderately bullish asset allocation

Lastly we consider the *bottoming out* phase of the business cycle, where returns along with correlations are rising again in anticipation of higher company earnings. Volatilities are now back to their previous high levels, since as the economic upturn approaches asset managers aggressively rotate their portfolios to gain the most advantageous sector exposure. The model inputs for this phase appear as in Table 13.7.

The naive asset allocation appears in Figure 13.14. Using the QRMCSAA technique we obtain the chart shown in Figure 13.15. Defensive sectors are now much less prominent, but offensive sectors again feature strongly across the risk spectrum, although not as much as in Phase 1. All sectors are included in what seems an intuitive set of asset allocations along the efficient frontier/ridge. The financials and industrials sectors appear prominently now, which makes sense since they will benefit from the increased economic activity. Relative to the other phases of the business cycle, the materials sector now also seem to be favoured by the asset allocation.

Table 13.7 Phase 4 – model Inputs

	Return	Volatility	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
Energy	2.9%	4.4%	1.00	0.41	0.56	0.57	0.60	0.47	0.47	0.47	0.56	0.57
Materials	5.2%	7.8%	0.41	1.00	0.60	0.58	0.34	0.42	0.59	0.50	0.32	0.27
Industrials	7.6%	11.4%	0.56	0.60	1.00	0.87	0.66	0.51	0.67	0.72	0.60	0.49
Cons. Disc.	6.6%	9.9%	0.57	0.58	0.87	1.00	0.73	0.47	0.64	0.74	0.68	0.47
Cons. Staples	5.4%	8.1%	0.60	0.34	0.66	0.73	1.00	0.46	0.35	0.48	0.70	0.54
Health Care	6.5%	9.8%	0.47	0.42	0.51	0.47	0.46	1.00	0.56	0.42	0.31	0.40
Financials	7.3%	11.0%	0.47	0.59	0.67	0.64	0.35	0.56	1.00	0.61	0.38	0.41
IT	6.7%	10.1%	0.47	0.50	0.72	0.74	0.48	0.42	0.61	1.00	0.54	0.44
Telecomm.	5.8%	8.7%	0.56	0.32	0.60	0.68	0.70	0.31	0.38	0.54	1.00	0.58
Utilities	4.9%	7.4%	0.57	0.27	0.49	0.47	0.54	0.40	0.41	0.44	0.58	1.00

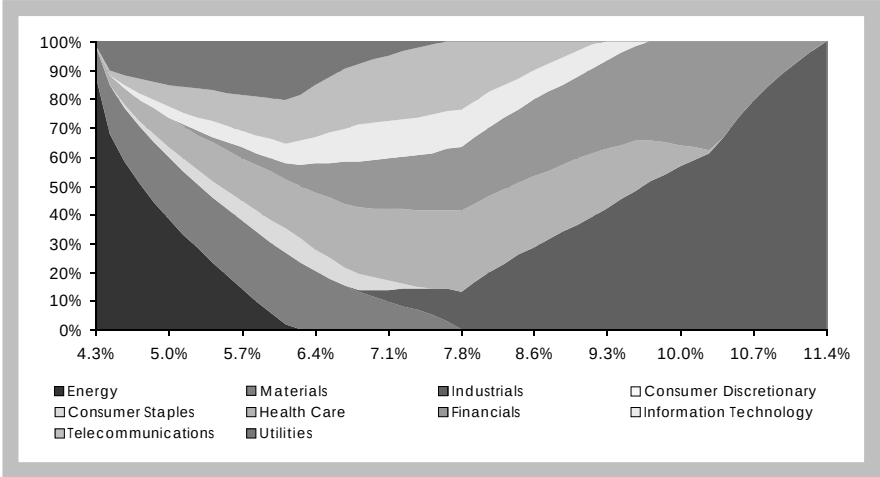


Figure 13.14 Asset allocation – Phase 4: MSCI Global Sector Model

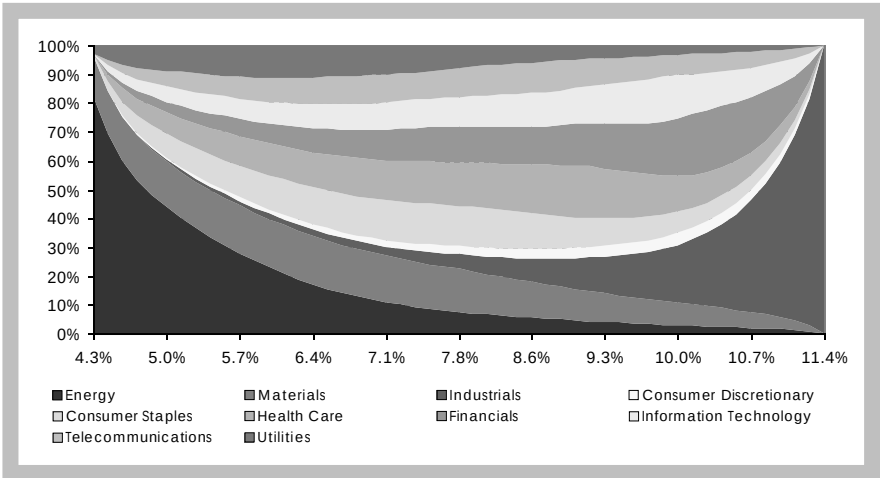


Figure 13.15 QRMCSAA – Phase 4: MSCI Global Sector Model

SUMMARY

In this chapter we have demonstrated that, provided with suitable model inputs, the QRMCSAA technique can be adapted for tactical asset allocation purposes, with a time horizon of six to twelve months. It is important to emphasise that the appropriate sector asset allocation that results from a Sector

Rotation Model such as that which has been outlined in this chapter does not magically produce results that asset managers should blindly follow. Suggested or recommended asset allocations are still very much the result of the assumptions made, particularly with regard to the expected levels of sector returns. Furthermore, one can argue both for and against using a simplified approach to determining expected volatilities such as the one applied here, or whether to attempt more direct forecasts of both returns and volatilities.

Having established this Sector Rotation Model, it seems evident that most of the challenge in applying such a model lies in the process of estimating or forecasting inputs, and then in calibrating the model to produce viable results. We use the word 'calibrate' quite deliberately to emphasise that an asset allocation model such as this does not necessarily produce workable and intuitive results from the outset. It will most likely need to be fine-tuned for a specific task at hand. Finally, we again emphasise that quantitative models do not generate information that is not already present in the input data. It merely extrapolates and refines the data, to produce asset allocations that make investment sense. At the end of the day, however, it is the responsibility of the asset manager to produce suitable model inputs.

Note

- 1 Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, Heiko Ebens (2000).

PART IV

QUANTITATIVE RISK MANAGEMENT

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TRACKING ERROR AND INFORMATION RATIO

DEFINITIONS OF TRACKING ERROR

Absolute vs. relative risk measures

In this chapter we will formally introduce and examine *portfolio tracking error*. As discussed in earlier chapters, risk is quantified by some measure of variance of asset returns. Assessing risk in absolute terms is usually done by calculating the volatility of an asset's or a portfolio's return, which in turn is defined as the annualised standard deviation of returns, or volatility. As demonstrated, the standard deviation of an asset's or portfolio's return is a simple measure of variation around its mean. It is defined as the square root of the variance, which in turn is defined as the sum of the squared differences between the returns and the mean of the returns.

We recall that the volatility for portfolio P over N time periods is defined as:

$$\sigma_P = \sqrt{\frac{1}{N-1} \sum_{n=1}^N (r_{P,n} - \bar{r}_P)^2} \cdot T \quad [\text{Eq. 14.1}]$$

This measure of risk is the one most commonly employed by asset managers, as it is relatively easy to calculate, and as it goes well with the conventional Markowitz-inspired mean-variance optimisation set-up that most asset managers employ to some extent. One could argue that investors ought to be concerned only with downside risk, but the volatility approach has gained

widespread use in the financial sector because of its relative ease of use. Methods to quantify downside risk, such as semi-variance, have not caught on. However, making the usual assumptions of asset returns normally and symmetrically distributed around a mean, and bearing in mind that all risk is relative, the use of the standard deviation is quite defensible.

As noted, in addition to an absolute measure of risk such as portfolio return volatility, asset managers also require some measure of relative risk in order to demonstrate their ability to meet clients' requirements in terms of risk management and control. This is mainly relevant for *passive management* or *indexing*, in which an asset manager attempts to design a so-called *replicating portfolio* in order to track a benchmark. However, it is also quite informative for *active* managers, for two main reasons. Firstly, the investment mandate of the client typically involves the definition of some appropriate benchmark. Typically the client will want to ensure that even though the manager has an active mandate and is therefore allowed and indeed expected to deviate from the benchmark, there is still some limit to the extent to which he can deviate from that benchmark over time. Secondly, in terms of performance measurement, the active manager is often measured by the risk he has taken on in order to reach the realised return over a certain period. The client will also want to know what contribution to risk the deviations from the benchmark have brought with them, in order to assess the manager's ability to control the risk of his active mandate. Simply stated, the client will not be very happy with a manager who has contributed a few percentage points of extra return by deviating from the benchmark, if this deviation has meant a large amount of extra risk carried by the client. In other words, what we need is some measure of the difference between the portfolio's performance and the benchmark performance.

An obvious measure of this is *tracking error*, more accurately called *active risk*, which we have alluded to earlier in this book. The notion of active risk stems from the insight that any portfolio can be viewed as a benchmark plus an active portfolio. Tracking error is defined as the volatility of the difference D between the return on a portfolio and the return on the benchmark. Such a benchmark will typically be some form of comparable index, such as the MSCI World, the FTSE100, the S&P500 or some tailor-made index constructed artificially for a specific purpose. Expressed in mathematical terms, the tracking error of a portfolio P relative to some benchmark B over N time periods is:

$$\psi_P = \frac{1}{N-1} \sum_{n=1}^N (D_n - \bar{D})^2 \cdot T \quad [\text{Eq. 14.2}]$$

where $D_n = (r_{P,n} - r_{B,n})$, the difference between the (excess) return on the portfolio at time n and the (excess) return on the benchmark at time n . $\bar{D}$ is the mean of the return differences. T is the sample frequency.

In order to move on we need to introduce another useful measure of relative risk called the *residual risk* – also called *diversifiable*, *unsystematic* or *non-benchmark-related* risk. This is the portfolio risk *orthogonal* to the systematic risk of the portfolio. The concept can be illustrated by looking at the relationship between portfolio (excess) return and benchmark (excess) return:

$$r_P = \beta_P \cdot r_B + \theta_P \quad [\text{Eq. 14.3}]$$

Portfolio return can thus be divided into two components – a benchmark-related (systematic) component and a component that is independent (non-systematic or residual) of the benchmark return. The strength of the relation between the portfolio return and the benchmark return is determined by Beta:

$$\beta_P = \frac{\text{Cov}(r_P, r_B)}{\text{Var}(r_B)} \quad [\text{Eq. 14.4}]$$

This is the factor that quantifies *systematic risk* relative to the benchmark – also called the *un-diversifiable* or *benchmark-related risk*.

The first term in Equation 14.3 expressed the dependency of portfolio return on benchmark return. Note that Equation 14.3 expresses a linear relationship, so for any given level of Beta, the return deviations from that linear relationship will show up as the residual, which are described by the second term in the equation. The term *residual* thus refers to the fact that in a linear relationship between portfolio returns and benchmark returns, the part of the portfolio return that is unexplained by the benchmark (orthogonal) is called the *residual term*. The residual risk of the portfolio relative to the benchmark is thus a measure of variance of that residual return. The mathematical expectation of the residuals is the residual return also called *alpha* – that is, $E[\theta_P] = \alpha$. It is the return above (or below) that which would be expected by the relationship between the portfolio and benchmark returns. The standard deviation of the residuals is the residual risk: $\text{Std}[\theta_P] = \omega$. This means that the portfolio's risk can be expressed as:

$$\sigma_P^2 = \beta_P^2 \cdot \sigma_B^2 + \omega_P^2 \quad [\text{Eq. 14.5}]$$

where σ_P and σ_B are the volatilities of the portfolio and the benchmark respectively. Note that these are just squared standard deviations. Equation 14.5 states that portfolio volatility is a product of the benchmark-related risk and the non-benchmark-related residual risk caused by the deviations of the portfolio from the benchmark. Note that if the portfolio replicates the benchmark completely, the residual risk term will be equal to zero, and portfolio volatility will be equal to benchmark volatility. This residual risk can consequently be expressed as:

$$\omega_P^2 = \sigma_P^2 - \beta_P^2 \cdot \sigma_B^2 \quad [\text{Eq. 14.6}]$$

which, after taking the square root of both sides, can be rewritten:

$$\omega_p = \sigma_p^2 - \beta_p^2 \cdot \sigma_B^2 \quad [\text{Eq. 14.7}]$$

This indicates that the residual risk of the portfolio is equal to the portfolio's volatility minus some measure of the effect of the benchmark on portfolio risk. This will prove a very useful attribute later on in this chapter.

RISK GEOMETRY

The benchmarking triangle: Markowitz meets Pythagoras

In order to throw more light on the nature of tracking error and the way it is related to portfolio and benchmark risk, we turn to a geometrical illustration. Pythagoras believed that all relations could be reduced to number relations. This generalisation stemmed from observations in music, mathematics and astronomy that revealed consistent patterns when examined. The most widely known discovery of this school was the fact that the diagonal of a square is not a rational multiple of its side. This result showed the existence of irrational numbers. Pythagoras' well known theorem can be written $c^2 = a^2 + b^2$, where c is the side of a triangle opposite the right angle, and a and b are the remaining sides. This of course implies that a can be expressed as $a = \sqrt{c^2 - b^2}$. Although this theorem now known as Pythagoras' Theorem was apparently known to the Babylonians a thousand years earlier, Pythagoras may well have been the first to prove it.

Since the relationships between the different definitions and concepts of risk presented above can seem a bit abstract, we may usefully illustrate them geometrically using the Pythagorean properties of right-angled triangles. The reader may at this point have noted that the geometrical relationship presented above is analogous to the relationship between 'residual risk', 'portfolio risk' and 'benchmark risk' in Equations 14.5–7. These relationships appear geometrically as in Figure 14.1. Note that the triangle in Figure 14.1 is made up of two right-angled triangles. We will use this fact below in the decomposition of total portfolio risk and active risk. Note that the relationship between portfolio risk and benchmark risk is related to the angle C of the triangle to the left. We will investigate this further below.

Decomposing tracking error

Figure 14.1 illustrates the orthogonality of residual risk to benchmark risk, and shows that the residual risk is uncorrelated with benchmark risk.¹ Furthermore it becomes clear that the residual risk of a portfolio can be defined as the

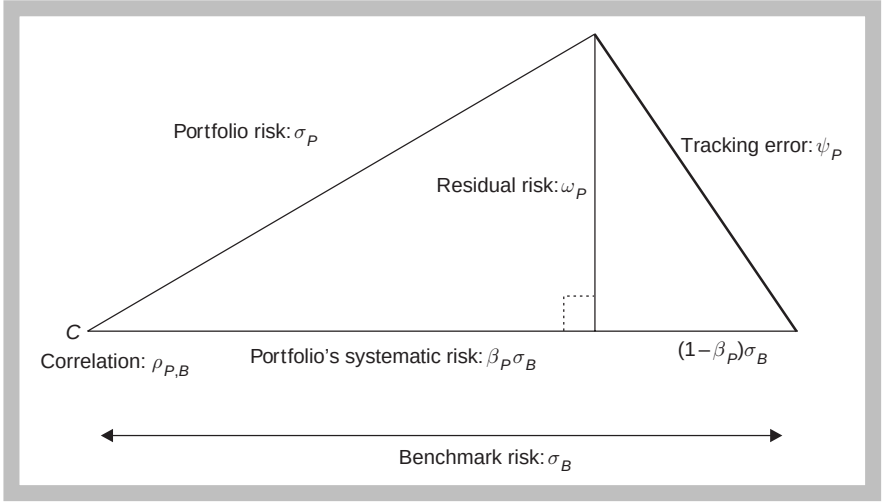


Figure 14.1 Geometric relations between portfolio and benchmark risk

portion of tracking error that is uncorrelated with benchmark risk. This can be seen from the fact that the residual risk-line is orthogonal to the benchmark risk-line. It is also apparent that portfolio P 's tracking error with respect to the benchmark B can be expressed as:

$$\Psi_P = ((1 - \beta_P) \cdot \sigma_B)^2 + \omega_P^2 \quad [\text{Eq. 14.8}]$$

we will need the following reformulation:

$$\omega_P^2 = \Psi_P^2 - (1 - \beta_P)^2 \cdot \sigma_B^2 \quad [\text{Eq. 14.9}]$$

The term $(1 - \beta_P) \cdot \sigma_B$ in Equation 14.8 requires a bit of thought. It is the risk orthogonal to the portfolio's residual risk. Therefore, $(1 - \beta_P) \cdot \sigma_B$ can be thought of as the residual risk of the active portfolio, or from Equation 14.8 the portfolio's active risk or tracking error, less its residual risk. It is thus the part of tracking error that is unrelated to or uncorrelated with the portfolio's residual risk. This is also apparent from the orthogonality in Figure 14.1 of the $(1 - \beta_P) \cdot \sigma_B$ line with respect to the ω_P line. Therefore, Equation 14.8 states that tracking error is equal to the portfolio's residual risk plus the active portfolio's residual risk – that is, the sum of the portfolio's passive and active residual or unsystematic risks.

Since the residual risk of a portfolio can also be defined as the portion of tracking error that is uncorrelated with benchmark risk, $(1 - \beta_P) \cdot \sigma_B$ can then be defined as the portion of tracking error that is related to benchmark risk: that is, the active portfolio's benchmark-related risk. This also becomes apparent when considering that in geometric terms, $(1 - \beta_P) \cdot \sigma_B$ is the *projec-*

tion of the tracking error line onto the benchmark risk line, and thus that the two lines are coincident, making them perfectly correlated.

There is one more illuminating interpretation of Equation 14.8 left. As demonstrated above, tracking error is the active portfolio's benchmark-related risk, plus the portfolio's residual risk (or unsystematic risk). To demonstrate this, assume that benchmark risk is equal to 1. This means that the active portfolio's benchmark-related risk is equal to $(1 - \beta_p) \cdot \sigma_B$. If one then *slides* the benchmark risk line to the left by exactly the amount $(1 - \beta_p)$, then *sticking out* on the other side of C is a portion of benchmark risk exactly equal in length to $-(1 - \beta_p) = (\beta_p - 1)$, which just happens to be the definition of the portfolio's *active Beta* – the Beta of the active portfolio. For a Beta of 1, $(1 - \beta_p) \cdot \sigma_B$ then becomes 0 and portfolio tracking error then equals residual risk – which is quite intuitive, since portfolio risk is then equal to benchmark risk, which in turn stems from the fact that the replicating portfolio is then matching the index perfectly in terms of risk.

Bearing in mind the relationship between correlation and tracking error, we return to Figure 14.1 and the composition of tracking error. From the two expressions of residual risk presented in Equations 14.6 and 14.9 we obtain:

$$\Psi_p^2 - (1 - \beta_p)^2 \cdot \sigma_B^2 = \sigma_p^2 - \beta_p^2 \cdot \sigma_B^2 \quad [\text{Eq. 14.10}]$$

This relation is another Pythagorean relation that can be seen clearly from Figure 14.1. Rearranging terms yields a measure of the active portfolio variance (note that we use the term 'variance' here, as the term is squared):

$$\Psi_p^2 = (1 - \beta_p)^2 \cdot \sigma_B^2 + \sigma_p^2 - \beta_p^2 \cdot \sigma_B^2 \quad [\text{Eq. 14.11}]$$

After a bit of algebra this can be reformulated to a surprisingly intuitive definition of portfolio tracking error:

$$\Psi_p = \sqrt{\frac{1}{2} \cdot \sigma_B^2 + \frac{1}{2} \cdot \sigma_p^2 - \beta_p \cdot \sigma_B^2} \quad [\text{Eq. 14.12}]$$

Equation 14.12 essentially states that portfolio tracking error depends on the average of portfolio and benchmark risks, less some measure of the covariance between portfolio risk and benchmark risk. This also demonstrates that, contrary to intuition, there is no need to explicitly consider the correlation coefficient between the portfolio and the benchmark, since this information is provided implicitly by the portfolio Beta.

The factor model variance–covariance approach

We can dig deeper into active and residual risk by restating the return relationship from Equation 14.3 in matrix form:

$$\vec{r} = X \cdot \vec{b} + \vec{u} \quad [\text{Eq. 14.13}]$$

where $\vec{r}$ is an N vector of excess returns, X is a $N \cdot K$ matrix of factor exposures, $\vec{b}$ is a K vector of factor returns, and $\vec{u}$ is an N vector of asset-specific returns. Similarly to Equation 14.4, letting F denote the factor covariance matrix and Σ denote the asset-specific variance matrix, we can now express the risk in terms of a covariance matrix of this factor model:

$$\Sigma = X^T \cdot F \cdot X + \Delta \quad [\text{Eq. 14.14}]$$

The risk is thus divided into asset or portfolio factor exposures and asset-specific exposures.²

In a portfolio context, a portfolio P 's factor exposure is given by the N vector:

$$\vec{x}_p = X^T \cdot \vec{w}_p \quad [\text{Eq. 14.15}]$$

where $\vec{w}_p$ is the N vector of portfolio weights. Portfolio risk (variance) can now be expressed as:

$$\sigma_p^2 = \vec{x}_p^T \cdot F \cdot \vec{x}_p + \vec{w}_p^T \cdot \Delta \cdot \vec{w}_p \quad [\text{Eq. 14.16}]$$

Again, the portfolio risk is divided into risk stemming from factor exposures and asset-specific risk, both naturally depending on the portfolio weights. This can of course be rewritten as the familiar:

$$\sigma_p^2 = \vec{w}_p^T \cdot \Sigma \cdot \vec{w}_p \quad [\text{Eq. 14.17}]$$

Since tracking error is defined as the portfolio active risk relative to the benchmark, or equivalently the risk of the active portfolio determined by the active portfolio weights, we need to determine these weights:

$$\vec{w}_{PA} = \vec{w}_P - \vec{w}_B \quad [\text{Eq. 14.18}]$$

The factor exposures of the active portfolio can be expressed as:

$$\vec{x}_{PA} = \vec{x}_P - \vec{x}_B \quad [\text{Eq. 14.19}]$$

Portfolio P 's active risk or tracking error relative to its benchmark B is thus simply given by:

$$\Psi_P^2 = \vec{x}_{PA}^T \cdot F \cdot \vec{x}_{PA} + \vec{w}_{PA}^T \cdot \Delta \cdot \vec{w}_{PA} \quad [\text{Eq. 14.20}]$$

which is equivalent to:

$$\Psi_P^2 = \vec{w}_{PA}^T \cdot \Sigma \cdot \vec{w}_{PA} \quad [\text{Eq. 14.21}]$$

This of course entails that portfolio tracking error is given by:

$$\Psi_P = \sqrt{\vec{w}_{PA}^T \cdot \Sigma \cdot \vec{w}_{PA}} \quad [\text{Eq. 14.22}]$$

Equation 14.22 is in fact just what we were looking for: the risk of the active portfolio. The calculation of active risk is thus analogous to the calculation of total portfolio risk. The only difference is that when calculating active risk we use the active portfolio weight vector instead of the total portfolio weight vector. The covariance matrix applied is the same.

Similarly the residual portfolio – the portion of the active portfolio that is unrelated to the benchmark – can be expressed as:

$$\vec{w}_{PR} = \vec{w}_P - \beta_P \cdot \vec{w}_B \quad [\text{Eq. 14.23}]$$

where β_P is portfolio P 's Beta relative to the benchmark. Combining this with the fact that the factor exposures of the residual portfolio can be expressed as:

$$\vec{x}_{PA} = \vec{x}_P - \vec{x}_B \quad [\text{Eq. 14.24}]$$

the portfolio's 'residual risk' is then given by:

$$\omega_P^2 = \vec{w}_{PR}^T \cdot \Sigma \cdot \vec{w}_{PR} \quad [\text{Eq. 14.25}]$$

INFORMATION RATIO

Definitions

The *information ratio* of a portfolio is an attempt at measuring the performance of the active manager by relating return above that of the benchmark to the risk incurred, above that of the benchmark. It is closely related to tracking error discussed above, and is mainly applied in a benchmarking context. Unfortunately there seems to be some confusion in the industry as to the proper definition of information ratio, which may stem from several similar ratios that each provide interesting information regarding managers' performance in terms of return and risk.

In this section we will present the most appropriate definition of information ratio and discuss why the definition presented here is the most appropriate for assessing active manager performance relative to a benchmark. Previously, we demonstrated that portfolio (excess) return can be described by:

$$r_P = \beta_P \cdot r_B + \theta_P \quad [\text{Eq. 14.26}]$$

with $E[\theta_P] = \alpha$ and $Std[\theta_P] = \omega$. The information ratio is defined as the annualised ratio of the portfolio's residual return to its residual risk. Formally we write the information ratio as follows:

$$IR_1 = \frac{\alpha}{\omega} \quad [\text{Eq. 14.27}]$$

As the names indicate, residual return and residual risk are defined as the portions of active portfolio return and risk that are unexplained by the return on the benchmark. In a regression analysis of benchmark return on portfolio return, α is the intercept in the regression or simply the mean of residual returns:

$$\alpha = r_p - \beta_p \cdot r_B \quad [\text{Eq. 14.28}]$$

and ω is the standard deviation of alpha. So, running a regression of benchmark returns on portfolio returns yields a series of residual returns equal to $\alpha_i = r_{pi} - \beta_p \cdot r_{Bi}$, where i denotes the observation number. The mean of these residual returns is thus the expectation of return, based on benchmark return. This expected return is called α ,³ and its standard deviation is called ω . The information ratio thus expresses expected residual or abnormal return relative to the risk incurred in order to obtain this return.

One could argue that when evaluating active management, using active return in the numerator and active risk (that is, tracking error) in the denominator of the information ratio would be more appropriate since we would then have a measure of actually obtained active return relative to actually incurred active risk. This would result in an information ratio defined as:

$$IR_2 = \frac{r_p - r_B}{\Psi_p} \quad [\text{Eq. 14.29}]$$

This would actually be quite intuitive, and also akin to the much-used Sharpe Ratio, which of course measures total portfolio return relative to total portfolio risk. However, the information ratio is calculated using residual return and residual risk and not active return and active risk because here we are interested in isolating the performance of the manager, not the benchmark. Active return and active risk *do*, and residual return and residual risk *do not*, include the portion of the return and risk that is caused by the benchmark. Thus, using residual return and residual risk instead of active return and active risk to calculate the information ratio means that the manager's performance relative to the benchmark is measured only in terms of the portion of active return and active risk that is *not* due to benchmark behaviour – that is, the active performance that is due to the manager's skill and/or luck.

Another way of putting this argument is that using active return and risk instead of residual return means that the implicit assumption is made that the portfolio's Beta is exactly equal to one, which is hardly ever the case. This follows easily from Equation 14.4 and Equation 14.8.

Residual return and residual risk are thus the more appropriate measures for the purpose of assessing active manager performance in terms of return relative to risk. While the ratio of total active return to total active risk may be informative in its own right, it is not the information ratio. The former includes bench-

mark effect on the active portfolio, which interferes with our goal of evaluation of manager skill. There seems to be some disparity in the industry between definitions of information ratio, but if the purpose of the exercise is to determine the manager’s ability to add value – in terms of return relative to risk – by deviating from the benchmark, the above definition of the information ratio is the appropriate measure. In other words, the information ratio measures the quality of the active manager’s *activeness* relative to the benchmark, but independent of the return on the benchmark itself.

Information ratio along the efficient frontier

Similar to the chart in Figure 9.7 illustrating the expected Sharpe Ratio of the portfolios along the efficient frontier, we can determine the expected information ratio along the active return efficient frontier presented in Chapter 11 – see Figure 14.2. To the far left of the chart the information ratio is naturally zero, since this is the minimum active risk portfolio: the portfolio that entails no deviations from the benchmark, and thus neither active return nor active risk. After shooting up to its highest level for a portfolio very close to the benchmark portfolio, the information ratio declines relatively smoothly as we increase the target tracking error level towards its maximum given the constraints on sector deviations or bets. This illustrates that while we may be able to squeeze out more active return by deviating more from the benchmark, this will happen at an increasing price as measured by active risk or tracking error. In order to

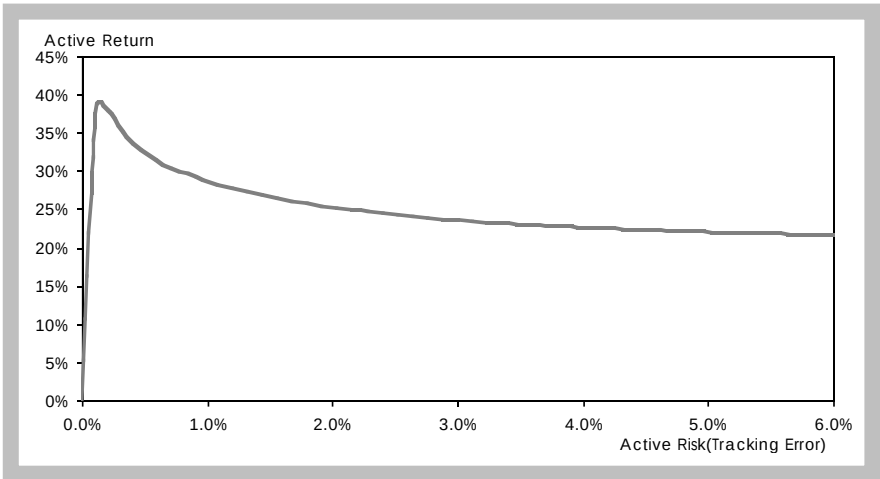


Figure 14.2 Expected information ratio along the efficient frontier

Table 14.1 Empirical distribution of information ratios

IR	Percentile	Interpretation
1.0	90	Exceptional
0.5	75	Good
0.0	50	Fair/Average
-0.5	25	Lacking
-1.0	10	Poor

calculate the ex post information ratio of a portfolio or manager, we need the time series of returns on the portfolio and on the benchmark.

Based on research on US mutual funds,⁴ Table 4.1 can be used as a rule of thumb for evaluating information ratios. Of course, the typical values in the table presented above do not mean that information ratios cannot lie above or below 1.0 or -1.0. However this is quite unusual, except for very small data series where statistical uncertainty may dominate results.

In reality we would need several years of performance to properly evaluate a manager. This of course also applies to the evaluation of 'tracking error'. Information ratios greater than 0.22 for 60-month results and greater than 0.28 for 36-month results are statistically different from an information ratio of zero at the 95% confidence level.

ACTIVE MANAGEMENT VALUE ADDED

The goal of the active manager – once more

The active manager is in the business of selling tracking error to clients. This is done with the expectation that the manager will be able to add value to the client by deviating from the benchmark, thereby creating tracking error that should, it is hoped, result in a higher return than that of the benchmark. If the active manager does not hold superior information, she should not be active: rather she should be passive and simply replicate the benchmark. Whether an active manager is capable of producing superior returns under strict risk control is exactly what the information ratio attempts to discover. On a risk-adjusted basis, is the manager able to add value to the client by deviating from the benchmark? If she is not, she should not deviate: the client would be better served with a passive portfolio.

Value added, the information coefficient and breadth

From the definition of the information ratio it is clear that the manager can only add value for the client by increasing residual return relative to residual risk.

$$IR = \frac{\alpha}{\omega} \quad \alpha = IR \cdot \omega \quad [\text{Eq. 14.30}]$$

However, this also means that the expected residual return or alpha can be increased by raising the residual risk, but this does not necessarily mean increased value added for the client. The value added can only be increased by boosting the information ratio. This will increase the expected residual return for a given level of residual risk. Following Grinold and Kahn, the value added by the manager can thus be quantified by:

$$VA = \alpha - \lambda \cdot \omega^2 \quad [\text{Eq. 14.31}]$$

where λ is a measure of the client's risk aversion. Using the above definition of alpha we can rewrite this:

$$VA = IR \cdot \omega - \lambda \cdot \omega^2 \quad [\text{Eq. 14.32}]$$

The value added by the manager is thus a concave function of residual risk – see Figure 14.3. Strictly speaking, it is thus the job of the active manager to choose the level of residual risk that maximises value added for the client

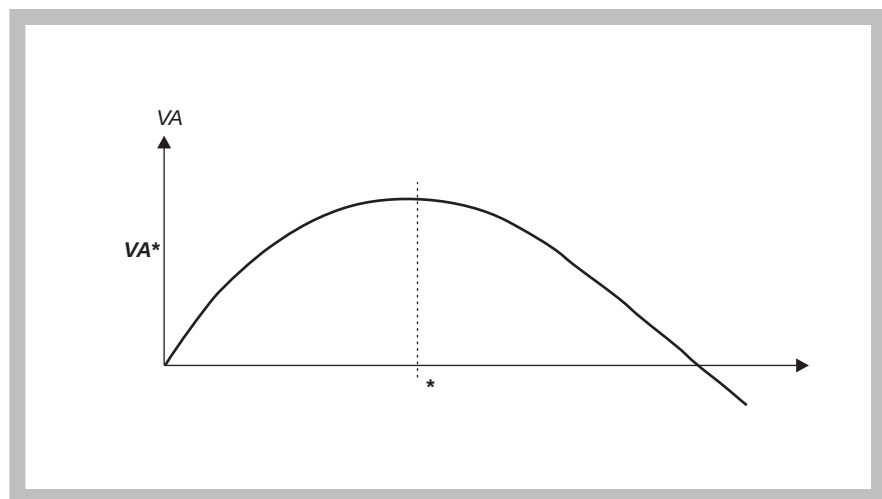


Figure 14.3 Value added as a function of residual risk

depending on the client's risk aversion. The level of residual risk that maximises value added is found by taking the partial derivative of VA with respect to residual risk and setting it equal to zero:

$$\frac{\partial VA}{\partial \omega} = IR - 2\lambda \cdot \omega = 0 \quad [\text{Eq. 14.33}]$$

This gives us an optimal level of residual risk:

$$\omega^* = \frac{IR}{2\lambda} \quad [\text{Eq. 14.34}]$$

The optimal level of information ratio is thus inversely related to risk aversion, and directly related to information ratio. This in turn means that the value added in the optimum is given by:

$$VA^* = \frac{(IR)^2}{4\lambda} \quad [\text{Eq. 14.35}]$$

Identifying the client's risk aversion λ is not an exact science, which is why the identification of the level of residual risk that will maximise client value added is not as easy as it might seem. The optimal level of value added as indicated by Equation 14.34 may thus be more elusive than it appears. However, there is an important point to be made here. Regardless of the client's risk aversion, the value added for the client is higher for higher levels of the information ratio. So whether or not the client has an identifiable risk aversion, and even regardless of whether the client is *aware* of his or her own risk aversion, the goal of the manager is to seek to boost the information ratio – all investors will agree that managers with higher information ratios are better managers. This is because per unit of residual risk the manager delivers higher residual return for higher information ratios. From Equation 14.35 we see that the increase in value added is proportional to the square of the information ratio. This is why high information ratios are a much sought-after *commodity*. A manager with consistently high information ratios is likely to have a relatively easy time selling tracking error or residual risk to clients.

In what has become known as the *fundamental law of active management*,⁵ the information ratio is expressed in terms of two components. The first is the *information coefficient* (IC), which measures the manager's forecasting ability or skill. It is determined by the correlation between forecast and realised alphas or residual returns. The second is *breadth* (BR), defined as the number of independent bets per year:

$$IR = IC \cdot \overline{BR} \quad \text{Eq[14.36]}$$

The information coefficient is thus a measure of the forecasting ability of the manager. More precisely it is the correlation between forecast alphas (or residual returns) and realised alphas. 'Breadth' is defined as the number of independent bets: by 'independent' we mean separate bets that are based on perceived superior information regarding an asset or an asset class, which does not pertain to other assets or asset classes.

SUMMARY

Tracking error is the most widely used measure for controlling (relative) risk in an active manager's portfolio. It is defined as the volatility of the active return on the portfolio relative to the benchmark. However, when measuring the value added by the manager, it makes more sense to use the residual return and risk. This is because these measures correct the active return and risk for the movement of the benchmark.

Although illustrating portfolio characteristics using geometry may seem very abstract, doing so can actually be quite useful, as we have demonstrated in this chapter. Particularly in terms of interpreting the sometimes non-intuitive portfolio risk expressions, this can provide some insight into portfolio and benchmark risk dynamics. Asset managers and the people who watch over them are surprisingly one-sided in their approach to describing portfolio performance, in that return performance receives perhaps 90 per cent of the attention, whereas risk is seen as an unpleasant residual that one must learn to live with. Naturally, at the end of the day obtaining a return is the main reason why people invest to start with. Without returns people would not invest. But in order to deliver high active returns on a consistent basis, asset managers need to manage risk as actively as they manage return. Bearing in mind that most asset managers employ some form of mean-variance technique for portfolio analysis, one may even argue that return and risk are inseparable, as their interaction with each other is by now internalised in almost all quantitative analysis.

For this reason it is vital that asset managers (including portfolio managers as well as risk managers) understand the risk dynamics of their portfolio in relation to their benchmark. Failure to do so may lead to inefficient asset allocations and to unintentional risk-taking, thereby undermining the quality of service provided to paying clients.

Notes

- 1 This is a practical property since it allows us to determine residual return and risk by running a regression of benchmark return on portfolio return, and to obtain the residual risk from the annualised standard deviation of the residuals of the regression itself.
- 2 Here we assume that the specific returns are uncorrelated with factor returns, and that asset i 's specific return is uncorrelated with asset j 's specific return. This means that all elements that are off the diagonal of the $N \times N$ are zero.
- 3 This is also sometimes referred to as the 'abnormal return' as it is the difference between the actual portfolio return and that which would be expected on the basis of benchmark return.
- 4 Kahn (1997).
- 5 Grinold and Kahn (1995), 'Active Portfolio Management'.

SECTOR RISK MODEL

THE GLOBAL PERSPECTIVE

As demonstrated in previous chapters, quantitative portfolio optimisation and asset allocation are closely tied to the risk management process. This is true whether we work with *total* return and risk or *active* return and risk. When choosing an appropriate asset allocation, we do so in accordance with our preferences for the trade-off that exists between expected return and expected risk. Many of the investment decisions being made in asset management organisations today increasingly depend on a global sector perspective. Previously the focus was mainly on stocks versus bonds, and subsequently on domestic versus international equity. Country considerations were very much at the heart of the asset allocation process, because large diversification benefits can be obtained from placing assets in equity markets in a range of countries globally. This still holds true, since there are considerable differences between the economic environment in different countries and regions. However the focus is increasingly shifting to global sectors, as more and more sectors become part of a global marketplace and become subject to global competition. Many sectors are thus becoming global, in the sense that the companies that comprise these sectors (such as the information technology sector, the automobile sector or the pharmaceutical sector) compete internationally, and operate, buy and sell in several international markets and/or economic regions.

This results in the country factor losing much of its significance, but at the same time the sector factor *increases* in importance. Hence, the return on

the stock of a pharmaceutical company based in Germany may well be more highly correlated with the returns on a similar company in the US than with the German equity market. On the other hand, the correlation between these two particular sectors may be low on a global scale, making it more sensible to diversify according to sector or business activity than according to geographical location. As more and more sectors undergo this change from geographically localised sectors to truly global sectors, the risk-management process should be adapted to reflect this. It follows that a model for analysing and assessing sector risk exposure, as opposed to country risk, may increasingly be of use to asset management organisations in the future.

In this chapter we construct such a Sector Risk Model, and examine its potential impact on investment and risk management.

RISK CHARACTERISATION

Time horizon

A model for assessing sector risk is best built in a global setting, since we are using the argument that more and more sectors are increasingly becoming global sectors. This by no means precludes the establishment of a similar regional sector model for say the US, Europe or Japan, but in this chapter we will continue working with the MSCI Global Sector Model inputs that were employed in the QRMCSAA methodology established in previous chapters. There is a clear advantage to operating on a sector basis, in that correlations between returns on sectors as well as sector volatilities tend to be relatively stable over time. Correlations between two individual stocks can vary quite a lot over time, making it difficult to use estimates of correlation coefficients in a quantitative model. The correlation between the return in two sectors, on the other hand, is typically much more stable over time, because the basic characteristics of the business models and the sensitivity to general economic activity of the two sectors will not change substantially over shorter time periods. However, we employ this model as a short-term model intended to give a snapshot of the current environment in terms of portfolio sector risk. We will thus estimate the required model parameters based on observations spanning just six months. This may be statistically shaky, but it ensures that the model reflects to a reasonable degree the current equity market dynamics. However, it also entails that the model should be used with a time horizon of, say, one month. Further than that, the chances are that market dynamics will have changed sufficiently to make the model unreliable.

Structure

The aim of this chapter is to establish a quantitative portfolio risk model, and we shall do so by again employing the framework established in Modern Portfolio Theory, using our previous definitions of portfolio risk. As in the asset allocation framework, portfolio risk is defined as the volatility of portfolio returns:

$$\sigma_p = \sqrt{\vec{w}_p^T \bullet \Sigma \bullet \vec{w}_p} \quad [\text{Eq. 15.1}]$$

Similarly, the portfolio tracking error or active risk is defined as the volatility of the portfolio's active returns:

$$\psi_p = \sqrt{\vec{w}_{pA}^T \bullet \Sigma \bullet \vec{w}_{pA}} \quad [\text{Eq. 15.2}]$$

So again the centrepiece of the model is the covariance matrix, constructed from the correlation matrix and the sector volatilities. Building on the MSCI Global Sector Model for asset allocation, using the same 10 sector definitions, we shall construct a model that enables us to examine portfolio risk and tracking error within the same sector framework employed in previous chapters.

Analysing returns series

Before we proceed to construct the Sector Risk Model, it is useful to contemplate the development over time of the components that will make up the model. This will help us appreciate the quantitative variables that we are going to work with, and how changes in these are likely to impact on the model and the results it generates. In an effort to make the model reflect current market conditions, we employ volatilities and correlations that are estimated over just 60 days. The idea behind this is that such a period is long enough to solidly describe the actually observed market dynamics, but still short enough to reflect short-term changes in the variables employed. Taking as an example the information technology sector during the period from October 2000 to May 2002, the MSCI World IT Index and its volatility appears in Figure 15.1.

The volatility of the index is clear to see, both from the index itself and from the 60-day moving volatility. Note how the volatility of the index was very high as the market had topped out and began heading down, but as information technology universally fell out of favour among investors during late 2001 and early 2002, the volatility level actually dropped off, reflecting the fact that the markets had by then formed a consensus that the information technology sector was not a desirable place to invest, and so the index kept tanking lower for several months. Note also that the volatility level tends to

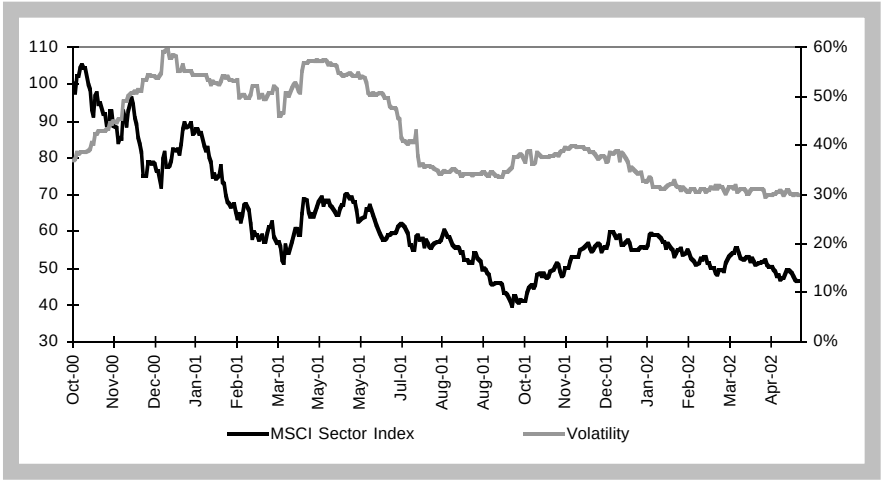


Figure 15.1 Information technology index and moving volatility

rise as the index reaches a temporary bottom and begins to rebound. As of January 2002, although the index has generally speaking continued its slide, the volatility level seems to have stabilised at around 30%. However, this is still very high, considering the number of stocks that comprise this index, thus generating some measure of diversification. Typically observed volatility levels for equity indices are in the region of 15%–25%, so the level of 30% reflects the continuous unease among investors about the earnings prospects of the sector as a whole. A similar chart for all 10 MSCI World sectors in the MSCI World Index can be found in the chapter appendix.

Alternatively, the volatility of returns can be illustrated by a simple bar chart of daily returns, with two bands, each of two standard deviations overlaid to indicate the changing level of volatility (see Figure 15.2). This is essentially the same information as that provided in the previous chart, but here we obtain a better sense of the changes in the level of volatility, and also how the previously illustrated level of volatility translates into actually observed sector returns.

Again we can see how the general variability of returns has faded somewhat over this particular time period, but that they still remain relatively large. Similar charts for all ten sectors can be found in the chapter appendix.

It may also be of interest to the investment or risk manager to observe how widely dispersed returns are over the period which we employ for estimating model parameters. As demonstrated, this can be achieved using the standard deviation or the volatility, but to illustrate further we can plot observed returns in a frequency chart, as shown in Figure 15.3. The chart illustrates the relat-

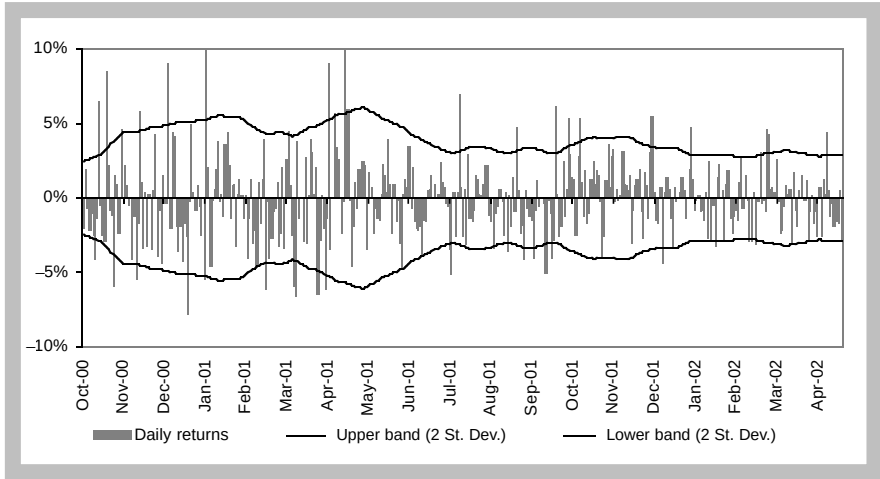


Figure 15.2 Information technology daily returns

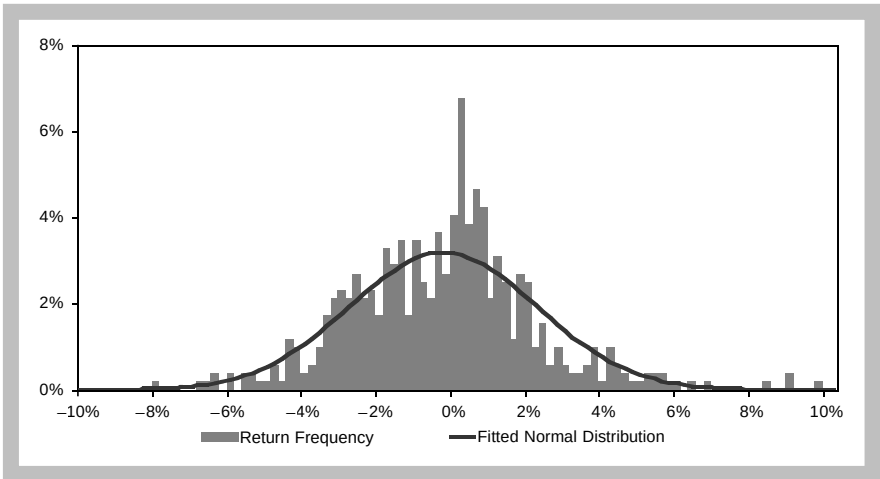


Figure 15.3 Information technology return frequency

ively large dispersion of returns ranging from -8% to $+10\%$ in one day. We have overlaid a normal distribution that has been fitted to the actually estimated standard deviation of the return series, and as indicated earlier it becomes clear that returns in the information technology sector are not normally distributed. There are many more returns in the extremes of the chart than a normal distribution would indicate. These are often labelled as

X-Sigma events, sigma being the Greek letter used to denote risk or standard deviation, and X being the number of standard deviations. A 4-Sigma event is thus a return that falls beyond 4 standard deviations from the mean in either a positive or a negative direction.

Figure 15.3 illustrates the danger of assuming that returns are normally distributed, since this assumption can lead to asset and risk managers seriously underestimating the probability of extreme events, which is after all what portfolio risk management should focus on. Furthermore it is a strong argument for employing a non-normal distribution, such as the logistic distribution presented above, as the basis for Monte Carlo Simulation in an asset allocation context. Even the fact that the distribution of returns for the information technology sector seems to exhibit two *peaks*, around -3% and 1% , could be incorporated into the model, provided that we have a good argument for doing so. Such arguments might involve the types of news that move the markets, and the resultant size of the returns. We shall return to this in Chapter 16 on Value-at-Risk.

Correlation analysis

Finally, in order to work meaningfully with portfolio risk we must of course determine correlations between sector returns over time. The core of the Sector Risk Model constructed in this chapter is the covariance matrix, determined by the correlation matrix and the volatility matrix presented in earlier chapters. A thorough understanding of the state of this matrix, as well as its dynamics over time, is essential in order to understand and apply the results produced by the Sector Risk Model. Figure 15.4 illustrates for each sector the average sector correlation with the remaining nine sectors. For the energy sector, for example, we can plot the development over time of its average correlation with the other nine sectors. This has been done for each sector, and an average of the correlation averages has also been plotted.

While this chart may appear a bit disordered, it does provide an indication of the level of correlation coefficients in the investment universe over time. In particular, we observe that the general correlation level between sector returns has risen during the period in question. Specifically, the average level of correlation coefficients has risen from around 0.2 at the end of 2000, to as much as 0.6 in May 2002. This is a significant increase, and one that will have been felt by asset and risk managers. We discuss this in more detail below.

The rise in correlation coefficients indicates that the opportunities for diversification have decreased during this period. Another interesting feature is the fact that average correlations between sectors are relatively stable over time. At

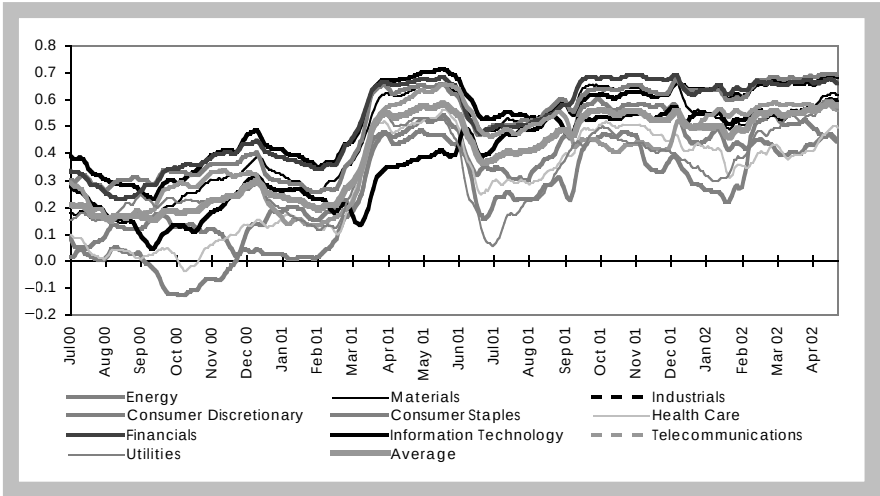


Figure 15.4 Sector correlations with other sectors

least they don't appear to change over night, indicating that to some extent we can assume that correlations do not change violently across all sectors in a very short amount of time. This is of use to the risk manager, since the practical use of the calculations of portfolio total risk and active risk to be performed here are to some extent dependent on these correlations being relatively stable over shorter periods of time.

Obviously the picture is a little more complex than this, since for every sector there exist nine individual correlation coefficients. But the average of a sector's correlation with the other nine sectors provides a reasonably good impression of changes in the opportunities for diversification in the portfolio. Note also that an overall average of the correlation coefficient averages has been included in the chart. As indicated, this is the average of the average correlation coefficients, and in terms of actual risk control or active risk management it does not say a lot and does not lend itself to immediate risk-management applications, but what it does do is provide a clear indication of the overall level of correlation coefficients in the portfolio, and thus the general level of opportunity for diversification.

In this particular example it is interesting to note that the average of the average correlation coefficients has risen steadily over the period, which generally speaking indicates that the opportunities for diversification across global sectors have been decreasing over this particular time period. This is important to be aware of, since a move to higher overall levels of correlation coefficients will increase both total portfolio risk and tracking error, even in a

situation where portfolio weights, benchmark weights and also volatilities stay constant. In other words it results in the tracking error increasing *even if the asset manager refrains from changing the portfolio*. Taking limitations on tracking error or total portfolio risk into consideration, it thus tends to give the manager less room for deviations from the benchmark.

As mentioned above, the focal point of the Sector Risk Model is the covariance matrix, generated from the observed volatilities and the observed correlation matrix. The latter is the same that is applied in the QRMCSAA framework and it appears as in Figure 15.5. This *heat map* illustrates what we

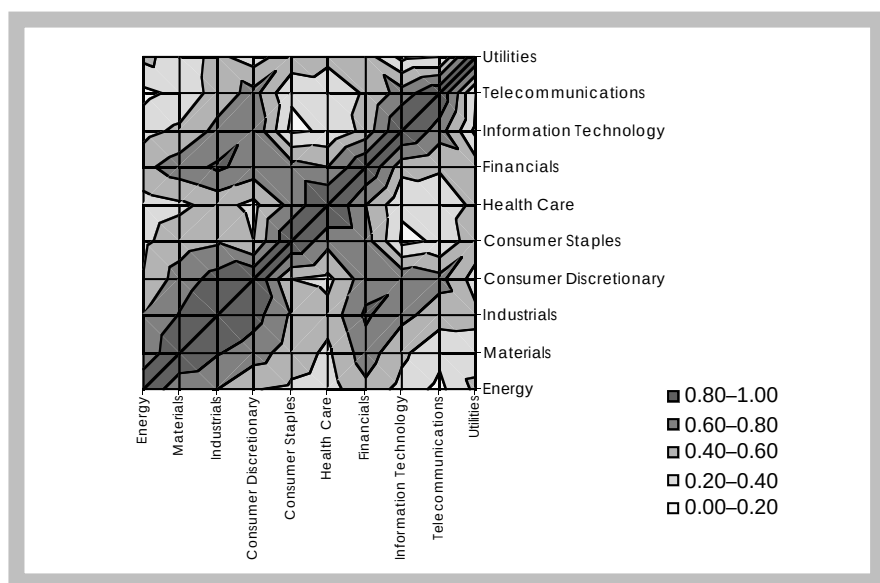


Figure 15.5 Sector correlation matrix

have observed already in connection with the results of the QRMCSAA methodology. There are quite obviously significant differences in the correlation coefficients between sector returns. Specifically, returns on the industrial, consumer discretionary, financials, information technology and telecommunications sectors seem to be relatively high; whereas the correlation of returns between this group and the remaining sectors, such as the utilities, energy, health care and consumer staples sectors, seem to be relatively low. It is these characteristics that we will exploit in the Sector Risk Model for controlling and tailoring the risk exposures of the portfolio, both on a total risk level and on a benchmark-relative tracking error level.

CONSTRUCTING THE MODEL

Risk and marginal contributions to risk

Portfolio managers constantly need to make decisions regarding changes to their portfolios that have implications for the sector weightings of the portfolio. Most often these changes are carried out by the manager, based on more or less subjective evaluations performed either by the manager himself or by an equity analyst or strategist, regarding the probably future appreciation of various alternative stocks in various sectors. Stocks are then bought or sold, often based solely on return expectations. However, since the return on an investment is only part of the performance – especially for portfolio managers who are measured relative to a benchmark, for example using the information ratio – ideally the manager should also consider the implications of portfolio rebalancing in terms of risk. It should be noted that most asset managers do this, since they are usually very aware of the current level of the portfolio's expected and/or realised tracking error. Often an asset manager will be told that his or her portfolio is close to the maximum tracking error level stipulated by the client in the investment mandate. At that point the manager will then need to alter the composition of the portfolio by changing a number of stock or sector weights in order to tilt the portfolio towards stocks that decrease tracking error – that is, make the return on the portfolio more similar to that of the benchmark.

It is therefore essential for the portfolio manager to be able to analyse and decompose portfolio tracking error, in order to be able to change its risk exposures in a structured way. If a decrease in tracking error is desired and the portfolio does not include all stocks in the benchmark – and this will often be the case – the manager must then decide if he might benefit from either switching to other stocks or including more stocks from the benchmark, as this will naturally mean a better *track* of the benchmark and thus a lower tracking error. So if the manager is forced to change the composition of the portfolio due to tracking error considerations, how does he evaluate the *riskiness* of sectors and stocks relative to each other and relative to the benchmark? The framework centred around the covariance matrix, as presented in previous chapters, provides a good tool for this sort of analysis because it allows the extraction of sector risk exposures on a marginal, thus enabling a more sophisticated risk analysis and control process. This is where a decomposition of tracking error becomes useful as a practical risk-management tool. That is, we can determine marginal contributions to risk that will give us a hint as to which sectors' weights we might consider changing in order to adjust expected portfolio risk and/or tracking error.

As we have discussed, in most asset management organisations portfolios are managed relative to some benchmark, which means that tracking error becomes the *de facto* tool for risk measurement and management. However,

since clients are ultimately interested in total return and total risk – at least over the long run – it is also of use to monitor total portfolio risk. This is an important point of this chapter, since in effect this means that it becomes important for the manager to be aware of *both* total portfolio risk and active risk, and also to know how changes to portfolio composition – in this case sector weights – may affect both the expected total portfolio risk and the expected active risk.

Marginal sector contributions to total risk

Having defined portfolio total risk and portfolio active risk in Equation 15.1 and Equation 15.2, we can determine the marginal contributions to both types of risk. We start off by determining the *marginal sector contribution to total risk* (MSCTR): the effect on expected total portfolio risk of a marginal change in the weight of a sector. This entails finding the partial derivative of total risk from Equation 15.1, with respect to changes in the portfolio weights, represented by the portfolio weight vector.

Using the definition of return variance (squared volatility) established in previous chapters, we know that portfolio return variance and portfolio volatility respectively can be expressed in matrix form thus:

$$\sigma_p^2 = \vec{w}_p^T \cdot \Sigma \cdot \vec{w}_p \quad \sigma_p = \frac{\vec{w}_p^T \cdot \Sigma \cdot \vec{w}_p}{\sigma_p} \quad [\text{Eq. 15.3}]$$

Taking the partial derivative of portfolio volatility in Equation 15.3 with respect to the portfolio weight vector, we obtain an expression for the change in volatility resulting from a marginal change in sector weightings:

$$MSCTR = \frac{\partial \sigma_p}{\partial \vec{w}_p^T} = \frac{\partial \vec{w}_p^T \cdot \Sigma \cdot \vec{w}_p}{\partial \vec{w}_p^T} = \frac{\Sigma \cdot \vec{w}_p}{\sigma_p} = \begin{bmatrix} \frac{\partial \sigma_p}{\partial w_{p,1}} \\ \vdots \\ \frac{\partial \sigma_p}{\partial w_{p,n}} \\ \vdots \\ \frac{\partial \sigma_p}{\partial w_{p,N}} \end{bmatrix} \quad [\text{Eq. 15.4}]$$

In other words, the resultant vector indicates the marginal contributions to total portfolio risk, given a marginal increase in the weight of one of the sectors. What constitutes a ‘marginal’ increase is for the risk manager to decide, but usually one interprets the result as the effect of a 1 percentage point change in the weight of the sector.¹ Each number in the vector in Equation 15.4 thus represents the marginal contributions to total portfolio risk – the increase or decrease in portfolio volatility that can be expected from increasing a sector weight by a single percentage point.

Marginal sector contributions to active risk

The same technique can be applied to tracking error, since the *marginal sector contribution to active risk* (MSCAR) is given by the partial derivative of the portfolio’s active risk with respect to the portfolio’s active weight vector:

$$\begin{aligned}
 MSCAR &= \frac{\partial \sigma_{PA}}{\partial \vec{w}_{PA}^T} = \frac{\partial \overline{\vec{w}_{PA}^T \cdot \Sigma \cdot \vec{w}_{PA}}}{\partial \vec{w}_{PA}^T} \\
 &= \frac{\Sigma \cdot \vec{w}_{PA}}{\sigma_{PA}} = \frac{\Sigma \cdot \vec{w}_{PA}}{\psi_P} = \begin{bmatrix} \frac{\partial \psi_P}{\partial w_{PA,1}} \\ \vdots \\ \frac{\partial \psi_P}{\partial w_{PA,n}} \\ \vdots \\ \frac{\partial \psi_P}{\partial w_{PA,N}} \end{bmatrix} \quad [\text{Eq. 15.5}]
 \end{aligned}$$

As with the total risk analysis, the manager will now be able to determine how much of an impact a change in sector weightings is likely to have on portfolio tracking error.² As was the case for the analysis of marginal sector contributions to total risk, the interpretation of the vector is the effect on active risk or tracking error as a result of an increase of 1 percentage point in the weighting of an individual sector. This is potentially very useful since, as already pointed out, much of the risk management that takes place in asset management organisations takes the form of tracking error limitations – that is, a set of minimum and maximum limits are established for the portfolio’s tracking

error with respect to an appropriate benchmark. Understanding how to reduce or increase expected tracking error in a structured manner, and also gaining an appreciation of the sensitivity of the portfolio's tracking error with respect to individual sectors bets, can thus potentially be of great help to investment managers. We shall illustrate this with an example.

Beta once more

The covariance matrix can also be used to determine the expected portfolio Beta: the degree to which the portfolio will expectedly covary with the market or the chosen benchmark. This is useful when managers need to assess the overall risk of the portfolio, relative to the market.

Recall that the covariance between the return on the portfolio and the return on the benchmark is given by:

$$\text{Cov}(r_P, r_B) = \frac{1}{T-1} \sum_{t=1}^T (r_{P,t} - \bar{r}_P)(r_{B,t} - \bar{r}_B) \quad [\text{Eq. 15.6}]$$

Calculated for a portfolio with a weight vector $\vec{w}_P$ and a benchmark with a weight vector $\vec{w}_B$, and using matrix notation, this covariance term can be written:

$$\text{Cov}(r_P, r_B) = \vec{w}_P^T \cdot \Sigma \cdot \vec{w}_B \quad [\text{Eq. 15.7}]$$

Note that this is identical to the definition of the variance of a portfolio or a single asset, except that now we employ portfolio weights *and* benchmark weights. This is of course because, as demonstrated earlier, an asset's covariance with itself is simply its variance.

We know that the Beta of a portfolio can be written:

$$\beta_P = \frac{\text{Cov}(r_P, r_B)}{\text{Var}(r_B)} = \frac{\text{Cov}(r_P, r_B)}{\sigma_B^2} \quad [\text{Eq. 15.8}]$$

Using matrix notation, the portfolio Beta is thus:

$$\beta_P = \frac{\vec{w}_P^T \cdot \Sigma \cdot \vec{w}_B}{\sigma_B^2} = \frac{\vec{w}_P^T \cdot \Sigma \cdot \vec{w}_B}{\vec{w}_B^T \cdot \Sigma \cdot \vec{w}_B} \quad [\text{Eq. 15.9}]$$

We also know that Beta is additive, and since in this context the portfolio's Beta is a weighted average of sector Betas, it then follows that the portfolio Beta can also be expressed by:

$$\beta_P = \vec{w}_P^T \cdot \vec{\beta} \quad \vec{\beta} = \frac{\beta_P}{\vec{w}_P^T} \quad [\text{Eq. 15.10}]$$

where $\vec{\beta}$ is the undetermined vector of sector Betas. But using Equation 15.9 and Equation 15.10 allows us to determine the vector of sector Betas, since:

$$\vec{\beta} = \frac{\Sigma \cdot \vec{w}_B}{\vec{w}_B^T \cdot \Sigma \cdot \vec{w}_B} = \frac{\Sigma \cdot \vec{w}_B}{\sigma_B^2} \quad [\text{Eq. 15.11}]$$

This vector thus indicates the Beta of each of the 10 MSCI sectors with respect to the benchmark, giving the manager an indication of relative risk among the sectors that comprise the benchmark.³

Attribution of total and active risk

The analysis above presents us with the opportunity to determine which sectors contribute to total and active risk, given the current portfolio and benchmark weights. Given the marginal sector contributions to total risk, and a portfolio with weight vector w_p , we know that total portfolio risk is given by:

$$\sigma_p = \vec{w}_p^T \cdot \frac{\Sigma \cdot \vec{w}_p}{\sigma_p} = \vec{w}_p^T \cdot \text{MSCTR}$$

$$= \begin{bmatrix} w_{p,1} \\ \vdots \\ w_{p,n} \\ \vdots \\ w_{p,N} \end{bmatrix}^T \cdot \begin{bmatrix} \text{MSCTR}_{p,1} \\ \vdots \\ \text{MSCTR}_{p,n} \\ \vdots \\ \text{MSCTR}_{p,N} \end{bmatrix} \quad [\text{Eq. 15.12}]$$

This provides us with an attribution of total portfolio risk to individual sector weights, since the amount of total portfolio risk that we can attribute to sector n is:

$$w_{p,n} \cdot \text{MSCTR}_{p,n} \quad [\text{Eq. 15.13}]$$

That is, for each sector n we can attribute the amount of total portfolio risk that comes from our holding in that particular sector. We can refine this by determining the percentage of total risk that can be attributed to sector n , simply by dividing Equation 15.12 by the total portfolio volatility. This yields:

$$\frac{w_{PA,n} \cdot MSCTR_{PA,n}}{\Psi_P} = 1 \quad [\text{Eq. 15.14}]$$

Since the left side of the equation is a sum of n weights times n marginal sector contributions to total risk, and the right side is equal to 100%, for each n , $w_{P,n} \cdot MSCTR_{P,n}$ must then necessarily be the *relative contribution* from the weight of asset n to total risk – that is, the percentage of total risk that can be attributed to the holding in sector n .

We can easily perform the same type of risk attribution analysis to the active portfolio w_{PA} and thus to tracking error. That is, we can attribute the amount of active risk contributed by each sector n by defining:

$$\frac{w_{P,n} \cdot MSCTR_{P,n}}{\sigma_P} = 1 \quad [\text{Eq. 15.15}]$$

Analogously to Equation 15.13, the amount of active portfolio risk that can be attributed to the weighting in sector n is thus given by:

$$w_{PA,n} \cdot MSCTR_{PA,n} \quad [\text{Eq. 15.16}]$$

This relative portfolio risk analysis setting allows the manager to see easily how much each sector contributes to total and to active risk, relative to each other. Note that these relative contributions change non-linearly as the portfolio's composition is changed, since the marginal contributions to sector risk change in a non-linear fashion. This means that there is potentially a danger of interpreting the marginal contribution to risk of a sector as the effect of a 1 per cent change in that sector's weight. However, this is mostly an issue when sector weights are very close to the benchmark weights. The investment manager needs to be aware of the implications of this, and we shall demonstrate the implications of this below.

PORTFOLIO RISK-MANAGEMENT IMPLICATIONS

This type of marginal risk analysis is potentially a very powerful risk control and management tool, provided that the risk dynamics of the portfolio – that is, the correlation matrix and the sector volatilities – remain relatively constant over time. We use the term 'potentially' to underscore the fact that the resultant marginal risk vectors are not to be interpreted as being the ultimate truth. As we mentioned above, the model inputs are estimated over just 60 days – some three months' worth of trading days. What is required in order for us to use the results provided by the model with a reasonable degree of confidence, and to trust their results as being valid over the next, say, one month, is simply

that the coming month be similar to the previous three months. Whether this will always be the case is uncertain at best, but nevertheless, one can apply this type of analysis and rely on the results if one has already decided with oneself that current and expected market dynamics are indeed very similar to those prevailing in the preceding three-month period. Asset managers are unlikely to agree on this subject, but none the less there seems to be general consensus that broad market trends persist for considerable lengths of time. Furthermore, the visibility of the underlying economic factors and company fundamentals is often less than perfect, resulting in rather slow changes to market sentiments and thus market trends. Another factor that amplifies this tendency is the reluctance of most equity analysts or portfolio managers to change their recommendations or portfolio positions by very much over a very short time period. It takes time for human beings to become convinced of something other than, and perhaps even diametrically opposite to, what they previously believed. This will also tend to make market sentiment slow in changing. In any case, the investment manager needs to be aware of these issues before employing models such as the one constructed in this chapter.

If for expositional purposes we assume that, to an acceptable degree, the coming month will be similar to the previous three months, we can now start to examine closer the implications of this type of Sector Risk Model. With a model of this kind, the manager will be able to evaluate or to simulate the effects on expected tracking error by prospective changes to the portfolio composition, whether these changes entail a *reshuffling* of existing stocks or sectors, or the inclusion or exclusion of stocks or even entire sectors from the portfolio. One of the more often used application of techniques such as those presented above is the determination of the portfolio changes required to bring down tracking error towards a maximum limit stipulated by the client. This type of application will typically occur when overall market volatility increases or when correlations between sectors or stocks increase across the board, which leads to an increase in tracking error without there necessarily having been any changes to the portfolio during that time period. In this way the manager can actually tailor her portfolio to meet certain requirements in terms of risk in a fairly accurate manner.

Instead of mindlessly pouring more stocks into the portfolio in order to reduce tracking error, or in a sector context eliminating all sector bets, the adjustment of the portfolio ought to be carried out in a structured manner taking into account the volatility of and correlations between individual sectors. This is in fact one of the most important points of structured risk models such as the one presented here. *An active position is not just an active position.* Its impact on expected tracking error is very much dependent on the nature of the sector that is being over- or under-weighted, as well as its correlation with other sectors. So an active bet of 1 percentage point in *one* sector

may have very different implications for portfolio tracking error from an active bet of 1 percentage point in *another* sector.

An additional and very important point to remember is that the investment manager should always be aware of the trade-off between return and risk, whether it is total risk or active risk. Taking an active position in a stock or a sector should only be considered if the expected increase in tracking error and the likely trade-off between the resultant expected active return and active risk are acceptable. Furthermore, one can perform a sort of comparative analysis in this context, whereby each active position is considered relative to its active risk but also relative to other active positions and their respective active risks. This enables the portfolio manager to gauge the expected impact on total risk, as well as tracking error, from changes in active positions. We shall elaborate further on this below. Putting it briefly, an example might involve active positions of equal size in the information technology sector and in the industrials sector: the two bets may well be of equal size, but since the information technology sector is much more volatile than the industrials sector, in terms of tracking error the bet in the information technology sector is perhaps as much as four times as large as that in the industrials sector. And this forces the portfolio manager to ask herself the following very important question: 'Do I really believe that my active return from the bet in the information technology sector will be at least four times as large as that expected to result from the bet in the industrials sector?'. Alternatively, one could ask, 'Is the level of information on the information technology sector is four times as good or as accurate as that on the industrials sector? Am I four times as certain about the relative performance of that sector?' This is potentially a very useful exercise for any investment management organisation. Working with risk characterisation and portfolio risk decomposition in this manner tends to force the portfolio manager to focus the decision-making process on return versus risk, and thereby on the information ratio expectations that are implied by the active positions in the portfolio. On a related topic, the relationship between the expectations an investment manager may have regarding asset return and risk can be quantified by calculating the *implied returns* on a portfolio. This will be demonstrated later in this book.

MSCTR AND MSCAR FOR THE GLOBAL SECTOR MODEL

With the techniques presented above we can now analyse the structure of total and active risk in the MSCI Global Sector Model presented earlier. This will enable us to better understand these risks, and how they may change as we change the asset allocation or portfolio composition.

The numbers and the interpretations

For the analysis presented here we use the optimally shrunk covariance matrix, the benchmark weights employed in the previous chapters, and finally an arbitrary, slightly conservative asset allocation or portfolio composition (see Table 15.1).

The general active risk level of this portfolio is relatively low, since the deviations from the benchmark specified by the active weights are relatively small. The biggest overweight positions are the financial and health care sectors. The largest negative bets are the consumer discretionary and information technology sectors. The resultant portfolio and benchmark volatilities are 13.40% and 14.71% respectively. Beta comes out at 0.91 again, indicating the defensive nature of the portfolio, and the expected tracking error is 1.73%.

Relative to the expected portfolio volatility of just 13.40%, the tracking error appears quite high at first glance for such a conservative portfolio. But this is to a large extent caused by the large underweight position in the information technology sector. Given the high volatility in this sector, a deviation from the benchmark in this sector will have a large impact on expected tracking error. In

Table 15.1 Portfolio and benchmark weights

	Portfolio weights	Benchmark weights	Active weights	Volatilities	MSCTR	MSCAR	Beta
Energy	5.8%	4.2%	1.7%	20.0%	0.09%	0.01%	0.54
Materials	8.0%	6.6%	1.4%	14.0%	0.11%	−0.09%	0.77
Industrials	10.6%	13.3%	−2.7%	17.1%	0.16%	−0.12%	1.07
Cons. Disc.	12.7%	17.3%	−4.6%	17.0%	0.16%	−0.13%	1.09
Cons. Staples	8.9%	7.9%	1.0%	8.7%	0.05%	−0.02%	0.35
Health Care	11.1%	7.6%	3.6%	10.6%	0.06%	−0.01%	0.39
Financials	18.3%	15.1%	3.2%	16.2%	0.15%	−0.10%	1.02
IT	10.0%	14.2%	−4.2%	30.5%	0.26%	−0.27%	1.84
Telecomm.	6.9%	8.0%	−1.1%	22.3%	0.18%	−0.16%	1.25
Utilities	7.7%	5.9%	1.8%	10.9%	0.07%	−0.02%	0.42
Sum	100.0%	100.0%	0.0%				
Portfolio Vol.	13.40%						
Benchmark Vol.	14.71%						
Beta	0.91						
Tracking error	1.73%						

Table 15.1 MCSTR and MSCAR, the marginal sector contributions to total and active risk, are also presented, along with an expected portfolio Beta. All three columns demonstrate the very large differences between the effects on total and active risk from deviations from the benchmark weights. This point is demonstrated quite forcefully by the fact that the Beta of the information technology sector is several times that of most other sectors. The Beta of the information technology sector is thus as high as 1.84, whereas the Beta of the consumer staples sector is as low as 0.35. The sector that seems to trade most in line with the overall market is the financials sector, with a Beta of 1.02.

MSCTR analysis

Using Equation 15.4 we can determine the marginal sector contributions to total risk for the Global Sector Model – see Figure 15.6. This figure demonstrates nicely the relative magnitude of impacts on total portfolio volatilities that can be expected from a *ceteris paribus* increase of 1 percentage point of the respective sectors. It is clear from Figure 15.6 that by far the biggest marginal contributor to total risk in the portfolio is the information technology sector, primarily because of its very high level of volatility. Changing the weight of the information technology sector by 1 percentage point will change the expected total portfolio volatility by 0.26%. The telecommunications sector is also a relatively large contributor, and the consumer staples and utilities sectors have the lowest marginal contribution to total portfolio risk.

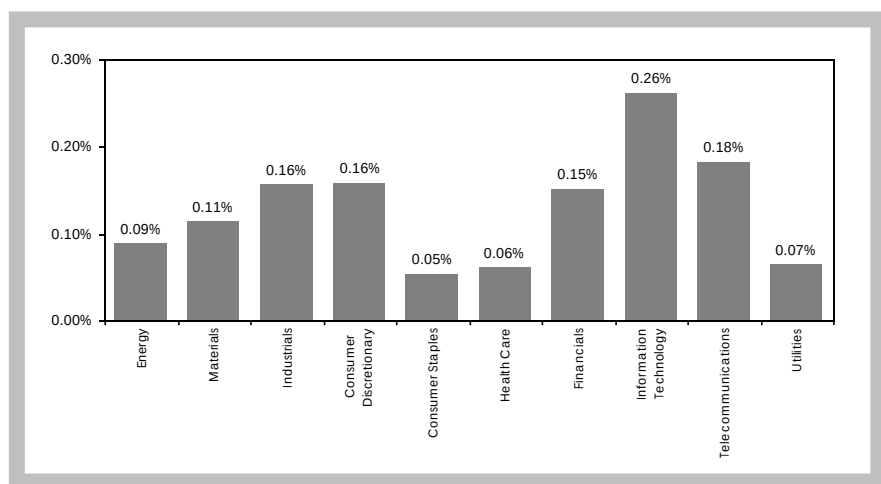


Figure 15.6 Marginal sector contributions to total risk

Using this set of numbers the portfolio manager can relatively easily assess the impact on portfolio volatility of prospective changes to sector weights. Bringing the portfolio weight in the information technology sector up to the benchmark weight of 14.2% would mean an increase of 4.2%, thus implying an expected increase in portfolio volatility of 1.10% (4.2 times 0.26%). Expected portfolio volatility would thus be raised from 13.40% to 14.50%. Using the information provided by the marginal sector contributions to total risk, we can thus exploit the covariance characteristics or risk structure of the portfolio to perform sector rotations that have a limited impact on total portfolio volatility.

MSCAR Analysis

We can apply a similar analysis to active risk, investigating the prospective impact of portfolio changes on tracking error, using Equation 15.5. Figure 15.7 illustrates the marginal sector contributions to active risk.

Figure 15.7 illustrates the marginal sector contributions to active risk or tracking error, and it is again clear that the sector with the biggest marginal impact is information technology. This is again because it is the most volatile sector over the estimation period, but also because its active weight is among the largest of the 10 sectors. An increase in the weight of the information technology sector will decrease the expected tracking error by 0.27% since it brings the sector’s weight closer to the benchmark weight. Raising the sector’s weight to neutral relative to the benchmark by increasing by 4.2% will, *ceteris paribus*, mean a decrease in tracking error of 1.13% to a level as low as 0.60%.

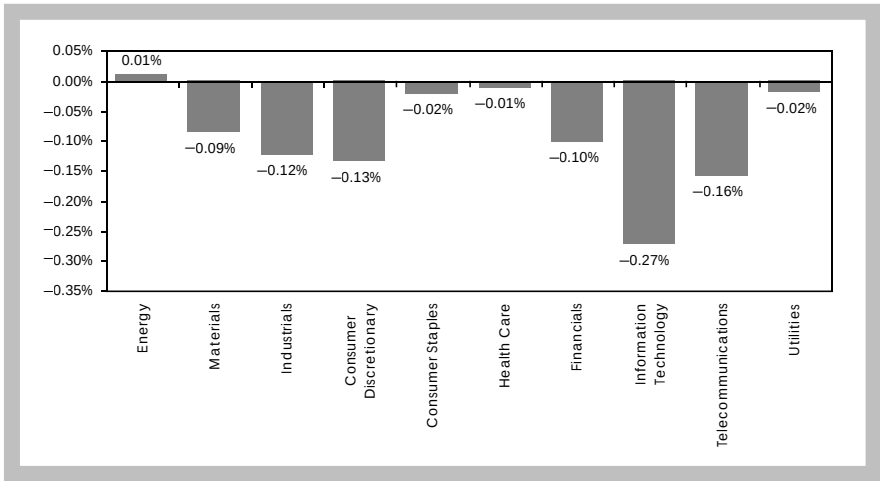


Figure 15.7 Marginal sector contributions to active risk

Again we stress that this under the assumption of everything else being equal, which is problematic for changes as large as this. Note here what initially seems to be a dichotomy between the two measures of marginal risk. The marginal total risk analysis indicates that an increase in the information technology sector will result in a higher total portfolio volatility, whereas marginal active risk analysis indicates a prospective drop in tracking error.

When examining the numbers it becomes evident that this is perfectly logical, but it does demonstrate a crucial point that asset managers need to be keenly aware of. One can obtain a lower tracking error by rotating between sectors, but this may well result in a higher total portfolio risk. Provided with this information, it is then up to the portfolio manager to decide what actually constitutes risk for the client. In other words, there may be instances where the trade-off between total and active risk must be examined and where more weight should be placed temporarily on one rather than the other. For example, it is conceivable that the client has stipulated for the asset manager a fixed target level for tracking error. Theoretically the asset manager must then adhere to this level of tracking error regardless of market conditions. However, there are times when equity markets behave so erratically that maintaining a strict adherence to a certain level of tracking error would inevitably end up destroying the client's wealth, in which case the asset manager might seek to preserve this instead of mindlessly sticking to the tracking error. Such instances might be covered in the investment policy statement, possibly in the form of a contingency framework, where special circumstances can be defined under which the asset manager must contact the client and possibly arrange for a temporary deviation from normal guidelines.

The MSCTR-vector and the MSCAR-vector also open up the possibility of considering the approximate effects of more extensive sector rotations. We use the term 'approximate' since the marginal contributions are calculated on a *ceteris paribus* basis, meaning that we assume a change in one sector weight holding all else unchanged. So naturally changing several portfolio weights will give results that are very slightly off target, but the effect is minuscule. For expositional purposes we will thus apply this technique. In order to illustrate the effects of a sector rotation, let us assume that we are contemplating increasing the weight in the information technology by 2%, decreasing the weight in the financials sector by 2%, decreasing the weight in the materials sector by 1% and increasing the weight in the telecommunications sector by 1%. This will make the portfolio more risky, since larger positions are placed in the information technology and telecommunications sectors. The combined effect of this sector rotation on total portfolio risk as well as on active risk is accordingly expected to be as shown in Table 15.2.

The rotation thus results in an increase in the total portfolio volatility, where the changes in the information technology and the telecommunications sectors

Table 15.2 Effect on MSCTR and MSCAR of sector rotation

	MSCTR	MSCAR	Change Pct. points	MSCTR Result	MSCAR Result
Energy	0.09%	0.01%	0	0.00%	0.00%
Materials	0.11%	−0.09%	−1	−0.11%	0.09%
Industrials	0.16%	−0.12%	0	0.00%	0.00%
Cons. Disc.	0.16%	−0.13%	0	0.00%	0.00%
Cons. Staples	0.05%	−0.02%	0	0.00%	0.00%
Health Care	0.06%	−0.01%	0	0.00%	0.00%
Financials	0.15%	−0.10%	−2	−0.30%	0.21%
IT	0.26%	−0.27%	2	0.52%	−0.55%
Telecomm.	0.18%	−0.16%	1	0.18%	−0.16%
Utilities	0.07%	−0.02%	0	0.00%	0.00%
Sum			0	0.29%	−0.41%

effect an increase and the materials and financials sectors effect a decrease. The effects are relatively straightforward since the sectors whose weights are increased are relatively high volatility sectors, and the sectors that are decreased are relatively low volatility sectors. The effect on the benchmark-relative or active risk is a decrease in expected tracking error of 0.41%, with the change in the weight in the information technology sector having the largest impact as its weight is moved closer to the benchmark weight, principally because of its relatively high level of volatility. As already pointed out, with an increase in total portfolio risk and a decrease in active risk, there exists an obvious trade-off between the two, which we can easily illustrate.

Below we simulate an increase in the weight of the information technology sector of between 1% and 10%, for a proportionate decline in the remaining nine sector weights. Given the initial sector weight of 10.0% and the resultant active weight of −4.2%, the effect on portfolio volatility and tracking error can be expected to be as in Figure 15.8. It is evident that tracking error can be brought down considerably by increasing the weight in the information technology sector, and thereby eliminating the active weight or bet. Note however that the expected tracking error does not bottom out until the proportion of the portfolio invested in the information technology sector is well above 16%. This is due to the fact that as the information technology sector is increased, the other sectors are decreased proportionately, so as to ensure a sum of portfolio weights of 100% and thereby a sum of active weights of 0%. This has the effect of reducing

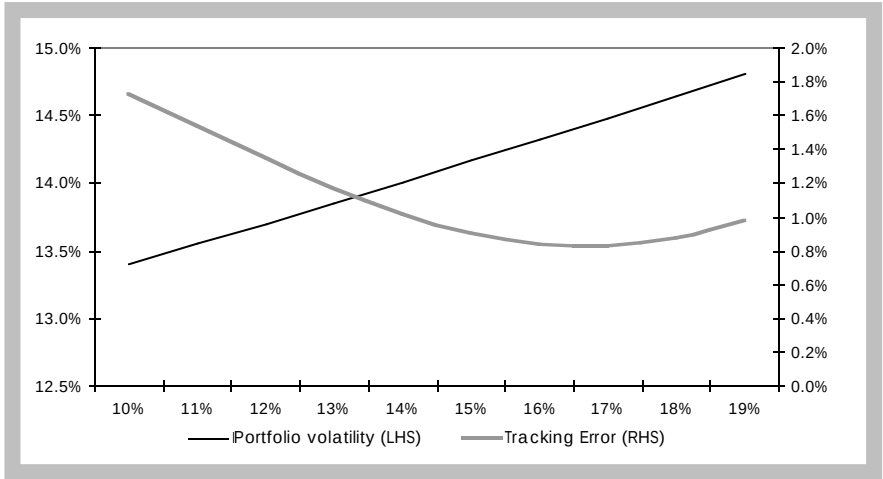


Figure 15.8 Total and active risk for changing information technology sector

many of the positive sector bets in the portfolio towards zero, thus effecting a continued reduction in tracking error, even as the weight of the information technology sector moves past and slightly above the benchmark weight.

We can thus reduce the tracking error of the portfolio in a calculated manner, but the obvious price paid for this decrease in active risk is a substantial increase in total portfolio volatility, since a much larger proportion of the assets are now invested in a much riskier asset class. The asset manager must decide what is preferable for the client in that situation, given expectations about market developments. He can stick with the existing sector allocation, thereby obtaining a lower tracking error at the cost of higher total volatility; or he can decide to reduce total portfolio risk and accept a higher tracking error. This trade-off is important to consider, and the result of such considerations may change over time, depending on markets and clients' preferences.

MSCTR and MSCAR in absolute and relative terms

The estimates for prospective consequences on the risk of sector rotations do contain an amount of uncertainty, which is why the relative sizes of the marginal contributions to risk are probably the most important element to consider in this analysis. Managers often have a good sense of where their (active) returns come from, but another question is equally important for the management of assets: what component of portfolio risk is the most influential

in the current situation? Figure 15.9 illustrates the relative marginal sector contributions to risk. In other words, of the total amount of risk present in the portfolio at this particular time, what proportion stems from exposures to which sector? Using Equation 15.14 this can be illustrated in Figure 15.9.

Not surprisingly, the vast majority of total portfolio volatility comes from the highly volatile information technology sector, as well as from the financials sector and telecommunications sector. Note that the energy sector contributed very little to total portfolio volatility, even though its volatility on a sector level is high. This illustrates the fact that its average correlation coefficient with the remaining nine sectors is very low, providing strong diversification effects in the portfolio. The consumer staples sector also contributes very little to total portfolio volatility, but this is because its sector volatility is very low. An analysis such as this enables the asset manager to assess the risk exposures in the portfolio relative to each other, and to evaluate whether these are in line with the investment strategy and current market expectations.

A similar chart can be produced for active risk, using Equation 15.15 (see Figure 15.10). Because of its relatively large and negative active sector weight, as well as its relatively high level of volatility, the vast majority of the portfolio's tracking error is caused by the bet in the information technology sector. Notably, the consumer discretionary sector also contributes heavily to tracking error. However, even though its active weight is similar in size to that of the information technology sector, its contribution to tracking error is only about half the size of that of the information technology sector. Again, this is directly related to the sector-specific levels of volatility.

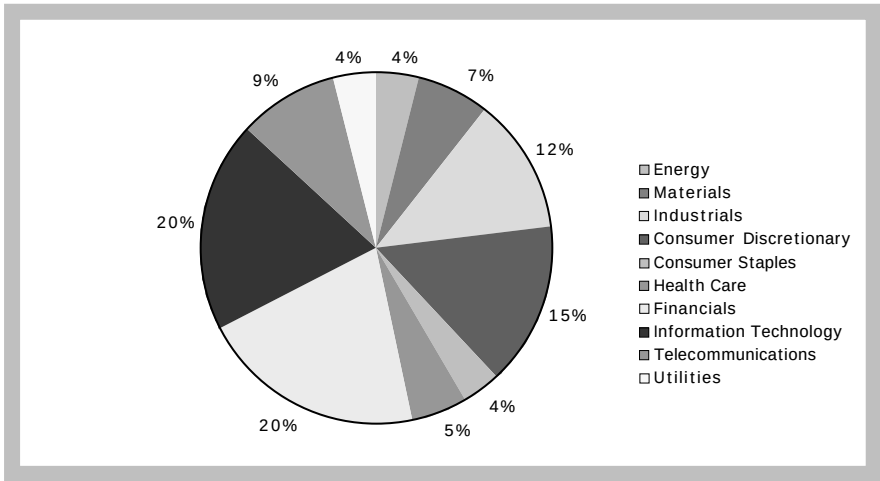


Figure 15.9 Relative marginal sector contributions to total risk

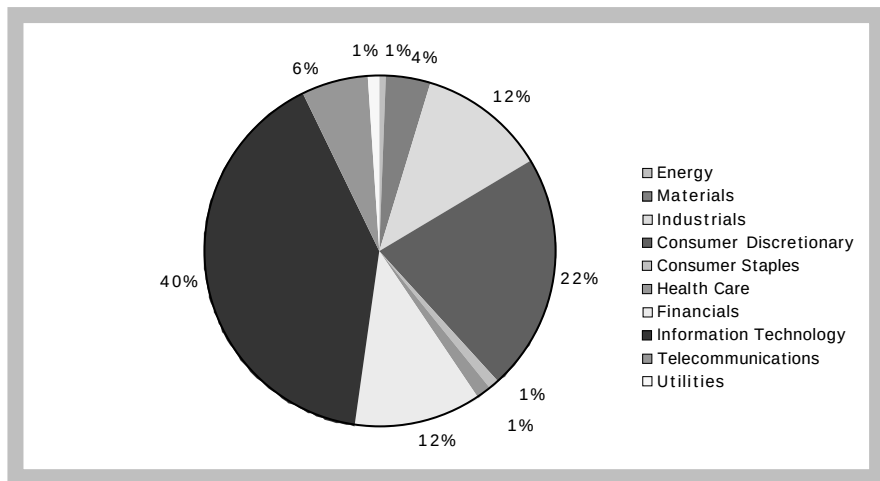


Figure 15.10 Relative marginal sector contributions to active risk

Returning to the point made above about the trade-off between active risk and active return, in the context of the portfolio manager being forced to consider whether he is more certain about one active position than another, we can use Figure 15.10 to evaluate exactly that issue. From the chart it is apparent that the largest contributor to tracking error is the active position in the information technology sector. When evaluating this result, the asset manager must ask himself the following question: ‘Is this commensurate with the level of confidence that I have in this particular active position’s likelihood of resulting in outperformance relative to the benchmark?’ If this is not the case, then the manager might consider bringing down the proportion of expected tracking error arising from this particular sector bet, and subsequently making sure that the expected contribution to tracking error of each sector is at least approximately proportional to the confidence the manager has in each sector’s likelihood of outperforming the benchmark. In other words, the size of the active weight is not important in itself. What matters is the impact on tracking error that results from this active weight; and, as demonstrated here, two active weights of similar size may well have very different impacts on expected tracking error. Not only is it important from a risk-management perspective, for the asset manager to know this, but it may be argued that the size of the expected contribution to tracking error must be proportional to the confidence the asset manager has in the bet that generates this contribution. If this were not the case, the portfolio would not properly reflect the expectations of the portfolio manager. Detailed analysis of the marginal as well as absolute contributions to portfolio tracking error such as that presented here thus provides a

useful tool for checking whether there is consistency in the way the portfolio manager has designed his portfolio, in terms of the relative size of sector bets versus the confidence in the bets as far as outperformance is concerned.

Beta

In Equation 15.9 and Equation 15.10 we derived the portfolio Beta and the sector components of Beta. We can apply this to the current MSCI Global Sector portfolio, to investigate each sector’s riskiness relative to the benchmark. The vector of component sector Betas appears as shown in Figure 15.11. For the estimation period employed in this analysis it becomes quite clear that the information technology sector is extremely volatile relative to the other sectors, which is also very clear from the appendix to this chapter, which shows the index and its volatility over the time period. Most investors would be extremely hesitant to invest heavily in a sector stock with a Beta close to two, unless they were completely certain that they were in a bull market. Note however that the period includes both the rise of the NASDAQ during the spring of 2000, and the subsequent collapse of almost everything information technology-related in the autumn of 2000 and during 2001. Regardless of the sustainability of the absolute levels of these component Betas, the overall picture is clear. Relative to the benchmark, the information technology sector has been extremely risky during this period. The telecommunications and consumer discretionary sectors have also been quite risky in a benchmark-relative perspective.

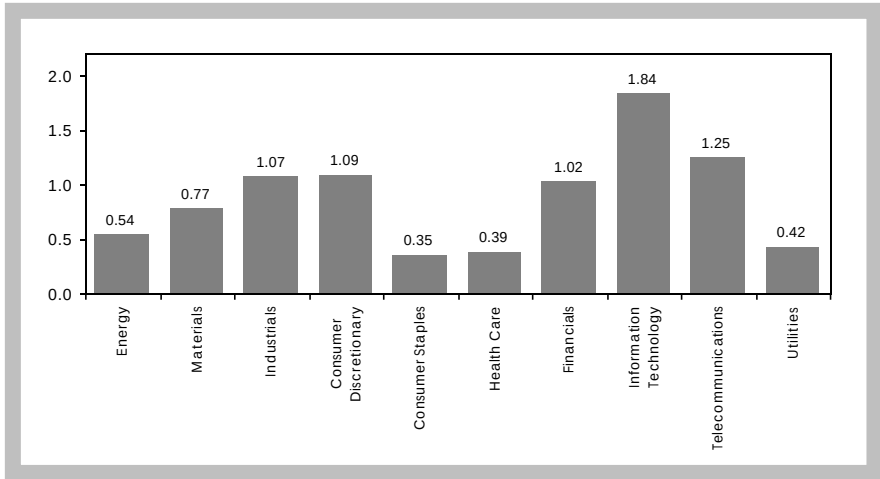


Figure 15.11 Sector component Betas

As with the calculations on the changes in portfolio volatility and tracking error expected to result from an increase in the weight in one sector against the others, we can illustrate the effect on Beta resulting from such a change. Figure 15.12 shows a chart indicating the change in portfolio Beta that may be expected from an increase in the weight in the information technology sector by between 1% and 10%.

Because of the relatively high level of volatility experienced in the information technology sector, an increase of the weight in this sector of 4.2%, thereby bringing the portfolio to a neutral weighting relative to the benchmark, would result in an increase in the expected portfolio Beta from 0.91 to

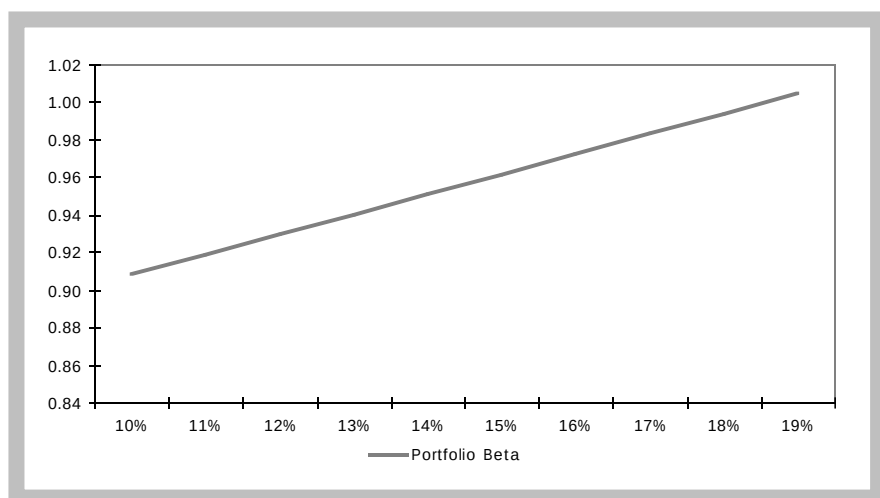


Figure 15.12 Beta for the information technology sector

0.95. This type of information may also prove useful to a manager attempting to assess the overall riskiness of the portfolio, should he decide to change its composition. Similar charts can easily be produced for the remaining nine sectors, giving the manager a comprehensive view of the individual sector contributions to non-diversifiable risk. Clearly the information technology sector is by far the biggest contributor to portfolio Beta given the current portfolio weights, but increasing the weight in other sectors, such as the energy, materials, consumer staples, health care and utility sectors will lower Beta quite significantly. Analysis such as this thus provides an additional tool for the asset manager to assess and potentially to adjust the benchmark-relative risk of the portfolio.

MSCTR over time

In addition to looking at snapshots of marginal sector contribution to total and active risk, it may be of use to the asset manager to observe the trends in these numbers over time, thereby gaining a better indication of the dynamics of the risk structure in the portfolio. Figure 15.13 illustrates how the MSCTR has developed over the estimation period, for the portfolio weights presented above.

Not surprisingly, the MSCTR of the information technology sector has remained the most significant over the whole period, and the fact that it has remained high reflects the fact that the volatility of this sector has always been high relative to that of the other sectors. We already knew that the volatility of the information technology sector has been high for the entire period, but this chart enables us to see how this has impacted on the marginal sector contributions to risk over that period. Note how, even with constant portfolio weights, it falls off slightly towards the end of the period, as sector volatility decreases. The MSCTR of the consumer staples sector has gone from being negative to being positive, indicating that this sector’s ability to offset portfolio volatility generated by such sectors as the information technology and telecommunications sectors has decreased during the period in question. This may have to do with the fact that correlation coefficients between sectors have risen over the period, eliminating many of the diversification effects that were present initially. Most other sectors are basically unchanged at their respective levels during the period, which is quite remarkable considering the general volatility of the markets. Even

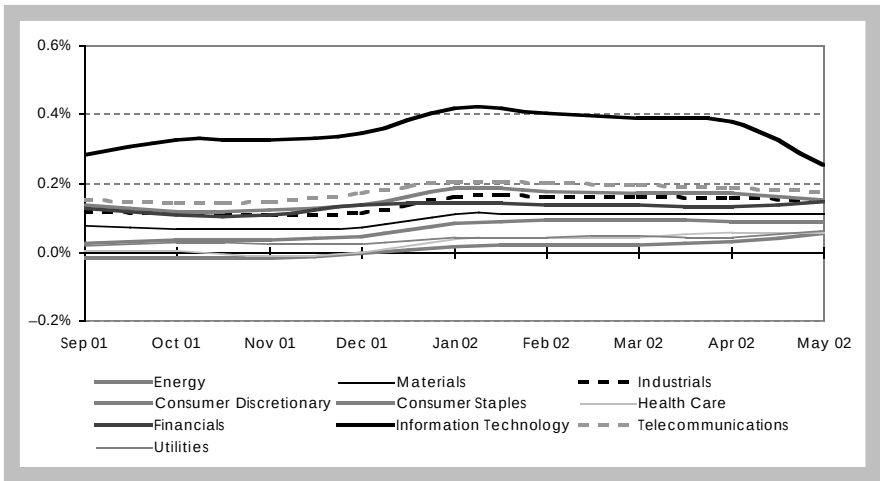


Figure 15.13 MSCTR during the estimation period

during periods of very high levels of volatility in certain sectors, the overall risk structure of the model does not change much. This information is potentially quite helpful for the portfolio or risk manager, since it demonstrates the relative firmness of model output.

MSCAR over time

We can produce a similar chart for how MSCAR has developed over time – see Figure 15.14. Again, starting off with the information technology sector its marginal contribution to active risk has decreased further into negative territory as the sector’s volatility has increased and the underweight position has been maintained. Towards the end of the period the MSCAR from the information technology sector decreases as the volatility of this sector decreases. The same type of pattern is apparent from the telecommunications sector. Interestingly the MSCAR from the consumer staples sector has fallen during the period, indicating that despite the relatively defensive nature of the sector the rise in overall correlation levels has caused its diversification effects on active risk to diminish.

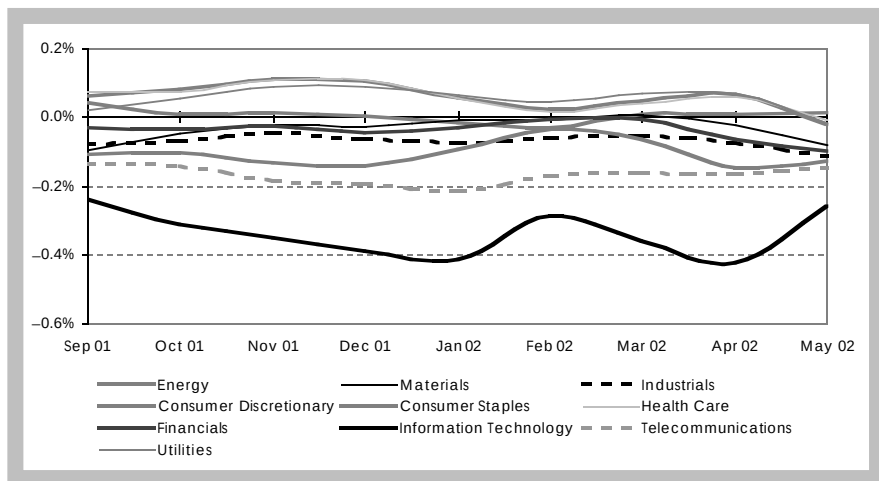


Figure 15.14 MSCAR during the estimation period

THE EFFICIENT RIDGE REVISITED

From Chapters 10 and 11 we recall that the efficient frontier was actually an efficient ridge, due to the fact that the return estimates entering into the opti-

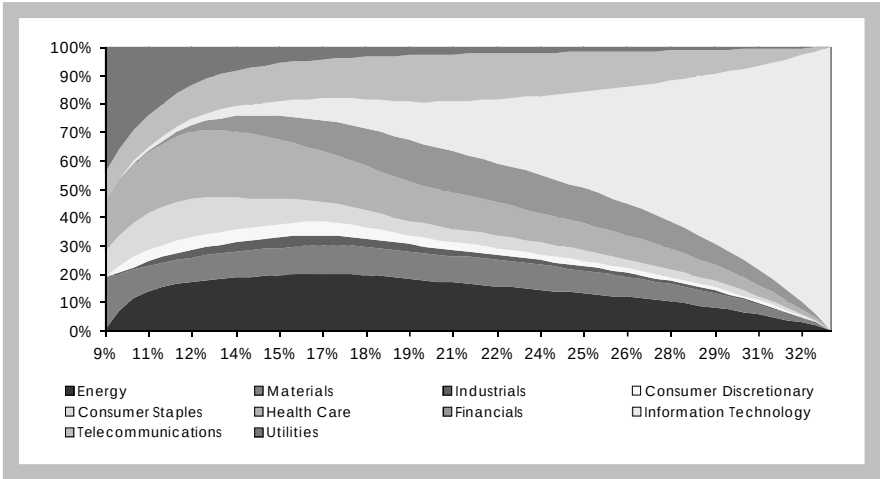


Figure 15.15 QRMCSAA – Ledoit: MSCI Global Sector Model

misation could best be described in terms of probability distributions. Recall that the QRMCSAA with the optimally shrunk covariance matrix for the Global Sector Model appeared as in Figure 15.5. We can now use this asset allocation along the efficient ridge to analyse the risk characteristics of the portfolio at different points along the risk spectrum. This can be done because the QRMCSAA illustrated in Figure 15.15 corresponding to the efficient ridge implies a set of marginal sector contributions to total and active risk – one set for each asset allocation along the risk spectrum – spanning from the minimum to the maximum variance portfolio. These are generated by taking each of the 51 asset allocation vectors along the efficient ridge and inputting them into Equations 15.4 and 15.5. The resulting vectors of marginal sector contribution to risk can then be illustrated in a chart to give an indication of the marginal risk sensitivity of the portfolios to changes in its composition, depending on the current composition of the portfolio.

MSCTR

We start out by looking at total portfolio risk. Calculating the MSCTR for all 51 portfolios along the efficient ridge asset allocation illustrated in Figure 15.8 results in the chart shown in Figure 15.16. Note the marginal sector contribution to total risk from the information technology sector. It actually starts out as only the second largest marginal contributor, but as we move along the efficient ridge and the proportion invested in this sector increases, its marginal contribution to total risk increases at an increasing rate. As we approach the

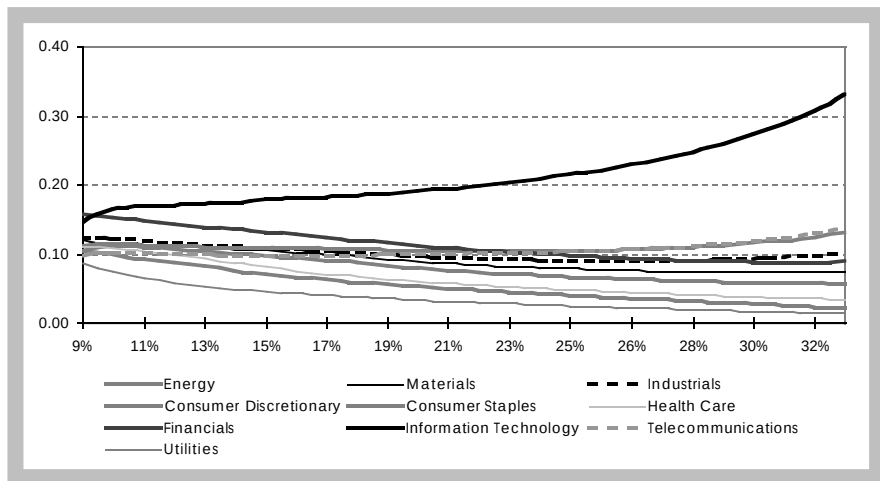


Figure 15.16 MSCTR along the efficient ridge

high-risk area to the far right in the chart, with close to 100% invested in the information technology sector, most other marginal contributions approach zero, except for the telecommunications, consumer discretionary and industrials sectors. Note, however, that the argument works in reverse: for very high portfolio volatility levels, a decrease in the information technology sector will have a very large negative impact on portfolio volatility, particularly if this sector is substituted with sectors such as health care or consumer staples.

MSCAR

Subtracting the benchmark weights from the weights generated by QRMCSAA for the Global Sector Model, we obtain the active weight for the portfolio as we move along the efficient ridge (see Figure 15.17). For the minimum-variance portfolio to the far left of the chart, defensive sectors naturally dominate quite heavily. The portfolio is overweight the health care, consumer staples and utilities sectors, and underweight the information technology, telecommunications, consumer discretionary and industrials sectors. The information technology sector starts out being heavily underweighted relative to the benchmark, but then increases exponentially as the desired risk is increased. Interestingly, the consumer discretionary sector remains significantly underweight over the entire risk spectrum, indicating that, relative to other similar sectors, its diversification potential is relatively small. The same is true to a lesser extent for the financials and the industrials sectors.

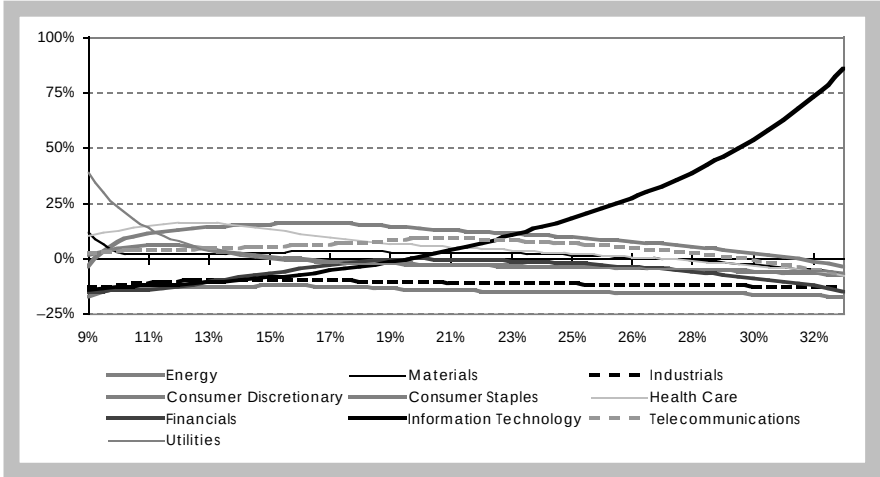


Figure 15.17 Active weights along the efficient ridge

The active weights illustrated in Figure 15.17 generate the tracking error levels along the efficient ridge shown in Figure 15.18. Using Equation 15.5 we can then generate the MSCAR along the efficient ridge (Figure 15.19). Note that the x -axis is not linear, since the active weights represented in Figure 15.17 generate a tracking error that first drops – as illustrated in Figure 15.18 – and then rises quite significantly as all of the portfolio’s assets are invested in the information technology sector.

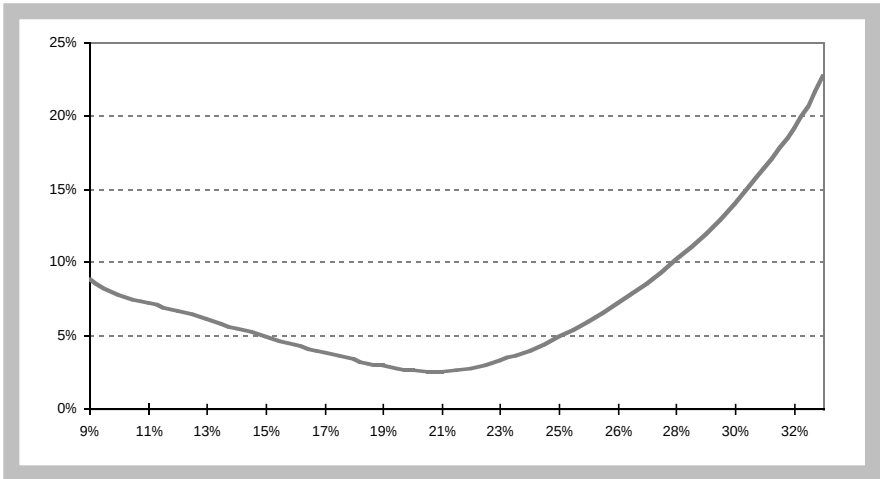


Figure 15.18 Tracking error along the efficient ridge

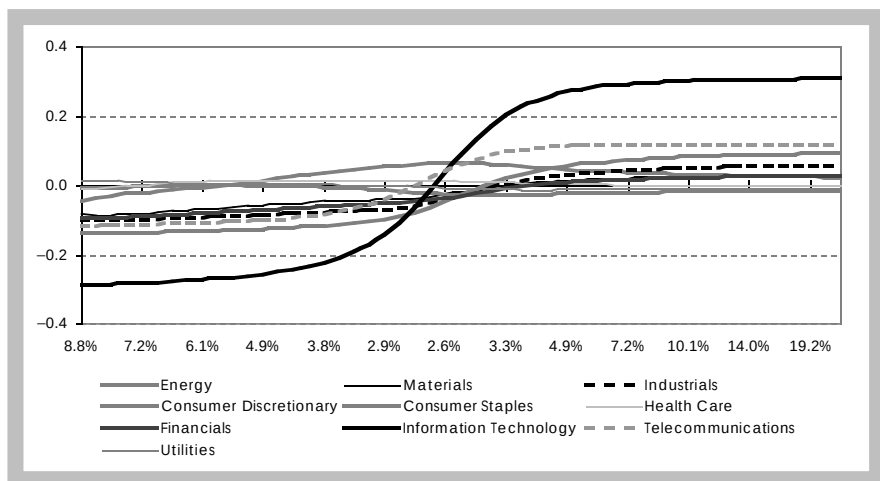


Figure 15.19 MSCAR along the efficient ridge

Figure 15.19 demonstrates very clearly how the marginal contributions to active risk change along the efficient ridge. Again the most eye-catching feature is the information technology sector, which moves from a large negative marginal contribution to a large positive marginal contribution. This is a simple function of the active weight moving from a large negative value to a large positive value as we move along the efficient ridge. Around the middle variance portfolio most active weights are close to zero, which means that both the tracking error and the marginal sector contributions to active risk are low. Because of similar characteristics, the telecommunications and consumer discretionary sectors see the same change in marginal contribution to active risk along the efficient ridge, except for the absolute size of the marginal contributions that are smaller due to the lower volatility of the sector.

Beta

Using Equations 15.10 and 15.11 we can also calculate Beta along the efficient frontier (Figure 15.20). Not surprisingly, as we move along the efficient ridge the portfolio's Beta rises steadily and reaches 1 near the middle variance portfolio where the average active weight is smallest, after which it accelerates as more and more sectors are substituted for the one with the highest Beta, the information technology sector.

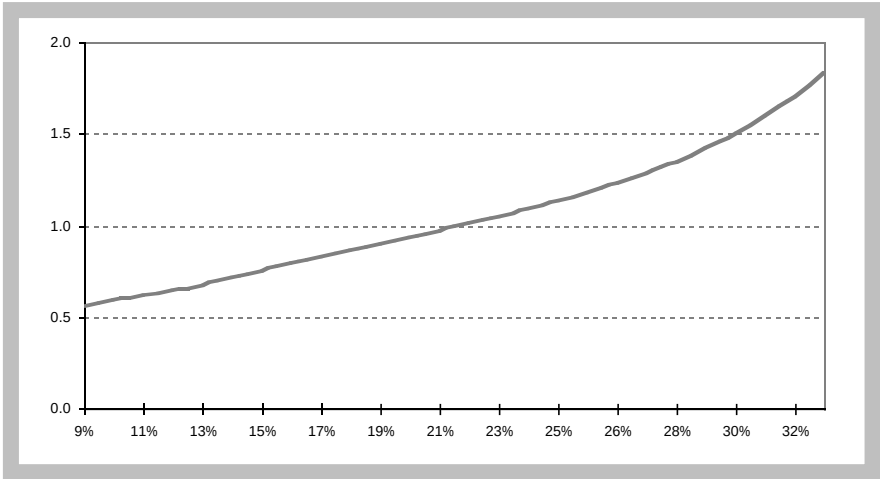


Figure 15.20 Beta along the efficient ridge

Forecast versus realised tracking errors

This brings us to the issue of forecast versus realised tracking error. To some extent it may not be particularly important whether the model is a hundred per cent correct all the time. If the mandate stipulates that the manager must ensure that the expected or forecast (factor-based) tracking error remains below 4%, then the manager will be safe as long as the portfolio meets this requirement on the day of reporting tracking error to the client. However, it may well be the case that for a considerable amount of time the portfolio is actually contributing more realised active risk than 4%. In other words, if the factor-based risk model is not a perfect forecast of covariances as well as portfolio and benchmark risk, the realised tracking error may well be above that which has been forecast by the model. Therefore it may be of interest to the manager to investigate the risk model’s ability to forecast tracking error by simply calculating it using realised portfolio and benchmark returns. This will be a benefit to the asset manager’s risk-management process, as well as to the portfolio manager’s understanding of the risk dynamics of his or her portfolio.

GENERAL THOUGHTS ON ACTIVE RISK MANAGEMENT

As demonstrated in previous chapters, quantitative portfolio optimisation and asset allocation is closely tied to the risk management process. This is true whether we work with total return and risk or active return and risk. When

choosing an appropriate asset allocation, we do so in accordance with our preference regarding the trade-off that exists between expected return and expected risk. However, many of the investment decisions being made in the asset management industry today have more to do with return expectations than with risk expectations, and to the extent that risk is explicitly considered in the active portfolio management process, its management is rarely implemented in a coherent and targeted fashion, and it often represents little more than an inconvenient measure that one must take in order to appease a client. Below we attempt to provide some thoughts on the practical implementation of active risk management into the active portfolio management process.

When attempting to apply risk management as an integrated part of the active portfolio management process, the portfolio manager must at all times make sure that the portfolio is representative of all his or her views on the market, locally as well as globally, and along all the dimensions present in the portfolio – sectors, factors, stocks, and so on. Generally speaking, when designing a portfolio – in this case a single-country, multi-sector portfolio – the following three issues should be considered:

- Every sector bet must be *explainable* on the basis of expectations about economic fundamentals, valuation relative to the market as a whole, or extraordinary issues that have arisen in the equity markets. In addition, all sector bets must be *internally consistent*. That is, they must represent the same overall expectation about future market, sector and stock returns.
- Every factor tilt and bias must be *intentional* and *explainable*, with regards both to its 'direction' and its size. These too must be internally consistent.
- Every stock bet must reflect the view of the portfolio manager on that particular stock *relative to an appropriate peer group*, and the expected impact on tracking error from each bet should be *proportional to the confidence* the manager has in that particular bet.

The three bullet points above represent the three levels on which the portfolio should be designed. The common denominators are *explainability*, *consistency* and *proportionality*. Below we provide a few more thoughts on each of these three aspects of the active portfolio risk management process.

Sector bet explainability and consistency

Whether a sector bet is explainable from an economic, company fundamental or valuation point of view is to some extent subjective, since the evaluation of the three depends on the eye of the beholder. It is the task of the portfolio

manager to gather and process relevant information that will enable him or her to make these decisions on the sector level.

However, as far as internal consistency between individual sector bets at a given point in time is concerned, we can turn to more quantitative, and thus more objective, assessment techniques. Let us look at an example. During the past decade there has been a high and fairly stable correlation between daily returns on the global information technology sector and those on the global financials sector. In fact, the correlation coefficient calculated on the basis of daily return observations over three months, over a time period spanning from January 1995 to May 2002, was 0.52 on average, was not negative at any time, and during the latter part of the period averaged 0.65. The rise in the correlation coefficient can be explained, at least as far as the equity markets of the late 1990s are concerned, by the fact that the booming or inflated information technology sector brought with it a large increase in the number of new listings of companies on the stock exchange, thereby generating fees and commission for the investment banks and brokers. The opposite argument explains the high correlation level as the equity market bubble burst and deflated, and the financial sector's investment banking and brokerage revenues collapsed.

Even without an equity market bubble, a 'normal' rising equity market will generally affect a rise in financial sector income. Thus, while the tendency for this particular correlation coefficient to be high was perhaps artificially amplified during the late 1990s, there still remains a sound economic argument why it should continue to be high in the future. Therefore, this insight provides us with a fairly dependable tool for evaluating the consistency of the portfolio's active sector weights as far as these two sectors are concerned. We demonstrate this below.

One way of determining whether the portfolio is internally consistent – on a sector level, say – is to examine the portfolio's active sector bets along with the correlation matrix. If the portfolio has a negative bet on the information technology sector and a positive bet on the financials sector, the manager should be able to explain why this is the case, since from a historical perspective, keeping correlation coefficients in mind, it seems to be an inconsistent sector allocation. Specifically, if the portfolio manager is bullish on the market as a whole, it would make sense to have positive sector bets in both the financials and the information technology sectors: if this is not the case, this apparent inconsistency must be investigated. A similar analysis can of course be applied to all the other sector bets and correlation coefficients.

In practice, an analysis such as this can be carried out by examining the historical levels for correlation coefficients, along with the bet structure of the portfolio. We wish to determine whether the portfolio's sector bets are internally consistent, so we need to know the 'direction' of each bet, as well as the historical level of *each* correlation coefficient – which in this case means 45 unique coefficients.

Instead of assigning actual historical levels to each correlation coefficient, it is more manageable simply to designate each coefficient as being either high or low *relative* to the average coefficient over the estimation period. This greatly simplifies the analysis, but at the same time retains the information needed to evaluate sector consistency.

Having determined these two sets of information we can begin with the consistency analysis. Table 15.3 shows a stylised portfolio with active sector weights (bets) and the appropriate correlation matrix, transformed to display only whether correlation coefficients are high (H) or low (L) relative to the average of coefficients. Note that we have left out the top right half of the matrix since this is identical to the bottom left half; and also the diagonal, which is a series of ten 1s indicating each sector's correlation with itself.

Using the information in Table 15.3 we can search for inconsistencies in the sector allocation based on the current bet structure and the correlation levels. Generated using simple logic functions, Table 15.4 demonstrates how this particular portfolio is mostly internally consistent, but also that it has two inconsistencies. The light grey cells indicate consistency and the dark grey cells indicate inconsistency. Let us look at some examples of internal sector consistencies and inconsistencies. Examining the bets on the energy and the materials sectors, we see that the historical correlation coefficient is low, and therefore having a negative bet on the energy sector and a positive bet on the materials sector represents an internally consistent allocation. On the other

Table 15.3 Historical correlation coefficient levels (relative)

	Active weight	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
	%	-0.90	0.60	-1.20	1.30	-0.80	0.30	0.80	1.20	0.30	-1.60
Energy	-0.90										
Materials	0.60	L									
Industrials	-1.20	L	H								
Cons. Disc.	1.30	L	H	H							
Cons. Staples	-0.80	H	L	L	L						
Health Care	0.30	H	L	L	L	H					
Financials	0.80	L	H	H	H	L	L				
IT	1.20	L	H	H	H	L	L	H			
Telecomm.	0.30	L	H	H	H	L	L	H	H		
Utilities	-1.60	H	L	L	L	H	H	L	L	L	

Table 15.4 Historical correlation coefficient levels (relative)

	Active weight	Energy	Materials	Industrials	Cons. Disc.	Cons. Staples	Health Care	Financials	IT	Telecomm.	Utilities
	%	-0.90	0.60	-1.20	1.30	-0.80	0.30	0.80	1.20	0.30	-1.60
Energy	-0.90										
Materials	0.60										
Industrials	-1.20										
Cons. Disc.	1.30										
Cons. Staples	-0.80										
Health Care	0.30										
Financials	0.80										
IT	1.20										
Telecomm.	0.30										
Utilities	-1.60										

hand, examining the industrials sector, we see that we have a negative bet on this sector in a portfolio that is otherwise more appropriate for rising markets. The correlation coefficient between returns on the industrials and the energy sectors is historically low, which indicates that they tend not to move in the same direction, and yet the portfolio's bets on those two sectors are both negative. Hence the dark cell indicating sector allocation inconsistency.

Similarly, the industrials sector's correlation with the materials sector is historically high, and yet the portfolio's bets on those two sectors 'point' in opposite directions, indicating inconsistency. Another example of why the negative bet on the industrials sector is inconsistent with the rest of the portfolio is that this sector's correlation with the information technology sector is historically high, yet the two bets 'point' in opposite directions.

Similar arguments apply to the health care sector, on which the portfolio has a positive bet. This sector has historically had a low correlation with the consumer discretionary sector, and yet the portfolio has a positive bet on this sector as well. Similar examples for the health care sector are obvious. Note, however, that there appears to be internal consistency between the health care sector and the industrials sector. This is of course due to the fact that they are both inconsistent with the *rest* of the portfolio, and are thus consistent with each other.

A sector bet structure such as that presented here is thus both non-intuitive and not in accordance with the historical correlation structure of the portfolio's

sectors. But importantly it may still be perfectly reasonable and explainable on the basis of extraordinary factors coming into play at that particular point in time. The point to be remembered here is that apparent internal inconsistencies in the structure of portfolio sector bets may not necessarily warrant a *change* to that structure. At the end of the day, it is the portfolio manager's job to ensure that no *unintentional* internal inconsistencies are present in the portfolio. This example demonstrates that there is no right or wrong allocation per se, but rather that this technique can assist the portfolio manager in ensuring that if he or she is taking a firm view on the markets, the structure of the portfolio is at least internally consistent.

An alternative and less sophisticated way of examining this issue, and thus a shortcut to this type of insight into portfolio risk dynamics, is to see whether a portfolio has positive sector bets on some high Beta sectors and negative sector bets on other high Beta sectors. Figure 15.21 demonstrates an example in which the portfolio manager has a positive view on the market as a whole, and has positioned the portfolio accordingly, with the largest bets in the sectors with the highest Betas.

Note again that this technique requires that the portfolio manager take a firm view on the direction of the market. Generally speaking, the *size* of the sector bets in the portfolio should be proportional to the *confidence* the manager has in those bets. Importantly, by 'size' we refer not to the percentage over- or under-weight relative to the benchmark, but rather to the expected impact on tracking

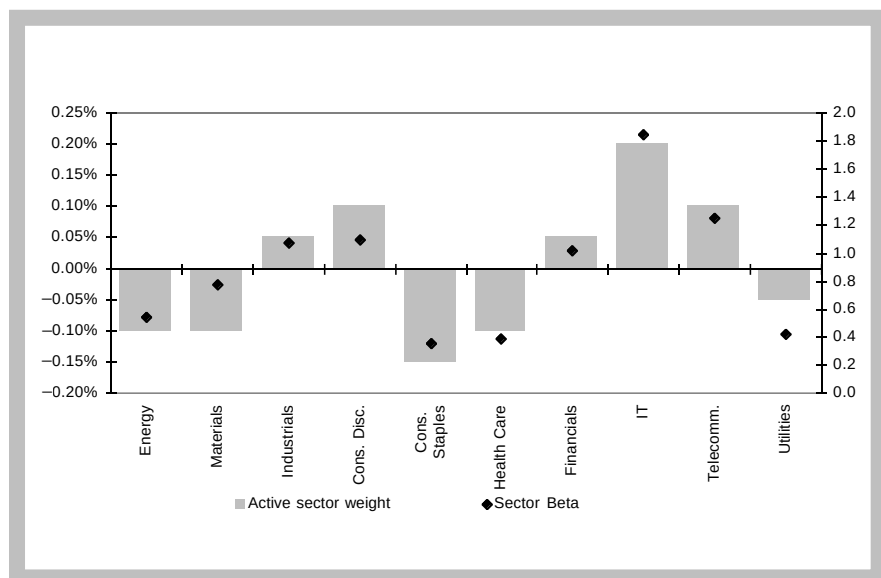


Figure 15.21 Size and direction of sector bets and Betas

error from each bet. It may well be that two bets of similar size have very different impacts on tracking error because of differing correlation coefficients and volatility levels; thus if one bet is expected to have twice the impact on tracking error as another bet, then ideally the portfolio manager should have twice as much confidence in the former bet as in the latter, even though they are of the same size in percentage terms. We shall illustrate this point further below.

Factor tilts and biases

We have discussed factor models previously in this book, but we shall touch briefly on them again here. The most common factor models applied in the asset management industry are fundamental factor models that attempt to capture and assign a range of factors to individual stocks, based on their individual factor exposures, so as to determine to what extent the portfolio has a bias towards one or several factors. Figure 15.22 shows a stylised example of a factor exposure breakdown for a portfolio, relative to its benchmark.

This particular portfolio is biased or tilted away from large capitalisation stocks, which means that by contrast it has a bias towards small capitalisation stocks: as such it is expected to outperform the benchmark when small capitalisation stocks outperform and large capitalisation stocks underperform.

We have also included a number of valuation factors, and based on those we can see that the portfolio has a somewhat ambiguous bias towards value

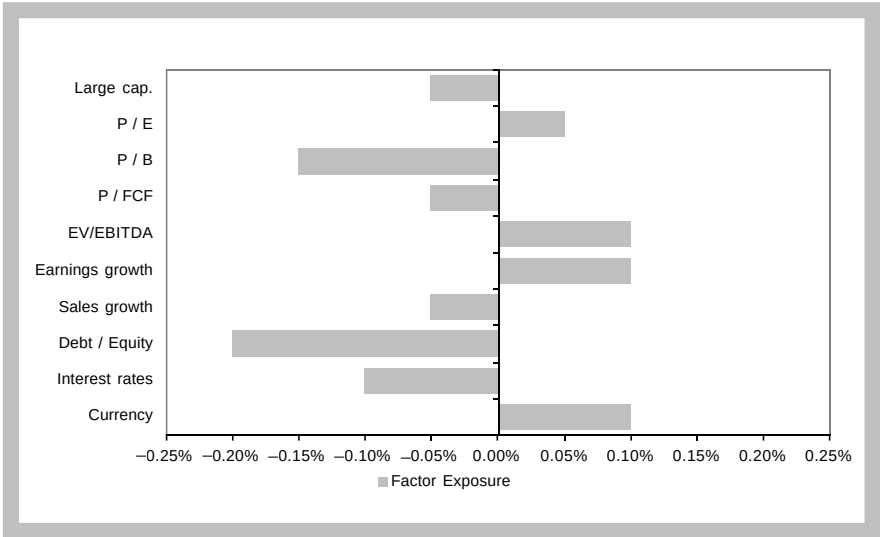


Figure 15.22 Portfolio factor exposures relative to the benchmark

stocks. It has a higher (market cap-weighted) average price-to-earnings ratio (P/E), but a lower price-to-book ratio (P/B) and also a lower price-to-free-cash-flow ratio (P/FCF). The portfolio's EV/EBITDA ratio, however, is higher than that of the benchmark. While this may appear inconsistent with the other valuation metrics, it is quite possible that it is intentional. Quite simply, the portfolio manager may decide to put more emphasis on certain valuation metrics than others, or perhaps to completely disregard some metrics if for extraordinary reasons they are nonsensical or if they have proven themselves to be unreliable predictors of share price performance.

The portfolio is tilted towards companies with above-average expected earnings growth and tilted away from companies with sales growth, revealing the manager's preference for companies that are able to generate earnings growth by restructuring the business and bringing down costs, rather than having to depend on the external economic environment and sales growth in order to grow earnings. In addition, the portfolio is strongly biased towards companies that have a healthy balance sheet, since the portfolio's exposure to debt relative to equity is lower than that of the benchmark.

Finally, an interest rate increase is expected to have a negative impact on the portfolio, revealing that the manager is implicitly expecting interest rates to remain stable or to come down; and that an increase in the exchange rate is expected to have a moderately positive impact on the portfolio, revealing the portfolio manager's preference for companies dependent on overseas demand.

The point to be underscored here is that, just as was the case with sector bets, both the 'direction' and the size of every factor bias must be intentional and explainable by the portfolio manager. Factor tilts and biases can potentially be just as important for active performance as sector bets, since equity markets regularly ignore the sectors in which stocks happen to be classified, and focus almost exclusively on stock characteristics such as those presented in Figure 15.22.

Peer groups, stock picks and risk matching

When the sector allocation has been determined, the portfolio manager must decide which stocks to include in the portfolio, and by how much they should be included; and by contrast, which stocks not to include. This is best carried out on a peer group basis, since two stocks being placed in the same sub-sector classification may run very different kinds of businesses. Every stock is thus best evaluated relative to an appropriate peer group of stocks. Once this task has been completed and the manager has decided which stocks to include in the portfolio, he must also consider the active risk that is being introduced. In other words, he must decide how large an active position (positive or negative) he is

going to establish in each stock, based on each stock’s individual potential for outperforming the benchmark. But this is not all, since the portfolio manager must also consider risk when designing his portfolio. He should thus consider the likely impact on the tracking error of the portfolio, and this depends on the size of each bet, the volatility of the stocks, and the correlation of each stock with the benchmark. As noted above, the absolute size of a bet in percentage terms is not always a good indicator of its contribution to tracking error, so the portfolio manager must examine the size of the bets and their volatility levels, and must also include the correlation structure of the portfolio, in order to arrive at a reasonably precise estimate of each bet’s contribution to tracking error.

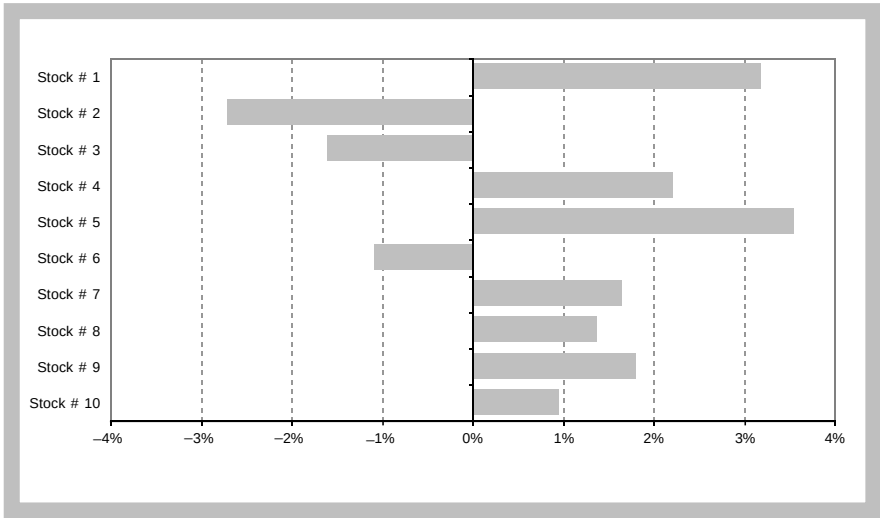


Figure 15.23 Size of the most ‘risky’ bets

In Figure 15.23 we illustrate the size of what will end up being the most ‘risky’ bets in the portfolio on a stock level, ranked from 1 to 10. Note that both positive and negative bets are included, since it makes no difference, as far as tracking error is concerned, whether the portfolio manager decides to overweight or underweight a stock. Note also that even though the largest bets seem to be at the top of the chart, they are not ranked according to bet size. This is clearly demonstrated by bet #5, which is much larger than bet #3 but lower on the ranked list of the ten largest contributors to expected tracking error. Similarly bet #6 is smaller than bets #7, #8 and #9, but it appears higher on the list. This is partially a function of the volatility levels for each stock, which are illustrated in Figure 15.24.

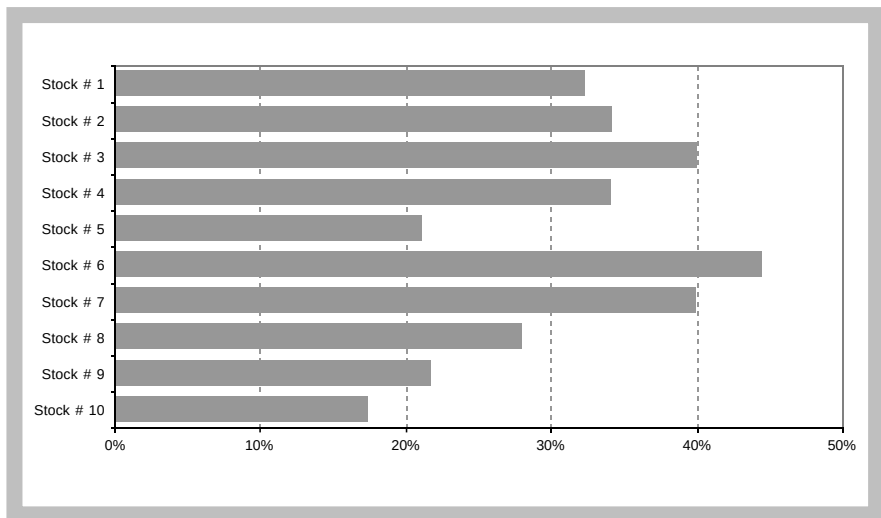


Figure 15.24 Volatility of the most ‘risky’ bets

Here we see that bet #3 is in a stock that is much more volatile than that in bet #5, and that thus has a greater impact on expected tracking error.

In Figure 15.25 – dubbed ‘hope and confidence’ – we take the correlation structure of the portfolio into account, thereby incorporating any beneficial or

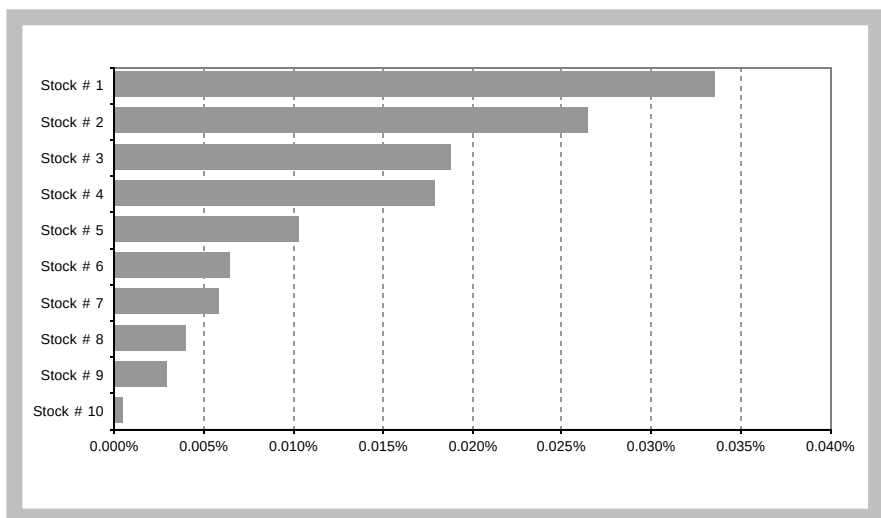


Figure 15.25 Hope and confidence: expected contribution to tracking error

adverse effects on portfolio tracking error from correlations between individual stock returns. Figure 15.25 thus outlines the expected contribution to tracking error from each of the ten most influential stock specific bets in the portfolio. But obviously this information is only valuable if the portfolio manager is able to determine whether these expected contributions to active risk are commensurate with her expectations as far as active return is concerned. We therefore use the phrase ‘hope and confidence’ to convey that when the portfolio manager is presented with a chart such as this, she should be able to recognise and explain it.

Is the portfolio manager really confident that the stocks that are responsible for the largest contribution to tracking error will generate the highest active return, or does she just really hope that they will? Ideally the manager should be able to evaluate the active return potential of each stock-specific bet relative to the other bets, and the size of each stock’s expected contribution to tracking error should ideally be proportional to the expected outperformance of that stock. Figure 15.26 illustrates a stylised example of this. Note that all the black dots – representing expected active return versus expected active risk for each of the ten bets – are nicely aligned around the straight line, except for the bet ranked #3. This bet has a *higher* expected tracking error than bet #4, yet it is ranked *lower* on the list as far as the (manager-specific and subjective) expected active return ranking is concerned. The solution to this inconsistency is to decrease bet #3, thereby reducing its expected contribution to tracking error, or to increase bet #4, thereby increasing its contribution to tracking error.

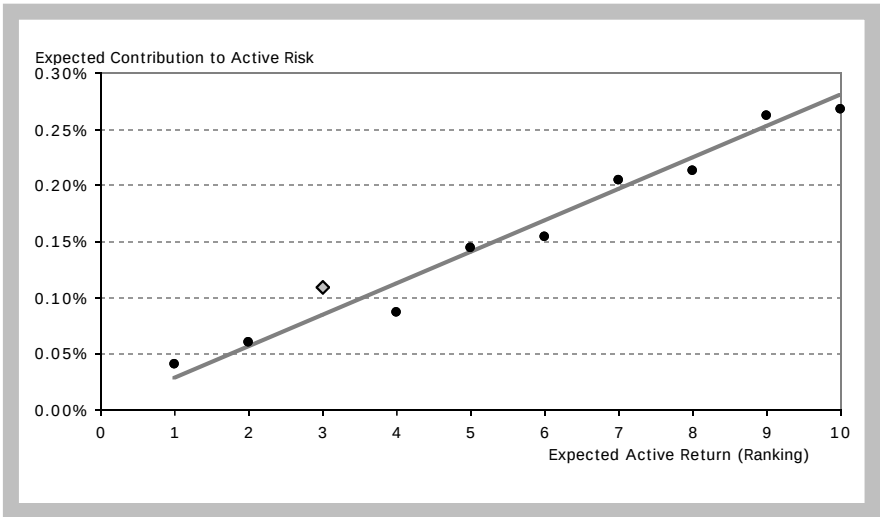


Figure 15.26 Matching active risk with active return

We use a ranking of expected active return to carry out this analysis, as a quantifiable estimate of the expected outperformance of each stock relative to the benchmark is difficult, to say the least. However, the manager should be able to rank stocks based on his expectations about the relative performance of each stock within the group of stocks being examined. Subsequently, any deviation from a linear return vs. risk relationship should be adjusted by changing the size of that particular bet to bring the portfolio more in line with the manager's expectations.

Implementation

The portfolios must be pointed in exactly the right direction, so that to the extent that we are right about the market this is reflected in the portfolio's composition, and the performance of the portfolio. In addition, every bet – and more importantly, its expected contribution to tracking error – must be proportional to the expected outperformance of that bet. This must be the case both on the sector and on the stock level. In order to properly evaluate the performance of a portfolio, the portfolio manager must employ quantitative attribution analysis, and in the process focus not solely on attribution of the active returns, but also on attribution of the active risk. At every stage of the portfolio management process – both in the ex ante portfolio construction process, and in the ex post performance attribution process – the evaluation of portfolio performance should be carried out by examining the cost that has been incurred, in terms of active risk, in order to generate any active return on the portfolio. Ideally this should be carried out on the sector level, as well as on the individual stock level.

SUMMARY

In this chapter we have constructed a global Sector Risk Model, that allows us to analyse the risk dynamics and exposures of the portfolio in terms of both total portfolio risk and active risk or tracking error. Using matrix algebra we have determined marginal contributions to total portfolio risk and active risk, thereby enabling a fairly precise assessment of the impact of prospective sector changes. This type of analysis can serve as a useful ex ante tool for determining how the risk dynamics of the portfolio might be changed, given a new set of expectations about the future investment environment. We have also demonstrated how both total and active risk can be decomposed so as to enable attribution to individual sectors. This is potentially a very powerful tool for the asset manager, allowing an assessment of whether the amount of active

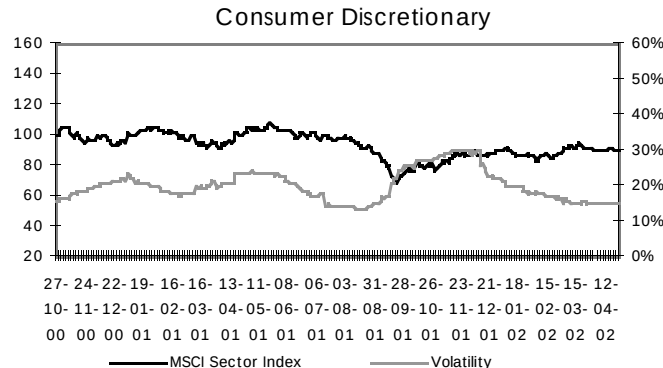
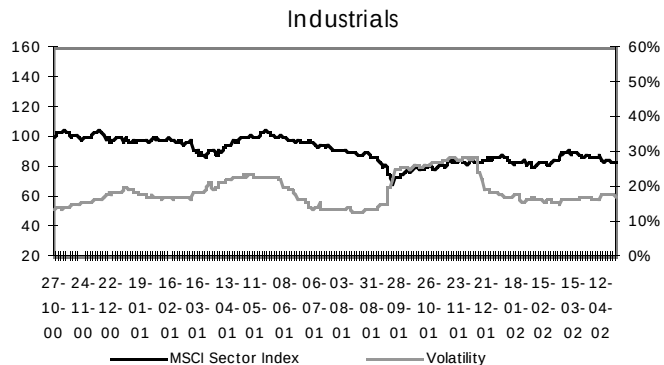
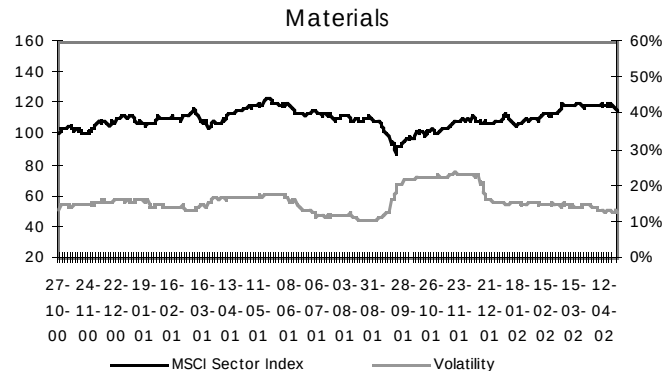
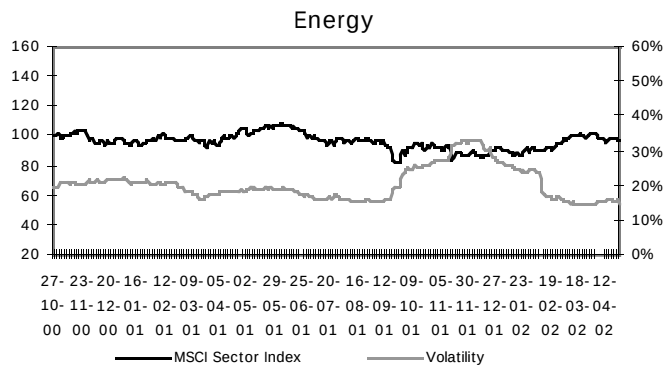
risk coming from a particular sector is commensurate with the expectation of the manager with regard to the likelihood of outperformance. Ideally, the expected impact on tracking error of an active weight, should be proportional to the confidence the asset manager has in that particular bet. If this is not the case, it could be argued that the structure of the active portfolio is inconsistent. The portfolio's Beta can also be broken down into sector components, allowing an assessment of likely effects on portfolio aggressiveness as a result of a prospective sector rotation. Finally we have demonstrated how these analysis tools evolve as we move along the efficient ridge.

Risk management is about more than just estimating the expected tracking error of the portfolio. The portfolio manager must be aware of and actively manage several aspects of the portfolio's active risk, specifically the size of sector bets (measured in terms of expected contribution to tracking error), the factor tilts and biases in the portfolio relative to the benchmark, and also the individual stock bets relative to each other. The portfolio manager work to ensure that internal consistency within the portfolio's bet structure is maintained, to ensure that all factor tilts and biases are intentional, and also to see to it that all individual stock bets are of an appropriate size (in terms of expected contribution to tracking error) and that they can be ranked according to the trade-off between expected active return and expected active risk.

Notes

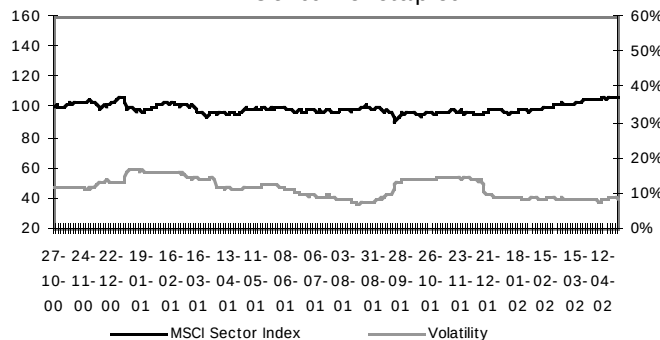
- 1 Implicit in this is that as we increase one of the ten sector weights by 1%, each of the remaining nine sector weights is decreased proportionally so as to ensure that weights sum to 100%.
- 2 Note that this analysis easily extends to *marginal sector contributions to residual risk* (MSCRR). Instead of using the active portfolio weights, we simply use the residual portfolio weights.
- 3 It is interesting to note that this definition of sector Betas is actually identical to the marginal contributions to portfolio Beta given an increase of 1% in the weight of one sector. This can be verified by taking the partial derivative of Equation 15.9 with respect to the portfolio weight vector.

APPENDIX 15A: SECTOR INDICES AND VOLATILITIES

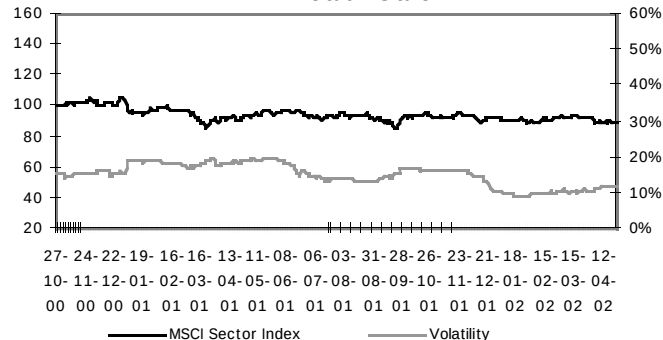


APPENDIX 15A (continued)

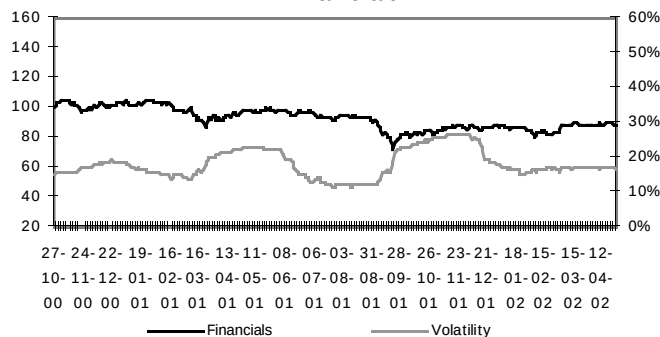
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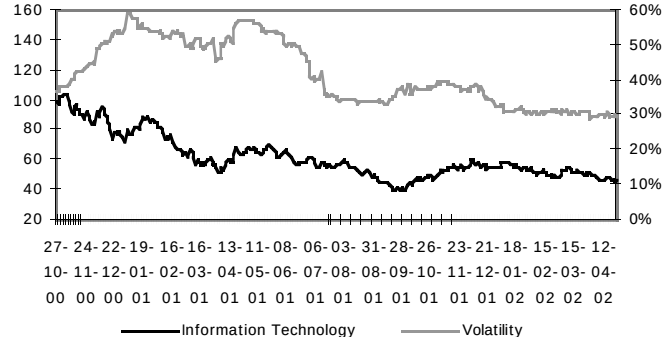
Health Care



Financials

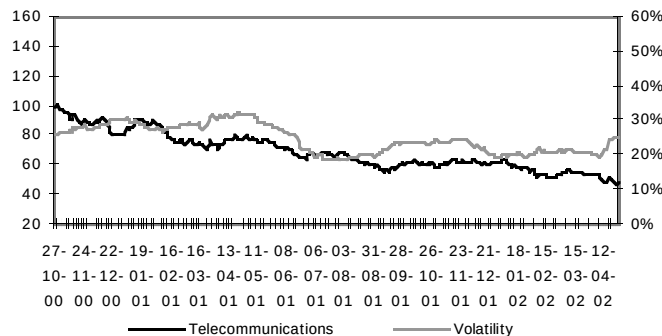


Information Technology

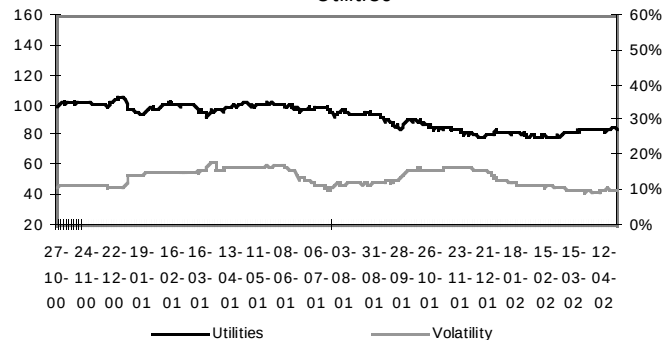


APPENDIX 15A (continued)

Telecommunications

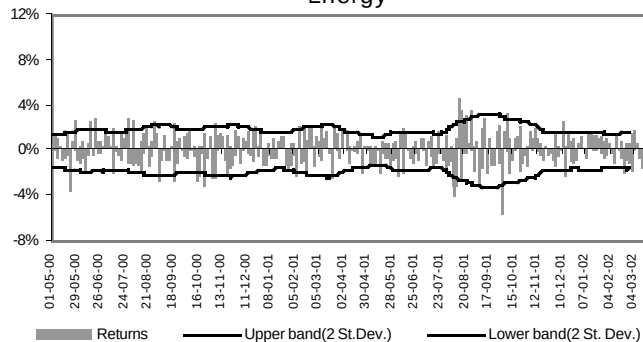


Utilities

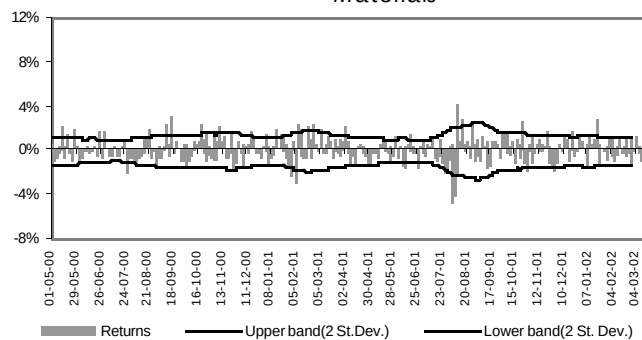


APPENDIX 15B: SECTOR RETURNS

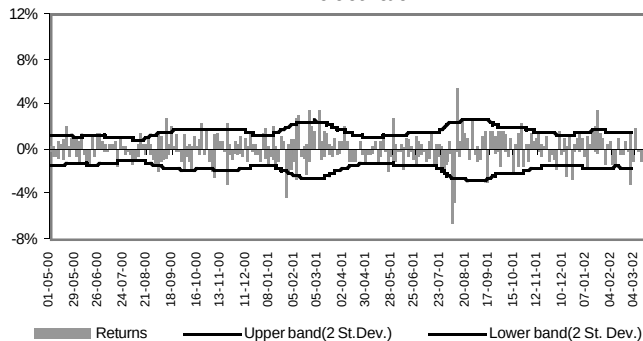
Energy



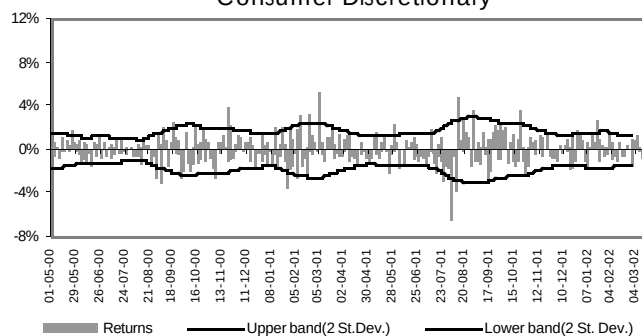
Materials



Industrials

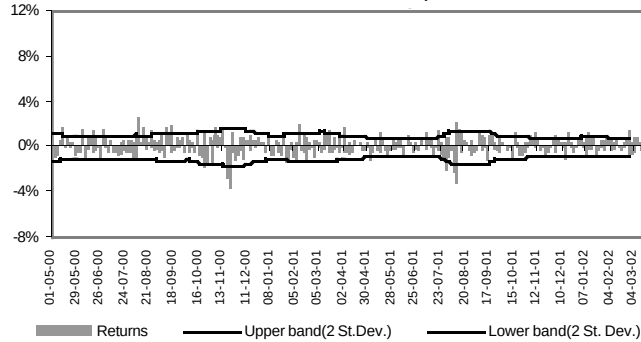


Consumer Discretionary

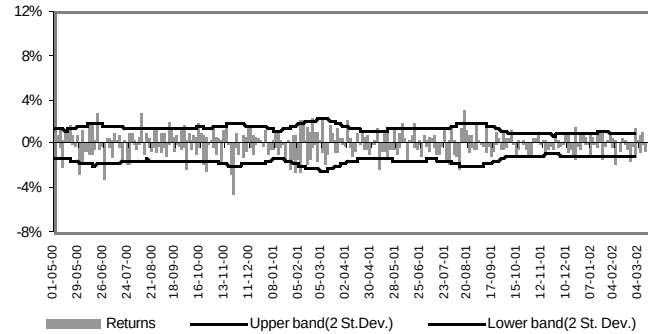


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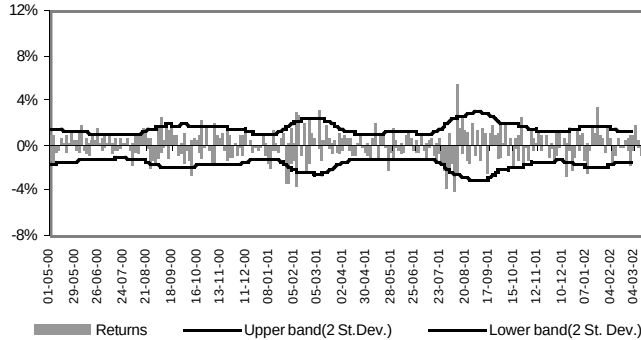
Consumer Staples



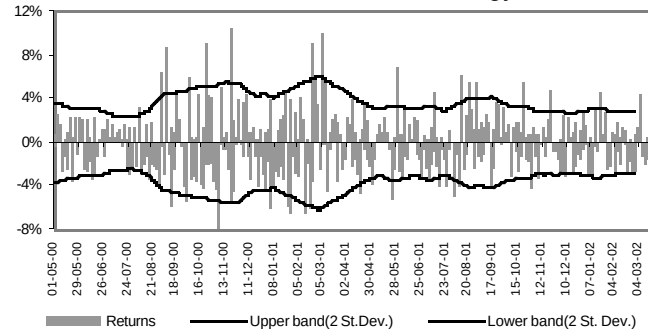
Health Care



Financials

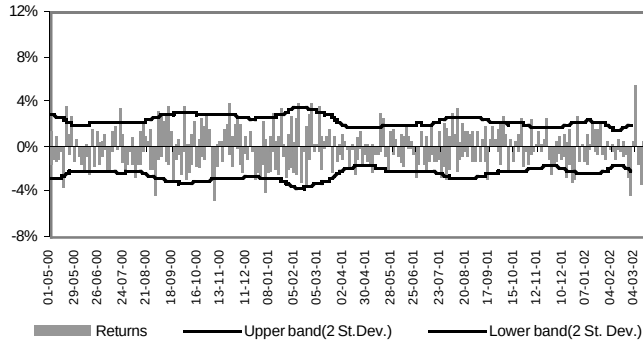


Information Technology

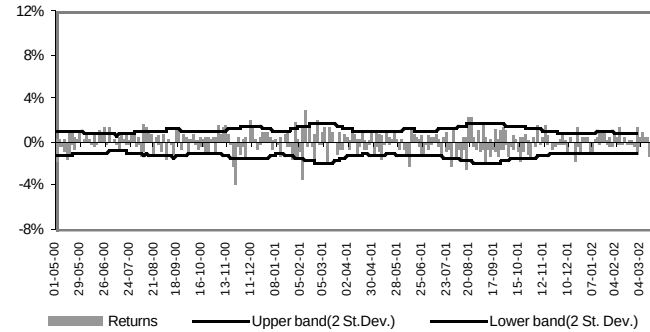


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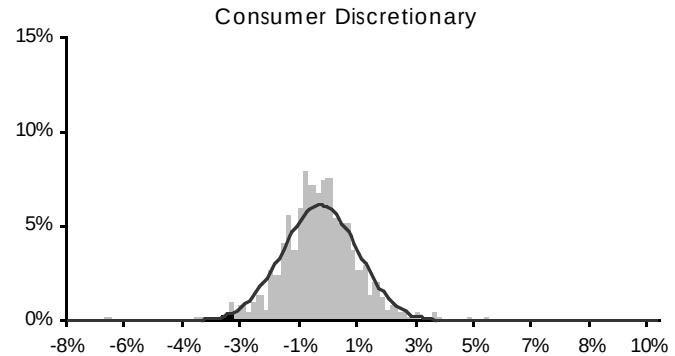
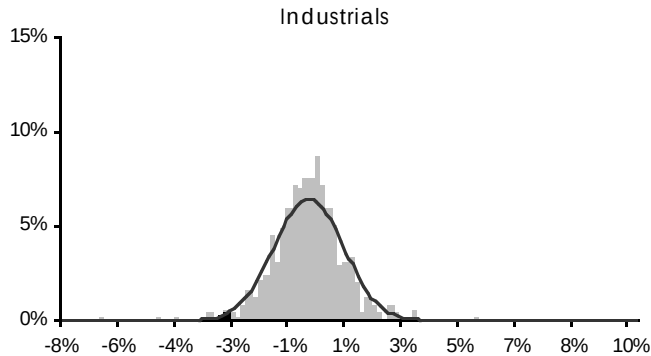
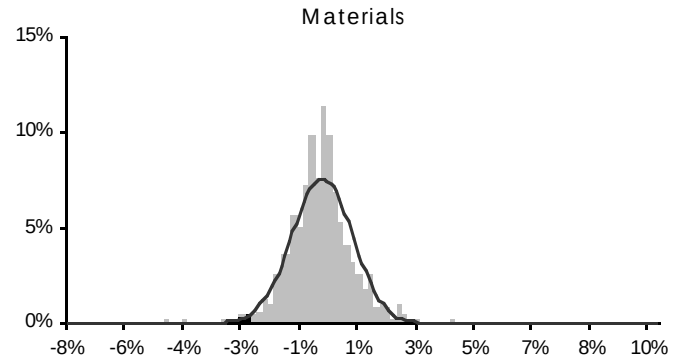
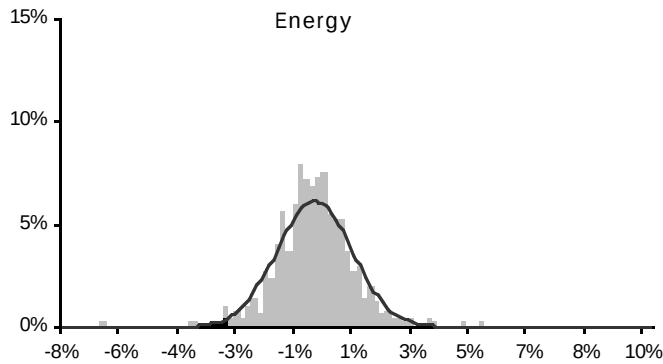
Telecommunications



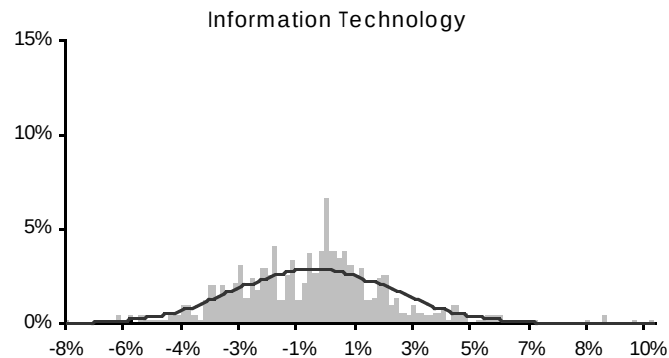
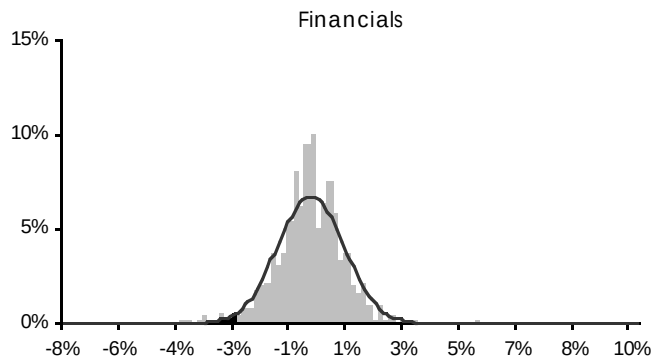
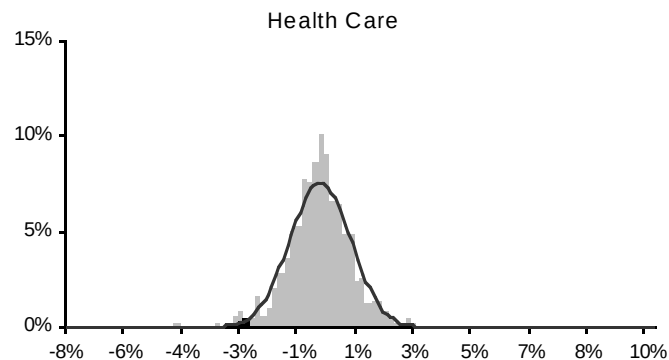
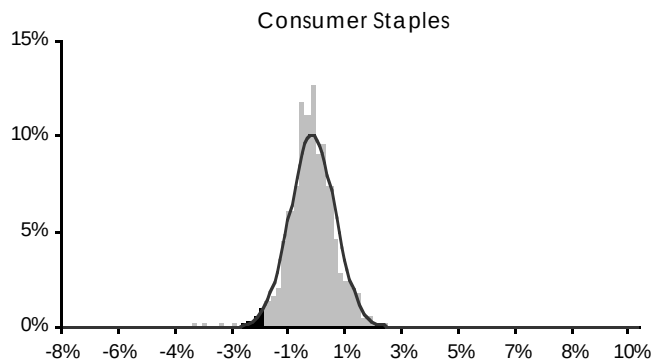
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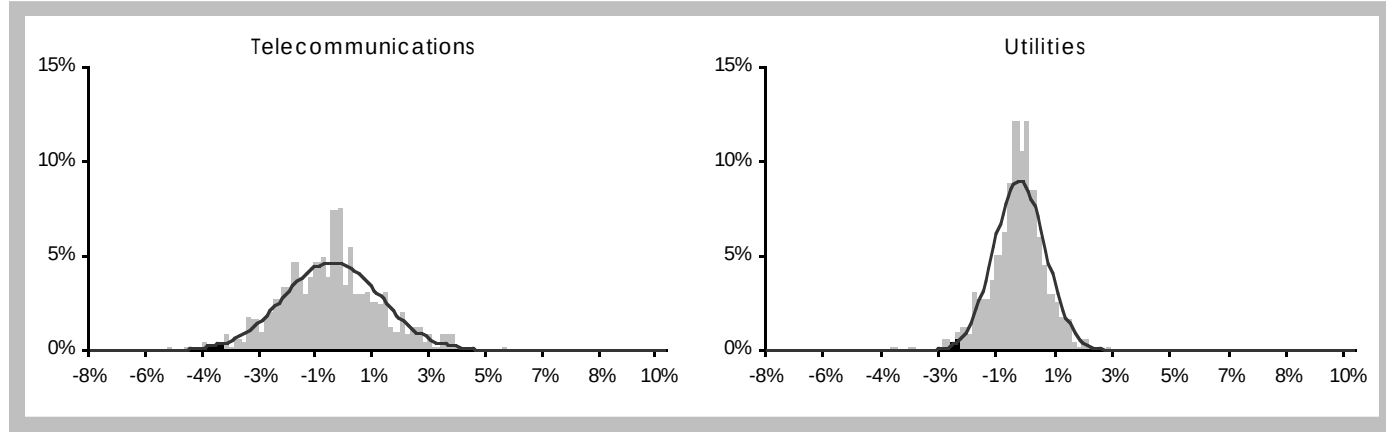
APPENDIX 15C: SECTOR RETURN DISTRIBUTIONS



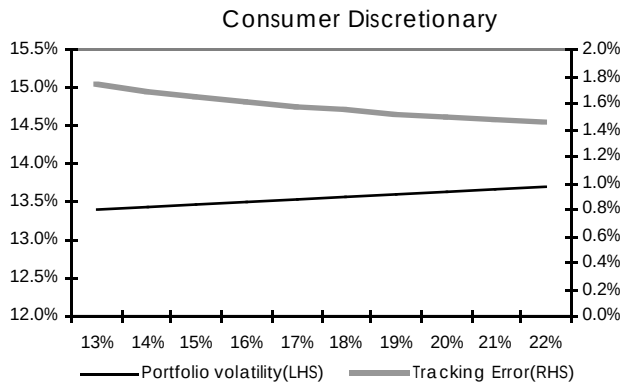
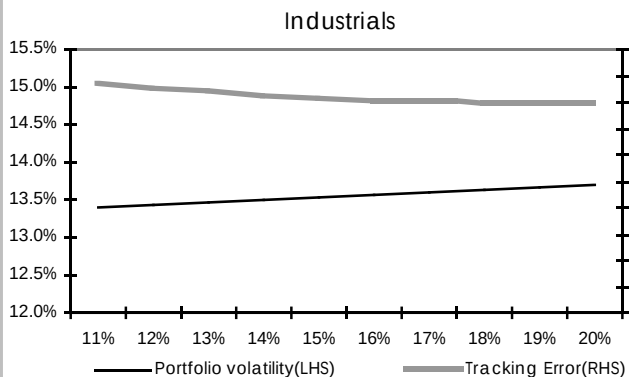
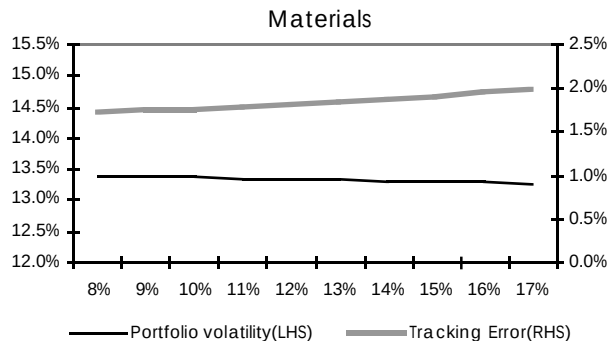
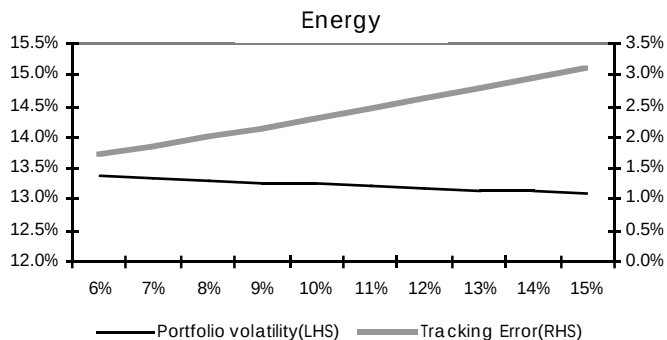
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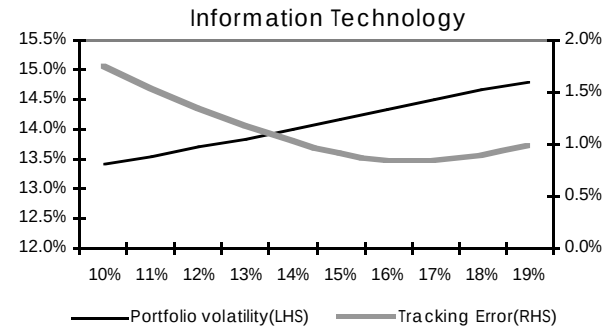
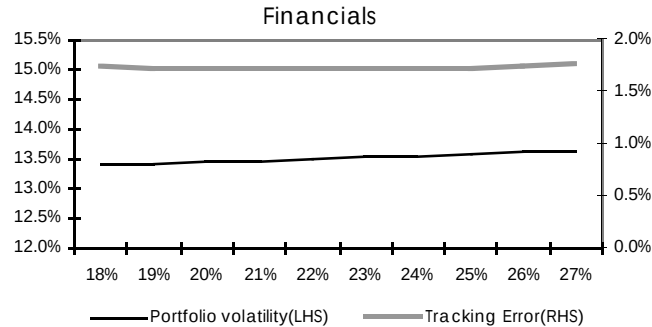
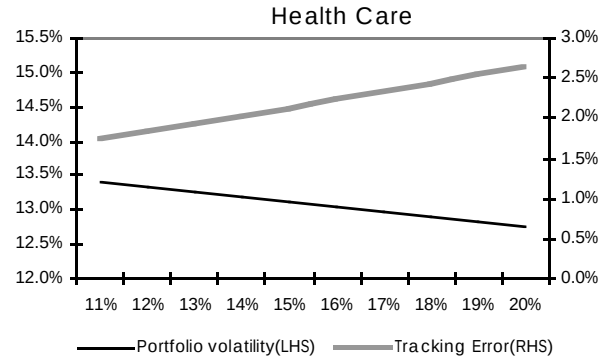
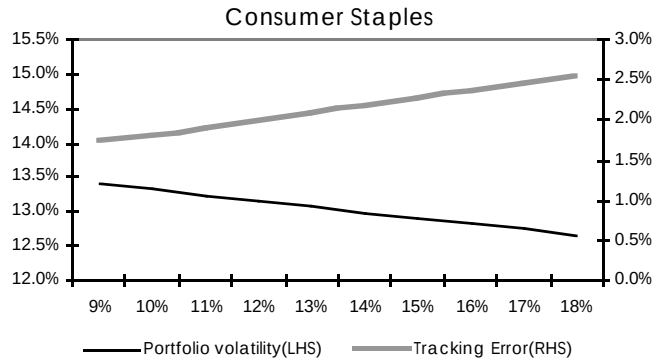
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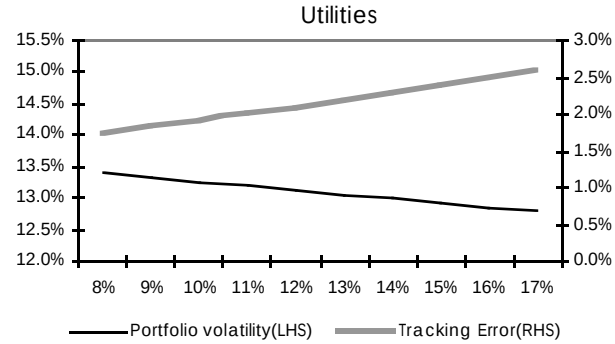
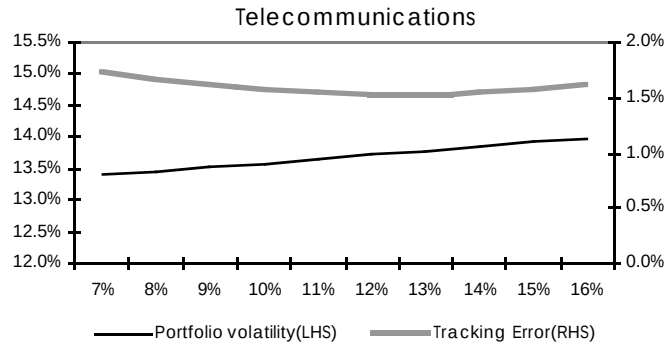
APPENDIX 15D: PORTFOLIO VOLATILITY AND TRACKING ERROR



APPENDIX 15D (continued)

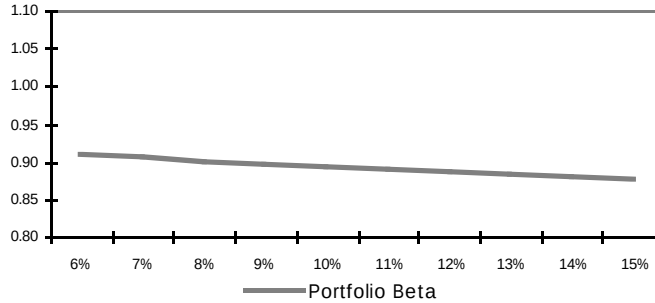


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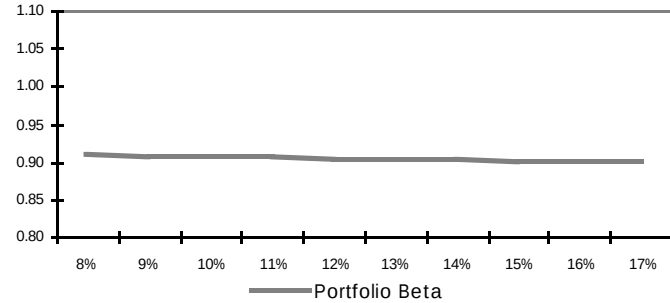


APPENDIX 15E: PORTFOLIO BETA

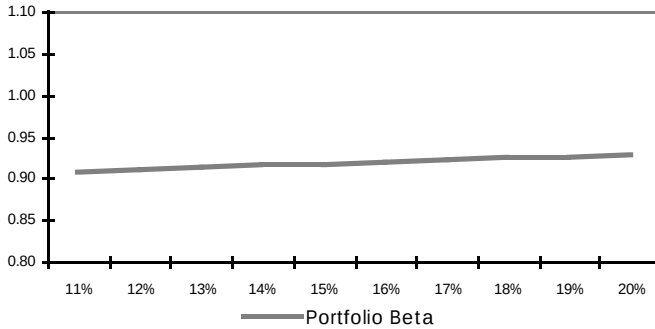
Energy



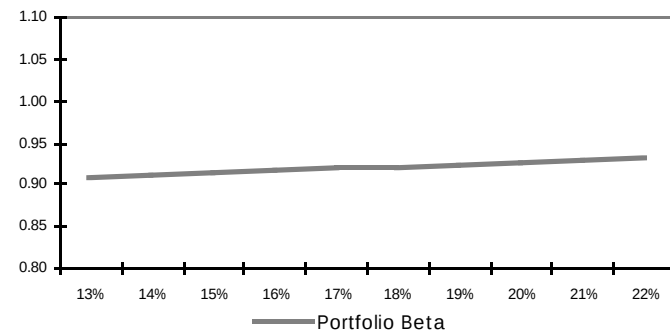
Materials



Industrials

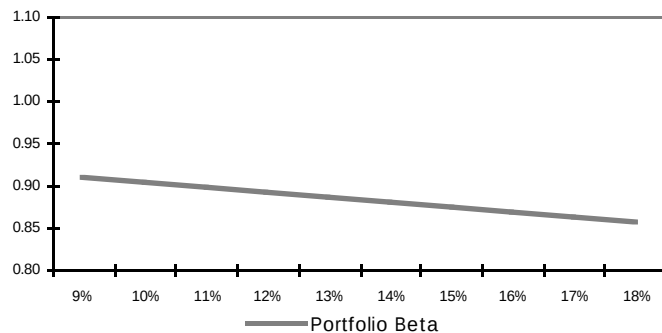


Consumer Discretionary

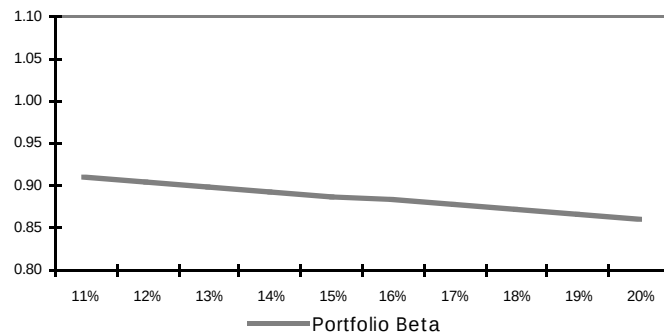


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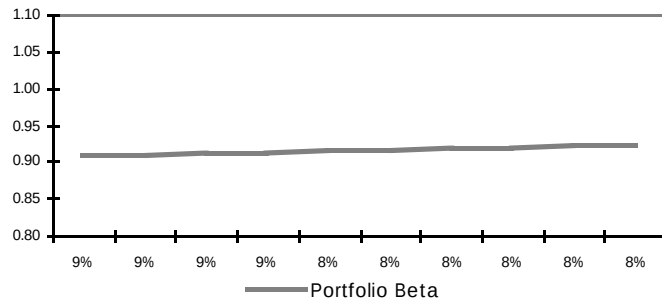
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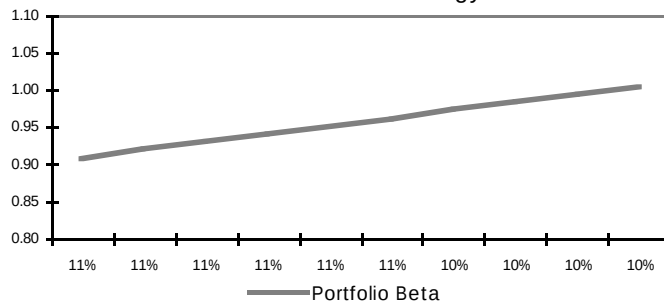
Health Care



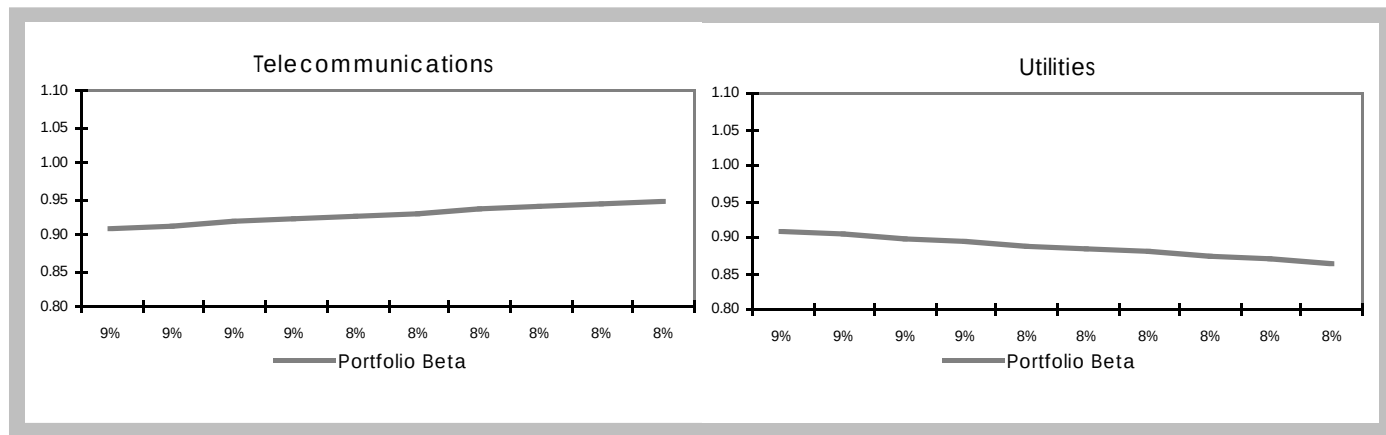
Financials



Information Technology



APPENDIX 15E (continued)



VALUE-AT-RISK (VAR) AND EXTREME VALUE THEORY (EVT)

THE BASICS

An alternative risk measurement tool

Over the past decade or so the concept of *Value-at-Risk* (VaR) as a risk-management tool has steadily become more and more prominent in the asset-management community. As the concept has increased in sophistication over the years, it has developed from an academic exercise to a viable risk management tool. Originally proposed by the investment bank J.P. Morgan in the early 1990s, the concept of VaR in its simplest form is about assessing the potential change in portfolio value within a particular timeframe at a particular level of certainty.

VaR works on the same principle as that presented in earlier chapters on asset allocation and characterisation of risk. Risk is best described by a probability distribution, and not merely by a single number such as standard deviation or volatility. The focal point of the VaR concept is thus estimation of the probability distribution of returns, most often portfolio returns since this is ultimately what the asset manager should pay attention to. However, it is actually slightly more complex than that, because what we are really interested in

is the *future* probability distribution for portfolio returns. That is, not only are we interested in the expected level of risk, but we are also interested in the *characteristics* of that particular expected risk, because these must be properly described if VaR is to deliver on its full potential. This is a highly challenging task, which is why risk managers spend so much time investigating the characteristics of individual assets' risks, in order to better understand the nature of risk for an entire portfolio.

The idea of measuring the amount of the portfolio that could be considered at risk within a given timeframe was originally proposed in J.P. Morgan's proprietary RiskMetrics™ framework for assessing the risk of bond portfolios in terms of expected absolute changes in portfolio value over time. Of course, the principle extends equally to equities and to other types of assets, as we shall see in this chapter.

Actually establishing a workable framework for VaR that may be employed usefully in the investment decision-making and portfolio construction process can be quite complicated and somewhat of a challenge. However, since VaR is concerned with probabilities of events, we establish the first building block by defining the probability of a change in portfolio value occurrence by saying that:

$$P(\Delta V_{p,T} > X) = c \quad [\text{Eq. 16.1}]$$

where P indicates the probability, V_p the change in portfolio value, X some predetermined limit, and c a desired level of certainty. Equation 16.1 states that with probability c , the value of the portfolio V_p will not change by more than X , the latter typically being a percentage or an absolute value. Alternatively, the VaR concept can be phrased as the following statement: 'We are c per cent certain that V_p will not lose more than X in the next T days.'

Any statistician will recognise that some predefined probability distribution has been established *a priori*, and implicitly in most cases this usually means that asset or portfolio returns are assumed to be normally distributed. This assumption is made primarily because it makes it very easy to calculate the VaR, as the risk element can be described by a clearly defined probability function – the normal distribution characterised only by a mean and a standard deviation. The value c indicating the probability level is thus 2.33 for a 99% confidence level, since in a normal distribution only 1% of the observation will lie more than 2.33 standard deviations from the mean. Basically this means that we would expect to observe portfolio returns that lie beyond 2.33 standard deviations from the mean in just 1 out of every 100 trading days. These are typically called *2.33Sigma events*, since they lie at or beyond 2.33 standard deviations. In risk-management contexts, terms such as *2Sigma* or *3Sigma events* are often applied to describe the magnitude of a return relative to the distribution to which it belongs.

This point can be illustrated by a fictional return frequency plot of an asset whose returns are normally distributed with a mean of 1% and a standard deviation of 2%. The dark grey area represents the level below which 5% of observations (returns) are expected to lie (see Figure 16.1). In the same manner, assuming normally distributed returns, 5% of the observations will lie 1.65 standard deviations from the mean, thus implying a 95% confidence level. This indicates that 5% of the returns are expected to be of an approximate magnitude of -2.4% or lower. Alternatively, 1% of the observations will lie 2.33 standard deviations from the mean, which implies that only 1% of the time would we expect to observe returns lower than approximately -6.2% . This is represented by the light grey area to the far left of the distribution.

This is very convenient for quick statistical analysis of return probabilities both on an asset and on a portfolio level, but unfortunately it is also at odds with reality, as we have discussed earlier in this book. Asset returns are usually not normally distributed, but rather exhibit skew as well as *leptokurtosis*, meaning that the distribution tends to ‘tilt’ to one side and also tends to have so-called *fat tails*. These non-normal characteristics of asset return distributions present the most important challenge in working with VaR as a reliable risk-management tool. For the VaR estimate to be reliable, it must be based on accurate estimates of return distributions. We shall examine the impact of using different probability distribution assumptions later in this chapter.

We now go into more detail on VaR as a tool, by introducing the variance-covariance approach, the historical simulation approach, and lastly the Monte Carlo Simulation-based approach.

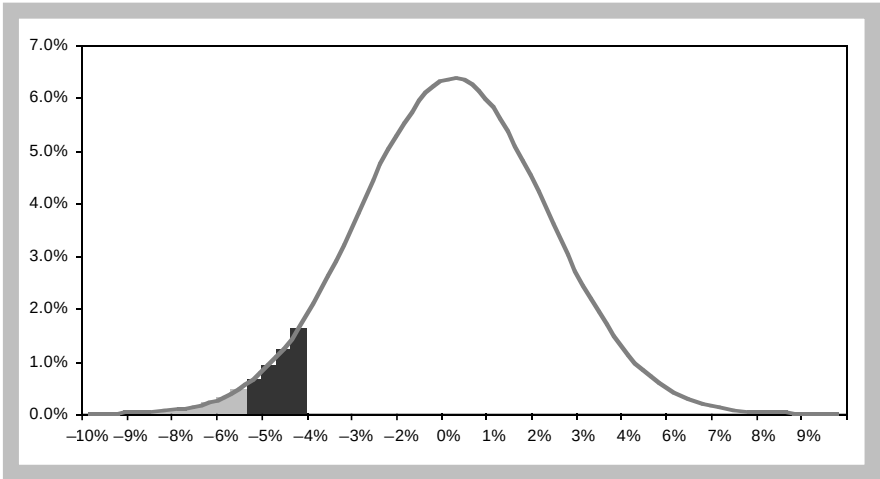


Figure 16.1 Normal distribution, 95% and 99% confidence levels

VARIANCE-COVARIANCE VAR

Approach and calculation

This approach is the most commonly applied for portfolio risk management, since it is relatively easy to work with, and in addition it fits well with the theoretical foundation for modern portfolio theory. However, in order to facilitate the understanding of the approach, we begin with an alternative way of expressing the essence of VaR: 'VaR is the estimate of the potential loss on a portfolio within some period of time at some probability of occurrence.' Its core is thus very simple: determine the current value of the portfolio, calculate the volatility of the portfolio, and finally determine the horizon over which the VaR estimate must span.

Formally the VaR number for a given portfolio at the 99% confidence level can thus be determined by the following equation:

$$VaR_p = 2.33 \cdot V_p \cdot \sqrt{w^T \cdot \Sigma \cdot w} \cdot \sqrt{T} \quad [\text{Eq. 16.2}]$$

The number 2.33 is a result of 1% (we chose a 99% confidence level) of returns being expected to lie beyond 2.33 standard deviations from the mean. Again, this of course requires returns to be normally distributed. Assuming this, the VaR for this portfolio P is thus given by the value of the portfolio times the risk of the portfolio (as measured by the volatility of its expected returns), times the time horizon over which we need to assess the portfolio's risk. As was illustrated for the calculation of asset and portfolio volatility, in order to expand the forecast or estimate over a time period, we must multiply by the square root of the time horizon or estimation period, since the risk measure used – volatility – has been transformed from a measure of variance to a measure of standard deviation.

Calculating VaR in this manner thus necessitates the characterisation of three elements: the current value of the portfolio, its composition in terms of assets, and the characteristics of these assets as measured by their variances and covariances and also by the nature of each asset's expected returns distribution. The first two elements are relatively easily determined. The most important element is the characterisation of the portfolio's assets. The estimate or forecast of variances and covariances as well as the estimates or forecasts of each asset's return probability distribution can potentially have a significant impact on VaR estimates, particularly when very high levels of confidence are required.

In this very basic form, the VaR of a portfolio is simply its volatility multiplied by a factor that is in turn determined by the desired level of confidence, the value of the portfolio, and the desired time horizon over which we need to

evaluate the risk of the portfolio. The impact of changing the time horizon can be illustrated by examining the so-called ‘square root of time’ rule.

Square root of time

The *square root of time* rule can be illustrated by looking at the degree to which uncertainty increases or ‘expands’ as the time horizon is extended. Figure 16.2 shows a normal distribution as we extend the time horizon from 1 day to 20 days. Every extra day adds to the uncertainty, and as a result the probability distribution becomes expanded and gets wider and wider, implying an ever-greater likelihood of observing returns far from the original mean. This chart thus serves to illustrate how quickly the VaR estimate becomes more uncertain as we increase the time horizon by just a few days, since the expected probability distribution widens quite quickly.

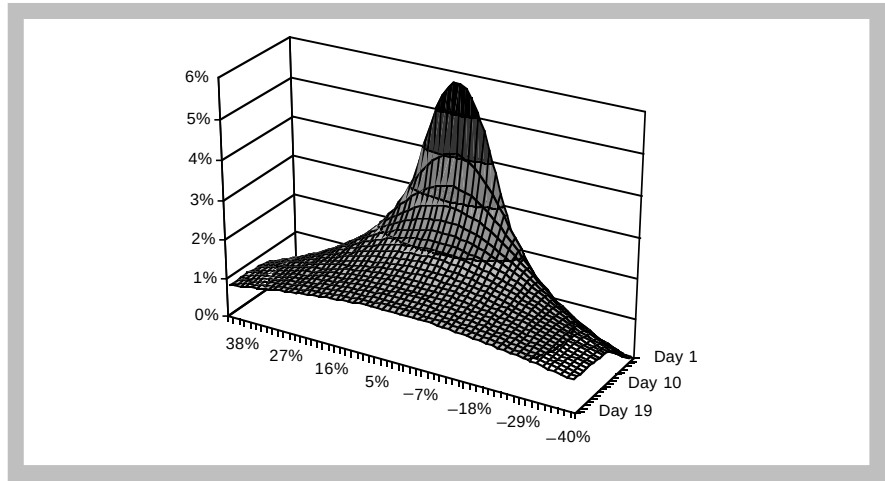


Figure 16.2 Normal distribution at varying time horizons

HISTORICAL SIMULATION OF VAR

Instead of relying on returns following a well-defined probability distribution such as the normal distribution, one can choose to employ historical simulations. This entails running the portfolio through a set of historically observed asset returns, to see how the portfolio returns behave. One such example might

be a period of currency crisis, another might be a prolonged period of companies reporting results much worse than expected. These types of events will typically result in large changes in asset values day by day, and may thus provide some insight into the vulnerability of the portfolio to such events. The major advantage of historical simulations of VaR is that it captures the third and fourth moment of returns. That is, if returns have exhibited skew and kurtosis in the past, then the VaR estimate will accurately reflect these characteristics.

A small study of equity returns

We have already presented some of the results of this small study in previous chapters when we discussed the estimation and characterisation of parameters for the portfolio optimisation and asset allocation model. Examining 100 stocks chosen at random from the MSCI World over 10 years from 1992 to 2002 yielded approximately 260,000 trading days' worth of data. It is obvious that strictly speaking these observations are not independent, but we will treat them as such for analytical simplicity. In Table 16.1 we present the expected distribution of these returns, assuming that they were normally distributed, as well as the actually observed distribution, both in terms of standard deviations from the mean.

If stock returns did follow a normal distribution, we would expect to observe 177,499 out of the 260,000 returns within 1 standard deviation of the mean. The fact that we actually observe 198,378 observations indicates that the distribution of returns is much more peaked than a normal distribution – more

Table 16.1 Normal distribution vs. actually observed equity returns

	Expected (normal distribution)	Actually observed
0 < std < 1	177,499	198,378
1 < std < 2	70,671	48,223
2 < std < 3	11,128	9,968
3 < std < 4	685	2,321
4 < std < 5	16	679
5 < std < 6	0.2	257
6 < std < 7	0.0005	109
7 < std < 8	0.0000006	43
8 < std < 9	0.000000006	18
9 < std < 10	0.000000000006	5

of the probability mass is found close to the mean. We find just 48,223 versus an expected 70,671 between 1 and 2 standard deviations from the mean, and just 9,968 versus an expected 11,128 between 2 and 3 standard deviations from the mean. This again reveals the peakedness of the distribution, with less than the expected probability mass a few standard deviations from the mean. The leptokurtosis of the distribution is even more evident when looking further away from the mean. The normal distribution would tell us to expect 685 out of the 260,000 observations to be found between 3 and 4 standard deviations from the mean. However the actual number is 2,321. This is 3.4 times more than the normal distribution would have us expect. This feature becomes even more evident when we look between 4 and 5 standard deviations from the mean. Here according to the normal distribution we would expect just 16 observations, but the actual number is 679, almost 42 times the expected value.

The problem of realised returns falling significantly outside of what we would expect becomes ever worse as we look further and further away from the mean. 9Sigma events – events that fall more than nine standard deviations from the mean – practically do not happen, at least not according to the normal distribution, which would have us believe that such an event occurs on average for an individual stock once every 1.4 billion trading days or roughly once every 38 million years. This means that according to such a probability we should expect to see a 9Sigma event $6 \cdot 10^{-13}$ out of the 260,000 trading days examined here. In fact such an event occurred five times during the period from 1992 to 2002 for the 100 randomly chosen stocks we examined. This is almost seven billion times more often than we would have expected. The probability analysis is summarised in the Table 16.2.

Table 16.2 Normal distribution vs. actually observed equity returns

	Expected (normal distribution)	Actually observed
0 < std < 1	68.269%	76.299%
1 < std < 2	27.181%	18.547%
2 < std < 3	4.280%	3.834%
3 < std < 4	0.264%	0.893%
4 < std < 5	0.006%	0.261%
5 < std < 6	0.000%	0.099%
6 < std < 7	0.000%	0.042%
7 < std < 8	0.000%	0.017%
8 < std < 9	0.000%	0.007%
9 < std < 10	0.000%	0.002%

This number juggling may seem rather academic, but the fact remains that events such as the five 9Sigma events discovered here, or the 432 5Sigma events, are the ones that really matter to asset managers. These are the events that can seriously affect performance, and hence these are the ones that warrant attention. We all know and accept that stock returns vary from day to day, and that sometimes we observe returns that are rather large, such as 2Sigma or 3Sigma events. This is trivial and not particularly interesting because it happens as often as it does. But to asset management organisations, what really matters is the ability to accurately describe and assess the consequences on portfolios of events that occur very rarely but that have very large portfolio impacts. Ignoring them just because of their rarity would be like ignoring new and more earthquake-resistant building techniques because earthquakes practically never happen anyway.

Considering Tables 16.1 and 16.2 one might argue that having five 9Sigma events out of 260,000 represents such a remote possibility that it is not worth taking into account. In other words, why make such a big fuss about something that apparently happens more rarely than the appearance of Halley's Comet? Firstly, one should remember that these are actually observed and experienced returns, and that they indicate that returns of a similar magnitude will surely occur again in the future. Secondly, most asset management organisations have a very large number of portfolios with quite a few individual stocks in them. It is thus not unusual for an asset management organisation to have invested in as many as 500 individual stocks or more at any given time. With approximately 260 trading days a year, this makes for 125,000 individual daily stock returns in just one year: so these very volatile returns, which on paper seem only remotely possible, are actually very much part of the everyday life of asset managers. In fact, on an individual stock basis, the average asset manager should expect to experience at least two of these events every year.

One should of course keep in mind that rarely do 9Sigma events for individual stocks cause a catastrophe, since these stocks are usually part of a large portfolio with anywhere from 50 to 100 stocks, providing for considerable diversification. Although one would expect there to be a certain correlation between the returns on the stocks in a given portfolio, a single 9Sigma event in one stock is unlikely to seriously hurt the portfolio, though it might well do a lot of damage. However, the problem of very large returns (positive and negative) persists, even at the portfolio level.

We can approximate portfolios by examining the returns on the 10 MSCI World Sector Indices from 1995/1 to 2002/5 (see Table 16.3). The 10 MSCI World indices are somewhat less prone to exhibiting high-Sigma events. This is because indices that consist of a large number of stocks are inherently less volatile, and so the frequency of observed large events is much lower than

Table 16.3 Distribution of returns for the 10 MSCI World Sectors: 1995/1–2002/5

[illegible]

Table 16.4 Distribution of returns for major stock indices

	Norm. Dist	Dow Jones	S&P500	NASDAQ	FTSE	DAX	TSE	Hang Seng	Average
0 < std < 1	68.23%	75.38%	75.50%	76.07%	74.50%	74.08%	73.77%	78.73%	75.43%
1 < std < 2	27.18%	19.59%	19.16%	18.85%	19.37%	20.73%	21.31%	16.29%	19.33%
2 < std < 3	4.28%	4.14%	4.03%	3.46%	5.08%	4.14%	3.51%	3.56%	3.99%
3 < std < 4	0.26%	0.58%	0.89%	1.05%	0.89%	0.52%	1.05%	0.94%	0.85%
4 < std < 5	0.01%	0.16%	0.26%	0.42%	0.10%	0.42%	0.21%	0.31%	0.27%
5 < std < 6	0.00%	0.10%	0.05%	0.10%	0.05%	0.05%	0.16%	0.05%	0.08%
6 < std < 7	0.00%	0.05%	0.10%	0.00%	0.00%	0.05%	0.00%	0.00%	0.03%
7 < std < 8	0.00%	0.00%	0.00%	0.05%	0.00%	0.00%	0.00%	0.10%	0.02%
8 < std < 9	0.00%	0.05%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.01%
9 < std < 10	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

for individual stocks. Note again that there is more probability mass close to and very far from the mean than would be expected according to the normal distribution. Note also that according to the normal distribution, 5Sigma events and above practically do not occur. Yet, even for indices such as these, which can be regarded as proxies for fairly well-diversified portfolios, we observe several of these and even as much as 7Sigma events.

When applying the same analysis to the indices on the major stock exchanges in the US, the UK, Germany, Japan and Hong Kong, we obtain the results shown in Table 16.4. Note again that there is more probability mass close to and very far from the mean than would be expected according to the normal distribution.

An alternative way of examining the distribution characteristics of returns on stock indices is to illustrate them in a chart. Figure 16.3 illustrates the distribution of returns for the MSCI World Index. It is quite evident from this figure that the returns on the MSCI World from January 1995 to May 2002 exhibit a large degree of kurtosis, but as indicated above the risk manager should focus her attention on the tails of the distribution. The dark grey area in the tail of the distribution on the left-hand side of the chart indicates the probability mass in the normal distribution that makes up the first 1% of the entire probability mass. In other words, it indicates the limit above which we would expect to observe returns only once in every one hundred trading days, if the return were normally distributed.

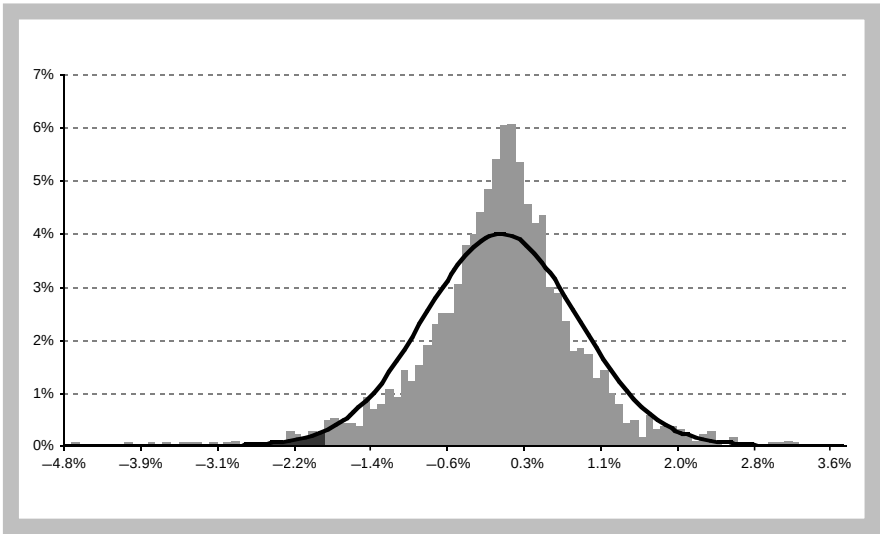


Figure 16.3 Return frequency on the MSCI World, 1995/1–2002/5

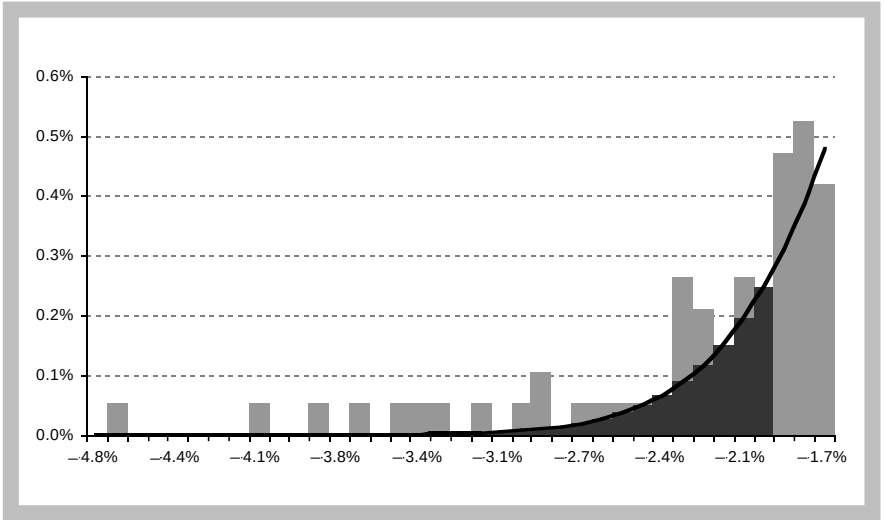


Figure 16.4 Tail return frequency on the MSCI World, 1995/1–2002/5

From Figure 16.4 it is clear that the number of returns that fall outside what would be expected from normally distributed returns is very high. This means that a risk manager who assumes normally distributed returns will seriously underestimate the portfolio's VaR, since extreme events are much more frequent than would then be assumed.

For a portfolio worth \$1,000,000 that is designed to replicate perfectly the MSCI World, this would imply a 1-day VaR of \$23,600 with a 99% level of confidence. Under the assumption of normally distributed returns, the VaR would come out at just \$19,200. This assessment is simply made by observing where the 1% limit is reached for the actually observed distribution and the fitted normal distribution. For the actually observed distribution this entails adding up the whole probability mass (light grey bars) from left to right until 1% is reached. This occurs at -2.36% on the x -axis, indicating that only 1% of actually observed returns have fallen below this limit, implying that the VaR for a \$1,000,000 portfolio at a 1% confidence level is \$23,600. Performing the same analysis for the fitted normal distribution results in a limit of 1.92% which is indicated by the dark grey bars.

From the analysis above it is thus clear that leptokurtotic return distributions will cause the estimate of VaR to be too low if returns are assumed to be normally distributed. It thus appears that applying a different distribution might be of value, particularly if the empirical data for actually observed distributions is scarce or of low quality.

MULTIVARIATE NORMAL DISTRIBUTIONS

An alternative approach to solving the problem of asset returns not following normal distributions is to generate a weighted average of two normal distributions, since this yields a distribution with a kurtosis higher than 3, indicating a leptokurtosis (fat tails). The combination (weighted average) of two normal distributions thus enables us to more accurately describe an observed return distribution, since the outcome is *not* normally distributed, but rather a distribution that has more probability mass very close to and very far from the average.

This approach has been proposed by Hull and White (1997), and the methodology is as follows. An observed return series is analysed to determine its mean and its standard deviation. Two new normal distributions are then defined by two standard deviations, which are scaled from the observed standard deviation to achieve the best fit. These are generated using scaling factors f_1 and f_2 . In addition, a weighting factor is determined, to yield the smallest deviation of the new combined distribution from the actually observed distribution. The fitting of the combined (weighted) distributions to the observed distribution is achieved with a simple OLS (Ordinary Least Square) estimator, whereby the squared deviations are minimised. Denoting q as the weighting factor, the probability function for the combination of the two normal distributions thus appears as follows:

$$g_i(e_i) = q \cdot \frac{1}{2 \cdot f_1 \cdot \sigma_i} \exp\left(-\frac{e_i^2}{2 \cdot f_1^2 \sigma_i^2}\right) + (1 - q) \cdot \frac{1}{2 \cdot f_2 \cdot \sigma_i} \exp\left(-\frac{e_i^2}{2 \cdot f_2^2 \sigma_i^2}\right) \quad [\text{Eq. 16.3}]$$

The density function for the new *bivariate distribution* can thus be expressed as follows:

$$G(e_i) = q \cdot N\left(\frac{e_i}{f_1 \cdot \sigma_i}\right) + (1 - q) \cdot N\left(\frac{e_i}{f_2 \cdot \sigma_i}\right) \quad [\text{Eq. 16.4}]$$

The bivariate distribution is thus a simple weighted average of two normal distributions, each with their standard deviation resulting from a linear scaling of the originally observed distribution that we attempt to fit.

Fitting technique

The fitted bivariate distribution is determined with an OLS-estimator of the squared deviations between the fitted distribution and the actually observed distribution. For each point along the horizontal x -axis we calculate the difference between the originally observed distribution and the fitted distribution. Each deviation is squared, and an algorithm is employed to determine the weighting factor, and the two scaling factors to obtain the best fit – the smallest sum of the squared deviations. Figure 16.5 illustrates the two normal distributions that generate the bivariate distribution. Note that the two normal distributions are quite different from each other. One has a standard deviation of 0.34% (that is, a volatility of 5.6%), and the other has a standard deviation of 0.92% (that is, a volatility of 14.9%). The actually observed standard deviation for the MSCI World return distribution for this particular period was 0.84%, (that is, a volatility of 13.6%). The scaling factors were thus 0.41 and 1.10 respectively. The weighting factor q was 0.32, implying that the bivariate distribution is constructed from 32% of the normal distribution with a standard deviation of 0.34%, and from 68% of the normal distribution with a standard deviation of 0.92%.

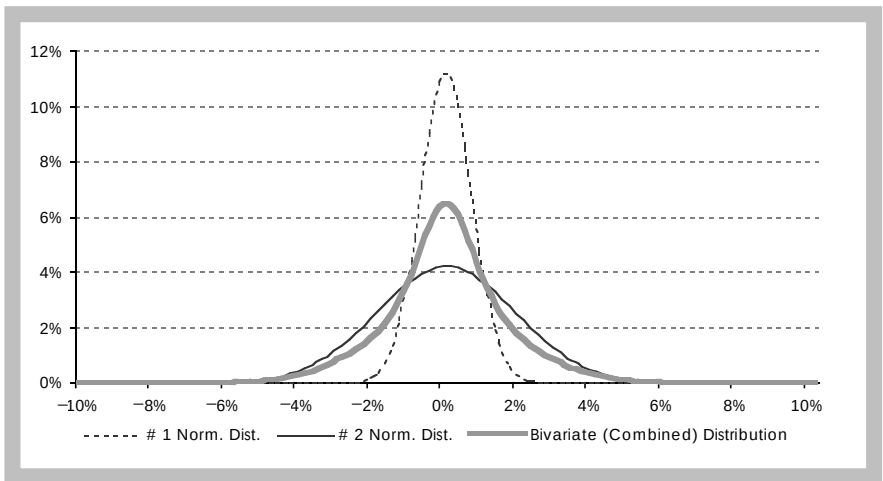


Figure 16.5 Two normal distributions and a bivariate (combined) distribution

Real-world example

We now turn to an example using real return data from the equity markets. Figure 16.6 illustrates the daily returns on the MSCI World from 1995/1 to 2002/5.

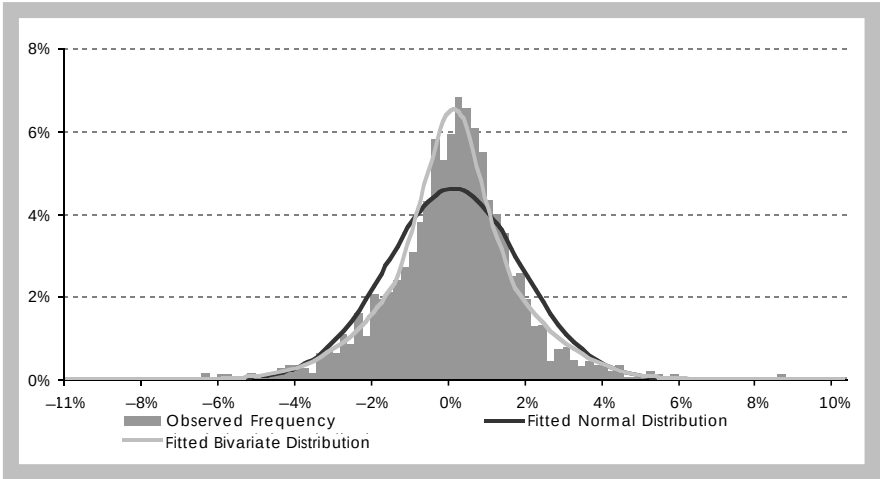


Figure 16.6 Normal and bivariate distributions,
MSCI World 1995/1–2002/5

It is clear that the bivariate (combined) distribution fits the data much better than the simple normal distribution, being both closer to the mean and further away in the tails of the distribution. In particular, note how the simple normal distribution underestimates the probability of observing returns that are close to the mean, overestimates the probability of returns one or two standard deviations from the mean, and finally – but from a risk-management perspective most importantly – how it underestimates the probability of observing returns in the tails of the distribution. The bivariate distribution, on the other hand, gives us a much better fit in all three regions, and in particular it provides a relatively close fit in the tails of the distribution.

We can take a closer look at the left tail of the distribution to get a clearer view of the degree of kurtosis, and also the closeness of the two fitted distributions (Figure 16.7). The distribution is obviously leptokurtotic, with quite a few observations falling very far to the left of the mean. The bivariate distribution represented by the grey line is clearly a better fit, both closer to and further from the mean, than the normal distribution represented by the black line. Importantly, this chart also shows that using the 99% confidence level captures this kurtosis effect in the tail, whereas a 95% confidence level would generate interference in the interpretation of the result, since the two fitted distributions would appear more similar over such a wide return interval. To put this into a VaR perspective, the solid grey bars represent the first 1% of the actually observed distribution, and the solid black bars represent the first 1% of the normal distribution. Remember that the VaR estimate according to the normal distribution was \$19,200, but the VaR estimate of the bivariate distrib-

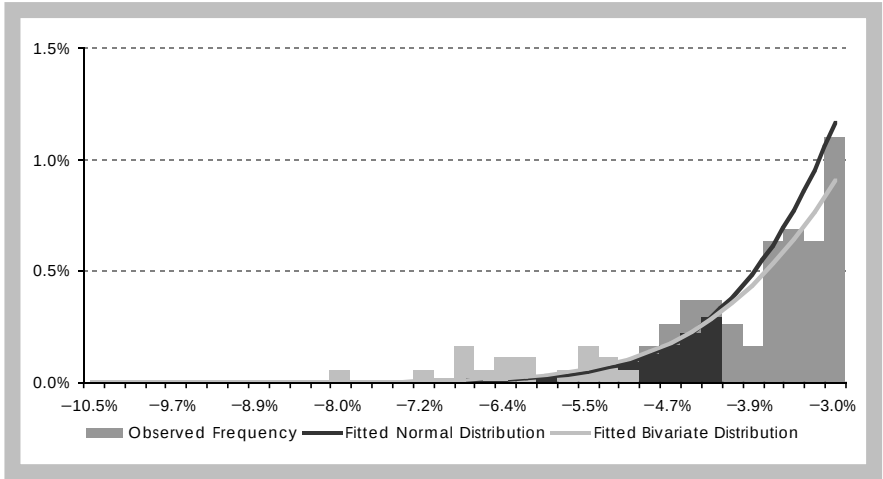


Figure 16.7 Normal and bivariate distributions,
MSCI World 1995/1–2002/5

ution is \$19,900. Again we see that the normal distribution underestimates the actual risk of the index in VaR terms, and although the bivariate distribution offers only a little improvement in this particular case, it is an improved risk estimate nonetheless.

Examining international equity indices

For the MSCI World examined above the OLS fit – that is, the sum of the squared differences – for the bivariate distribution fitted to the actually observed distribution is 0.12, whereas the sum of the squared differences for the fitted normal distribution is 0.26. The bivariate distribution generated by the optimal weighting of two normal distributions, as illustrated in Figure 16.4, thus provides a fit that is roughly twice as good as that of the normal distribution. This method is thus worth considering when we wish to accurately characterise return frequencies and work with them for risk-management purposes.

Table 16.5 illustrates the magnitude of improvement in the OLS fit when using a bivariate distribution as opposed to a simple normal distribution for the 10 MSCI World Sector Indices.

It is clear from the column to the far right-hand side of the table that using the bivariate distribution estimation approach yields much better results than is the case when using simple normal distributions. According to the OLS estimator, the squared differences between the bivariate estimator and the actually

Table 16.5 Bivariate and normal distributions, MSCI Sectors, 1995/1–2002/5

	Standard Deviation	Skew	Kurtosis	OLS – Norm. Dist.	OLS – Bivariate Dist.	Improvement %
Energy	1.11%	–0.04	0.66	0.22	0.13	43%
Materials	0.85%	–0.09	3.50	0.25	0.10	59%
Industrials	0.89%	–0.40	2.89	0.28	0.11	60%
Cons. Disc.	0.93%	–0.18	1.51	0.25	0.11	57%
Cons. Staples	0.78%	–0.18	1.40	0.24	0.12	49%
Health Care	0.99%	–0.13	1.30	0.22	0.10	54%
Financials	1.04%	–0.04	4.42	0.26	0.10	62%
IT	1.79%	0.10	0.90	0.26	0.14	47%
Telecomm.	1.18%	–0.13	1.52	0.27	0.13	51%
Utilities	0.65%	–0.34	1.96	0.24	0.12	49%
Average	1.02%	0.14	2.01	0.25	0.12	53%

Table 16.6 Bivariate and normal distributions, country indices, 1995/1–2002/5

	Standard Deviation	Skew	Kurtosis	OLS – Norm. Dist.	OLS – Bivariate Dist.	Improvement %
Dow Jones	0.98%	−0.51	1.91	0.19	0.11	41%
S&P500	1.12%	−0.20	2.64	0.23	0.11	53%
NASDAQ	1.91%	0.13	3.95	0.31	0.14	54%
UK	1.06%	−0.15	0.17	0.22	0.13	41%
DAX	1.44%	−0.38	1.44	0.20	0.14	32%
TSE	1.29%	0.16	4.61	0.19	0.13	34%
Hang Seng	1.91%	0.45	4.76	0.29	0.11	61%
Average	1.39%	−0.07	2.78	0.23	0.12	45%

observed distribution is an average of 53% smaller than the squared differences between the normal distribution and the actually observed distribution.

We can perform the same type of analysis to the set of country indices already introduced (Table 16.6). Again the bivariate estimator seems to provide a much better fit to the actually observed return distributions. On average we achieve a fit that is approximately 50% better when using the bivariate distribution instead of the normal distribution. However, in some cases we achieve more than a 60% improvement in the fit – a substantial improvement in the accuracy of our estimate. Again we point out that the effect from fitting to a bivariate distribution is likely to be even larger when applied to single stocks, since their returns usually exhibit even greater degrees of leptokurtosis than do indices.

Implications for VaR estimates

Using the example of the MSCI World Index we can now examine how the different techniques presented above compare in their estimate of VaR. Beginning with the simple variance–covariance approach and applying Equation 16.2, the 99% confidence level, one-day VaR estimate for a representative \$1,000,000 MSCI World portfolio is:

$$VaR_p = 2.33 \cdot \$1,000,000 \cdot 0.84\% \cdot \sqrt{1} = \$19,554$$

The interpretation is that in one out of every one hundred trading days we would expect to see a negative return or loss on the portfolio above \$19,554 in just one day. Using the bivariate approach, we can calculate exactly above what level of negative return the cumulative probability exceeds 1%. This negative level of return is then multiplied by \$1,000,000, since this is the level below which we would expect to see just 1% of all observations. This yields a VaR of approximately \$19,800, which is a little higher than the simple variance–covariance estimate. Naturally this is caused by the kurtosis or ‘fat tails’ characteristic of equity returns that is intentionally incorporated into the bivariate distribution. The fitted bivariate distribution is thus kurtotic, and hence has more probability mass in the tails of the distribution, causing the 99% confidence level to be further away from the mean.

The VaR level we obtained above from using the actually observed distribution was approximately \$23,600, which is almost 20% higher than the estimate obtained from the simple variance–covariance approach, assuming normally distributed returns. One might say that even though the portfolio’s

standard deviation indicates a daily standard deviation of 0.84%, equalling a volatility of approximately 13.5%, its actual riskiness is in fact as much as 20% higher than that, since the expected loss is around 20% higher in the tails of the distribution.

Table 16.7 illustrates the results of the four VaR estimators presented in this chapter. It is clear that the actually observed VaR for the ten MSCI World Sector Indices over the period from January 1995 to May 2002 is significantly higher than VaR suggested by the normal distribution. Not surprisingly, the variance-covariance approach (VCV) yields estimates of VaR that are more or less in line with those of the normal distribution. The bivariate approach seems to provide somewhat better results – closer to the actually observed VaR – than the other measures, but it is still underestimating the impact of kurtosis in the return distributions, particularly when levels of both the index volatility and its kurtosis are high.

Table 16.8 summarises a similar analysis of major equity indices. Not surprisingly, the equity indices that exhibit the highest degree of leptokurtosis are also those for which the VaR estimate based on actually observed returns is highest relative to the estimate based on the assumption of normally distributed returns. Note also how high levels of standard deviations and high levels of kurtosis are correlated, with large differences between the VaR estimate based on the normal distribution and the VaR estimate based on either the bivariate or the actually observed distributions.

The implication of this is that if an asset management organisation selects VaR as one of its key risk-management tools, it will be ill advised to simply assume that equity returns are normally distributed. We have already illustrated that the presence of leptokurtosis results in extreme events occurring significantly more often than we would expect under an assumption of normally distributed returns. But here we also demonstrate that the actual losses, when they do occur, are likely to be significantly higher than we would expect.

When interpreting these results, one should keep in mind that the problem of leptokurtosis is often much more pronounced for individual stocks than it is for indices. This underlines the severity of the problem of managing risk on the basis of the assumption of normally distributed returns. Putting aside the issues of the bivariate distribution, the most important result of this analysis is that equity indices can be said to be roughly 20% riskier than they would appear from looking at the calculated standard deviation or volatility, since the expected losses are approximately 20% larger than the normal distribution indicates.

Table 16.7 VaR estimates using three estimation techniques

	Standard Deviation	Kurtosis	Skew	VaR, VCV	VaR, Normal	VaR, Bivariate	VaR, Actual	Normal vs. Actual %
Energy	1.11%	−0.04	0.66	25,765	25,218	25,407	28,140	11%
Materials	0.85%	−0.09	3.50	19,709	19,578	21,903	22,702	4%
Industrials	0.89%	−0.40	2.89	20,692	20,442	22,246	26,225	18%
Cons. Disc.	0.93%	−0.18	1.51	21,722	21,414	21,450	26,928	26%
Cons. Staples	0.78%	−0.18	1.40	18,278	17,855	18,742	21,155	13%
Health Care	0.99%	−0.13	1.30	23,035	22,349	24,247	26,851	11%
Financials	1.04%	−0.04	4.42	24,253	23,846	29,027	30,519	5%
IT	1.79%	0.10	0.90	41,819	41,166	43,383	49,619	14%
Telecomm.	1.18%	−0.13	1.52	27,461	27,253	27,616	33,225	20%
Utilities	0.65%	−0.34	1.96	15,078	14,853	15,754	18,242	16%
Average	1.02%	−0.14	2.01	23,781	23,397	24,978	28,361	14%

Table 16.8 VaR estimates using three estimation techniques

	Standard Deviation	Kurtosis	Skew	VaR, VCV	VaR, Normal	VaR, Bivariate	VaR, Actual	Normal vs. Actual %
Dow Jones	0.98%	-0.51	1.91	22,825	22,332	23,027	27,781	21%
S&P500	1.12%	-0.20	2.64	26,082	25,548	27,403	30,312	11%
NASDAQ	1.91%	0.13	3.95	44,612	44,007	44,322	58,205	31%
UK	1.06%	-0.15	0.17	24,679	24,316	26,426	31,454	19%
DAX	1.44%	-0.38	1.44	33,551	33,057	33,849	40,368	19%
TPX	1.29%	0.16	4.61	30,089	30,161	31,209	40,775	31%
HSI	1.91%	0.45	4.76	44,391	44,128	44,243	51,921	17%
Average	1.39%	-0.07	2.78	32,318	31,936	32,926	40,117	21%

MONTE CARLO SIMULATED VaR

The previous VaR techniques were mostly discussed in a single asset or equity index context. However, when we are dealing with a portfolio consisting of many assets, the accurate calculation of volatility as well as VaR becomes a bit more complicated. As discussed at length in previous chapters, the advantage of using Monte Carlo Simulation techniques is the ability to perform calculations on a large set of data with many different sets of assumptions about the characteristics of the assets involved, without having to explicitly set up a complete model. This is particularly true when asset returns do not follow any well-defined probability distribution. We can model one factor at the time, and have the simulation combine all the characteristics of the model.

Just as we could produce a more accurate asset allocation result with Monte Carlo Simulation, we can obtain a very accurate estimate of VaR by characterising each asset in the portfolio in terms of risk, and then determining the risk characteristics of the entire portfolio. Each asset is assigned an expected return distribution characterised by a mean, a standard deviation (volatility), skew and kurtosis. As before, the manner in which returns behave relative to each other is then modelled in a correlation matrix, and the portfolio's expected return and volatility can then be determined. What we end up with is a portfolio return distribution similar to that illustrated in Figure 16.1, and the portfolio's VaR can then be determined at the 95% confidence level, the 99% confidence level, or any other level that might suit the risk manager.

The crucial part of this operation is naturally the accurate characterisation or forecast of portfolio asset returns. This will ultimately decide the character of the portfolio's expected returns distribution, and thereby the estimate of VaR. As an example we will construct an equal-weighted portfolio from the ten MSCI World Sectors used in the previous chapters. For simplicity we will assume that their respective standard deviations, skew and kurtosis are fairly constant over time, so that we may use them as a forecast (Figure 16.8). Again we emphasise that this is simplistic, but also acceptable since we are not trying to construct a crystal ball but rather an illustrative example.

Note that the return scale is much wider here, since in the MSCI Global Sector Model we operate with annual returns. Again, the dark grey area indicates the level below which we would only expect to observe a return value one in every one hundred years. This Monte Carlo Simulated distribution of returns for the equal weighted portfolio translates into an annual VaR of approximately \$210,000.

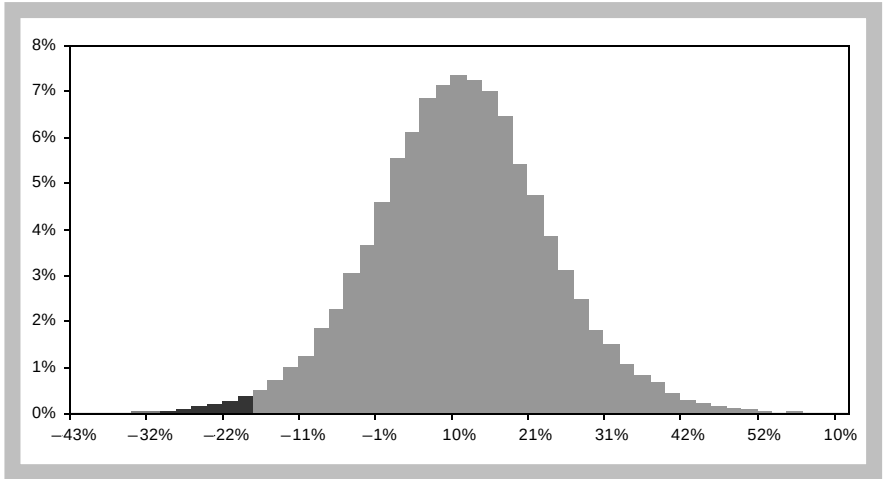


Figure 16.8 VaR of equal-weighted portfolio for the MSCI Global Sector Model

VAR ALONG THE EFFICIENT FRONTIER

Given the emphasis on the accurate characterisation of expected return distributions in both the QRMCSAA and the general risk-management framework, it seems obvious to combine the asset allocation results with VaR. Since each point on the efficient ridge can be described by expected mean, volatility, skew and kurtosis, it is relatively straightforward to apply the VaR methodology to asset allocation. For a given level of certainty we can calculate the 99% VaR estimate, given a portfolio of say \$100,000,000. As each point along the efficient ridge is essentially a weighted average of asset-specific probability distributions, the VaR is not just a simple scaling of the VaR obtained in the minimum variance portfolio. One cannot simply linearly increase the VaR estimate according to the expected volatility, because the nature of volatility of any specific point on the efficient ridge will be different from all others in terms of skew and kurtosis, and thereby ultimately also in terms of the magnitude of the VaR that results. VaR is not the answer to all prayers in relation to monitoring and controlling risk. Just like all other risk-management tools, it is only as good as its inputs, and VaR in particular only works if the probability distributions are accurately estimated or predicted. It serves as an interesting supplement to established tools such as tracking error, but it should not be left to stand alone as the be-all and end-all of risk management.

MARGINAL CONTRIBUTIONS TO VaR

Extending the framework of VaR

Returning briefly to the variance–covariance framework for calculating VaR, we can integrate this with the risk analysis carried out in Chapter 15 on the Sector Risk Model. Since VaR in the variance–covariance framework is essentially just a volatility number multiplied by the portfolio’s value, the level of desired statistical significance, and the square root of the time horizon over which we would like to estimate VaR, it follows that we can use this approach to calculate sector-specific contributions to VaR as well as marginal sector contributions to VaR. Using the data provided in Table 15.1 we can now calculate sector VaR numbers for the MSCI Global Sector portfolio.

From Equation 15.12 recall that portfolio volatility is:

$$\sigma_p = \vec{w}_p^T \cdot \frac{\Sigma \cdot \vec{w}_p}{\sigma_p} = \vec{w}_p^T \cdot MSCTR$$

From this follows that:

$$\begin{aligned} VaR_p &= 2.33 \cdot V_p \cdot \sigma_p \cdot \sqrt{T} \\ &= 2.33 \cdot V_p \cdot \vec{w}_p^T \cdot MSCTR \cdot \sqrt{T} \end{aligned} \quad [\text{Eq. 16.5}]$$

We can thus easily calculate the portfolio’s sector contributions to VaR. In Table 16.9 we present these values for a portfolio based on the Global Sector Model and worth \$1,000,000. The largest contributors to portfolio VaR are the financials and information technology sectors, because of their large proportion and high MSCTR respectively. The smallest contributors are the energy, consumer staples and utilities sectors. Since the portfolio worth \$1,000,000 has a volatility of 13.40%, we can calculate portfolio VaR at the 99% confidence level using Equation 15.2:

$$VaR_p = 2.33 \cdot \$1,000,000 \cdot 13.40\% \cdot \sqrt{1} = \$312,265$$

which happens to be the same as the sum of component sector VaRs. Note that the sector contributions to total portfolio VaR are proportional to the relative marginal sector contributions to risk presented in Chapter 15, since they basically use the same input except for the portfolio value and confidence level which enters into Equation 16.5.

Table 16.9 Sector contributions to total portfolio VaR

	Portfolio weights	MSCTR	Component VaR
Energy	5.8%	0.09%	11,960
Materials	8.0%	0.11%	21,168
Industrials	10.6%	0.16%	38,862
Cons. Disc.	12.7%	0.16%	46,897
Cons. Staples	8.9%	0.05%	11,090
Health Care	11.1%	0.06%	15,871
Financials	18.3%	0.15%	64,539
IT	10.0%	0.26%	60,963
Telecomm.	6.9%	0.18%	29,196
Utilities	7.7%	0.07%	11,720
Total VaR			312,265

Marginal contributions To VaR

Using Equations 15.1 and 15.4 we can easily determine the marginal sector contributions to portfolio VaR or MSCTVaR, since we know from Chapter 15 that the marginal sector contribution to total risk is given by:

$$MSCTR = \frac{\partial \sigma_p}{\partial \vec{w}_p^T}$$

From this it follows that the marginal sector contribution to total portfolio VaR is:

$$\begin{aligned}
 MSCTVaR &= 2.33 \cdot V_p \cdot \frac{\partial \sigma_p}{\partial \vec{w}_p^T} \cdot \overline{T} \\
 &= 2.33 \cdot V_p \cdot \frac{\partial \overline{\vec{w}_p^T \cdot \Sigma \cdot \vec{w}_p}}{\partial \vec{w}_p^T} \cdot \overline{T}
 \end{aligned}$$

$$= 2.33 \cdot V_p \cdot \begin{bmatrix} \frac{\partial \sigma_p}{\partial w_{p,1}} \\ \bullet \\ \bullet \\ \bullet \\ \frac{\partial \sigma_p}{\partial w_{p,n}} \\ \bullet \\ \bullet \\ \bullet \\ \frac{\partial \sigma_p}{\partial w_{p,N}} \end{bmatrix} \cdot \overline{T}$$

[Eq. 16.6]

Table 16.10 presents the results of Equation 16.6 when applied to the global sector portfolio worth \$1,000,000. This indicates that by far the largest increase in portfolio VaR as a result of a sector weight increase of 1 percentage point is expected from the information technology sector. However, the telecommunications sector also features prominently. Similarly to the analysis carried out in Chapter 15 on marginal sector contributions to risk, we find that the largest and most volatile sectors contribute most to VaR. This is not particularly surprising, but it is useful to see the exact portfolio VaR breakdown so that this may be taken into consideration if the risk profile of the portfolio needs to be changed.

Table 16.10 Marginal sector contributions to VaR

	Portfolio weights	MSCTR	MSCTVaR
Energy	5.8%	0.09%	2,062
Materials	8.0%	0.11%	2,646
Industrials	10.6%	0.16%	3,666
Cons. Disc.	12.7%	0.16%	3,693
Cons. Staples	8.9%	0.05%	1,246
Health Care	11.1%	0.06%	1,430
Financials	18.3%	0.15%	3,527
IT	10.0%	0.26%	6,096
Telecomm.	6.9%	0.18%	4,231
Utilities	7.7%	0.07%	1,522

EXTREME VALUE THEORY (EVT)

Extending the framework of VaR

The essence of *Extreme Value Theory* is the quantitative exploration and characterisation of extreme events, in the sense of observed asset returns that fall markedly out of place relative to what we would normally expect. A useful yardstick would be to concentrate on 3Sigma events or above – that is, asset returns that lie three or more standard deviation from the observed mean.

The first step in analysing any phenomenon is to decide on the most appropriate model and yardsticks. An ordinary bathroom scale is not going to be of much use in weighing a sumo wrestler. Nor would it be much good for weighing a newborn infant. Phenomena that deviate substantially from the norm require special measurement techniques. This idea applies to the economy as well. Economic turning points that come along once every 50 or 100 years can only be studied in terms of the laws of history. Conversely, analysing a three-year business cycle requires nothing more than data for the previous 10–15 years. In short, the choice of model and yardstick makes all the difference in diagnosing how the economy is doing and where it is headed. Unless an observer is prepared to use different yardsticks for different situations, his observations are unlikely to be correct.

We can also usefully examine the extreme events in the return series. This area of research was originally employed in the analysis of insurance claims. Insurance companies, obviously, are in the business of assessing the likelihood of extreme events, and so naturally such a company would like to be able to produce a reliable forecast for such extreme events. The basic idea is to illustrate how returns are distributed, comparing that to a hypothesised extreme value distribution such as an exponential distribution, and then using the pattern to attempt to predict the likelihood of extreme events, prediction of which is difficult because of insufficient empirical evidence. There is a plethora of literature on this subject, but we can analyse it using a simple approach. We simply plot daily sector returns against the logarithm of the likelihood (here equalling the observed frequency) of extreme events.

The logarithmic chart in Figure 16.9 is for the information technology sector. The dots represent the likelihood – based on historical observations – of observing a return of 0% to 9%. According to the empirical evidence available here, the likelihood of observing a return of between 4% and 5% would be around 1%. The grey line indicates return frequencies, assuming normally distributed returns. If we were also to superimpose a straight line over the dots, and thus to assume an exponential relationship between the size of the return and its likelihood, we would obtain some indication of the likelihood of a return of say 9%, even if that has never occurred. We explain this in greater detail below.

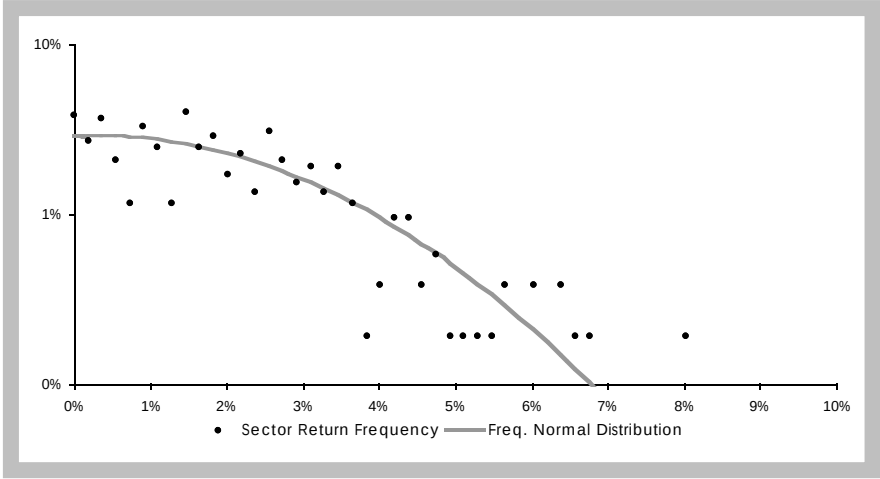


Figure 16.9 Extreme value theory and the information technology sector

The basic idea behind this type of chart originates from the empirical observation that the logarithm of the frequency tends to plot in a straight line against the size of the returns. This in turn suggests that one can predict with a reasonable degree of certainty the likelihood of observing a return of a particular size. For example, based on the data available here for the information technology sector, the likelihood of observing a return (positive or negative) of 6% in one day is around 0.21% or once every 476 trading days, according to the normal distribution; but empirically the likelihood is actually as high as 0.38% or once every 263 trading days, almost twice as high. Since the return distribution is leptokurtotic – that is, it exhibits ‘fat tails’ – this underestimation of extreme events becomes more severe for higher return values. According to the normal distribution, the likelihood of observing a return of 8% in one day is 0.02% or once every 5000 trading days, but empirically the actual likelihood seems to be as high as 0.19% or once every 526 trading days – more than nine and a half times more often. These conclusions are similar to those presented in earlier chapters, but the real strength of Extreme Value Theory is its predictive capabilities. Accepting the notion of linearity between the logarithm of return frequencies and daily returns enables us to estimate the likelihood of observing a daily return of any size, without actually ever having observed it.

Let us turn to the telecommunications sector for another example (Figure 16.10). According to the EVT chart for the telecommunications sector, as estimated over the period from January 1995 to May 2002, we can assess the likelihood of a daily return of say 8%. If returns were normally distributed, the

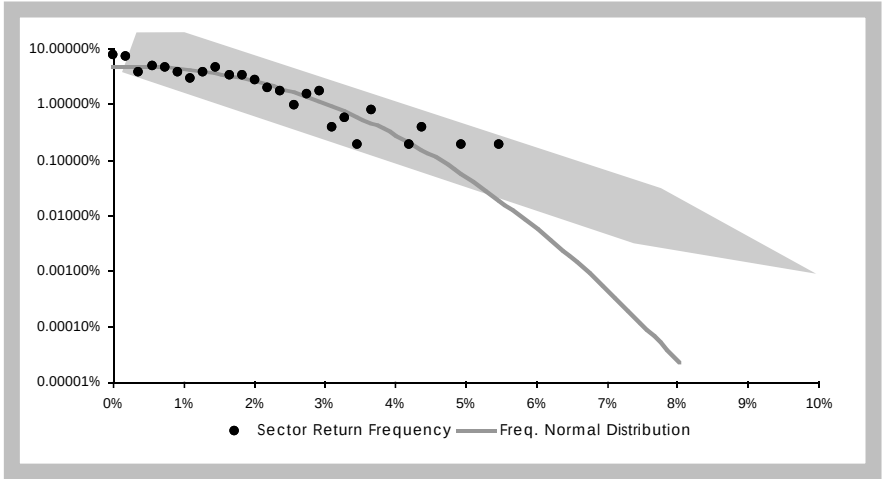


Figure 16.10 Extreme value theory and the telecommunications sector

likelihood of observing such a return would be approximately 0.00002%, or once every 4.6 million trading days. Applying the EVT estimate, we end up with a likelihood in the neighbourhood of 0.01% or once every 10,000 trading days. This amounts to a 460,000-fold difference in likelihoods between the two estimates. Even a 6% daily return would be seriously underestimated, since the normal distribution would suggest a likelihood of 0.01% or once every 10,000 trading days, whereas the EVT estimator would suggest a likelihood of around 0.06% or roughly once every 1700 trading days. This amounts to a 6-fold difference in likelihoods between the two estimates.

Putting this into a practical asset management perspective, it will mean quite a difference for the portfolio's performance – either in absolute or in relative terms – if extreme events are underestimated 6-fold and if very extreme events are underestimated 460,000-fold. Therefore EVT techniques such as that presented here can potentially provide asset and risk managers with valuable insights into the nature and likelihood of events that might seriously impact on the portfolio. EVT charts for all MSCI World Sectors can be found in the chapter appendix.

SUMMARY

VaR has gained in influence as a risk-management tool over the past decade. Its strengths lie in its flexibility in being able to accurately describe risk as prob-

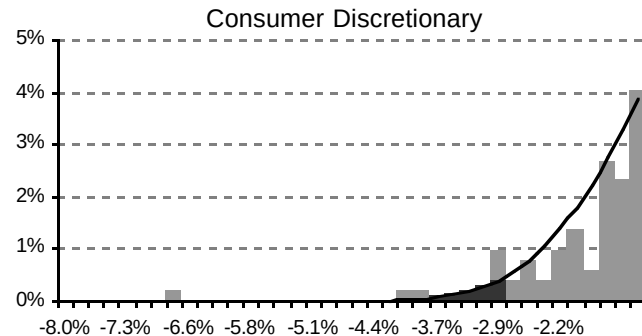
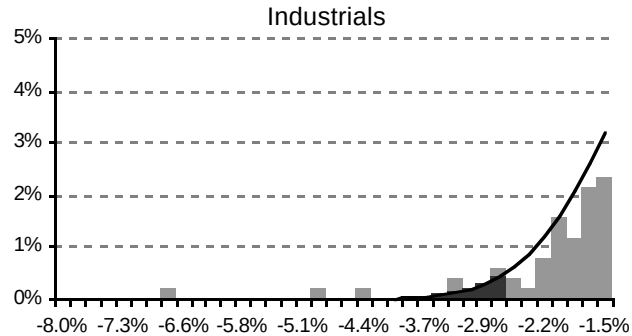
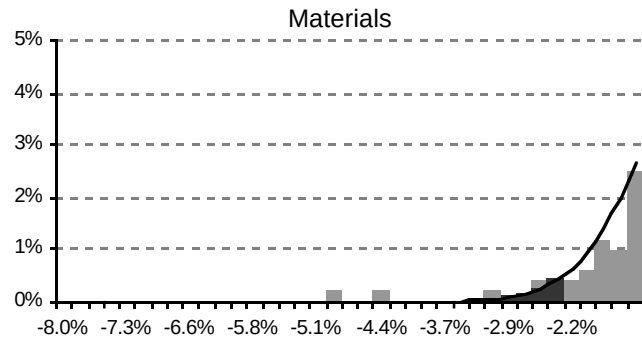
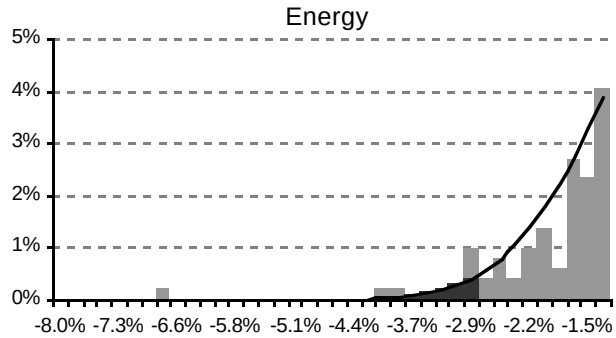
ability distributions, and not just single numbers such as volatility. VaR enables the asset manager to assess the probability of incurring a loss of a given size over a given time horizon. Its inputs are either arbitrarily chosen probability distributions for individual assets – stocks, bonds or even highly non-linear instruments such as derivatives – or it can use historical numbers for assessing the likely impact on portfolio performance during severe market conditions.

We have presented variance–covariance VaR, historical VaR, VaR using a bivariate return distribution estimator and Monte Carlo Simulation-based VaR, and we have compared the outcomes when these methods are applied to a number of sector and country indices. We have also examined the decomposition of VaR using the framework established in Chapter 15 for general portfolio risk analysis.

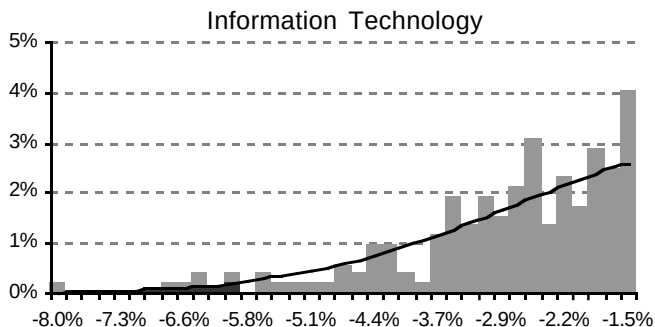
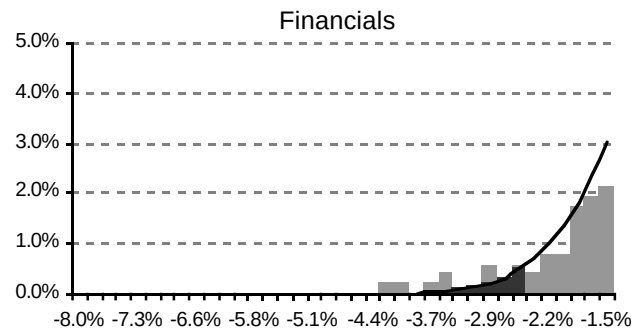
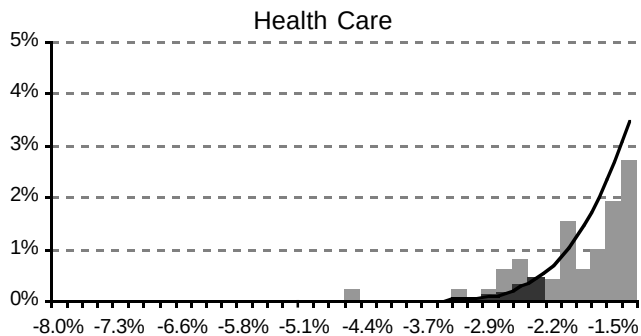
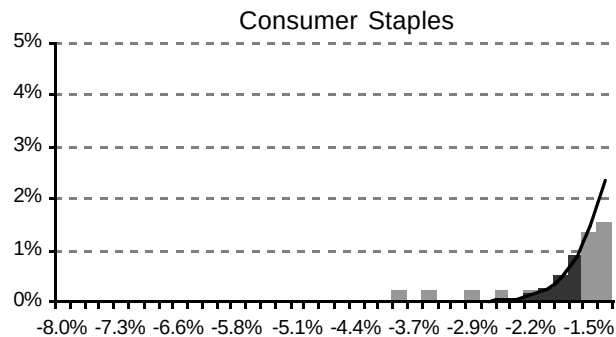
There are several reasons why information about component VaR is valuable. Firstly, when presented with a VaR number, the asset manager needs to know where the bulk of portfolio VaR stems from. It thus reveals localised areas of disproportionate portfolio risk which the manager may want to address. Secondly, if portfolio VaR appears too high for the asset manager's liking, she must obtain knowledge as to what action she should take in order to bring down the VaR. This argument is identical to that applied in the discussion about attribution of portfolio risk or volatility to individual sectors.

We introduced a simple but illuminating version of Extreme Value Theory (EVT), which allows us to assess the likelihood of extreme events, whether they have actually occurred in the past or not. This allows asset and risk managers to assess the risk of extreme events on the portfolio. Ideally, risk managers should spend more time looking at the likelihood and probable impact on the portfolio of extreme events rather than simply measuring and reporting day-to-day portfolio risk.

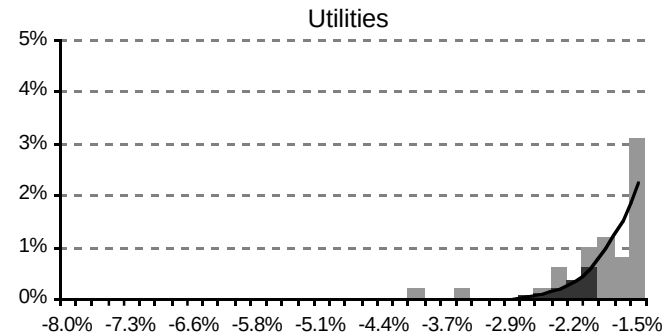
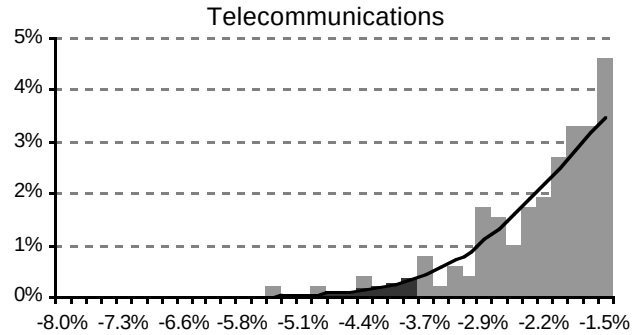
APPENDIX 16A: SECTOR TAIL RETURN FREQUENCIES



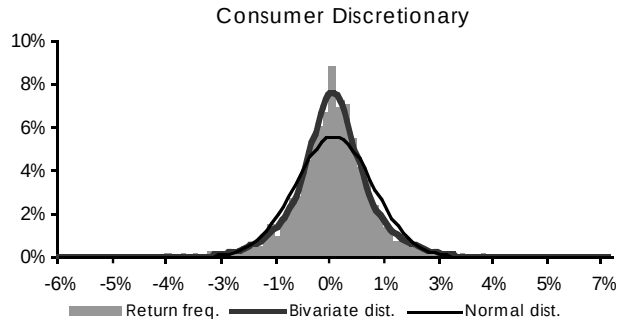
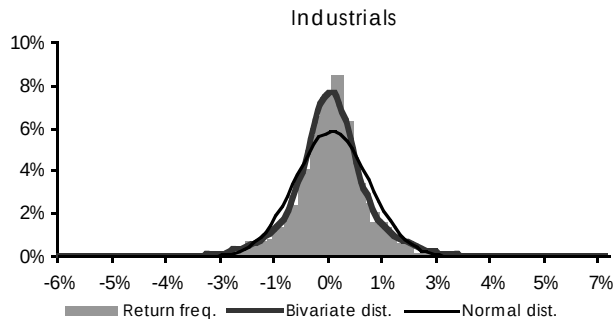
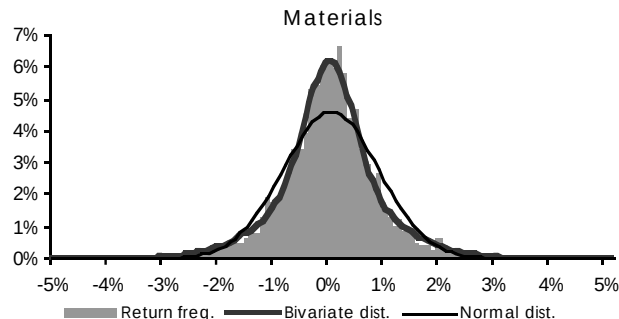
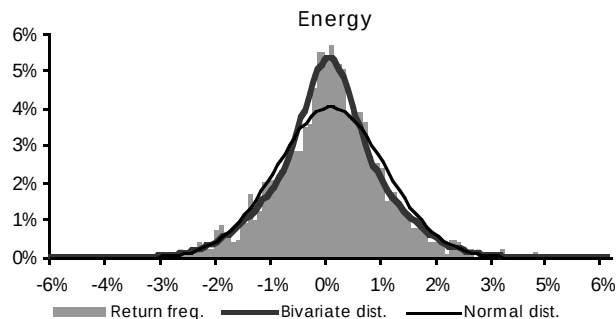
APPENDIX 16A (continued)



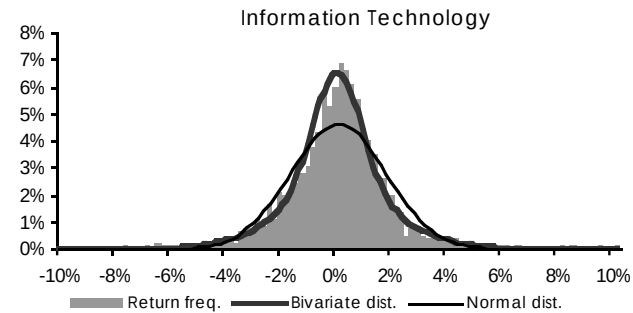
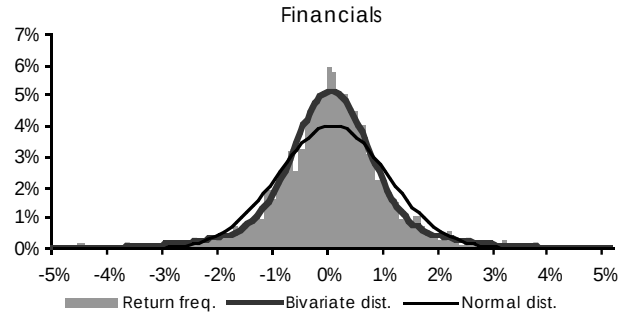
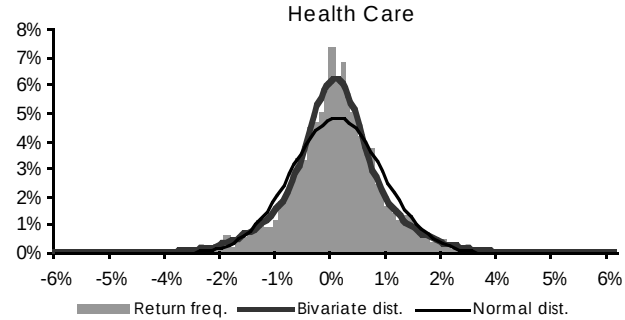
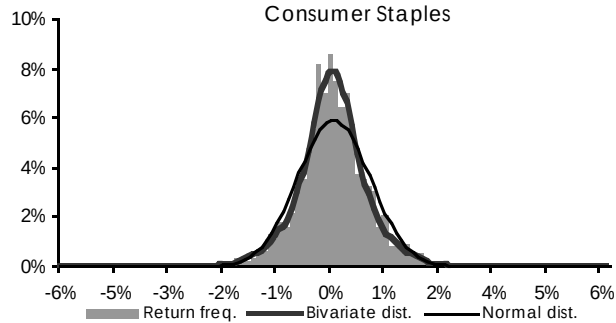
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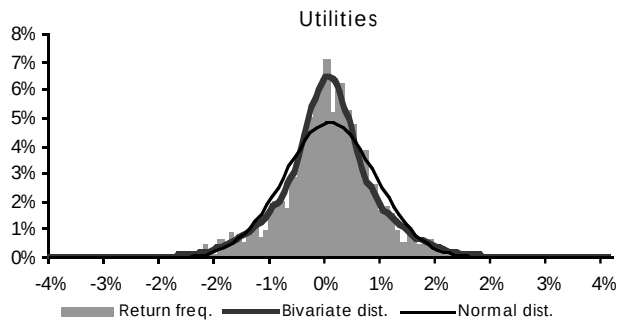
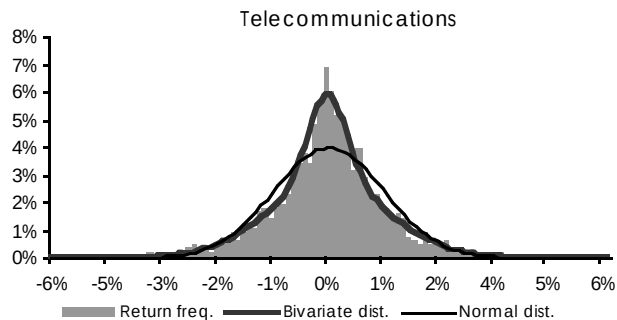
APPENDIX 16B: SECTOR MULTIVARIATE NORMAL DISTRIBUTION



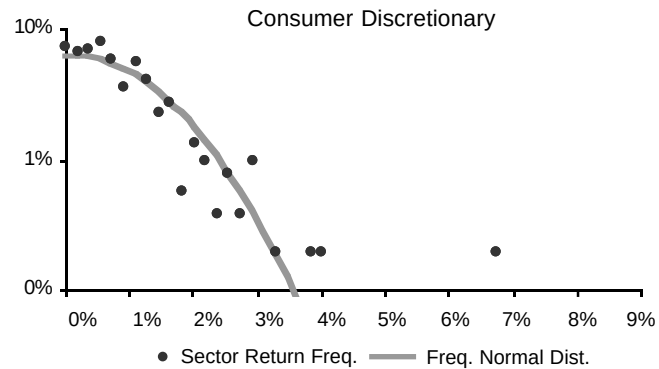
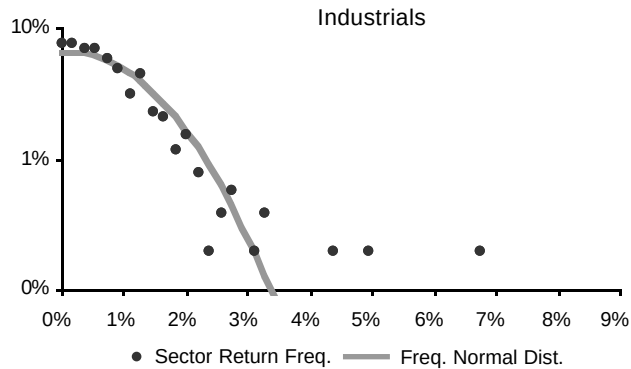
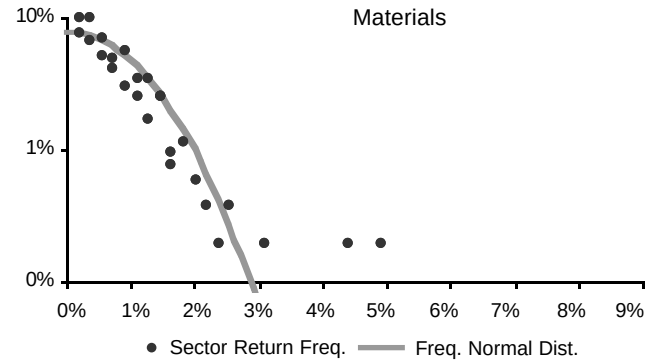
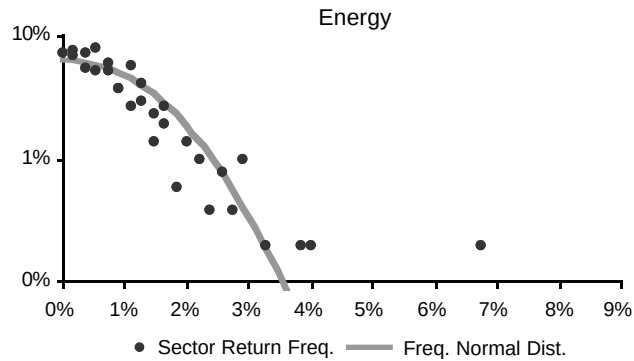
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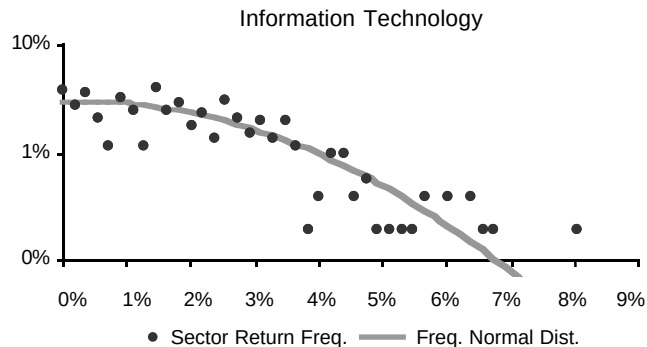
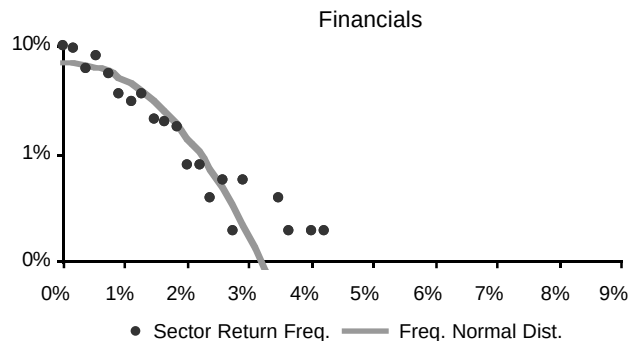
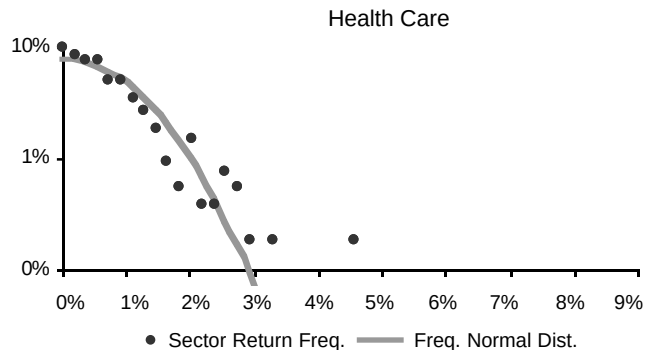
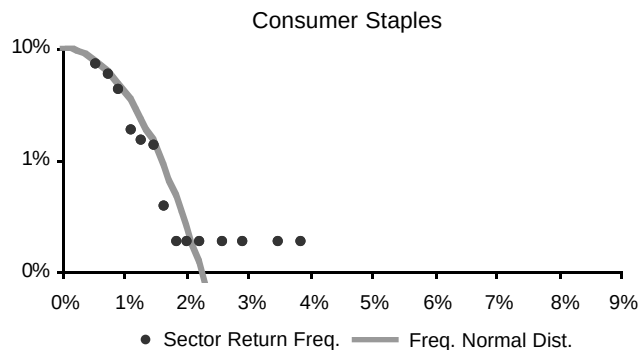
APPENDIX 16B (continued)



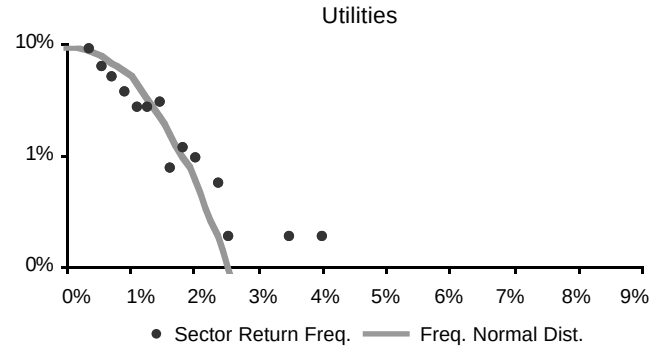
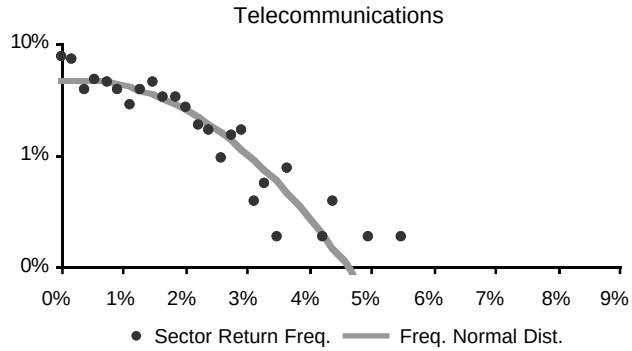
APPENDIX 16C: SECTOR EXTREME VALUE CHARTS



APPENDIX 16C (continued)



APPENDIX 16C (continued)



NOTATION

In order to be able to clearly present the contents of this book, we have attempted to be as consistent as possible in the use of notation. The notation below applies to all chapters of this book. Generally, plain text indicates scalars, and **bold** text indicates vectors or matrices.

CALCULUS

$\cdot$	Indicates standard multiplication
$\bullet$	Indicates vector or matrix multiplication
$\forall$	Indicates 'for all observations' within a set of data
$\sum_{n=0}^N f(n)$	The sum of the values produced by a function f by inputting all n from $n=0$ to $n=N$
$\prod_n^N f(n)$	The product of the values produced by a function f by inputting all n from $n=0$ to $n=N$

RETURN, RISK AND EXPOSURES

$R_{A,n}$	The total return on an asset A at time period n
$r_{A,n}^0$	The rate of return on an asset A at time period n

r_f	The risk-free rate of return
$r_{A,n}$	The rate of excess return on an asset A at time period n
$r_{A,n}^{\times}$	The expected rate of excess return on an asset A at time period n
i	The non-factor related or asset specific (residual) return on asset i
$f_{k,t}$	The return on risk factor k at time t
$i_{i,t}$	The residual return on asset i at time t
σ_i	The volatility of asset i
w_i	The portfolio weight of asset A
$\vec{w}$	A vector of portfolio asset weights
$\vec{w}^T$	The transpose of the vector of portfolio asset weights
$\vec{w}_{PA}$	A vector of portfolio asset active weights
$\vec{w}_{PA}^T$	The transpose of the vector of portfolio asset active weights
$\vec{r}$	A vector of portfolio returns
σ	The diagonal matrix of portfolio volatilities
σ^T	The transpose of the diagonal matrix of portfolio volatilities
β_A	The beta of asset A
$\vec{\beta}$	The vector of sector (or portfolio asset) betas
ψ_p	Portfolio active risk (tracking error)
ω_p	Portfolio residual risk

PORTFOLIO CORRELATION AND COVARIANCE

$\rho_{A,B}$	The correlation coefficient between asset A and asset B
P	The portfolio's Correlation Matrix
Σ	The portfolio's Variance–Covariance Matrix
Σ^{-1}	The inverse of the portfolio's Variance–Covariance Matrix
$\vec{e}$	The unit vector of 1s.
$\vec{e}^T$	The transpose of the unit vector of 1s.
I	The Identity Matrix of diagonal 1s.

GLOSSARY OF TERMS

Active Portfolio Management	The pursuit of returns above that of a predefined benchmark, by actively managing a portfolio.
Active Return	The portfolio's return above the return of the benchmark.
Active Risk	The volatility of the portfolio's active returns.
Active Weight	The portfolio holding or weight (in percentage terms) exceeding the holding or weight of the portfolio's benchmark.
Agent	The legal entity (person or company) that is employed by a Principal to perform a task.
AIMR	Association for Investment Management and Research.
Alpha	The statistical outperformance of a portfolio relative to a benchmark. Determined as the intercept, when regressing observed benchmark returns onto observed portfolio returns.
Arbitrage	The ability to make a profit on trading an asset while incurring no risk.
Arbitrage Pricing Theory (APT)	A multi-factor version of the CAPM, which includes a number of explanatory variables.
Asset Allocation	The intentional distribution of a portfolio across the available range of investable assets.
Asset Manager	A person or a firm that has as its business the investment and management of client assets.
Autoregressive Conditional Heteroscedasticity	A statistical forecasting technique that builds on the assumption that future returns are to some extent a function of past returns.

Bayesian Priors	Values or numbers that are applied to an estimation technique in order to serve as an anchor for the end result. Usually reflects some <i>a priori</i> knowledge about the variable being estimated.
Behavioural Finance	The science of explaining the behaviour of financial agents, based on human nature.
Benchmark	An index against which portfolio return and risk are measured, providing the basis for calculating Active Weights, Active Return and Active Risk.
Benchmark Related Risk	The proportion of Active Risk or Tracking Error related to the market (or benchmark).
Benchmark Risk	The volatility (annualised standard deviation) of the return on the benchmark.
Bet	See Active Weight.
Beta	Mathematical measure of the co-variation of the return on an asset with the return on the market (or benchmark). Determined by dividing the covariance of the asset and the market with the variance of the market.
Bivariate Return Distribution	Non-normal return distribution that is a weighted sum of two normal distributions.
CAC40	French Equity Index.
Capital Asset Pricing Model (CAPM)	A single factor model for explaining asset returns as a function of the return on the market.
Characteristic Line	The empirical relationship between the return on an asset and the return on a portfolio or a benchmark.
Compounding Effect	The arithmetic procedure of calculating interest on payable interest.
Compound Interest	Interest calculated for each period on interest accrued in previous periods.
Consensus Efficient Frontier	The efficient frontier that is implied by the consensus expectations about return, risk and correlations for a portfolio of assets.
Constraints	Limitations imposed on a quantitative optimisation in order to restrict the resultant asset allocation to a desired sub-space of results.
Continuously Compounded Interest	Interest calculated continuously on interest accrued in previous periods.
Correlation	The degree to which two variables move in similar direction over time.
Correlation Coefficient	The coefficient measuring the degree of correlation between two variables. By definition the correlation coefficient ranges between -1 and +1.

Covariance	The degree and magnitude to which two variables move together over time.
DAX30	German Equity Index.
Debt to Equity (D/E)	The ratios of debt to equity on a company's balance sheet. Applied along with other measures to measure the financial quality of a company.
Discount	The amount less that an investor is willing to pay for an asset which is believed to be inferior relative to an appropriate peer group.
Discount Factor	The factor used to discount future payments back to the present, to adjust for the opportunity cost of receiving the payment later as opposed to immediately.
Discounted Free Cash Flow	The cash flow available in a company after operating, investing and financial activities. Used to assess the value of a company or other cash generating (portfolios of) assets.
Discounting	The mathematical procedure of determining the present value of future payments using the discount factor. Also, the inclusion of the effects of relevant events into the assessment of the value of a stock, on the part of investors.
Diversification	The property of a portfolio of assets exhibiting less risk than the (weighted) average of asset volatilities. A result of correlation coefficients being less than 1.
Dividend	The payment by a company to its shareholders as a form of return on their investment.
Dividend Discount Model (DDM)	A series of valuation models using the discounted future stream of dividends paid, to determine the value of a company.
Dividend Yield	The ratio of dividends to current share price. Comparable to the yield on a bond.
Dow Jones Industrial Average	US Equity Index.
Earnings Yield	The ratio of earnings to current share price. Comparable to the yield on a bond.
EBIT	Earnings Before Interest and Tax.
Efficient Frontier	The two-dimensional representation of the return/risk space in which efficient portfolios are plotted.
Efficient Portfolios	Portfolios or asset allocations that generate the highest attainable return for a given level of risk, or the lowest attainable level of risk for the highest level of return.
Efficient Ridge	The three-dimensional representation of the return/risk/probability space, in which efficient portfolios are plotted.

Equal Weighting	Weighting scheme applied to the calculation of standard deviation. Also, the action of ensuring that all types of assets in a portfolio are of equal value.
Equity Risk Premium	The required additional return on equity over the risk-free return (approximated by the return on a long-term government bond). Reflect the premium the investor requires to hold a risky asset such as equity.
EVA	Economic Value Added. A corporate management tool for assessing the efficiency or profitability of resources employed.
EV/EBITDA	The ratio of Enterprise Value (market capitalisation less debt) to Earnings Before Interest, Tax, Depreciation and Amortisation. Applied along with other measures to measure the value of a company.
Exponential Weighting	The process of calculating a weighted average while placing an exponentially increasing weight on observations the more recent they are. Exponentially Weighted Moving Average (EWMA) volatility is an example.
Factor Exposures	The factor model exposure of a company's stocks to a specific factor.
Factor Models	Quantitative multiple regression models that attempt to explain return and risk characteristics of individual assets based on each asset's exposure to a number of predefined global or universal factors.
Factors	The explanatory variables that enter into factor models. Examples are interest rates, GDP growth, sector classification, balance sheet leverage and so on.
Fund Manager	See Portfolio Manager.
Gearing	When a fund or portfolio has borrowed to invest more than the value of its assets.
Geometric Compounding	See Compounding Effect.
GIPS	Global Investment and Performance Standards.
Hang Seng	Hong Kong Equity Index.
Historical VaR Simulation	Risk assessment technique that employs past return distributions to infer future portfolio risk.
Identity Matrix	A matrix that has 1s along the diagonal and 0s in all other cells.
Implied Growth	The short-term growth implied by a share price as a function of its current level, an assumed long term growth rate and an assumed discount rate.
Implied Volatility	The volatility implied by the price of an option.
Information Coefficient	A measure of portfolio manager skill, this is the correlation between forecast and realised returns.

Information Ratio	The ratio of active return to active risk, or the ratio of residual return to residual risk.
Investable Universe	The universe of stocks or other assets from which the asset manager is not restricted when investing.
Investment Manager	See Asset Manager.
LaGrange Estimator	A technique for determining a model optimum by use of partial derivatives.
Leverage	The degree to which a company has borrowed to finance its operational activities.
Long Position	A net bought position in an asset.
Marginal Contribution to Risk	The net increase in either total portfolio or active risk, as a result of a marginal change in the (active) weight of a portfolio asset.
Market Efficiency	The property of assets reflecting completely all the information available about them.
Maximum Variance Portfolio	The portfolio which exhibits the greatest attainable degree of risk.
Mean	Average of a sample.
Mean-Variance Efficiency	See Efficient Portfolios.
Mean-Variance Optimisation	See Quantitative Portfolio Optimisation.
Minimum Variance Portfolio	The portfolio which exhibits the lowest attainable degree of risk.
Modern Portfolio Theory	Mathematical framework for describing and assessing the return and risk of a portfolio of assets, using only returns, volatilities and correlations.
Modified Dietz	Return calculation method designed to account for inter-temporal errors that may occur if an asset is traded between two time measurements.
MSCI	Morgan Stanley Capital International.
Multivariate Density Estimation	Mathematical forecasting technique, whereby a forecast is made based on the similarity of the present state with previous states, across a range of arbitrarily chosen variables.
NASDAQ Composite	US Equity Index.
Net Present Value	The present value of a discounted future stream of cash flows.
OLS	Ordinary Least Squared. A statistical technique for assessing the fit of a model with a set of observed data.
Opportunity Cost	The unrealised cost forgone by not participating in an investment or similar.
Optimal Portfolios	See Portfolio Efficiency.

Orthogonal	Geometrically, the diagonal. Implies zero correlation between two variables.
Portfolio Efficiency	See Efficient Portfolios.
Portfolio Manager	A person employed by an asset manager to manage individual portfolios of assets, often relative to a benchmark.
Portfolio Optimisation	See Quantitative Portfolio Optimisation.
Portfolio Risk	The volatility (annualised standard deviation) of the return on the portfolio.
Portfolio Tracking Error	The annualised standard deviation of active returns on a portfolio relative to a benchmark.
Premium	The additional amount an investor is willing to pay for an asset which is believed to be superior relative to an appropriate peer group.
Price to Book Ratio	The price of a stock relative to its book value.
Price to Cash Ratio	The price of a stock relative to the company's free cash flow per share.
Price to Earnings Ratio	The price of a stock relative to the company's earnings per share.
Principal	The legal entity (person or company) that employs an Agent to perform a task.
Principal Component Analysis (PCA)	A statistical technique for decomposing and characterising correlation between factors.
Quantitative Portfolio Optimisation	The process of maximising the expected return on a portfolio relative to the expected volatility, using asset returns, volatilities and correlations.
R Squared	A statistical measure of the explanatory power of a variable in a regression analysis. Conveys how much of the observed behaviour of the variable under examination can be explained by the explanatory variable.
Regression	A statistical data analysis technique which optimally fits a model based on the squared differences between data points and the model fitted points.
Residual Return	Portfolio return independent of the benchmark. Defined as the return relative to Beta times the benchmark return.
Residual Risk	The volatility (annualised standard deviation) of the residual return.
Reverse Optimisation	A mathematical technique whereby the implied asset or sector returns in a portfolio can be computed using volatilities and correlations.

Risk-Free Asset	An asset assumed to have zero risk attached to it, typically a long term government bond, although a market implied risk-free rate can be determined.
Risk-Free Rate	The return on the risk-free asset.
Risk Management	The process of actively managing the risk of a portfolio along all its dimensions.
Risk Measurement	The process of attaining the level and nature of risk of an asset or in a portfolio.
Risk Spectrum	The spectrum from zero to maximum volatility, in an absolute asset allocation framework. The spectrum from zero to maximum tracking error, in a relative asset allocation framework.
Security Market Line	The line through the plot of asset returns against their covariances or Betas with respect to the benchmark.
Sharpe Ratio	The ratio of return to risk (volatility).
Short Position	A net sold position in an asset.
Short Selling	Selling assets that are borrowed and not owned.
Standard&Poors	A credit rating agency.
Standard&Poors 500	US Equity Index.
Standard Deviation	A statistical measure of variation around a sample mean. The square root of variance.
Stein Estimators	Techniques applying Bayesian priors in order to shrink observed returns towards a mean determined a priori.
Strategic Asset Allocation	The long-term asset allocation of an asset manager.
Stress Testing	The procedure of subjecting a portfolio to a number of severe scenarios, possibly reflecting previously observed extreme events, in order to gauge its stability.
Tactical Asset Allocation	The short-term asset allocation of an asset manager.
Target Return Level	The desired level of return set prior to a quantitative portfolio optimisation.
Target Volatility Level	The desired level of risk or volatility set prior to a quantitative portfolio optimisation.
Time Weighted Return	See Modified Dietz.
Topix	Japanese Equity Index.
Tracking Error	See Active Risk.
Variance	A statistical measure of variation around a sample mean. The square of volatility.
Volatility	The annualised standard deviation of asset or portfolio returns.

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